The Quasi-Riemann Hypothesis
A Zero-Free Half-Plane $\Re s > 7/8$

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Abstract

We prove that all finite-order Hecke L-functions over $\mathbb{Q}(\sqrt{-3})$ and all Dirichlet L-functions are zero-free in the half-plane $\Re s > 7/8$, with the principal pole at $s = 1$ allowed. In particular, the Riemann zeta function is zero-free in this half-plane, proving the quasi-Riemann hypothesis. The proof compares cubic-theta reflection with a Poisson expansion of the same character sum, whose principal term contains the reciprocal of the target L-function. A balanced comparison first gives the boundary 11/12. Inserting prime factors to cancel a local Euler contribution, together with two character-moment estimates, then gives 7/8.

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1 Introduction

Let $F = \mathbb{Q}(\sqrt{-3})$, let $\mathcal{O}$ be its ring of integers, and write $N\mathfrak{a} = |\mathcal{O}/\mathfrak{a}|$ for the norm of a nonzero integral ideal. A finite-order Hecke character η modulo a nonzero integral ideal $\mathfrak{f}$ is a character of the ray class group modulo $\mathfrak{f}$, extended by zero to ideals not coprime to $\mathfrak{f}$. This agrees with the usual finite-order idelic definition: the complex place contributes no nontrivial finite-order continuous character [21, Chapter VI, Section 1]. For $\Re s > 1$, put

$$L_F(s,\eta) = \sum_{\mathfrak{a}\neq 0} \frac{\eta(\mathfrak{a})}{(N\mathfrak{a})^s} = \prod_{\mathfrak{p}} \bigl(1 - \eta(\mathfrak{p})(N\mathfrak{p})^{-s}\bigr)^{-1}.$$

The same notation denotes its meromorphic continuation. For a Dirichlet character χ modulo q, extended by zero on nonunits, we likewise write

$$L(s,\chi) = \sum_{n\geq 1} \frac{\chi(n)}{n^s} \qquad (\Re s > 1)$$

and then continue meromorphically; the character modulo 1 gives the Riemann zeta function $\zeta(s)$. The problem here is uniform exclusion: to find a constant $\sigma_0 < 1$, independent of the character, its conductor, and the height, such that these functions have no zeros in $\Re s > \sigma_0$.

For ζ alone, the existence of a fixed $\sigma_0 < 1$ for which $\zeta(s) \neq 0$ in $\Re s > \sigma_0$ is called the quasi-Riemann hypothesis [3, Section 1]. It asks for one gap valid at every height. The corresponding uniform formulation over all Dirichlet characters appears in [8].

Theorem 1.1. *Every finite-order Hecke L-function over $F = \mathbb{Q}(\sqrt{-3})$ has no zero in $\Re s > 7/8$. The same holds for every Dirichlet L-function, including $\zeta(s)$. A pole at $s = 1$ for a principal character is allowed.*

In particular, Theorem 1.1 resolves the quasi-Riemann hypothesis affirmatively: the supremum of the real parts of the nontrivial zeros of ζ, namely its zeros in $0 < \Re s < 1$, is at most $7/8$. The boundary $\Re s = 7/8$ is not included, while any exceptional real zero in $(7/8, 1)$ is excluded.

Least nonresidues and square roots over prime fields

The fixed Dirichlet zero-free half-plane gives a polylogarithmic bound for the least quadratic nonresidue and a deterministic square-root algorithm over prime fields.

Corollary 1.2. *There are absolute constants $A, C > 0$ such that, for every odd prime p, the least positive quadratic nonresidue $n(p)$ satisfies*

$$n(p) \leq C(\log p)^A.$$

In particular, $n(p) \ll_\delta p^\delta$ for every $\delta > 0$. Given an odd prime p and $a \in \mathbb{F}_p$ in binary representation, there is a deterministic algorithm, with running time polynomial in $\log p$, that returns a square root of a or reports that none exists.

The subpower bound in Corollary 1.2 proves Vinogradov's least quadratic nonresidue conjecture [25, Conjecture 1.1].

Proof. By Theorem 1.1 and the functional equation, the nontrivial zeros of every primitive Dirichlet L-function lie in $1/8 \leq \Re s \leq 7/8$. They therefore lie in the strictly larger strip $1/16 < \Re s < 15/16$. This supplies the weak-GRH hypothesis of Bhargava, Ivanyos, Mittal, and Saxena [2, Conjecture 6.3 and Theorem 6.7] with $\epsilon = 7/16$. Their bound for the least quadratic nonresidue gives the asserted inequality, for example with $A = 32$. The assertion for each $\delta > 0$ follows because every fixed power of $\log p$ is $O_\delta(p^\delta)$.

For the algorithm, first handle $a = 0$ and test a nonzero a by Euler's criterion. If a is a square, scan $2, 3, \ldots$ until a quadratic nonresidue is found, testing each candidate by its Legendre symbol. The bound just proved makes this a polynomial-time deterministic search; its stopping rule does not require knowing C. The resulting nonresidue is the input to the Tonelli–Shanks algorithm [9, Section 2.9, Algorithm 3 and Lemma 2.9.5]. All remaining steps are deterministic and polynomial in $\log p$. Writing $p - 1 = 2^e u$ with u odd only requires removing factors of two, not factoring u. □

Background and methods

The relation between zeros and prime distribution provides the background for uniform exclusion. Character L-series separate residue classes in Dirichlet's proof of the infinitude of primes in reduced arithmetic progressions [6]. Riemann related the zeros of ζ to prime distribution and formulated the critical-line conjecture [24]; nonvanishing on $\Re s = 1$ gave the prime number theorem through the work of Hadamard and de la Vallée Poussin [14, 27]. Hecke developed the corresponding character zeta functions over number fields [17].

Classical zero-free regions for Hecke L-functions approach the line $\Re s = 1$ as the conductor or height increases and can allow one simple real exceptional zero; a precise form for abelian extensions is given by [26]. Such regions do not give uniform exclusion in a fixed half-plane. Zero-density estimates address a different question: they bound the number of zeros to the right of a given vertical line. For example, Guth and Maynard's large-value estimates improve zero-density bounds for ζ [13], but these zero-density bounds do not exclude every zero.

Kubota's metaplectic theory and Patterson's cubic theta series provide the automorphic setting [19, 23]. Proposition 4.1 derives the reflection used here from the unconditional explicit cusp expansions recorded by Dunn and Radziwiłł [7], retaining the characters' zero-on-nonunit restrictions.

Quadratic and cubic large-sieve estimates control the reflected character sums: we use the Hecke-family estimate of Goldmakher and Louvel [12] and Heath-Brown's cubic estimate [16]. For the sextic characters, Lemma 8.1 carries out the higher-order norm recursion of Blomer, Goldmakher, and Louvel [4] with paired residue characters in the primary-generator convention; Gao and Zhao also record an Eisenstein-integer form [11]. The later moment recursions are related to Heath-Brown's quadratic method [15]. The ordinary Hecke functional equation [10] and prime counting in a fixed ray class [26] supply the analytic reflection and prime normalizations.

The planar additive large sieve used in the first stage is classical; compare Huxley's multivariable and number-field inequality [18] and the Poisson proof in [1]. We include a direct Eisenstein-lattice proof to record the normalization needed for the reduced-fraction expansion. David, de Faveri, Dunn, and Stucky combine Patterson's coefficients, the cubic large sieve, and mollified moments to prove nonvanishing at $s = 1/2$ for a positive proportion of a cubic Hecke family [5]. The zero detector below uses the classical truncated-inverse mechanism; compare [20], and for a higher-order-character density application [4].

The unconditional first-moment asymptotic for cubic Gauss sums is treated in the separate manuscript [22].

The proof has two stages, each comparing two representations of a completed cubic-theta sum but using a different normalized sum in the continuation argument. Part I proves the corresponding 11/12 assertion in Theorem 3.1, already obtaining a fixed zero-free half-plane for both families. Part II starts from that conclusion and introduces prime compensation, asymmetric scales, and two additional moment estimates. The contribution developed here is the construction of these compatible reflected and Poisson comparisons, including the principal residues, with positive exponent margins chosen independently of the target character.

Proof overview

The analytic obstacle is uniformity over the entire character family. The quantity β_* in (2.1) records the rightmost possible real part of a zero in the primitive Hecke family, with $1/2$ as a lower bound. If β_* exceeds a proposed boundary σ_0, we seek a holomorphic continuation of every target reciprocal across one common positive distance to the left of β_*. Imprimitive functions have the same possible zeros in $\Re s > 0$, since the finitely omitted Euler factors are nonzero there.

For each target η, we form a smoothed character sum in a large real scale Z. Cubic-theta reflection bounds this sum directly. Poisson summation gives a second expression whose principal term is a Mellin integral containing $1/L_F(s,\eta)$, multiplied by a holomorphic factor bounded away from zero and with finitely many Euler factors deleted. Power savings in the direct bound and in the remaining Poisson terms give the continuation by Proposition 2.1. The two parts use different normalized sums, with affine powers

$$C_{\mathrm{I}}(s) = s - \frac{2}{3}, \qquad C_{\mathrm{II}}(s) = s - \frac{11}{16}.$$

The arithmetic construction retains every cubic factor in a theta coefficient's index and evaluates its character factors on that whole index. This completion supplies the reflection identity in Proposition 4.1. The support and zero-extension conventions are given with the construction in Sections 3.1, 4.1, and 5.1.

1. *A first fixed half-plane.* Part I uses equal averaging scales $Z^{1/2}$. Reflection separates the sum into a theta factor and an additive factor. A quadratic large sieve bounds the first mean square; separation of reduced fractions in $\mathbb{C}/\mathcal{O}$ gives a planar large-sieve bound for the second. Cauchy–Schwarz combines them in Proposition 5.3.

   The nonprincipal terms in the Poisson expansion involve finite-order Hecke twists of the target. The zero detector of Section 7 turns the relevant zeros into large values of truncated reciprocal polynomials. The sextic large sieve then bounds how many terms can contribute at that size; Proposition 8.2 gives the resulting count. The principal term is evaluated separately through its residues and Euler identity. Section 10 combines these estimates in the continuation criterion at $11/12$, proving Theorem 3.1.

2. *The seven-eighths refinement.* Part II starts from the first half-plane, inserts selected prime factors to cancel an unwanted local Euler contribution, and uses unequal averaging scales. Reflection and an additive mean-square estimate again give the direct bound, now in Section 14. The principal residue and scalar normalization yield the affine power C_{II} above.

   The sharper estimate for the other terms uses both polynomials supplied by the zero detector. These are a truncated reciprocal polynomial, whose coefficients include the Möbius function, and a character polynomial without that factor. Their normalizations are given in Section 3.1. The detector gives simultaneous large values with one common character and twist height. Using moment bounds for both gives the refined count in Section 18.

   The two required moment estimates are Lemmas 16.1 and 17.2. The first controls an inverse polynomial with disjoint prime factors; the second controls a product of two plain polynomials with permitted prime factors. Each uses two finite Poisson transformations in a separate induction. The inverse induction ends with a reflected mean-square bound, while ordinary Hecke reflection reduces the lengths in the plain induction. The character exclusions and exceptional terms are treated with the estimates, including Section 17.9.

   Section 19 combines the refined counts with the compensated expansion, local errors, contour tails, and principal residue. After normalization, this verifies the second comparison in Proposition 2.1 at $7/8$.

3. *Transfer to Dirichlet characters.* Composing a Dirichlet character χ with the ideal norm gives a Hecke character over F. Away from finitely many Euler factors, quadratic factorization expresses its L-function as the product attached to χ and to $\chi\chi_{-3}$, where χ_{-3} is the quadratic character of $F/\mathbb{Q}$. The omitted factors are nonzero in $\Re s > 0$, and the principal pole is handled separately. Proposition 10.3 therefore transfers each Hecke half-plane to all Dirichlet characters.

2 Mellin continuation and a zero-free half-plane

Both parts of the proof use the same analytic principle. A character sum is bounded directly and is also compared with a Mellin integral containing the reciprocal of a target L-function. A power saving in both comparisons then continues that reciprocal across the rightmost possible zeros. We prove this principle for a variable boundary, so that it can be applied at $11/12$ and at $7/8$ without repeating the argument.

Recall that $F = \mathbb{Q}(\sqrt{-3})$. It suffices to consider primitive target characters. Indeed, a character induced from a primitive character η has L-function differing from $L_F(s,\eta)$ by finitely many factors $1 - \eta(\mathfrak{p})N\mathfrak{p}^{-s}$, all nonzero for $\Re s > 0$. Define

$$\beta_* = \sup\left(\{1/2\} \cup \left\{\Re\rho : \begin{array}{l} 1/2 \le \Re\rho \le 1,\ L_F(\rho,\eta) = 0 \\ \text{for some primitive finite-order Hecke character } \eta \end{array}\right\}\right). \tag{2.1}$$

Poles are not included. Absolute convergence of the Euler product excludes zeros in $\Re s > 1$, so $1/2 \le \beta_* \le 1$.

For a finite set S of prime ideals, a superscript S means that the corresponding Euler factors have been deleted:

$$L_F^S(s,\eta) = L_F(s,\eta)\prod_{\mathfrak{p}\in S}(1 - \eta(\mathfrak{p})N\mathfrak{p}^{-s}).$$

The local value is zero at a ramified prime. Each displayed factor is nonzero for $\Re s > 0$, so deleting finitely many factors neither creates nor removes a zero there.

Proposition 2.1 (A Mellin continuation criterion). *Let β_* be defined by (2.1). Fix $\sigma_0 \in (1/2, 1)$ and suppose $\Delta_0 = \beta_* - \sigma_0 > 0$. Let $C(s) = s + c$, where $c \in \mathbb{R}$ is fixed. Suppose there exist ω, σ such that*

$$0 < \omega < \Delta_0, \qquad \sigma > 0$$

and, for every primitive finite-order Hecke character η, there exist a finite set S, a holomorphic function H_η on $\Re s > \sigma_0$, and a function $J_\eta(Z)$ defined for all sufficiently large real Z, such that

$$\sup_{\Re s > \sigma_0} |H_\eta(s) - 1| \le \frac{1}{2}. \tag{2.2}$$

For $Z > 0$, define

$$f_\eta(Z) = \frac{1}{2\pi i}\int_{\Re s = 2} Z^{C(s)} e^{(s-5/6)^2}\frac{H_\eta(s)}{L_F^S(s,\eta)}\,ds. \tag{2.3}$$

Assume that, as $Z \to \infty$,

$$|J_\eta(Z)| \ll_\eta Z^{C(\sigma_0)+\omega}, \tag{2.4}$$

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\sigma}. \tag{2.5}$$

The implied constants, lower thresholds, excluded set, and function H_η may depend on η. Then these assumptions contradict $\beta_* > \sigma_0$.

Proof. Endpoint bounds. Put

$$\epsilon_* := \min\{\Delta_0 - \omega, \sigma\} > 0.$$

Because C has slope one, the triangle inequality gives

$$|f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-\epsilon_*} \qquad (Z \geq Z_{0,\eta}). \tag{2.6}$$

Moreover $\epsilon_* \leq \Delta_0 - \omega < \Delta_0$, so $\beta_* - \epsilon_* > \sigma_0$.

We also need control as $Z \downarrow 0$. By (2.2), H_η is bounded on $\Re s > \sigma_0$. The reciprocal Euler product is absolutely and uniformly bounded on $\Re s \geq 2$. On every fixed strip $2 \leq \Re s \leq B$, the Gaussian in (2.3) is $O_B(e^{-(\Im s)^2})$. Cauchy's theorem on rectangles therefore moves the contour to any fixed $B > 2$, with horizontal integrals tending to zero. Hence

$$|f_\eta(Z)| \ll_{\eta,B} Z^{B+c} \qquad (0 < Z \leq 1).$$

Since B is arbitrary, f_η has arbitrarily rapid power decay at zero.

The Mellin transform. This bound and (2.6) show that

$$F_\eta(s) := \int_0^\infty f_\eta(Z) Z^{-C(s)} \frac{dZ}{Z}$$

converges locally uniformly on $\Re s > \beta_* - \epsilon_*$. On each compact subset, choose B larger than all occurring real parts for the integral near zero, and use (2.6) near infinity. Thus F_η is holomorphic on that half-plane.

Identification on the initial line. Fourier inversion identifies this Mellin transform on $\Re s = 2$. Set

$$A_\eta(s) = e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s,\eta)} \qquad (\Re s > 1).$$

The function $A_\eta(2+it)$ is continuous and integrable in t. Writing $Z = e^u$, (2.3) becomes

$$e^{-(2+c)u} f_\eta(e^u) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{itu} A_\eta(2+it)\, dt.$$

The left side is integrable in u by the two endpoint estimates. Ordinary Fourier inversion therefore gives $F_\eta(2+it) = A_\eta(2+it)$ for every real t. Both sides are holomorphic on $\Re s > 1$, so the identity theorem gives

$$F_\eta(s) = e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s,\eta)} \qquad (\Re s > 1).$$

Continuation and contradiction. The bound (2.2) implies $|H_\eta(s)| \geq 1/2$ on $\Re s > \sigma_0$. Consequently

$$e^{-(s-5/6)^2} \frac{F_\eta(s)}{H_\eta(s)} \qquad (\Re s > \beta_* - \epsilon_*)$$

is a holomorphic continuation of $1/L_F^S(s,\eta)$. The number ϵ_* was chosen independently of η. By the definition of β_*, some target has a zero ρ with $\Re\rho > \beta_* - \epsilon_*$. Its reciprocal has a pole at ρ, and the deleted factors are nonzero there. This contradicts the continuation. □

Proposition 2.1 compares the direct bound at exponent $C(\sigma_0) + \omega$ with the approximation error at exponent $C(\beta_*) - \sigma$. A row contribution may reach the direct-bound scale while retaining a positive power saving relative to the hypothetical rightmost zero. Part I uses

$$\sigma_0 = \frac{11}{12}, \qquad C(s) = s - \frac{2}{3}, \qquad C(\sigma_0) = \frac{1}{4},$$

whereas Part II uses

$$\sigma_0 = \frac{7}{8}, \qquad C(s) = s - \frac{11}{16}, \qquad C(\sigma_0) = \frac{3}{16}.$$

Only the positive power margins must be uniform in the target. Fixed-character constants, excluded sets, and sufficiently large lower thresholds may depend on it in both applications.

Part I
The eleven-twelfths zero-free half-plane

3 The first half-plane and preliminary estimates

We first prove a fixed zero-free half-plane using the basic completed cubic-theta sum. This isolates the mechanism that excludes zeros before the additional estimates needed for the sharper boundary are introduced.

Theorem 3.1 (The 11/12 half-plane). *Every finite-order Hecke L-function over $F = \mathbb{Q}(\sqrt{-3})$ has no zero in $\Re s > 11/12$. The same holds for every Dirichlet L-function, including $\zeta(s)$. A pole at $s = 1$ for a principal character is allowed.*

For the Hecke assertion, suppose for contradiction that

$$\Delta_1 := \beta_* - \frac{11}{12} > 0. \tag{3.1}$$

Proposition 5.3 gives a direct bound for the completed character sum. Proposition 10.2 compares its normalized form with the Mellin integral in (2.3), while the Euler-product estimate in Lemma 6.2 and (9.2) supply the holomorphic, nonzero correction. Together these estimates verify Proposition 2.1 with $C(s) = s - 2/3$, $\omega = \Delta_1/2$, and $\sigma = 1/4800$, contradicting (3.1). The full deduction appears at the end of Section 10, where Proposition 10.3 also transfers the Hecke conclusion to Dirichlet L-functions.

The fixed family in (2.1) is essential even when the desired consequence concerns ζ: the Poisson representation introduces finite-order Hecke twists of the target. Proving the family-wide assertion also supplies the precise input used at the beginning of Part II. The estimates supporting this first half-plane will also be used in Part II. We first fix the arithmetic conventions and prove the Gauss and reciprocity identities. The subsequent analytic tools separate smooth norm profiles and control a Hecke L-function on a disk known to be zero-free.

3.1 Arithmetic notation and coefficient classes

Recall the field $F = \mathbb{Q}(\sqrt{-3})$. Let $\omega = e^{2\pi i/3}$, and put

$$\mathcal{O} = \mathbb{Z}[\omega], \qquad \lambda = \sqrt{-3} = \omega - \omega^2 = 1 + 2\omega.$$

For an element a write $q_a = |a|^2 = N_{F/\mathbb{Q}}(a)$, and for $a \neq 0$ write $\alpha(a) = a/|a|$. The same notation $q_{\mathfrak{a}}$ denotes the norm of a nonzero ideal. The ring $\mathcal{O}$ is Euclidean for this norm: a point of $\mathbb{C}$ is at distance at most $1/\sqrt{3} < 1$ from the triangular lattice $\mathcal{O}$, which gives Euclidean division. In particular every ideal is principal. The six units are $\{\pm 1, \pm\omega, \pm\omega^2\}$, and their images are the six units of $\mathcal{O}/3\mathcal{O}$. Consequently every ideal coprime to 3 has a unique generator congruent to 1 (mod 3); we call this generator primary. Products of primary generators are primary.

In the character-polynomial estimates, a row is an outer index for a character polynomial, while a column is an ideal index, or a tuple of ideal indices, summed inside it. A label is an auxiliary index distinguishing parts of the family. A label may be averaged or held fixed in the current sum; holding it fixed there does not make it part of the arithmetic datum.

For the arithmetic datum in any given invocation, fix a finite set S of prime ideals containing the primes above 6 and the prime supports of the defining moduli of every finite-order character presentation in that datum and of the fixed finite ray group used to present them. For a character

presentation, its defining-modulus support consists exactly of the primes at which it is extended by zero, including any redundant primes of an imprimitive presentation.

Write $\mathcal{A}$ for the arithmetic datum: the complete zero-extended presentations of the fixed finite characters, their defining moduli, the finite ray group used to present them, the excluded set S, and any arithmetic normalization. These data are independent of Z, the current rows, and the averaged labels; they may depend on a previously fixed target. In particular every such character has modulus one at every prime outside S.

We call primes outside S good, and call an ideal good if all its prime divisors lie outside S. Unless a different support is specified, ideal sums exclude S and use primary generators. We write "sf" for squarefree and μ for the ideal Möbius function. Element rows, in contrast, may have arbitrary prime powers and unit factors.

For $z \in F$ define

$$e(z) = \exp\bigl(2\pi i \operatorname{Tr}_{F/\mathbb{Q}}(z/\lambda)\bigr).$$

Its extension to $\mathbb{C}$ is $e(z) = \exp(4\pi i \operatorname{Im} z/\sqrt{3})$. The measure $d\mu(z) = (2/\sqrt{3})\,dx\,dy$ makes $\mathcal{O}$ self-dual for the pairing $(z, y) \mapsto e(zy)$. Indeed, writing $y = u + v\omega$ with real u, v, the conditions $e(y) = e(\omega y) = 1$ say $v, u - v \in \mathbb{Z}$, hence $y \in \mathcal{O}$; and the covolume of $\mathcal{O}$ for $d\mu$ is one. Lattice point counting in a disk, followed by division by the six units, gives the unrestricted ideal count

$$\#\{\mathfrak{a} : q_{\mathfrak{a}} \le H\} = \frac{\pi H}{3\sqrt{3}} + O(\sqrt{H} + 1) \qquad (H \ge 0).$$

For example, the error follows by covering the boundary of the disk with $O(\sqrt{H} + 1)$ fixed fundamental parallelograms.

If $\mathfrak{p} \notin S$ is prime, its residue field has order $P = q_{\mathfrak{p}} \equiv 1 \pmod 6$: the six roots of unity remain distinct in that field. Define the sextic symbol $\chi_{\mathfrak{p}}(a) = (a/\mathfrak{p})_6$ to be the unique sixth root of unity satisfying

$$\chi_{\mathfrak{p}}(a) \equiv a^{(P-1)/6} \pmod{\mathfrak{p}} \quad (\mathfrak{p} \nmid a), \qquad \chi_{\mathfrak{p}}(a) = 0 \quad (\mathfrak{p} \mid a).$$

For a primary $c = \prod p^{v_p(c)}$ outside S, define $\chi_c(a) = \prod_{p|c} \chi_p(a)^{v_p(c)}$, and put $\chi_1(a) = 1$ for every a. For every integer j, including $j = 0$ and negative j, the notation $\chi_c(a)^j$ means its usual power when $(a, c) = 1$ and means zero otherwise. Thus an exponent divisible by six is the function $1_{(a,c)=1}$, not the constant function one. The square χ_c^2 is the cubic symbol. For squarefree c set

$$\gamma_j(c) = q_c^{-1/2} \sum_{v \bmod c} \chi_c(v)^j e(v/c), \qquad \gamma_j(1) = 1.$$

Lemma 3.2 (Fixed numerators give ray characters). *Fix $0 \ne a \in \mathcal{O}$. At a prime ideal $\mathfrak{p} \nmid 6a$, let $(a/\mathfrak{p})_6$ be the unique sixth root of unity congruent to $a^{(q_{\mathfrak{p}}-1)/6} \pmod{\mathfrak{p}}$; at a prime outside S this is $\chi_{\mathfrak{p}}(a)$. The extension $F(a^{1/6})/F$ is finite abelian and unramified outside the primes dividing $6a$, and*

$$\frac{\operatorname{Frob}_{\mathfrak{p}}(a^{1/6})}{a^{1/6}} = (a/\mathfrak{p})_6,$$

where $\operatorname{Frob}_{\mathfrak{p}}$ is arithmetic Frobenius. The multiplicative extension of this symbol to ideals coprime to $6a$ is a finite-order ray character whose conductor is supported on the primes dividing $6a$. The identification is on ideals coprime to $6a$.

In particular, choose one generator $\pi_{\mathfrak{p}}$ for each $\mathfrak{p} \in S$. On primary ideals outside S, the characters

$$A \longmapsto \chi_A\left(u \prod_{\mathfrak{p} \in S} \pi_{\mathfrak{p}}^{v_{\mathfrak{p}}}\right), \qquad u \in \mathcal{O}^\times, \quad v_{\mathfrak{p}} \ge 0,$$

belong to one finite family of ray characters with a common modulus supported on S; this family is determined by u and the residues $v_{\mathfrak{p}} \pmod 6$.

The prescribed zero of $\chi_A(a)$ remains when a good ideal A meets a.

Proof. Let $\xi^6 = a$. Since F contains all sixth roots of unity, all roots of $X^6 - a$ lie in $F(\xi)$, and $\sigma \mapsto \sigma(\xi)/\xi$ embeds its Galois group into μ_6. The extension is therefore abelian. A prime outside $6a$ is unramified; for example this follows from the discriminant of $X^6 - a$, which is supported on $6a$. At such a prime, arithmetic Frobenius is characterized on the residue field by $x \mapsto x^{q_\mathfrak{p}}$. Consequently its quotient on ξ reduces to $a^{(q_\mathfrak{p}-1)/6}$. Reduction is injective on μ_6 outside 6, proving the displayed identity. Multiplicativity of the Artin map gives the assertion for ideals. Artin reciprocity makes this map factor through a ray group with modulus supported on the primes dividing $6a$; these are the power-residue and ray-group assertions in [21, Chapter VIII, (5.3) and (5.5)].

For the last statement, an ideal A outside S is coprime to every $\pi_\mathfrak{p}$ and to every unit. Its symbol therefore has no zero there, and the power of each $\chi_A(\pi_\mathfrak{p}) \in \mu_6$ depends only on $v_\mathfrak{p} \pmod 6$. There are at most $6^{|S|+1}$ possible displayed numerators after this reduction. The first assertion supplies a ray modulus for each numerator, and a common multiple serves the whole family. Their prime supports are all contained in S because S contains the primes above 6. □

Remark. The conductor assertion specifies its prime support; it gives no bound on the exponents at those primes.

For a numerator containing a moving good prime, both the character and its natural zero support remain moving data. Lemma 3.2 places only the fixed numerators in a ray group independent of Z.

On the full ideal domain, the two character-polynomial forms are, for smooth compactly supported W on $(0,\infty)$, scale $D > 0$, and finite-order Hecke character ψ,

$$D^{-1/2} \sum_{\mathfrak{a}\neq 0} \mu(\mathfrak{a})\psi(\mathfrak{a})W(N\mathfrak{a}/D), \qquad D^{-1/2} \sum_{\mathfrak{a}\neq 0} \psi(\mathfrak{a})W(N\mathfrak{a}/D),$$

where μ is the ideal Möbius function. The first is a truncated reciprocal, and the second is a plain character polynomial.

For a norm scale $D > 0$, the centrally normalized character polynomial is

$$S_\psi(D; W) = D^{-1/2} \sum_{l} \psi(l)W(q_l/D), \tag{3.2}$$

and the centrally normalized inverse polynomial is

$$M_\psi(D; W) = D^{-1/2} \sum_{n} \mu(n)\psi(n)W(q_n/D). \tag{3.3}$$

A scale $D = Z^r$ has log-length r at base $Z > 1$; the same convention applies with U as the base. Every coprimality restriction in the complete character presentation ψ is retained in both definitions.

An annular test is a smooth function with support in a fixed compact subinterval of $(0,\infty)$. Families of annular or coupled profiles are used only with fixed logarithmic support and uniform bounds for every logarithmic derivative that is invoked. The precise separation norm is given in Lemma 3.6. In a product of two plain factors, both factors have the same row character, including the same fixed finite-ray twist. Conjugating a whole factor inside its absolute value permits the opposite orientation; conjugating only part of its coefficients does not.

We use the following uniformity convention. Log-lengths range over fixed bounded sets, and every estimate allows any specified positive power loss. The exponents of the norm scales depend only on those real parameter ranges, the strict margins, and the specified losses. The dependence on the arithmetic datum is the one specified above.

Finite seminorm orders, polynomial height orders, implied constants, and lower thresholds may depend on $\mathcal{A}$. They are uniform over the declared moving moduli and outer labels in their stated ranges, even when an outer label is fixed during one row sum.

For a nonzero ideal $\mathfrak{a}$, let $d_{\mathcal{O}}(\mathfrak{a}) = \#\{\mathfrak{d} : \mathfrak{d} \mid \mathfrak{a}\}$, where the count includes all integral ideal divisors, including those meeting S, and for $0 \neq f \in \mathcal{O}$ put $d_{\mathcal{O}}(f) = d_{\mathcal{O}}((f))$. A nonnegative multiplicity $w(f)$ is called divisor-bounded only if

$$w(f) \leq C\, d_{\mathcal{O}}(f)^C$$

for one fixed $C \geq 1$, independent of Z, the current rows, and the averaged labels. The constant C may depend on $\mathcal{A}$. This convention does not relax any separate condition that w depend only on f. For every $\delta > 0$ it implies $w(f) \ll_{C,\delta} q_f^{\delta}$ uniformly. Indeed, for prime ideals $\mathfrak{p}$ of sufficiently large norm, $(e+1)^C \leq q_{\mathfrak{p}}^{\delta e}$ for every integer $e \geq 0$, by $e + 1 \leq 2^e$ for $e \geq 1$. For each of the finitely many remaining prime ideals, $\sup_{e \geq 0} (e+1)^C q_{\mathfrak{p}}^{-\delta e} < \infty$; multiplying these bounds over the prime factorization of f proves the assertion.

3.2 Gauss sums and the finite reciprocity phase

We first evaluate the prime Gauss sums. This also fixes the orientation of the cubic symbol in all subsequent formulas.

Lemma 3.3 (Prime Gauss identities). *Let p be the primary generator of a prime ideal outside S, and put $P = q_p$. Then*

$$\gamma_2(p)^3 = -\alpha(p), \qquad \gamma_1(p)\gamma_2(p) = \overline{\chi_p(4)}\gamma_3(p)\gamma_2(p)^3.$$

For $j \not\equiv 0 \pmod 6$, one has $|\gamma_j(p)| = 1$.

Proof. Finite-field identities. Let $k = \mathcal{O}/(p)$ and let $\psi(x) = e(x/p)$ be its nontrivial additive character. For a multiplicative character A of $k^\times$, extended by zero, put

$$\tau(A) = \sum_{x \in k} A(x)\psi(x), \qquad J(A,B) = \sum_{x \in k} A(x)B(1-x).$$

If A is nonprincipal, the change of variables $x = ty$ gives

$$|\tau(A)|^2 = \sum_{t \in k^\times} A(t) \sum_{y \in k^\times} \psi((t-1)y) = P.$$

The inner sum is $P - 1$ for $t = 1$ and -1 otherwise, and $\sum_{t \in k^\times} A(t) = 0$. Conjugation also gives $\tau(A)\tau(\overline{A}) = A(-1)P$. When AB is nonprincipal, grouping the product $\tau(A)\tau(B)$ by $x + y$ gives

$$\tau(A)\tau(B) = J(A,B)\tau(AB).$$

The group with $x + y = 0$ vanishes because AB is nonprincipal; for a nonzero sum, division by $x + y$ gives the displayed Jacobi factor. In particular $|J(A,B)| = \sqrt{P}$ if A, B, AB are all nonprincipal.

The number of solutions of $4x(1-x) = y$ is $1 + \chi_p^3(1-y)$. Summing $\chi_p(y)$ times this identity and using $\sum_y \chi_p(y) = 0$ gives $J(\chi_p, \chi_p^3) = \chi_p(4)J(\chi_p, \chi_p)$. Write $\tau_j = \tau(\chi_p^j)$. The Gauss–Jacobi identity and $\tau_2\tau_4 = P$ now give

$$\frac{\tau_1\tau_3}{\tau_4} = \chi_p(4)\frac{\tau_1^2}{\tau_2}, \qquad \tau_1\tau_2 = \overline{\chi_p(4)}\frac{\tau_3\tau_2^3}{P}.$$

Here $\chi_p^2(-1) = 1$ because -1 is a cube.

The cubic Jacobi sum and its unit. It remains to fix the cubic Jacobi sum, including its unit. Put $m = (P-1)/3$. The definition of χ_p^2 gives, in k,

$$J(\chi_p^2, \chi_p^2) \equiv \sum_{x \in k} x^m(1-x)^m = 0 \pmod p.$$

Indeed the polynomial has degree $2m < P-1$, and the sum over k of each monomial of degree less than $P-1$ is zero in k (the constant case is $P=0$ in k). The Jacobi sum belongs to $\mathcal{O}$ and has absolute value $\sqrt{P}$.

For $x \neq 0,1$, choose $j_x \in \{0,1,2\}$ such that $\chi_p^2(x(1-x)) = \omega^{j_x}$. The products of x and of $1-x$ over these x are both -1, so $\prod_{x\neq 0,1} x(1-x) = 1$. Hence $\sum_{x\neq 0,1} j_x \equiv 0 \pmod 3$. Since

$$\omega^j \equiv 1 + j(\omega-1) \pmod{(\omega-1)^2}, \qquad ((\omega-1)^2) = (3),$$

we obtain $J(\chi_p^2,\chi_p^2) \equiv P-2 \equiv -1 \pmod 3$. Divisibility by p and equality of norms imply $J(\chi_p^2,\chi_p^2) = up$ for a unit u. As p is primary and the six units have distinct residues modulo 3, the congruence forces $u=-1$. Finally, $\tau_2^3 = J(\chi_p^2,\chi_p^2)\tau_2\tau_4 = -pP$. Dividing the two Gauss identities by the appropriate powers of $\sqrt{P}$ proves the lemma. □

The remaining phase is quadratic. Its evaluation below is valid even for odd elements that are not squarefree or primary.

Lemma 3.4 (Quadratic four-term formula). *For a nonzero odd $c \in \mathcal{O}$, define*

$$\Gamma(c) = |c|^{-1} \sum_{x \bmod c} e(x^2/c).$$

Then

$$\Gamma(c) = \frac{1}{2} \sum_{y \bmod 2\mathcal{O}} e(-cy^2/4). \tag{3.4}$$

For $c = a+b\omega$ the right side is $(1+i^{-b}+i^a+i^{b-a})/2$. In particular Γ depends only on $c \bmod 4\mathcal{O}$ and $\Gamma(cv^2) = \Gamma(c)$ for every odd v. The units modulo 4 have square subgroup $\{1,\omega,\omega^2\}$ and square-class representatives $1,-1,\lambda,-\lambda$. The function Γ never vanishes on these units, and $\mathfrak{r}(a,b) = \Gamma(ab)/(\Gamma(a)\Gamma(b))$ satisfies

$$\begin{array}{c|cccc} c & 1 & -1 & \lambda & -\lambda \\ \hline \Gamma(c) & 1 & 1 & i & -i \end{array}$$
$$\mathfrak{r}((-1)^e\lambda^f,(-1)^g\lambda^h) = (-1)^{eh+fg+fh}, \qquad e,f,g,h \in \{0,1\}. \tag{3.5}$$

One has $|\Gamma(c)| = 1$ and $\mathfrak{r}$ is a symmetric $\{\pm 1\}$-valued bicharacter of the square-class group.

Proof. We regularize the quadratic phase by a Gaussian and compare the two sides of Poisson summation as the regularization tends to zero. The resulting four-term sum then determines the square-class values.

The regularized transform. With Fourier kernel $e(-zy)$ and measure $d\mu$, Poisson summation applies to $f_\varepsilon(z) = e(z^2/c)e^{-\pi\varepsilon|z|^2}$ for $\varepsilon > 0$. Direct Gaussian integration gives, with $r_\varepsilon = 1 + 3q_c\varepsilon^2/16$,

$$\widehat{f_\varepsilon}(y) = \frac{|c|}{2\sqrt{r_\varepsilon}} e^{-\pi\varepsilon q_c|y|^2/(4r_\varepsilon)} e\big(-cy^2/(4r_\varepsilon)\big).$$

To check the normalization, rotate z by half the argument of c. The quadratic matrix of the resulting real two-variable Gaussian is

$$\begin{pmatrix} \varepsilon & -4i/(\sqrt{3}|c|) \\ -4i/(\sqrt{3}|c|) & \varepsilon \end{pmatrix}, \qquad \det = \frac{16r_\varepsilon}{3q_c}.$$

Its eigenvalues are conjugates with positive real part. The positive square root of the determinant, the factor $2/\sqrt{3}$ in $d\mu$, and completion of the square give the formula above.

The lattice means. Put $I_\varepsilon = \int_{\mathbb{C}} e^{-\pi\varepsilon|z|^2}\, d\mu(z) = 2/(\sqrt{3}\varepsilon)$. For a fixed ideal (b), a fixed class v mod b, and $a > 0$, Gaussian Poisson summation on $b\mathcal{O}$ gives

$$\frac{1}{I_\varepsilon} \sum_{z \in v + b\mathcal{O}} e^{-\pi a\varepsilon|z|^2} \longrightarrow \frac{1}{aq_b} \qquad (\varepsilon \downarrow 0).$$

The zero dual vector gives the limit and every nonzero dual vector is exponentially small. The phase $e(z^2/c)$ is periodic modulo c. Consequently the normalized left side of Poisson's identity tends to $\Gamma(c)/|c|$.

On the right side one may replace r_ε by 1. For fixed c, the total error from the phase is at most

$$C_c\varepsilon^2 \sum_{y \in \mathcal{O}} |y|^2 e^{-C_c^{-1}\varepsilon|y|^2} = O_c(1),$$

because the sum is $O_c(\varepsilon^{-2})$. The amplitude error is $O_c(\varepsilon)$, as is the damping error, by the same lattice estimate. All three are $o(I_\varepsilon)$. The phase $e(-cy^2/4)$ is periodic modulo $2\mathcal{O}$. In the preceding Gaussian mean take $b = 2$ and $a = q_c/4$; each of its four classes has normalized mean $1/q_c$. The normalized right side therefore tends to $(2|c|)^{-1} \sum_{y \bmod 2} e(-cy^2/4)$, which proves (3.4).

The finite square classes. The representatives $0, 1, \omega, \omega^2$ modulo 2 give the asserted four-term expression. Multiplication by an odd v permutes these classes, so the same formula gives $\Gamma(cv^2) = \Gamma(c)$. The group of units modulo 4 has order 12. Squaring a lift modulo 4 depends only on its class modulo 2, and its three possible squares are $1, \omega, \omega^2$. The four representatives in the statement are distinct modulo this subgroup; also $\lambda^2 = -3 \equiv 1 \pmod 4$. Evaluating the four-term expression on them gives the table. Evaluating $\Gamma(ab)/(\Gamma(a)\Gamma(b))$ on the two generators $-1, \lambda$ gives the displayed exponent, proving the last claims. □

On all primary pairs outside S, including noncoprime pairs, define $\mathcal{R}$ by the bicharacter $\mathfrak{r}$ of (3.5). Define, on every primary index outside S,

$$G(c) = \overline{\chi_c(4)}\Gamma(c).$$

Lemma 3.5 (Sextic reciprocity and the fixed Gauss phase). *For coprime primary a, b outside S,*

$$\chi_b(a) = \mathcal{R}(a,b)\chi_a(b), \qquad \mathcal{R}(a,b) = \mathfrak{r}(a,b).$$

The function G has modulus one and factors through a fixed ray group supported at $2, 3$. For all primary v, w outside S,

$$G(vw) = G(v)G(w)\mathcal{R}(v,w), \qquad G(1) = 1. \tag{3.6}$$

Moreover $G(v^2) = \chi_v(4)$, and at good primes

$$G(p^3) = \gamma_3(p), \qquad \mathcal{R}(p,p) = \gamma_3(p)^2 = \chi_p(-1). \tag{3.7}$$

On every primary n outside S, one has $\chi_n(-1) = \mathcal{R}(n,n)$. The diagonal $t \mapsto \mathcal{R}(t,t)$ is a character of the fixed ray group. In the square-class notation of (3.5), it is $\mathfrak{r}(-1,n)$, where -1 denotes its residue square class modulo 4, not the ideal ray class of the unit ideal (-1). For squarefree c the complete Gauss identities are

$$\gamma_2(c)^3 = \mu(c)\alpha(c), \quad \gamma_1(c)\gamma_2(c) = \mu(c)\alpha(c)G(c), \quad G(c) = \overline{\chi_c(4)}\gamma_3(c), \quad |G(c)| = 1. \tag{3.8}$$

On nonsquarefree indices, G is the finite-ray function defined above; it is not defined by a nonsquarefree Gauss sum.

Proof. Reciprocity. At a good prime, counting square roots in the residue field gives

$$\sum_{x \bmod p} e(x^2/p) = \sum_{y \bmod p} (1+\chi_p(y)^3)e(y/p) = \sum_{y \bmod p} \chi_p(y)^3 e(y/p).$$

The additive sum without the character is zero. Hence $\Gamma(p) = \gamma_3(p)$, and replacing x^2/p by ux^2/p for a unit u multiplies the sum by $\chi_p(u)^3$. For squarefree c, representing a residue as $\sum_{p|c}(c/p)x_p$ gives, in both sums,

$$\Gamma(c) = \prod_{p|c} \chi_p(c/p)^3\Gamma(p), \qquad \gamma_3(c) = \prod_{p|c} \chi_p(c/p)^3\gamma_3(p).$$

The cross terms in the square are integral in the additive character. Thus $\Gamma(c) = \gamma_3(c)$ for squarefree c. In particular, for distinct good primes p, q,

$$\frac{\Gamma(pq)}{\Gamma(p)\Gamma(q)} = \chi_p(q)^3\chi_q(p)^3.$$

We use cubic reciprocity in the following precise form: for coprime primary a, b one has $(a/b)_3 = (b/a)_3$; see [7, Equation (1.4)]. Thus the quotient $\chi_b(a)/\chi_a(b)$ is a sign. At p, q its cube is $\chi_q(p)^3\chi_p(q)^3$, so this sign equals $\mathfrak{r}(p, q)$. Multiplicativity in both arguments and the bicharacter property extend the equality to every coprime primary pair. The definition by $\mathfrak{r}$ on noncoprime pairs involves no division of zero symbols.

The finite-ray refinement. The element -2 is primary and -1 is a cube. Cubic reciprocity therefore gives

$$\chi_c(4) = (2/c)_3 = (-2/c)_3 = (c/(-2))_3.$$

This is a character of c mod 2. Together with the dependence of Γ on c mod 4, this proves that G is a function on a fixed ray group (for example the ray group modulo 12 suffices here). Indeed, if two primary generators represent the same ray class modulo 12, their quotient differs from an element congruent to one modulo 12 by a unit; reduction modulo 3 forces that unit to be one. The generators therefore have the same residue modulo 4. Its multiplicative refinement is the definition of $\mathfrak{r}$. Since $\Gamma(v^2) = 1$ and $\chi_v(4)$ has order dividing three, $G(v^2) = \overline{\chi_v(4)}^2 = \chi_v(4)$. The table also gives

$$\Gamma(p)^2 = (-1)^{(q_p-1)/2} = \chi_p(-1).$$

For the last equality use $\chi_p(-1) = (-1)^{(q_p-1)/6}$ and the equality of these parities. Now $\Gamma(p^3) = \Gamma(p)$, and $\overline{\chi_p(4)}^3 = 1$, proving $G(p^3) = \gamma_3(p)$. Finally $\mathcal{R}(p, p) = 1/\Gamma(p)^2 = \Gamma(p)^2$, as this square is a sign.

For a symmetric sign-valued bicharacter its diagonal is multiplicative:

$$\mathcal{R}(vw, vw) = \mathcal{R}(v, v)\mathcal{R}(v, w)^2\mathcal{R}(w, w) = \mathcal{R}(v, v)\mathcal{R}(w, w).$$

The prime identity consequently gives $\mathcal{R}(n, n) = \chi_n(-1)$ for every primary n outside S. On the square class $(-1)^e\lambda^f$, the table gives both this diagonal and $\mathfrak{r}(-1, n)$ the value $(-1)^f$. This use of -1 is in the residue square-class group; the ideal (-1) is the identity in an ideal ray class group and is not being substituted there.

Squarefree Gauss identities. For coprime squarefree primary a, b outside S, the Chinese remainder theorem gives

$$\gamma_j(ab) = \chi_a(b)^j\chi_b(a)^j\gamma_j(a)\gamma_j(b).$$

One obtains this by representing a residue as $bx+ay$ with x mod a and y mod b. Cubing the factor for $j = 2$ gives one. The product of the factors for $j = 1$ and $j = 2$ is $(\chi_a(b)\chi_b(a))^3 = \mathcal{R}(a, b)$. The prime identities from Lemma 3.3, followed by (3.6), therefore prove (3.8) by induction on the number of prime factors. □

For later element rows, fix the generators $\pi_{\mathfrak{p}}$ from Lemma 3.2. Every $0 \neq m \in \mathcal{O}$ has a unique expression

$$m = u m_S m_{\mathrm{good}}, \qquad m_S = \prod_{\mathfrak{p} \in S} \pi_{\mathfrak{p}}^{v_{\mathfrak{p}}(m)},$$

where u is a unit and m_{good} is the primary generator of the part of (m) outside S. On every primary A outside S, multiplicativity and sextic reciprocity give the zero-preserving identity

$$\chi_A(m) = \chi_A(u m_S)\mathcal{R}(A, m_{\mathrm{good}}) \prod_{p \mid m_{\mathrm{good}}} \chi_p(A)^{v_p(m)}. \tag{3.9}$$

For $(A, m_{\mathrm{good}}) = 1$ this is the reciprocity formula just proved; if they meet, both sides are zero and $\mathcal{R}$ is evaluated as its separately defined bicharacter. Once the good part's ray class and the S-valuations modulo six are fixed, the first two factors on the right range over a fixed finite family of characters of A. The product over good primes retains the moving character factors and all their zeros.

Every function on a fixed finite abelian ray group has a finite Fourier expansion in its characters. Explicitly, if $\phi : T \to \mathbb{C}$, then

$$\phi(v) = \sum_{\theta \in \widehat{T}} a_\theta \theta(v), \qquad a_\theta = \frac{1}{|T|} \sum_{u \in T} \phi(u)\overline{\theta(u)}.$$

Parseval and Cauchy–Schwarz bound $\sum_\theta |a_\theta|$ by $|T|^{1/2} \max_T |\phi|$. Applying the same statement on $T \times T$ separates $\mathcal{R}(a, b)$ as a finite sum of products of characters. These expansions remain valid at noncoprime primary pairs because $\mathcal{R}$ there is the fixed-ray bicharacter, not a symbolic quotient. For example, for a good prime p and every primary n outside S,

$$\chi_n(p)\overline{\chi_p(n)} = \mathcal{R}(p, n) 1_{(n,p)=1}.$$

On coprime pairs this is sextic reciprocity, and on the remaining pairs both sides vanish. Thus cancellation of the moving local characters can leave both a coprimality indicator and a fixed-ray phase; neither may be discarded.

3.3 Smooth norm profiles

The estimates below separate smooth norm variables, control suprema over scale and height parameters, and bound Fourier and Mellin tails. Their derivative orders record the dependence on the test independently of the polynomial growth in spectral height. This separation of orders is needed when a tail order is chosen after the height range.

3.3.1 Separation and parameter suprema

For a profile w on $(0, \infty)^d$ with logarithmic support in a fixed compact set $\Omega \subset \mathbb{R}^d$, put

$$w_{\log}(\boldsymbol{u}) = w(e^{u_1}, \dots, e^{u_d}), \qquad p_j(w) = \sum_{|\alpha| \le j} \sup_{\boldsymbol{u} \in \mathbb{R}^d} |\partial^\alpha w_{\log}(\boldsymbol{u})|. \tag{3.10}$$

We always understand that $w_{\log}$ is smooth and supported in Ω. Define

$$\widehat{w}(\boldsymbol{t}) = \int_{\mathbb{R}^d} w_{\log}(\boldsymbol{u}) e^{-i\boldsymbol{t}\cdot\boldsymbol{u}}\, d\boldsymbol{u}, \qquad \|w\|_{J,\mathrm{sep}} = \int_{\mathbb{R}^d} |\widehat{w}(\boldsymbol{t})|(1 + |\boldsymbol{t}|)^J\, d\boldsymbol{t}. \tag{3.11}$$

For a fixed finite tuple $\boldsymbol{w} = (w_1, \dots, w_r)$, possibly of different fixed dimensions, we use $p_j(\boldsymbol{w}) = 1 + \sum_{i=1}^r p_j(w_i)$.

Lemma 3.6 (Fourier separation of norm profiles). *Let w be a smooth profile with logarithmic support in the fixed compact set $\Omega \subset \mathbb{R}^d$, and let $p_j(w)$ and $\|w\|_{J,\mathrm{sep}}$ be the seminorms in (3.10) and (3.11). For integers $J \geq 0$,*

$$\|w\|_{J,\mathrm{sep}} \ll_{J,d,\Omega} p_{J+d+2}(w).$$

Logarithmic Fourier inversion separates any fixed norm monomial. More precisely, if $x_j = c_j \prod_{\ell=1}^{r} y_\ell^{a_{j\ell}}$ with $c_j, y_\ell > 0$ and fixed real exponents $a_{j\ell}$, then

$$w(x_1, \dots, x_d) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \widehat{w}(\boldsymbol{t}) \prod_j c_j^{it_j} \prod_\ell y_\ell^{i\sum_j a_{j\ell} t_j} \, d\boldsymbol{t}.$$

If the coefficient measure in this formula is common to a collection of rows, Minkowski's inequality passes any separated row ℓ^2 bound through the integral using $\|w\|_{J,\mathrm{sep}}$ whenever the separated bound has height growth at most $(1+|\boldsymbol{t}|)^J$.

Proof. For an integer m with $2m > J + d$, integration by parts gives

$$(1+|\boldsymbol{t}|^2)^m \widehat{w}(\boldsymbol{t}) = \int_{\mathbb{R}^d} (1-\Delta)^m w_{\log}(\boldsymbol{u}) e^{-i\boldsymbol{t}\cdot\boldsymbol{u}} \, d\boldsymbol{u}.$$

Its absolute value is bounded by the volume of Ω times $C_{m,d} p_{2m}(w)$. The weighted integral is finite because $2m > J + d$; one can choose $2m \leq J + d + 2$. Fourier inversion gives the monomial formula. For row vectors $B(\boldsymbol{t})$ in ℓ^2, the precise inequality used there is

$$\left\| \frac{1}{(2\pi)^d} \int \widehat{w}(\boldsymbol{t}) B(\boldsymbol{t}) \, d\boldsymbol{t} \right\|_{\ell^2} \leq \frac{1}{(2\pi)^d} \int |\widehat{w}(\boldsymbol{t})| \|B(\boldsymbol{t})\|_{\ell^2} \, d\boldsymbol{t}.$$

□

Remark. A kernel depending on a common product, such as $y_1 y_2$, gives the same norm power on both variables in the monomial formula; this preserves an equal-product-scale difference. The smoothness hypothesis applies after any nonsmooth arithmetic restrictions have been resolved.

Lemma 3.7 (A parameter supremum bound). *For a unit box $I \subset \mathbb{R}^d$ and a smooth scalar function F on $I + [-1,1]^d$,*

$$\sup_I |F|^2 \ll_d \sum_{\alpha \in \{0,1\}^d} \int_{I+[-1,1]^d} |\partial^\alpha F|^2. \tag{3.12}$$

Proof. In one dimension, the fundamental theorem of calculus, averaging a base point over the enlarged interval, and Cauchy–Schwarz give $\sup_I |F|^2 \ll \int_{I+[-1,1]} (|F|^2 + |F'|^2)$. Applying this successively in each coordinate proves (3.12). □

Remark. For a row family $(F_i)_i$, summing (3.12) over the original row index before integrating controls rowwise parameter suprema, since all terms are nonnegative. The required inputs are squared row-norm bounds for every derivative family $(\partial^\alpha F_i)_i$ with $\alpha \in \{0,1\}^d$, including $\alpha = 0$, uniformly at each parameter value. A logarithmic scale range of length $O(\log Z)$ needs $O(\log Z)$ unit intervals; a height range $[-T_1, T_1]$ needs $O(1+T_1)$ intervals. If each of these bounds is $O((1+|\boldsymbol{t}|)^H)$ for d_t height parameters, with constants uniform on the enlarged unit boxes, covering the height range $|t_j| \leq T_1$ and integrating gives at most a fixed multiple of $(1+T_1)^{H+d_t}$. For scale parameters in a polynomial range, the analogous covering in logarithmic coordinates introduces powers of $\log Z$. Derivatives with respect to a logarithmic scale insert $yW'(y)$, together with the constant derivative of a central normalization. Derivatives of a normalized twist y^{it} insert powers of $\log y$. These are again annular profiles with finite seminorm bounds.

3.3.2 Fourier and Mellin tails

Lemma 3.8 (Weighted Fourier tails). *Let w be a smooth profile with logarithmic support in a fixed compact set, with separation norms as in Lemma 3.6. For $T_1 \geq 1$ and integers $J, N \geq 0$,*

$$\int_{|\boldsymbol{t}|>T_1} |\widehat{w}(\boldsymbol{t})|(1+|\boldsymbol{t}|)^J \, d\boldsymbol{t} \leq T_1^{-N} \|w\|_{J+N,\mathrm{sep}}. \tag{3.13}$$

Proof. On $|\boldsymbol{t}| > T_1$ one has $(1+|\boldsymbol{t}|)^{-N} \leq T_1^{-N}$, proving (3.13). □

Remark. For a separated integral, write its integrand as $\widehat{w}(\boldsymbol{t})A_Z(\boldsymbol{t})$ after translating any pure Mellin twist as in Lemma 3.9. Fix an exponent B and an integer $J \geq 0$ for which $|A_Z(\boldsymbol{t})| \leq Z^B(1+|\boldsymbol{t}|)^J$. For every integer $N \geq 0$, (3.13) then bounds the absolute tail by $Z^B T_1^{-N}\|w\|_{J+N,\mathrm{sep}}$. Here the same bound for A_Z, with the same B, J, is required for all choices of N. Increasing the derivative order of the uniformly smooth profile changes its separation norm, while this arithmetic height bound remains fixed.

Lemma 3.9 (Mellin decay). *With the Mellin convention $\mathcal{M}W(s) = \int_0^\infty W(y)y^s \, dy/y$, a pure twist obeys*

$$\mathcal{M}[W(y)y^{i\omega}](s) = \mathcal{M}W(s+i\omega).$$

Let $D_y = y\partial_y$, let $I \subset \mathbb{R}$ be compact, and let $N \geq 0$ be an integer. Suppose the integrals below are finite and the boundary terms in N logarithmic integrations by parts vanish. Then, uniformly for $\sigma \in I$ and $T \in \mathbb{R}$,

$$|\mathcal{M}W(\sigma+iT)| \ll_{I,N} (1+|T|)^{-N} \sum_{j=0}^{N} \int_0^\infty y^\sigma |D_y^j W(y)| \frac{dy}{y}. \tag{3.14}$$

These hypotheses hold for annular W on every such I. They also hold when W is smooth at zero and Schwartz at infinity and I is a compact subset of $(0,\infty)$.

Proof. The Mellin shift is immediate from its definition. To prove (3.14), put $F_\sigma(u) = e^{\sigma u}W(e^u)$. For $|T| \geq 1$, integration by parts gives

$$\mathcal{M}W(\sigma+iT) = (-iT)^{-N} \int_{\mathbb{R}} F_\sigma^{(N)}(u) e^{iTu} \, du, \qquad F_\sigma^{(N)}(u) = e^{\sigma u} \sum_{j=0}^{N} \binom{N}{j} \sigma^{N-j} D_y^j W(e^u).$$

Taking absolute values gives the bound on a compact real strip. For $|T| < 1$, the absolute integral gives it after increasing the constant. The two stated classes of W have the required endpoint decay: compact support suffices in the first case, while $e^{\sigma u}$ at $-\infty$ and Schwartz decay at $+\infty$ suffice in the second. □

Remark. For a horizontal join, the smooth factor to which (3.14) applies is $\mathcal{M}W(\sigma+iT)$. Write the other factors as $A_Z(\sigma+iT)$. Suppose the join has a bounded real interval, $|T|$ comparable to $T_1 \geq 1$, and $|A_Z(\sigma+iT)| \leq Z^B(1+|T|)^J$ there for fixed B, J. For every integer $N \geq 0$, assume the test has the finite weighted derivatives and vanishing boundary terms in (3.14) at order $N + \lceil J \rceil$. Multiplying that pointwise estimate by the bound for A_Z and integrating over the real interval gives $Z^B T_1^{-N}$ times the corresponding derivative norm of W. It is $O(Z^B T_1^{-N})$ when those norms are uniform. The same arithmetic bound for A_Z is required for every N; only the smooth-test order increases.

For integrated tails, let $\boldsymbol{t} = (t_1, t_2, t_3)$ be the original heights. Fix a nonnegative integer k, separating frequencies $\boldsymbol{\xi} \in \mathbb{R}^k$, and real linear forms ℓ_1, ℓ_2, ℓ_3 on $\mathbb{R}^k$. Write

$$\boldsymbol{x} = (t_1 + t_2 + \ell_1(\boldsymbol{\xi}),\ t_2 + \ell_2(\boldsymbol{\xi}),\ t_3 + \ell_3(\boldsymbol{\xi})). \tag{3.15}$$

Here $k = 0$ gives the three-height transformation, with the separating density and translations omitted.

Lemma 3.10 (Joint Gaussian and Mellin tails). *For measurable functions $a(v), b(w)$ for which the right side is finite, and integers $J, N \geq 0$,*

$$\begin{aligned}\int_{\mathbb{R}^2} e^{-(T+v)^2}|a(v)b(w)|(1+|T|+|v|+|w|)^J\,dv\,dw \\ \ll_{J,N} (1+|T|)^{-N}\sup_v(1+|v|)^{J+N}|a(v)|\int_{\mathbb{R}}(1+|w|)^J|b(w)|\,dw.\end{aligned} \tag{3.16}$$

The product of a Gaussian in the sum of two heights and rapidly decreasing Mellin transforms in the other two heights has rapid pointwise decay on a fixed-height slice. For the integrated product, let a, b be the rapidly decreasing Mellin factors, let $c(\boldsymbol{\xi})$ be the separating density, and use (3.15). For integers $J, N \geq 0$ and $T_1 \geq 1$, whenever the right-hand integral is finite,

$$\begin{aligned}\int_{\|(\boldsymbol{t},\boldsymbol{\xi})\|_\infty > T_1} e^{-x_1^2}|a(x_2)b(x_3)c(\boldsymbol{\xi})|(1+\|\boldsymbol{t}\|_1+\|\boldsymbol{\xi}\|_1)^J\,d\boldsymbol{t}\,d\boldsymbol{\xi} \\ \ll_{J,N,k,\ell_1,\ell_2,\ell_3} T_1^{-N}\int_{\mathbb{R}^{3+k}} e^{-x_1^2}|a(x_2)b(x_3)c(\boldsymbol{\xi})|(1+\|\boldsymbol{x}\|_1+\|\boldsymbol{\xi}\|_1)^{J+N}\,d\boldsymbol{x}\,d\boldsymbol{\xi}.\end{aligned} \tag{3.17}$$

For $k = 0$, the factors $c(\boldsymbol{\xi})$ and $d\boldsymbol{\xi}$ and all $\boldsymbol{\xi}$-norms are omitted. Finiteness of the displayed moments for every N gives every prescribed power of T_1^{-1}.

Proof. For (3.16), use

$$1+|T| \leq (1+|T+v|)(1+|v|), \qquad 1+|T|+|v|+|w| \leq 2(1+|T+v|)(1+|v|)(1+|w|).$$

After $(1+|T|)^N$ is moved to the left, the weighted $a(v)$ is bounded pointwise and the remaining polynomial weight is integrated against the Gaussian in $T+v$. The remaining w integral is the one displayed. This proves the joint bound. For the integrated assertion, the map $(\boldsymbol{t}, \boldsymbol{\xi}) \mapsto (\boldsymbol{x}, \boldsymbol{\xi})$ in (3.15) is block triangular with determinant one. Its inverse has fixed operator norm, so for a constant $C_\ell \geq 1$ depending only on the fixed dimension and linear forms,

$$1+\|\boldsymbol{t}\|_1+\|\boldsymbol{\xi}\|_1 \leq C_\ell(1+\|\boldsymbol{x}\|_1+\|\boldsymbol{\xi}\|_1).$$

On the complement of the box, the original weight of order J is at most T_1^{-N} times that weight of order $J+N$. The displayed comparison and the change of variables therefore prove (3.17), with constant at most C_ℓ^{J+N}. The transformed moment is controlled by the separate weighted moments of the Gaussian, the two Mellin factors and the separating density. For $k = 0$ this is the original three-height argument; the appended frequencies enter only through the fixed block triangular map. □

3.3.3 Gaussian profiles and transformed tests

Define the Gaussian

$$W_G(y) = \frac{1}{2\sqrt{\pi}}\exp\bigl(-\tfrac{1}{4}(\log y)^2\bigr). \tag{3.18}$$

This function is not annular. Choose a fixed $\chi \in C_c^\infty(\mathbb{R})$ such that $\sum_{k\in\mathbb{Z}}\chi(u-k) = 1$, and define its annular profiles by

$$W_{G,k}(y) = W_G(y)\chi(\log y - k), \qquad w_k(x) = W_G(e^k x)\chi(\log x). \tag{3.19}$$

Lemma 3.11 (Gaussian annular decomposition). *Let W_G be the Gaussian in (3.18). For a fixed $\chi \in C_c^\infty(\mathbb{R})$ satisfying $\sum_{k\in\mathbb{Z}}\chi(u-k) = 1$, let $W_{G,k}$ and w_k be the profiles in (3.19). The profiles w_k have one fixed compact logarithmic support and, for every fixed $A \geq 0$ and integer $j \geq 0$,*

$$\sum_{k\in\mathbb{Z}} e^{A|k|}p_j(w_k) < \infty. \tag{3.20}$$

For fixed $\kappa > 0$, $A, B, C_0 \geq 0$, and $M > 0$ one also has

$$Z^B \sum_{|k| > \kappa \log Z - C_0} e^{A|k|} p_j(w_k) \ll Z^{-M} \qquad (Z \to \infty).$$

The Mellin transform satisfies $\mathcal{M}W_G(s) = e^{s^2}$, and the weighted logarithmic derivatives in (3.14) are finite uniformly on every compact real strip.

Proof. Such a partition is obtained by normalizing the integer translates of a nonnegative compactly supported smooth function positive on $[-1/2, 1/2]$. Let $\operatorname{supp} \chi \subset [-C, C]$. Each logarithmic derivative of $W_G(e^k x)$ is a polynomial in $k + \log x$ times the same Gaussian. The product rule therefore gives

$$p_j(w_k) \leq C_j (1 + |k|)^j \exp\bigl(-\tfrac{1}{4}((|k| - C)_+)^2\bigr).$$

This is summable after multiplication by $e^{A|k|}$ for every fixed A. The first assertion follows, and the separation norms follow from Lemma 3.6. In the profile's logarithmic Fourier transform, rescaling an annulus inserts only the unit phase e^{-ikt}, so it does not change these norms; any real normalization of a surrounding polynomial remains explicit. On the indicated tail, the logarithm of the last bound after multiplication by $Z^B e^{A|k|}$ is at most $-c\kappa^2 (\log Z)^2 + O(\log Z)$ for some fixed $c > 0$; summing the Gaussian tail proves the stated estimate for every M. Finally, set $u = \log y$ and complete the square in its Gaussian integral to obtain $\mathcal{M}W_G(s) = e^{s^2}$. The same Gaussian bounds every $e^{\sigma u}(d/du)^j W_G(e^u)$ in $L^1(\mathbb{R})$, uniformly for σ in a compact interval. □

The next two lemmas give finite-order continuity for Fourier kernels and Mellin multipliers. In an arithmetic application, the transform must supply the kernel and the available real lines required by the corresponding lemma.

Lemma 3.12 (Finite seminorms for Fourier kernels). *For a Schwartz function f on $\mathbb{R}^d$, put*

$$s_j(f) = \sum_{|\alpha|, |\beta| \leq j} \|x^\alpha \partial^\beta f(x)\|_{L^1(\mathbb{R}^d)}.$$

For every $A \geq 0$ and multi-index β, the ordinary Fourier transform satisfies

$$\sup_\xi (1 + |\xi|)^A |\partial^\beta \widehat{f}(\xi)| \ll_{A,\beta,d} s_{\lceil A \rceil + |\beta|}(f).$$

The same conclusion, with a fixed change in the constant, holds for any fixed nondegenerate linear Fourier pairing. For a radial transform written as $\widehat{f}(\xi) = F(|\xi|^2)$, any prescribed bound

$$\sup_{r \geq 0} (1 + r)^A |(r\partial_r)^j F(r)|$$

is controlled by finitely many Schwartz seminorms of f.

Proof. Differentiating $\widehat{f}$ inserts x^β, and multiplication by a monomial in ξ differentiates $x^\beta f$ before Fourier transformation. The L^1 bound for the Fourier transform, the product rule, and the bound of $(1 + |\xi|)^A$ by a finite sum of monomials of degrees at most $\lceil A \rceil$ give the first assertion. For a radial function, $r\partial_r$ corresponds to $(\xi \cdot \partial_\xi)/2$. Expanding its jth power gives finitely many polynomial multiples of Fourier derivatives, so the first assertion gives the radial one. A fixed linear change of coordinates changes only its constants. □

Lemma 3.13 (Finite seminorms for Mellin kernels). *Let $m(s)$ be holomorphic on a fixed closed vertical strip and suppose, uniformly there, $|m(\sigma + it)| \leq C(1 + |t|)^h$ for a fixed $h \geq 0$. On an available line $\Re s = \sigma$ in that strip define*

$$K_W(x) = \frac{1}{2\pi i} \int_{(\sigma)} \mathcal{M}W(-s) m(s) x^{-s}\, ds,$$

where W has the finite weighted logarithmic derivatives needed to make the integral absolutely convergent, with the boundary terms in the logarithmic integrations by parts vanishing. For every integer $j \geq 0$,

$$|(x\partial_x)^j K_W(x)| \ll x^{-\sigma} \int_{\mathbb{R}} |\mathcal{M}W(-\sigma - it)|(1+|t|)^{h+j}\, dt.$$

The integral on the right is bounded by finitely many integrals of $y^{-\sigma}|(y\partial_y)^k W(y)|\, dy/y$, uniformly for σ in the fixed strip. For $W(y)y^{i\omega}$ the same bound has at most an additional factor $(1+|\omega|)^{h+j}$. After multiplication by a fixed annular cutoff, these conclusions also bound every fixed logarithmic seminorm of $y \mapsto K_W(Ry)$, with the factor $R^{-\sigma}$.

Proof. Differentiation under the absolutely convergent integral inserts $(-s)^j$. On the fixed strip this is bounded by a constant times $(1+|t|)^j$. Write $y = e^u$ and integrate the Fourier transform of $e^{-\sigma u}W(e^u)$ by parts more than $h+j+1$ times. The product rule bounds the resulting L^1 derivatives by the stated weighted logarithmic derivatives, because σ ranges over a fixed set. For a pure twist the Mellin integrand is $\mathcal{M}W(-\sigma - it + i\omega)$. Substitute $v = t - \omega$ and use $(1+|v+\omega|)^{h+j} \leq (1+|v|)^{h+j}(1+|\omega|)^{h+j}$. Finally the product rule for a fixed annular cutoff and the chain rule for Ry give the last statement. □

3.4 Growth and logarithmic control for Hecke functions

The disk estimate will be applied only after zeros have been excluded from that disk. The following elementary growth bound provides the input for the complex-analytic argument and makes clear which constants are uniform in the conductor. In this subsection $\Gamma(s)$ denotes Euler's gamma function; the finite quadratic function $\Gamma(c)$ was confined to the arithmetic identities above.

Lemma 3.14 (Hecke strip growth). *Let ψ be a primitive nonprincipal finite-order Hecke character of F, with conductor norm Q. Then*

$$|L_F(\sigma + it, \psi)| \ll Q^{3/5}(3+|t|)^2, \qquad -1/10 \leq \sigma \leq 11/10. \tag{3.21}$$

For the principal character, the same bound with $Q = 1$ holds for $(s-1)\zeta_F(s)/(s+1)$, with the removable value used at $s = 1$.

Proof. The proof establishes boundary bounds, verifies qualitative growth inside the strip, and then applies a damped maximum principle to obtain a constant uniform in the conductor.

Boundary bounds. A finite-order character has trivial infinite type, since $\mathbb{C}^\times$ is connected. The primitive Hecke functional equation in this case is

$$\Lambda(s,\psi) = (3Q)^{s/2}(2\pi)^{-s}\Gamma(s)L_F(s,\psi), \qquad \Lambda(s,\psi) = \varepsilon(\psi)\Lambda(1-s,\overline{\psi}), \qquad |\varepsilon(\psi)| = 1.$$

This is the functional equation for primitive characters of trivial infinite type stated in [10, Equation (1.1)]; a nonprincipal $L_F(s,\psi)$ is entire. Absolute Euler convergence bounds $L_F(11/10 + it, \psi)$ uniformly in Q, t. Applying the functional equation and Stirling's formula on $\Re s = -1/10$ gives

$$|L_F(-1/10 + it, \psi)| \ll Q^{3/5}(3+|t|)^{6/5}.$$

The estimate for bounded t follows by continuity of the gamma quotient on that line, so the constant is uniform there as well.

Growth inside the strip. A lattice theta integral supplies the growth hypothesis needed for the strip principle. Let $\mathfrak{f} = (f)$ be the conductor and define the periodic residue character $\phi(a) = \psi((a))$ when $(a, \mathfrak{f}) = 1$, with value zero otherwise. For the principal conductor (1) put $\phi(a) = 1$ for all a, including $a = 0$. Since the ideal class group is trivial, conductor one has no

nonprincipal character. Periodicity modulo $\mathfrak{f}$ follows from the ray character property; ϕ is trivial on the six units because $(u) = (1)$ for a unit. Thus, for $\Re s > 1$,

$$6\pi^{-s}\Gamma(s)L_F(s,\psi) = \int_0^\infty \big(\Theta_\phi(v) - \phi(0)\big)v^{s-1}\,dv, \qquad \Theta_\phi(v) = \sum_{a\in\mathcal{O}} \phi(a)e^{-\pi v q_a}.$$

The factor six counts the generators of each ideal. If $\widehat{\phi}(h) = Q^{-1}\sum_{a \bmod f}\phi(a)e(-ha/f)$, finite Fourier inversion and Gaussian Poisson summation give

$$\Theta_\phi(v) = \frac{2}{\sqrt{3}v}\sum_{h \bmod f}\widehat{\phi}(h)\sum_{b\in\mathcal{O}}\exp\left(-\frac{4\pi|b - h/f|^2}{3v}\right).$$

Changing the generator f only reindexes the finite sum. For each fixed conductor, the nonzero dual vectors have a positive minimum length. With $A_\phi = 2\widehat{\phi}(0)/\sqrt{3}$, this proves

$$\Theta_\phi(v) = A_\phi v^{-1} + O_\phi(v^{-1}e^{-c_\phi/v}) \quad (0 < v \le 1), \qquad \Theta_\phi(v) = \phi(0) + O_\phi(e^{-c_\phi v}) \quad (v \ge 1)$$

for some $c_\phi > 0$. Splitting the Mellin integral at one consequently gives

$$\begin{aligned} 6\pi^{-s}\Gamma(s)L_F(s,\psi) = \frac{A_\phi}{s-1} - \frac{\phi(0)}{s} + \int_0^1 \big(\Theta_\phi(v) - A_\phi/v\big)v^{s-1}\,dv \\ + \int_1^\infty \big(\Theta_\phi(v) - \phi(0)\big)v^{s-1}\,dv. \end{aligned}$$

Both integrals are entire and bounded in height on each bounded real strip, with constants that may depend on the fixed conductor. Stirling's formula for $1/\Gamma(s)$ therefore gives at most exponential height growth for $L_F(s,\psi)$ on that strip. In the principal case the same conclusion holds after multiplication by $(s-1)/(s+1)$. Only this qualitative growth, for each fixed conductor, is used in the strip principle.

A uniform strip bound. To isolate the boundary constant from the fixed-conductor growth constants, put

$$H_Q(s) = Q^{-3/5}\frac{L_F(s,\psi)}{(s+2)^2}.$$

It is holomorphic in the closed strip and is bounded by one absolute constant C on both vertical boundary lines, by the bounds already proved. For $\delta > 0$, apply the maximum principle on the rectangle of height T to $H_Q(s)\exp(\delta(s-1/2)^2)$. On the vertical sides the exponential has modulus at most $e^{9\delta/25}$, because $|\Re s - 1/2| \le 3/5$. On the horizontal sides its modulus is at most $e^{\delta(9/25 - T^2)}$, which tends to zero faster than the qualitative fixed-Q exponential growth as $T \to \infty$. First let $T \to \infty$ for this fixed Q, δ, and then let $\delta \downarrow 0$. The result is $|H_Q(s)| \le C$ throughout the strip, with the same constant for every Q. Multiplication by $Q^{3/5}|s+2|^2$ proves (3.21).

For the principal function, the factor $(s-1)/(s+1)$ removes its only pole in the strip and is bounded on the boundary. Its functional equation gives the same boundary bounds with $Q = 1$. This equation also follows from the preceding Poisson formula with $\phi = 1$: after writing $v = 2t/\sqrt{3}$, the theta relation is $\Theta_\phi(2t/\sqrt{3}) = t^{-1}\Theta_\phi(2/(\sqrt{3}t))$, whose split Mellin integral is invariant under $s \mapsto 1 - s$. The same damped-rectangle argument applied to $(s-1)\zeta_F(s)/((s+1)(s+2)^2)$ proves the principal bound. □

Lemma 3.15 (Logarithmic control). *Let ψ be a primitive nonprincipal finite-order Hecke character of conductor norm Q, and put $\mathcal{L}(s) = L_F(s,\psi)$. For the principal character put $Q = 1$ and $\mathcal{L}(s) = (s-1)\zeta_F(s)/(s+1)$, with its removable value at one. Fix $a \in [1/2, 1]$ and $0 < e < 10^{-3}$. Suppose $\mathcal{L}$ has no zero in the open disk of radius $2 - a - 2e$ centered at $2 + it$. Uniformly on the closed concentric disk of radius $2 - a - 6e$,*

$$|\mathcal{L}(s)| + |\mathcal{L}(s)^{-1}| \ll_{e,\epsilon} \{2Q(3 + |t|)^2\}^\epsilon. \tag{3.22}$$

On the disk of radius $2-a-8e$ one also has

$$|\mathcal{L}'(s)/\mathcal{L}(s)| \ll_e \log\{2Q(3+|t|)^2\}.$$

In particular, with β_* defined in (2.1), for every fixed $b \geq \beta_*$ and $v > 0$, these bounds hold on $\Re s \geq b+v$, with constants depending on v, ϵ (and with the height of s in place of t). In the principal case the reciprocal bound passes to $1/\zeta_F$; the upper and logarithmic-derivative bounds are for the regularized function $\mathcal{L}$.

Proof. The strip bound first controls a holomorphic logarithm. Interpolation with its bounded value on a smaller Euler-product disk then gives the arbitrarily small power in the conclusion. Write $R_j = 2-a-je$ and $\mathcal{C} = 2Q(3+|t|)^2$. All the disks used here lie in $\Re s > 1/2$, where the principal regularizer is holomorphic. On the zero-free disk choose a holomorphic logarithm $g = \log \mathcal{L}$ whose value near the center is the Euler logarithm, together with the logarithm of the regularizing factor in the principal case. The value $g(2+it)$ is uniformly bounded. Lemma 3.14 and Euler convergence to the right of $11/10$ give

$$\Re g(s) = \log|\mathcal{L}(s)| \ll \log \mathcal{C} \qquad (|s-(2+it)| \leq R_3).$$

The possible heights differ from t by at most $3/2$, which is included in $\mathcal{C}$. Borel–Carathéodory on radii R_3, R_4, whose difference is e, now gives $|g| \ll_e \log \mathcal{C}$ on the disk of radius R_4.

On the disk of fixed radius $r_0 = 49/100$, one has $\Re s \geq 151/100 > 1$. The absolutely convergent Euler logarithm is uniformly bounded there. For the principal function the logarithm of $(s-1)/(s+1)$ is also bounded on this disk, using its branch in $\Re s > 1$. Since $r_0 < R_6 < R_4$, Hadamard's three-circles theorem applied to g gives

$$\max_{|s-(2+it)| \leq R_6} |g(s)| \ll_e (\log \mathcal{C})^\theta, \qquad \theta = \frac{\log(R_6/r_0)}{\log(R_4/r_0)} < 1.$$

For fixed e, continuity on the compact interval $1/2 \leq a \leq 1$ makes θ uniformly smaller than one. For every $\epsilon > 0$, $\exp(C_e(\log \mathcal{C})^\theta) \ll_{e,\epsilon} \mathcal{C}^\epsilon$. Applying this to both e^g and e^{-g} proves (3.22). Cauchy's estimate between radii R_6 and R_8 gives the stated bound for $g' = \mathcal{L}'/\mathcal{L}$.

For the global assertion first suppose $\beta_* \leq b \leq 1$. Choose $0 < e < \min(10^{-3}, v/8)$ and put $a = b$. The disk of radius R_2 is contained in $\Re s > b + 2e > \beta_*$, so it is zero-free by the definition of β_* and absolute Euler convergence beyond one. The disk of radius R_8 covers the points $b+v \leq \Re s \leq 2$ at its center's height; Euler convergence covers $\Re s \geq 2$. The preceding constants are uniform in b. If $b > 1$, Euler convergence on $\Re s \geq 1+v$ suffices. Finally

$$\frac{1}{\zeta_F(s)} = \frac{s-1}{s+1}\mathcal{L}(s)^{-1}, \qquad \left|\frac{s-1}{s+1}\right| \leq 1 \quad (\Re s \geq 0),$$

which proves the principal reciprocal assertion. □

Lemma 3.16 (Deleted Euler factors). *Let R be a squarefree ideal and let $|a_p| \leq 1$ for $p \mid R$. Put $D_R(s) = \prod_{p|R}(1 - a_p q_p^{-s})$. For fixed $\sigma_0 > 0$ and every $\epsilon > 0$,*

$$|D_R(s)| + |D_R(s)^{-1}| \ll_{\sigma_0,\epsilon} q_R^\epsilon \qquad (\Re s \geq \sigma_0).$$

On a bounded real strip one has, uniformly in the height,

$$|D_R(s)| \ll_\epsilon q_R^{(-\Re s)_+ + \epsilon}, \qquad \left|\frac{D_R'(s)}{D_R(s)}\right| \ll_{\sigma_0} \log(2q_R) \qquad (\Re s \geq \sigma_0).$$

In particular, if $q_R \leq Z^B$ with bounded B, both $|D_R(s)|$ and $|D_R(s)^{-1}|$ are bounded by a constant times any prescribed positive power of Z on a fixed positive real half-plane.

Proof. Every factor is nonzero on $\Re s > 0$. For $\Re s \geq \sigma_0$, both its absolute value and the absolute value of its inverse are at most $(1 - q_p^{-\sigma_0})^{-1}$. For sufficiently large q_p, depending on σ_0, ϵ, $-\log(1 - q_p^{-\sigma_0}) \leq \epsilon \log q_p$. The finitely many smaller primes contribute a fixed constant. This proves the first estimate. For any real σ,

$$\prod_{p|R}(1 + q_p^{-\sigma}) \leq q_R^{(-\sigma)_+} 2^{\#\{p:p|R\}} \ll_\epsilon q_R^{(-\sigma)_+ + \epsilon}.$$

The last inequality follows by separating the finitely many primes with $q_p < 2^{1/\epsilon}$. Finally, logarithmic differentiation gives

$$\left|\frac{D_R'}{D_R}(s)\right| \leq \sum_{p|R} \frac{\log q_p}{q_p^{\sigma_0} - 1} \ll_{\sigma_0} \log(2q_R).$$

If $q_R \leq Z^B$, choose the exponent in the first estimate smaller than the desired exponent of Z divided by the bounded value of B; the case $B = 0$ is immediate. This proves the polynomial-support consequence. □

Remark. For an imprimitive function $L_F(s, \psi^*)D_R(s)$, logarithmic control therefore uses both the primitive conductor and the deletion radical R.

4 Completed cubic reflection and row mean squares

This section first transforms a completed sum of Gauss coefficients while retaining every zero extension in its character. It then combines that identity with the quadratic large sieve to bound its mean square over arbitrary nonzero element rows. At equal row and completed lengths, the result is the bound $Z^{1+\epsilon}$ needed in the balanced argument.

The theta function and its Fourier coefficients at three fixed cusps are the unconditional input from Dunn and Radziwiłł [7, Section 5 and Appendix A]. The original coefficient and cusp calculations are in Patterson [23, Theorem 8.1 and Sections 7–8]; the formal input here remains the formulas in [7]. The finite transform below derives the formula with all coprimality restrictions, including its local factors and phase. The later mean-square argument uses this phase only after fixing its finite cusp sectors.

4.1 The reflection formula and its data

Recalling the character e of Section 3.1, define $\check{e}$ by

$$\check{e}(z) = \exp\bigl(2\pi i(z + \bar{z})\bigr), \qquad e(z) = \check{e}(z/\lambda) \quad (z \in \mathbb{C}).$$

On F, the exponent in $\check{e}$ is $2\pi i \operatorname{Tr}_{F/\mathbb{Q}} z$. Recall the zero-extension convention of Section 3.1: in particular, $\chi_p(x)^0 = \mathbf{1}_{p\nmid x}$.

Fixed and moving character data. Fix a finite family $\mathcal{X}$ of finite-ray characters whose conductors are supported on S. Choose one fixed integral modulus $\mathfrak{E}$ with prime support exactly S, divisible by every conductor in this family. On primary elements extend each member of $\mathcal{X}$ by zero away from the $\mathfrak{E}$-units. Choose a single $L \in \mathcal{O} \setminus \{0\}$ with prime support S, with $(18) \mid (L)$ and $\mathfrak{E} \mid (L)$, large enough that, for every $\Psi_0 \in \mathcal{X}$, the function on $\mathcal{O}$

$$\phi(x) = \begin{cases} \chi_x(\lambda)^2 \Psi_0(x), & x \equiv 1 \pmod 3,\ (x, \mathfrak{E}) = 1, \\ 0, & \text{otherwise}, \end{cases} \qquad \widehat{\phi}(h) = q_L^{-1} \sum_{x \bmod L} \phi(x) e(-hx/L)$$

is L-periodic. The zero branch is evaluated directly. The proof below establishes the existence of L from the cubic supplementary law and ray periodicity. Both $\mathfrak{E}$ and L are fixed for the entire

family before any moving prime is chosen. Fix a member $\Psi_0 \in \mathcal{X}$ and put $\mathfrak{L} = (L)$. Let $\mathcal{P}$ be any finite set of distinct primary primes not dividing $\mathfrak{L}$, choose $j_p \in \{0, 1, \dots, 5\}$ for each $p \in \mathcal{P}$, and set, on primary elements,

$$\Psi(n) = \Psi_0(n) \prod_{p \in \mathcal{P}} \chi_p(n)^{j_p}.$$

The completed sum. The variables in the following product may share prime factors:

$$D(s, \Psi) = \sum_{\substack{n \equiv 1 \ (3) \\ (n, \mathfrak{E})=1}}^{\mathrm{sf}} \gamma_2(n) \bar{\alpha}(n) \Psi(n) q_n^{-s},$$

$$L_0(s, \Psi) = \sum_{\substack{b \equiv 1 \ (3) \\ (b, \mathfrak{E})=1}} \bar{\alpha}(b)^3 \Psi(b)^3 q_b^{-3s+1/2}, \qquad T(s, \Psi) = L_0(s, \Psi) D(s, \Psi).$$

Let $X > 0$. Suppose that $V \in C^\infty(0, \infty)$ and that, for every $C > 0$ and every integer $j \geq 0$, $(x\partial_x)^j V(x)$ is $O_{C,j}(x^C)$ as $x \to 0$ and $O_{C,j}(x^{-C})$ as $x \to \infty$. Define

$$\widehat{V}(t) = \int_0^\infty V(x) x^t \frac{dx}{x}.$$

The smoothed expression to be transformed is the completed sum

$$\frac{1}{2\pi i} \int_{(a)} \widehat{V}(s - 1/2) X^{s-1/2} T(s, \Psi)\, ds = \sum_{\substack{n \text{ sf},\ b \\ n, b \equiv 1 \ (3) \\ (nb, \mathfrak{E})=1}} \frac{\gamma_2(n) \overline{\alpha(nb^3)} \Psi(nb^3)}{\sqrt{q_n}\, q_b} V\left(\frac{q_n q_b^3}{X}\right), \qquad a > 1.$$

The character acts on the whole index nb^3, with its zero values retained; n and b may share prime factors. Absolute convergence and Mellin inversion give the displayed equality.

The reflected test. Define

$$K = \frac{(2\pi)^4}{27}, \qquad R(t) = \prod_{\pm} \frac{\Gamma(1 + t \pm 1/6)}{\Gamma(1 - t \pm 1/6)},$$

$$V^\sharp(x) = \frac{1}{2\pi i} \int_{(\sigma)} \widehat{V}(-t) R(t) (Kx)^{-t}\, dt \qquad (\sigma \geq 0).$$

Local branch data. A branch is indexed by $h_0 \bmod L$ and a subset of active primes in $\mathcal{P}$. Every prime with $j_p \neq 0$ is active; a prime with $j_p = 0$ may be active or inactive. Write $r = \prod_{p \text{ active}} p$. Let c_F be the reduced denominator of $\lambda^2 h_0 / L$, with the normalizing unit specified in the proof, and put $c = c_F r$ and $M = \lambda^{12} L^4$. Define the local column factors by

$$B_p(x) = \begin{cases} \chi_p(x)^{-j_p - 2}, & j_p \neq 0, 4, \\ q_p^{-1/2}(-1 + q_p \mathbf{1}_{p|x}), & j_p = 4, \\ q_p^{-1/2} \chi_p(x)^{-2}, & j_p = 0 \text{ and } p \text{ is active.} \end{cases}$$

For $m \not\equiv 0 \pmod 6$, the normalized Gauss phases are

$$\tau_{p,m}^{\pm} = q_p^{-1/2} \sum_{y \bmod p} \chi_p(y)^m e(\pm y/p), \qquad |\tau_{p,m}^{\pm}| = 1.$$

For each active p, define the units modulo p

$$\sigma_p = \lambda^2 c / p, \qquad \epsilon_p = -\big((\lambda^3 c/p)\sigma_p\big)^{-1},$$

and the local phase

$$\omega_{p,j} = \begin{cases} \tau^-_{p,j}\tau^+_{p,j+2}\chi_p(\epsilon_p)^{-j-2}, & j \neq 0, 4, \\ \tau^-_{p,4}, & j = 4, \\ -\tau^+_{p,2}\chi_p(\epsilon_p)^{-2}, & j = 0. \end{cases}$$

The indices of τ are read modulo six. For a complex scalar κ_F of absolute value one, define

$$\zeta = -\frac{i}{81}\bar{\alpha}(c)^2\widehat{\phi}(h_0)\bar{\kappa}_F \prod_{p \text{ inactive}} (1 - q_p^{-1}) \prod_{p \text{ active}} \chi_p(\sigma_p)^{-2}\omega_{p,j_p}. \tag{4.1}$$

Proposition 4.1 (Completed cubic reflection). *Let Ψ be the zero-extended product character of Section 4.1, formed from a member Ψ_0 of the fixed finite family $\mathcal{X}$ and the moving local exponents $(j_p)_{p\in\mathcal{P}}$. Let V satisfy the smoothness and endpoint-decay conditions stated there, let $X > 0$, and use the completed product $T(s, \Psi)$ and reflected test $V^\sharp$ defined there. Both series defining T and their product converge absolutely for $\Re s > 1$. The integral defining $V^\sharp$ is absolutely convergent and is independent of $\sigma \geq 0$.*

The product T has an entire continuation and polynomial growth on each fixed vertical strip, with constants allowed to depend on $L, \Psi_0, \mathcal{P}, (j_p)$ and the strip. There is one fixed triple of cusp coefficient functions from $\lambda^{-4}\mathcal{O} \setminus \{0\}$ to $\mathbb{C}$, constructed from the theta expansions in the proof and used for all arithmetic data, scales, and tests. For each fixed arithmetic choice, there is a finite family of branches with the local data above, independent of X, V and $a > 1$. On each branch, d is selected from that triple and ϑ is an additive character of $\mathcal{O}$ modulo a fixed modulus depending only on L. For every $a > 1$,

$$\begin{aligned} &\frac{1}{2\pi i}\int_{(a)} \widehat{V}(s - 1/2)X^{s-1/2}T(s, \Psi)\, ds \\ &\quad = \sum_{\text{terms}} \zeta \sum_{0\neq\mu\in\lambda^{-4}\mathcal{O}} \frac{d(\mu)\alpha(\mu)\vartheta(\lambda^4\mu)}{\sqrt{q_\mu}} \prod_{p \text{ active}} B_p(\lambda^4\mu)V^\sharp\left(\frac{q_\mu X}{q_c^2}\right). \end{aligned} \tag{4.2}$$

Every inner sum is absolutely convergent. The coefficient bounds and simultaneous choices are specified in Proposition 4.2.

Proposition 4.2 (Local factors and compatible branches). *For the fixed and moving character data of Section 4.1, use the local branch definitions. The branches and coefficients in (4.2) can be chosen with the following properties.*

1. *The denominators c_F range over a fixed finite set with prime divisors dividing L. There are $O_L(2^{|\mathcal{P}|})$ terms, or $O_L(4^{|\mathcal{P}|})$ after splitting each $j_p = 4$ Ramanujan factor into its two summands.*

2. *The local column factors are the functions B_p defined above. For each of the three fixed cusp coefficient functions d, writing u for an Eisenstein unit, the support and size satisfy*

$$\operatorname{supp} d \subset \{u\lambda^k nb^3 : k \geq -4,\ n, b \equiv 1 \pmod 3,\ n \text{ sf}\}, \qquad |d(\mu)| \leq 27\, 3^{k/6}|b|. \tag{4.3}$$

   The bound refers to the displayed representation of a supported index. It permits common primes of n and b, and it imposes no exclusion at primes of L other than the displayed restriction at λ. The character ϑ factors through a quotient of $\mathcal{O}$ by a fixed modulus depending only on L.

3. *The normalized Gauss phases $\tau^\pm_{p,m}$ have absolute value one. In (4.1), the scalar κ_F has absolute value one and depends only on h_0 and the class of r modulo M^2. The resulting ζ is independent of μ and satisfies $|\zeta| \leq 1/81$. In each of these finitely many sectors, d and*

ϑ are fixed. Apart from the scale q_c^2, all moving-prime dependence outside the B_p is in ζ, which may depend on the whole active set.

These choices are simultaneous for all local prime sets with the same L and Ψ_0. Adding a good prime p with exponent zero and declaring it inactive leaves $c_F, c, d, \vartheta, \kappa_F$, the data at every other active prime, and the kernel unchanged from the branch in which that prime is absent. Its scalar is multiplied by $1 - q_p^{-1}$. This compatibility holds for every h_0, including nonunit classes modulo L.

Within a fixed sector, the independence in Proposition 4.2(3) makes d and ϑ common column factors in the later row norm.

4.2 Transformation at rational cusps

Proof of Propositions 4.1 and 4.2. We first express the characters with their zero extensions through finitely many translated theta values. We then transform their rational cusps and compute the resulting automorphy multipliers and local Fourier factors; the final Mellin comparison gives the reflection identity. Lemma 4.3 then proves the bounds for its transformed test.

The fixed theta input. Let $\Gamma = \mathrm{SL}_2(\mathcal{O})$, $\Gamma_1(3) = \{g \in \Gamma : g \equiv I \pmod 3\}$, and $\Gamma_2 = \langle \mathrm{SL}_2(\mathbb{Z}), \Gamma_1(3)\rangle$. The unconditional formulas of [7, Equations (5.4), (5.7), (5.9), and (5.12)–(5.17)] give a smooth cubic theta function θ on $\mathbb{H}^3 = \mathbb{C} \times \mathbb{R}_{>0}$ with

$$\theta(gw) = \kappa(g)\theta(w) \quad (g \in \Gamma_2), \qquad \kappa\begin{pmatrix} a & b \\ c & \delta' \end{pmatrix} = \left(\frac{c}{a}\right)_3 \quad \left(\begin{pmatrix} a & b \\ c & \delta' \end{pmatrix} \in \Gamma_1(3),\ c \neq 0\right).$$

Here κ is one on $\mathrm{SL}_2(\mathbb{Z})$ and is one on an element of $\Gamma_1(3)$ with lower left entry zero. The subscript 3 denotes the ordinary cubic residue symbol; for a good prime $p \notin S$, one has $(x/p)_3 = \chi_p(x)^2$. These automorphy and coefficient formulas hold unconditionally.

Put

$$\mathrm{L}(t) = \begin{pmatrix} 1 & 0 \\ t & 1 \end{pmatrix}, \qquad (t_0, t_1, t_2) = (0, \omega^2, \omega), \qquad \Theta_b(w) = \theta(\mathrm{L}(t_b)w).$$

If $v \in \mathbb{Z} + 3\mathcal{O}$, then $\mathrm{L}(v)$ is a product of an element of $\mathrm{SL}_2(\mathbb{Z})$ and a lower translation in $\Gamma_1(3)$ with multiplier one. The analogous assertion holds for upper translations. Therefore

$$\theta(\mathrm{L}(\lambda)w) = \Theta_1(w), \qquad \theta(\mathrm{L}(-\lambda)w) = \Theta_2(w),$$

because $\lambda - \omega^2 = 2 + 3\omega$ and $-\lambda - \omega = -1 - 3\omega$. If $u_0 = s + t\omega$ with $0 \le s, t < 3$, and $E = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, then

$$\begin{pmatrix} u_0 & -1 \\ 1 & 0 \end{pmatrix} = E\mathrm{L}(-u_0), \qquad -u_0 - t_t \in \mathbb{Z} + 3\mathcal{O},$$

and hence $\theta\left(\left(\begin{smallmatrix} u_0 & -1 \\ 1 & 0 \end{smallmatrix}\right) w\right) = \Theta_t(w)$. These are the only cusp representatives needed below.

For any of them, define d_H by

$$\bar{\theta}(H(z, v)) = C_H(v) + \sum_{\mu \neq 0} d_H(\mu) v K_{1/3}(4\pi|\mu|v)\check{e}(\mu z). \tag{4.4}$$

Here $C_H(v)$ is independent of z. The three functions $\Theta_0, \Theta_1, \Theta_2$ are respectively F_1, F_{19}, F_{10} in [7]. Its Appendix A, rows 1, 19, 10, expresses their coefficients in terms of τ, τ_1, τ_2 in its Equations (5.7), (5.13), and (5.14). In their notation the conjugate expansion uses $d_j(-\mu)$, as in their Equation (5.16). Those formulas give the support inclusion in (4.3). For completeness, the two possible magnitudes from τ at $\mu = u\lambda^k n b^3$ are

$$3^{k/6+8/3}|b| \quad \text{and} \quad 3^{k/6+3}|b|,$$

and τ_1, τ_2 have magnitude $9|b|$ and $k = -4$. Here the Gauss sums use only the cubic symbol, including at primes in S. To make their domain explicit, for every primary squarefree n define, as in [7, Equation (1.7)],

$$\widetilde{g}_3(n) = q_n^{-1/2} \sum_{v \bmod n} \left(\frac{v}{n}\right)_3 \check{e}(v/n), \qquad \widetilde{g}_3(1) = 1. \tag{4.5}$$

The ordinary cubic symbol is defined at every prime other than λ and is extended by zero on nonunits and multiplicatively in the denominator. Every primary n is prime to λ. At each prime divisor of n the cubic character is nontrivial, and $\check{e}(v/n) = e(\lambda v/n)$ is primitive because λ is a unit there. The finite-field Gauss identity and the Chinese remainder theorem therefore give $|\widetilde{g}_3(n)| = 1$ for every such squarefree n. This includes the primary prime -2 of norm 4; no sextic character with that denominator is used. Also, whenever $(t, n) = 1$, the change of variable $y = tv$ gives

$$q_n^{-1/2} \sum_{v \bmod n} \left(\frac{v}{n}\right)_3 \check{e}(tv/n) = \left(\frac{t}{n}\right)_3^{-1} \widetilde{g}_3(n).$$

This covers the twists $t = \lambda^2, \omega\lambda^2, \omega^2\lambda^2$ in the other source coefficient families. Thus the unnormalized Gauss sums occurring in all three cusp families have magnitude $|n|$. Consequently each coefficient magnitude displayed above is at most $27\,3^{k/6}|b|$, including $k = -4$. The formulas require only that n be squarefree and that n, b be primary; they do not require $(n, b) = 1$. Prime valuations of nb^3 also show that its representation by such n, b, when it exists, is unique. In particular the same bound is valid when the variables share primes or contain primes excluded from the primal sums.

At infinity the condition $x = \lambda^3\mu \equiv 1 \pmod 3$ isolates exactly $\mu = \lambda^{-3}nb^3$ with n, b primary and n squarefree. The infinity support has $\lambda^3\mu \in \mathcal{O}$; every other ramified exponent in Equation (5.7) of [7] makes x divisible by λ, and the negative sign in the remaining pair gives $x \equiv -1 \pmod 3$. At the isolated index the coefficient in $\bar{\theta}$ is

$$d_I(\lambda^{-3}nb^3) = 3^{5/2}|b|\widetilde{g}_3(n). \tag{4.6}$$

This is the source formula before any excluded-prime restriction, valid for every primary squarefree n and every primary b, with no coprimality condition between them. Only when $(n, S) = 1$ may it be expressed in the global sextic notation: substituting $y = \lambda v$ in $\gamma_2(n)$ gives

$$\widetilde{g}_3(n) = \left(\frac{\lambda}{n}\right)_3^{-1} \gamma_2(n) = \chi_n(\lambda)^{-2}\gamma_2(n) \qquad ((n, S) = 1). \tag{4.7}$$

The cubic supplementary laws in [7, Equation (1.5)] make $(\lambda/x)_3$ periodic on all primary x modulo 9. On primary elements prime to S it equals $\chi_x(\lambda)^2$. A character in $\mathcal{X}$ is periodic on primary elements prime to $\mathfrak{E}$ modulo its conductor, and the zero condition $(x, \mathfrak{E}) \neq 1$ is periodic modulo $\mathfrak{E}$. Thus a sufficiently divisible L with prime support S works for every member of the fixed finite family. This verifies its choice before the moving local primes are selected.

Finite Fourier inversion with the zero extensions. For $j \in \{0, \dots, 5\}$ define

$$C_{p,j}(h) = q_p^{-1} \sum_{x \bmod p} \chi_p(x)^j e(-hx/p).$$

The character pairing furnished by e is nondegenerate on $\mathcal{O}/(p)$. Changing variables in a Gauss sum, and using the orthogonality of a nontrivial multiplicative character, gives

$$C_{p,j}(h) = \begin{cases} q_p^{-1/2}\tau_{p,j}^{-}\chi_p(h)^{-j}, & j \neq 0,\ h \neq 0, \\ 0, & j \neq 0,\ h = 0, \\ -q_p^{-1}, & j = 0,\ h \neq 0, \\ 1 - q_p^{-1}, & j = 0,\ h = 0. \end{cases} \tag{4.8}$$

Here and below $h = 0$ means $h \equiv 0 \pmod{p}$. Each nontrivial power of χ_p is primitive modulo the prime p, so the usual finite-field Gauss-sum identity gives $|\tau^{\pm}_{p,m}| = 1$.

Fourier inversion on $\mathcal{O}/(L)$ and on each $\mathcal{O}/(p)$ now gives

$$\phi(x)\prod_p \chi_p(x)^{j_p} = \sum_h C(h)e\left(x\left(h_0/L + \sum_p h_p/p\right)\right), \qquad C(h) = \widehat{\phi}(h_0)\prod_p C_{p,j_p}(h_p).$$

This inversion includes nonunit h_0 and zero local frequencies. These multipliers can fail the unit-Fourier-support hypothesis of [7, Lemma 5.2 and Corollary 5.1], so the finite theta sum below supplies the required transform directly. The theta function with this multiplier on its infinity coefficient $x = \lambda^3\mu \in \mathcal{O}$ is the finite sum

$$\mathcal{F}(z,v) = \sum_h C(h)\bar{\theta}(z + z_h, v), \qquad z_h = \lambda^2\left(h_0/L + \sum_p h_p/p\right). \tag{4.9}$$

For an isolated infinity index $x = nb^3$, the piecewise definition of ϕ first discards every term for which $(nb, \mathfrak{E}) \neq 1$. This restriction applies to the whole completed index: $(nb^3, \mathfrak{E}) = 1$ if and only if both n and b are $\mathfrak{E}$-units. Only on that remaining domain do we use complete multiplicativity, including $\chi_b(\lambda)^6 = 1$, to write

$$\phi(nb^3)\prod_p \chi_p(nb^3)^{j_p} = \chi_n(\lambda)^2\Psi(n)\Psi(b)^3 \qquad ((nb, \mathfrak{E}) = 1).$$

It remains true when n, b share a moving prime, since the corresponding local powers retain their zero values, including for $j_p = 0$. For the discarded terms the completed coefficient is defined to be zero directly; no value of $\chi_n(\lambda)$ is evaluated there. Combining (4.6) and (4.7) on the retained domain now identifies the direct Mellin series with the product T. The dual coefficient functions d_H keep the full source support in (4.3); they are not restricted to indices prime to S.

Fix h_0 and declare p active precisely when $h_p \neq 0$. The identity (4.8) forces all primes with $j_p \neq 0$ to be active. Choose once for each h_0 mod L a representative and a reduced expression $\lambda^2 h_0/L = a_F/c_F$. Multiplying numerator and denominator by one unit, normalize the numerator to be $1 \pmod 3$ if $\lambda \mid c_F$ and the denominator to be $1 \pmod 3$ otherwise. Since every active prime is primary, this unit depends only on h_0. For $r = \prod_{p \text{ active}} p$, (4.9) then gives

$$z_h = \frac{a}{c}, \qquad c = c_F r, \qquad a = a_F r + \lambda^2 c_F \sum_{p \mid r}(r/p)h_p. \tag{4.10}$$

At a prime dividing c_F, the numerator is congruent to $a_F r$ and is a unit. At $p \mid r$, it is congruent to $\lambda^2 c_F (r/p) h_p$ and is again a unit. Thus this fraction is reduced, even when h_0 is a nonunit modulo L. The empty product $r = 1$ is allowed. Replacing any lift by another changes z_h by an element of $\lambda^2\mathcal{O} = 3\mathcal{O}$, a period of θ.

Choice of the cusp matrix and its multiplier. For each reduced cusp fraction, we choose a matrix whose residues at the fixed primes are constant within a sector. This isolates the moving-prime part of its automorphy multiplier and reflected additive phase. Set $M = \lambda^{12}L^4$. For every active p, the Chinese remainder theorem permits the chosen nonzero class h_p mod p to be lifted with $h_p \equiv 0 \pmod{M^2}$. Restrict r to one class modulo M^2. The identity (4.10) shows that a is then fixed modulo Mc_F; in fact its moving terms are divisible by $M^2 c_F$, and $a_F r$ is fixed modulo M^2, which is divisible by Mc_F because c_F divides L up to a unit. It also shows $a \equiv 1 \pmod 3$ if $\lambda \mid c$, and $c \equiv 1 \pmod 3$ otherwise.

Choose δ' by the following compatible local congruences:

$$\begin{aligned}
&\delta' \equiv a^{-1} \pmod{Mc_F} && \text{at every prime dividing } c_F,\\
&\delta' \equiv 0 \pmod{M} && \text{at every prime dividing } M \text{ but not } c_F,\\
&\delta' \equiv a^{-1} \pmod{p} && (p \mid r).
\end{aligned}$$

The first line means the indicated prime-power parts of Mc_F. All inverses exist because a/c is reduced. Put $b = (a\delta' - 1)/c$ and $g = \left(\begin{smallmatrix} a & b \\ c & \delta' \end{smallmatrix}\right)$. Then $g \in \Gamma$, and division of $a\delta' - 1$ by c gives

$$b \equiv 0 \pmod{M} \text{ at primes of } c_F, \qquad b \equiv -c^{-1} \pmod{M} \text{ at the other primes of } M.$$

Consequently a, b, c, δ' have fixed residues at the needed powers of every prime dividing L, independent of the classes h_p. Whenever a zero entry would occur in a residue-symbol calculation, one may first translate z_h by $\lambda^2 M^2 \mathcal{O}$ and then add a multiple of the combined congruence modulus to δ'. These changes preserve all displayed congruences and avoid the finitely many values making an entry zero. They do not change the translated theta function.

There are three cases, distinguished by $v_\lambda(c)$. If $3 \mid c$, put $H = I$. Then $G = g \in \Gamma_1(3)$, and cubic reciprocity for the primary a and the primary primes of r gives

$$\kappa(G) = \left(\frac{c_F}{a}\right)_3 \left(\frac{a}{r}\right)_3. \tag{4.11}$$

If $v_\lambda(c) = 1$, take the unique $u_0 \in \{\lambda, -\lambda\}$ with $u_0 \equiv c \pmod 3$ and put $H = \mathrm{L}(u_0)$; these are the two possibilities because $\mathcal{O}/(\lambda) \simeq \mathbb{F}_3$. The congruences above show that $G = gH^{-1} \in \Gamma_1(3)$. Write $A = a - u_0 b$. Both A and a are primary, and they are coprime: $(a, b) = 1$ and $(a, u_0) = 1$. The determinant equation gives

$$a(c - u_0\delta') = cA - u_0, \qquad bc \equiv -1 \pmod{a}.$$

It follows, by cubic reciprocity for a, A, that

$$\begin{aligned} \kappa(G) &= \left(\frac{-u_0}{A}\right)_3 \left(\frac{a}{A}\right)_3^{-1} \\ &= \left(\frac{-u_0}{A}\right)_3 \left(\frac{c/u_0}{a}\right)_3 = \left(\frac{-u_0}{A}\right)_3 \left(\frac{c_F/u_0}{a}\right)_3 \left(\frac{a}{r}\right)_3. \end{aligned}$$

For the second equality, reduce A modulo a and use $(A/a)_3 = (-u_0 b/a)_3 = (u_0/a)_3 (c/a)_3^{-1}$; the cubic symbol of -1 is one. Finally suppose $(\lambda, c) = 1$. Choose $u_0 = s + t\omega \equiv a \pmod 3$ with $0 \le s, t < 3$ and put $H = \left(\begin{smallmatrix} u_0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Here $c \equiv 1$, $\delta' \equiv 0$, and $b \equiv -1 \pmod 3$, so

$$G = gH^{-1} = \begin{pmatrix} -b & a + bu_0 \\ -\delta' & c + \delta' u_0 \end{pmatrix} \in \Gamma_1(3).$$

Both $-b$ and c are primary. Moreover $-bc = 1 - a\delta' \equiv 1 \pmod 9$. Factor $\delta' = u\lambda^k \delta_0$ with δ_0 primary. The supplementary laws make $(u\lambda^k/(1 - a\delta'))_3 = 1$; cubic reciprocity makes $(\delta_0/(1 - a\delta'))_3 = 1$ because the numerator after reciprocity is $1 \pmod{\delta_0}$. Thus

$$\left(\frac{\delta'}{-bc}\right)_3 = 1, \qquad \kappa(G) = \left(\frac{\delta'}{-b}\right)_3 = \left(\frac{\delta'}{c}\right)_3^{-1} = \left(\frac{a}{c}\right)_3 = \left(\frac{a}{c_F}\right)_3 \left(\frac{a}{r}\right)_3.$$

The last inverse uses $a\delta' \equiv 1 \pmod c$.

In all three cases we have proved

$$\kappa(G) = \kappa_F \prod_{p \mid r} \chi_p(a)^2, \qquad |\kappa_F| = 1. \tag{4.12}$$

The factor κ_F is constant as the active h_p vary in the fixed sector. If $3 \mid c$, it is $(c_F/a)_3$: the unit and λ-power factors are determined by a mod 9, and reciprocity determines the remaining fixed-prime factors from the residue of a at the primes of c_F. In the middle case it is $(-u_0/A)_3((c_F/u_0)/a)_3$, determined in the same way by the fixed residues of $A = a - u_0 b$ and a. In the last case it is $(a/c_F)_3$, which is determined directly by the fixed denominator c_F and $a \bmod c_F$. The

displayed congruences for a, b, c, δ' therefore determine κ_F using only h_0 and r mod M^2. They also determine the chosen one of the three functions Θ_b.

The reflected additive phase. Write $x = \lambda^4\mu$ and $D_F = \lambda^3 c_F$. At each $p \mid r$, (4.10) gives $a \equiv \sigma_p h_p \pmod p$. Hence $\delta' \equiv \sigma_p^{-1} h_p^{-1} \pmod p$. Additive Chinese remaindering for the coprime factors of $D_F r$ gives the exact identity

$$\begin{aligned} \check{e}(-\delta'\mu/c) &= e(-\delta' x/(D_F r)) \\ &= \vartheta(x) \prod_{p\mid r} e(\epsilon_p h_p^{-1} x/p), \qquad \vartheta(x) = e(-\delta'_F r^{-1} x/D_F). \end{aligned} \tag{4.13}$$

Here δ'_F is the class of δ' modulo D_F, and r^{-1} in the formula for ϑ is taken modulo D_F. For example, the local numerator at p is $-\delta'(D_F r/p)^{-1} = \epsilon_p h_p^{-1}$; this verifies the signs and the powers of λ. The chosen congruences fix δ'_F, since D_F divides the corresponding local moduli, and the sector fixes r^{-1} mod D_F. A common multiple of the finitely many D_F is therefore a fixed modulus for all the characters ϑ.

Conjugating (4.12) contributes $\bar\kappa_F \prod_{p\mid r} \chi_p(\sigma_p)^{-2}\chi_p(h_p)^{-2}$. For $j \neq 0$, multiplication by (4.8) leaves the local sum

$$q_p^{-1/2}\tau^-_{p,j} \sum_{h\neq 0} \chi_p(h)^{-j-2} e(\epsilon_p h^{-1} x/p).$$

Upon putting $y = h^{-1}$, the sum is a Gauss sum of exponent $j+2$. If $j \neq 4$, its value is $q_p^{1/2}\tau^+_{p,j+2}\chi_p(\epsilon_p x)^{-j-2}$, also at $p \mid x$ because both sides then vanish. If $j = 4$, it is the Ramanujan sum

$$\sum_{y\neq 0} e(\epsilon_p xy/p) = -1 + q_p \mathbf{1}_{p\mid x}.$$

For an active $j = 0$, the same substitution gives

$$-q_p^{-1} \sum_{h\neq 0} \chi_p(h)^{-2} e(\epsilon_p h^{-1} x/p) = -q_p^{-1/2}\tau^+_{p,2}\chi_p(\epsilon_p x)^{-2}.$$

These are exactly $\omega_{p,j} B_p(x)$; an inactive zero exponent contributes $1 - q_p^{-1}$. This proves the local factors of Proposition 4.2, including their zero extensions. The choices of h_0 and the active subset give at most $q_L 2^{\#\{p:j_p=0\}}$ terms. Splitting the $j_p = 4$ factors multiplies this by at most $2^{\#\{p:j_p=4\}}$, proving the stated counts.

The denominator and the fixed-sector data constructed before these local sums depend on the local prime set only through h_0 and the active radical r; the remaining active-frequency dependence is exactly the local dependence just summed. We use the same representatives and normalized fractions for each h_0, and the same fixed-sector choices of cusp function, fixed additive character, and κ_F whenever those data recur. An absent prime and an inactive zero-exponent prime have the same active radical; their other active σ_p, ϵ_p are therefore also identical. The inactive coefficient $1 - q_p^{-1}$ is the only change. This proves the simultaneous branch compatibility in Proposition 4.2.

The marked reflection in Section 13.2 will use the following dependence on pairs of active primes. Since

$$\epsilon_p = -\lambda^{-5}(c/p)^{-2} \pmod p,$$

there is a scalar $\xi_{p,j}$ of absolute value one, depending on p, j but on no other active prime such that

$$\chi_p(\sigma_p)^{-2}\omega_{p,j} = \xi_{p,j}\chi_p(c/p)^{2j+2}. \tag{4.14}$$

For $j = 4$ this uses $2j + 2 \equiv -2 \pmod 6$; for the other exponents it follows immediately by inserting the expression for ϵ_p. Thus the contribution of another active prime to the phase at p has exponent exactly $2j + 2$ modulo six. In particular an active exponent $j = 1$ has column coupling $\chi_p(x)^{-3} = \chi_p(x)^3$.

Mellin normalization and continuation. Let

$$J(s) = \int_0^\infty \partial_{\bar z}\mathcal{F}(z,v)|_{z=0}\, v^{2s-1}\, dv.$$

The Bessel identity used in [7, Lemma 5.3] is

$$\int_0^\infty K_{1/3}(y) y^{w-1}\, dy = 2^{w-2}\Gamma((w-1/3)/2)\Gamma((w+1/3)/2) \qquad (\Re w > 1/3).$$

On $\Re s > 1$ the coefficient sums converge absolutely, so we may differentiate and integrate their expansions term by term. Since $\partial_{\bar z}\check e(\mu z) = 2\pi i\bar\mu \check e(\mu z)$, the identity with $w = 2s+1$ gives

$$J(s) = \frac{i}{4}(2\pi)^{-2s}\Gamma(s+1/3)\Gamma(s+2/3) \sum_{0\neq\mu\in\lambda^{-3}\mathcal{O}} d_I(\mu)\phi(\lambda^3\mu)\prod_p \chi_p(\lambda^3\mu)^{j_p}\bar\alpha(\mu) q_\mu^{-s}.$$

The whole-index restriction first gives $(nb, \mathfrak{E}) = 1$, the domain on which (4.6) and (4.7) identify the coefficient. Using $q_{\lambda^{-3}} = 27^{-1}$ and $\bar\alpha(\lambda^{-3}) = -i$ gives

$$J(s) = \frac{3^{5/2}}{4}\left(\frac{27}{(2\pi)^2}\right)^s \Gamma(s+1/3)\Gamma(s+2/3)T(s,\Psi). \tag{4.15}$$

For one translated term $z_h = a/c$, use the matrix $g = GH$ constructed above. The coordinate action of g^{-1} on hyperbolic space, from [7, Equation (5.1)], gives

$$g^{-1}(z + a/c, v) = \left(-\frac{\delta'}{c} - \frac{\bar z}{c^2(v^2+|z|^2)}, \frac{v}{q_c(v^2+|z|^2)}\right).$$

At $z = 0$, the derivative of its first coordinate with respect to $\bar z$ is $-(cv)^{-2}$; the derivatives of its conjugate coordinate and of its vertical coordinate with respect to $\bar z$ are zero. The chain rule and $\bar\theta(gw) = \bar\kappa(G)\bar\theta(Hw)$ therefore give

$$\partial_{\bar z}\bar\theta(z+a/c,v)\big|_{z=0} = -\frac{\bar\kappa(G)}{c^2v^2}\,\partial_z\bar\theta(H(z,1/(q_cv)))\big|_{z=-\delta'/c}.$$

In particular the derivative removes $C_H(v)$ from every cusp expansion in (4.4). Substituting that expansion and then $v \mapsto 1/(q_cv)$ in the integral gives, for $\Re s < 0$, the contribution

$$\begin{aligned} -\frac{i}{4}(2\pi)^{2s-2}\bar\alpha(c)^2 q_c^{1-2s}\bar\kappa(G)\prod_{\pm}\Gamma(3/2-s\pm 1/6) \\ \times \sum_{\mu\neq 0} d_H(\mu)\alpha(\mu)\check e(-\delta'\mu/c) q_\mu^{-(1-s)}. \end{aligned} \tag{4.16}$$

To verify the scalar directly, before applying the Bessel integral the factor is $-2\pi i c^{-2}q_c^{2-2s}$ and the remaining integral is $\int_0^\infty u^{2-2s}K_{1/3}(4\pi|\mu|u)\,du$. The Bessel identity with $w = 3-2s$, together with $c^{-2} = \bar\alpha(c)^2/q_c$, gives exactly the display.

We justify both the continuation and the contour operations here. The coefficient bound in (4.3) implies absolute convergence of the reflected series when $\Re(1-s) > 1$: the separate majorants are

$$\sum_n{}^{\mathrm{sf}} q_n^{-(1-\Re s)}, \qquad \sum_b q_b^{1/2-3(1-\Re s)}, \qquad \sum_{k\geq -4} 3^{k(1/6-(1-\Re s))}.$$

Each converges in that region. The fixed support has a positive lower bound for $|\mu|$. The exponential decay of $K_{1/3}$ and of all its derivatives, the coefficient bound, and the direct expansion imply exponential decay at $v\to\infty$ for every logarithmic derivative of each translated

derivative. The chain-rule formula just obtained implies exponential decay at $v \to 0$ as well, with constants depending on the fixed translated term. Repeated integration by parts in $\log v$ now shows that J is entire and decreases faster than every power of $|\Im s|$ on each fixed strip.

The identity (4.15), with reciprocal gamma functions on its right, continues T to an entire function. Its direct series is bounded on any line to the right of 1. The identity (4.16) and Stirling's formula give a polynomial bound on any line to the left of 0; the exponential parts of the two gamma products cancel. On a strip between such lines, the rapid bound for J and the reciprocal gamma factors first give $|T(\sigma+it)| \ll (1+|t|)^C e^{\pi|t|}$ for some C. Divide T by a sufficiently large power of $B+s$, with B chosen so that $B+s$ has no zero on the strip, and multiply by $e^{\varepsilon s^2}$. The horizontal sides of a growing rectangle tend to zero for each $\varepsilon > 0$. The maximum principle, followed by $\varepsilon \to 0$, transfers the polynomial bounds on the two vertical sides to the strip. This proves the stated polynomial strip growth.

The entire continuation, polynomial strip bound, and rapid Mellin decay permit shifting the integral on the left of (4.2) from $\Re s = a$ to $\Re s = 1/2 - \sigma < 0$ with $\sigma > 1/2$: (3.14) applied to the weighted logarithmic derivatives of V makes the horizontal joins tend to zero. Inserting (4.16) divided by (4.15), and putting $t = 1/2 - s$, gives the gamma quotient $R(t)$ and the factor

$$-\frac{i}{81}\bar{\alpha}(c)^2 q_\mu^{-1/2}\left(\frac{Kq_\mu X}{q_c^2}\right)^{-t}.$$

Indeed $(2\pi)^{4s-2}27^{-s}3^{-5/2} = 3^{-4}K^{-t}$ and $q_c^{1-2s}q_\mu^{s-1} = q_\mu^{-1/2}(q_\mu/q_c^2)^{-t}$. The reflected coefficient series is absolutely convergent on this line, so it may be interchanged with the integral. Summing its finite local factors using (4.13) and the computed Gauss sums proves (4.2) and (4.1). Since $|\phi| \le 1$, we have $|\widehat{\phi}(h_0)| \le 1$; every other factor in the scalar has absolute value at most one. This proves $|\zeta| \le 1/81$.

The absolute convergence of the dual sums and the independence of the kernel contour follow from the transformed-test estimates proved next. □

4.3 Bounds for the reflected test

For the same class of tests V, for $A \ge 0$ and an integer $B \ge 0$, define

$$\mathfrak{M}_{A,B}(V) = \sup_{-A \le \eta \le 1/4} \int_{\mathbb{R}} (1+|u|)^B |\widehat{V}(\eta+iu)|\, du.$$

Lemma 4.3 (Bounds for the reflected test). *Let V satisfy the smoothness and endpoint-decay conditions of Proposition 4.1, and let $V^\sharp$ be its reflected test. For every $A \ge 0$ and every integer $j \ge 0$,*

$$|(x\partial_x)^j V^\sharp(x)| \ll_{A,j} \min\{x^{1/4}, (1+x)^{-A}\}\mathfrak{M}_{A,\lceil 4A\rceil+j+2}(V). \tag{4.17}$$

If $\chi \in C_c^\infty(0,\infty)$ is fixed, $Y > 0$, and $J \ge 0$ is an integer, then

$$\int_{\mathbb{R}} (1+|u|)^J \left|\int_0^\infty \chi(y) V^\sharp(Yy) y^{-iu}\frac{dy}{y}\right| du \ll_{A,J,\chi} \min\{Y^{1/4}, (1+Y)^{-A}\}\mathfrak{M}_{A,\lceil 4A\rceil+J+4}(V). \tag{4.18}$$

These are finite smooth seminorms. More precisely,

$$\mathfrak{M}_{A,B}(V) \ll_{A,B} \int_0^\infty (1+x^{-A}+x^{1/4})\big(|V(x)| + |(x\partial_x)^{B+2}V(x)|\big)\frac{dx}{x}. \tag{4.19}$$

Replacing $V(x)$ by $x^{iu_0}V(x)$ multiplies $\mathfrak{M}_{A,B}$ by at most $(1+|u_0|)^B$.

Proof. The first negative pole of the numerator of $R(t)$ is $-5/6$. The reciprocal denominator gamma functions are entire. Hence the integrand defining $V^\sharp$ is holomorphic for $\Re t > -5/6$. On $-1/4 \le \sigma \le A$, Stirling's formula gives

$$|R(\sigma+iu)| \ll_A (1+|u|)^{4\sigma}.$$

The estimate is uniform also for bounded u, since this closed strip has no numerator pole. The bound (3.14) makes $\widehat{V}$ rapidly decreasing pointwise on every vertical strip. Rectangular contour shifts are therefore valid within $-1/4 \le \Re t \le A$, including between any two nonnegative lines. Applying $(x\partial_x)^j$ contributes $(-t)^j$. The three lines $-1/4, 0, A$ bound the resulting integral respectively by constant multiples of

$$x^{1/4}\mathfrak{M}_{A,\lceil 4A\rceil+j+2}(V), \qquad \mathfrak{M}_{A,\lceil 4A\rceil+j+2}(V), \qquad x^{-A}\mathfrak{M}_{A,\lceil 4A\rceil+j+2}(V).$$

Combining them proves (4.17). The rapid large-x bound, with A arbitrarily large, also makes every dual sum in (4.2) absolutely convergent when $V^\sharp$ is represented on the zero line. Thus the shift of the kernel contour does not require a termwise shift of a conditionally convergent Dirichlet series.

For (4.18), set $f_Y(v) = \chi(e^v)V^\sharp(Ye^v)$. This is supported on a fixed compact interval. Integrating its Fourier transform by parts $J+2$ times and using the trivial bound for $|u| \le 1$ gives

$$\int_{\mathbb{R}} (1+|u|)^J |\widehat{f_Y}(u)|\, du \ll_J \sum_{j=0}^{J+2} \|f_Y^{(j)}\|_{L^1(\mathbb{R})}.$$

Leibniz's rule and (4.17) bound each norm by the right side of (4.18), because e^v stays in a fixed compact subinterval of $(0,\infty)$. Finally, for $|u| \ge 1$, $B+2$ integrations by parts give

$$|\widehat{V}(\eta+iu)| \le |\eta+iu|^{-B-2} \int_0^\infty |(x\partial_x)^{B+2}V(x)|x^\eta \frac{dx}{x}.$$

For $|u| \le 1$ use the corresponding undifferentiated integral. On $-A \le \eta \le 1/4$, $x^\eta \le 1+x^{-A}+x^{1/4}$, proving (4.19). The identity $\widehat{x^{iu_0}V}(\eta+iu) = \widehat{V}(\eta+i(u+u_0))$ and $1+|u| \le (1+|u+u_0|)(1+|u_0|)$ give the asserted norm-twist bound. The Gaussian of Lemma 3.11 satisfies these test hypotheses: its logarithmic derivatives are polynomials times the Gaussian, giving arbitrary power decay at both endpoints. Its Mellin transform is the one recorded there, and $\mathfrak{M}_{A,J}$ is finite for every finite A, J. Thus it is directly admissible in Proposition 4.1. □

4.4 Element rows and fixed sectors

To apply the reflection to a character $A \mapsto \chi_A(m)$, we separate the fixed supplementary phases from the good prime divisors of the row. The separation retains the zero at every shared good prime.

Lemma 4.4 (Fixed ray sectors for nonzero rows). *Let Ψ_{base} range over a fixed finite family of finite-ray characters with conductors supported on S, extended by zero off the primary elements prime to S. With the generators $\pi_{\mathfrak{l}}$ for $\mathfrak{l} \in S$ from Lemma 3.2, recall the unique factorization of a row $0 \ne m \in \mathcal{O}$:*

$$m = um_S m_{\mathrm{good}}, \qquad m_S = \prod_{\mathfrak{l}\in S} \pi_{\mathfrak{l}}^{v_{\mathfrak{l}}(m)}, \qquad m_{\mathrm{good}} = \prod_{p\notin S} p^{v_p(m)},$$

where the good prime generators are primary and u is a unit. On every primary element A prime to S, (3.9) reads

$$\chi_A(m) = \chi_A(um_S)\mathcal{R}(A, m_{\mathrm{good}}) \prod_{p\mid m_{\mathrm{good}}} \chi_p(A)^{v_p(m)}. \tag{4.20}$$

All powers on the right retain their zero values, including when $v_p(m) \equiv 0 \pmod 6$. Fix u, the valuations $v_{\mathfrak{l}}(m)$ modulo six, and the class of m_{good} in the fixed ray group through which $\mathcal{R}$ factors. In such a sector define

$$\Psi_0^{[m]}(A) = \begin{cases} \Psi_{\mathrm{base}}(A)\chi_A(um_S)\mathcal{R}(A, m_{\mathrm{good}}), & A \equiv 1 \pmod 3,\ (A,S)=1, \\ 0, & \text{otherwise.} \end{cases} \tag{4.21}$$

The nonzero branch is a member of a fixed finite family of finite-ray characters with conductors supported on S. There exist a single $\mathfrak{E}$ and a single L satisfying the conditions of Proposition 4.1 simultaneously for all these sectors. The local prime set contains every $p \mid m_{\mathrm{good}}$, with exponent $v_p(m)$ modulo six, even when this exponent is zero. Every moving good prime remains in the local prime set, outside the fixed modulus L.

Proof. The identity (4.20) is (3.9), which proves the identity both on coprime pairs and at shared good primes, with the same zero convention.

On this domain $\chi_A(um_S)$ depends only on u and the S-valuations modulo six, because every numerator factor is a unit modulo A. For each of the finitely many representatives $d = u\prod_{\mathfrak{l}\in S}\pi_{\mathfrak{l}}^{e_{\mathfrak{l}}}$, $0 \le e_{\mathfrak{l}} < 6$, Lemma 3.2 identifies $A \mapsto \chi_A(d)$ with a finite-ray character whose conductor is supported at primes over $6d$, all in S. Once the good ray class is fixed, $A \mapsto \mathcal{R}(A, m_{\mathrm{good}})$ is one of a fixed finite family of characters supported at the primes over $2, 3$. Their products with the finite family of base characters give the asserted fixed family. Choosing a common multiple of its conductors and the modulus enforcing coprimality to S gives the common $\mathfrak{E}$ and then L. Additional good local residue-symbol factors in a completed coefficient are placed in $\mathcal{P}$ and combined at a shared prime with the same zero convention; they do not alter this fixed modulus. The row conversion applies to $m \neq 0$. □

4.5 Common profiles and lattice tails

The reflected series has a scale depending on its row. We keep that dependence inside one joint smooth profile until after Fourier inversion; this gives one coefficient measure for the whole row norm.

Lemma 4.5 (Common annular kernel profile). *Let V satisfy the test conditions of Proposition 4.1, and let $V^\sharp$ be the kernel bounded in Lemma 4.3. Fix an integer $d \ge 1$, a compact set $\Omega \subset \mathbb{R}^d$, and real exponents $a_1, \dots, a_d$. Let W be one smooth profile with logarithmic support in Ω, common to all rows in a given block, and let $Y > 0$ be fixed within that block. Put*

$$h(\mathbf{y}) = \prod_{i=1}^{d} y_i^{a_i}, \qquad F_Y(\mathbf{y}) = W(\mathbf{y})V^\sharp(Yh(\mathbf{y})), \qquad \mathrm{m}_A(Y) = \min\{Y^{1/4}, (1+Y)^{-A}\}.$$

For $A \ge 0$ and integers $j, J \ge 0$,

$$p_j(F_Y) \ll_{A,j,d,\Omega,\mathbf{a}} \mathrm{m}_A(Y)p_j(W)\mathfrak{M}_{A,\lceil 4A\rceil+j+2}(V), \tag{4.22}$$

$$\|F_Y\|_{J,\mathrm{sep}} \ll_{A,J,d,\Omega,\mathbf{a}} \mathrm{m}_A(Y)p_{J+d+2}(W)\mathfrak{M}_{A,\lceil 4A\rceil+J+d+4}(V). \tag{4.23}$$

Here p_j is the homogeneous seminorm of a single profile. The constants are independent of Y and of the norm labels within the block.

Proof. On the support of W, the monomial h is bounded above and below by positive constants depending only on Ω and the exponents. Each operator $y_i\partial_{y_i}$ acting on $V^\sharp(Yh(\mathbf{y}))$ is a_i times the Euler derivative of $V^\sharp$ at $Yh(\mathbf{y})$. It produces no additional power of Y. Leibniz's rule and (4.17) prove (4.22). Applying Lemma 3.6 with $j = J + d + 2$ gives (4.23). □

Remark. *The common profile in the row norm.* Let $\mathcal{I}$ be the original row set, and suppose a row vector $\mathcal{U}$ is linear in the whole profile F_Y. The profile includes the cutoffs for the full fixed logarithmic boxes of its row and column norm variables. Assume that simultaneous logarithmic Fourier inversion expresses the vector using one density common to every row. After the relevant arithmetic coefficient independences have been verified, write $F_Y = \mathrm{m}_A(Y)\widetilde{F}_Y$. Minkowski gives

$$\|\mathcal{U}\|_{\ell^2(\mathcal{I})} \le \frac{\mathrm{m}_A(Y)}{(2\pi)^d}\int_{\mathbb{R}^d} |\widehat{\widetilde{F}_Y}(\mathbf{t})|\|B(\mathbf{t})\|_{\ell^2(\mathcal{I})}\,d\mathbf{t}. \tag{4.24}$$

Here $B(\mathbf{t})$ is the separated row vector, whose norm powers all have absolute value one. The identity underlying this inequality is a Bochner integral in $\ell^2(\mathcal{I})$; the finite separation norm above ensures its absolute integrability whenever the separated norm has the stated polynomial height growth.

The nonnegative norm on the right may then be enlarged, using the same separated formula for B on the added rows. The kernel comparison with Y is used on the original row set only.

Squaring this inequality gives the factor $\mathrm{m}_A(Y)^2$. If only the small-argument bound is being used, the same argument permits the weaker scalar $\min\{1, Y^{1/4}\}$, whose square is $\min\{1, Y^{1/2}\}$. The normalized profile has bounded tuple seminorms. Their definition includes a constant one, so the extracted smallness belongs to the scalar. A kernel occurring once in an already expanded quadratic expression instead contributes its scalar once, not automatically its square.

Compact logarithmic support in Lemma 4.5 is required of the joint factor W. For the Gaussian itself, the annular decomposition of Lemma 3.11 supplies summable separation norms.

Lemma 4.6 (Lattice kernel tails). *Let Λ be either $\mathcal{O}$ or $\lambda^{-4}\mathcal{O}$. Suppose $K : (0,\infty) \to \mathbb{C}$ satisfies $|K(x)| \le C_A x^{-A}$ for $x \ge 1$, where $A > 1$. Then, for $a > 0$ and $U \ge 1$,*

$$\sum_{\substack{0 \neq \mu \in \Lambda \\ aq_\mu > U}} |K(aq_\mu)| \ll_A C_A(1 + a^{-1})U^{1-A}. \tag{4.25}$$

For $K = V^\sharp$ one may take $C_A \ll_A \mathfrak{M}_{A,\lceil 4A\rceil+2}(V)$.

Proof. The shell $2^j U < aq_\mu \le 2^{j+1}U$ contains $O(1 + 2^j U/a)$ points of either fixed lattice. Its contribution is at most $C_A(1 + 2^j U/a)(2^j U)^{-A}$. Summing the two geometric series gives $O_A(C_A(U^{-A} + a^{-1}U^{1-A}))$, which implies the display because $U \ge 1$. The last assertion follows from (4.17). □

4.6 A quadratic norm for completed indices

The next estimate bounds the quadratic character left by reflection on a squarefree factor times a cube. We begin with the imported sieve. For squarefree primary ideals outside a fixed set containing the primes over 6, the quadratic kernel $\chi_a(b)^3$ satisfies

$$\sum_{\substack{a \text{ sf} \\ q_a \le U}} \left| \sum_{\substack{b \text{ sf} \\ q_b \le V}} c_b \chi_a(b)^3 \right|^2 \ll_\epsilon (UV)^\epsilon (U + V) \sum_b |c_b|^2. \tag{4.26}$$

This is Goldmakher–Louvel's quadratic large sieve [12, Definition 1 and Theorem 1.1]. To verify its family hypotheses, let $\vartheta_a((x))$ denote the quadratic residue symbol evaluated at the primary generator of (x). The adjustment of x to its primary generator contributes $\epsilon_\lambda(x)^{e(a)}$, where ϵ_λ is the nontrivial character modulo λ and $e(a) = (q_a - 1)/2 \bmod 2$. The powers of ω contribute nothing because they are squares. CRT therefore gives exact primitive conductor $a\lambda^{e(a)}$ and trivial infinite type. We use the primitive inducing character: the displayed formula first specifies its restriction to the λ-units, and the factor at λ is absent when $e(a) = 0$. This does not change any value on the indices in (4.26). In a fixed primary square class modulo 4, two indices have the same $e(a)$; for coprime such indices the primitive product has conductor exactly their product. The reciprocity factor is the fixed bicharacter $\mathcal{R}$ of (3.5). These are precisely the family conditions of the cited theorem. A finite ray partition and transpose duality give (4.26) in the displayed orientation.

A fixed restriction on the row set decreases the positive outer sum. A fixed restriction on the coefficient support is implemented by setting the omitted coefficients to zero. A coprimality

restriction depending on both a row and a column requires a separate reduction. The next lemma resolves the restriction produced by square factors of a completed index.

Definition 4.7 (Data for the quadratic norm). Let $K, N, B \geq 1$. Let $\mathcal{I}$ be any set of squarefree primary good ideals k with $q_k \ll K$. Let n range over squarefree primary ideals with $q_n \asymp N$, and let b range over primary ideals with $q_b \asymp B$. The ideals n, b may contain primes of S other than λ. Let $\beta(n, b)$ be arbitrary complex coefficients independent of k. Fixed restrictions on $\mathcal{I}$ and fixed restrictions on the (n, b)-support are allowed.

Factorization and ranges. Write uniquely $b = gt^2$ with g squarefree, and set $c = (n, g)$, $n = cm$, $g = ch$. The factors c, m, h are pairwise coprime and squarefree. Fix one set of dyadic ranges

$$q_c \asymp C, \qquad q_g \asymp G, \qquad q_t \asymp T, \qquad B \asymp GT^2,$$

where all comparison constants are fixed.

Let $\mathcal{Q}_{C,G,T}(\beta)$ be the sum over $k \in \mathcal{I}$ of the squared absolute value of the (n, b)-sum restricted to this set of ranges, with summand $\beta(n, b)\chi_k(nb)^3$.

For a squarefree good ideal r and a fixed t, let $\mathcal{D}_r(t)$ be the set of triples (c', m, h) for which

$$c = rc', \qquad n = rc'm, \qquad b = rc'ht^2$$

belongs to the original support and to these ranges, with the stated squarefreeness and pairwise coprimality of c, m, h.

Lemma 4.8 (Quadratic reduction for completed indices). *Let K, N, B, the row set $\mathcal{I}$, the coefficient array $\beta(n, b)$, and the dyadic factors c, g, t satisfy Definition 4.7, including its squarefreeness, coprimality, support and coefficient-independence conditions. For $\mathcal{Q}_{C,G,T}$ and $\mathcal{D}_r(t)$ in that definition,*

$$\mathcal{Q}_{C,G,T}(\beta) \ll_\epsilon (KNB)^\epsilon T \sum_{q_t \asymp T} \sum_{\substack{r \text{ sf, } (r,S)=1 \\ q_r \ll \min(K,C)}} \left(\frac{K}{q_r} + \frac{NG}{C^2}\right) \frac{C}{q_r} \sum_{(c',m,h)\in\mathcal{D}_r(t)} |\beta(rc'm, rc'ht^2)|^2. \tag{4.27}$$

Empty quotient ranges contribute zero; all bounded ranges, including the unit ideal, are included by the fixed comparison constants. Fixed ray sectors and bounded row scalars are permitted. In particular, if $|\beta(n, b)| \leq 1$, then

$$\sum_{k\in\mathcal{I}} \left| \sum_{n,b} \beta(n, b)\chi_k(nb)^3 \right|^2 \ll_\epsilon (KNB)^\epsilon (K + NB)NB. \tag{4.28}$$

The constants may depend on the fixed support comparisons and on S, but are uniform in the coefficients and in the permitted support restrictions.

Proof. We first expand the collision condition, then apply the quadratic sieve to the remaining squarefree index. Counting the grouped coefficients gives the refined inequality and its bounded-coefficient consequence.

Resolving the coprimality restriction. The factorization $b = gt^2$ permits g and t to share primes. Since $nb = c^2mht^2$, the zero convention gives the exact identity

$$\chi_k(nb)^3 = \chi_k(mh)^3 \mathbf{1}_{(k,ct)=1}. \tag{4.29}$$

Indeed a square has sixth power under the displayed quadratic character; this is one on units and zero at a collision. The identity therefore also holds when n or g shares a prime with t.

For the chosen ranges there are $O(T)$ possible t. Hilbert-space Cauchy gives

$$\left\| \sum_{q_t \asymp T} F_t \right\|_2^2 \ll T \sum_{q_t \asymp T} \|F_t\|_2^2.$$

Fix t in the positive sum. The condition $(k,t)=1$ is now a fixed row restriction. For the remaining condition use the full identity

$$\mathbf{1}_{(k,c)=1} = \sum_{r \mid (k,c)} \mu(r).$$

For each k the number of possible r is at most $d_{\mathcal{O}}(k)$. Rowwise Cauchy followed by summation over k bounds the norm by $(KNB)^\epsilon$ times a positive sum over fixed squarefree good r. Write $k = rk'$ and $c = rc'$. Squarefreeness gives the fixed row restriction $(k', r) = 1$ and the coefficient restriction $(c', r) = 1$; terms with $(r,t) > 1$ vanish, and otherwise $(k', t) = 1$ is another fixed row restriction. The full expansion has already resolved this coprimality restriction, so the individual r-summand has no remaining (k', c') condition. A fixed ray sector for k becomes a fixed sector for k' once r is fixed.

The squarefree quadratic sieve. The factor $\chi_r(mh)^3$ is a bounded column factor. Split the squarefree parts of m, h supported on S from their good parts: $m = m_S m_g$ and $h = h_S h_g$. There are only finitely many choices of m_S, h_S. Their symbols are bounded row factors, while $j = m_g h_g$ is squarefree and good, of norm $O(NG/C^2)$. For fixed r, t, m_S, h_S, group the remaining columns by j. If C_j is the resulting coefficient, the quadratic large sieve gives

$$\sum_{k'} \left| \sum_j C_j \chi_{k'}(j)^3 \right|^2 \ll_\epsilon (KNB)^\epsilon \left(\frac{K}{q_r} + \frac{NG}{C^2} \right) \sum_j |C_j|^2.$$

The outer sum may retain all its fixed restrictions. On a nonempty range, both quotient lengths in the last display are bounded below by a fixed positive constant; replacing them by their maxima with one changes only the comparison constant.

For each j, the number of factorizations into m_g, h_g is divisor-bounded. There are $O(C/q_r)$ possible c' in its range. Cauchy over these two choices therefore gives

$$\sum_j |C_j|^2 \ll_\epsilon (KNB)^\epsilon \frac{C}{q_r} \sum_{(c', m, h) \in \mathcal{D}_r(t)} |\beta(rc'm, rc'ht^2)|^2.$$

The bounded column factor from χ_r has been discarded only in this positive sum. The fixed bad-part symbols were row factors of unit modulus and were removed from the outer modulus before applying the sieve. The ideals m, h here are the original ideals, reconstructed from their fixed S-parts; the S-part of c' is untouched. Combining the displays proves (4.27).

Bounded coefficients. Suppose now $|\beta| \le 1$. Ideal counting gives

$$\#\mathcal{D}_r(t) \ll \frac{C}{q_r} \frac{N}{C} \frac{G}{C}.$$

For $q_r \asymp r_0$, its product with the preceding factor C/q_r is $O(NG/r_0^2)$. There are $O(r_0)$ possible r in this range and $O(T)$ possible t. Since $B \asymp GT^2$, their contribution to (4.27) is at most

$$(KNB)^\epsilon \frac{NB}{r_0} \left(\frac{K}{r_0} + \frac{NG}{C^2} \right) \ll (KNB)^\epsilon NB(K + NB).$$

Here r_0, C, T are bounded below on a nonempty range and $G \ll B$. The number of dyadic choices for C, G, T, r_0 is logarithmic in the norm ranges. Hilbert-space triangle over the C, G, T blocks, followed by a rescaling of ϵ, proves (4.28). □

4.7 The completed sum and the equal-scale example

For a fixed finite-ray character ν, extended by zero away from primary elements prime to S, and for $m \neq 0$, define the central completed sum

$$\mathcal{C}_V(X; m, \nu) = \sum_{\substack{c \text{ sf},\ n \\ c,n\equiv 1\ (3) \\ (cn,S)=1}} \frac{\gamma_2(c)\overline{\alpha(cn^3)}\nu(cn^3)\chi_{cn^3}(m)}{\sqrt{q_c}\, q_n} V\left(\frac{q_c q_n^3}{X}\right). \tag{4.30}$$

Here c, n may share primes. The defining series is absolutely convergent for every $X > 0$ when V satisfies the hypotheses of Proposition 4.1. Complete multiplicativity, with the zeros retained, gives $\chi_{cn^3}(m) = \chi_c(m)\chi_n(m)^3$. Thus for every $a > 1$,

$$\mathcal{C}_V(X; m, \nu) = \frac{1}{2\pi i}\int_{(a)} \widehat{V}(s - 1/2) X^{s-1/2} T(s, \nu\chi_\bullet(m))\, ds. \tag{4.31}$$

Indeed, absolute convergence permits termwise Mellin inversion on that line.

We bound the mean square of this completed sum over nonzero element rows. At equal completed and row norm scales, the goal is

$$\sum_{0<q_m\le C_m Z} |\mathcal{C}_V(Z; m, \nu)|^2 \ll Z^{1+\epsilon}\mathfrak{M}_{A,J}(V)^2$$

for every fixed $C_m \geq 1$ and $\epsilon > 0$, with suitable finite orders A, J. Lemma 4.11 below gives the more general estimate that tracks the repeated prime factors of the row.

The reason for the quadratic sieve is already visible for squarefree rows. Take $m = R$ primary, squarefree and good, with $q_R \asymp Z$, and set $X = Z$. Fix a row sector, a cusp sector, a dual unit u, and one integer $k \geq -4$ in the dual support. The row contributes only its valuation-one primes in this case. Every prime of R enters reflection with exponent one, so for $\mu = u\lambda^k nb^3$ its column factor gives

$$\chi_R(\lambda^4\mu)^3 = \chi_R(u\lambda^{k+4})^3\chi_R(nb)^3.$$

This identity retains all zeros, and n, b may share primes. The first factor depends only on R; the cusp coefficient and additive character are common throughout the sector. Their coefficient bound leaves the column normalization $q_n^{-1/2}q_b^{-1}$, up to a constant depending on the fixed k. Thus the only arithmetic interaction between the row and the two column indices is the kernel of Lemma 4.8.

On dual dyads $q_n \asymp U$, $q_b \asymp B$, the reflected kernel has argument comparable to UB^3/Z. Its rapid decay permits discarding whole blocks with $UB^3 > Z^{1+\tau}$ for any fixed $\tau > 0$; the tail argument below makes this restriction uniform when the ramified exponent and the other parameters vary. On the retained blocks, Lemma 4.5 separates the kernel with one common coefficient measure while retaining the original row annulus. Write $\mathcal{U}_{U,B,k}(R)$ for this dyadic piece of the reflected sum. For a retained block, the quadratic norm and the squared column normalization give

$$\sum_R |\mathcal{U}_{U,B,k}(R)|^2 \ll_{V,k,\tau,\epsilon} Z^\epsilon \frac{(Z + UB)UB}{UB^2} = Z^\epsilon \frac{Z + UB}{B} \ll_{V,k,\tau,\epsilon} Z^{1+\tau+\epsilon}.$$

Here $B \geq 1$ and $UB^3 \leq Z^{1+\tau}$ give the last inequality. This explains the equal-scale mean-square bound. The general argument must also sum the ramified exponents, retain the kernel saving at unequal scales, and treat repeated prime factors of the row. We hold the prime powers of the row whose exponents are at least two fixed and apply the same quadratic norm to its squarefree residual factor. The construction below records these local terms. Lemma 4.10 bounds a single block, and Lemma 4.11 sums the blocks over the repeated prime factors of the row.

4.8 Constructing the reflected blocks

Row and character data. Fix $Z \geq 2$, $X = Z^N$, one of the fixed characters Ψ_0 and its common modulus L from Proposition 4.1, and put $M_{\mathrm{ref}} = \lambda^{12} L^4$. Let V satisfy the test conditions of Proposition 4.1. For the current norm over R, hold fixed a finite set $\mathcal{F}$ of good primes with exponents $j_p \in \{0, \dots, 5\}$. Let R range over squarefree primary good ideals coprime to every prime of $\mathcal{F}$, in one fixed class modulo M_{ref}^2 and with any further fixed row restrictions. These restrictions are independent of the dual variables. On primary elements prime to S set

$$\Psi_R(A) = \Psi_0(A) \prod_{p \in \mathcal{F}} \chi_p(A)^{j_p} \prod_{p \mid R} \chi_p(A).$$

Every local power retains its zero. In the reflection for Ψ_R, fix h_0, the active or inactive decision at every $j_p = 0$ prime of $\mathcal{F}$, and one of the two summands of every $j_p = 4$ Ramanujan factor. All primes of R are active and have exponent one. Let F_{act} be the product of the active primes in $\mathcal{F}$.

Local assignments. For each chosen divisibility summand at $j_p = 4$, use the exact partition

$$\mathbf{1}_{p \mid n_0 b_0^3} = \mathbf{1}_{p \mid n_0} + \mathbf{1}_{p \nmid n_0} \mathbf{1}_{p \mid b_0}.$$

Choose one term. Let N_F be the product of the primes assigned to the squarefree ideal n_0, and let B_F be the product assigned to b_0 but not n_0. Write $n_0 = N_F n$ and $b_0 = B_F b$. In the first assignment only the occurrence in n_0 is extracted, even if b_0 contains that prime; in the second exactly one occurrence is extracted from b_0. All remaining source restrictions are retained on n, b. Let F_{sm} be the product of the small $j_p = 4$ summands and the active $j_p = 0$ primes, and put

$$A_0 = \log_Z q_{F_{\mathrm{act}}}, \qquad S_0 = \log_Z q_{F_{\mathrm{sm}}}, \qquad N_0 = \log_Z q_{N_F}, \qquad B_0 = \log_Z q_{B_F}.$$

Empty products have log-norm zero.

Norm coordinates. Fix a unit u, an integer $k \geq -4$, and dual ranges

$$\mu = u \lambda^k N_F n (B_F b)^3, \qquad q_n \asymp Z^v, \qquad q_b \asymp Z^{\ell_b}, \qquad q_{\lambda^k} = Z^{e_\lambda},$$

where $v, \ell_b \geq 0$. The squarefree n and the primary b retain the full source support of (4.3), including permitted primes of S and shared primes. Let $H \geq 0$, and choose a common smooth profile $W_1(y_R, y_n, y_b)$ on a fixed compact logarithmic box, where

$$y_R = q_R / Z^H, \qquad y_n = q_n / Z^v, \qquad y_b = q_b / Z^{\ell_b}.$$

Choose fixed smooth annular cutoffs $\chi_{\mathrm{row}}, \chi_n, \chi_b$ for these three ranges, and put

$$W_0(\mathbf{y}) = \chi_{\mathrm{row}}(y_R) \chi_n(y_n) \chi_b(y_b) W_1(\mathbf{y}).$$

The supports and invoked seminorms are fixed uniformly over the block. The primes in $\mathcal{F}$ may vary between invocations and remain outside the arithmetic modulus L. They are held fixed during the current norm over R.

Definition 4.9 (A reflected block). For the row character Ψ_R, the local prime assignments, and the norm coordinates in Section 4.8, denote by $\mathcal{U}_{v,\ell_b,e_\lambda}(R)$ the specified term of the right side of (4.2), restricted to this dual representation and multiplied by $W_0(y_R, y_n, y_b)$. The original row set $\mathcal{I}_H$ consists of the allowed R for which $\chi_{\mathrm{row}}(q_R/Z^H) \neq 0$. This definition is made separately for every fixed choice above.

The parameters have three roles. The row and primal scales are H, N; A_0, S_0, N_0, B_0 record the prime products in the local assignments; and v, ℓ_b, e_λ record the dual norms. The branch denominator is $c = c_F F_{\mathrm{act}} R$. With the other branch data held fixed, the row is the squarefree factor R and the dual indices are (n, b).

4.9 The mean square of a reflected block

Lemma 4.10 (The mean square of a reflected block). *Consider the block $\mathcal{U}_{v,\ell_b,e_\lambda}(R)$ of Definition 4.9, with squarefree row R and dual variables (n,b) on their stated support. Set*

$$T_0 = 2H + 2A_0 - N - N_0 - 3B_0, \qquad \nu_0 = \frac{v}{2} + \ell_b + \frac{e_\lambda}{3} + \frac{S_0 + B_0}{2}.$$

Let all these log-lengths range over fixed bounded sets. Put

$$Y = \frac{q_{\lambda^k} N_F B_F^3 X Z^v Z^{3\ell_b}}{q_{c_F F_{\mathrm{act}}}^2 Z^{2H}}, \qquad W(\mathbf{y}) = y_n^{-1/2} y_b^{-1} W_0(\mathbf{y}), \qquad F_Y(\mathbf{y}) = W(\mathbf{y}) V^\sharp(Y y_n y_b^3 y_R^{-2}).$$

There are functions $\rho(R)$ and $\beta(n,b)$, with absolute values at most one, such that β is independent of R and

$$\mathcal{U}_{v,\ell_b,e_\lambda}(R) = \frac{1}{3} Z^{-\nu_0} \rho(R) \sum_{n,b} \beta(n,b) \chi_R(nb)^3 F_Y(y_R, y_n, y_b). \tag{4.32}$$

The sum retains all the fixed dual restrictions from the definition. The only arithmetic factor here that depends simultaneously on R and the dual variables is the displayed zero-extended quadratic symbol. Moreover,

$$\frac{q_\mu X}{q_c^2} = Y y_n y_b^3 y_R^{-2}, \qquad c = c_F F_{\mathrm{act}} R. \tag{4.33}$$

This is an exact identity on the original support, including when N_F shares primes with b.

For every $\epsilon > 0$ and every fixed $A \geq 0$,

$$\begin{aligned} \sum_{R \in \mathcal{I}_H} |\mathcal{U}_{v,\ell_b,e_\lambda}(R)|^2 &\ll_{\epsilon,A} Z^{E_0+\epsilon} p_5(W)^2 \mathfrak{M}_{A,\lceil 4A \rceil + 7}(V)^2, \\ E_0 &= \max(H, v + \ell_b) - S_0 - B_0 - \ell_b - \frac{2e_\lambda}{3} - \frac{1}{2}(T_0 - v - 3\ell_b - e_\lambda)_+. \end{aligned} \tag{4.34}$$

The constants may depend on the kernel order A, the arithmetic datum, the bounded log-length ranges, and the support box, and are uniform in the locally fixed set $\mathcal{F}$. Norm twists in W_0 multiply the bound by at most the polynomial factor given by its finite seminorms.

Proof. We isolate the quadratic row–column character, separate the remaining joint profile, and then apply Lemma 4.8. The kernel amplitude is extracted before the row norm is squared.

Arithmetic factorization. For the fixed h_0 and the fixed class of R modulo M_{ref}^2, the class of $F_{\mathrm{act}} R$ is fixed. Proposition 4.2 therefore gives one common d and one common additive character ϑ for the whole row set. Write $x_F = u\lambda^{k+4} N_F B_F^3$, so that $\lambda^4 \mu = x_F n b^3$. The residual local factors are exactly

$$\prod_{p \mid R} B_p(\lambda^4 \mu) = \chi_R(x_F)^3 \chi_R(nb^3)^3 = \chi_R(x_F)^3 \chi_R(nb)^3.$$

Both equalities include zeros; no unit symbol has been divided. In particular, any collision with x_F is a restriction depending only on R. Every remaining local factor is fixed and depends only on the dual index. The scalar ζ_R of the reflection depends on R but not on μ.

Define on the specified dual support

$$\delta(n,b) = \frac{d(\mu)\alpha(\mu)\vartheta(\lambda^4\mu)}{27\, 3^{k/6} |B_F b|}.$$

The coefficient bound gives $|\delta(n,b)| \leq 1$. Its denominator is nonzero, and the definition remains valid when the numerator is zero. Using $q_\mu = 3^k q_{N_F n} q_{B_F b}^3$ gives the exact factorization

$$\frac{d(\mu)\alpha(\mu)\vartheta(\lambda^4\mu)}{\sqrt{q_\mu}} = 27 \delta(n,b) q_{\lambda^k}^{-1/3} q_{N_F n}^{-1/2} q_{B_F b}^{-1}.$$

A small Ramanujan factor or an active exponent-zero factor contributes $q_p^{-1/2}$. A divisibility factor contributes $q_p^{1/2}$; for a prime assigned to N_F it cancels $q_p^{-1/2}$ from the squarefree denominator, and for a prime assigned to B_F it leaves $q_p^{-1/2}$ after extraction. When a prime assigned to N_F also divides b, its entire occurrence in b remains in q_b^{-1} and in $\delta(n,b)$. These facts give exactly the common power $Z^{-\nu_0} y_n^{-1/2} y_b^{-1}$.

The product of δ, the signs of small Ramanujan terms, the bounded fixed local characters, and the fixed dual indicators defines $\beta(n,b)$. It is bounded by one and independent of R. Put $\rho(R) = 81\zeta_R \chi_R(x_F)^3$; it is bounded by one because $|\zeta_R| \le 1/81$. The factor $27/81 = 1/3$ proves (4.32). Multiplicativity of the norm proves (4.33), including when N_F and b share primes.

The common profile and its amplitude. By the definitions of the log-norms,

$$Y = q_{c_F}^{-2} Z^{v+3\ell_b+e_\lambda-T_0}.$$

The finite set of possible c_F depends only on L. Consequently $\mathrm{m}_A(Y) \ll_L Z^{-(T_0-v-3\ell_b-e_\lambda)_+/4}$. Lemma 4.5 in dimension three and (4.24) separate the profile in the structural identity with this amplitude factor. The single density is common to all R because W uses the row and both dual norms as coordinates. At each Fourier mode the norm powers have absolute value one and can be absorbed into the bounded row scalar and the common dual coefficient. The separated positive row norm can now be enlarged from $\mathcal{I}_H$ to squarefree good R with $q_R \ll Z^H$, retaining any fixed restrictions. The kernel estimate has already been used on the original support.

The quadratic norm. The bound (4.28), with $K = Z^H$, $N_{\mathrm{col}} = Z^v$, and $B_{\mathrm{col}} = Z^{\ell_b}$, bounds the separated squared norm by

$$Z^{\epsilon+\max(H,v+\ell_b)+v+\ell_b}.$$

The squared common coefficient in the structural identity has exponent $-v-2\ell_b-2e_\lambda/3-S_0-B_0$. The square of the extracted profile scalar contributes $-(T_0-v-3\ell_b-e_\lambda)_+/2$. Combining these exponents and the dimension-three separation norm proves (4.34). □

4.10 Truncation and summation over element rows

Truncating whole dual blocks. For later summation, call a dual block retained when

$$v+3\ell_b+e_\lambda \le T_0+\tau_{\mathrm{ref}}, \qquad \tau_{\mathrm{ref}} > 0.$$

On the fixed support box, (4.33) places the kernel argument in $[C_{\mathrm{ker}}^{-1}, C_{\mathrm{ker}}]$ times $Z^{v+3\ell_b+e_\lambda-T_0}$, for one fixed C_{ker}. If $\log Z \ge 2\log C_{\mathrm{ker}}/\tau_{\mathrm{ref}}$, every unretained whole block has kernel argument greater than $Z^{\tau_{\mathrm{ref}}/2}$ throughout its support. These blocks can be removed before Fourier inversion with arbitrary power saving when the other log-lengths lie in bounded ranges. The lattice bound gives this saving without restricting the discarded dual lengths.

On the full source support,

$$\frac{|d(\mu)|}{\sqrt{q_\mu}} \le 27 q_{\lambda^k}^{-1/3} q_{n_0}^{-1/2} q_{b_0}^{-1} \le 27 q_\lambda^{4/3}.$$

After its indicator is discarded, each local factor is bounded by $q_p^{1/2}$, uniformly in μ. The product of these bounds has a fixed polynomial size when the active log-norm is bounded. For a fixed row put $a = X/q_c^2$. Its factor $1+a^{-1}$ also has a fixed polynomial bound on the stated row and conductor ranges. The finite local choices and bounded row counts contribute a further polynomial factor; let B_{tail} be the sum of these norm-scale exponents. The profile seminorms and their polynomial height factors remain multiplicative factors outside this exponent. It is chosen before the kernel order and does not count discarded dual indices. The remaining dual sum is the unrestricted lattice sum in Lemma 4.6, with $U = Z^{\tau_{\mathrm{ref}}/2}$. For any $D > 0$, choosing

$$A > 1 + \frac{2(B_{\mathrm{tail}}+D)}{\tau_{\mathrm{ref}}}$$

makes the discarded contribution $O(Z^{-D})$ times a finite seminorm of V and the fixed profile bounds. The assertion holds either for the absolute total or, after increasing B_{tail} to include the bounded row count, for the row norm. The dyadic cutoffs have bounded overlap, so this lattice estimate controls their whole sum.

The residual-row support. Summing the reflected blocks for an element row uses the following consequence of its support. Suppose an element row has an ideal factorization $(m) = m_{\mathrm{pow}} m_{\mathrm{sup}} R$, with integral factors, and the fixed support constants give

$$q_m \le C_m Z^M, \qquad q_{m_{\mathrm{pow}}} \ge Z^O / C_O, \qquad q_R \ge Z^H / C_H.$$

Since $q_{m_{\mathrm{sup}}} \ge 1$, multiplicativity gives

$$H \le M - O + \frac{\log C_{\mathrm{res}}}{\log Z}, \qquad C_{\mathrm{res}} = C_H C_m C_O. \tag{4.35}$$

Only the lower bound $q_{m_{\mathrm{sup}}} \ge 1$ enters this inequality; it applies with arbitrary larger norm and need not be an equality. We will use it with the three factors on disjoint prime supports and with the unit residual ideal in the bounded $H = 0$ dyad.

Lemma 4.11 (A mean square over element rows). *Let ν range over a fixed finite family of multiplicative finite-ray characters with conductors supported on S, all extended by zero away from the primary elements prime to S. Fix bounded ranges for $M \ge 0$ and $N \in \mathbb{R}$, a constant $C_m \ge 1$, and fixed dyadic support comparisons. For a nonzero element m, let*

$$m_{\mathrm{pow}} = \prod_{v_{\mathfrak{p}}(m) \ge 2} \mathfrak{p}^{v_{\mathfrak{p}}(m)}$$

be its maximal powerful ideal factor. The dyadic centers O are nonnegative, with center zero for the unit range.

For every $\epsilon > 0$ there exist a finite $A \ge 0$ and an integer $J \ge 0$ such that, for every test V satisfying Proposition 4.1, every supported O-dyad, and all sufficiently large Z,

$$\sum_{\substack{0 < q_m \le C_m Z^M \\ q_{m_{\mathrm{pow}}} \asymp Z^O}} |\mathcal{C}_V(Z^N; m, \nu)|^2 \ll Z^{O/2 + \max\{M-O,\, 2M-O-N\} + \epsilon} \mathfrak{M}_{A,J}(V)^2. \tag{4.36}$$

The finite orders, constant, and lower threshold may depend on the fixed arithmetic data, the parameter ranges, C_m, the dyadic support comparisons, and ϵ. They are uniform in the element rows and in ν in the fixed family. The test V may have noncompact support. Replacing $V(x)$ by $x^{it_0} V(x)$ multiplies the bound by at most a fixed polynomial in $1 + |t_0|$.

Proof. The proof partitions the rows by their repeated prime factors, applies the block estimate on the residual annuli, and sums the factors held fixed in each block. The powerful-ideal count supplies the factor $Z^{O/2}$.

Ray classes and repeated prime factors. Fix one character ν. Lemma 4.4, with $\Psi_{\mathrm{base}} = \nu$, supplies a fixed finite family of characters for the reflection of $\nu(A)\chi_A(m)$. Choose their common $\mathfrak{E}, L$ before any good row prime varies. Partition the rows into the finitely many unit, S-valuation residue, and good-ray sectors from that lemma. This keeps every good local prime, including those whose row valuation is divisible by six.

For a row in the prescribed O-dyad, split its valuation-one primes into m_{sup} supported on S and the squarefree good product R. Thus $(m) = m_{\mathrm{pow}} m_{\mathrm{sup}} R$ on disjoint prime supports. Hold the unit and the ideals $m_{\mathrm{pow}}, m_{\mathrm{sup}}$ fixed, and partition R into smooth annuli $q_R \asymp Z^H$, with $H \ge 0$ and the unit in the bounded zero dyad. Keep this row cutoff in each reflected vector. The support bound (4.35) gives

$$H \le M - O + O_{\mathrm{fixed}}(1/\log Z). \tag{4.37}$$

The fixed row restrictions, including coprimality with the fixed factors, are independent of the dual variables.

After the row-sector conversion, take $\mathcal{F}$ in Definition 4.9 to be the good primes of m_{pow}, with their valuations reduced modulo six. The primes of R have exponent one. Fix h_0, a class of R modulo M_{ref}^2, and the local active and Ramanujan choices at the primes held fixed. Every active good prime held fixed divides m_{pow} to exponent at least two. Consequently

$$q_{F_{\mathrm{act}}}^2 \le q_{m_{\mathrm{pow}}}, \qquad 2A_0 \le O + O_{\mathrm{fixed}}(1/\log Z). \tag{4.38}$$

This uses the powerful dyad's upper comparison. The possible S-prime factors of m_{pow} only increase its norm.

The reflection formula (4.2) at $X = Z^N$ transforms the Mellin representation (4.31) into an absolutely convergent dual sum. Smooth dyadic partitions in the residual squarefree and cube norms, with the unit and ramified exponent of the dual index fixed, express it as the blocks of Definition 4.9. Their profiles W have uniformly bounded seminorms: the row and dual cutoffs, together with the normalized inverse roots, supply compact support while the full test remains in $V^\sharp$. Thus the construction applies directly to the given V, including nonannular tests.

Truncation of the dual sum. Fix a small $\tau_{\mathrm{ref}} > 0$. The row lengths H, O, the active length A_0, and N lie in fixed bounded ranges. The same is true of S_0, N_0, B_0, since their prime products divide F_{act}. The tail argument following Lemma 4.10 therefore removes all unretained whole dual blocks with any prescribed power saving, with a finite seminorm of V. The polynomial factor includes the row and local counts below; the unrestricted lattice bound has already summed all discarded dual indices. Choose its kernel order first, and increase A, J in the statement so that $\mathfrak{M}_{A,J}(V)$ dominates this tail seminorm and the block seminorm. This is possible because increasing A widens the supremum defining $\mathfrak{M}_{A,J}$ and increasing J increases its weight.

The source restriction $k \ge -4$ gives $e_\lambda \ge -4\log 3/\log Z$. For a retained block,

$$v + 3\ell_b + e_\lambda \le T_0 + \tau_{\mathrm{ref}}.$$

Since $v, \ell_b \ge 0$ and T_0 is bounded, this also bounds every retained dual length. There are only logarithmically many retained choices for the dual annuli and for k. Choose a nonnegative $\epsilon_Z = O_{\mathrm{fixed}}(1/\log Z)$ large enough to cover the support offsets in (4.37) and (4.38), and to ensure $e_\lambda \ge -\epsilon_Z$.

The exponent on a retained block. The exponent in (4.34) satisfies

$$E_0 \le \max\{H, T_0 - S_0 - B_0\} + \frac{5}{3}\epsilon_Z + \tau_{\mathrm{ref}}.$$

To see this, use $\max(H, v+\ell_b) - \ell_b = \max(H - \ell_b, v)$ and discard the nonpositive kernel term. The first branch is at most $H + 2\epsilon_Z/3$, since $\ell_b, S_0, B_0 \ge 0$. In the second branch, retention gives $v \le T_0 - 3\ell_b - e_\lambda + \tau_{\mathrm{ref}}$, so it is at most $T_0 - S_0 - B_0 + 5\epsilon_Z/3 + \tau_{\mathrm{ref}}$. Moreover,

$$\begin{aligned} T_0 - S_0 - B_0 &= 2H + 2A_0 - N - N_0 - S_0 - 4B_0 \\ &\le 2M - O - N + O(\epsilon_Z), \end{aligned}$$

by (4.37) and (4.38). Thus every retained block, with its row factors held fixed, has exponent at most

$$\max\{M - O, 2M - O - N\} + \tau_{\mathrm{ref}} + O(\epsilon_Z).$$

Summing the repeated prime factors. It remains to count the fixed row factors and the finite expansions. Every powerful ideal is uniquely x^2y^3 with y squarefree: at a prime, the exponent of y is the parity of the powerful exponent, and an odd positive exponent is at least three. Ideal counting and $\sum_y q_y^{-3/2} < \infty$ therefore give

$$\#\{m_{\mathrm{pow}} : q_{m_{\mathrm{pow}}} \ll Z^O\} \ll Z^{O/2}.$$

The squarefree m_{sup} is a divisor of the fixed radical of S and has only finitely many choices. The local branches at the good primes of m_{pow} have divisor-bounded multiplicity, as do their Ramanujan assignments. Rowwise divisor Cauchy followed by summation of the local choices introduces at most an arbitrarily small power of Z. There are finitely many choices of h_0, ray sector, and unit, while the row and retained dual annuli contribute only powers of $\log Z$. This counts the powerful contribution $O/2$ exactly once.

Choose τ_{ref} and the local small-power exponents with sum less than the prescribed ϵ. For a sufficiently large threshold depending on the fixed data, the additional $O(\epsilon_Z)$ terms keep the total below ϵ. The block bound, the preceding counts, and the already chosen tail saving prove (4.36). The twist assertion follows from the norm-twist bound for $\mathfrak{M}_{A,J}$ in Lemma 4.3. □

At equal lengths, take the Gaussian W_G of Lemma 3.11. Using its Mellin transform from Lemma 3.11, at $X = Z$ the completed sum has the exact form

$$\mathcal{C}_{W_G}(Z; m, \nu) = \frac{1}{2\pi i} \int_{(4)} Z^t e^{t^2} T(t + 1/2, \nu\chi_\bullet(m))\, dt.$$

In the defining sum the coefficients remain normalized by $q_c^{-1/2} q_n^{-1}$; the scale occurs in $W_G(q_c q_n^3/Z)$.

Finally take $M = N = 1$. The exponent in a supported O-dyad is $1 - O/2 + \epsilon$. The nonnegative O-dyads are logarithmically many, and their unit dyad is included. Summing them, with a smaller preliminary loss, gives the equal-length conclusion for every admissible V:

$$\sum_{0<q_m\le C_m Z} |\mathcal{C}_V(Z; m, \nu)|^2 \ll Z^{1+\epsilon} \mathfrak{M}_{A,J}(V)^2. \tag{4.39}$$

5 The base average and its direct bound

We now place the completed sums of the preceding section inside one average $I_\eta(X, Y, Z)$. Its original representation separates into an additive polynomial and a completed theta row. At the balanced scales $X = Y = Z^{1/2}$, their mean-square estimates give the direct bound $I_\eta \ll_\eta Z^{1/4+\epsilon}$. We first define the average for independent positive scales X, Y, Z; only the final proposition in this section specializes them. The next section applies Poisson summation to this same average and identifies the reciprocal of the target L-function in its principal row.

5.1 Definition and separation of the two factors

Fixed arithmetic data. Choose a finite ray group T, independently of the target character, through which the functions $\mathcal{R}$, G, and the fixed primary and supplementary phases supported over $2, 3$ in Lemma 3.5 factor. The target-dependent fixed-numerator characters below need not factor through T. For a finite-order Hecke character η, write

$$\Theta = \langle \eta, \widehat{T} \rangle.$$

Use one excluded set S for the finite family $\{\eta\theta : \theta \in \widehat{T}\}$. It contains the primes over 6, the prime supports of the defining moduli of the fixed zero-extended presentations of η and the characters in $\widehat{T}$, and every prime of norm at most a fixed P_0. These full presentations belong to the fixed arithmetic data. They are fixed before X, Y, Z and the varying row and prime parameters are chosen. Enlarging S will not change T.

The character and phase corrections in the average have two purposes: the finite transform must exclude frequencies meeting S, including zero, and its surviving coefficients must factor prime by prime.

Character and phase corrections. Let b_* generate the squarefree product of the primes in S. Choose a residue character ξ modulo b_* whose restriction to each prime factor is nonprincipal

and whose order divides six. Such a choice exists: at a prime of odd residue characteristic one may use the quadratic character, and the residue field at the prime over 2 has order four and has a character of order three. The Chinese remainder theorem makes ξ primitive modulo b_*. All residue characters below are extended by zero on nonunits. Put

$$g_\psi(c,k)=\sum_{d \bmod c}\psi(d)e(kd/c),\qquad \tau=q_{b_*}^{-1/2}g_\xi(b_*,1),\qquad \Xi(a)=\xi(a)\chi_a(b_*),$$

$$\Psi_{m,s,\eta}(a)=\eta(a)\Xi(a)^{-1}\mathcal{R}(a,s)\chi_a(m)\qquad((a,S)=1).$$

Finite character orthogonality at each prime gives $|\tau|=1$. By Lemma 3.2, the fixed-numerator symbol $a\mapsto\chi_a(b_*)$ is a finite ray character whose conductor is supported on S. Thus Ξ is a fixed finite character for this target, although it need not factor through T.

The primitive character ξ at the primes in S will force the Poisson frequency to be prime to S. The accompanying normalizations cancel the unit factors introduced by the auxiliary modulus b_*. For the completed index $A=cn^3$, the factor $\mathcal{R}(A,s)$ cancels the cross-prime reciprocity phases between A and s, while $\overline{G(A)}$ cancels the remaining pair phases within A. We verify these cancellations coefficientwise in Section 6.2.

The row and its average. Define the corrected row at spectral parameter $t+1/2$ by

$$T_{m,s,\eta}(t)=\sum_{\substack{c\ \mathrm{sf}\\ n}}\gamma_2(c)\overline{\alpha(cn^3)}\Psi_{m,s,\eta}(cn^3)\overline{G(cn^3)}q_c^{-1/2-t}q_n^{-1-3t}.$$

Here and below the ideal variables avoid S. In particular every factor in $\Xi(a)^{-1}$ is evaluated on a unit. The value $G(cn^3)$ is the finite-ray extension in (3.6), also when cn^3 is not squarefree.

Fix nonnegative, nonzero functions $W_0,W_1\in C_c^\infty(0,\infty)$ and put $\Phi(t)=e^{t^2}$. Define

$$\begin{aligned}I_\eta(X,Y,Z)=\frac{1}{Y}\sum_s\frac{W_1(q_s/Y)\chi_s(b_*)}{\tau\xi(s)\sqrt{q_{b_*}q_sX}}\\ \cdot\sum_{m\in\mathcal{O}}\xi(m)q_s^{-1/2}g_{\chi_s}(s,-m)W_0\Big(\frac{q_m}{q_{b_*}q_sX}\Big)\frac{1}{2\pi i}\int_{(4)}Z^t\Phi(t)T_{m,s,\eta}(t)\,dt.\end{aligned}\tag{5.1}$$

The masks in this definition remove nonunits at S. All sums in (5.1) are absolutely convergent on the displayed line; the m and s sums are finite because the weights are annular.

Separating the two row factors. The fixed finite Fourier expansion of the correction is

$$\overline{G(A)}=\sum_{\theta\in\widehat{T}}a_\theta\theta(A),\qquad a_\theta=\frac{1}{|T|}\sum_{v\in T}\overline{G(v)}\,\overline{\theta(v)}.$$

Parseval and Cauchy–Schwarz give $\sum_\theta|a_\theta|\le|T|^{1/2}$. For a ray class $\sigma\in T$ choose a representative s_σ and put

$$\nu_\sigma(a)=\eta(a)\Xi(a)^{-1}\mathcal{R}(a,s_\sigma)\qquad((a,S)=1),$$

$$B_{m,\sigma}(Z)=\sum_{\theta\in\widehat{T}}a_\theta\frac{1}{2\pi i}\int_{(4)}Z^t\Phi(t)T(t+1/2,\nu_\sigma\theta\chi_\bullet(m))\,dt.$$

Extend ν_σ by zero away from the primary elements prime to S, without evaluating $\Xi(a)^{-1}$ there. The function T here is the completed product in (4.2), with its original zero masks. Since $\mathcal{R}(A,s)=\mathcal{R}(A,s_\sigma)$ for $s\in\sigma$, the displayed Mellin integral of the corrected row equals $B_{m,\sigma}(Z)$ throughout that class.

Use the Gaussian W_G of Lemma 3.11. For $m\neq0$, the central Mellin identity (4.31) gives the specialization

$$B_{m,\sigma}(Z)=\sum_{\theta\in\widehat{T}}a_\theta\mathcal{C}_{W_G}(Z;m,\nu_\sigma\theta).$$

The characters $\nu_\sigma\theta$ form one fixed finite family whose conductor and defining-modulus supports are contained in S; they need not factor through T. Their fixed zero extensions are away from the primary elements prime to S, as required in Lemma 4.11. In the completed row of (4.30), the moving symbol $\chi_{cn^3}(m)$ supplies the zeros at shared good primes. The row-sector conversion preserves those zeros, including when a row valuation is divisible by six. The triangle inequality in the row Hilbert space introduces the fixed factor $\sum_\theta |a_\theta|$.

Write $Q = q_{b_*}XY$. Choose a smooth compactly supported function Ω on $(0,\infty)$ equal to one on every value of q_m/Q that can occur on the support of $W_1(q_s/Y)W_0(q_m/(q_{b_*}q_sX))$. For $v \in \mathbb{R}$, set

$$A_{m,\sigma,v}(Y) = \frac{1}{Y}\sum_{s\in\sigma} \frac{W_1(q_s/Y)\chi_s(b_*)}{\tau\xi(s)}(q_s/Y)^{-1/2+iv}q_s^{-1/2}g_{\chi_s}(s,-m). \tag{5.2}$$

Our Mellin convention for W_0 is $\widehat{W_0}(iv) = \int_0^\infty W_0(r)r^{iv}\,dr/r$. Since $q_m/(q_{b_*}q_sX) = (q_m/Q)/(q_s/Y)$, Mellin inversion gives the identity

$$\begin{aligned} I_\eta(X,Y,Z) = \frac{Q^{-1/2}}{2\pi}\sum_{\sigma\in T}\int_{\mathbb{R}} \widehat{W_0}(iv)\sum_{m\neq 0}\Omega(q_m/Q)\xi(m)(q_m/Q)^{-iv} \\ \cdot A_{m,\sigma,v}(Y)B_{m,\sigma}(Z)\,dv. \end{aligned} \tag{5.3}$$

The omitted $m = 0$ term vanishes by the annular support of W_0. The factor $(q_m/Q)^{-iv}$ has absolute value one. In A, the arithmetic coefficient is fixed and bounded independently of m, v; all dependence on v occurs in a norm power of the annular variable.

5.2 The balanced direct estimate

For the balanced estimate, the completed-row moment already bounds the mean square of $B_{m,\sigma}(Z)$. It remains to bound the additive factor $A_{m,\sigma,v}(Y)$. Its Gauss expansion gives distinct reduced fractions in $\mathbb{C}/\mathcal{O}$; their separation and coefficient mass give the required mean square. The planar additive large sieve is classical; compare Huxley's multivariable and number-field inequality [18] and the Poisson proof in [1, Section 3, Theorem 3]. The lattice proof in Appendix A records the normalization used here.

Lemma 5.1 (Planar additive large sieve). *Let $0 < \delta \le 1$, let $Q \ge 1$, and let $z_1,\dots,z_J$ be a finite set of points in $\mathbb{C}/\mathcal{O}$ satisfying*

$$\inf_{n\in\mathcal{O}} |z_j - z_k - n| \ge \delta \qquad (j \neq k).$$

Then, for arbitrary complex numbers $a_1,\dots,a_J$,

$$\sum_{\substack{m\in\mathcal{O}\\ q_m\le Q}} \left|\sum_{j=1}^J a_j e(mz_j)\right|^2 \ll (Q+\delta^{-2})\sum_{j=1}^J |a_j|^2.$$

The implied constant is absolute for the lattice, additive character, and self-dual measure fixed above. A fixed multiple of Q in the row ball is allowed by changing this constant.

The proof appears in Appendix A.

We apply this estimate to the full residue classes in the Gauss sums. The zero extension makes the fractions reduced with respect to the full displayed modulus, including any redundant primes of an inducing presentation.

Lemma 5.2 (The balanced additive norm). *Fix the arithmetic data of (5.1), the annular weight W_1, a ray class $\sigma \in T$, and a row-ball constant $C > 0$. For $Y, Q \ge 1$ and every real v, the*

polynomial $A_{m,\sigma,v}(Y)$ defined in (5.2) satisfies

$$\sum_{\substack{m\in\mathcal{O}\\ q_m\le CQ}} |A_{m,\sigma,v}(Y)|^2 \ll_{\mathcal{A},W_1,C} \frac{Q+Y^2}{Y}.$$

The bound is uniform in v. In particular, when $Q = q_{b_*}Y^2$,

$$\sum_{q_m\le CQ} |A_{m,\sigma,v}(Y)|^2 \ll_{\mathcal{A},W_1,C} \frac{Q}{Y}.$$

Proof. The Gauss expansion reduces the estimate to the separation of its frequencies and the square mass of its coefficients.

The reduced fractions. For a primary s outside S, put $r_s = q_s/Y$ and

$$c_\sigma(s) = \mathbf{1}_{s\in\sigma}\frac{\chi_s(b_*)}{\tau\xi(s)}.$$

The inverse is evaluated only on these elements prime to S. Here $|\tau| = 1$, $|\xi(s)| = 1$, and $|\chi_s(b_*)| = 1$, because b_* is supported on S. Thus $|c_\sigma(s)| \le 1$. The definition of A and the Gauss expansion give the exact identity

$$A_{m,\sigma,v}(Y) = Y^{-3/2}\sum_s c_\sigma(s)W_1(r_s)r_s^{-1+iv}\sum_{d \bmod s}\chi_s(d)e(-md/s).$$

The sum is over primary ideals outside S. For every such s, the character $\chi_s(d)$ is zero exactly when $(d,s) > 1$, and has absolute value one otherwise. This includes nonsquarefree s: a local exponent divisible by six is the indicator of the units at that prime, not the constant function one. For $s = 1$ the single residue class is included and $\chi_1 = 1$.

We verify that the fractions d/s for the contributing pairs are distinct modulo $\mathcal{O}$. Suppose that $(d,s) = (d',s') = 1$ and $d/s - d'/s' \in \mathcal{O}$. Then $ds' - d's$ is divisible by ss'. Reducing this divisibility modulo s and using the inverse of d modulo s gives $s \mid s'$. The symmetric argument gives $s' \mid s$. The primary generator of an ideal outside S is unique, so $s = s'$; the original congruence then gives $d = d' \pmod s$. The same conclusion includes the unit modulus. Changing a residue representative only translates its fraction by an element of $\mathcal{O}$.

Frequency separation. Let $0 < r_- < r_+$ be fixed with $\operatorname{supp} W_1 \subset [r_-, r_+]$. For two distinct contributing fractions and every $n \in \mathcal{O}$, the Eisenstein integer $ds' - d's - nss'$ is nonzero. Hence

$$\left|\frac{d}{s} - \frac{d'}{s'} - n\right| = \frac{|ds' - d's - nss'|}{|ss'|} \ge \frac{1}{|ss'|} \ge \frac{1}{r_+Y}.$$

The last inequality uses $q_s, q_{s'} \le r_+Y$. Thus these fractions are δ_Y-separated in $\mathbb{C}/\mathcal{O}$, where $\delta_Y = \min(1, r_+^{-1})/Y \le 1$.

Coefficient mass. Writing the expanded polynomial as $\sum_{s,d} a_{s,d}e(-md/s)$, with only the unit residue classes retained, its coefficient square mass is

$$\sum_{s,d}|a_{s,d}|^2 = Y^{-3}\sum_s |c_\sigma(s)W_1(r_s)|^2 r_s^{-2}\varphi_{\mathcal{O}}(s),$$

$$\varphi_{\mathcal{O}}(s) := \#(\mathcal{O}/s\mathcal{O})^\times, \qquad \varphi_{\mathcal{O}}(1) := 1.$$

There is no dependence on v in this expression. Since $\varphi_{\mathcal{O}}(s) \le q_s \le r_+Y$, $r_s \ge r_-$ on the support, and there are $O(Y)$ ideals with $q_s \le r_+Y$, this mass is $O(Y^{-1})$. Lemma 5.1, applied to the points $-d/s$ with separation δ_Y, now proves the first bound. Finally $q_{b_*} \ge 1$, so $Q = q_{b_*}Y^2$ implies $Q + Y^2 \le 2Q$, proving the second. □

The additive norm and the unmarked completed-row norm combine by Cauchy–Schwarz to bound the original average.

Proposition 5.3 (Balanced direct estimate). *Fix the arithmetic data and smooth weights used to define I_η in (5.1). For every $\epsilon > 0$, as $Z \to \infty$,*

$$|I_\eta(Z^{1/2}, Z^{1/2}, Z)| \ll_{\mathcal{A},\epsilon} Z^{1/4+\epsilon}.$$

The exponent is independent of the target character; the implied constant and lower threshold may depend on its fixed arithmetic data.

Proof. Set $X = Y = Z^{1/2}$ and $Q = q_{b_*}XY = q_{b_*}Z$. The function Ω in (5.3) is bounded and supported in a fixed compact subinterval of $(0, \infty)$. Thus all rows in that identity lie in $0 < q_m \le C_\Omega Q$ for a fixed constant C_Ω.

The specialization $M = N = 1$ of Lemma 4.11, followed by the finite ray decomposition defining $B_{m,\sigma}$, gives for every $\epsilon_1 > 0$

$$\sum_{0<q_m\le C_\Omega Q} |B_{m,\sigma}(Z)|^2 \ll_{\mathcal{A},\epsilon_1} Z^{1+\epsilon_1}.$$

Here q_{b_*} is fixed, so the row ball is a fixed multiple of the $M = 1$ ball, and the completed scale Z gives $N = 1$. The finite-family decomposition of B above retains the fixed and moving zero masks just described.

Cauchy–Schwarz bounds the row sum in (5.3) by the product of the two row norms. The factor $(q_m/Q)^{-iv}$ has absolute value one, $|\xi(m)| \le 1$ with its zero extension, and Ω is bounded. Lemma 5.2 is uniform in v, while B is independent of v. Since W_0 is smooth and annular, $\int_{\mathbb{R}} |\widehat{W_0}(iv)|\, dv < \infty$. The finite sum over $\sigma \in T$ therefore gives

$$|I_\eta(X, Y, Z)| \ll_{\mathcal{A},\epsilon_1} Q^{-1/2} \left(\frac{Q}{Y}\right)^{1/2} Z^{1/2+\epsilon_1/2} = Z^{1/4+\epsilon_1/2}.$$

Taking $\epsilon_1 = 2\epsilon$ proves the proposition. □

6 The Poisson representation and its Euler factors

The direct estimate is now available. To compare the average with the Mellin signal in Proposition 2.1, we return to independent positive scales X, Y, Z and apply Poisson summation in the element variable m. The resulting rows are indexed by the sixth-power-free part of the frequency. We will factor each row into Hecke L-functions and a holomorphic Euler product; the row $u = 1$ will contain the reciprocal of the target function.

6.1 The exact Poisson series

The correction $\overline{G(A)}$ is independent of m, and the varying character in the completed row is precisely $\chi_A(m)$, including its zeros. Consequently Poisson summation uses the full modulus sb_*A, even if s and A share primes. Expand the s-Gauss sum and first sum the lifts of a residue modulo b_*A. The lifts impose $H \equiv b_*Ad \pmod{s}$ and give the coefficient

$$F(s, A, H) = \sum_{\substack{d \bmod s \\ H\equiv b_*Ad \pmod{s}}} \chi_s(d) g_{\xi\chi_A}\left(b_*A, \frac{H - b_*Ad}{s}\right). \tag{6.1}$$

Indeed, the finite Fourier transform of $\xi(m)\chi_A(m)q_s^{-1/2}g_{\chi_s}(s, -m)$ modulo sb_*A is $q_s^{1/2}F(s, A, H)$. The Poisson prefactor for the row scale $q_{b_*}q_sX$ is X/q_A. After the outer square-root normalization

in (5.1), leaving its separate Y^{-1} outside, their product is $X^{1/2}/(q_{b_*}^{1/2}q_A)$. This also verifies all factors of q_s in the transformation.

At the primes of b_* the primitive Gauss sum vanishes unless $(H,S)=1$. It therefore also removes $H=0$. Every remaining element has a unique expression $H=ua^6$, where u is a sixth-power-free element, including its unit factor, and a is the primary generator of an ideal. Write $\sum_u^{(6)}$ for the sum over such nonzero u with $(u,S)=1$.

Let $\widetilde{W_0}(|y|^2)$ be the planar Fourier transform of $W_0(|y|^2)$ for the self-dual measure and character fixed in Section 3. The radial Fourier argument after Poisson is Xq_H/q_A. Put

$$M(z)=\int_0^\infty \widetilde{W_0}(r)r^{z-1}\,dr, \qquad \widehat{W_1}(w)=\int_0^\infty W_1(r)r^{w-1}\,dr.$$

Smoothness at zero and Schwartz decay show that M is holomorphic for $\Re z>0$. On every compact positive real strip it has arbitrary polynomial decay in $\Im z$: every derivative of $e^{(\Re z)v}\widetilde{W_0}(e^v)$ is integrable in v, uniformly on that strip, so repeated integration by parts applies. The same argument gives arbitrary polynomial decay for $\widehat{W_1}$ on every fixed real strip.

We will need the strict positivity

$$M(1/6)>0. \tag{6.2}$$

A Gaussian integral expresses this Mellin value in terms of the nonnegative original weight. For $0<\sigma<1$,

$$|y|^{2\sigma-2}=\frac{1}{\Gamma(1-\sigma)}\int_0^\infty t^{-\sigma}e^{-t|y|^2}\,dt.$$

The pairing of the left side with $\widetilde{W_0}(|y|^2)$ is absolutely integrable. Fubini and Fourier duality express it as

$$\frac{1}{\Gamma(1-\sigma)}\int_0^\infty t^{-\sigma}\int_{\mathbb{C}} W_0(|x|^2)\frac{2\pi}{\sqrt{3}t}\exp\left(-\frac{4\pi^2|x|^2}{3t}\right)d\mu(x)\,dt>0.$$

The inner integral is strictly positive for every $t>0$, since W_0 is nonnegative and nonzero. Absolute integrability follows either from the displayed representation on the Fourier side or from the annular support of W_0 on this side. Polar integration identifies the original pairing with $(2\pi/\sqrt{3})M(\sigma)$, proving (6.2).

After the change $H=ua^6$, Mellin inversion introduces z for the radial Fourier weight and w for the s-weight. Set $x=t+1-z$; this makes x the completed-index variable in the coefficient series. For a nonzero sixth-power-free u with $(u,S)=1$, define

$$\mathcal{F}_{\eta,u}(x,w,z)=\sum_{\substack{c\ \mathrm{sf}\\ n,s,a}} K_\eta(c,n,s,a;u)q_c^{-1/2-x}q_n^{-1-3x}q_s^{-w}q_a^{-6z}, \tag{6.3}$$

where $A=cn^3$ and the coefficient, with all normalizations retained, is

$$K_\eta(c,n,s,a;u)=\frac{F(s,A,ua^6)\chi_s(b_*)}{\sqrt{q_{b_*}}\tau\xi(s)\overline{\xi(u)}}\gamma_2(c)\overline{\alpha(A)G(A)}\eta(A)\Xi(A)^{-1}\mathcal{R}(A,s). \tag{6.4}$$

For convenience define the common Mellin weight

$$\mathcal{W}(X,Y,Z;x,w,z)=X^{1/2-z}Z^{x+z-1}Y^{w-1}\Phi(x+z-1)M(z)\widehat{W_1}(w).$$

For example, the lines $\Re x=3$, $\Re w=3$, $\Re z=2$ correspond to the original line $\Re t=4$ and lie in absolute convergence. The complete high identity is

$$\begin{aligned} I_\eta(X,Y,Z)=\frac{1}{(2\pi i)^3}\int_{(3)}\int_{(3)}\int_{(2)}&\mathcal{W}(X,Y,Z;x,w,z)\\ &\cdot\sum_u^{(6)} q_u^{-z}\overline{\xi(u)}\mathcal{F}_{\eta,u}(x,w,z)\,dz\,dw\,dx. \end{aligned} \tag{6.5}$$

One may justify all interchanges on these lines by the elementary bound $|F(s,A,H)| \le q_s q_{b_* A}$ and by the annular weights. Thus (5.3) and (6.5) are two identities for the average I_η defined in (5.1).

6.2 Coefficient factorization and reciprocity phases

The coefficient in (6.4) factors prime by prime after the unit factors and reciprocity phases cancel. The resulting local series separates the terms with positive valuation of the completed index A at a given prime. Its summation in Section 6.3 and the convergence estimates in Section 6.4 will give the scalar Euler product for the target η.

Fix a nonzero sixth-power-free row u with $(u,S)=1$ and a prime $p \notin S$. The first line below records the row residue data, the next two give the coefficient phases, and the last gives the geometric ratios used in the local summation. Set

$$\begin{gathered} Q = q_p, \qquad j = v_p(u) \in \{0,\dots,5\}, \qquad \rho = \chi_p(u/p^j), \qquad \omega_p = \chi_p(-1), \\ a_p = \overline{\alpha(p)}^{\,3}\eta(p)^3, \qquad b_p = \overline{\alpha(p)}^{\,2}\eta(p)^2\overline{\chi_p(4)}, \qquad b_p^3 = a_p^2, \\ V = Q^{-6z}, \qquad R = a_p^2 Q^{4-6x-6z}, \qquad W_{\mathrm{loc}} = \rho Q^{-w}. \end{gathered} \tag{6.6}$$

The values ρ, a_p, b_p have absolute value one, and $\omega_p \in \{1,-1\}$. The prime p ranges over all primes outside S.

The finite scalar and its unit factors. Write $G_r = g_{\chi_p^r}(p,1)$, with $G_0 = -1$, and $\gamma_r = Q^{-1/2}G_r$ for $1 \le r \le 5$. For $t \ge 1$ and $j' \ge 0$, summing the lifts of a residue modulo p gives

$$g_{\chi_p^r}(p^t, p^{j'}) = \begin{cases} Q^{t-1}G_r \mathbf{1}_{j'=t-1}, & 6 \nmid r, \\ Q^t \mathbf{1}_{j' \ge t} - Q^{t-1}\mathbf{1}_{j' \ge t-1}, & 6 \mid r. \end{cases}$$

The second line is the Ramanujan sum for the principal character extended by zero. Orthogonality also gives $G_1 G_{-1} = \omega_p Q$.

Let $e_0 = v_p(c) \in \{0,1\}$, $l = v_p(n)$, $t = e_0 + 3l$, $k = v_p(s)$, and $m = v_p(a)$, so $v_p(H) = j' = j + 6m$. The part of (6.1) at p, before its unit factors are restored, is

$$C_p(t,k,j') = \sum_{\substack{d \bmod p^k \\ p^{j'} \equiv p^t d \pmod{p^k}}} \chi_p^k(d)\, g_{\chi_p^t}\left(p^t, \frac{p^{j'} - p^t d}{p^k}\right). \tag{6.7}$$

At modulus one both factors in this formula mean one. Its complete evaluation is

$$C_p(t,k,j') = \begin{cases} 1, & t = k = 0, \\ \mathbf{1}_{j'=0}, & t = 0,\ k > 0, \\ g_{\chi_p^t}(p^t, p^{j'}), & t > 0,\ k = 0, \\ \omega_p^{-1} G_1 g_{\chi_p^{t-1}}(p^t, p^{j'-1}), & t > 0,\ k = 1,\ j' \ge 1, \\ 0, & \text{otherwise.} \end{cases} \tag{6.8}$$

For $t = 0$ and $k > 0$, the congruence fixes $d = p^{j'}$ modulo p^k, whose zero-extended character is nonzero exactly when $j' = 0$. For the fourth line the congruence first forces $j' \ge 1$. Expanding the inner Gauss sum, the sum over d mod p is

$$\sum_{d \bmod p} \chi_p(d) e(-vd/p) = \omega_p^{-1} G_1 \chi_p(v)^{-1} \qquad (p \nmid v),$$

which changes the Gauss character from χ_p^t to χ_p^{t-1}. If $t > 0$ and $k \geq 2$, the permitted lifts $d = d_0 + p^{k-1}v$ do not change $\chi_p^k(d)$, but their Gauss phase is $e(-yv/p)$ with the Gauss variable y a unit. Their sum is zero. This proves (6.8), including the cases where the Gauss character is principal and zero-extended.

For the original units write $h_p = H/p^{j'}$, $s_p = s/p^k$, and $A_p^\circ = A/p^t$. Chinese remaindering and the substitution $d = h_p(b_* A_p^\circ)^{-1} d'$ give the additional unit factor

$$U_p = \chi_p^{k-t}(h_p)\chi_p^t(s_p)\chi_p^{t-k}(A_p^\circ)\chi_p^{t-k}(b_*). \tag{6.9}$$

More explicitly, the outer residue character gives $\chi_p^k(h_p)\chi_p^{-k}(b_* A_p^\circ)$, while the Chinese remainder factor and the rescaling of the Gauss argument give $\chi_p^t(b_* A_p^\circ)\chi_p^{-t}(h_p/s_p)$. Their product is (6.9). Since $h_p = (u/p^j)(a/p^m)^6$, one has $\chi_p(h_p) = \rho$.

At the primes of b_*, one has $(H - b_* A d)/s \equiv H/s \pmod{b_*}$. The local factor in the complete Gauss sum is $\sqrt{q_{b_*}}\tau\xi(A)\overline{\xi(H/s)}$. Here $\overline{\xi(H/s)} = \overline{\xi(u)}\xi(s)$ because $\xi(a)^6 = 1$. This cancels every displayed ξ factor in (6.4). The product of the last factors in (6.9) cancels $\chi_s(b_*)\chi_A(b_*)^{-1}$. This cancellation is coefficientwise. For any fixed enlargement of S, the same calculation uses the same ray group T.

Cancellation of the remaining pair phases. With $t_p = v_p(A)$ and $k_p = v_p(s)$, reciprocity at each pair of distinct primes gives

$$\mathcal{R}(A, s)\prod_p \chi_p^{t_p}(s_p)\chi_p^{-k_p}(A_p^\circ) = \prod_p \mathcal{R}(p, p)^{t_p k_p}.$$

Indeed, for $p \neq r$ the two orientations have opposite symbol exponents, and their quotient is $\mathcal{R}(p, r)$; this cancels the corresponding pair in the bicharacter $\mathcal{R}(A, s)$. Only its diagonal remains.

We now cancel the phases involving two prime factors of A. Equation (3.8) gives $\gamma_2(c) = \mu(c)\alpha(c)G(c)/\gamma_1(c)$. The Chinese remainder formula

$$\gamma_1(c) = \prod_{p|c}\gamma_1(p)\prod_{p<r,\ p,r|c}\chi_p(r)\chi_r(p)$$

removes the squarefree part of the remaining pair product. If $e_p = v_p(c)$ and $l_p = v_p(n)$, its exponent at a pair $p < r$ is $t_p t_r - e_p e_r = 3(e_p l_r + e_r l_p + 3 l_p l_r)$. The cube of $\chi_p(r)\chi_r(p)$ is $\mathcal{R}(p, r)$, whose square is one. Thus the pair product left after this division is

$$\prod_{p<r}\mathcal{R}(p, r)^{e_p l_r + e_r l_p + l_p l_r} = \mathcal{R}(c, n)G(n^3)\prod_p \overline{G(p^3)}^{l_p}\mathcal{R}(p, p)^{-e_p l_p - l_p(l_p-1)/2}.$$

The equality follows by expanding the quadratic refinement (3.6) on the prime factors of n^3. The same refinement gives $G(c)\mathcal{R}(c, n)G(n^3) = G(cn^3)$, which is cancelled by the inserted $\overline{G(cn^3)}$. Finally (3.7) gives $G(p^3) = \gamma_3(p)$ and $\mathcal{R}(p, p) = \omega_p$. These identities account for all pair phases, with no restriction on the ray class of p.

The coefficient in (6.4) has therefore separated into prime factors. In absolute convergence, its factor at p is the series

$$\begin{aligned} P_p = \sum_{\substack{e_0=0,1\\ l,k,m\geq 0}} & (-1)^{e_0}\gamma_1^{-e_0}\eta(p)^{e_0}(a_p\overline{\gamma_3})^l \rho^{k-t}\omega_p^{tk-e_0 l - l(l-1)/2} C_p(t, k, j+6m) \\ & \cdot Q^{-(x+1/2)e_0-(1+3x)l-wk-6zm}, \qquad t = e_0 + 3l. \end{aligned} \tag{6.10}$$

Let P_p^* be the same sum restricted to $t > 0$, equivalently to positive valuation of the completed index A.

6.3 Summing the local series

Lemma 6.1 (The complete local identity). *Fix a nonzero sixth-power-free u with $(u,S)=1$ and a prime $p\notin S$. With the local parameters of (6.6), let P_p be the series in (6.10) and let P_p^* be its restriction to $t>0$. Then*

$$\begin{aligned} P_p^* &= \frac{1}{1-R}\left\{\frac{R(1-Q^{-1})-\eta(p)(Q-1)Q^{-x-w}V^{\mathbf{1}_{j\le 1}}}{1-V}+J_j\right\},\\ P_p &= \frac{1}{1-V}+\mathbf{1}_{j=0}\frac{W_{\mathrm{loc}}}{1-W_{\mathrm{loc}}}+P_p^*, \end{aligned} \tag{6.11}$$

where the six values of J_j are

j	J_j
0	$-\eta(p)\rho^{-1}Q^{-x}+W_{\mathrm{loc}}R$
1	$\eta(p)Q^{-x-w}$
2	$a_p\rho^{-3}Q^{3/2-3x}$
3	$-\eta(p)b_p\rho^{-2}Q^{2-3x-w}+b_p^2\rho^{-4}Q^{2-4x}$
4	$-\eta(p)a_p\rho^{-3}Q^{5/2-4x-w}$
5	$-a_p^2Q^{3-6x}$.

Proof. The summation separates into four families according to the parity of l and the squarefree exponent e_0. Within each family the repeated terms form geometric series in R and V.

The nonzero local terms. There are four families with $t>0$: $(e_0,l)=(0,2r+2),(1,2r),(0,2r+1),(1,2r+1)$, where $r\ge 0$. In each family, increasing r by one and the first permitted m by one multiplies the summand of (6.10) by R. In fact the phase ratio is a_p^2: the remaining ratio is one because $\gamma_3^2=\omega_p$, $\omega_p^2=1$, and $\rho^6=1$. Increasing m further multiplies a summand by V.

The following table lists every nonzero summand after division by R^r. In its first row j is arbitrary; all other conditions are as displayed. Every omitted case is zero by (6.8).

(e_0,l)	k	condition on (j,m)	R^{-r} times summand
$(0,2r+2)$	0	$m\ge r+1$	$R(1-Q^{-1})V^{m-r-1}$
$(0,2r+2)$	0	$j=5,\ m=r$	$-a_p^2Q^{3-6x}$
$(0,2r+2)$	1	$j=0,\ m=r+1$	$W_{\mathrm{loc}}R$
$(1,2r)$	0	$j=0,\ m=r$	$-\eta(p)\rho^{-1}Q^{-x}$
$(1,2r)$	1	$j=1,\ m=r$	$\eta(p)Q^{-x-w}$
$(1,2r)$	1	$m\ge r+\mathbf{1}_{j\le 1}$	$-\eta(p)(Q-1)Q^{-x-w}V^{m-r}$
$(0,2r+1)$	0	$j=2,\ m=r$	$a_p\rho^{-3}Q^{3/2-3x}$
$(0,2r+1)$	1	$j=3,\ m=r$	$-\eta(p)b_p\rho^{-2}Q^{2-3x-w}$
$(1,2r+1)$	0	$j=3,\ m=r$	$b_p^2\rho^{-4}Q^{2-4x}$
$(1,2r+1)$	1	$j=4,\ m=r$	$-\eta(p)a_p\rho^{-3}Q^{5/2-4x-w}$.

(6.12)

Here is a direct check of its entries. In the first family $t=6r+6$. For $k=0$, the principal Gauss lift is $Q^t(1-Q^{-1})$ for $m\ge r+1$; its extra boundary value at $(j,m)=(5,r)$ is $-Q^{t-1}$. For $k=1$, the product $\omega_p^{-1}G_1G_{-1}=Q$ allows only $(j,m)=(0,r+1)$. These give the first three rows. In the second family $t=6r+1$. For $k=0$ the nonprincipal equality $j'=t-1$ gives the fourth row. For $k=1$, the principal Gauss lift has its negative boundary at $j'=t$, giving the fifth row, and its value $Q^{t-1}(Q-1)$ for $j'>t$, giving the sixth row. The latter condition is exactly $m\ge r+\mathbf{1}_{j\le 1}$.

In the two odd families, $t=6r+3$ and $t=6r+4$. Their $k=0$ lines require $j'=t-1$ and their $k=1$ lines require $j'=t$. The phase reductions are

$$\gamma_1\gamma_2=-\alpha(p)\overline{\chi_p(4)}\gamma_3,\qquad \frac{\gamma_4}{\gamma_1}=-\frac{\overline{\alpha(p)}}{G(p)}.$$

The first is (3.8) at p; the second follows from the first and $\gamma_2\gamma_4 = 1$. Substitution gives the last four rows, including both $j = 3$ terms. Thus the table covers all five ramified valuations as well as $j = 0$.

Summing the geometric series. Summing $r \geq 0$ gives $(1-R)^{-1}$, and the two unbounded m ranges give $(1-V)^{-1}$. These sums are precisely the formula for P_p^* in (6.11). If $t = 0$, then $e_0 = l = 0$. The $k = 0$ terms give $(1-V)^{-1}$, while the $k > 0$ terms require $j = m = 0$ and give $W_{\mathrm{loc}}/(1-W_{\mathrm{loc}})$. This proves the formula for P_p as well. □

6.4 The holomorphic Euler product

The local identity now isolates the Euler factors of the scalar zeta and Hecke functions. Put $W = \chi_p(u)Q^{-w}$ and $D = \eta(p)\overline{\chi_p(u)}Q^{-x}$, with the original zero extension at $p \mid u$, and define

$$H_p = P_p \frac{(1-V)(1-W)}{1-D}, \qquad \mathcal{H}_{\eta,u} = \prod_{p \notin S} H_p. \tag{6.13}$$

Write $x_r = \Re x$, $w_r = \Re w$, and $z_r = \Re z$.

Lemma 6.2 (Euler factorization and holomorphic correction). *For a nonzero sixth-power-free row u with $(u, S) = 1$, use the local factors H_p and their product $\mathcal{H}_{\eta,u}$ defined in (6.13). Then the series in (6.3) has the scalar factorization*

$$\mathcal{F}_{\eta,u}(x,w,z) = \frac{\zeta_F^S(6z)L^S(w,\chi_\bullet(u))}{L^S(x,\eta\overline{\chi_\bullet(u)})}\mathcal{H}_{\eta,u}(x,w,z). \tag{6.14}$$

For every fixed $\epsilon_0 > 0$, the correction product converges normally on a neighborhood of every point in each of the following regions, and hence defines a holomorphic function there:

$$x_r \geq 51/100, \quad z_r \geq 17/50, \quad w_r \geq -1/100, \quad x_r + w_r \geq 1 + \epsilon_0; \tag{6.15}$$

$$x_r \geq 7/8, \qquad z_r \geq 33/200, \qquad w_r \geq 19/20. \tag{6.16}$$

Uniformly in imaginary parts and unit phases, $\mathcal{H}_{\eta,u} \ll_\epsilon q_u^\epsilon$, with the constant also depending on ϵ_0 in the first region. For $u = 1$ in the second region, $\mathcal{H}_{\eta,1} = 1 + O(P_0^{-c_H})$ with, for example, $c_H = 4/5$. Equation (6.14) begins in absolute convergence and supplies a meromorphic continuation through these regions.

The first region will contain the buffered contours for nonprincipal rows, including the reflected numerator line. The second contains the principal residue point $w = 1$, $z = 1/6$ and the contours used to reach it.

Proof. We first rewrite the correction so that it has no $1 - W$ denominator. Bounds for the resulting local terms then give normal convergence, including where w_r is negative.

Cancellation of the local denominator. Put $\mathcal{E}_p = P_p^* + D$. The contribution from $t = 0$ is $(1-V)^{-1} + W/(1-W)$, so simplification gives

$$H_p - 1 = \frac{D(V + W - VW) - VW + (1-V)(1-W)\mathcal{E}_p}{1-D}. \tag{6.17}$$

This expression has no $1 - W$ denominator. At $p \mid u$, where $D = W = 0$, it reduces to $(1-V)\mathcal{E}_p$. In both stated regions $|R| < 1$, $|V| < 1$, and $|D| < 1$, uniformly away from one, so these are holomorphic local expressions.

Bounds in the two regions. For $p \nmid u$ and $\vartheta = (-w_r)_+$, the $j = 0$ row gives

$$|\mathcal{E}_p| \ll Q^{4-6x_r-6z_r+\vartheta} + Q^{1-x_r-w_r-6z_r}.$$

In the first region $4-6x_r-6z_r\le -11/10$ and $\vartheta\le 1/100$. The extra factor $1-W$ in (6.17) therefore leaves this contribution at most $Q^{-27/25}$. The term DW is at most $Q^{-1-\epsilon_0}$; the other terms in that equation are smaller. For $p\mid u$, the strict second-family term has exponent $1-x_r-w_r\le -\epsilon_0$. The closest other exponents in the table are

$$3/2-3x_r\le -3/100,\qquad 2-3x_r-w_r\le -1/50-\epsilon_0,$$

and all remaining ones are no larger than $-3/100$. Consequently, with $\epsilon_H=\min(\epsilon_0,1/50)$,

$$H_p-1=O(Q^{-1-\epsilon_H})\quad (p\nmid u),\qquad H_p-1=O(Q^{-\epsilon_H})\quad (p\mid u).$$

In the second region the same comparison gives the stronger bounds

$$H_p-1=O(Q^{-363/200})\quad (p\nmid u),\qquad H_p-1=O(Q^{-33/40})\quad (p\mid u).$$

The closest good-prime term here is $Q^{1-x_r-w_r-6z_r}$, and the closest ramified term is $Q^{1-x_r-w_r}$.

The Euler product. The sum over primes not dividing u converges normally, and the finite product over primes of u is $O_\epsilon(q_u^\epsilon)$ by the divisor-product bound. The estimates are uniform in all imaginary parts and unit phases. When $u=1$, the good-prime tail above P_0 is $O(P_0^{-163/200})$ by the ideal count, which implies the asserted $O(P_0^{-4/5})$ estimate for the product. Finally extracting $(1-V)^{-1}(1-W)^{-1}(1-D)$ at every prime gives (6.14). The normal convergence of the remaining product proves its claimed continuation. □

For $u=1$, (6.14) contains $1/L^S(x,\eta)$, and its numerator has the two principal factors $\zeta_F^S(6z)$ and $\zeta_F^S(w)$. Their residues will produce the target Mellin signal. For intermediate row norms, the next section assigns one zero-free rectangle to the finite family of twists associated with each row and produces simultaneous polynomial witnesses from a selected zero. The principal and bounded rows, together with the outer norm ranges, are treated in Section 9.

7 A zero detector with simultaneous polynomial bounds

The high expansion contains a sum over sixth-power-free rows u. We associate each nonprincipal row with a buffered zero-free rectangle for the finite family of character presentations that it determines. Whenever the resulting bin lies above a fixed floor, a zero produces an inverse polynomial and a plain polynomial with simultaneous lower bounds. These witnesses can then be counted using different mean estimates. The analytic inputs here are Lemma 3.15, the smooth calculus of Lemmas 3.6–3.10, and the global growth estimate in (3.21), together with the deleted Euler-factor bounds of Lemma 3.16.

Throughout this section, assume $\beta_*>51/100$, where β_* is the global supremum in (2.1). The detector will be used at both boundaries of the continuation criterion.

7.1 Buffered rectangles and pointwise bounds

Fix the arithmetic data $\mathcal{A}$ independently of Z, as in Section 3. Recall that the finite group

$$\Theta=\langle \eta,\widehat{T}\rangle$$

contains the target and the fixed ray twists used in the high-row family, and every conductor prime of a member of Θ belongs to S. For a sixth-power-free element u, define the finite collection of presentations

$$\mathcal{X}_u=\{\psi_{u,\nu,\varsigma}(n)=\nu(n)\chi_n(u)^\varsigma:\nu\in\Theta,\ \varsigma\in\{-1,1\}\}.$$

At a nonunit the value of $\chi_n(u)^{-1}$, like every other power of the symbol, is zero. The group property of Θ shows that $\mathcal{X}_u$ is closed under conjugation. It contains the numerator character

$\chi_\bullet(u)$, the denominator character $\eta\overline{\chi_\bullet(u)}$, and every zero-extended presentation obtained by multiplying either orientation of the sextic symbol by a member of Θ.

Let ψ^* be the primitive character inducing a presentation $\psi \in \mathcal{X}_u$, and let Q_ψ be its conductor norm. Reciprocity and the fixed conductor primes give

$$Q_\psi \ll_{\mathcal{A}} q_u, \qquad L_{\mathrm{orig}}(s,\psi) = L(s,\psi^*) \prod_{p\in E_u} (1-\psi^*(p)q_p^{-s}), \qquad E_u \subset S \cup \{p : p \mid u\}.$$

The radical of the deleted product has norm $O_S(q_u)$. The deleted factors are those of the given presentation, so their zero extensions remain tied to the row u.

If $v_p(u) = j \in \{1,\dots,5\}$ at a prime $p \notin S$, the local character of $\chi_\bullet(u)$ on units at p has order $6/\gcd(6,j) > 1$. A character in Θ is unramified there, so it cannot cancel this local character. Consequently a presentation in $\mathcal{X}_u$ can induce the principal character only when u is supported on S. There are finitely many such sixth-power-free rows, including unit factors. Remove them for the present detector; each application must estimate these bounded rows supported on S separately.

The detector uses a row-norm scale, a height cutoff, and a grid for real parts of zeros. Fix $0 < d_{\min} < d_{\max} < \infty$ and write $U = Z^d$, where $d_{\min} \le d \le d_{\max}$. The rows in the present dyad satisfy $q_u \asymp U$, with fixed comparison constants. Let

$$0 < \tau \le d_{\min}/100, \qquad T_1 = Z^\tau > 2, \qquad 0 < e < 10^{-3}, \qquad I = \lceil 1/e \rceil + 2.$$

The value of τ will be chosen after the fixed height orders are known. For now it is fixed independently of Z. In particular $T_1 \le U^{1/100}$.

For each retained row and $j = 1,\dots,I$, define its running zero maxima by

$$M_j(u) = \max\left(\{51/100\} \cup \left\{\Re\rho : \begin{array}{l} \psi \in \mathcal{X}_u,\ L(\rho,\psi^*) = 0, \\ \Re\rho \ge 51/100,\ |\Im\rho| \le 3jT_1 \end{array}\right\}\right).$$

The floor $51/100$ makes this definition apply even when the rectangle contains no zero.

Lemma 7.1 (Buffered zero-free bins). *Assume $\beta_* > 51/100$ and use the arithmetic data, retained nonprincipal rows, and parameters of Section 7.1. Thus $U = Z^d$ is the row-norm scale, $T_1 = Z^\tau$ is the height, and e is the grid spacing. For every retained row u, its running maxima $M_j(u)$ admit an index $\iota \in \{1,\dots,I-1\}$ and a grid point*

$$a \in (51/100 + e\mathbb{Z}_{\ge 0}) \cap [51/100, 1]$$

such that

$$a \le M_\iota(u) < a + e, \qquad M_{\iota+1}(u) < a + 2e.$$

If $a > 51/100$, some $\psi \in \mathcal{X}_u$ has a zero $\rho = \sigma + i\gamma$ with

$$a \le \sigma < a + e, \qquad |\gamma| \le 3\iota T_1.$$

For every $\epsilon_1 > 0$, every $\psi \in \mathcal{X}_u$, and sufficiently large Z, uniformly in the retained row,

$$|L_{\mathrm{orig}}(s,\psi)| + |L_{\mathrm{orig}}(s,\psi)^{-1}| \ll_{\mathcal{A},e,\epsilon_1} U^{\epsilon_1} \qquad \big(\Re s \ge a + 6e,\ |\Im s| \le (3\iota+2)T_1\big).$$

On the reflected line in the same height range,

$$|L_{\mathrm{orig}}(1-a-6e+it,\psi)| \ll_{\mathcal{A},e,\epsilon_1} U^{a-1/2+12e+\epsilon_1}(3+T_1)^C,$$

where C depends only on the fixed real strip and the field. There are $O_e(1)$ possible pairs (ι, a), independently of the row and its conductor.

Proof. Each maximum exists: the collection is finite, the zeros of each nonprincipal primitive L-function are discrete in compact rectangles, and absolute Euler convergence excludes zeros with real part greater than one. The sequence $M_j(u)$ is nondecreasing and lies in $[51/100, 1]$. If every one of its $I-1$ increments exceeded e, then

$$M_I(u) - M_1(u) > (I-1)e > 1,$$

which is impossible. Choose an index with $M_{\iota+1}(u) - M_\iota(u) \le e$, and round $M_\iota(u)$ down on the stated grid. This gives the two inequalities. When $a > 51/100$, the maximum $M_\iota(u)$ is attained by an actual zero, giving ρ. This reasoning allows a zero on the line one.

For $|t| \le (3\iota+2)T_1$, the closed disk centered at $2+it$ with radius $2-a-2e$ has real part at least $a+2e$. Its radius is less than $3/2 < T_1$, so every point in it has imaginary part of absolute value less than $3(\iota+1)T_1$. The bound $M_{\iota+1}(u) < a+2e$ therefore excludes every zero of every primitive function in this disk. Lemma 3.15, with center $2+i\Im s$, controls the concentric disk of radius $2-a-6e$; absolute Euler convergence supplies the bound farther right. Since $Q_\psi \ll_{\mathcal{A}} U$ and $(3+|t|)^2 \ll_e U^{1/50}$, the arbitrarily small power in that lemma can be chosen so that its contribution is $U^{\epsilon_1/2}$. The estimates for polynomial-size deleted Euler products following that lemma supply the other $U^{\epsilon_1/2}$.

For the last assertion, the primitive functional equation recalled in the proof of Lemma 3.14 [10, Equation (1.1)] relates the value at $1-a-6e+it$ to the conjugate primitive function at $a+6e-it$. The conjugate presentation is in $\mathcal{X}_u$. The conductor and gamma quotient contribute at most

$$Q_\psi^{a-1/2+6e}(3+|t|)^C.$$

The reflected primitive value has an arbitrarily small U-power by the preceding disk bound. Finally, $1-a-6e \ge -6e$, so the upper bound for the deleted product is $U^{6e+\epsilon_1}$, after reducing the preliminary losses. This proves the reflected estimate. The ranges of ι and a give $O_e(1)$ pairs. □

We call (ι, a) the bin of the row and put

$$\delta = 2a-1, \qquad 1/50 \le \delta \le 1.$$

Every primitive character inducing a presentation in $\mathcal{X}_u$ is a finite-order Hecke character. The definition of β_*, together with $\beta_* > 51/100$, therefore gives $M_\iota(u) \le \beta_*$. In particular, the bin satisfies the inequalities

$$a \le \beta_*, \qquad \delta \le 2\beta_* - 1.$$

The bin $a = 51/100$ will be bounded by the trivial row count. For $a > 51/100$, the zero in Lemma 7.1 produces large polynomials to which the moments can be applied.

Lemma 7.2 (Pointwise dyadic estimates). *Fix the arithmetic data $\mathcal{A}$, bounded nonnegative ranges for r, m, and a family of untwisted profiles with uniformly bounded smooth seminorms on a fixed annulus. For every $\epsilon > 0$, there exist $e_0 > 0$, depending only on ϵ and the bounded length ranges, and a finite height order $A_{\mathcal{A}}$ with the following property. For every e and every fixed positive τ in the range of Section 7.1, the estimates below hold uniformly in the retained rows and profiles, for sufficiently large Z, whenever*

$$0 < e < e_0, \qquad (1+T_1)^{A_{\mathcal{A}}} \le U^{\epsilon/10}.$$

Let u be such a row of Lemma 7.1, with bin (ι, a), and put $\delta = 2a-1$, $D = U^r$ and $N = U^m$. The polynomials

$$M_\psi(D; W_M), \qquad S_\psi(N; W_S)$$

are those of (3.3) *and* (3.2) *at these norm scales, for* $\psi \in \mathcal{X}_u$. *Every pure norm twist has height at most* $(3\iota+1)T_1$, *and all additional Mellin frequencies entering the same* L*-argument have total absolute value at most* $T_1/2$. *Then*

$$|M_\psi(D; W_M)|^2 \ll_{\mathcal{A},\epsilon} U^{\delta r+\epsilon}, \tag{7.1}$$

$$|S_\psi(N; W_S)|^2 \ll_{\mathcal{A},\epsilon} U^{\delta \min(m,1-m)+\epsilon}. \tag{7.2}$$

For each such τ, *there exists a finite external tail order for which these estimates hold with the same* $A_{\mathcal{A}}$.

Proof. The bin controls a finite-height rectangle. Shifting the central Mellin integral inside that rectangle bounds the inverse polynomial. The plain polynomial also admits a reflected shift, and the better of its two bounds gives the minimum in the exponent. All remaining tails are estimated on an absolute line. Write K for the external tail order in this proof.

The inverse polynomial. For a fixed annular W, a bounded real number σ, and a pure twist $y^{i\omega}$, Mellin inversion on a line $c+\sigma > 1$ gives

$$\begin{aligned} &U^{-r/2}\sum_n \mu(n)\psi(n)W(q_n/U^r)(q_n/U^r)^{-\sigma+i\omega} \\ &\quad = \frac{1}{2\pi i}\int_{(c)} \mathcal{M}W(s)\, U^{r(s+\sigma-i\omega-1/2)} L_{\mathrm{orig}}(s+\sigma-i\omega,\psi)^{-1}\, ds. \end{aligned}$$

Here $\mathcal{M}W$ is the Mellin transform defined in Lemma 3.9. The twist occurs in the L-argument, while the tail estimates apply to the transform of the untwisted profile. Only the portion with added frequency at most the assigned fraction of T_1 is shifted to $\Re(s+\sigma) = a+6e$. The horizontal joins and the shifted portion remain in the zero-free rectangle of Lemma 7.1. Its bound gives $U^{(a-1/2+6e)r+\epsilon_1}$ for the central portion.

The remaining tails stay on the absolute line, which we may take to be $\Re(s+\sigma) = 2$. There the reciprocal Euler product is absolutely bounded. On the inverse joins the reciprocal remains inside the buffered rectangle. Since the real join interval and the length range are bounded, the remaining arithmetic and scale factors on a join are $O(U^B(1+|\Im s|)^J)$ for fixed B, J chosen before the external tail order. The pointwise Mellin bound in (3.14), applied on that compact real interval with derivative order $K+\lceil J\rceil+1$, bounds each horizontal join by $O(U^B T_1^{-K})$. The separate vertical tails have the same bound by (3.13). Both estimates apply to the untwisted profile, uniformly in the stated family. Because $T_1 = Z^\tau$, $\tau > 0$, and all lengths lie in a fixed bounded interval, a sufficiently large fixed K makes these errors smaller than the claimed bound. Taking e and ϵ_1 small in the prescribed ϵ proves (7.1).

The plain polynomial. For the plain polynomial the same calculation uses L_{orig} instead of its reciprocal. Shifting to $\Re(s+\sigma) = a+6e$ gives

$$|S_\psi(N; W_S)| \ll U^{(a-1/2+6e)m+\epsilon_1} + O(U^B T_1^{-K}).$$

Alternatively shift its central portion to $\Re(s+\sigma) = 1-a-6e$. The function is entire because the row is nonprincipal. The reflected estimate of Lemma 7.1 gives

$$|S_\psi(N; W_S)| \ll U^{a-1/2+12e+\epsilon_1} U^{(1/2-a-6e)m}(3+T_1)^C + O(U^B T_1^{-K}).$$

The joins here need only the global upper strip bound and the negative-strip bound for the deleted product, since no reciprocal is present. These give fixed B, J as above, so the pointwise Mellin estimate bounds the joins and the integrated Fourier estimate bounds the absolute-line tails. Taking the better central bound, choosing K to dominate also the possibly negative exponent when $m > 1$, and using the displayed height hypothesis proves (7.2). A primitive conductor smaller than U reduces the reflected conductor factor. The original deleted Euler factors have already been included in Lemma 7.1.

Height and smoothness orders. There are only finitely many separated variables in any one L-argument. Each receives a fixed fraction of the single total allowance $T_1/2$, shared across all successive shifts. The profile windows and the finite height orders used in the row estimates are fixed before K. Lemma 3.8 differentiates only the untwisted separating profiles when increasing K, which proves the last assertion. □

7.2 Simultaneous lower bounds for the two polynomials

At a selected zero, the truncated-inverse construction below produces a product with squared size at least $U^{\delta(r+m)-\epsilon}$. The pointwise estimates bound the same product by $U^{\delta(r+\min(m,1-m))+\epsilon}$. Since $\delta \ge 1/50$, these two bounds force $m \le 1/2 + O(\epsilon)$. Each factor then attains its own pointwise exponent up to the prescribed loss, as asserted in the next proposition.

Proposition 7.3 (Simultaneous polynomial lower bounds). *Let u be a retained row of Lemma 7.1 with bin (ι, a), where $a > 51/100$, and put $\delta = 2a - 1$. For every $t \in [1, 3/2]$ and every prescribed $\epsilon > 0$, there exists a positive preliminary loss ϵ_{pre} such that, whenever the loss and height hypotheses of Lemma 7.2 hold with loss ϵ_{pre}, the following conclusions hold for sufficiently large Z. There exist $\psi \in \mathcal{X}_u$, a zero $\rho = \sigma + i\gamma$ as in Lemma 7.1, dyadic lengths $D = U^r$, $N = U^m$, and two polynomials $M_\psi(D; W_M), S_\psi(N; W_S)$ for the common row character ψ such that*

$$\begin{gathered} r \le t + O(1/\log U), \qquad r + m \ge t - O(1/\log U), \\ |M_\psi(D; W_M) S_\psi(N; W_S)|^2 \gg_{\mathcal{A},\epsilon} U^{\delta(r+m)-\epsilon}. \end{gathered} \tag{7.3}$$

Their lengths and individual values also satisfy

$$\begin{aligned} &t - \tfrac{1}{2} - O(\epsilon) \le r \le t + O(1/\log U), \qquad 0 \le m \le \tfrac{1}{2} + O(\epsilon), \\ &|M_\psi(D; W_M)|^2 \gg_{\mathcal{A},\epsilon} U^{\delta r - \epsilon}, \qquad |S_\psi(N; W_S)|^2 \gg_{\mathcal{A},\epsilon} U^{\delta m - \epsilon}. \end{aligned} \tag{7.4}$$

The constants in the $O(\epsilon)$ terms are absolute on the stated parameter ranges. Both polynomials have the same twist height $\gamma - \nu$, where $|\nu| \le cT_1$ for a fixed $c > 0$ chosen within the cumulative frequency allowance. Their untwisted profiles form a uniformly smooth annular family.

For each row, both witnesses use one presentation $\psi_{u,\vartheta,\varsigma}$, selected by a label (ϑ, ς) in the fixed finite set $\Theta \times \{-1, 1\}$. The presentation itself varies with u. The dyadic pair can likewise be selected separately for each row from $O((\log U)^2)$ possibilities.

Proof. The truncated-inverse and Gamma-integral construction is a form of the classical zero detector; compare [20, Appendix C] and [13, Section 13.1]. We first use the zero to obtain a nonzero tail of the truncated inverse, then separate a pair of dyadic blocks and compare their product with Lemma 7.2.

The detector integral. Choose the zero ρ supplied by the bin and set

$$D_* = U^t, \qquad Y_* = U^{20}.$$

The scale D_* sets the inverse truncation length; the larger scale Y_* will make the shifted Gamma integral tend to zero. Let $V_\le$ be a fixed smooth function equal to one on $[0, 1]$ and zero on $[2, \infty)$. Define

$$C_\psi(s) = \sum_l \mu(l)\psi(l) V_\le(q_l/D_*) q_l^{-s}, \qquad J = \frac{1}{2\pi i} \int_{(2)} Y_*^z \Gamma(z) L_{\mathrm{orig}}(\rho + z, \psi) C_\psi(\rho + z)\, dz.$$

The row character is nonprincipal, so L_{orig} is entire. Moving the line to $\Re z = -1/4$ crosses only the pole of $\Gamma(z)$ at zero, and its residue vanishes because $L_{\mathrm{orig}}(\rho, \psi) = 0$. The global strip bound in (3.21), together with the deleted Euler factors, bounds the new line by

$$U^{-5} U^2 D_*^{1-\sigma+1/4+o(1)} T_1^2.$$

Indeed $\Re(\rho - 1/4) \geq 26/100 > 0$, so the deleted product has an arbitrarily small U-power; the deliberately weaker U^2 includes the primitive conductor bound. The finite polynomial is bounded by the displayed power of D_* using the ideal count. Finally, the fixed polynomial in the added height is integrable against $\Gamma(-1/4 + iv)$, since $(1 + |\gamma + v|)^2 \leq (1 + |\gamma|)^2(1 + |v|)^2$. Uniformly for $1 \leq t \leq 3/2$,

$$-3 + t(5/4 - \sigma) \leq -3 + \frac{3}{2}\left(\frac{5}{4} - \frac{51}{100}\right) = -\frac{189}{100}.$$

Thus $J = o(1)$. This Gamma integration uses global upper bounds, so the reciprocal height allowance remains available for the witness estimates.

On the original line, absolute convergence and the Mellin formula for the exponential give

$$J = \sum_{l,k} \mu(l)\psi(lk)V_{\leq}(q_l/D_*)(q_l q_k)^{-\rho}e^{-q_l q_k/Y_*}.$$

For $1 < q_n \leq D_*$, the coefficient of $\psi(n)q_n^{-\rho}$ is $\sum_{l|n} \mu(l) = 0$. For $n = 1$ it is one. When $q_n > D_*$, $V_{\leq}(2q_n/D_*) = 0$. Consequently insertion of $1 - V_{\leq}(2q_l q_k/D_*)$ removes exactly the unit contribution $e^{-1/Y_*} = 1 + o(1)$. The resulting tail has absolute value $1 + o(1)$. Multiplying it by a fixed smooth terminal cutoff equal to one for $q_l q_k \leq U^{21}$ and zero for $q_l q_k \geq 2U^{21}$ changes it by $o(1)$; this follows from $Y_* = U^{20}$, exponential decay, and the ideal count.

A pair of dyadic blocks. Partition $q_l \asymp D$, $q_k \asymp N$ by a fixed smooth dyadic partition. There are $O((\log U)^2)$ relevant pairs, and their support satisfies

$$D \ll D_*, \qquad DN \gg D_*, \qquad DN \ll U^{21}.$$

With $x = q_l/D$, $y = q_k/N$, the profile for a pair is

$$W_1(x)V_{\leq}(Dx/D_*)\, W_2(y)\, H_{D,N}(xy),$$

where Ω is a fixed annular cutoff equal to one on the product of the supports of W_1, W_2, and

$$H_{D,N}(s) = \Omega(s)[1 - V_{\leq}(2DNs/D_*)]e^{-DNs/Y_*}V_{\mathrm{term}}(DNs/U^{21}).$$

Every fixed logarithmic derivative is bounded uniformly in D, N, U. On a cutoff transition its argument is bounded, and a logarithmic derivative of the exponential is a polynomial in its argument times that exponential. Logarithmic Fourier inversion therefore gives

$$H_{D,N}(xy) = \frac{1}{2\pi}\int_{\mathbb{R}} \widehat{H}_{D,N}(\nu)x^{i\nu}y^{i\nu}\, d\nu,$$

$$\int_{|\nu|>cT_1} |\widehat{H}_{D,N}(\nu)|(1 + |\nu|)^J\, d\nu \ll_{J,N_0,c} T_1^{-N_0}$$

for every fixed J, N_0. The coefficient L^1-norm on the full line is also uniformly bounded. Trivial bounds for the two finite polynomials have a fixed U-power because their lengths are bounded by $U^{21+o(1)}$. After $\tau > 0$ is fixed, choose N_0 so that the discarded Fourier tail is $o(1)$.

The absolute value of the remaining sum of integrals is bounded below by a positive constant. The number of dyadic pairs and the uniform L^1-norm show that, for one pair and one $|\nu| \leq cT_1$, the product of the two unnormalized blocks has absolute value $\gg (\log U)^{-2}$. Both profiles have the same height $\gamma - \nu$:

$$W_M(x) = W_1(x)V_{\leq}(Dx/D_*)x^{-\sigma - i(\gamma-\nu)},$$
$$W_S(y) = W_2(y)y^{-\sigma - i(\gamma-\nu)}.$$

The first extra cutoff remains part of the inverse annular profile and creates no second frequency. Removing the factors $D^{1/2-\sigma}$ and $N^{1/2-\sigma}$ by central normalization gives

$$|M_\psi(D; W_M)S_\psi(N; W_S)|^2 \gg (\log U)^{-4}(DN)^{2\sigma-1} \geq U^{\delta(r+m)-\epsilon}$$

for sufficiently large U. The support inequalities give the two length bounds in (7.3).
Individual lower bounds. Lemma 7.2 bounds each selected block from above. Choose its loss smaller than the present ϵ. Comparison of the product lower bound with the two upper bounds gives

$$\delta\{m - \min(m, 1-m)\} = 2\delta(m - 1/2)_+ \ll \epsilon.$$

Since $\delta \geq 1/50$, this implies $m \leq 1/2 + O(\epsilon)$ with an absolute constant. The support inequality then gives $r \geq t - 1/2 - O(\epsilon)$. Dividing the product lower bound in turn by each individual upper bound gives the two individual lower bounds in (7.4). Only the dyadic support errors are $O(1/\log U)$; all other losses are arbitrarily small fixed powers. Lemma 3.7 permits the stated rowwise choices with a fixed polynomial height factor. □

8 A sextic-sieve row count

The zero detector supplies a large inverse polynomial for each row above the floor bin. We now turn that lower bound into a count of sixth-power-free element rows. We first prove the required squarefree sextic large sieve in the primary convention, using the norm-recursion method of Blomer, Goldmakher, and Louvel [4, Section 3]. We then fix the part of an element row whose prime valuations are at least two and apply the sieve to its squarefree factor. The size of the fixed part also gives an independent upper bound for the number of rows; the better of the two bounds produces the required exponent.

8.1 The sextic large sieve in the primary convention

We continue to use the fixed finite set S of prime ideals, containing the primes above 6 and all fixed conductor primes. All ideal indices in the next lemma are prime to S and are represented by their primary generators. In particular, "squarefree" refers to ideals of $\mathcal{O} = \mathbb{Z}[\omega]$, not to their rational norms.

Lemma 8.1 (Sextic large sieve). *Let S and the primary representatives of ideals be as in Section 8.1. Let $K, D \geq 1$, let $\varsigma \in \{-1, 1\}$, and let (c_n) be any sequence of complex numbers indexed by the squarefree ideals outside S. The sequence is fixed independently of the row k. For every $\epsilon > 0$,*

$$\sum_{\substack{k \text{ sf} \\ q_k \leq K}} \left| \sum_{\substack{n \text{ sf} \\ q_n \leq D}} c_n \chi_n(k)^{\varsigma} \right|^2 \ll_{S,\epsilon} (KD)^{\epsilon}\{K + D + (KD)^{2/3}\} \sum_{\substack{n \text{ sf} \\ q_n \leq D}} |c_n|^2. \tag{8.1}$$

Every symbol in this formula has its original zero value on a nonunit. A fixed restriction of the row set is allowed. A fixed restriction of the coefficient support is allowed by extending that coefficient sequence by zero. Such a restriction may depend on an object fixed before the row sum, but the coefficients may not otherwise depend on k.

Proof. We follow the higher-order large-sieve recursion of Blomer, Goldmakher, and Louvel [4, Section 3], giving the local verification for the present primary normalization. In the finite Fourier calculation the individual residue characters act on elements; their unit-trivial quotient is the character on ideals to which we apply Poisson summation.

For good ideals a, b, represented by their primary generators, put

$$F_b(a) = \chi_a(b), \qquad J = \{1, 2, 4\}.$$

This uses the original zero-extended symbol, also when a is not squarefree. It is multiplicative in both a and b. In every power below, including a multiple of six, a nonunit still has value zero.

For $M > 0$ let

$$\mathcal{I}_M = \{a \text{ good} : M < q_a \le 2M\}, \qquad \mathcal{A}_M = \{a \in \mathcal{I}_M : a \text{ squarefree}\}.$$

For a sequence supported on $\mathcal{A}_N$ write $\|\lambda\|_2^2 = \sum_{b\in\mathcal{A}_N} |\lambda_b|^2$, and define

$$D_j(M,N) = \sup_{\|\lambda\|_2=1} \sum_{a\in\mathcal{A}_M} \left| \sum_{b\in\mathcal{A}_N} \lambda_b F_b(a)^j \right|^2,$$

$$E_j(M,N) = \sup_{\|\lambda\|_2=1} \sum_{a\in\mathcal{I}_M} \left| \sum_{b\in\mathcal{A}_N} \lambda_b F_b(a)^j \right|^2.$$

An empty support gives norm zero. These norms include all finite ray classes. We will prove the symmetric bound for D_1; the matrix in the lemma is its transpose with the two lengths exchanged.

The squarefree norm D_j is our target. We also need E_j, since Poisson summation produces arbitrary evaluation ideals. Opening the square in its defining sum and pairing the column characters will replace a row length M by a dual length of order N^2/M. A power decomposition then returns the unrestricted row sum to squarefree norms. It selects either the first-power or the second-power factor, so the exponents $j \in \{1, 2, 4\}$ are kept together: this set is closed under $j \mapsto 2j \pmod 6$. We will use this cycle to improve a provisional exponent in the bound for D_j.

The paired summation formula. Use the fixed finite group

$$\mathcal{T} = \ker\big((\mathcal{O}/36\mathcal{O})^\times \longrightarrow (\mathcal{O}/3\mathcal{O})^\times\big), \qquad t(b) = b \bmod 36\mathcal{O}.$$

The primary normalization identifies these with the ray classes modulo $36\mathcal{O}$: each unit orbit has a unique representative equal to one modulo 3. In particular t is multiplicative. This subdivision is fixed, since the primes above 2 and 3 belong to S.

For a global unit u and a good prime p, reduction modulo p gives $\chi_p(u) = u^{(q_p-1)/6}$. Since $q_p \equiv 1 \pmod 6$, multiplication over the prime factors of b gives

$$\chi_b(u) = u^{(q_b-1)/6}. \tag{8.2}$$

The exponent is read modulo six: expanding a product of integers $1 + 6h$ proves the equality of exponents. Thus equal t-classes have the same restriction to the six global units. They also have the same residue modulo 4, which controls the reciprocity factor $\mathcal{R}$ of Lemma 3.5.

Let b, c be coprime squarefree good ideals with $t(b) = t(c)$ and let $j \in J$. The raw character on elements

$$\Theta_{b,c}^{(j)}(z) = \chi_b(z)^j \overline{\chi_c(z)^j} \qquad (z \bmod bc)$$

is trivial on global units by (8.2). It therefore defines a finite-order ray character on ideals by

$$\Psi_{b,c}^{(j)}((z)) = \Theta_{b,c}^{(j)}(z).$$

For fractional ideals prime to bc, the residue symbols are evaluated on local unit fractions; the same unit cancellation makes this definition independent of the generator. At each good prime the sextic residue character has exact order six, since the residue multiplicative group is cyclic. Hence its powers j and $-j$ are nontrivial. CRT shows that $\Theta_{b,c}^{(j)}$ has raw conductor bc, and the ray character has the same conductor: omitting a prime would contradict this nontriviality by varying only its residue. It is extended by zero on ideals meeting bc. For the pair $b = c = 1$ define $\Psi_{1,1}^{(j)}$ to be principal.

Sextic reciprocity, with its factor fixed by $t(b) = t(c)$, gives for every good ideal a

$$\Psi_{b,c}^{(j)}(a) = F_b(a)^j \overline{F_c(a)^j}. \tag{8.3}$$

For coprime inputs the two reciprocity factors cancel; on the other inputs both sides are zero. With the Gauss sums already defined in the arithmetic preliminaries, CRT using $z = cv + bw$ gives

$$\begin{aligned}\Gamma_{b,c}^{(j)} &:= q_{bc}^{-1/2} \sum_{z \bmod bc} \Theta_{b,c}^{(j)}(z) e(z/(bc)) \\ &= \chi_b(c)^j \overline{\chi_c(b)^j} \gamma_j(b)\gamma_{-j}(c) = \eta_{t(b),j}\gamma_j(b)\gamma_{-j}(c),\end{aligned} \tag{8.4}$$

where $\eta_{t(b),j} = \mathcal{R}(b,c)^j$ depends only on the common class. Every $\gamma_{\pm j}$ here has modulus one by prime Gauss orthogonality and CRT. Formula (8.4) retains γ_{-j} itself, including its factor at -1 under conjugation.

Choose once a nonnegative $W \in C_c^\infty((0,\infty))$ with $W \geq 1$ on $[1,2]$. For the Fourier transform and self-dual measure in the arithmetic preliminaries, put $w(z) = W(|z|^2)$ and define $\mathscr{W}$ by

$$\widehat{w}(y) = \mathscr{W}(|y|^2).$$

The transform is radial, smooth at zero, and rapidly decreasing. Thus

$$\mathscr{W}(x) \ll_A (1+x)^{-A}, \qquad \int_{\mathbb{R}} |\mathcal{M}\mathscr{W}(\sigma + it)|\, dt < \infty \quad (\sigma > 0). \tag{8.5}$$

Here $\mathcal{M}$ denotes the Mellin transform. The second assertion follows by integration by parts in the Mellin integral; all logarithmic derivatives are bounded at zero and rapidly decreasing at infinity. Scaling the Fourier transform gives $M\mathscr{W}(M|y|^2)$ for $z \mapsto W(|z|^2/M)$.

For $bc \neq 1$, finite Fourier inversion for the primitive raw character is

$$\sum_{v \bmod bc} \Theta_{b,c}^{(j)}(v) e(vy/(bc)) = \overline{\Theta_{b,c}^{(j)}(y)} \sqrt{q_{bc}}\, \Gamma_{b,c}^{(j)}.$$

For a nonunit y the left side vanishes because one nontrivial local character is summed against the trivial additive character. For the stated e and $d\mu$, the trace pairing on the basis $(1,\omega)$ has matrix $\left(\begin{smallmatrix} 0 & 1 \\ 1 & -1 \end{smallmatrix}\right)$, of determinant -1, and the $d\mu$-covolume of $\mathcal{O}$ is one. Thus the dual of $bc\mathcal{O}$ is $(bc)^{-1}\mathcal{O}$, while the covolume of $bc\mathcal{O}$ is q_{bc}. Coset Poisson summation, followed by division by the six generators of each ideal, therefore gives

$$\sum_{a \neq 0} W(q_a/M)\Psi_{b,c}^{(j)}(a) = \frac{M\Gamma_{b,c}^{(j)}}{\sqrt{q_{bc}}} \sum_{a \neq 0} \mathscr{W}(Mq_a/q_{bc}) \overline{\Psi_{b,c}^{(j)}(a)}. \tag{8.6}$$

Both sums here are over all nonzero integral ideals. The division by six uses unit-triviality of $\Theta_{b,c}^{(j)}$ and radiality of the weight. The zero frequency vanishes since this character is nontrivial.

Poisson transformation of the squared norm. For λ supported on one t-class in $\mathcal{A}_N$, set

$$Q_j(M,N;\lambda) = \sum_{\substack{b,c \in \mathcal{A}_N \\ (b,c)=1}} \lambda_b \overline{\lambda_c} \sum_{a \neq 0} W(q_a/M)\Psi_{b,c}^{(j)}(a).$$

We claim, for every $\varepsilon > 0$,

$$\begin{aligned}|Q_j(M,N;\lambda)| &\ll_{S,\varepsilon} (2+MN)^\varepsilon \|\lambda\|_2^2 \\ &\quad \times \left\{ M + \frac{M}{N} \max_{\substack{1 \leq H \\ H \ll_{S,\varepsilon} (2+MN)^\varepsilon N^2/M}} E_j(H,N) \right\}.\end{aligned} \tag{8.7}$$

The maximum runs over dyadic shells and is zero if empty. If $N \leq 1$, only the unit good ideal can occur and the annular ideal count proves the M term. Also $Q_j = 0$ when M is below a fixed

positive constant, because W has a fixed upper support endpoint and nonzero ideals have norm at least one.

Suppose $N > 1$ and normalize $\|\lambda\|_2 = 1$. Then every pair in Q_j is nontrivial and $q_{bc} \asymp N^2$. Equation (8.6) expresses the sum in terms of dual ideals, to which the triangle inequality applies. For a small $\delta > 0$ put $Z = (2 + MN)^\delta$. Cauchy gives $(\sum_b |\lambda_b|)^2 \ll N$. By (8.5), the part with $q_a > ZN^2/M$ is, for any $A > 1$, at most

$$M \sum_{q_a > ZN^2/M} (Mq_a/N^2)^{-A} \ll N^2 Z^{1-A}.$$

The same bound holds if the cutoff is below one. Taking A large makes this an arbitrary negative power error.

In the remaining sum write $a = a_0\mathfrak{s}$, with a_0 good and $\mathfrak{s}$ supported on S, and choose one generator s_0 for each fixed $\mathfrak{s}$. Equations (8.3) and (8.4) give the separated factors

$$\overline{\Psi_{b,c}^{(j)}(a_0\mathfrak{s})} = \overline{F_b(a_0)^j} F_c(a_0)^j \overline{\chi_b(s_0)^j} \chi_c(s_0)^j.$$

All $\chi_b(s_0)$ have modulus one. Splitting a_0 into shells $a_0 \in \mathcal{I}_H$, where $H \ll ZN^2/(Mq_\mathfrak{s})$, and using Mellin inversion for $\mathscr{W}$ on $\Re w = \delta$ separates the weight into row and coefficient factors. Apart from a bounded factor depending on $M, q_{a_0}, q_\mathfrak{s}$, the two coefficient sequences are

$$A_b = \lambda_b \gamma_j(b) \overline{\chi_b(s_0)^j} q_b^{-1/2+\delta+it}, \qquad B_b = \overline{\lambda_b} \gamma_{-j}(b) \chi_b(s_0)^j q_b^{-1/2+\delta+it}.$$

They are fixed across the a_0 sum and have absolute values $\ll N^{-1/2+\delta}|\lambda_b|$. Detecting $(b, c) = 1$ by Möbius inversion and applying Cauchy in a_0 gives

$$\sum_{a_0 \in \mathcal{I}_H} \left| \sum_{\substack{b,c \in \mathcal{A}_N \\ (b,c)=1}} A_b B_c \overline{F_b(a_0)^j} F_c(a_0)^j \right|$$
$$\le E_j(H,N) \sum_\mathfrak{d} \left(\sum_{\mathfrak{d}|b} |A_b|^2 \right)^{1/2} \left(\sum_{\mathfrak{d}|b} |B_b|^2 \right)^{1/2} \ll N^{-1+3\delta} E_j(H,N).$$

Here the divisor bound absorbs the sum over $\mathfrak{d}$. Conjugating the entire first polynomial uses the same E_j norm. The factors with $H < 1$ have just the unit evaluation ideal and satisfy $E_j(H, N) \ll N$. The number of choices of $\mathfrak{s}$ and of the dyadic shell is bounded by a fixed power of $\log(2 + ZN^2)$, because S is fixed and a nonzero Q_j has M bounded below. Equation (8.5) integrates the t variable. Choosing δ sufficiently small in terms of ε proves (8.7), including the allowed global loss on its M term.

The paired estimate bounds E_j after partitioning its inner sum by t, inserting W, and opening the square. Write the two squarefree indices as db, dc, where d is their gcd, $(b, c) = 1$, and $(d, bc) = 1$. Their residual classes agree because t is multiplicative. Index multiplicativity and (8.3) give on every good a

$$F_{db}(a)^j \overline{F_{dc}(a)^j} = 1_{(a,d)=1} \Psi_{b,c}^{(j)}(a).$$

Let $\mathfrak{s}_S = \prod_{\mathfrak{p} \in S} \mathfrak{p}$. Möbius inversion for $(a, \mathfrak{s}_S d) = 1$ writes $a = \mathfrak{r}\mathfrak{a}$ with $\mathfrak{r} \mid \mathfrak{s}_S d$. Choose one generator r_0 of each fixed $\mathfrak{r}$. The value of Ψ at $\mathfrak{r}$ separates as $\chi_b(r_0)^j \overline{\chi_c(r_0)^j}$, so the common residual sequence is

$$\lambda'_b = \lambda_{db} 1_{(b,d)=1} \chi_b(r_0)^j, \qquad |\lambda'_b| \le |\lambda_{db}|.$$

It is supported on $\mathcal{A}_{N/q_d}$ in one class. Equation (8.7) applies with lengths $M/q_\mathfrak{r}$ and N/q_d. The summed coefficient mass is at most

$$\sum_d \sum_{\mathfrak{r} | \mathfrak{s}_S d} \sum_b |\lambda_{db}|^2 \le d_\mathcal{O}(\mathfrak{s}_S) \sum_m d_\mathcal{O}(m)^2 |\lambda_m|^2 \ll_{S,\delta} (2 + N)^\delta \|\lambda\|_2^2.$$

Writing $\Delta_{\mathfrak{r}} = q_{\mathfrak{r}}$ and $\Delta_d = q_d$, we have $1 \le \Delta_d \le 2N$ and $1 \le \Delta_{\mathfrak{r}} \le q_{\mathfrak{s}_S}\Delta_d$. We obtain the recursive interface

$$E_j(M,N) \ll_{S,\varepsilon} (2+MN)^{\varepsilon} \max_{\substack{1\le\Delta_d\le 2N\\ 1\le\Delta_{\mathfrak{r}}\le q_{\mathfrak{s}_S}\Delta_d}} \Bigg\{ \frac{M}{\Delta_{\mathfrak{r}}} + \frac{M\Delta_d}{N\Delta_{\mathfrak{r}}} \max_{\substack{1\le H\\ H\ll_{S,\varepsilon}(2+MN)^{\varepsilon}N^2\Delta_{\mathfrak{r}}/(M\Delta_d^2)}} E_j(H, N/\Delta_d) \Bigg\}. \tag{8.8}$$

The maxima may be restricted to nonempty ranges. This is the combination of the gcd and Poisson steps in [4, Lemmas 3.2–3.3]; every sequence used here is common within its norm sum, and S remains fixed.

Initial norm bounds. Row inclusion gives $D_j \le E_j$. Hilbert space duality and sextic reciprocity, splitting one matrix variable by its finite t-class, give

$$D_j(M,N) \ll_S D_j(N,M). \tag{8.9}$$

The phases are unit-valued and the noncoprime entries are zero in both orientations. We also have the initial estimate

$$D_j(M,N) \ll_S M^2 + N \qquad (M,N \ge 1). \tag{8.10}$$

Indeed, for the squarefree row a, finite Fourier inversion gives on every b, including nonunits,

$$\chi_a(b)^j = \frac{1}{\sqrt{q_a}\gamma_{-j}(a)} \sum_{x \bmod a} \overline{\chi_a(x)^j} e(xb/a).$$

The factor from Cauchy–Schwarz is at most $\#(\mathcal{O}/(a))^{\times}/q_a \le 1$. The reduced fractions x/a from $a \in \mathcal{A}_M$ are distinct modulo $\mathcal{O}$ and separated by at least $1/(2M)$: a nonzero numerator of $x/a - y/c - h$ has absolute value at least one, while $|ac| \le 2M$; equality modulo $\mathcal{O}$ forces the same primary denominator and residue. The dual of Lemma 5.1, with the coefficient generators in the ball $q_b \le 2N$, proves (8.10).

Enlarging the row scale. The recursion will compare row lengths with a threshold determined by the column length. The following near-monotonicity permits enlargement to that threshold; it is also used in the large-sieve recursions of [12, Lemma 4.4 and Remark 5] and [4, Lemma 3.1]:

$$D_j(M_1,N) \ll_S D_j(M_2,N) \quad \text{if } M_2 \ge CM_1\log(2M_1N), \quad M_1,M_2,N \ge 1. \tag{8.11}$$

Here is a direct verification. The assertion is immediate for a zero norm. Otherwise choose a maximizing unit coefficient vector for $D_j(M_1,N)$, and split the row shell at $3M_1/2$; one part carries at least half its mass. Put $L = M_2/M_1$. For the first part use good prime ideals of norms in $(L, 4L/3]$, and for the second use norms in $(2L/3, L]$. Then $ap \in \mathcal{A}_{M_2}$ when $p \nmid a$. The fixed-field prime ideal theorem supplies $P \gg_S L/\log L$ such primes. An ideal of norm at most $2x$ contains at most $d_x = \log(2x)/\log(2L/3)$ of them. For $S_a(v) = \sum_b v_b F_b(a)^j$ and $p \nmid a$, multiplicativity gives

$$S_a(\lambda) = S_{ap}\big((\lambda_b\overline{F_b(p)^j})_b\big) + S_a\big((\lambda_b 1_{p|b})_b\big).$$

The first coefficient is zero at $p \mid b$ and cancels the factor at p otherwise. Squaring and summing over the chosen rows and primes gives

$$\frac{P - d_{M_1}}{2} D_j(M_1,N) \le 2PD_j(M_2,N) + 2d_N D_j(M_1,N).$$

For the stated threshold with C sufficiently large, $d_{M_1} + 4d_N \le P/2$, proving (8.11).

Power decomposition. To estimate the unrestricted norm E_j in terms of squarefree norms, use the power decomposition and recursion of [4, Lemmas 3.4–3.5]. Suppose, simultaneously for $j \in J$, that for every positive loss

$$D_j(M,N) \ll_{S,\varepsilon} (MN)^{\varepsilon}\{M^{\alpha} + N + (MN)^{2/3}\}, \qquad \alpha > 4/3, \tag{8.12}$$

for $M, N \geq 1$; the unit shells satisfy the same bound directly. The initial estimate gives this with $\alpha = 2$. Write a good evaluation ideal uniquely as

$$a = a_1 a_2^2 a_3^3 a_4^4 a_5^5 a_6^6,$$

where $a_1, \dots, a_5$ are pairwise coprime and squarefree, and a_6 is arbitrary. Dyadic subdivision places $a_i \in \mathcal{I}_{X_i}$; put $X = \prod_i X_i$, where $X_i \geq 1/2$ and $\prod_i X_i^i \asymp M$. Choose $\ell \in \{1,2\}$ with $X_\ell = \max(X_1, X_2)$. After the other factors are fixed, the coefficient

$$\lambda_b' = \lambda_b \prod_{i \neq \ell} F_b(a_i)^{ij}$$

has absolute value at most $|\lambda_b|$, and the remaining kernel is exactly that of $D_{j\ell \bmod 6}(X_\ell, N)$. Enlarging its fixed positive row restriction gives this norm bound. This includes the zero of the sixth power factor. The set J is closed under $j \mapsto j\ell \bmod 6$. Counting the fixed ideals and forgetting their coprimalities in positive sums gives

$$E_j(M,N) \ll_{S,\varepsilon} (MN)^{\varepsilon} \max_X \min\{M^{1/3}, (M/X_\ell)^{2/3}\} D_{j\ell \bmod 6}(X_\ell, N). \tag{8.13}$$

To verify the factor, the number of fixed choices is $O(X/X_\ell)$. For $\ell = 1$ it is at most a constant times both $(M/X_1)^{1/2} \ll (M/X_1)^{2/3}$ and $(MX_2/X_1)^{1/3} \leq M^{1/3}$; for $\ell = 2$ it is at most a constant times $(MX_1^2/X_2^2)^{1/3} \leq M^{1/3} \ll (M/X_2)^{2/3}$. The number of boxes is a fixed power of $\log(2+M)$.

Improving the exponent. Substitution of the simultaneous squarefree bounds (8.12) into the power decomposition (8.13) gives the unrestricted norm bound

$$E_j(M,N) \ll_{S,\varepsilon} (MN)^{\varepsilon}(M^{\alpha} + M^{1/3}N).$$

The three terms before simplification are bounded by M^{α}, $M^{1/3}N$, and $(MN)^{2/3}$; the last is bounded by one of the first two according as $N \leq M$ or $N \geq M$. Inserting this bound for the dual norm in (8.8) returns an estimate at the original row and column lengths. All auxiliary nonempty scales are bounded by fixed powers of $2 + MN$: here $N/\Delta_d \geq 1/2$ and $H(N/\Delta_d) \ll_{S,\varepsilon} (2+MN)^{3+\varepsilon}$. The unit column shell satisfies the same estimate directly. Thus preliminary losses can be chosen smaller to give any specified final loss. The two scale factors are

$$\Delta_{\mathfrak{r}}^{\alpha-1}\Delta_d^{1-2\alpha} \ll_S \Delta_d^{-\alpha} \leq 1, \qquad (\Delta_{\mathfrak{r}}\Delta_d)^{-2/3} \leq 1.$$

Consequently, for $M, N \geq 1$,

$$D_j(M,N) \leq E_j(M,N) \ll_{S,\varepsilon} (MN)^{\varepsilon}\{M + N^{2\alpha-1}M^{1-\alpha} + (MN)^{2/3}\}. \tag{8.14}$$

Put $h = 2 - 3/(3\alpha-1) = (2\alpha - 5/3)/(\alpha - 1/3)$ and $\beta = 2 - 2/(3\alpha-1)$. If $M \geq N^h$, the middle term in (8.14) is at most $(MN)^{2/3}$. If $M < N^h$, apply (8.11) with a fixed multiple of $N^h \log(2MN)$ in place of M. Equation (8.14) there gives $(MN)^{\varepsilon}N^{\beta}$, since $2(h+1)/3 = \beta$ and $h < \beta$. Thus in both cases

$$D_j(M,N) \ll_{S,\varepsilon} (MN)^{\varepsilon}\{M + N^{\beta} + (MN)^{2/3}\}.$$

Symmetry gives (8.12) with β in place of α, simultaneously for every $j \in J$. Starting from 2, the map $\alpha \mapsto 2 - 2/(3\alpha-1)$ decreases to $4/3$. For a requested loss take finitely many iterations

until the remaining exponent gap is smaller than a fixed fraction of that loss, and absorb it into $(MN)^{\varepsilon}$. Combining the bound and its symmetric form, using $M^{4/3} \le (MN)^{2/3}$ when $M \le N$ and the analogous inequality when $N \le M$, proves

$$D_j(M,N) \ll_{S,\varepsilon} (MN)^{\varepsilon}\{M+N+(MN)^{2/3}\} \qquad (j \in J). \tag{8.15}$$

Returning to norm balls. Dyadic subdivision of both norm balls, with Cauchy across the column shells, converts (8.15) to the same bound for the ball matrix with entries $F_b(a) = \chi_a(b)$. The logarithmic factors are absorbed by a smaller preliminary loss, and the unit shells are bounded directly. The matrix in (8.1) for $\varsigma = 1$ is the transpose of this matrix at lengths (D,K). A complex matrix and its transpose have the same operator norm, by the adjoint identity followed by whole-vector conjugation. The bound is symmetric in the two lengths, so it gives precisely $K+D+(KD)^{2/3}$. Conjugating the entire inner sum gives $\varsigma = -1$, with unchanged zeros. Finally, a fixed row restriction removes nonnegative terms, and a fixed coefficient restriction is imposed by zero extension. This proves all assertions of the lemma. □

8.2 Counting sixth-power-free rows

With the sieve proved, it remains to apply it to the detector witnesses. We apply the lemma only to a squarefree factor of the element row. The separation of squarefree and powerful row factors is also used in the proof of Corollary 1.4 of [4]; here we keep the actual norm of the powerful factor because it controls both the moment and the number of rows.

Proposition 8.2 (Row count from the sextic sieve). *Fix the arithmetic data $\mathcal{A}$ and the bounded row-norm range $d_{\min} \le d \le d_{\max}$ from the retained-row setting of Section 7. Assume the height and loss hypotheses of Lemmas 7.1 and 7.2. Put $U = Z^d$ and retain the notation T_1 of those lemmas. Let $\mathcal{B}$ be any set of retained sixth-power-free element rows u with $q_u \asymp U$, all in one bin (ι, a) of Lemma 7.1 with $a > 51/100$. Put $\delta = 2a-1$, so $1/50 < \delta \le 1$.*

For every $\epsilon > 0$, there exist positive preliminary detector and sieve losses and a finite order $A_{\mathcal{A}}$ such that, for every fixed positive τ satisfying the height hypotheses above, with the preliminary losses in Proposition 7.3 and Lemma 8.1 equal to these values, the following bound holds for all sufficiently large Z:

$$\#\mathcal{B} \ll_{\mathcal{A},\epsilon} U^{R(\delta)+\epsilon}(1+T_1)^{A_{\mathcal{A}}}, \qquad R(\delta) = \min\left\{1, \max\left(1-\frac{\delta}{2}, \frac{4}{3}-\delta\right)\right\}. \tag{8.16}$$

The same $A_{\mathcal{A}}$ applies to every such τ in $T_1 = Z^{\tau}$, every row-norm dyad, every bin, and all moving rows. The estimate is valid for each fixed retained tuple of external parameters and for the original row restrictions. The implied constant may depend on the fixed comparison constants and finitely many of the shared profile seminorms. The rowwise profile choices are precisely those permitted by Proposition 7.3 and Lemma 3.7; no arbitrary row-dependent coefficients are allowed.

The floor bin $a = 51/100$ is excluded from the witness assertion. For that bin the elementary bound $\#\mathcal{B} \ll_{\mathcal{A}} U$ applies; it agrees with the numerical value $R(1/50) = 1$.

Proof. Selecting inverse witnesses. Fix a small preliminary loss $\epsilon_0 > 0$. Proposition 7.3 with $t = 1$ supplies a witness for every row in $\mathcal{B}$. Subdivision by the presentation labels (ν, ς) and by the dyadic pair of polynomial lengths makes these labels and lengths common within each group. There are a fixed finite number of presentation labels and $O_{\mathcal{A}}((\log U)^2)$ pairs. Within one such subdivision, the inverse length $D = U^r$ and the pair (ν, ς) are fixed. Let

$$\psi_u(n) = \nu(n)\chi_n(u)^{\varsigma}.$$

The full zero-extended presentation of ν and the sign are common to the subdivision. The inverse polynomial in (3.3) specializes to

$$M_{\psi_u}(D;W) = D^{-1/2}\sum_n \mu(n)\nu(n)\chi_n(u)^{\varsigma}W(q_n/D).$$

The witness gives

$$\frac{1}{2} - O(\epsilon_0) \le r \le 1 + O(1/\log U), \qquad |M_{\psi_u}(D; W_u)|^2 \gg_{\mathcal{A},\epsilon_0} U^{\delta r - \epsilon_0}. \tag{8.17}$$

The polynomial retains the displayed presentation and its original zero extension. The notation W_u records the permitted rowwise norm-profile parameters. We first bound the moment for fixed profile parameters; a parameter supremum will then cover their rowwise choice.

Factoring and counting rows. For each sixth-power-free row, factor its ideal uniquely as

$$(u) = \mathfrak{s}\mathfrak{w}\mathfrak{k}, \quad \mathfrak{s} = \prod_{p \in S} p^{v_p(u)}, \quad \mathfrak{w} = \prod_{\substack{p \notin S \\ 2 \le v_p(u) \le 5}} p^{v_p(u)}, \quad \mathfrak{k} = \prod_{\substack{p \notin S \\ v_p(u) = 1}} p.$$

Thus $\mathfrak{k}$ is squarefree, $(\mathfrak{k}, \mathfrak{w}) = 1$, and $\mathfrak{w}$ is powerful, meaning that each of its positive prime valuations is at least two. There are only finitely many possibilities for $\mathfrak{s}$, because its valuations lie in $\{0, \dots, 5\}$ on the fixed set S. Choose one generator $s_{\mathfrak{s}}$ for each of these ideals. Writing w, k for the primary generators of $\mathfrak{w}, \mathfrak{k}$, there is a unique unit υ such that

$$u = \upsilon s_{\mathfrak{s}} w k.$$

We fix $\mathfrak{s}$ and υ whenever a row sum is taken.

We group the powerful factors in dyadic ranges $q_{\mathfrak{w}} \asymp V$, including the unit ideal in the bounded range $V = 1$. Only $1 \le V \ll_{\mathcal{A}} U$ can occur, and there are $O_{\mathcal{A}}(\log U)$ such ranges. The original annulus $q_u \asymp U$ gives the actual residual annulus

$$q_{\mathfrak{k}} = \frac{q_u}{q_{\mathfrak{s}} q_{\mathfrak{w}}} \asymp_{\mathcal{A}} \frac{U}{V}. \tag{8.18}$$

We keep this restriction when forming the sum. It will be enlarged to the ball $q_k \ll_{\mathcal{A}} U/V$ only when applying a positive row bound. Let $\mathcal{U}_V$ denote the rows of the current witness subdivision in this powerful-part dyad, with all original restrictions on the element rows retained.

The number of powerful ideals of norm at most V is $O(V^{1/2})$. Indeed, every powerful ideal has a unique expression $\mathfrak{v}^2\mathfrak{y}^3$ with $\mathfrak{y}$ squarefree; the two factors need not be coprime. The ideal count from the arithmetic preliminaries gives

$$\#\{\mathfrak{w} : q_{\mathfrak{w}} \le V,\ \mathfrak{w} \text{ powerful}\} \ll V^{1/2} \sum_{\mathfrak{y}} q_{\mathfrak{y}}^{-3/2} \ll V^{1/2}.$$

The last series converges by the same ideal count. For each fixed $\mathfrak{w}$ in its dyad, the ball containing (8.18) has $O_{\mathcal{A}}(U/V)$ ideals. A nonempty annulus has U/V bounded below by a fixed positive constant, which also covers the unit residual ideal. Since the factorization of (u) and its unit are unique, the number of rows in this powerful-part dyad satisfies

$$\#\mathcal{U}_V \ll_{\mathcal{A}} UV^{-1/2}. \tag{8.19}$$

Discarding any of the restrictions defining $\mathcal{U}_V$ in this upper count can only add rows.

Applying the sieve. For fixed $\mathfrak{w}, \mathfrak{s}, \upsilon$, put $v = \upsilon s_{\mathfrak{s}} w$. Multiplicativity in the numerator gives the identity

$$\chi_n(u)^{\varsigma} = \chi_n(v)^{\varsigma} \chi_n(k)^{\varsigma}. \tag{8.20}$$

It includes the zero extensions: if n meets either factor, both sides are zero, and otherwise it is ordinary multiplicativity. In particular, the inverse polynomial for a fixed profile W is the row sum in Lemma 8.1 with coefficients

$$c_n = D^{-1/2} \mu(n) \nu(n) \chi_n(v)^{\varsigma} W(q_n/D).$$

The factor $\mu(n)$ keeps the columns squarefree, and $\chi_n(v)^\varsigma$ retains every original zero at $(n,v)>1$. For each fixed $\mathfrak{w}$, these coefficients are fixed across the residual row k; they may vary with $\mathfrak{w}$. The only remaining zero depending on both n and k is the zero of the sieve kernel. The restriction $(k,w)=1$ and the inherited restrictions on the element rows for fixed external parameters are fixed row restrictions. The primes of w remain in the coefficient mask, while S stays unchanged. Thus the sieve constant is independent of w, even when that mask has large norm.

The common annular support and ideal counting give, uniformly for the fixed profile parameters,

$$\sum_n |c_n|^2 \le D^{-1} \sum_{q_n \asymp D} |W(q_n/D)|^2 \ll_{\mathcal{A}} 1.$$

The same statement holds for each of the finitely many profile derivatives used below, with the corresponding fixed seminorm. For each fixed $\mathfrak{w}$, Lemma 8.1 applies after enlarging the positive residual row sum from its actual annulus to $q_k \ll_{\mathcal{A}} U/V$. A fixed enlargement makes the row bound at least one on every nonempty quotient range and includes the annular column support. The three sieve terms therefore have residual row scale U/V and column scale D. Summing these positive bounds over the $O(V^{1/2})$ choices of $\mathfrak{w}$ and the finitely many choices of $\mathfrak{s}, \upsilon$ gives

$$\sum_{u\in\mathcal{U}_V} |M_{\psi_u}(D;W)|^2 \ll_{\mathcal{A},\epsilon_1} (UD)^{\epsilon_1} V^{1/2} \left\{ \frac{U}{V} + D + \left(\frac{UD}{V}\right)^{2/3} \right\}. \tag{8.21}$$

Taking the profile supremum. The fixed-profile moment now controls the rowwise choice of W_u in (8.17) through the parameter Sobolev inequality of Lemma 3.7. Here the detector was used with $t=1$, so $D_*=U$. After the presentation and dyadic pair are fixed, its inverse profiles have the form

$$W_{\mathrm{dyad}}(x) V_{\le}(Dx/U) x^{-\sigma-i\omega}, \qquad \omega = \gamma - \nu_F,$$

where W_{dyad} is the fixed smooth dyadic cutoff and ν_F is the Fourier frequency from the detector. The factor $V_{\le}(Dx/U)$ is common to the row sum; the varying parameters are $\sigma \in [51/100, 1]$ and $|\omega| \le (3\iota+1)T_1$. Differentiating in either parameter inserts a power of $\log x$, which is bounded on the fixed annulus. Cover these two parameter ranges by unit boxes. There are at most a fixed power of $1+T_1$ such boxes. The parameter Sobolev inequality in Lemma 3.7 bounds the supremum on a box by a finite sum of integrals of squared profile derivatives. Summing over the rows before these integrals keeps the coefficients common at each parameter value. Differentiation then changes only the fixed profile coefficient, inserting the allowed logarithmic weights or profile derivatives. Equation (8.21) therefore applies to every such integral. This proves

$$\sum_{u\in\mathcal{U}_V} |M_{\psi_u}(D;W_u)|^2 \ll_{\mathcal{A},\epsilon_1} (UD)^{\epsilon_1} (1+T_1)^{A_{\mathcal{A}}} V^{1/2} \left\{ \frac{U}{V} + D + \left(\frac{UD}{V}\right)^{2/3} \right\}. \tag{8.22}$$

The derivative order and hence $A_{\mathcal{A}}$ are fixed before choosing τ. The supremum is restricted to the displayed shared profile family, with coefficients fixed across k at every parameter value. The contours, height intervals and single total allowance for additional frequencies remain those of the detector estimates.

Combining the two row bounds. The witness lower bound converts the moment (8.22) into a count of rows; the independent count (8.19) gives a second bound for the same set. Write

$$V = U^o, \qquad D = U^r, \qquad e_o(r) = \frac{o}{2} + \max\left\{1-o, r, \frac{2(1-o+r)}{3}\right\}.$$

These are the parameters of the actual dyads. A nonempty dyad has $0 \le o \le 1 + O_{\mathcal{A}}(1/\log U)$; the unit dyad has $o=0$. Since the witness lengths stay bounded, the arbitrarily small power of UD in (8.22) can be written as an arbitrarily small power of U. Dividing that equation by

the witness lower bound (8.17) and taking the smaller of this count and (8.19) gives, for this subdivision,

$$\#\mathcal{U}_V \ll U^{\min\{1-o/2,e_o(r)-\delta r\}+O(\epsilon_0)+\epsilon_1}(1+T_1)^{A_{\mathcal{A}}}, \tag{8.23}$$

after decreasing the sieve loss denoted by ϵ_1 if necessary.

Optimizing the exponent. We compute the largest exponent in this expression first on $0 \le o \le 1$ and $1/2 \le r \le 1$. Put $x = 1-o$ and $A = (1+x)/2$. Then

$$e_o(r) - \delta r = \max\left\{A - \delta r, \frac{1-x}{2} + (1-\delta)r, \frac{1}{2} + \frac{x}{6} + \left(\frac{2}{3} - \delta\right) r\right\}.$$

The minimum of A with a maximum of three real numbers is the maximum of its minima with those numbers. The first such minimum is $A - \delta r \le 1 - \delta/2$. For fixed r, the second is the minimum of an increasing and a decreasing affine function of x. They cross at $x = (1-\delta)r \in [0,1]$, so its maximum over x is

$$\frac{1+(1-\delta)r}{2} \le 1 - \frac{\delta}{2}.$$

Both functions in the third minimum increase with x. Its maximum over x is therefore

$$\min\left\{1, \frac{2}{3} + \left(\frac{2}{3} - \delta\right) r\right\}.$$

The affine expression in r takes the values $1 - \delta/2$ and $4/3 - \delta$ at the two endpoints. It follows, and equality is attained at one of those endpoints, that

$$\max_{\substack{0\le o\le 1\\ 1/2\le r\le 1}} \min\{1 - o/2, e_o(r) - \delta r\} = \min\left\{1, \max\left(1 - \frac{\delta}{2}, \frac{4}{3} - \delta\right)\right\}. \tag{8.24}$$

This computation is uniform for $0 \le \delta \le 1$.

The functions just used are maxima and minima of affine functions whose slopes are uniformly bounded on a fixed neighborhood of these compact ranges. Replacing the actual o by its nearest point in $[0,1]$, and the actual witness r by its nearest point in $[1/2,1]$, changes the exponent by at most $O_{\mathcal{A}}(\epsilon_0 + 1/\log U)$, by (8.17) and the actual dyad bounds. This projection is confined to the final optimization; the row sum and its sieve application retain the actual dyads. Choose ϵ_0 and the preliminary sieve loss small enough in terms of the prescribed ϵ. The $O((\log U)^2)$ witness pairs and $O(\log U)$ powerful-part dyads are absorbed into another arbitrarily small power. Summing (8.23) proves (8.16).

Finally, the bin $a = 51/100$ need not contain an actual zero, so no use of Proposition 7.3 is made there. Counting all elements in its row-norm annulus gives $O_{\mathcal{A}}(U)$ rows. Since $1/50 < 1/3$, the displayed formula for R gives $R(1/50) = 1$, as claimed. □

For use in the high estimate, the row-count exponent has the explicit form

$$R(\delta) = \begin{cases} 1, & 0 \le \delta \le 1/3, \\ 4/3 - \delta, & 1/3 \le \delta \le 2/3, \\ 1 - \delta/2, & 2/3 \le \delta \le 1. \end{cases}$$

The inverse witness alone supplies this row count.

9 Analytic estimates for the high expansion

The row-count bound now enters the Poisson representation of the base average. For the first-stage application, retain the supposition $\Delta_1 = \beta_* - 11/12 > 0$ and set $X = Y = Z^{1/2}$. Write s

for the first Mellin variable called x in Section 6.2. Equations (6.5) and (6.14) give the actual integral to be estimated:

$$I_\eta(Z^{1/2},Z^{1/2},Z)=\frac{1}{(2\pi i)^3}\int_{(3)}\int_{(3)}\int_{(2)}\mathcal{W}(Z^{1/2},Z^{1/2},Z;s,w,z)\\ \cdot\sum_u{}^{(6)}\,q_u^{-z}\overline{\xi(u)}\frac{\zeta_F^S(6z)L^S(w,\chi_\bullet(u))}{L^S(s,\eta\overline{\chi_\bullet(u)})}\mathcal{H}_{\eta,u}(s,w,z)\,dz\,dw\,ds.$$

The Mellin weight $\mathcal{W}$ is defined in Section 6.1. The rows are the nonzero sixth-power-free elements, with their unit factors and the coprimality restriction $(u,S)=1$. Both character presentations retain their original zeros at primes dividing u. The coefficientwise calibration of the base average is already included in this identity.

The row $u=1$ contains the reciprocal of the target function. For the intermediate row norms we will keep one buffered bin fixed, move its integral to

$$\Re s=a+16e,\qquad \Re w=1-a-6e,\qquad \Re z=17/50,$$

and apply the row-count exponent $R(2a-1)$ there. Apart from small real losses and a fixed height factor, the numerator on this contour is bounded by $U^{a-1/2}$ for a row norm $q_u\asymp U$; the row-count bound controls how many such terms occur. The principal-row normalization identifies the signal. Lemmas 9.3 and 9.4 then reduce the intermediate rows to an aggregate bound on the retained contour, where the row count applies. Lemma 9.5 extracts the principal residues, and Lemmas 9.6 and 9.7 bound the outer row norms.

9.1 The principal row and the normalization

We identify the row whose numerator is principal. An element row with a prime factor outside S has nonprincipal numerator by the ramification argument in Section 7. Because the row domain imposes $(u,S)=1$, the remaining rows are units. If a unit $u\neq 1$ had a sixth root in F, that root would have valuation zero at every prime, hence would be a unit of F; every unit of F has sixth power one. Thus $F(u^{1/6})/F$ is nontrivial. It is finite Galois because F contains the sixth roots of unity. Chebotarev applied to a nonidentity Frobenius class supplies a prime outside the fixed set S where the associated sextic character is nontrivial [26, Theorem 1.1]. By the Kummer description in Lemma 3.2, this is $\chi_\bullet(u)$. Therefore $u=1$ is the sole principal numerator row. A denominator attached to a bounded unit row may be principal; its reciprocal has a zero at one and will be kept on a global line below.

Define

$$H_\eta(s)=\mathcal{H}_{\eta,1}(s,1,1/6),\qquad c_S=\widehat{W}_1(1)M(1/6)\left(\operatorname*{Res}_{v=1}\zeta_F^S(v)\right)^2/6. \tag{9.1}$$

Lemma 6.2 makes H_η holomorphic on $\Re s>7/8$. Its uniform estimate $H_\eta=1+O(P_0^{-4/5})$ permits a choice of P_0 before the target, because the local bound is uniform in the target unit phases. Fix P_0 large enough that

$$\sup_{\Re s>7/8}|H_\eta(s)-1|\le\frac{1}{2}. \tag{9.2}$$

Enlarging S by these primes leaves the fixed ray group T unchanged, as the calibration in the local identity holds coefficientwise. The excluded set and hence the particular function H_η may differ between applications, but within one application they are exactly those of the exact high representation being used.

The constant c_S is positive. Indeed $\widehat{W}_1(1)>0$ because W_1 is nonnegative and nonzero, and $M(1/6)>0$ by (6.2). The simple pole of ζ_F^S has positive residue: deletion multiplies the positive residue of ζ_F by $\prod_{p\in S}(1-q_p^{-1})>0$.

For the first stage put

$$C_{\mathrm{I}}(s) = s - \frac{2}{3}, \qquad J_{\mathrm{I},\eta}(Z) = \frac{I_\eta(Z^{1/2}, Z^{1/2}, Z)}{c_S}.$$

Let $f_{\mathrm{I},\eta}$ be the integral in (2.3) with $C = C_{\mathrm{I}}$, the same excluded set S, and the function H_η just defined. The two scalar poles at $w = 1$ and $z = 1/6$ will give $c_S f_{\mathrm{I},\eta}$; our remaining objective is a common power saving for every other term, measured relative to $Z^{C_{\mathrm{I}}(\beta_*)}$.

9.2 The exact identity used in contour moves

The contour proof applies to a full holomorphic correction in the displayed integral. The compensated average uses the same scalar quotient with a different correction, so both applications require the following identity and bounds. In the present application the correction is $\mathcal{H}_{\eta,u}$, independent of Z, and the parameters below are $l_x = l_y = h = 1/2$ and $\ell = 0$.

The scalar quotient in (6.14) is the part of the high expansion that determines both the contour move and the principal residues. We keep that quotient fixed and state explicitly what may be changed around it. Put

$$\begin{aligned}\mathcal{D}_1(\epsilon_0) = \{(s,w,z) : \Re s \ge 51/100, \quad \Re z \ge 17/50, \quad \Re w \ge -1/100,\\ \Re(s+w) \ge 1+\epsilon_0\}, \qquad \epsilon_0 > 0,\\ \mathcal{D}_2 = \{(s,w,z) : \Re s \ge 7/8, \quad \Re z \ge 33/200, \quad \Re w \ge 19/20\}.\end{aligned}$$

These are the two regions in Lemma 6.2. A real box below means a compact rectangular subset of $\mathbb{R}^3$ for $(\Re s, \Re w, \Re z)$. Recall the common Mellin weight from Section 6.1:

$$\mathcal{W}(X,Y,Z;s,w,z) = X^{1/2-z} Z^{s+z-1} Y^{w-1} e^{(s+z-1)^2} M(z)\widehat{W_1}(w).$$

Fix real numbers $l_x, l_y > 0$ and $\ell \ge 0$ such that

$$X = Z^{l_x}, \quad Y = Z^{l_y}, \quad h = 1 - l_x + \ell > 0, \quad C(s) = s + \frac{l_x}{2} - 1 + \frac{h}{6} = s - \frac{5}{6} + \frac{l_x}{3} + \frac{\ell}{6}. \tag{9.3}$$

These parameters describe the scales and the power of the principal signal.

Definition 9.1 (Data for an exact high representation). Let $l_x, l_y > 0$ and $\ell \ge 0$ satisfy (9.3), with h, C as defined there, and use the regions and Mellin weight of Section 9.2. For every primitive finite-order target η, let S, ξ, W_0, W_1 be the data of Section 5, independent of Z. An exact high representation for these data satisfies the following three requirements.

1. $\mathscr{I}_\eta(Z)$ is an independently specified complex-valued quantity for every sufficiently large real Z. In each application it is given by a finite linear combination, for that Z, of the completed sums in (5.1), allowing specified restrictions on the completed index A and specified rescalings of its three positive scales. The number of terms and coefficients in this finite combination may depend on Z.

2. For every nonzero sixth-power-free element u with $(u,S) = 1$ and every such Z, there is a function $\mathfrak{H}_{\eta,u,Z}(s,w,z)$. It is holomorphic on a neighborhood of each point of $\mathcal{D}_2$ and of every $\mathcal{D}_1(\epsilon_0)$.

   For every $D > 0$ and every real box $\mathcal{K}$ contained in one of these regions, there are finite constants $B_{\eta,\mathcal{K},D}$ and $J_{\eta,\mathcal{K},D}$ such that, uniformly for $1 \le q_u \le Z^D$, all imaginary parts, and real parts in $\mathcal{K}$,

   $$|\mathfrak{H}_{\eta,u,Z}(s,w,z)| \ll_{\eta,\mathcal{K},D} Z^{B_{\eta,\mathcal{K},D}} (1 + |\Im s| + |\Im w| + |\Im z|)^{J_{\eta,\mathcal{K},D}}. \tag{9.4}$$

   The constants and exponents are independent of Z, u and of any later order of integration by parts.

3. The following identity holds for the independently specified $\mathscr{I}_\eta(Z)$, with absolute convergence of the sum and integrals on the displayed lines:

$$\begin{aligned}\mathscr{I}_\eta(Z) = \frac{1}{(2\pi i)^3}\int_{(3)}\int_{(3)}\int_{(2)} \mathcal{W}(X,Y,Z;s,w,z)\sum_u{}^{(6)}\, q_u^{-z}\overline{\xi(u)} \\ \cdot \frac{\zeta_F^S(6z)L^S(w,\chi_\bullet(u))}{L^S(s,\eta\overline{\chi_\bullet(u)})}\mathfrak{H}_{\eta,u,Z}(s,w,z)\,dz\,dw\,ds,\end{aligned} \tag{9.5}$$

where the sum has the row restrictions of (6.5) and $\mathcal{W}$ is the weight specified in Section 9.2.

Remark. The correction $\mathfrak{H}$ and its individual local factors may vanish. Definition 9.1 requires the stated holomorphy and growth bounds for the full correction.

Equation (9.5) is the identity to be proved for the specified sum. Its direct estimate is taken from that expression. For the base average, the defining sum is $I_\eta(Z^{l_x}, Z^{l_y}, Z)$, $\ell = 0$, and $\mathfrak{H}_{\eta,u,Z} = \mathcal{H}_{\eta,u}$. Equations (6.5) and (6.14) prove the required identity, and Lemma 6.2 gives the holomorphy and (9.4). The function $\mathfrak{H}$ is the full correction after removal of the displayed scalar quotient. If it is a sum of products of local factors, those products must be holomorphic in the stated regions, including at zeros of individual factors. The decompositions used to estimate a retained contour are specified separately below.

9.3 External tails on vertical lines and horizontal joins

The product of our three Mellin tests decays in three independent height directions. The following estimate turns that decay into bounds both off a large box and on the horizontal sides of a contour rectangle.

Lemma 9.2 (External integrated and trace tails). *Let $m \ge 2$, let $\mathcal{K}$ be a compact set of real contour parameters, and, for $\boldsymbol{r} \in \mathcal{K}$ and $Z \ge 2$, let $K_{\boldsymbol{r},Z}$ be a nonnegative Borel measurable kernel on $\mathbb{R}^m$. Suppose it is jointly Borel measurable in $(\boldsymbol{r},\boldsymbol{t})$ and, for every integer $L \ge 0$,*

$$K_{\boldsymbol{r},Z}(\boldsymbol{t}) \ll_L (1+|\boldsymbol{t}|)^{-L}$$

uniformly in $\boldsymbol{r}, Z, \boldsymbol{t}$. Let $\mathcal{D} \subseteq \mathbb{R}^m$ be Borel measurable, and suppose a Borel measurable factor F_Z satisfies on $\mathcal{D}$

$$|F_Z(\boldsymbol{t})| \le CZ^B(1+|\boldsymbol{t}|)^J, \tag{9.6}$$

where B, J, C are fixed independently of Z and of the integer N below. Write $\widehat{\boldsymbol{t}}_j$ for the vector obtained by omitting the jth coordinate of $\boldsymbol{t}$. Uniformly for $T \ge 1$ and $\boldsymbol{r} \in \mathcal{K}$,

$$\int_{\mathcal{D}\cap\{\max_j |t_j|>T\}} K_{\boldsymbol{r},Z}(\boldsymbol{t})|F_Z(\boldsymbol{t})|\,d\boldsymbol{t} \ll_N Z^BT^{-N}, \tag{9.7}$$

$$\int_{\mathcal{D}\cap\{t_j=T\}} K_{\boldsymbol{r},Z}(\boldsymbol{t})|F_Z(\boldsymbol{t})|\,d\widehat{\boldsymbol{t}}_j \ll_N Z^B(1+|T|)^{-N}. \tag{9.8}$$

The second assertion holds for either sign of T and every j, using coordinate Lebesgue measure on $t_j = T$. Both assertions remain valid after integration over a real contour interval of bounded length when the hypotheses are uniform there and the kernels, domain indicators and arithmetic factors are jointly Borel measurable in that real parameter and the remaining coordinates.

Proof. The kernel bound and (9.6) bound the integrand by a constant times $Z^B(1+|\boldsymbol{t}|)^{J-L}$. Integrating $(1+|\boldsymbol{t}|)^{J-L}$ outside the box proves (9.7) when $L > N+J+m$. On $t_j = T$, integration in the other $m-1$ coordinates gives

$$\int_{\mathbb{R}^{m-1}} (1+|T|+|\widehat{\boldsymbol{t}}_j|)^{J-L}\,d\widehat{\boldsymbol{t}}_j \ll (1+|T|)^{J-L+m-1}.$$

Taking $L > N+J+m-1$ proves (9.8). Restricting to $\mathcal{D}$ decreases these positive integrals, and a bounded real interval contributes only its length. □

For our triple integral, take

$$(y_1, y_2, y_3) = (\Im(s+z), \Im z, \Im w), \quad K_{\boldsymbol{r},Z}(\boldsymbol{t}) = e^{-y_1^2} |M(r_z + iy_2)|\, |\widehat{W_1}(r_w + iy_3)|.$$

The change of height variables has determinant one. On a fixed real box with r_z in a compact subinterval of $(0,\infty)$, the Mellin estimates of Section 6 give arbitrary polynomial decay of M and $\widehat{W_1}$. The Gaussian has the same property. Their product is therefore $O_L((1+|\boldsymbol{y}|)^{-L})$ for every L, hence also $O_L((1+|\boldsymbol{t}|)^{-L})$. The omitted factor $e^{(\Re(s+z)-1)^2}$ is bounded on that real box.

After the w residue only s, z remain. The same proof applies to the kernel

$$K^{(2)}_{\boldsymbol{r},Z}(\boldsymbol{t}) = e^{-y_1^2} |M(r_z + iy_2)|, \qquad (y_1, y_2) = (\Im(s+z), \Im z). \tag{9.9}$$

This supplies the trace bound needed for the subsequent z-join.

The same kernel test handles additional external separating variables. For $m \geq 3$, let $\boldsymbol{y} = A\boldsymbol{t}$ with a fixed $A \in \mathrm{GL}_m(\mathbb{R})$, and multiply the three-factor kernel by a Borel density $|v_{\boldsymbol{r},Z}(y_4, \dots, y_m)|$, jointly Borel in $(\boldsymbol{r}, \boldsymbol{v})$, satisfying

$$\sup_{\boldsymbol{r},Z,\boldsymbol{v}} (1+|\boldsymbol{v}|)^L |v_{\boldsymbol{r},Z}(\boldsymbol{v})| < \infty \quad \text{for every integer } L \geq 0. \tag{9.10}$$

For $m = 3$ the density is one on the zero-dimensional space. The product is again rapidly decreasing in all m coordinates, since A and its inverse are fixed. For the two-factor kernel, the identical argument uses a density on $\mathbb{R}^{m-2}$ and $m \geq 2$, with density one when $m = 2$. Thus both product-kernel forms have the integrated and trace bounds above.

Uniform annular profiles separated by logarithmic Fourier inversion satisfy (9.10) by repeated integration by parts, as in Lemma 3.6. After every pure norm twist has first been translated into its Mellin argument, the additional height coordinates append a block triangular matrix with fixed inverse to the three-dimensional change of variables.

An internal coefficient measure controlled by a fixed weighted L^1 norm remains integrated at its fixed moment order inside F_Z. The trace estimate instead requires the pointwise density hypothesis (9.10) for each additional external coordinate on which it is used.

The domain $\mathcal{D}$ records the range on which the arithmetic bound holds. Suppose a reciprocal is bounded only while the imaginary part of its argument $\gamma + \sum_j a_j t_j$ lies in a buffered interval. A coordinate with $a_j \neq 0$ may be integrated only over the range that preserves this condition; the lemma does not extend the reciprocal bound beyond it. Coordinates with $a_j = 0$ may be extended when the other factors satisfy (9.6) there. All factors on an extended axis must use their global or absolute estimates, not an estimate valid only on a retained interval. A horizontal join uses the trace estimate (9.8), in addition to the integrated estimate for vertical tails.

Increasing N raises only the decay orders of the external smooth tests, with F_Z unchanged. Thus a fixed moment-profile order or a fixed arithmetic height exponent in F_Z is unchanged when N is chosen later. This is the same order distinction made in Lemma 3.8.

9.4 Moving a fixed bin

The full row integral is moved while the bin remains fixed. The row-count bound applies only after this transformation, on the retained contour.

Lemma 9.3 (Contour transformation for a fixed bin). *Let $\sigma_0 \in [7/8, 1)$ with $\beta_* > \sigma_0$, $0 < d_{\min} \leq d_{\max} < \infty$, and $0 < e < 10^{-3}$, and put $z_0 = 17/50$. For every primitive finite-order target η with the data of Definition 9.1, let $0 < \tau \leq d_{\min}/100$ be fixed and put $T_1 = Z^\tau > 2$. For $U = Z^d$ with $d_{\min} \leq d \leq d_{\max}$, let $\mathcal{B}$ be a finite set of retained element rows $q_u \asymp U$ in one bin (ι, a) of Lemma 7.1. The comparison constants in this annulus are fixed, and $\mathcal{B}$ is fixed before any Mellin variable is moved.*

For this target, fix positive constants $c_s, c_w, c_z \leq 1/2$, independently of Z and the rows. Retain the original heights $|\Im s| \leq c_s T_1$, $|\Im w| \leq c_w T_1$ and $|\Im z| \leq c_z T_1$. The integrand uses the full

holomorphic correction in (9.5). There exists a finite exponent B_η such that, for every fixed N and sufficiently large Z, the contribution of $\mathcal{B}$ on the starting lines in that equation equals its integral on these retained segments of

$$\Re s = a + 16e, \qquad \Re w = 1 - a - 6e, \qquad \Re z = z_0, \tag{9.11}$$

up to $O_{\eta,N}(Z^{B_\eta} T_1^{-N})$. An empty row set contributes zero.

Proof. Global contour moves. Section 7 gives $a \le \beta_*$, including the floor bin. For the finite sum over $\mathcal{B}$ on the absolute lines in (9.5), first move z from 2 to z_0, keep $w = 3$, and then move s from 3 to $\beta_* + 20e$. These moves stay in $\mathcal{D}_1(\epsilon_0)$ for a fixed positive ϵ_0, the two numerator arguments remain to the right of their poles, and the reciprocal stays in $\Re s > \beta_*$. Its global bound follows from Lemmas 3.15 and 3.16, using the conductor and deletion radical bounds in Section 7. The correction obeys (9.4). Lemma 9.2 with $T \to \infty$ therefore justifies these moves.

Now move w from 3 to $1 - a - 6e$ while s stays on that global line. Throughout the move,

$$\Re(s + w) \ge 1 + (\beta_* - a) + 14e \ge 1 + 14e.$$

The numerator character of every retained row is nonprincipal, so its L-function is entire. Also $\Re w \ge -6e > -1/100$ and $\Re z = z_0$. Thus the correction remains holomorphic in $\mathcal{D}_1(\epsilon_0)$, for example with $\epsilon_0 = 10e$, and no pole is crossed. On a w-join the reciprocal is still global. The global upper strip bound for the numerator, the deleted-factor bound and (9.4) give a majorant of the form (9.6) on the axes integrated there. The trace estimate makes the joins tend to zero.
Truncation on the global line. Before moving s farther, restrict the three original Mellin heights to the stated box. On the discarded part keep s on $\Re s = \beta_* + 20e$. There the global reciprocal bound, the global numerator bound, the deleted factors and the all-height correction bound give Z^{B_η} times a fixed polynomial in the heights: the row range is bounded by $q_u \ll Z^{d_{\max}}$, and the number of rows is $O(U)$. The integrated tail in Lemma 9.2 gives $O_{\eta,N}(Z^{B_\eta} T_1^{-N})$.
The buffered shift. Move only the retained s segment to $a + 16e$. On this rectangle,

$$\Re(s + w) \ge 1 + 10e,$$

and the other inequalities defining $\mathcal{D}_1(10e)$ still hold. The real part of the reciprocal argument is at least $a + 16e > a + 6e$. The stated height box places both scalar arguments strictly inside $|\Im v| \le (3\iota + 2)T_1$ for sufficiently large Z. Lemma 7.1 therefore excludes its zeros throughout the retained rectangle. Since $6z_0 > 1$, the scalar zeta factor has no pole on this move. Thus no pole is crossed.

For an s-join, its imaginary coordinate is fixed at a constant multiple of T_1. The scalar denominator in (9.5) depends only on s, so its buffered estimate remains valid when the $\Im z$ and $\Im w$ integrations are extended to their whole axes. On those extended axes use the global numerator estimate and (9.4). The trace estimate then gives $O_{\eta,N}(Z^{B_\eta} T_1^{-N})$. The w- and z-joins are handled while the s-axis is still global; the buffered s-axis is used only for the last move. □

The contour transformation acts on the full correction and the bin fixed before the move. Decomposition of the correction and subdivisions depending on a contour point enter only in the subsequent estimate of the retained integrand. Auxiliary separating variables in that estimate obey the height and tail conditions below.

9.5 Central estimates for a correction split into pieces

The quantity to be bounded is the fixed-bin triple integral in (9.5), on the retained contours (9.11). The bin controls its scalar reciprocal. An aggregate absolute bound for the numerator times the correction then controls the remaining row sum; multiplying by the outside Mellin powers and integrating the tests gives the exponent below.

Let $\sigma_0 \in [7/8, 1)$ with $\beta_* > \sigma_0$, $0 < d_{\min} < d_{\max} < \infty$, $0 < e < 10^{-3}$, and $\varepsilon_c \geq 0$. Fix a bounded range for (R, g) and put $z_0 = 17/50$. These are the common parameters in the following setup, which applies to every primitive finite-order target η with the data of Definition 9.1.

The fixed integral. For the target η, let $0 < \tau \leq d_{\min}/100$ be fixed, put $T_1 = Z^\tau > 2$, and take $U = Z^d$ with $d_{\min} \leq d \leq d_{\max}$. Let $\mathcal{B}$ be a finite set of element rows u with $q_u \asymp U$ in one retained bin (ι, a) of Lemma 7.1, and put $\delta = 2a - 1$. The annulus comparison constants are fixed. The bin $\mathcal{B}$ is fixed before any Mellin variable is moved. The integral contains the full correction $\mathfrak{H}_{\eta,u,Z}$ of (9.5).

The aggregate row bound. On the retained central contours of (9.11), suppose that there is a decomposition

$$\mathfrak{H}_{\eta,u,Z}(s, w, z) = \sum_{\lambda \in \Lambda} \mathfrak{H}^{(\lambda)}_{\eta,u,Z}(s, w, z),$$

where Λ is finite, nonempty and fixed before Z. This equality is required only on the retained contours; the summands need not be holomorphic away from them.

For each λ and each retained (s, w, z), suppose $\mathcal{B}$ is partitioned into at most $C(1 + \log Z)^D$ sets $\mathcal{B}_{\lambda,\nu}(s, w, z)$, with fixed C, D. For each set at each retained point let (R, g) lie in the common bounded range. Suppose that, for some finite $A_\eta \geq 0$,

$$\sum_{u \in \mathcal{B}_{\lambda,\nu}(s,w,z)} \left| L^S(w, \chi_\bullet(u)) \mathfrak{H}^{(\lambda)}_{\eta,u,Z}(s, w, z) \right| \ll_\eta U^{R+\delta/2+\varepsilon_c} Z^{\ell(z_0-1/2)+g+\varepsilon_c} (1 + T_1)^{A_\eta}. \tag{9.12}$$

The bound is uniform in the retained point, row set and moving labels. The dependence of each pair (R, g) is restricted to d, a, the pointwise set and the global number β_*; it has no additional dependence on η.

If extra Mellin integrals are used to establish (9.12), that inequality is required for its full left side after those integrations. Any discarded parts must have been left on their global or absolute lines, bounded by Lemma 9.2 or Lemmas 3.8–3.10, and included in the displayed bound before the hypothesis is asserted. The single allowance $T_1/2$ applies across all successive estimates.

Heights and the exponent supremum. For the fixed bin and height just specified, let $v_1, \dots, v_m$ comprise all external coordinates that enter a buffered L-value, reciprocal, or logarithmic derivative in the central row bound, together with all three original Mellin heights. A coordinate has coefficient zero in each argument where it does not occur. After pure-twist translations, every such argument has imaginary part $\gamma_j + \sum_{k=1}^m a_{jk} v_k$, where the finite matrix (a_{jk}) is fixed and $|\gamma_j| \leq (3\iota + 1)T_1$. The retained heights satisfy

$$|v_k| \leq c_k T_1, \qquad c_k = \frac{1}{2m(1 + \max_j |a_{jk}|)}. \tag{9.13}$$

Coordinates with zero coefficients may be restricted by the same rule. Then every added height is at most $T_1/2$ in total. Fixed bounded enlargements are included by increasing the lower threshold for Z.

The exponent associated with a pair (R, g) is

$$E_{\sigma_0}(d; R, g) = a - \sigma_0 + h(z_0 - 1/6) - a l_y - \ell/2 + g + d(R + \delta/2 - z_0). \tag{9.14}$$

Let $\mathcal{P}_Z$ be the set of all pairs (R, g) that occur at any retained point in the decomposition. When $\mathcal{B}$ is nonempty, define

$$E^{\max}_{\sigma_0,Z}(d) = \sup_{(R,g) \in \mathcal{P}_Z} E_{\sigma_0}(d; R, g). \tag{9.15}$$

Lemma 9.4 (Central integral bound). *For the common parameters $\sigma_0, d_{\min}, d_{\max}, e, \varepsilon_c$ and bounded pair range specified above, consider every primitive finite-order target η with an exact high representation as in Definition 9.1. Let $\mathcal{B}$ and the decomposition satisfy the fixed-integral and aggregate-bound conditions of Section 9.5, including (9.12) after any auxiliary integrations,*

and impose the single height allocation (9.13). *Use the exponent and supremum defined in* (9.14) *and* (9.15).

If $\mathcal{B}$ is empty, its contribution is zero. Otherwise, for every $\varepsilon_d, \varepsilon_p > 0$, the total contribution of $\mathcal{B}$ on the central contours is, for sufficiently large Z,

$$\begin{aligned} &\ll_\eta Z^{C(\sigma_0)+E^{\max}_{\sigma_0,Z}(d)+(16-6l_y)e} \\ &\quad \cdot Z^{(1+d)\varepsilon_c+d\varepsilon_d+\varepsilon_p}(1+T_1)^{A_\eta}. \end{aligned} \tag{9.16}$$

This estimates the original fixed-bin integral. There exists a finite B_η such that, for every fixed N, all discarded portions and horizontal joins in its contour move are $O_{\eta,N}(Z^{B_\eta}T_1^{-N})$.

Proof. Lemma 9.3, applied to the original triple integral with its three box constants equal to the corresponding c_k in (9.13), gives the contour transformation. These constants are positive and at most $1/2$, as required for the retained integral and the asserted errors. The aggregate-bound hypothesis already includes any auxiliary integrations and their discarded portions.

The central decomposition gives the pointwise sets $\mathcal{B}_{\lambda,\nu}$. At each retained point, the triangle inequality bounds the original row sum by the sum over these sets, and (9.12) bounds each of their contributions. There are at most a fixed multiple of $(1+\log Z)^D$ such sets pointwise. The bounded range of pairs makes the supremum over $\mathcal{P}_Z$ finite. Each exponent is bounded by this supremum, and only the original holomorphic row sum is integrated. Thus neither measurability of an individual pointwise subdivision nor one common label set across contour points is needed. On the retained contours, Lemma 7.1 bounds the reciprocal by $O_{\eta,e,\varepsilon_d}(U^{\varepsilon_d})$. The tests have a uniformly bounded joint L^1 norm, since the transformation $(\Im s, \Im z, \Im w) \mapsto (\Im(s+z), \Im z, \Im w)$ is invertible. The scalar factor $\zeta_F^S(6z_0)$ is absolutely bounded. It remains to compute the real power of Z.

The outside factor, including $q_u^{-z_0}$ and the real displacements in (9.11), contributes

$$l_x(1/2-z_0)+a+z_0-1-al_y-dz_0+(16-6l_y)e.$$

Equation (9.12) adds $d(R+\delta/2+\varepsilon_c)+\ell(z_0-1/2)+g+\varepsilon_c$. The reciprocal adds $d\varepsilon_d$. Subtracting $C(\sigma_0)=\sigma_0+l_x/2-1+h/6$ and using $h=1-l_x+\ell$ gives (9.14) and the remaining terms in (9.16). The pointwise logarithmic multiplicity is bounded by Z^{ε_p} for sufficiently large Z. This proves the claim. □

Local divisions used to define the pieces in (9.12) require proved nonvanishing on the retained region and supply estimates only there. For the base correction there is one piece; Lemma 7.1 supplies its reflected numerator factor $U^{\delta/2+12e+\epsilon}$.

9.6 Applying the row-count bound

For the base correction, Lemma 6.2 verifies the hypotheses of Lemma 9.3. Indeed $|\mathcal{H}_{\eta,u}| \ll_\epsilon q_u^\epsilon$ uniformly in all three heights in both Euler regions; for $q_u \le Z^D$, taking $\epsilon = 1$ gives (9.4) with $B = D$ and height order zero. We now estimate the retained integral, using $l_x = l_y = h = 1/2$ and $\ell = 0$ throughout this subsection.

Fix

$$d_{\min} = \frac{1}{63}, \qquad \zeta = \frac{1}{1000}, \qquad d_{\max} = h+\zeta = \frac{501}{1000}.$$

The lower endpoint will give a saving for the small rows in Section 9.8; the extension ζ above h is controlled by the positive slope computed below. For a dyad $U = Z^d$ in this range, fix one dynamic bin (ι, a) before moving any Mellin variable, and put $\delta = 2a-1$. Proposition 8.2, with requested loss $\epsilon_r > 0$, gives its cardinality at most $U^{R(\delta)+\epsilon_r}(1+T_1)^{A_\eta}$ up to a fixed constant. For the floor this is the direct count $O_\eta(U)$, since $R(1/50) = 1$; it uses no witness.

The numerator $L^S(w, \chi_\bullet(u))$ is precisely the original zero-extended presentation with $\nu = 1$ in Section 7. On the retained line $\Re w = 1-a-6e$, Lemma 7.1 bounds it by $U^{\delta/2+12e+\epsilon_n}(3+T_1)^C$

for any $\epsilon_n > 0$. This also holds in the floor bin. The central point lies in $\mathcal{D}_1(10e)$, so the stronger local estimate in Lemma 6.2 gives $|\mathcal{H}_{\eta,u}| \ll_{e,\epsilon_H} U^{\epsilon_H}$ there, for any $\epsilon_H > 0$. Consequently

$$\sum_{u\in\mathcal{B}} |L^S(w,\chi_\bullet(u))\mathcal{H}_{\eta,u}(s,w,z)| \ll_\eta U^{R(\delta)+\delta/2+12e+\epsilon_r+\epsilon_n+\epsilon_H}(1+T_1)^{A'_\eta}.$$

Here A'_η is finite and is fixed before the final external tail order.

Put $\varepsilon_c = 12e+\epsilon_r+\epsilon_n+\epsilon_H$. The buffered reciprocal is bounded by U^{ε_d} for any $\varepsilon_d > 0$, and $\zeta_F^S(6z_0)$ is absolutely bounded. The tests have bounded joint L^1 norm in the three independent height coordinates above. The outside powers, including $q_u^{-z_0}$, are

$$Z^{a/2-3/4+z_0/2+13e}U^{-z_0}.$$

Multiplying these bounds gives, for every $\varepsilon_p > 0$, a retained bin contribution at most

$$\ll_\eta Z^{1/4+E_{\mathrm{I}}(d)+13e+(1+d)\varepsilon_c+d\varepsilon_d+\varepsilon_p}(1+T_1)^{A'_\eta}. \tag{9.17}$$

We have increased the bound by $Z^{\varepsilon_c+\varepsilon_p}$ to use the loss notation of Lemma 9.4. With the current bin δ fixed, the exponent is

$$E_{\mathrm{I}}(d) = -\frac{3}{4} + \frac{\delta+R(\delta)}{2} + \left(d-\frac{1}{2}\right)\left(R(\delta)+\frac{\delta}{2}-\frac{17}{50}\right).$$

The three original Mellin integrals are the only ones in this application. Their scalar buffered arguments have heights $\Im s$ and $\Im w$. The witness frequencies and profile choices stay within the detector's already assigned allowance in the row count and do not enter either scalar argument.

At $d = h = 1/2$, the three pieces of the row-count exponent give, respectively,

$$E_{\mathrm{I}}(h) = \begin{cases} -1/4+\delta/2, & 0\le\delta\le 1/3,\\ -1/12, & 1/3\le\delta\le 2/3,\\ -(1-\delta)/4, & 2/3\le\delta\le 1. \end{cases}$$

The first two pieces are at most $-1/12$. In the third, the bin ceiling from Section 7 is

$$\delta \le 2\beta_* - 1 = \frac{5}{6} + 2\Delta_1.$$

It follows in every piece that

$$E_{\mathrm{I}}(h) - \Delta_1 \le -\frac{1}{24} - \frac{\Delta_1}{2}.$$

This is the comparison with $C_{\mathrm{I}}(\beta_*)$ required here; $E_{\mathrm{I}}(h)$ itself need not be negative. The frequency slope has the three forms

$$R(\delta)+\frac{\delta}{2}-\frac{17}{50} = \begin{cases} 33/50+\delta/2, & 0\le\delta\le 1/3,\\ 149/150-\delta/2, & 1/3\le\delta\le 2/3,\\ 33/50, & 2/3\le\delta\le 1. \end{cases}$$

It therefore lies in $[33/50, 62/75]$. Positivity controls all $d \le h$, and the extension to $d \le h+\zeta$ increases the exponent by at most $(62/75)\zeta$. Thus, before adjustable losses, every central dyad has saving at least

$$m_c = \frac{1}{24} - \frac{62}{75}\zeta = \frac{1021}{25000}$$

relative to $C_{\mathrm{I}}(\beta_*)$.

9.7 Extracting the principal signal

The intermediate-row estimate leaves the principal row and the two outer norm ranges. We first cross the two scalar poles in the principal row. The following formulation also records the precise condition under which a different correction has the same target signal.

Lemma 9.5 (Extraction of the principal signal). *Assume Definition 9.1. For every primitive finite-order target η, let H_η and c_S be the correction and normalizing constant in (9.1), and assume (9.2). Fix $\sigma_0 \in [7/8, 1)$ with $\beta_* > \sigma_0$ and $0 < e < 10^{-3}$. Suppose that for every $\epsilon > 0$ there exists a finite $J_{\eta,e,\epsilon}$ such that, for every external tail order, uniformly in all imaginary parts on*

$$\Re s = \beta_* + e, \qquad 19/20 \le \Re w \le 1 + e, \qquad 33/200 \le \Re z \le 1/6 + e,$$

one has

$$|\mathfrak{H}_{\eta,1,Z}(s, w, z)| \ll_{\eta,e,\epsilon} Z^{\ell\Re z+\epsilon}(1 + |\Im s| + |\Im w| + |\Im z|)^{J_{\eta,e,\epsilon}}, \tag{9.18}$$

with the same $J_{\eta,e,\epsilon}$ for every such order. Suppose also that $A_\eta(Z) \neq 0$ for every sufficiently large real Z, that $|A_\eta(Z)|^{-1} \ll_{\eta,\epsilon} Z^\epsilon$ for every $\epsilon > 0$, and that on $\Re s = \beta_* + e$, uniformly in $\Im s$,

$$\mathfrak{H}_{\eta,1,Z}(s, 1, 1/6) = H_\eta(s) Z^{\ell/6} A_\eta(Z)(1 + \mathcal{R}_{\eta,Z}(s)). \tag{9.19}$$

Suppose there exists a number $\mu > 0$ such that, for every primitive finite-order target η, either $\mathcal{R}_{\eta,Z}$ is identically zero or $|\mathcal{R}_{\eta,Z}(s)| \ll_\eta Z^{-\mu}$ on that entire line for every sufficiently large Z. The implied constant and lower threshold in this hypothesis may depend on η.

Let $\mathscr{P}_\eta(Z)$ be the $u = 1$ term of (9.5), and define

$$f_\eta(Z) = \frac{1}{2\pi i} \int_{(2)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s, \eta)}\, ds.$$

Then $c_S > 0$, and, for every $\epsilon > 0$,

$$\begin{aligned} \frac{\mathscr{P}_\eta(Z)}{c_S A_\eta(Z)} - f_\eta(Z) \ll_{\eta,e,\epsilon} Z^{C(\beta_*)+(1+h)e-l_y/20+\epsilon} + Z^{C(\beta_*)+e-h/600+\epsilon} \\ + Z^{C(\beta_*)+e-\mu+\epsilon}. \end{aligned} \tag{9.20}$$

The last term is omitted when $\mathcal{R}_{\eta,Z}$ is identically zero. The implied constants and lower thresholds may depend on the target, but the displayed real exponents do not.

Proof. The two scalar residues. For $u = 1$, the scalar factor in the high identity is

$$\frac{\zeta_F^S(6z)\zeta_F^S(w)}{L_F^S(s, \eta)}.$$

Starting with this term on the absolute contours, move to $\Re s = \beta_* + e$, $\Re w = 1 + e$ and $\Re z = 1/6 + e$. The reciprocal stays in $\Re s > \beta_*$, and both zeta arguments stay to the right of one. The paths lie in $\mathcal{D}_2$, so the correction is holomorphic. The all-height majorant and Lemma 9.2 justify the horizontal limits.

Moving w to $19/20$ crosses its simple pole at $w = 1$. Moving z in that residue to $33/200 = 1/6 - 1/600$ then crosses its simple pole at $z = 1/6$. The unresidued w integral keeps $\Re z = 1/6 + e$. All these paths lie in $\mathcal{D}_2$, and $M(z)$ is holomorphic for $\Re z > 0$. Thus the two stated scalar poles are the only poles crossed. The reciprocal remains on its global line throughout, even when η is principal.

For the horizontal limits, every extended vertical w- or z-axis used to bound a tail lies strictly on one side of its scalar pole; $w = 1$ and $z = 1/6$ occur only as residues, and the horizontal joins have large nonzero height. On those axes the zeta functions have fixed polynomial bounds. Consequently the integrated and trace estimates of Lemma 9.2 apply without an unremoved

pole in their majorants. After the w residue, its Mellin factor has become the constant $\widehat{W_1}(1)$. The remaining heights use the two-dimensional kernel in (9.9), with $(y_1, y_2) = (\Im(s+z), \Im z)$; its trace estimate justifies the subsequent z-join.

Remainder estimates. On the principal contours, (9.18) and the outside powers give the raw exponent

$$\frac{l_x}{2} + \Re s - 1 + h\Re z + l_y(\Re w - 1).$$

For the unresidued w integral this is $C(\beta_*) + (1+h)e - l_y/20$; for the leftover z integral in the w residue it is $C(\beta_*) + e - h/600$. The reciprocal on $\Re s = \beta_* + e$ has an arbitrarily small power of its fixed target conductor and a fixed polynomial in height by Lemmas 3.15 and 3.16. The tests integrate those height powers. Requesting the small powers in the correction and in A_η^{-1} to be sufficiently small gives the first two terms of (9.20).

The product of the scalar residues is c_S: the residue of $\zeta_F^S(6z)$ at $1/6$ is one sixth of the residue of $\zeta_F^S(v)$ at one. Positivity of c_S was proved when the normalization was defined.

Identification of the signal. At the double residue the outside power is

$$X^{1/3} Z^{s-5/6} Z^{\ell/6} = Z^{C(s)}, \qquad \Phi(s + 1/6 - 1) = e^{(s-5/6)^2}.$$

Equation (9.19) therefore makes the normalized double residue

$$\frac{1}{2\pi i} \int_{(\beta_*+e)} Z^{C(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s,\eta)} (1 + \mathcal{R}_{\eta,Z}(s))\, ds.$$

When present, the error factor is estimated on this original line, where the global reciprocal bound and Gaussian give $O_{\eta,e,\epsilon}(Z^{C(\beta_*)+e-\mu+\epsilon})$. Only the main integral will be shifted.

Only the main integral is now moved right to $\Re s = 2$. The reciprocal is holomorphic for $\Re s > \beta_*$ by the definition of β_* and absolute Euler convergence beyond one. At a principal pole its reciprocal has a zero, so that case adds no residue. The function H_η is holomorphic there by Lemma 6.2, and is bounded there by (9.2). The global reciprocal estimate is polynomial in height uniformly on the fixed real strip, while the Gaussian is $O(e^{-(\Im s)^2})$. Rectangular contours therefore have vanishing horizontal sides, and the shifted integral is exactly $f_\eta(Z)$. This also covers $\beta_* = 1$. The three remainders give the asserted bound. □

The function H_η is the correction of the unmodified scalar Euler product at the double residue. A different full correction must verify (9.19) with a nonzero normalizer. In every application the normalized sum is $\mathscr{I}_\eta/(c_S A_\eta)$; the same normalizer must be used for its direct estimate and for this principal comparison. The raw central and outer-row bounds acquire only an arbitrarily small additional power after this division, by the hypothesis on A_η^{-1}.

The base correction. Take $l_x = l_y = h = 1/2$ and $\ell = 0$. On the principal rectangle of Lemma 9.5, the local correction is bounded uniformly in every height because that rectangle lies in $\mathcal{D}_2$. Thus (9.18) holds with $\ell = 0$ and height order zero. At the double residue it satisfies exactly

$$\mathfrak{H}_{\eta,1,Z}(s, 1, 1/6) = H_\eta(s) = H_\eta(s) Z^{\ell/6} \cdot 1 \cdot (1 + 0).$$

Hence $A_\eta = 1$ and the residue error is identically zero. In the row $u = 1$, the outside factor $q_u^{-z}\overline{\xi(u)}$ is one; the sole factor $1/6$ is already in c_S from the residue of $\zeta_F^S(6z)$. Lemma 9.5 therefore has just its two remainder terms. Their savings before losses are

$$m_w = \frac{l_y}{20} = \frac{1}{40}, \qquad m_z = \frac{h}{600} = \frac{1}{1200}.$$

9.8 Small and large row norms

The buffered bin estimate is needed only on a bounded interval of positive row exponents. The following direct bounds cover its two complements. They use absolute estimates on the full correction.

Lemma 9.6 (Small row norms). *Assume Definition 9.1, with the row domain and full correction in (9.5). Fix $\sigma_0 \in [7/8, 1)$ with $\beta_* > \sigma_0$, $0 < e < 10^{-3}$, $z_0 = 17/50$, and $d_{\min} > 0$. For a dyadic number $U \geq 1$, let $\mathscr{R}_\eta(U; Z)$ denote the contribution in (9.5) of element rows $U \leq q_u < 2U$, omitting $u = 1$.*

Suppose that for every $\epsilon > 0$, for all such rows with $q_u \leq 2Z^{d_{\min}}$, and uniformly in all imaginary parts on $(\Re s, \Re w, \Re z) = (\beta_* + e, 1/2, z_0)$,

$$|\mathfrak{H}_{\eta,u,Z}(s, w, z)| \ll_{\eta,e,\epsilon} Z^{\ell z_0 + \epsilon} q_u^{\epsilon} (1 + |\Im s| + |\Im w| + |\Im z|)^{J_{\eta,e,\epsilon}}. \tag{9.21}$$

Then, for every $\epsilon > 0$ and $1 \leq U \leq Z^{d_{\min}}$,

$$|\mathscr{R}_\eta(U; Z)| \ll_{\eta,e,\epsilon} Z^{C(\beta_*) + h(z_0 - 1/6) - l_y/2 + e + \epsilon} U^{63/50 + \epsilon}. \tag{9.22}$$

In particular the sum of these dyads is

$$\ll_{\eta,e,\epsilon} Z^{C(\beta_*) + h(z_0 - 1/6) - l_y/2 + e + (63/50) d_{\min} + \epsilon}. \tag{9.23}$$

Proof. Every row under consideration has nonprincipal numerator by the classification in Section 9.1. The denominator may be principal for a bounded unit row; its reciprocal is nevertheless holomorphic and bounded on $\Re s = \beta_* + e$ by the principal specialization of Lemma 3.15. For all rows, the conductor and deletion radical bounds in Section 7, together with Lemma 3.16, give an arbitrarily small power of q_u and a fixed height polynomial for that reciprocal.

For a small dyad move its finite sum to $(\Re s, \Re w, \Re z) = (\beta_* + e, 1/2, z_0)$. The reciprocal stays global. The numerator is entire, and the scalar zeta argument stays to the right of one. These paths may be taken in $\mathcal{D}_1(3/8)$: at the final point $\Re(s + w) = \beta_* + e + 1/2 > 1 + 3/8$, and the other inequalities are immediate. The all-height correction bound and Lemma 9.2 justify the moves. On the final lines, (3.21) and the deletion bound give $U^{3/5+\epsilon}$ times a fixed height polynomial for the nonprincipal numerator. There are $O(U)$ element rows in the dyad, including unit factors, and $q_u^{-z_0} \ll U^{-z_0}$. The correction is bounded by (9.21). The tests integrate all fixed height powers. The resulting exponent outside the row power, relative to $C(\beta_*)$, is

$$h(z_0 - 1/6) - l_y/2 + e,$$

and the row power is $1 + 3/5 - z_0 = 63/50$. Requesting the component small powers to sum to the displayed ϵ proves (9.22). The row exponent is positive, so dyadic summation up to $Z^{d_{\min}}$ gives (9.23), after decreasing the preliminary losses. □

Lemma 9.7 (Large row norms). *Assume Definition 9.1, with the row domain and full correction in (9.5). Fix $\sigma_0 \in [7/8, 1)$ with $\beta_* > \sigma_0$, $0 < e < 10^{-3}$, $z_0 = 17/50$, and $d_{\min} > 0$. Recall from Lemma 9.6 that, for a dyadic number $U \geq 1$, $\mathscr{R}_\eta(U; Z)$ denotes the contribution in (9.5) of element rows $U \leq q_u < 2U$, omitting $u = 1$. The denominator may be principal for a bounded unit row.*

Suppose that for every fixed $v > 2$ and $\epsilon > 0$, for every element row in the sum, uniformly in all imaginary parts on $(\Re s, \Re w, \Re z) = (2, 2, v)$,

$$|\mathfrak{H}_{\eta,u,Z}(s, w, z)| \ll_{\eta,v,\epsilon} Z^{\ell v + \epsilon} q_u^{\epsilon} (1 + |\Im s| + |\Im w| + |\Im z|)^{J_{\eta,v,\epsilon}}. \tag{9.24}$$

Write $B_0 = l_x/2 + 1 + l_y$. Then, for every dyad U,

$$|\mathscr{R}_\eta(U; Z)| \ll_{\eta,v,\epsilon} Z^{B_0 + hv + \epsilon} U^{1 + \epsilon - v}. \tag{9.25}$$

For every fixed $\zeta > 0$, if $\epsilon < v - 1$, its sum over $U > Z^{h+\zeta}$ is

$$\ll_{\eta,v,\epsilon} Z^{B_0+(h+\zeta)(1+\epsilon)-\zeta v+\epsilon}. \tag{9.26}$$

Thus a fixed sufficiently large v makes this last contribution smaller than any prescribed power of Z.

Proof. For each fixed Z and large dyad, move its finite row sum to the absolute lines $(2, 2, v)$. The path from $(3, 3, 2)$ remains in $\Re s, \Re w \geq 2$ and $\Re z \geq 2$, so the scalar factors are holomorphic there. For this fixed dyad choose D large enough to contain its rows in $q_u \leq Z^D$; (9.4) and the external trace bound make its horizontal limits vanish. Constants used only to justify this equality may depend on the fixed dyad. The estimate on the final lines is uniform in the dyad by (9.24). The scalar L-factors are absolutely bounded there, as is $\zeta_F^S(6z)$, and (9.24) applies. The outside power, including the correction but not the row count, is

$$l_x(1/2 - v) + 1 + v + l_y + \ell v = B_0 + hv.$$

The $O(U)$ rows and q_u^{-v} give U^{1-v}; the tests integrate the fixed height polynomial. This proves (9.25). Because $1 + \epsilon - v < 0$, summing its geometric dyadic tail gives (9.26). For fixed $\zeta > 0$, the coefficient of v in that exponent is $-\zeta$, proving the last assertion. The original row series is absolutely convergent on its starting lines, and the displayed bounds give an absolutely summable final tail, so the individual dyadic contour identities may be summed. The number v and the finite test orders it requires are fixed before Z tends to infinity. □

The first-stage outer ranges. Use $l_x = l_y = h = 1/2$, $\ell = 0$, $d_{\min} = 1/63$ and $\zeta = 1/1000$, as in the intermediate-row estimate. For the small rows, the line $(\beta_* + e, 1/2, z_0)$ lies in $\mathcal{D}_1(3/8)$, since $\beta_* > 11/12$. The local bound $\mathcal{H}_{\eta,u} \ll_\epsilon q_u^\epsilon$, uniform in all heights, verifies (9.21) for all the prescribed rows, with height order zero. Lemma 9.6 gives (9.23), whose relative exponent before its losses is:

$$h(z_0 - 1/6) - \frac{l_y}{2} + \frac{63}{50} d_{\min} = \frac{13}{150} - \frac{1}{4} + \frac{1}{50} = -\frac{43}{300}.$$

For the large rows, $(2, 2, v)$ with $v > 2$ lies in $\mathcal{D}_2$. The same all-height local bound verifies (9.24) for every element row in the sum. Here $B_0 = l_x/2 + 1 + l_y = 7/4$, so Lemma 9.7 gives (9.26) with exponent

$$\frac{7}{4} + \frac{501}{1000}(1 + \epsilon) - \frac{v}{1000} + \epsilon.$$

It tends to $-\infty$ as the fixed number v increases. For example, $v = 4000$ makes it less than -1 when $\epsilon \leq 10^{-6}$, whereas $C_{\mathrm{I}}(\beta_*) \geq 1/4$. This more than supplies the saving needed below. The three ranges $U \leq Z^{d_{\min}}$, $Z^{d_{\min}} < U \leq Z^{h+\zeta}$, and $U > Z^{h+\zeta}$ cover every row-norm dyad. The row $u = 1$ was already assigned to the principal term.

10 The 11/12 conclusion

For the normalized sum $J_{\mathrm{I},\eta}$ the central estimate, principal extraction, and outer-row estimates now give the following margins before adjustable losses:

contribution	saving relative to $C_{\mathrm{I}}(\beta_*)$
intermediate rows	1021/25000
unresidued principal w integral	1/40
remaining principal z integral	1/1200
small rows	43/300
large rows ($v = 4000$)	> 1

The principal z remainder determines the smallest margin. Proposition 10.2 retains a common positive margin after the real losses; Lemma 10.1 then absorbs the height factors for each target. The sum $J_{\mathrm{I},\eta}$ and the signal $f_{\mathrm{I},\eta}$ are independent of that height.

10.1 Choosing the final height

The preceding estimates separate real powers from finite powers of the analysis height T_1. The following elementary step records the order of choices needed to obtain one power saving for every target.

Lemma 10.1 (Height selection). *Fix $\sigma_0 \in (1/2, 1)$ with $\Delta_0 = \beta_* - \sigma_0 > 0$ and an affine function $C(s) = s + c$. Fix the real parameters and finite structural data of the application, including $d_{\min} > 0$. Let J_η, f_η be functions independent of T_1, indexed by the primitive finite-order targets η. Suppose there exist $m > 0$ and $\omega \in (0, \Delta_0)$ such that, for every target η,*

$$|J_\eta(Z)| \ll_\eta Z^{C(\sigma_0)+\omega}.$$

Suppose also that, for every target η, there exist finite profile orders, finite numbers $A_\eta \geq 0$, B_η, and $\tau_{0,\eta} > 0$ with the following property: for every integer $N \geq 0$ and every $0 < \tau \leq \min\{d_{\min}/100, \tau_{0,\eta}\}$, the finite sum of estimates gives

$$|J_\eta(Z) - f_\eta(Z)| \ll_{\eta,N} Z^{C(\beta_*)-m}(1+T_1)^{A_\eta} + Z^{B_\eta} T_1^{-N}, \tag{10.1}$$

for $T_1 = Z^\tau$ and all Z above a threshold that may depend on η, N, τ. The finite profile orders, A_η, B_η, $\tau_{0,\eta}$ and the real powers in (10.1) are the same for every N and τ; among test seminorms, only the external ones may change with N.

Then for every target η there exist $\tau_\eta \in (0, \min\{d_{\min}/100, \tau_{0,\eta}\}]$ and an integer $N_\eta \geq 0$ such that, for all sufficiently large Z,

$$|J_\eta(Z) - f_\eta(Z)| \ll_\eta Z^{C(\beta_*)-m/2}.$$

Proof. For the fixed target choose

$$0 < \tau_\eta \leq \min\left\{\frac{d_{\min}}{100}, \tau_{0,\eta}, \frac{m}{4(A_\eta+1)}\right\}, \qquad T_1 = Z^{\tau_\eta}.$$

Then $(1+T_1)^{A_\eta} \ll_\eta Z^{m/4}$, so the first term in (10.1) is $O_\eta(Z^{C(\beta_*)-3m/4})$. Next choose a fixed integer N such that

$$B_\eta - N\tau_\eta < C(\beta_*) - m/2.$$

This is possible because $\tau_\eta > 0$ and B_η is finite. The second term is then bounded by the required power. After these choices, the bound holds above a possibly larger lower threshold for Z. Both m and ω were fixed before the target, whereas τ_η, N and the threshold may depend on it. Since J_η and f_η do not contain T_1, this proves the asserted family-wide exponent without changing either function. □

The ceiling $\tau_{0,\eta}$ can enforce a preceding height condition, such as that in Lemma 7.2. The common saving $\sigma = m/2$ and the low estimate then have the target quantifiers required by Proposition 2.1.

Proposition 10.2 (Balanced high estimate). *Assume (3.1), so $\Delta_1 = \beta_* - 11/12 > 0$. Let C_{I}, $J_{\mathrm{I},\eta}$ and $f_{\mathrm{I},\eta}$ be the functions defined in Section 9.1. For every primitive finite-order target η and all sufficiently large Z,*

$$|J_{\mathrm{I},\eta}(Z) - f_{\mathrm{I},\eta}(Z)| \ll_\eta Z^{C_{\mathrm{I}}(\beta_*)-1/4800}.$$

The displayed saving is independent of the target; the implied constant and lower threshold may depend on it.

Proof. Take $l_x = l_y = h = 1/2$, $\ell = 0$ and $C = C_{\mathrm{I}}$. The identity and local correction at the start of Section 9 put $J_{\mathrm{I},\eta}$ in the form required for the high estimate. The intermediate rows satisfy (9.17); the principal and outer specializations there give the other margins in the table above. We first choose common real losses, then verify the detector height ceiling and the external tail bound needed for Lemma 10.1.

Common real losses. Put

$$m_0 = \frac{1}{1200}, \qquad \epsilon_* = 10^{-6}, \qquad m = \frac{1}{2400}, \qquad \omega = \frac{\Delta_1}{2}.$$

The smallest unadjusted saving is m_0. We use ϵ_* for the component losses and m for the common saving before the height factor; ω is the loss in the low estimate. Fix the displayed geometry, row ranges, and $v = 4000$ before the target. Each row count, numerator bound, local correction, buffered reciprocal, central multiplicity bound, and principal or outer estimate is taken with loss ϵ_*. An additional loss ϵ_* absorbs the $O(\log Z)$ dyads of the original row norm. The witness-pair and powerful-part logarithms are already included in the requested row-count loss; the $O_e(1)$ bins and fixed presentation choices contribute constant factors.

To obtain the requested row-count loss, first choose its preliminary witness, dyadic, and sieve losses sufficiently small in terms of ϵ_*. These choices are independent of the target. Indeed the witness length error has an absolute coefficient, and the affine slopes in the optimization of Proposition 8.2 are uniformly bounded. The target-dependent constants in the actual dyadic endpoints multiply only $1/\log U$, which changes a fixed constant or lower threshold. Lemma 7.2 applies on the enclosing ranges $0 \le r, m \le 22$ before saturation: the detector's terminal product cutoff is $O(U^{21})$ before it proves the shorter witness lengths. Its number e_0 depends only on the requested dyadic loss and these bounded ranges. We may thus fix, still before the target,

$$0 < e < \min\{10^{-3}, \epsilon_*, e_0\}.$$

With $\epsilon_r = \epsilon_n = \epsilon_H = \epsilon_*$, this gives $\varepsilon_c \le 15\epsilon_*$. Since $d \le 501/1000 < 1$, the explicit central losses in (9.17), together with the loss from summing those row dyads, are at most

$$13\epsilon_* + 2(15\epsilon_*) + \epsilon_* + \epsilon_* + \epsilon_* = 46\epsilon_* < \frac{1}{4800} = \frac{m_0}{4}.$$

The two principal losses are at most $(3/2+1)\epsilon_*$ and $2\epsilon_*$; the small-row loss is at most $2\epsilon_*$. Each is less than $m_0/4$. The displayed large-row estimate already uses its requested loss. Therefore all these terms, after division by the fixed c_S, retain at least $3m_0/4 = 1/1600$ before the height factor. We use $m = 1/2400$, which is smaller by $1/4800$ than this retained saving.

The detector height ceiling. To verify the height hypotheses, let $\epsilon_{\mathrm{det}} > 0$ be the fixed dyadic loss chosen above, and, after fixing a target, let $A_{\mathrm{det},\eta}$ dominate the finitely many orders in its uses of Lemma 7.2. Set

$$\tau_{0,\eta} = \frac{d_{\min}\epsilon_{\mathrm{det}}}{20(A_{\mathrm{det},\eta}+1)} > 0.$$

For $0 < \tau \le \tau_{0,\eta}$ and sufficiently large Z,

$$(1+Z^{\tau})^{A_{\mathrm{det},\eta}} \le Z^{d_{\min}\epsilon_{\mathrm{det}}/10} \le U^{\epsilon_{\mathrm{det}}/10} \qquad (d \ge d_{\min}).$$

This verifies the detector height condition uniformly over the original row range, where $d \ge d_{\min}$. The fixed factor from $1 + Z^{\tau} \le 2Z^{\tau}$ is absorbed by the lower threshold. We also impose $\tau \le d_{\min}/100$ as required by the bins. The internal tail orders used to establish the witnesses are chosen after this τ and are separate from the final external order N below. Their increase does not change the retained profile or height orders. Any fixed constants introduced by those internal choices are absorbed into the unused power $1/4800$ by increasing the lower threshold, which may depend on η and τ.

External tails. For every final integer $N \geq 0$, Lemma 9.3 bounds the discarded high portions and joins by $O_{\eta,N}(Z^{B_\eta}T_1^{-N})$, with B_η fixed before N. The bounded real and row ranges and the finite set of bins permit one such B_η for all the central dyads. Summing the $O(\log Z)$ dyads can be absorbed by increasing B_η by one, again before N. Combining the preceding estimates gives

$$|J_{\mathrm{I},\eta}(Z) - f_{\mathrm{I},\eta}(Z)| \ll_{\eta,N} Z^{C_{\mathrm{I}}(\beta_*)-m}(1+T_1)^{A_\eta} + Z^{B_\eta}T_1^{-N}$$

for some finite A_η, B_η, whenever $0 < \tau \leq \min\{d_{\min}/100, \tau_{0,\eta}\}$ and Z is sufficiently large. Both functions are independent of T_1.

The low estimate and height choice. Proposition 5.3, applied with loss ω and divided by the same fixed c_S, supplies

$$|J_{\mathrm{I},\eta}(Z)| \ll_\eta Z^{1/4+\omega} = Z^{C_{\mathrm{I}}(11/12)+\omega}, \qquad 0 < \omega < \Delta_1.$$

All hypotheses of Lemma 10.1 are now verified. It gives the saving $m/2 = 1/4800$ asserted in the proposition. □

10.2 Transfer to Dirichlet L-functions

The continuation criterion concerns the Hecke family over F. We prove once that a strict zero-free half-plane for that family has the corresponding Dirichlet consequence.

Proposition 10.3 (Quadratic transfer). *Let $\sigma_0 \in [1/2, 1)$. Suppose every primitive finite-order Hecke L-function over $F = \mathbb{Q}(\sqrt{-3})$ is zero-free on $\Re s > \sigma_0$, with its principal pole at one allowed. Then every finite-order Hecke L-function over F and every Dirichlet L-function is zero-free on that same strict half-plane, again allowing the principal pole at one.*

Proof. The norm pullback of a Dirichlet character gives a Hecke function that factors into two Dirichlet functions after deletion of the Euler factors specified below. We first pass between primitive and induced characters, then check that this factorization transfers the strict zero-free half-plane, including the possible pole-zero cancellation at one.

Passing from a primitive Hecke character to one that it induces changes only finitely many factors $1 - \eta(\mathfrak{p})N\mathfrak{p}^{-s}$. They are nonzero for $\Re s > 0$, so the Hecke assertion extends to all finite-order characters. The same observation for factors $1 - \chi(p)p^{-s}$ reduces the Dirichlet assertion to a primitive Dirichlet character χ of conductor q.

Let χ_{-3} be the quadratic character of conductor three, and let η be the finite-order Hecke character given by $\mathfrak{a} \mapsto \chi(N\mathfrak{a})$ on ideals coprime to $3q$. It is a ray character: if $\alpha \equiv 1 \pmod{q\mathcal{O}}$, then $N\alpha \equiv 1 \pmod{q}$, so the norm character is trivial on the corresponding principal ray subgroup. Let $S_{\mathbb{Q}}$ be the rational primes dividing $3q$, and let $\mathcal{S}_{\mathbb{Q}}$ be the primes of F above them. Superscripts by these sets denote deletion of those Euler factors.

For $p \notin S_{\mathbb{Q}}$, the local factors agree as follows. If p splits in F, then $\chi_{-3}(p) = 1$, the two prime ideals have norm p, and the Hecke factor is $(1 - \chi(p)p^{-s})^{-2}$. This is the product of the Dirichlet factors for χ and $\chi\chi_{-3}$. If p is inert, then $\chi_{-3}(p) = -1$, the unique prime ideal has norm p^2, and its factor is

$$(1 - \chi(p)^2p^{-2s})^{-1} = (1 - \chi(p)p^{-s})^{-1}(1 + \chi(p)p^{-s})^{-1}.$$

These exhaust the primes outside $S_{\mathbb{Q}}$. Absolute Euler convergence for $\Re s > 1$, followed by uniqueness of meromorphic continuation, gives

$$L_F^{\mathcal{S}_{\mathbb{Q}}}(s, \eta) = L^{S_{\mathbb{Q}}}(s, \chi)L^{S_{\mathbb{Q}}}(s, \chi\chi_{-3}). \tag{10.2}$$

The product character on the right may be imprimitive; deletion of $S_{\mathbb{Q}}$ makes the identity independent of that choice. Every deleted factor is nonzero for $\Re s > 0$.

Both Dirichlet factors are holomorphic in $0 < \Re s < 1$. A zero of $L(s, \chi)$ there would therefore give a zero of the Hecke factor, with no cancellation by a pole, and the assumed Hecke half-plane

excludes it when $\Re s > \sigma_0$. Absolute Euler convergence handles $\Re s > 1$. On $s = 1 + it$ with $t \neq 0$, neither Dirichlet factor has a pole, so the same product argument applies.

At $s = 1$, a pole-zero cancellation could occur only if one primitive inducing character among χ and $\chi\chi_{-3}$ were principal. The other would then be χ_{-3}. Its Dirichlet series converges at one by bounded partial sums of the nonprincipal periodic character, and grouping consecutive terms gives

$$\begin{aligned} L(1, \chi_{-3}) &= \sum_{n \geq 0} \left(\frac{1}{3n+1} - \frac{1}{3n+2} \right) \\ &= \int_0^1 \frac{1-x}{1-x^3}\, dx = \int_0^1 \frac{dx}{1+x+x^2} = \frac{\pi}{3\sqrt{3}} > 0. \end{aligned}$$

The first integral follows by monotone convergence of the nonnegative paired integrands. Thus there is no zero in this last case either; a principal pole is allowed. This proves the asserted strict half-plane statement. □

Proof of Theorem 3.1. Suppose that $\beta_* > 11/12$. Lemma 6.2 and (9.2) give the required holomorphic, nonzero H_η on $\Re s > 11/12$ for every primitive target. The low and high bounds established in Proposition 10.2 and its proof verify Proposition 2.1 with

$$\sigma_0 = \frac{11}{12}, \qquad C = C_{\mathrm{I}}, \qquad \omega = \frac{\Delta_1}{2}, \qquad \sigma = \frac{1}{4800}.$$

These two positive losses are independent of the target, while the allowed constants, excluded sets, and thresholds may depend on it. The continuation criterion contradicts the supposition. Therefore $\beta_* \leq 11/12$. By its definition and absolute Euler convergence in $\Re s > 1$, every primitive finite-order Hecke L-function over F is zero-free in the strict half-plane $\Re s > 11/12$, with the principal pole allowed.

Proposition 10.3, applied at the same boundary, extends this assertion to all finite-order Hecke characters and all Dirichlet L-functions, including $\zeta(s)$. It preserves the strict half-plane and the allowed principal pole. This proves the theorem. □

Part II
The seven-eighths zero-free half-plane

11 A marked and rescaled character sum

Theorem 3.1 applies to every primitive character entering the supremum in (2.1), and therefore gives $\beta_* \leq 11/12$. Suppose for contradiction throughout Part II that $\beta_* > 7/8$, and put

$$\Delta := \beta_* - \frac{7}{8}, \qquad 0 < \Delta \leq \frac{1}{24}, \qquad \kappa := 2\beta_* - 1 = \frac{3}{4} + 2\Delta \leq \frac{5}{6}. \tag{11.1}$$

The exact bin ceiling in Section 7 applies, since $\beta_* > 7/8 > 51/100$. For every retained nonprincipal row and its bin (ι, a) it gives

$$a \leq \beta_*, \qquad \delta = 2a - 1 \leq \kappa \leq \frac{5}{6}. \tag{11.2}$$

Indeed, the finite set defining $M_\iota(u)$ consists of the floor $51/100$ and real parts of actual zeros, all at most β_*. The boundary case $\delta = 5/6$ remains part of the argument.

For the second application of Proposition 2.1, recall the affine normalization for Part II

$$C_{\mathrm{II}}(s) = s - \frac{11}{16}, \qquad C_{\mathrm{II}}(7/8) = \frac{3}{16}. \tag{11.3}$$

The finite modification below will define a normalized sum $J_{\mathrm{II},\eta}$, distinct from the balanced sum $J_{\mathrm{I},\eta}$. The reflected row norm of Section 13 and the additive Gram bound of Section 14 will give its low-side bound $|J_{\mathrm{II},\eta}(Z)| \ll_\eta Z^{3/16+\omega}$ for a common $0 < \omega < \Delta$. Its high side must satisfy the signal estimate in (2.5) with this C_{II}. Section 15 expresses the same finite modification through local Euler factors. The nonzero scalar normalization will be specified after the principal term is evaluated.

For each primitive target η, use an admissible fixed instance of the data $T, S, b_*, \xi, \tau, \Xi, W_0, W_1$ from Section 5. The choices of P_0 and the slot system will be made in the parameter order below. These data may differ from those used in Part I, but within this application they are independent of Z and are the same in the finite expression and its high representation.

The following geometry gives $M + \ell = 1$ and aligns the low exponent with $C_{\mathrm{II}}(7/8)$, as checked in (14.9):

$$\begin{gathered} b = \frac{1}{8}, \qquad h = \frac{13}{16}, \qquad \ell = \frac{1}{6}, \\ l_x = 1 + \ell - h = \frac{17}{48}, \qquad l_y = l_x + b = \frac{23}{48}, \\ X = Z^{l_x}, \qquad Y = Z^{l_y}, \qquad M = l_x + l_y = \frac{5}{6}, \qquad M + \ell = 1. \end{gathered} \tag{11.4}$$

Fix positive slot lengths $\ell_1, \dots, \ell_K$ of total length ℓ, and put $P_i = Z^{\ell_i}$. Each slot has a nonnegative, nonzero smooth annular weight $W_i(q_p/P_i)$. Its allowed prime set is

$$\mathcal{P}_i(Z) = \{p \text{ prime} : p \notin S,\ p \in 1_T,\ q_p/P_i \in \operatorname{supp} W_i\}.$$

The underlying window sets $\{p \text{ prime} : q_p/P_i \in \operatorname{supp} W_i\}$ for distinct slots are required to be disjoint before imposing the ray and S restrictions. In particular, the sets $\mathcal{P}_i(Z)$ are disjoint, and every tuple in $\prod_i \mathcal{P}_i(Z)$ consists of distinct primes. The number and lengths of the slots will be chosen later, but they are fixed independently of Z. A sum over $p \in 1_T$ for slot i below always retains this same exclusion and annular support.

For a squarefree product D of slot primes, let $I_{\eta;D}(X, Y, Z)$ denote (5.1) with the indicator $1_{D|cn^3}$ inserted in its completed row. Thus $I_{\eta;1} = I_\eta$. For a tuple $(p_1, \dots, p_K) \in \prod_i \mathcal{P}_i(Z)$ and $J \subseteq \{1, \dots, K\}$ write $p_J = \prod_{i \in J} p_i$, with $p_\varnothing = 1$. The subset J indexes the rescaled slots and J^c the marked slots; this subset notation is distinct from the normalized sum $J_{\mathrm{II},\eta}$. Define the modified sum by the finite identity

$$\begin{aligned} I_{\eta,\mathrm{modified}}(Z) = \sum_{(p_i) \in \prod_i \mathcal{P}_i(Z)} \prod_{i=1}^{K} W_i(q_{p_i}/P_i) \sum_{J \subseteq \{1,\dots,K\}} (-1)^{|J|} q_{p_J}^{-3/2} \overline{\eta(p_{J^c})} \\ \cdot I_{\eta;p_{J^c}}(X/q_{p_J}, Y/q_{p_J}, Z q_{p_{J^c}}). \end{aligned} \tag{11.5}$$

Equivalently, at each prime p the operation is the marked term $\overline{\eta(p)} I_{\eta;p}(X, Y, Zq_p)$ minus the rescaled term $q_p^{-3/2} I_\eta(X/q_p, Y/q_p, Z)$. Formula (11.5) specifies their composition: every slot window stays at its original scale P_i, including when another slot changes the completed scale. On 1_T one has $\theta(p) = 1$ for every $\theta \in \widehat{T}$. Thus marking commutes with the Fourier decomposition of G and preserves the same excluded set and zero masks in every summand. The subtraction is designed to cancel the scalar prime contribution on the high side, leaving the sextic-character prime factor used by the moment estimates. Section 15 proves the exact identity and bounds the remaining local errors.

12 Coefficient conventions and finite correlations

The prime factors require coefficient and support conditions beyond those of Part I. The fixed-ray prime normalizer below gives their size, and the full finite Fourier correlations retain the shared

prime powers needed in the additive Gram bound and the fourth moment. The residue-symbol and fixed-data conventions of Part I remain in force.

12.1 Additional coefficient conditions

We use the plain and inverse polynomials, annular profiles, and fixed arithmetic datum $\mathcal{A}$ from Section 3. The following conditions specify the extra prime factors and moving zero supports used in this part.

Prime-slot coefficients. A prime slot of log-length z_i is

$$Q_{\psi,i} = Z^{-z_i/2} \sum_{p \text{ prime}} \psi(p)\nu_i(p)W_i(q_p/Z^{z_i}). \tag{12.1}$$

Here ν_i is a fixed finite-ray character or a fixed finite linear combination of such characters. It is independent of the row and of the other selected primes. Distinct slots have disjoint underlying prime supports before any common mask or row zero extension is imposed. For the fourth moment of Section 17, a fixed finite group Θ of ray characters is part of the data, and every character component with nonzero coefficient in every positive-length slot must belong to Θ.

For the inverse moment of Section 16, we also allow a bounded coefficient $a_i(p)$ chosen independently for each slot and independently of the row and all other columns; the coefficient of a prime tuple is then the product of its individual slot coefficients. The Θ restriction does not apply to these separate inverse-moment weights.

Moving character support. For an auxiliary fourth moment the rows are elements $0 < q_k \ll Z^m$, and

$$\psi_k(n) = \tau(n)\chi_n(k), \qquad M = m + q. \tag{12.2}$$

The twist τ is common to the row sum after its outer labels are fixed, and its displayed factorization into fixed finite-ray and moving residue-symbol factors is part of the data. Let D_{mov} be the squarefree product of all good primes at which at least one displayed moving factor has its natural zero on nonunits. This includes a prime even when that factor has exponent divisible by six, including exponent zero, or when local characters cancel after multiplication. We require $q_{D_{\mathrm{mov}}} \le Z^q$, counting an overlapping prime once.

Common coprimality restrictions. An additional coprimality restriction is an indicator $1_{(n,R)=1}$ with R squarefree, $q_R \le Z^B$, and B in a prescribed bounded range. It must be common to the current row sum and to every plain and prime factor in that sum; it may depend on previously fixed outer labels.

Transferring a prime from D_{mov} to R requires an exact factorization of its displayed local factor into the coprimality indicator and the remaining local or fixed-ray character phases. The remaining phases stay in the factorization, the refactored zero is removed from the displayed moving factor, and the common coprimality restriction is deleted before the natural reflection in Section 17. Every zero prime must remain in at least one of the two supports. A prime at which a moving character is still ramified stays in D_{mov}; only a redundant zero may be transferred to R.

A fixed finite-ray character has its complete zero-extended presentation and ray group fixed in the preceding sense. A character with moving conductor remains moving data, including when its outer label is fixed for the current sum. Locally fixing a moving twist factor or redundant zero mask preserves the moving-support and coprimality restriction requirements; in particular, a Z-varying redundant mask remains outside the fixed arithmetic datum. The zero extension of $\chi_n(k)$ may vary naturally with k. Cancellation between that row factor and a fixed twist, however, does not permit a separately chosen coprimality restriction for each row. Externally chosen row-dependent coprimality restrictions and row-dependent column coefficients remain excluded. These support conditions govern every fourth-moment invocation in Section 17.

Uniformity. The common uniformity convention has the following additional clauses for these coefficient classes. Any required slot mesh depends only on the fixed real log-length ranges, the strict margins, and the specified positive power losses. Finite seminorm orders, polynomial height orders, implied constants, and lower thresholds may also depend on a specified fixed number of slots. These orders, constants, and thresholds are uniform over the moving moduli, the radicals included in M, the admissible coprimality restrictions, and the outer labels in their stated ranges, even when an outer label is fixed during one row sum.

The constant C in the definition of a divisor-bounded multiplicity may also depend on this fixed slot count. The independence of C from Z, the current rows, and the averaged labels, and any separate requirement that the multiplicity depend only on f, remain as in Section 3. The slot count is a separate fixed parameter, and moving labels remain outside $\mathcal{A}$.

12.2 Prime counting in a fixed ray class

Lemma 12.1 (Fixed-ray prime normalizer). *Let T be a fixed quotient of a ray class group of F, and write $p \in 1_T$ when the image of an unramified prime ideal is the identity. For a fixed nonnegative, nonzero smooth annular weight W,*

$$\sum_{p \in 1_T} W(q_p/P) q_p^{-5/6} \sim \frac{P^{1/6}}{|T| \log P} \int_0^\infty W(y) y^{-5/6}\, dy. \tag{12.3}$$

Deleting any further fixed finite set of primes does not change this asymptotic. Its lower threshold may depend on all the fixed data.

Proof. For the fixed abelian extension of F corresponding to T, the prime ideal theorem in a fixed Frobenius class gives

$$\pi_{1_T}(x) := \#\{p \in 1_T : q_p \le x\} \sim \frac{\mathrm{Li}(x)}{|T|}.$$

This is the fixed-extension consequence of Chebotarev in [26, Theorem 1.1]; the extension and its conductor are fixed as $x \to \infty$. If $\operatorname{supp} W \subset [a, b] \subset (0, \infty)$, the error $o(x/\log x)$ is uniform for $aP \le x \le bP$ as $P \to \infty$. Stieltjes integration by parts therefore changes the left side of (12.3) by $o(P^{1/6}/\log P)$ when $d\pi_{1_T}(x)$ is replaced by $dx/(|T| \log x)$. The resulting integral is

$$\frac{P^{1/6}}{|T| \log P} \int_a^b W(y) y^{-5/6} \frac{\log P}{\log(Py)}\, dy.$$

The last ratio tends uniformly to one. This proves the formula and its positivity. A fixed finite set is eventually outside the annular window. □

12.3 Full finite Fourier correlations

We now allow arbitrary prime powers in a modulus. The two-argument notation $G(a, k)$ below is a finite Fourier sum; it is distinct from the one-argument finite-ray function $G(a)$ of Lemma 3.5. Define

$$G(a, k) = q_a^{-1/2} \sum_{x \bmod a} \chi_a(x) e(kx/a), \qquad G(1, k) = 1. \tag{12.4}$$

All characters in this subsection have the zero extensions specified in Section 3, and we put $v_p(0) = +\infty$.

Lemma 12.2 (Prime-power Fourier sums). *Let p be the primary generator of a prime ideal outside S, with the sextic symbol and zero extension of Section 3.1. Let $k \in \mathcal{O}$, put $P = q_p$, and let G be the Fourier sum in (12.4). For every integer $a \ge 1$,*

$$\begin{aligned} |G(p^a, k)| &= P^{(a-1)/2} 1_{v_p(k) = a-1}, && 6 \nmid a, \\ G(p^a, k) &= P^{a/2} 1_{p^a \mid k} - P^{a/2-1} 1_{p^{a-1} \mid k}, && 6 \mid a. \end{aligned} \tag{12.5}$$

Proof. Write a residue modulo p^a as $x_0 + py$, with $x_0 \bmod p$ and $y \bmod p^{a-1}$. The sum over y is zero unless $p^{a-1} \mid k$, and equals P^{a-1} otherwise. In the latter case write $k = p^{a-1}k_0$. The remaining normalized sum is

$$P^{a/2-1} \sum_{x_0 \bmod p} \chi_p(x_0)^a e(k_0 x_0/p).$$

If $6 \nmid a$, its inner sum vanishes when $p \mid k_0$ and otherwise has absolute value $\sqrt{P}$. If $6 \mid a$, the inner sum is the sum of the additive character over the units, namely $P1_{p|k_0} - 1$. These are exactly the two cases in the statement. □

For primary moduli u, v outside S and $j \in \mathcal{O}$, define the full correlation by

$$F(u, v; j) = \sum_{\substack{x \bmod u,\ y \bmod v \\ vx - uy \equiv j \pmod{uv}}} \chi_u(x)\overline{\chi_v(y)}. \tag{12.6}$$

For Möbius inversion we will also need a bounded periodic extension to noncoprime residual columns. The local factors in Lemma 12.4 specify that extension.

Lemma 12.3 (Fourier transform and diagonal). *For the full correlation $F(u, v; j)$ in (12.6), with primary u, v outside S and $j \in \mathcal{O}$,*

$$F(u, v; j) = \frac{1}{\sqrt{q_u q_v}} \sum_{h \bmod uv} G(u, h)\overline{G(v, h)} e(-jh/(uv)).$$

At zero frequency, $F(u, v; 0) = 0$ unless $u = v$, and $F(u, u; 0) = \varphi(u)$, where $\varphi(u)$ is the number of units modulo u.

Proof. Expanding both Gauss sums in the displayed Fourier transform leaves

$$\frac{1}{q_u q_v} \sum_{x \bmod u, y \bmod v} \chi_u(x)\overline{\chi_v(y)} \sum_{h \bmod uv} e\big(h(vx - uy - j)/(uv)\big).$$

Additive orthogonality makes the inner sum $q_u q_v$ when the congruence holds and zero otherwise. If $j = 0$ and a summand is nonzero, x and y are units modulo u and v. Reducing the congruence modulo u gives $u \mid v$, and reducing modulo v gives $v \mid u$. Since the generators are primary, $u = v$. The congruence then says $x = y \bmod u$ and gives $\varphi(u)$. This argument uses the zero masks also when a local character power is principal. □

Lemma 12.4 (Common-factor correlation). *Let u, v be primary moduli outside S, let $j \in \mathcal{O}$, and let F be the correlation in (12.6). Put $C = (u, v)$ and write $u = Cn_1$, $v = Cn_2$, where $(n_1, n_2) = 1$. Then $F(u, v; j) = 0$ unless $C \mid j$. For $j = Ck$,*

$$\begin{aligned} F(Cn_1, Cn_2; Ck) &= \chi_{n_1}(k)\overline{\chi_{n_2}(-k)}\mathcal{R}(n_1, n_2) L_C(n_1, n_2; k), \\ L_C(n_1, n_2; k) &= \sum_{\substack{x, y \bmod C \\ n_2 x - n_1 y \equiv k \pmod C}} \chi_C(x)\overline{\chi_C(y)} = \prod_{p^c \| C} L_{p^c}. \end{aligned} \tag{12.7}$$

For $P = q_p$ and $p \nmid n_1 n_2$, the local factor is

$$L_{p^c} = P^{c-1}\chi_p(n_1/n_2)^c \begin{cases} P - 1, & p \mid k, \\ -1, & p \nmid k,\ 6 \nmid c, \\ P - 2, & p \nmid k,\ 6 \mid c. \end{cases} \tag{12.8}$$

If p divides exactly one of n_1, n_2, the local factor is $P^{c-1}(P-1)1_{6|c}1_{p\nmid k}$. Extend L_C to all residual pairs by the product of these local formulas, setting the local value to zero if $p \mid (n_1, n_2)$. The congruence sum on coprime residuals and this product extension both satisfy $|L_C| \le q_C$, and the extension is periodic modulo $\operatorname{rad} C$ *in each residual column.*

Remark. On $(n_1, n_2) = 1$, L_C denotes the congruence sum in (12.7). When used on all residual pairs, it denotes the product extension in Lemma 12.4. Outside the coprime residual locus this extension is not the original congruence sum.

Proof. Removing the common factor. The congruence implies $C \mid j$. After division by C, the congruence for $j = Ck$ is

$$n_2 x - n_1 y \equiv k \pmod{Cn_1n_2}.$$

Reduction modulo n_1 and n_2 gives, with zero values retained,

$$\chi_{n_1}(x) = \chi_{n_1}(k)\overline{\chi_{n_1}(n_2)}, \qquad \overline{\chi_{n_2}(y)} = \overline{\chi_{n_2}(-k)}\chi_{n_2}(n_1).$$

The product of the two unit factors is $\mathcal{R}(n_1, n_2)$. It remains to identify the multiplicity of lifts of the congruence modulo C. This can be checked at each prime. If $c = v_p(C)$ and neither residual modulus contains p, there is no additional lift. If, say, $d = v_p(n_1) > 0$ and $p \nmid n_2$, then $y \bmod p^c$ is free and the congruence uniquely determines $x \bmod p^{c+d}$ from it. Reduction modulo p^c is exactly the common congruence in (12.7). The case $p \mid n_2$ is symmetric, and this argument includes $c = 0$. The Chinese remainder theorem thus gives a bijection with the common solutions used in L_C, even when C meets one residual modulus.

Local factors. Put $k_p = \mathcal{O}/(p)$ and $A = \chi_p^c$, a character of $k_p^\times$ extended by zero. Since at least one of n_1, n_2 is a unit at p, each solution modulo p has P^{c-1} lifts modulo p^c. If both are units and $p \mid k$, the unit variables are proportional and the field sum is $A(n_1/n_2)(P-1)$. If $p \nmid n_1n_2k$, put $y = (k/n_1)t$ and $x = (k/n_2)(1+t)$. The field sum becomes

$$A(n_1/n_2) \sum_{t \ne 0,-1} A((1+t)/t) = A(n_1/n_2) \sum_{z \in k_p^\times \setminus \{1\}} A(z).$$

It is $-A(n_1/n_2)$ for nonprincipal A and $(P-2)A(n_1/n_2)$ for principal A. If $p \mid n_1$ and $p \nmid n_2$, the field equation forces $x = k/n_2$. For $p \mid k$ its character is zero. For $p \nmid k$ the remaining sum over y is zero unless A is principal, in which case it is $P-1$. The other case is symmetric. This proves all local formulas on the coprime residual locus. Extending them by the stipulated zero when both residuals meet p gives functions of the residual columns modulo p, each of absolute value at most P^c. Their product proves the bound and periodicity for the extension as well. The congruence-sum identity has been proved on the coprime residual locus. □

For some applications it is preferable to remove all primes common to the two full moduli, with their entire multiplicities. The next form leaves one common row character on each remaining product.

Lemma 12.5 (Complete-common-support correlation). *Suppose $u = Da$, $v = Eb$ are primary and outside S, with $(a, b) = 1$ and $(ab, DE) = 1$. Then, for every $j \in \mathcal{O}$,*

$$\begin{aligned} F(Da, Eb; j) = F(D, E; j)\mathcal{R}(a, E)\overline{\mathcal{R}(b, D)}\mathcal{R}(a, b) \\ \cdot \chi_a(j)\overline{\chi_b(-j)}. \end{aligned} \tag{12.9}$$

The moduli D, E may contain any prime powers, including shared auxiliary prime factors.

Proof. At primes of a and b, solving the congruence in (12.6) gives respectively $\chi_a(j)\overline{\chi_a(Eb)}$ and $\overline{\chi_b(-j)}\chi_b(Da)$. These statements remain valid when a character of j is zero; only the displayed unit factors are inverted. At primes of DE change variables $x' = bx \bmod D$, $y' = ay \bmod E$. The

congruence becomes $Ex' - Dy' \equiv j \pmod{DE}$ and its character factor changes by $\overline{\chi_D(b)}\chi_E(a)$. The remaining unit factor is

$$\frac{\chi_E(a)}{\chi_a(E)}\frac{\chi_b(D)}{\chi_D(b)}\frac{\chi_b(a)}{\chi_a(b)} = \mathcal{R}(a,E)\overline{\mathcal{R}(b,D)}\mathcal{R}(a,b).$$

All denominators here are symbols of units by the hypotheses. The Chinese remainder theorem and the definition of $F(D,E;j)$ complete the proof. □

The correlation identity (12.9) applies under both of its coprimality hypotheses. For Möbius inversion, extend its right side as follows. Fix D, E, j and define $\widetilde{F}_{D,E}(a,b;j)$ to be the right side of that equation for all primary a, b outside S with $(ab, DE) = 1$, retaining its zero symbols even when $(a,b) > 1$. For any finite coefficient array $c(a,b)$ on this locus,

$$\sum_{(a,b)=1} c(a,b)F(Da,Eb;j) = \sum_{a,b} c(a,b)\widetilde{F}_{D,E}(a,b;j)\sum_{t|a,\ t|b}\mu(t).$$

Indeed the full inner divisor sum is $1_{(a,b)=1}$, and the two correlation expressions agree on that locus. The value of $\widetilde{F}_{D,E}$ elsewhere is artificial, not a formula for $F(Da,Eb;j)$. In particular, the full divisor sum precedes every factorwise estimate that separates the two residual columns. Extra common primes introduced by an individual divisor term do not become primes of the original D, E correlation.

13 Marked completion and reflected row energy

The compensated low estimate and the inverse moment use the same marked completion. We establish its reflected mean square here. The low estimate will use the case with no additional squarefree label or coprimality restriction; the more general form allows the labels and masks introduced by the inverse moment's Poisson transformations.

We retain the notation of Section 3. In particular, ideal variables have their multiplicative primary generators, the fixed excluded set S contains the primes over 6, and every power of a residue symbol is zero on nonunits, even when its exponent is divisible by six. Element rows need not be squarefree or prime to S.

13.1 Completed sums and product-form marks

Fix a finite index set I and pairwise disjoint lists $\mathcal{P}_i$ of primes outside S, with $q_p \asymp Z^{z_i}$ for $p \in \mathcal{P}_i$. The lists and their bounded individual coefficients $a_i(p)$ are independent of the current row and squarefree label. Their nominal lengths z_i lie in fixed bounded ranges.

Here and below a fixed coprimality restriction means a function

$$\rho(n) = 1_{(n,r_\rho)=1},$$

where r_ρ is squarefree and is required to be fixed only within the indicated current row and label sums. A bounded product of such functions has this form with r_ρ equal to the radical of the product of their moduli. The norm of this radical will be bounded explicitly. It may depend on previously fixed outer ideals, but not on either current averaging variable.

Let f be a squarefree ideal outside S, with $q_f \asymp Z^V$ for a nonnegative V in a fixed bounded range, and let $k \in \mathcal{O}$. For a fixed multiplicative finite ray character ν whose full zero-extended defining modulus has prime support in S, define $a(n)$ only for squarefree primary n outside S, and define $\Psi_k(n)$ for every primary n outside S by

$$a(n) = \overline{\alpha}(n)\gamma_2(n)\nu(n)\rho(n), \qquad \Psi_k(n) = \nu(n)\rho(n)\chi_n(k)\chi_n(f)^4.$$

The finite character, coprimality restriction, and all slot lists are independent of k, f. A fixed function on a finite ray group is always expanded into group characters before this definition is used; thus ν is multiplicative even when an earlier step produced a finite linear combination of characters.

For a subcollection $I' \subset I$ of disjoint prime lists with bounded coefficients, define its mark by

$$\mathfrak{d}_{I'}(A) = \sum_{(p_i)\in\prod_{i\in I'}\mathcal{P}_i} \prod_{i\in I'} a_i(p_i) 1_{p_i\mid A}. \tag{13.1}$$

We omit I' from the notation when it is understood. Because the lists are disjoint, a tuple has a squarefree product. For every fixed $\epsilon > 0$, the number of tuples dividing A is $O_\epsilon(q_A^\epsilon)$. Assigning slots to an extracted factor uses the product form of the coefficient in (13.1), in addition to its pointwise divisor bound.

For a smooth annular W, put $V_*(y) = y^{1/2}W(y)$ and define the completed marked sum

$$\mathcal{T}_{\mathfrak{d}}(X;k) = \sum_{\substack{n\\ b\ \mathrm{sf}}} \frac{\overline{\alpha}(n)\gamma_2(n)\Psi_k(n)\,\overline{\alpha}(b)^3\Psi_k(b)^3}{\sqrt{q_n}\,q_b} \mathfrak{d}(nb^3) V_*(q_n q_b^3/X). \tag{13.2}$$

Both primal ideals avoid S; they may share primes. The mark belongs to the whole index nb^3. With $\mathfrak{d} = 1$, this is the Mellin completion on the left of (4.2). The Gaussian test of Lemma 3.11 is also permitted when only the completed estimate is used.

For the reflection formula we use Lemma 4.4, only for a nonzero row. Write $k = uk_S k_{\mathrm{good}}$ as in that Lemma. After fixing the unit, the S-valuations modulo six, and the good row ray class, its literal sector character is

$$\Psi_0^{[k]}(A) = \begin{cases} \nu(A)\chi_A(uk_S)\mathcal{R}(A, k_{\mathrm{good}}), & A \equiv 1 \pmod 3,\ (A,S) = 1, \\ 0, & \text{otherwise.} \end{cases}$$

The zero branch is evaluated before either character. The Lemma supplies one finite family and hence one common L, with prime support S, before the good primes vary. Write $M_{\mathrm{ref}} = \lambda^{12}L^4$; the full reflection sector modulus is M_{ref}^2.

All good primes of k, f, r_ρ remain in the local prime set, with their exponents combined modulo six and every zero retained. The fourth power at f has no extra reciprocity sign, since $\mathcal{R}(A, f)^4 = 1$. The coprimality restriction is the product of the local zero powers at its good primes. Thus the local presentation gives exactly $\Psi_k(n)\Psi_k(b)^3$ on the stated primal support, including at shared good primes; no moving good prime enters L.

13.2 Whole-index marked reflection

The product-form marks can be inserted by finite inclusion–exclusion in the reflection from Part I. Each resulting branch has its own active radical: an inactive marked prime is absent from its conductor. The local factors must therefore be assembled with that branch's conductor and kernel scale.

Fix an invocation of Proposition 4.1, with base local prime set P_0, powers j_p, fixed multiplier ϕ, and test V. Put $M_{\mathrm{ref}} = \lambda^{12}L^4$ for this invocation. Let Q be a finite set of distinct good primes disjoint from P_0. In the direct completed coefficient sum, insert the factor $\prod_{q\in Q} 1_{q\mid nb^3}$; denote this sum by $\mathcal{T}_Q(X;\Psi,V)$. It has the same normalization as (13.2), with Ψ_k, V_* replaced by Ψ, V. The primal squarefree and cube ideals may share primes.

A branch $\mathcal{B}$ specifies h_0 mod L, the base active and inactive sets $\mathcal{A}_0, \mathcal{I}_0$, and a partition $Q = \mathcal{A}_Q \sqcup \mathcal{I}_Q$. All base primes with $j_p \neq 0$ are active. Put

$$r = \prod_{p\in\mathcal{A}_0\cup\mathcal{A}_Q} p, \qquad c = c_F(h_0)r.$$

The base factors B_p and the canonical functions d, ϑ are those of Proposition 4.2 for this active radical. Every $\sigma_p, \epsilon_p, \omega_{p,j}$ is computed using this branch's whole c.

Corollary 13.1 (Completed reflection with whole-index marks). *For the completed sum $\mathcal{T}_Q(X;\Psi,V)$ obtained by inserting whole-index marks at distinct good primes Q disjoint from the base local set P_0 of Proposition 4.1, use the branch data in Section 13.2. Then*

$$\begin{aligned}\mathcal{T}_Q(X;\Psi,V) = \sum_{\mathcal{B}} \zeta_Q \sum_{0\neq\mu\in\lambda^{-4}\mathcal{O}} \frac{d(\mu)\alpha(\mu)\vartheta(\lambda^4\mu)}{\sqrt{q_\mu}} \prod_{p\in\mathcal{A}_0} B_p(\lambda^4\mu) \\ \cdot \prod_{q\in\mathcal{A}_Q} q_q^{-1/2}\chi_q(\lambda^4\mu)^{-2} V^\sharp\left(\frac{q_\mu X}{q_c^2}\right).\end{aligned} \tag{13.3}$$

The branch scalar is

$$\begin{aligned}\zeta_Q = -\frac{i}{81}\overline{\alpha}(c)^2\widehat{\phi}(h_0)\overline{\kappa}_F \prod_{p\in\mathcal{I}_0}(1-q_p^{-1}) \prod_{q\in\mathcal{I}_Q} q_q^{-1} \\ \cdot \prod_{p\in\mathcal{A}_0} \chi_p(\sigma_p)^{-2}\omega_{p,j_p} \prod_{q\in\mathcal{A}_Q} \chi_q(\sigma_q)^{-2}(-\omega_{q,0}).\end{aligned} \tag{13.4}$$

In particular $|\zeta_Q| \le 1/81$. The functions d, ϑ and the scalar κ_F have the dependence on h_0 and $r \bmod M_{\mathrm{ref}}^2$ specified in Proposition 4.2(3). Every dual sum is absolutely convergent. Its coefficient d has the support and size in (4.3): a supported index has the form $\mu = u\lambda^k nb^3$, where u is an Eisenstein unit, $k \ge -4$, $n, b \equiv 1 \pmod 3$, and n is squarefree. The ideals n, b may share primes, and there is no additional exclusion at primes of L beyond the displayed restriction at λ. All local factors in (13.3) retain their zero extensions. If a marked prime instead belongs to the base local set, the corresponding primal term is identically zero and is removed before this formula is used.

Proof. For every good prime and every $A \in \mathcal{O}$, the zero convention gives the pointwise identity $1_{q|A} = 1 - \chi_q(A)^0$. Expanding the product over Q writes the marked sum as finitely many completed sums to which Proposition 4.1 applies. For a fixed h_0 and active radical, Proposition 4.2 uses the same $c_F, c, d, \vartheta, \kappa_F$, other local phases, and kernel whether an absent prime is omitted from the local set or included with exponent zero in its inactive branch. Thus the two inactive contributions at q combine with coefficient $1-(1-q_q^{-1}) = q_q^{-1}$. The active contribution comes only from the subtracted zero-mask transform and has local factor $q_q^{-1/2}\chi_q(\lambda^4\mu)^{-2}$ and scalar $-\omega_{q,0} = \tau_{q,2}^+\chi_q(\epsilon_q)^{-2}$. Collecting these choices at all marked primes proves (13.3) and (13.4). The inactive prime is absent from r, hence from that branch's conductor and kernel scale. Thus finite inclusion–exclusion combines precisely the branches with compatible active radicals.

If a marked prime is a base local prime, the mark forces it to divide nb^3, while the zero-extended base factor at that prime then vanishes, including for exponent zero. This proves the last assertion without any division of a zero symbol. It also covers the case where the primal squarefree and cube ideals share the marked prime. □

To apply a row norm, we next separate the phases that depend on the residual row from those that depend on the marked tuple. For distinct active good primes p, q, (4.14) and cubic reciprocity give

$$\chi_p(q)^{2j_p+2}\chi_q(p)^{2j_q+2} = \left(\frac{q}{p}\right)_3^{j_p+j_q+2}. \tag{13.5}$$

For an active mark the exponent is $j_q = 0$; its changed sign occurs only in its one-prime scalar. A residual row prime with $j_p = 1$ and a marked prime therefore have pair factor $(q/p)_3^3 = 1$. A $j_p = 1$ row prime and a moving $j_q = 4$ prime would instead leave $(q/p)_3$, which need not be one.

All nonresidual base primes, especially every $j = 4$ prime, are therefore fixed before the inner row norm.

Write R for the product of the residual $j = 1$ row primes and P for the product of active marked primes. Primal row-mark collisions are removed first by the last assertion of the corollary, and are subsequently represented by $1_{(P,R)=1}$. If the fixed active base product is F_{act}, then $r = F_{\mathrm{act}}RP$. Fix the class of R and the class of P separately modulo the full M_{ref}^2. These separate full-modulus classes, which contain more information than their product or the smaller reciprocity classes, fix the cusp data without adding a pair-dependent restriction. Row-row phases are row scalars, mark-mark phases are tuple scalars, and their interactions with fixed primes depend on only one of those sets. The angular conductor scalar factors in the same way.

After fixing the dual unit, its λ-valuation, and extracted fixed factors, write the remaining dual part as nb^3. The moving columns are exactly

$$\chi_R(nb^3)^3 = \chi_R(nb)^3, \qquad \chi_P(nb^3)^{-2} = \chi_P(n)^{-2}1_{(P,b)=1}.$$

These identities include every zero. A zero caused by a row or a tuple meeting an extracted fixed factor is a fixed restriction on that row or tuple. The source coefficient, fixed additive factor, and all fixed local column factors are functions of the full extracted dual index alone. Hence, after common smooth separation, the tuple coefficient may be a general bounded function of the tuple independent of R, n, b, while the dual coefficient is independent of R, P. The earlier product form of the marks remains required when slots are assigned to extracted factors.

13.3 A quadratic–cubic norm estimate

Part I established the required orientation of Goldmakher–Louvel's quadratic large sieve and proved the zero-preserving completed reduction in Lemma 4.8. We add Heath–Brown's cubic large sieve:

$$\sum_{\substack{a\equiv 1\ (3)\\ q_a\le U}}^{*}\left|\sum_{\substack{b\equiv 1\ (3)\\ q_b\le V}}^{*} c_b\left(\frac{b}{a}\right)_3\right|^2 \ll_\epsilon (UV)^\epsilon\{U+V+(UV)^{2/3}\}\sum_b^{*}|c_b|^2, \tag{13.6}$$

where both stars mean squarefree in $\mathcal{O}$, and the coefficients are arbitrary complex numbers [16, Theorem 2]. The indices need not have squarefree rational norm, and there is no exclusion of rational prime factors. Conjugation and cubic reciprocity allow the opposite orientation. For this sieve and the quadratic sieve in (4.26), a fixed restriction on the row set decreases the positive outer sum. A fixed restriction on the coefficient support is implemented by setting the omitted coefficients to zero and applying the same theorem. These two operations concern fixed restrictions; the next lemma treats its two pair-dependent masks separately.

Lemma 13.2 (A quadratic–cubic norm bound). *Let $K, N, B, L \ge 1$. Let k, n, P be squarefree primary ideals with $q_k \ll K$, $q_n \asymp N$, and $q_P \asymp L$; let b be any primary ideal with $q_b \asymp B$. The ideals k, P avoid the current excluded set S; additional fixed exclusions are allowed. The source ideals n, b may contain primes of S other than λ. Let $a(P)$ and $\beta(n,b)$ satisfy*

$$|a(P)| \le 1, \qquad |\beta(n,b)| \le 1.$$

Here a is independent of k, n, b, and β is independent of k, P. Then

$$\sum_k\left|\sum_P \frac{a(P)}{\sqrt{q_P}}1_{(P,k)=1}\sum_{n,b}\beta(n,b)\chi_k(nb)^3\chi_P(n)^{-2}1_{(P,b)=1}\right|^2 \ll_\epsilon (KNBL)^\epsilon(K+NB)B\{N+L+(NL)^{2/3}\}. \tag{13.7}$$

Fixed restrictions on the k-set and fixed restrictions on the (n,b)- or P-supports are allowed, as are fixed ray sectors. All such restrictions must preserve the two coefficient-independence conditions above. Divisor-bounded multiplicities may be included provided they preserve those conditions.

Proof. We first expand the row–mark coprimality mask, then apply the completed quadratic reduction to the row variable. This leaves a positive sum to which the cubic sieve applies. The final divisor sums recover the stated bound. Multiplicities that do not preserve the statement's independence conditions are first removed by the divisor Cauchy inequalities below.

The row–mark mask. Write $\mathcal{H}$ for the left side of (13.7). The complete mask has the expansion

$$1_{(P,k)=1} = \sum_{D \mid (P,k)} \mu(D).$$

For each row the number of such divisors is at most $d_{\mathcal{O}}(k)$. Rowwise divisor Cauchy, followed by the positive sum over rows, therefore introduces a permitted factor $(KNBL)^{\epsilon}$. In a fixed D-summand write $P = DP'$ and $k = Dk'$. Squarefreeness gives the fixed restrictions $(D, P') = (D, k') = 1$. Define

$$a_D(P') = a(DP')1_{(D,P')=1},$$
$$\beta_D(n,b) = \beta(n,b)\chi_D(nb)^3\chi_D(n)^{-2}1_{(D,b)=1}.$$

All these factors retain their zeros and $|\beta_D| \le 1$. Factoring $q_P^{-1/2} = q_D^{-1/2}q_{P'}^{-1/2}$ gives

$$\mathcal{H} \ll (KNBL)^{\epsilon} \sum_D \frac{1}{q_D} \sum_{k'} \left| \sum_{n,b} \beta_D(n,b)\chi_{k'}(nb)^3 \sum_{P'} \frac{a_D(P')}{\sqrt{q_{P'}}} \chi_{P'}(n)^{-2}1_{(P',b)=1} \right|^2. \tag{13.8}$$

Here $q_{k'} \ll K/q_D$, $q_{P'} \asymp L/q_D$, and all inherited fixed restrictions are retained. The outer D's are the squarefree good divisors allowed by the split, so $q_D \ll \min(K,L)$. In particular, no (P',k') mask is reinstated inside an individual D-summand. Its cancellation belongs to the complete Möbius sum already bounded by rowwise Cauchy.

The quadratic reduction. For fixed D, the product of $\beta_D(n,b)$ and the inner P'-sum is an arbitrary complex coefficient independent of k'. Thus (4.27) applies with row length K/q_D. In its notation,

$$b = gt^2, \quad c = (n,g), \quad n = cm, \quad g = ch, \quad q_c \asymp C, \quad q_g \asymp G, \quad q_t \asymp T,$$

so $B \asymp GT^2$, and write $c = rc'$ for its complete c-mask divisor. The ideals c, m, h are pairwise coprime and squarefree; no coprimality between g and t is imposed. The cited reduction already includes the natural k'-t restriction and the complete Möbius expansion of the k'-c mask. It does not reinstate a mask between the remaining row and c'.

For one C, G, T block its positive output, inserted in (13.8), is at most

$$\begin{aligned}(KNBL)^{\epsilon} \sum_D \frac{1}{q_D} T \sum_{q_t \asymp T} \sum_{\substack{r \text{ sf},\ (r,S)=1 \\ q_r \ll \min(K/q_D, C)}} \left\{ \frac{K}{q_D q_r} + \frac{NG}{C^2} \right\} \frac{C}{q_r} \sum_{c',m,h} |\beta_D(rc'm, rc'ht^2)|^2 \\ \cdot \left| \sum_{P'} \frac{a_{D,r}(P')}{\sqrt{q_{P'}}} \chi_{P'}(c'm)^{-2}1_{(P',ht)=1} \right|^2,\end{aligned} \tag{13.9}$$

where the positive c', m, h-sum keeps the original support and

$$a_{D,r}(P') = a_D(P')\chi_{P'}(r)^{-2}.$$

The factorization used here is, including all zeros,

$$\chi_{P'}(rc'm)^{-2}1_{(P',rc'ht^2)=1} = \chi_{P'}(r)^{-2}\chi_{P'}(c'm)^{-2}1_{(P',ht)=1}.$$

The character already supplies the missing zeros at r, c', and m. Thus the r-factor is tuple-only, including its zero when $(P', r) \neq 1$. The fixed D-phases remain inside β_D until this positive bound. The quadratic reduction has already separated the finitely many squarefree S-parts of m, h before sieving their good product; m, h in (13.9) are the original ideals.

The cubic sieve. The positive bound now permits us to discard $|\beta_D|^2 \leq 1$. For fixed h, t, D, r, the P'-coefficients are independent of $j = c'm$, and ideal counting gives

$$\sum_{P'} \frac{|a_{D,r}(P')|^2 1_{(P',ht)=1}}{q_{P'}} \ll_\epsilon (KNBL)^\epsilon.$$

The product $j = c'm = n/r$ is squarefree primary and has norm $O(N/q_r)$. It may contain a permitted prime of S other than λ, since only the quadratic column was stripped of its fixed S-part. Grouping c', m by j introduces at most a divisor factor. Only this positive squarefree j-range is enlarged. Cubic reciprocity, conjugation, and (13.6), followed by the $O(G/C)$ choices of h, give, on $q_D \asymp D_0$, $q_r \asymp r_0$,

$$(KNBL)^\epsilon \frac{G}{C}\left\{\frac{N}{r_0} + \frac{L}{D_0} + \left(\frac{N}{r_0}\frac{L}{D_0}\right)^{2/3}\right\}. \tag{13.10}$$

The fixed mask at ht is a coefficient restriction in this cubic sieve, not a row-dependent mask.

Summing the dyads. The reduction supplies $T\sum_t$, with $O(T)$ possible t, and the factor C/q_r. Together with the h-count, these contribute

$$T^2(C/r_0)(G/C) \asymp B/r_0.$$

There are $O(r_0)$ possible r in its dyad, while the sum of q_D^{-1} over its dyad is $O(1)$, up to a permitted small power. Both sieve factors increase when D_0, r_0 are replaced by 1, and $NG/C^2 \ll NB$. Summing the logarithmically many dyads proves (13.7). Subunit quotient ranges are empty, and bounded ranges are included by changing the fixed annular constants. □

13.4 The reflected energy bound

We next quantify the reflection of (13.2). The bound separates the contributions of the residual row length, the extracted local factors, and the dual scale.

The block data have three roles: the row decomposition determines H and O, the extracted local factors determine A_0, S_0, N_0, B_0, and the dual coordinates determine v, ℓ_b, e_λ. The resulting exponent is given in (13.15).

Row decomposition. Fix f, ρ and put $Q = \log_Z q_{r_\rho}$. For an element row $0 < q_k \ll Z^M$, define the maximal powerful part of its ideal by

$$k_{\mathrm{pow}} = \prod_{v_p(k)\geq 2} p^{v_p(k)}.$$

Split the valuation-one primes into the squarefree product k_{sup} supported on $S \cup \mathrm{supp}(fr_\rho)$ and the squarefree product k_{res} outside that set. These three prime supports are pairwise disjoint, and their product is the ideal of k. A powerful ideal here means that every positive prime valuation is at least two. Fix the unit of k. Smooth dyadic partitions give the ranges

$$q_{k_{\mathrm{pow}}} \asymp Z^O, \qquad q_{k_{\mathrm{res}}} \asymp Z^H.$$

Choose these ideal centers nonnegative, with center zero for the unit dyad. In particular, an actual annular factor $\chi_{\mathrm{row}}(q_{k_{\mathrm{res}}}/Z^H)$ is kept in the row sum. The bounded dyad includes $k_{\mathrm{res}} = 1$. Choose fixed constants $C_k, C_O, C_H \geq 1$, from these supports, such that

$$q_k \leq C_k Z^M, \qquad q_{k_{\mathrm{pow}}} \geq Z^O/C_O, \qquad q_{k_{\mathrm{res}}} \geq Z^H/C_H.$$

Since $q_{k_{\mathrm{sup}}} \geq 1$, the general norm inequality (4.35) gives here

$$H \leq M - O + \frac{\log C_{\mathrm{res}}}{\log Z}, \qquad C_{\mathrm{res}} = C_H C_k C_O.$$

The inequality need not be an equality, because k_{sup} may have positive norm length.

Local factors. Fix the actual ideals $k_{\mathrm{pow}}, k_{\mathrm{sup}}$. At every prime outside S, combine the local exponents of $\rho(n)\chi_n(k)\chi_n(f)^4$ modulo six, always retaining the zero extension. The primes of k_{res} have exponent $j = 1$. All other such primes, including every non-slot prime of exponent $j = 4$, are now fixed. In the reflection formula call a prime active when it occurs in c. For a fixed choice of the active zero-exponent primes and a fixed splitting of each $j = 4$ Ramanujan factor, let:

A_0 be the log-norm of all active non-slot primes outside k_{res},

S_0 be the log-norm of the small Ramanujan terms and active zero-mask terms among them,

N_0 be the log-norm of the $j = 4$ divisibility terms assigned to the squarefree dual ideal,

B_0 be the log-norm of the $j = 4$ divisibility terms assigned to the cube ideal but not the squarefree ideal.

Here a log-norm is $\log_Z$ of the norm of the indicated product. The assignment uses $1_{p|nb^3} = 1_{p|n} + 1_{p\nmid n}1_{p|b}$. Let $z_a = \sum_{i \text{ active}} z_i$ be the sum of their nominal slot lengths. The product of the active primes has norm divided by Z^{z_a} in a fixed compact interval, including for zero nominal lengths or the empty list.

Dual coordinates. After extracting the forced squarefree and cube primes, restrict the remaining dual ideals to

$$q_n \asymp Z^v, \qquad q_b \asymp Z^{\ell_b}, \qquad q_{\lambda^{h_\lambda}} = Z^{e_\lambda}.$$

Choose the ideal centers v, ℓ_b nonnegative. These are source dual ideals: they retain every prime of S permitted by (4.3); no extra S-mask is placed on them. Only the displayed λ-valuation support is imposed. The unit and the integer $h_\lambda \geq -4$ are fixed in this dyad.

For these choices, let $\mathcal{U}_{v,\ell_b,e_\lambda}(k_{\mathrm{res}})$ denote the right side of (13.3), summed with the product-form tuple coefficients. It retains the specified local choices and dual restrictions, together with the actual factor $\chi_{\mathrm{row}}(q_{k_{\mathrm{res}}}/Z^H)$. Active slot tuples are still summed. This definition is made separately for each fixed powerful and supported row part and each sector obtained by fixing separate classes of the residual row product and active slot product modulo the full reflection modulus M_{ref}^2.

Fix a small $\tau_{\mathrm{ref}} > 0$ and write $X = Z^{N_*}$. Define the available dual length T_d and the retained dyads by

$$\begin{aligned} T_d &= 2H + 2A_0 + 2z_a - N_* - N_0 - 3B_0, \\ v + 3\ell_b + e_\lambda &\leq T_d + \tau_{\mathrm{ref}} \qquad \text{for a retained dyad.} \end{aligned} \tag{13.11}$$

The fixed-factor extraction in (4.32) supplies the common central scalar

$$c_{\mathrm{cent}} = Z^{-v/2-\ell_b-e_\lambda/3-(S_0+B_0)/2}. \tag{13.12}$$

The factors counted by N_0, B_0 have been extracted, while n, b are the remaining dual ideals. Their normalized inverse-root factors stay in the common smooth profile. An active tuple with

product $P = \prod_{i \text{ active}} p_i$ carries the additional factor $q_P^{-1/2}$ from (13.3), before the kernel factor is inserted. The power of Z in the bound for this scalar times the tuple factor is

$$-v/2 - \ell_b - e_\lambda/3 - (S_0 + B_0 + z_a)/2. \tag{13.13}$$

The quadratic–cubic saving and the resulting aggregate exponent are

$$u = \min\{v, z_a, (v + z_a)/3\}, \tag{13.14}$$

and

$$\begin{aligned} E_{\mathrm{ref}} = O/2 + \max(H, v + \ell_b) - S_0 - B_0 + z_a - u - \ell_b - 2e_\lambda/3 \\ - \tfrac{1}{2}(T_d - v - 3\ell_b - e_\lambda)_+. \end{aligned} \tag{13.15}$$

Lemma 13.3 (Reflected energy). *Let $Z \geq 2$, and suppose the reflected block log-lengths in Section 13.4 range over fixed bounded sets. Let $\mathcal{T}_{\mathfrak{d}}(Z^{N_*}; k)$ be the completed sum in (13.2), with residual row contribution $\mathcal{U}_{v,\ell_b,e_\lambda}$ defined by that block decomposition. The fixed character, coprimality restriction, and slot coefficients are independent of k, f, the slot supports are disjoint, and the slot coefficients are products as in (13.1). The fixed row restrictions after fixing $k_{\mathrm{pow}}, k_{\mathrm{sup}}$ must be independent of the active slots and dual variables, apart from the explicit zero mask $1_{(P,k_{\mathrm{res}})=1}$. Let T_d, u, E_{ref} be the dual-length bound, quadratic–cubic saving, and aggregate exponent in (13.11), (13.14), and (13.15).*

For every $D > 0$, terms with

$$v + 3\ell_b + e_\lambda > T_d + \tau_{\mathrm{ref}}$$

have total size $O(Z^{-D})$, with fixed polynomial dependence on height, whenever Z exceeds a threshold depending only on the fixed support intervals and τ_{ref}. For every retained dyad and active tuple on its original slot support, the central scalar in (13.12) satisfies $|c_{\mathrm{cent}}| q_P^{-1/2} \ll Z^{-v/2-\ell_b-e_\lambda/3-(S_0+B_0+z_a)/2}$, with the implied constant depending only on the fixed slot support intervals. The normalized inverse-root factors and the smooth tests remain in the common profile with kernel argument given by (13.16); its smooth dependence is through the profile seminorms p_j in (3.10) and the Mellin seminorms $\mathfrak{M}_{A,B}$ controlled in (4.19), with the common-profile bounds of (4.23).

For every $\epsilon > 0$, the contribution with fixed $k_{\mathrm{pow}}, k_{\mathrm{sup}}$ satisfies

$$\sum_{k_{\mathrm{res}}\ \mathrm{sf}} |\mathcal{U}_{v,\ell_b,e_\lambda}(k_{\mathrm{res}})|^2 \ll Z^{E_{\mathrm{ref}} - O/2 + \epsilon}.$$

Summing the fixed parts in their O-dyad and the local choices changes this to $O(Z^{E_{\mathrm{ref}}+\epsilon})$. For each prescribed D and ϵ, the finitely many seminorm orders and the polynomial orders in norm-twist heights may depend on the fixed arithmetic data, the fixed number of slots, the bounded log-length ranges, the fixed support intervals, and τ_{ref}. Their norm-twist dependence is the one in Lemma 4.3 and (4.23). These orders and the implied constants are uniform in the moving ideals and coprimality restrictions in the stated ranges. The negative value of e_λ, when present, is $O(1/\log Z)$. The residual-row dyads in this assertion are those of the original reflected sum: they retain their original annular supports and satisfy (4.35).

Proof. The proof has three quantitative inputs: the coefficient extracted from the reflection formula, the amplitude of its kernel on the actual annuli, and the quadratic–cubic norm estimate. We keep active tuples in the row norm throughout and sum the fixed row parts at the end.

Inactive slots and fixed data. The triangle inequality in the row Hilbert space removes the inactive slots. At such a slot the absolute coefficient mass is

$$\sum_{p \in \mathcal{P}_i} q_p^{-1} \ll_\epsilon Z^\epsilon,$$

uniformly under any fixed restriction, including for a bounded or zero-length list. It therefore suffices to prove a uniform bound with those primes fixed. Each such branch has its own conductor, cusp sector, and kernel scale, with the inactive prime absent from the conductor. Its earlier collision exclusion is now a fixed row restriction.

Fix $f, \rho, k_{\mathrm{pow}}, k_{\mathrm{sup}}$, the non-slot local choices, h_0, and separate classes of the residual row product and active slot product modulo the full M_{ref}^2. Also fix the unit and ramified exponent of the dual index. All non-slot $j = 4$ labels are fixed. Inside the original sum over active tuples, we use the algebraic fixed-base extraction in (4.32). Corollary 13.1 and the subsequent phase separation make that extraction simultaneous across the tuples: the base coefficient is common to the residual rows and tuples after their separate sectors are fixed, while the remaining scalar separates into a row factor and a tuple factor. The active tuples remain together in the row norm.

The reflected coefficient. The structural extraction gives the common central coefficient

$$Z^{-v/2-\ell_b-e_\lambda/3-(S_0+B_0)/2}$$

times normalized inverse-root factors and bounded coefficients; the displayed scalar is c_{cent}. Its N_0-assignment extracts a forced $j = 4$ prime only from the squarefree dual ideal, even if the cube ideal shares it; the entire shared cube part remains in its residual norm and coefficient. Its B_0-assignment extracts one occurrence from the cube ideal and leaves the factor $q_p^{-1/2}$. These are statements about the full source support, including the permitted primes of S. Every active marked column in (13.3) contributes $q_p^{-1/2}$. Together with c_{cent}, these factors give $c_{\mathrm{cent}} q_P^{-1/2}$ for each active tuple. Since q_P/Z^{z_a} lies in a fixed compact interval, its absolute value is bounded by the stated power of Z, whose exponent is (13.13).

The kernel on the actual annuli. Let $F_{\mathrm{act}}, N_F, B_F$ be the actual fixed products with log-norms A_0, N_0, B_0. Put $R = k_{\mathrm{res}}$, $P = \prod_{i\ \mathrm{active}} p_i$. The branch denominator and a supported dual index are

$$c = c_F F_{\mathrm{act}} RP, \qquad \mu = u\lambda^{h_\lambda} N_F n (B_F b)^3.$$

With the fixed block centers, define

$$y_R = q_R/Z^H, \qquad y_n = q_n/Z^v, \qquad y_b = q_b/Z^{\ell_b}, \qquad y_i = q_{p_i}/Z^{z_i},$$

$$Y = \frac{q_{\lambda^{h_\lambda} N_F B_F^3} X Z^v Z^{3\ell_b}}{q_{c_F^2 F_{\mathrm{act}}^2} Z^{2H} \prod_{i\ \mathrm{active}} Z^{2z_i}} = q_{c_F}^{-2} Z^{v+3\ell_b+e_\lambda-T_d}.$$

Multiplicativity gives the exact identity

$$\frac{q_\mu X}{q_c^2} = Y y_n y_b^3 y_R^{-2} \prod_{i\ \mathrm{active}} y_i^{-2}. \tag{13.16}$$

It remains true if N_F shares primes with b. All fixed factors in Y are their actual products, and every moving norm remains a coordinate. The actual row cutoff and the individual dual and slot cutoffs therefore place the kernel argument in

$$[C_{\mathrm{ker}}^{-1}, C_{\mathrm{ker}}] Z^{v+3\ell_b+e_\lambda-T_d}$$

for one $C_{\mathrm{ker}} \geq 1$ depending only on the fixed support intervals and the common fixed modulus. In particular this comparison has not used any shorter row added by positivity.

Put $D_{\mathrm{ker}} = T_d - v - 3\ell_b - e_\lambda$. On this full product of actual annuli, Lemma 4.5 permits

$$m_{\mathrm{ker}} = Z^{-(D_{\mathrm{ker}})_+/4}$$

to be factored from the entire homogeneous linear row vector, at a factor at most $C_{\mathrm{ker}}^{1/4}$ in its fixed seminorm constants. The normalized profile, including the inverse roots from the structural

extraction and every other common smooth norm window, has bounded fixed Euler seminorms. In the homogeneous estimate of (4.24), the scalar m_{ker} multiplies the one common Fourier density before the row norm is squared.

The tail is removed on the original, unseparated sums. A sufficiently far shift of the kernel contour gives the decay in Lemma 4.3, to which the lattice tail bound applies. If $\log Z \geq 2 \log C_{\mathrm{ker}}/\tau_{\mathrm{ref}}$, every whole dyad with center excess greater than τ_{ref} has actual kernel argument greater than $Z^{\tau_{\mathrm{ref}}/2}$. These dyads satisfy the tail condition of Lemma 4.6 with lattice $\lambda^{-4}\mathcal{O}$, $a = X/q_c^2$, and $U = Z^{\tau_{\mathrm{ref}}/2}$. On the full source support, the coefficient bound used in Part I gives

$$|d(\mu)|/\sqrt{q_\mu} \leq 27 q_\lambda^{4/3}.$$

Each base or marked local factor is at most $q_p^{1/2}$ after its indicator is discarded. The bounded row, tuple, and local log-length ranges, the local branch counts, and $1 + a^{-1}$ are bounded by a fixed power $Z^{B_{\mathrm{tail}}}$, with B_{tail} chosen independently of the kernel order. For any desired saving $D > 0$, the choice

$$A > 1 + \frac{2(B_{\mathrm{tail}} + D)}{\tau_{\mathrm{ref}}}$$

in that shell lemma makes the absolute total of these whole dyads $O(Z^{-D})$ times a finite test seminorm. Norm twists have the fixed polynomial height dependence of that seminorm. The shell sum counts all discarded dual indices, so no count at a retained dual length is used for this tail. It is removed before Fourier absolutization.

Separating the profile and applying the norm bound. For a retained dyad, the common-profile lemma and (4.24) separate the joint profile in (13.16). The coefficient density is common to the entire current row Hilbert space, including the active-tuple sum: the actual row, dual, and individual slot norms are its coordinates, not parameters of separately chosen measures. The full fixed support boxes govern every invoked weighted Fourier norm and its height dependence. At a separated mode the norm powers have absolute value one and preserve the row, tuple, and dual coefficient independences. Only inside this separated nonnegative row norm may the row set be enlarged from the original annulus to $q_R \ll Z^H$, using the same separated formula on the added rows. The kernel is never evaluated there.

After extracting the common coefficient but before using m_{ker}^2, the separated squared norm is bounded by

$$Z^{-v-2\ell_b-2e_\lambda/3-S_0-B_0+\epsilon}\mathcal{N},$$

where

$$\mathcal{N} = \sum_{\substack{k_{\mathrm{res}}\ \mathrm{sf} \\ q_{k_{\mathrm{res}}} \ll Z^H}} \left| \sum_{\boldsymbol{p}} \frac{a(\boldsymbol{p})}{\sqrt{q_P}} 1_{(P,k_{\mathrm{res}})=1} \sum_{\substack{n \\ b}}^{\mathrm{sf}} \beta(n,b) \chi_{k_{\mathrm{res}}}(nb)^3 \chi_P(n)^{-2} 1_{(P,b)=1} \right|^2, \tag{13.17}$$

$$P = \prod_{i\ \mathrm{active}} p_i, \qquad q_P \asymp Z^{z_a}.$$

The ideals n, b retain their stated dyadic norms and fixed coefficient restrictions. After extracting common bounds, $|a(\boldsymbol{p})|, |\beta(n,b)| \leq 1$.

We verify the two independence hypotheses before invoking the norm bound. Every residual prime has exponent $j = 1$, every active slot is a whole-index $j = 0$ mark, and all other active primes are fixed. The separate full M_{ref}^2 classes fix the cusp coefficient and additive factor. The reciprocity identity (13.5) cancels every row-slot phase, while the preceding marked-reflection discussion assigns all row-row, slot-slot, and fixed interactions to a row scalar or a tuple scalar. The angular conductor scalar separates in the same fashion. The remaining source and fixed local columns depend only on the full extracted dual index; their zeros at a fixed factor give fixed row, tuple, or dual restrictions. Finally the exact zero-preserving column identities leave only

$1_{(P,k_{\mathrm{res}})=1}$ and $1_{(P,b)=1}$ in addition to the character zeros. Hence a is independent of k_{res}, n, b, and β is independent of $k_{\mathrm{res}}, \boldsymbol{p}$, exactly as required in (13.17). The function a need not retain product form at this norm step.

The verified coefficient independences allow us to apply Lemma 13.2 with

$$K = Z^{H}, \qquad N = Z^{v}, \qquad B = Z^{\ell_b}, \qquad L = Z^{z_a}.$$

Aggregating tuples with the same P introduces only a divisor factor. Since

$$v + z_a - u = \max\{v, z_a, 2(v + z_a)/3\},$$

the hybrid bound and the outside coefficient give exponent

$$\max(H, v + \ell_b) - S_0 - B_0 + z_a - u - \ell_b - 2e_\lambda/3.$$

The scalar m_{ker} was extracted before squaring the row norm, so its square supplies the last term in (13.15). This proves the fixed-part assertion.

Summing the fixed row parts. Every powerful ideal is uniquely x^2y^3 with y squarefree. Ideal counting and $\sum_y q_y^{-3/2} < \infty$ give $O(Z^{O/2})$ such ideals in the O-dyad. The supported squarefree part divides the radical of Sfr_ρ, so it has $O(Z^{\epsilon})$ choices in the stated ranges; the local choices have the same divisor bound. Rowwise divisor Cauchy on the local expansion followed by the positive sum over these fixed parts therefore adds $O/2 + \epsilon$ once. The active tuples were already inside the hybrid norm and are not counted again. This proves the aggregate assertion. □

13.5 The unmarked bound in the original row length

The actual inequality for H allows us to bound the dependence on M, while allowing the supported squarefree row part to have positive norm length. At one active non-slot prime the contribution to $2A_0 - N_0 - S_0 - 4B_0$ is as follows:

local choice	coefficient of its log-norm
j odd, or $j = 2$	2
$j = 0$ active	1
$j = 4$ small or assigned to n	1
$j = 4$ assigned to b	-2
$j = 0$ inactive	0.

A residual row prime contributes two through $2H$. Prime by prime, these coefficients are bounded by the valuations in

$$(q_k^2/q_{k_{\mathrm{pow}}})q_f q_{r_\rho}.$$

Indeed, a powerful prime of row valuation $e \geq 2$ has valuation e in the first factor, a squarefree row prime has valuation two, and a prime belonging only to the squarefree f or to r_ρ has valuation one. Intersections only increase the valuation of this upper bound. Fixed excluded primes contribute only a fixed factor $C_S \geq 1$. Equivalently, summing over primes gives the actual norm inequality

$$q_{k_{\mathrm{res}}}^2 Z^{2A_0 - N_0 - S_0 - 4B_0} \leq C_S \frac{q_k^2}{q_{k_{\mathrm{pow}}}} q_f q_{r_\rho}.$$

Choose a fixed $C_f \geq 1$ with $q_f \leq C_f Z^V$, and put $C_{\mathrm{width}} = C_H^2 C_S C_k^2 C_O C_f$. The preceding inequality and the definition of T_d imply

$$T_d - S_0 - B_0 \leq 2M - O + Q + V - N_* + 2z_a + \frac{\log C_{\mathrm{width}}}{\log Z}. \tag{13.18}$$

For empty slot lists, $z_a = u = 0$. Choose $\eta > 0$ and then take Z so large that the two displayed constant errors are at most η and $e_\lambda \geq -\eta$; the latter follows from $\log Z \geq 4 \log q_\lambda/\eta$. On a retained dyad, (13.15) and $v + 3\ell_b + e_\lambda \leq T_d + \tau_{\mathrm{ref}}$ give

$$E_{\mathrm{ref}} \leq O/2 + \max\{H, T_d - S_0 - B_0\} + \tfrac{5}{3}\eta + \tau_{\mathrm{ref}}.$$

For the first branch, drop the nonpositive terms after using the lower bound for e_λ; for the second, use $v \leq T_d - 3\ell_b - e_\lambda + \tau_{\mathrm{ref}}$. The maximum is increasing in each argument. We may therefore use (4.35) for its first argument and (13.18) for its second, then sum the logarithmically many actual H-dyads. Taking $\eta, \tau_{\mathrm{ref}}$ and the local small-power losses sufficiently small in terms of ϵ proves

$$\sideset{}{'}\sum_{0<q_k \ll Z^M} |\mathcal{T}_1(Z^{N_*}; k)|^2 \ll Z^{O/2+\max\{M-O,\, 2M-O+Q+V-N_*\}+\epsilon}. \tag{13.19}$$

The prime on the sum restricts the powerful part to its O-dyad. A bounded dual range is included in the first term; an empty range is negligible. The deduction uses the upper bound for H in (4.35), including its fixed-width error.

14 The compensated low estimate

We now bound the modified character sum defined in (11.5). Its exact low separation has two factors. We first bound the completed row after summing the marked slots by applying Lemma 13.3 directly. We then prove a quantitative Gram bound for the additive polynomial from Section 5. Cauchy–Schwarz will combine these estimates to give the exponent 3/16.

14.1 The marked completed row

For a fixed rescaled subset J, with marked subset J^c, put

$$\begin{gathered} d = \sum_{i \in J} \ell_i, \qquad X' = X/q_{p_J} \asymp XZ^{-d}, \qquad Y' = Y/q_{p_J} \asymp YZ^{-d}, \\ M' = M - 2d, \qquad \ell' = \ell - d, \qquad 0 \leq d \leq \ell = 1/6. \end{gathered} \tag{14.1}$$

For a squarefree product D of slot primes, define the marked completed series

$$T_D(s_0, \psi) = \sum_{\substack{c \text{ sf} \\ n}} \gamma_2(c)\overline{\alpha(cn^3)}\psi(cn^3) 1_{D|cn^3} q_c^{-s_0} q_n^{-3s_0+1/2}.$$

The ideals c, n are the original ideals prime to S, represented by their primary generators, with c squarefree; they may share primes. All character powers retain their original zero extensions. For the empty product $D = 1$ this is the completed $T(s_0, \psi)$. For this fixed J, let

$$\begin{aligned} B^J_{m,\sigma}(Z) = \sum_{p_i \in \mathcal{P}_i(Z),\ i \notin J} \prod_{i \notin J} \overline{\eta(p_i)} W_i(q_{p_i}/P_i) \sum_{\theta \in \widehat{T}} a_\theta \frac{1}{2\pi i} \int_{(4)} (Zq_{p_{J^c}})^t \Phi(t) \\ \cdot T_{p_{J^c}}(t + 1/2, \nu_\sigma \theta \chi_\bullet(m))\, dt. \end{aligned}$$

Thus B^J is exactly the completed-row factor for the J summand of (11.5) after the marked slots have been summed. The fixed tuple p_J affects only X', Y' in its low separation.

Lemma 14.1 (The compensated completed-row norm). *Let $B^J_{m,\sigma}(Z)$ be the completed-row factor in Section 14.1, formed from the slot system of Section 11, and let X', Y', M', d be as in (14.1). For every $\epsilon > 0$, every fixed rescaled subset J, and every $\sigma \in T$, with $Q = q_{b_*} X'Y' \asymp Z^{M'}$,*

$$\sum_{0 < q_m \ll Q} |B^J_{m,\sigma}(Z)|^2 \ll Z^{M' + ((d-1/6)/4)_+ + \epsilon} = Z^{M'+\epsilon}. \tag{14.2}$$

The bound is uniform in the fixed rescaled tuple and has no loss depending on the mesh of the surviving slots.

Proof. We first partition the Gaussian test into annuli and separate each annular profile with one density common to the row and slot sums. Lemma 13.3 then bounds the row norm. A two-case comparison of the hybrid saving bounds its exponent by M' plus the prescribed small loss.

Gaussian annuli and their normalization. Put $A = cn^3$ and $\varrho_i = q_{p_i}/P_i$ for $i \notin J$. Since $q_A = q_c q_n^3$ even when c, n share primes, and $q_{p_{Jc}} = Z^{\ell'} \prod_{i\notin J} \varrho_i$, Mellin inversion of the displayed integral defining B^J gives the smooth factor in the completed sum

$$W_G\left(\frac{q_A}{Z q_{p_{Jc}}}\right) = W_G\left(\frac{r}{\prod_{i\notin J}\varrho_i}\right), \qquad r = \frac{q_A}{Z^{1+\ell'}}.$$

Here W_G is the Gaussian of Lemma 3.11. Empty products are one.

The fixed translate partition χ from that lemma partitions the r-variable into annuli. On $r = e^k x$, the single joint profile is

$$w_{k,J}(x,\varrho) = \chi(\log x) W_G\left(\frac{e^k x}{\prod_{i\notin J}\varrho_i}\right)\prod_{i\notin J} W_i(\varrho_i).$$

The logarithms of x and of all the ϱ_i lie in fixed compact sets on this support. Thus the logarithm of the Gaussian argument is $k + O_{\mathrm{fixed}}(1)$. The derivative estimate in Lemma 3.11, with this bounded shift and the fixed slot windows, gives for every fixed $\kappa_G > 0$, $A_{\mathrm{cnt}}, B_{\mathrm{cnt}}, C_{\mathrm{ann}} \ge 0$, $D_{\mathrm{tail}} > 0$, and fixed seminorm order j,

$$\sum_{k\in\mathbb{Z}} e^{A_{\mathrm{cnt}}|k|} p_j(w_{k,J}) < \infty, \qquad Z^{B_{\mathrm{cnt}}} \sum_{|k|>\kappa_G \log Z - C_{\mathrm{ann}}} e^{A_{\mathrm{cnt}}|k|} p_j(w_{k,J}) \ll Z^{-D_{\mathrm{tail}}}.$$

The constants may depend on the fixed slot system, but not on its moving prime labels. On an annulus $\log r = k + O(1)$, an absolute bound for the row ℓ^2 norm is at most $Z^{B_{\mathrm{cnt}}} e^{A_{\mathrm{cnt}}|k|} p_j(w_{k,J})$ for some fixed exponents, by the elementary row, slot, and ideal counts. Choose C_{ann} larger than the fixed logarithmic radius of χ and discard only whole annuli with $|k| > \kappa_G \log Z + C_{\mathrm{ann}}$. Summing their norm bounds and then squaring gives less than any prescribed power of Z^{-1}. Every annulus meeting $|\log_Z r| \le \kappa_G$ is retained.

For each remaining annulus, keep fixed individual annular cutoffs equal to one on the support of $w_{k,J}$ and apply Lemma 3.6 to this whole profile. Its logarithmic Fourier inversion uses one coefficient density common to every row and every surviving slot tuple. The normalized slot ratios remain variables of the joint profile until separation; they do not index separately chosen densities. For fixed Fourier variables, each resulting slot coefficient is a product of the original arithmetic coefficient $\overline{\eta(p_i)}$ and an individual annular cutoff and norm power. It remains bounded and independent of the row and of every other slot. The completed factor is an annular test at length

$$N_* = 1 + \ell' + \theta_N, \qquad \theta_N = \frac{k}{\log Z}, \qquad |\theta_N| \le \kappa_G + O_{\mathrm{fixed}}(1/\log Z).$$

These retained lengths lie in a fixed bounded range. The weighted Fourier norms have a summable total by the displayed Gaussian estimate. As in (4.24), the small annular weight stays in the homogeneous Fourier norm of the single common profile, and Minkowski's inequality passes the separated row norm through that density. The existing $q_c^{-1/2} q_n^{-1}$ normalization stays unchanged when the test scale is recentered, so no additional power of e^k is introduced. Choose κ_G within the final ϵ allowance.

Applying the reflected energy bound. For each fixed θ, Lemma 4.4 supplies the row-sector reduction of $\nu_\sigma \theta \chi_\bullet(m)$ with every original zero mask. It covers the unweighted row norm in (14.2) over all nonzero element rows, including those meeting S. The row ball depends only on the fixed rescaled tuple p_J, through $Q = q_{b_*} X'Y'$, and is independent of the surviving marked slots and dual variables. The original slot sets remain disjoint and their separated coefficients

are product-form. These facts supply the support and coefficient-independence hypotheses of Lemma 13.3.

The triangle inequality for the finite sum over θ, followed by Lemma 13.3, therefore gives the exponent in (13.15) at this actual N_*, taking the retention parameter τ_{ref} and the independently prescribed output and separation losses all at most a small $\epsilon_0 > 0$. The exponent E_{ref} applies after summation over the fixed row parts. Their count contributes the principal exponent $O/2$ exactly once in $E_{\mathrm{ref}} + \epsilon_0$.

Here $f = 1$ and the coprimality restriction is $\rho = 1$, so the only moving non-slot primes are those of the row. After the finite unit and reflection-sector splits in Lemma 13.3, freeze its powerful part of norm length O and its valuation-one part supported on S, and split its residual squarefree good product into actual dyads of norm $\asymp Z^H$. Let $\epsilon_Z = O_{\mathrm{fixed}}(1/\log Z)$ be nonnegative and large enough to contain all fixed annular and fixed-conductor logarithmic offsets in this calculation. Then

$$H \le M' - O + \epsilon_Z, \qquad \Delta_H = M' - O - H \ge -\epsilon_Z, \qquad O \ge -\epsilon_Z.$$

The actual row and dual annular cutoffs are kept in one joint profile while its common Fourier density is separated. The small kernel amplitude is extracted from that homogeneous density before the row norm is squared, and only then may the positive sieve sum be enlarged, as in (4.24). In particular the kernel saving in (13.15) uses this H; it is not reevaluated on any shorter rows added by the positive enlargement.

In the notation of that energy estimate, every moving non-slot prime counted in A_0 divides the powerful row part to exponent at least two. The comparison of its exact norm with the powerful dyad, and any fixed-conductor contribution, therefore gives

$$2A_0 \le O + \epsilon_Z, \qquad N_0 \le A_0, \qquad z_a \le \ell'.$$

The available dual length at the actual completed scale is

$$T_d = 2H + 2A_0 + 2z_a - 1 - \ell' - \theta_N - N_0 - 3B_0.$$

Since $M' + \ell' - 1 = -3d$, direct subtraction gives

$$T_d - (H - 3d) = H - M' + 2A_0 + 2(z_a - \ell') - N_0 - 3B_0 - \theta_N \le 2\epsilon_Z + |\theta_N|.$$

As in the proof of the reflected energy, dual ranges below a fixed negative length are negligible after a positive Mellin shift, and the bounded boundary range contributes an arbitrarily small power. The source dual ideals n, b in these dyads retain the support of (4.3): they may share primes and may contain the permitted primes of S; no additional S-mask or coprimality condition between n and b is imposed. On each remaining dual dyad, put $y = v + 3\ell_b + e_\lambda$. Retention gives $y \le T_d + \epsilon_0$, while the nonzero unit ranges give $v, \ell_b, e_\lambda \ge -\epsilon_Z$ after increasing its fixed constant. It follows that

$$\max(H, v + \ell_b) = H + O(\epsilon_0 + \kappa_G + \epsilon_Z).$$

Comparing the two exponent bounds. Recall that $u = \min\{v, z_a, (v + z_a)/3\}$ by (13.14). If $u = z_a$, then $-S_0 - B_0 + z_a - u \le 0$, the kernel term is nonpositive, and

$$-\ell_b - \frac{2e_\lambda}{3} \le \frac{5}{3}\epsilon_Z.$$

Here O outside the error notation denotes the powerful-row logarithmic length. It follows that

$$\begin{aligned} E_{\mathrm{ref}} &\le O/2 + H + O(\epsilon_0 + \kappa_G + \epsilon_Z) \\ &= M' - O/2 - \Delta_H + O(\epsilon_0 + \kappa_G + \epsilon_Z). \end{aligned}$$

Otherwise $u \ge v/2 - O(\epsilon_Z)$, and the bounded unit ranges imply

$$u + \ell_b + \frac{2e_\lambda}{3} + \frac{(T_d - y)_+}{2} \ge \frac{y}{4} + \frac{(T_d - y)_+}{2} - O(\epsilon_Z) \ge \frac{T_d}{4} - O(\epsilon_Z).$$

The last inequality holds separately for $y \le T_d$ and $y \ge T_d$. Here the squared kernel saving is obtained by first extracting $\min\{1, Z^{(y-T_d)/4}\}$ from the common Fourier density and only then squaring the row norm. Relative to M', the saving before the remaining $+z_a$ is exactly

$$\begin{aligned} O/2 + \Delta_H + S_0 + B_0 + T_d/4 = {} & \frac{2M' + 2z_a - 1 - \ell' + 2\Delta_H - \theta_N}{4} \\ & + \frac{2A_0 - N_0 + B_0 + 4S_0}{4}. \end{aligned} \tag{14.3}$$

The second numerator is nonnegative. Since $z_a \le \ell'$, the energy in this branch is at most

$$\begin{aligned} & M' + \frac{1 + 3\ell' - 2M' - 2\Delta_H + \theta_N}{4} + O(\epsilon_0 + \kappa_G + \epsilon_Z) \\ & \quad = M' + \frac{d - 1/6 - 2\Delta_H + \theta_N}{4} + O(\epsilon_0 + \kappa_G + \epsilon_Z). \end{aligned}$$

After using $O \ge -\epsilon_Z$ and absorbing θ_N, the two upper bounds are

$$M' - \Delta_H + O(\epsilon_0 + \kappa_G + \epsilon_Z), \qquad M' + \frac{d - 1/6 - 2\Delta_H}{4} + O(\epsilon_0 + \kappa_G + \epsilon_Z).$$

They are nonincreasing in Δ_H. Since $\Delta_H \ge -\epsilon_Z$, their formal $\Delta_H = 0$ values bound every actual dyad up to $O(\epsilon_Z)$, whether or not an endpoint dyad occurs. Their maximum is therefore at most $M' + ((d - 1/6)/4)_+ + O(\epsilon_0 + \kappa_G + \epsilon_Z)$. At $d = 1/6$ one has $\ell' = z_a = 0$ and the clipped unit range falls in the first branch; bounded negative unit dyads change only ϵ_Z. Choose ϵ_0 and κ_G within the prescribed ϵ allowance, and then increase the fixed-data lower threshold so that ϵ_Z also lies within it. Summing the logarithmically many row and dual dyads proves the lemma. □

14.2 The additive Gram bound

We next estimate the additive factor in the range $Y'^2/Q \ge 1$. The common-factor formula of Lemma 12.4 retains the collision zeros that determine both its cancellation and its exceptional frequencies.

Proposition 14.2 (A quantitative additive Gram bound). *Fix the arithmetic data S, T, b_*, ξ, the annular weight W_1, and a ray class $\sigma \in T$. Let $Q, Y' \ge 1$ be in fixed polynomial ranges in Z, and suppose*

$$P_a = Y'^2/Q \ge 1.$$

For a real height ν, put $A_m = A_{m,\sigma,\nu}(Y')$ as defined in (5.2). For every fixed row ball $q_m \ll Q$ and every $\epsilon > 0$, one has

$$\sum_{q_m \ll Q} |A_m|^2 \ll (1 + |\nu|)^{J_{\mathrm{Gr}}} \frac{Q}{Y'} \left(1 + P_a^{1/6} + \frac{P_a^2}{Y'}\right) Z^{\epsilon}. \tag{14.4}$$

Here J_{Gr} is a fixed seminorm order, independent of the moving scales and of ν. The same assertion holds when the arithmetic coefficient is any fixed finite linear combination of $1_{s \in \sigma}\chi_s(b_*)/\xi(s)$, chosen independently of the row m.

Proof. Poisson summation in the row variable produces the Fourier correlation of Lemma 12.3; Lemma 12.4 then separates its common factor. Its diagonal gives the first term in (14.4). At

nonzero frequencies, a vanishing complete mean gives the term P_a^2/Y', while the exceptional frequencies contribute $P_a^{1/6}$.

Normalization and full correlation. Majorize the row ball by a fixed nonnegative radial Schwartz function. Put $r_s = q_s/Y'$ and, on the primary elements prime to S, $c_\sigma(s) = 1_{s\in\sigma}\chi_s(b_*)/(\tau\xi(s))$, extended by zero elsewhere before the inverse character is evaluated. The exact annular rewriting of the polynomial is

$$A_m = Y'^{-3/2}\sum_s c_\sigma(s)W_1(r_s)r_s^{-1+i\nu}g_{\chi_s}(s,-m).$$

Thus the expanded square has the common normalization Y'^{-3} and the joint annular profile contains the factors $W_1(r_{s_1})r_{s_1}^{-1+i\nu}W_1(r_{s_2})r_{s_2}^{-1-i\nu}$. Write $s_1 = Cn_1$, $s_2 = Cn_2$, with $(n_1,n_2) = 1$.

Poisson in the row variable has factor $Q/(q_{s_1}q_{s_2})$, whereas the complete transform of the two unnormalized Gauss sums is $q_{s_1}q_{s_2}F(s_1,s_2;j)$. It therefore gives the exact prefactor Q/Y'^3 and the correlation $F(Cn_1,Cn_2;Ck)$ of (12.7). Here F again denotes the full correlation of (12.6), and is distinct from the local coefficient of (6.1). The first Fourier kernel has argument comparable to q_Cq_k/P_a.

On coprime residuals the factor L_C in that correlation has $|L_C| \le q_C$. For the subsequent signed extension use its specified bounded extension, periodic modulo $\operatorname{rad} C$ in both columns, including when C shares primes with a column. Its value is defined to be zero when a prime of C divides both columns; this is a definition of the extension, not a claim about the original correlation there. Before any factorwise estimate, the complete Möbius identity

$$1_{(n_1,n_2)=1} = \sum_{d\mid n_1,\ d\mid n_2}\mu(d)$$

is inserted, with $n_i = dm_i$. No coprimality condition on m_1,m_2 is then imposed. Nonzero terms have $(d,Ck) = 1$. Throughout this step use the fixed ray extension of $\mathcal{R}(n_1,n_2)$; a quotient of zero residue symbols would not be defined.

A common modulus and the zero mean. We now identify a fixed part of the coefficient modulus, so that its moving part is controlled by C and k. For $k\neq 0$ write $k = \zeta k_S k_{\mathrm{good}}$ using the unit convention and the fixed S-prime generators of Lemma 3.2, with $k_S = \prod_{\mathfrak{p}\in S}\pi_{\mathfrak{p}}^{e_{\mathfrak{p}}}$ and $k_{\mathrm{good}} = \prod_{p\notin S}p^{e_p}$. For a primary residual ideal m outside S, reciprocity gives

$$\chi_m(k) = \kappa_{\zeta,\boldsymbol{e}}(m)\mathcal{R}(k_{\mathrm{good}},m)\prod_{p\mid k_{\mathrm{good}}}\chi_p(m)^{e_p},$$
$$\kappa_{\zeta,\boldsymbol{e}}(m) = \chi_m(\zeta)\prod_{\mathfrak{p}\in S}\chi_m(\pi_{\mathfrak{p}})^{e_{\mathfrak{p}}\bmod 6}.$$

Every displayed good local power retains its zero on a shared prime, including when $6\mid e_p$. Lemma 3.2 places the finitely many $\kappa_{\zeta,\boldsymbol{e}}$ in a fixed ray group supported on S. Also $\mathcal{R}(k_{\mathrm{good}},m)$ belongs to a fixed finite family in m.

Choose $\mathfrak{l}_{\mathrm{fixed}}$ once to contain the primary-class modulus and the S zero masks, the moduli of every fixed ray coefficient in A_m, a period for $\mathcal{R}$ in each variable, and the conductors of all the preceding $\kappa_{\zeta,\boldsymbol{e}}$ characters. The extended coefficient is zero off the primary class or at a fixed nonunit before any inverse character is evaluated. This modulus is independent of C,d,k, and may have higher powers at S than $\operatorname{rad}(k)$. After reciprocity the joint arithmetic coefficient in m_1,m_2 is periodic modulo

$$\mathfrak{r} = \operatorname{lcm}(\mathfrak{l}_{\mathrm{fixed}},\operatorname{rad} C,\operatorname{rad}(k)),\qquad q_{\mathfrak{r}} \ll_{\mathrm{fixed}} q_Cq_k.$$

If k has a valuation e_p not divisible by six at a prime outside C and $\mathfrak{l}_{\mathrm{fixed}}$, call k nonexceptional. At that prime the first residual column has the nonprincipal zero-extended factor $\chi_p(m_1)^{e_p}$ up

to a unit scalar. The lift, the fixed phases, and the other prime factors are independent of m_1 modulo p. Its complete mean, and hence the complete joint mean by the Chinese remainder theorem, is zero. The substitution by d does not change this conclusion because $(d,k)=1$. Principal zero masks at valuations divisible by six remain present in the exceptional case.

The frequency $k=0$ contributes only the diagonal: the correlation forces $n_1=n_2=1$ and equals $\varphi(C)$. There are $O(Y')$ possible C of norm comparable to Y', each with $\varphi(C)\le q_C\ll Y'$. Its contribution is therefore $O(Q/Y')$.

Nonexceptional frequencies. For $k\neq 0$, split the Fourier argument into dyadic shells $q_Cq_k/P_a\asymp R$, with $R=1,2,4,\dots$ and the first shell including all arguments at most two. Choose dyad centers $C_0,D_0\in\{1,2,4,\dots\}$ for $q_C\asymp C_0$ and $q_d\asymp D_0$, with the first dyads containing the bounded unit ranges. The nominal positive residual scale is

$$N=Y'/(C_0D_0),\qquad \mathcal{Q}=q_{\mathfrak{r}}\ll_{\text{fixed}} RP_a.$$

The actual residual norms lie in fixed multiples of N; the nominal value N is allowed to be below one. Fix arbitrary $A>2$ and $B>2$. For each fixed C,d,k, the one joint annular profile W_R for the two columns has the finitely many homogeneous single-profile seminorms used below bounded by $O_{B,J_{\mathrm{Gr}}}((1+|\nu|)^{J_{\mathrm{Gr}}}R^{-B})$, with one fixed J_{Gr} chosen after these derivative orders and A,B. This follows by differentiating the first Fourier kernel, whose Schwartz decay absorbs every resulting power of R; derivatives of the norm powers contribute a fixed power of $1+|\nu|$. The factor R^{-B} remains in the single-profile Fourier estimate and occurs once in the linear lattice sum below.

Let $a_{C,d,k}$ be this periodic arithmetic coefficient. For a nonexceptional frequency its normalized complete Fourier transform, with normalization $\mathcal{Q}^{-2}$ on the two residue variables, has absolute value at most q_C and vanishes at $(0,0)$. Since every ideal of $\mathcal{O}$ is principal, the period lattice has linear scale $\mathcal{Q}^{1/2}$. Four-dimensional lattice Poisson therefore gives, for this A,

$$\left|\sum_{\boldsymbol{m}\in\mathcal{O}^2} a_{C,d,k}(\boldsymbol{m})W_R(\boldsymbol{m}/\sqrt{N})\right|\ll(1+|\nu|)^{J_{\mathrm{Gr}}}R^{-B}q_CN^2\min\{1,(\mathcal{Q}/N)^A\}. \tag{14.5}$$

For $N\ge\mathcal{Q}$, the zero frequency vanishes and the other dual vectors have linear scale $(N/\mathcal{Q})^{1/2}$. Fourier decay of order $2A>4$ bounds their sum by $(\mathcal{Q}/N)^A$. For $N<\mathcal{Q}$, use the point count in the two annuli. A nonempty subunit annulus has N bounded below by a fixed positive constant, so it still contains $O(N)$ lattice points; below that constant it is empty. This proves (14.5) without any primitivity or tensor-product assumption on $a_{C,d,k}$.

Put $K_R=RP_a/C_0$. The fixed shell comparison constants give $q_k\le C_{\mathrm{sh}}K_R$ for a fixed $C_{\mathrm{sh}}>0$. Hence the nonzero frequency range is empty when $K_R<C_{\mathrm{sh}}^{-1}$. On a nonempty range, elementary lattice counting gives

$$\#\{k:0<q_k\le C_{\mathrm{sh}}K_R\}\ll_{\text{fixed}} K_R+\sqrt{K_R}+1\ll_{\text{fixed}} K_R.$$

There are therefore $O(C_0D_0K_R)$ choices of C,d,k on these dyads. Multiplying (14.5) by this count and by the common prefactor gives at most

$$(1+|\nu|)^{J_{\mathrm{Gr}}}R^{-B}\frac{Q}{Y'}\frac{RP_a}{C_0D_0}\min\left\{1,\left(\frac{RP_aC_0D_0}{Y'}\right)^A\right\}.$$

For $\Lambda=Y'/(RP_a)$ the required positive dyadic sum is

$$\begin{aligned}\sum_{i,j\ge0}2^{-i-j}\min\{1,(2^{i+j}/\Lambda)^A\}&=\sum_{n\ge0}(n+1)2^{-n}\min\{1,(2^n/\Lambda)^A\}\\&\ll_A\frac{1+\log(2+\Lambda)}{\Lambda}.\end{aligned}$$

For $\Lambda \geq 1$, split the two geometric tails at $2^n \asymp \Lambda$; for $\Lambda < 1$, the left side is bounded and the displayed right side is larger than a positive constant. Thus the nonexceptional frequencies contribute

$$\ll (1+|\nu|)^{J_{\mathrm{Gr}}} \frac{Q}{Y'} \frac{P_a^2}{Y'} Z^{\epsilon}$$

after the R sum, on choosing a fixed $B > 2$.

Exceptional frequencies. It remains to count the exceptional nonzero frequencies. They have $(k) = a_1^6 e$, where e is sixth-power-free and supported on the primes of C and the fixed support. There are at most $6^{\omega(C)+O_S(1)} \ll_{\epsilon} q_C^{\epsilon}$ possible e. For each of them the shell bound $q_k \leq C_{\mathrm{sh}} K_R$ permits only $q_{a_1} \leq (C_{\mathrm{sh}} K_R / q_e)^{1/6}$. If this range is nonempty, its upper bound is at least one, and the ideal count gives $O_{\mathrm{fixed}}(K_R^{1/6})$ choices; if $q_e > C_{\mathrm{sh}} K_R$ it is empty. The unit choices contribute only a fixed factor. Hence there are $O_{\epsilon}(K_R^{1/6} q_C^{\epsilon})$ such elements k, including the bounded subunit range of K_R.

The bounds $K_R = RP_a/C_0$ and $|L_C| \leq q_C$, together with the trivial two-column point count, give the following contribution on these dyads before multiplication by Q/Y'^3:

$$\ll (1+|\nu|)^{J_{\mathrm{Gr}}} R^{-B} Z^{\epsilon} Y'^2 (RP_a)^{1/6} C_0^{-1/6} D_0^{-1}.$$

The sums over C_0, D_0 converge after reducing the arbitrary small power, and the R sum converges for the same fixed $B > 2$. This gives the term $(Q/Y')P_a^{1/6} Z^{\epsilon}$. Adding the diagonal and nonexceptional terms proves the proposition. □

14.3 Completion of the low bound

The two preceding estimates now apply to the two factors of the same rescaled summand in the exact low separation.

Proposition 14.3 (The compensated low estimate). *For the modified sum in (11.5), the geometry in (11.4), and every $\epsilon > 0$,*

$$|I_{\eta,\mathrm{modified}}(Z)| \ll Z^{l_x/2 + b/12 + \epsilon} = Z^{3/16+\epsilon}. \tag{14.6}$$

The exponent is independent of the target and the slot mesh; the constant and lower threshold may depend on the fixed arithmetic data, slot system, and smooth tests.

Proof. For a fixed rescaled subset and tuple, put $Q = q_{b_*} X'Y'$. Its separation is (5.3) with $A_{m,\sigma,v}(Y')$ and $B^J_{m,\sigma}(Z)$. Put $r_J = q_{p_J}/Z^d$; the fixed annular supports give $0 < c_J \leq r_J \leq C_J < \infty$ uniformly in the tuple. Exactly,

$$X' = Z^{l_x - d}/r_J, \qquad Y' = Z^{l_y - d}/r_J,$$
$$Q = q_{b_*} Z^{M'}/r_J^2, \qquad P_a = Y'^2/Q = q_{b_*}^{-1} Z^{1/8}.$$

In particular $P_a \geq 1$ for $Z \geq q_{b_*}^8$. Since $M' \geq 1/2$ and $l_y - d \geq 5/16$, the bounds $Q, Y' \geq 1$ also hold uniformly over the tuple ratios above a sufficiently large fixed-data threshold. The remaining length inequalities are

$$l_x - d \geq 3/16, \qquad l_y - d \geq 5/16, \qquad l_y - d - 11b/6 \geq 1/12.$$

The last one gives

$$\frac{Y'}{P_a^{11/6}} = q_{b_*}^{11/6} r_J^{-1} Z^{l_y - d - 11b/6} \gg_{\mathrm{fixed}} Z^{1/12},$$

and hence $P_a^2/Y' \ll P_a^{1/6}$. Proposition 14.2 therefore gives

$$\sum_{q_m \ll Q} |A_{m,\sigma,v}(Y')|^2 \ll (1+|v|)^{J_{\mathrm{Gr}}} (Q/Y') P_a^{1/6} Z^{\epsilon}. \tag{14.7}$$

Cauchy–Schwarz in the row sum, Lemma 14.1, and the integrability of $(1+|v|)^{J_{\mathrm{Gr}}/2}|\widehat{W_0}(iv)|$ now bound this tuple's unscaled separation by Z^{ϵ} times

$$\begin{aligned} Q^{-1/2}\big[(Q/Y')P_a^{1/6}\big]^{1/2}\big[Z^{M'}\big]^{1/2} &= \big[Z^{M'}/Y'\big]^{1/2}P_a^{1/12} \\ &= r_J(X')^{1/2}P_a^{1/12}. \end{aligned}$$

Since r_J is bounded, this is $O((X')^{1/2}P_a^{1/12}Z^{\epsilon})$.

For $d=\sum_{i\in J}\ell_i$ there are at most $Z^{d+\epsilon}$ rescaled tuples, the coefficient is $O(Z^{-3d/2})$, and $(X')^{1/2}\asymp X^{1/2}Z^{-d/2}$. The total additional exponent is consequently

$$f(d)=d-3d/2-d/2=-d\le 0. \tag{14.8}$$

Choose the small powers in the component estimates so that their sum lies within the prescribed ϵ. Summing the finitely many subsets proves (14.6). For the already defined C_{II} of (11.3), the normalization used for the principal signal satisfies

$$\begin{aligned} C_{\mathrm{II}}(s) &= l_x/2+s-1+h/6 = s-11/16, \\ C_{\mathrm{II}}(7/8) &= 3/16 = l_x/2+b/12. \end{aligned} \tag{14.9}$$

Thus, under the contradiction hypothesis in (11.1), this low bound is smaller than $Z^{C_{\mathrm{II}}(\beta_*)}$ by the exact power $\Delta=\beta_*-7/8$, apart from the arbitrarily prescribed ϵ. □

15 The local compensation and its errors

The low estimate in Proposition 14.3 concerns the finite modification of Section 11. Its high representation is (15.6), with the holomorphic correction in (15.4). The local cancellation then separates the main prime factors from errors controlled jointly with the numerator.

Throughout, the local notation $P_p, P_p^*, H_p, V, W, D, R$ is that of Lemma 6.1 and (6.13); in the shared analytic estimates its variable x is called s.

Fix a nonzero sixth-power-free row u with $(u,S)=1$. All slot primes belong to the sets $\mathcal{P}_i(Z)$ of Section 11; for each such prime put $Q=q_p$. As in Lemma 6.2, write $x_r=\Re x$, $w_r=\Re w$, and $z_r=\Re z$. The factor $W_{\mathrm{loc}}=\rho Q^{-w}$ inside P_p^* is distinct from $W=\chi_p(u)Q^{-w}$: at $p\mid u$, the latter and D vanish, while the former retains its unit phase.

15.1 The full holomorphic correction

For a fixed slot prime p, restricting the completed index to $p\mid A$ replaces P_p by P_p^* in the high series. Changing Z to ZQ multiplies its Mellin weight by Q^{x+z-1}. The marked term of (11.5) therefore has, at points where P_p is defined and nonzero, multiplier $\overline{\eta(p)}Q^{x+z-1}P_p^*/P_p$. The rescaled term has multiplier

$$Q^{-3/2}Q^{-(1/2-z)}Q^{-(w-1)}=Q^{z-w-1}.$$

Thus, on the same locus, the exact local multiplier of the two-term operation is

$$Q^{z-1}\mathcal{B}_p, \qquad \mathcal{B}_p=\overline{\eta(p)}Q^xP_p^*/P_p-Q^{-w}. \tag{15.1}$$

Define the combined local replacement by

$$G_p=\frac{(\overline{\eta(p)}Q^x-Q^{-w})(1-V)(1-W)P_p^*-Q^{-w}(1-VW)}{1-D}. \tag{15.2}$$

At points in the Euler regions where the raw quotient in (15.1) is defined, one has $G_p=H_p\mathcal{B}_p$. The displayed formula defines G_p also where that quotient is not defined. After inserting the

formula for P_p^*, the denominators on the right are $1-R$, $1-V$, and $1-D$. The bounds $|R|,|V|,|D|<1$ in the proof of Lemma 6.2, together with the formula for P_p^* in Lemma 6.1, show that G_p is holomorphic in both stated Euler regions, including at zeros of P_p or H_p.

At points in the Euler regions where the selected quotients are defined, write the full slot multiplier

$$\mathcal{B}_i(u;x,w,z)=\sum_{p\in\mathcal{P}_i(Z)} W_i(q_p/P_i)q_p^{z-1}\mathcal{B}_p. \tag{15.3}$$

The correction used for contour moves is defined without these quotients:

$$\mathfrak{H}_{\eta,u,Z}(x,w,z)=\sum_{(p_i)\in\prod_i\mathcal{P}_i(Z)}\prod_{i=1}^{K}\bigl[W_i(q_{p_i}/P_i)q_{p_i}^{z-1}G_{p_i}\bigr]\prod_{\substack{p\notin S\\ p\notin\{p_1,\dots,p_K\}}} H_p. \tag{15.4}$$

This is the full correction after the scalar quotient in (6.14) has been extracted. To verify this assertion, fix an allowed tuple and put $\mathcal{P}=\{p_1,\dots,p_K\}$. On the absolute starting lines the selected operation replaces P_p by

$$\overline{\eta(p)}Q^{x+z-1}P_p^*-Q^{z-w-1}P_p=\frac{1-D}{(1-V)(1-W)}Q^{z-1}G_p. \tag{15.5}$$

The equality follows by substituting $P_p-P_p^*=(1-V)^{-1}+W/(1-W)$ into (15.2). It extracts the same scalar local factor $(1-V)^{-1}(1-W)^{-1}(1-D)$ as at an unselected prime. Each selected prime occurs exactly once, because the slot supports are disjoint. The complete factor for this tuple is therefore the scalar quotient in (6.14) times its summand in (15.4). This reasoning involves no division by P_p or H_p. At points in the Euler regions where all raw selected quotients are defined, the same finite sum also factors as $\mathcal{H}_{\eta,u}\prod_i\mathcal{B}_i$.

Combining the exact finite operation with (6.5) and (6.14) now gives

$$\begin{aligned}I_{\eta,\mathrm{modified}}(Z)=\frac{1}{(2\pi i)^3}\int_{(3)}\int_{(3)}\int_{(2)}\mathcal{W}(X,Y,Z;x,w,z)\\ \cdot\sum_{u}^{(6)}q_u^{-z}\overline{\xi(u)}\frac{\zeta_F^S(6z)L^S(w,\chi_\bullet(u))}{L^S(x,\eta\overline{\chi_\bullet(u)})}\mathfrak{H}_{\eta,u,Z}(x,w,z)\,dz\,dw\,dx.\end{aligned} \tag{15.6}$$

This identity is absolutely convergent on the displayed lines, because for fixed Z it is obtained from finitely many absolutely convergent rescaled identities for I_η.

Holomorphy and bounds in height. The unselected product in each summand converges normally in both Euler regions. More quantitatively, in any fixed subregion of (6.15), put $\epsilon_H=\min(\epsilon_0,1/50)>0$. The primewise defect bounds in Lemma 6.2 give, uniformly in the selected tuple,

$$\prod_{\substack{p\notin S\\ p\notin\mathcal{P}}}|H_p|\le\prod_{\substack{p\notin S\\ p\nmid u}}(1+Cq_p^{-1-\epsilon_H})\prod_{p\mid u}(1+Cq_p^{-\epsilon_H})\ll_\epsilon q_u^{\epsilon}. \tag{15.7}$$

The first positive product converges by the ideal count, and the second satisfies the divisor-product bound. Omitting selected factors only removes factors from these positive majorants. In the second Euler region the same argument uses the respective defect exponents $-363/200$ and $-33/40$. Thus (15.7) holds there as well, with the corresponding positive majorants. These positive majorants bound the products even when an individual H_p vanishes.

For each fixed Z, (15.4) is a finite sum of products of holomorphic selected factors and normally convergent unselected products. It is therefore holomorphic on a neighborhood of every point in both Euler regions. It also satisfies the all-height requirement in Definition 9.1. Indeed, on a fixed real box in either region, the formula for P_p^* and the denominators bounded away from

zero give $|G_p| \ll Q^B$ for some fixed B, uniformly in all imaginary parts and unit phases. There are $O(P_i)$ ideals in a slot and $q_p \asymp P_i$ there. The finite tuple sum and (15.7) consequently give

$$|\mathfrak{H}_{\eta,u,Z}(x,w,z)| \ll_\epsilon q_u^\epsilon \prod_i P_i^{z_r+B} \ll_\epsilon q_u^\epsilon Z^{B'}. \tag{15.8}$$

The exponent B' is fixed for that real box. For $q_u \le Z^D$ this is (9.4), even with height exponent zero. All constants here are fixed before any later order of integration by parts.

The scalar quotient in (15.6) and $\mathfrak{H}_{\eta,u,Z}$ contain all Euler factors. Together with the holomorphy and height bound just proved, that identity verifies the exact high representation of Definition 9.1 for the finite expression $I_{\eta,\mathrm{modified}}$, with the geometry in (11.4). In particular, $l_x/2 - 1 + h/6 = 17/96 - 1 + 13/96 = -11/16$, as required by (11.3). It is an identity for the same modified sum bounded in Proposition 14.3. Every continuation uses the full correction in (15.4), not a globally defined product of individual quotients.

15.2 Local errors on the moving contours

At an identity-ray prime every fixed phase from T is one, although $\eta(p)$ may be any unit. Suppose first that $p \nmid u$ and put $v = \chi_p(u)$. Then $D = \eta(p)v^{-1}Q^{-x}$, and $\overline{\eta(p)}Q^x D = v^{-1}$. Using $v^{-1}W = Q^{-w}$ and $\mathcal{E}_p = P_p^* + D$ gives the exact normalized cancellation

$$\begin{aligned} G_p + v^{-1}H_p = \frac{(1-V)(1-W)}{1-D}\Big[&(v^{-1}-Q^{-w})\{V/(1-V) - D\} \\ &+ (\overline{\eta(p)}Q^x + v^{-1} - Q^{-w})\mathcal{E}_p\Big]. \end{aligned} \tag{15.9}$$

This is an identity of holomorphic local expressions in the Euler regions. Within these regions, on the raw quotient locus its left side is $H_p(\mathcal{B}_p + v^{-1})$. To see the cancellation directly on that locus, substitute $P_p^* = -D + \mathcal{E}_p$ and $P_p = (1-V)^{-1} + W/(1-W) + P_p^*$ into the left side before normalization. The constants $-v^{-1}$, $v^{-1} - Q^{-w}$, and $(v^{-1} - Q^{-w})W/(1-W) = Q^{-w}$ sum to zero. The resulting rational identity has only the denominators $1-R$, $1-V$, and $1-D$, which are nonzero in the Euler regions; hence it gives the displayed holomorphic identity there. For $p \mid u$, the main term is zero by the original zero extension, and (15.2) instead becomes

$$G_p = \overline{\eta(p)}Q^x(1-V)\mathcal{E}_p - Q^{-w}H_p.$$

The error bound below requires the reflected primitive numerator to be small at the retained height. To obtain this hypothesis, instantiate Section 7 with the current fixed data and $\Theta = \langle \eta, \widehat{T} \rangle$, using its zero-extended presentations $\nu(n)\chi_n(u)^\varsigma$ for $\nu \in \Theta$ and $\varsigma \in \{-1, 1\}$. The numerator $\chi_\bullet(u)$ and its conjugate belong to $\mathcal{X}_u$ with fixed multiplier $1 \in \Theta$, while the denominator $\eta\overline{\chi_\bullet(u)}$ belongs to $\mathcal{X}_u$ with fixed multiplier $\eta \in \Theta$.

At the points used below, $\Re(1-w) = a + 6e$ and $|\Im(1-w)| \le T_1$, so the reflected point lies in the buffered rectangle. The buffered estimates of Section 7 and the inverse deleted-factor bound of Lemma 3.16 therefore supply the reflected-numerator hypothesis at the retained heights. At each slot prime the identity-ray restriction sets every fixed T phase to one, while preserving the original row u and every nonunit zero.

For a row u in this sixth-power-free family, define the main prime factor of slot i by

$$\mathcal{Q}_i(u;z) = -\sum_{p \in \mathcal{P}_i(Z)} W_i(q_p/P_i) q_p^{z-1} \overline{\chi_p(u)}. \tag{15.10}$$

Its support retains the identity-ray condition and the original zero extension.

Proposition 15.1 (Local error products and the conductor bound). *Let* $51/100 \le a \le 1$, $0 < e \le 10^{-3}$, *and take*

$$x_r = a + 16e, \qquad w_r = 1 - a - 6e, \qquad z_r = 17/50.$$

Choose P_0 sufficiently large in terms of e and the fixed data. For $U, T_1 \geq 1$ and a sixth-power-free row u with $(u, S) = 1$ and $q_u \asymp U$, let ψ_u^* be the primitive character inducing $\chi_\bullet(u)$. Let $\mathcal{W}_u$ be any subset of $\{w : \Re w = w_r,\ |\Im w| \leq T_1\}$. Suppose ψ_u^* is nonprincipal and, uniformly for $w \in \mathcal{W}_u$,

$$|L(1-w, \overline{\psi_u^*})| \ll U^\epsilon$$

with any prescribed small power. For this row on the displayed region, define $\mathcal{B}_i$ by (15.3), interpreting each $\mathcal{B}_p$ as G_p/H_p, with G_p from (15.2). For the stated choice of P_0, $H_p \neq 0$ throughout this region for every slot prime; individual continuation is asserted only on this region. With $\mathcal{Q}_i$ as in (15.10), let the error part be

$$\mathcal{D}_i(u) = \mathcal{B}_i(u; x, w, z) - \mathcal{Q}_i(u; z).$$

Then, for every subset I of the slots and the same points $w \in \mathcal{W}_u$,

$$\left| L^S(w, \chi_\bullet(u)) \prod_{i \in I} \mathcal{D}_i(u) \right| \ll U^{a-1/2+O(e)+\epsilon} (3+T_1)^C \prod_{i \in I} P_i^{z_r - 1/2 + O(e) + \epsilon}. \tag{15.11}$$

The main parts retain the original row u, all zero masks, and the fixed identity-ray restriction. In particular, their central normalization is

$$\mathcal{Q}_i(u; z) = -P_i^{z-1/2} \left(P_i^{-1/2} \sum_{p \in \mathcal{P}_i(Z)} \overline{\chi_p(u)} W_i(q_p/P_i)(q_p/P_i)^{z-1} \right).$$

At the same points $w \in \mathcal{W}_u$, and on this region only, the full correction has the finite decomposition

$$\mathfrak{H}_{\eta,u,Z} = \mathcal{H}_{\eta,u} \prod_{i=1}^{K} (\mathcal{Q}_i + \mathcal{D}_i) = \mathcal{H}_{\eta,u} \sum_{I \subseteq \{1,\dots,K\}} \prod_{i \in I} \mathcal{D}_i \prod_{i \notin I} \mathcal{Q}_i. \tag{15.12}$$

Proof. We first obtain nonvanishing of the local correction and bound the errors at primes off the row. At ramified primes, the local expansion leaves one strict term; the conductor deficit controls all selected strict terms simultaneously.

Nonvanishing and errors off the row. In this region $\vartheta = (-w_r)_+ \leq 6e$. The estimates in the proof of Lemma 6.2 sharpen to $H_p - 1 = O(Q^{-1-10e})$ for $p \nmid u$ and $H_p - 1 = O(Q^{-10e})$ for $p \mid u$. A sufficiently large fixed P_0 therefore gives $|H_p - 1| < 1/2$ for all these primes. The holomorphic factor G_p in (15.2), divided by H_p, now defines $\mathcal{B}_p$ throughout this region. This lower bound is for H_p; the raw factor P_p may be singular when $w_r < 0$.

Consequently the tuple sum in (15.4) factors as $\mathcal{H}_{\eta,u} \prod_i \mathcal{B}_i$ on this region. Substituting $\mathcal{B}_i = \mathcal{Q}_i + \mathcal{D}_i$ proves (15.12). This factorization is not used to continue the correction outside the present region.

For $p \nmid u$, (15.9) and the estimate for $\mathcal{E}_p$ give

$$\begin{aligned} |H_p(\mathcal{B}_p + \overline{\chi_p(u)})| &\ll Q^{-x_r+2\vartheta} + Q^{-6z_r+2\vartheta} + Q^{4-5x_r-6z_r+2\vartheta} + Q^{1-w_r-6z_r+\vartheta} \\ &\ll Q^{-51/100}. \end{aligned}$$

The four exponents are bounded respectively by $-a-4e$, $-51/25+12e$, $49/25-5a-68e$, and $a-51/25+12e$. Division by the bounded H_p preserves this estimate. The number of primes in a slot is $O(P_i)$ by the ideal count; hence its total contribution from $p \nmid u$, including Q^{z-1}, is $O(P_i^{z_r-51/100+\epsilon})$.

Ramified local terms. For $p \mid u$ there are only divisor-many possible labels. A monomial in $\mathcal{E}_p$ is multiplied by Q^{z-1+x} in $Q^{z-1}\mathcal{B}_p$, so after separating Q^z its exponent is $x_r - 1$ plus the

exponent of that monomial. For the non-tail boundary terms of the six-valuation table, these exponents at $e = 0$ are

j	1	2	3	4	5
	$a-2$	$1/2-2a$	$-a,\ 1-3a$	$1/2-2a$	$2-5a.$

At positive e their changes are respectively $+6e$, $-32e$, $(-26e, -48e)$, $-42e$, $-80e$. They are all strictly below $-1/2$ in the stated range. The common R term is smaller still: its exponent after separating Q^z is at most $49/25 - 1 - 5a - 80e < -1/2$. Additional valuation pairs and m values have ratios $|R| \le Q^{-11/10}$ and $|V| = Q^{-51/25}$, so their geometric sums preserve these bounds. The strict second-family term with an additional V is also smaller than $Q^{-1/2}$ after this normalization.

The sole remaining term is the strict $(e_0, l, k, m) = (1, 0, 1, 0)$ term, which occurs for $j \ge 2$ and is $-\eta(p)(Q-1)Q^{-x-w}$. Its exponent after separating Q^z is $-w_r$. The explicit rescaling term $-Q^{-w}$ has exponent after separating Q^z $-1 - w_r < -1/2$, so it needs no further estimate.

The conductor and simultaneous allocation. Let $\mathfrak{f}_u$ be the conductor of ψ_u^*. Write $u = \epsilon_u \prod_{p|u} p^{j_p}$, with ϵ_u a unit. For primary n coprime to uS, reciprocity gives

$$\chi_n(u) = \epsilon_u^{(q_n-1)/6} \prod_{p|u} \mathcal{R}(p, n)^{j_p} \chi_p(n)^{j_p}.$$

The unit exponent is read modulo six. The prime formula extends to composite n: when $A \equiv B \equiv 1 \pmod 6$,

$$\frac{AB-1}{6} \equiv \frac{A-1}{6} + \frac{B-1}{6} \pmod 6.$$

Thus the unit factor depends only on q_n modulo 36, and the $\mathcal{R}$ product depends only on n modulo 4.

Together with the multiplicativity of $\mathcal{R}$ and the local symbols, the displayed norm congruence makes the right side multiplicative on integral ideals prime to $6u$; extend it to their fractional ideal group. A generator congruent to one modulo $(36)\,\mathrm{rad}(u)$ is already primary and makes every displayed factor one. This character therefore has a ray presentation with that fixed $2, 3$ modulus times $\mathrm{rad}(u)$. It agrees with the Kummer character $\chi_\bullet(u)$ of Lemma 3.2 on ideals avoiding S. These characters induce the same primitive character: in the principal ideal ring $\mathcal{O}$, the Chinese remainder theorem supplies a representative avoiding the additional finite set S in every ray class of a common modulus.

At a good prime $p \mid u$, hold the fixed class and all other good residues at one and vary the residue modulo p by the Chinese remainder theorem. The remaining local character $\chi_p^{j_p}$ has exact order $6/\gcd(6, j_p) > 1$. Its conductor exponent at p is therefore exactly one: the displayed ray modulus has only the first power of p, and the character cannot descend to a modulus omitting p. This is a local assertion; the ideal character may still need the fixed normalization modulus at $2, 3$.

It follows that, for any set J_0 of distinct selected ramified labels,

$$q_{\mathfrak{f}_u} \ll_S q_{\mathrm{rad}(u)} \le q_u \prod_{p \in J_0} q_p^{-(j_p-1)}. \tag{15.13}$$

The functional equation for the primitive numerator [10, Equation (1.1)], the assumed reflected bound, and Stirling give the bound $q_{\mathfrak{f}_u}^{A_*} U^{\epsilon}(3 + T_1)^C$, where

$$A_* = 1/2 - w_r = a - 1/2 + 6e > 0.$$

At a selected $p \mid u$ the primitive character already has value zero, so restoring the original local factor there multiplies by exactly one. The remaining deleted Euler factors have radical $O_S(U)$ and contribute a factor at most $U^{6e+\epsilon}$, because $w_r \ge -6e$.

Expanding a product of error slots separates the coprime-prime terms, the already bounded ramified terms, and the strict ramified terms. For each fixed tuple in this triangle expansion, let J_0 be precisely its strict ramified labels. They are distinct because the slot supports are disjoint. The single inequality (15.13) applies to the entire J_0 before the labels are summed. The numerator together with these local factors, including q_p^{z-1} at each selected prime, is then bounded by

$$U^{A_*+6e+\epsilon}(3+T_1)^C \prod_{p\in J_0} q_p^{z_r-w_r-(j_p-1)A_*} \le U^{a-1/2+12e+\epsilon}(3+T_1)^C \prod_{p\in J_0} q_p^{z_r-1/2},$$

because $j_p \ge 2$ and $-w_r - A_* = -1/2$. Thus different selected labels use different factors of the conductor deficit, and the bounds hold simultaneously. Summing the divisor-many ramified labels introduces the factor U^ϵ, while the coprime labels have the stronger exponent $z_r - 51/100$. This proves (15.11).

The triangle inequality has been applied only to error labels; the original row u and every main-slot coefficient and mask remain unchanged. The tuple-independent factor $\mathcal{H}_{\eta,u}$ is separately bounded by U^ϵ by Lemma 6.2. Relative to the central scale $P_i^{z_r-1/2}$ of a main slot, an error slot thus has no positive amplitude exponent in the later row count. □

15.3 The principal local factor

In the second Euler region of Lemma 6.2, take $u = 1$ and $p \in 1_T$. Then $v = 1$ in (15.9), and H_p is bounded away from zero after the same fixed enlargement of P_0. For this principal row on this region, define $\mathcal{B}_p = G_p/H_p$ and define $\mathcal{B}_i$ by (15.3). The lower bound and the disjoint supports give the separate principal factorization

$$\mathfrak{H}_{\eta,1,Z}(x,w,z) = \mathcal{H}_{\eta,1}(x,w,z) \prod_{i=1}^{K} \mathcal{B}_i(1;x,w,z) \qquad \text{in the second Euler region.} \tag{15.14}$$

This principal-row factorization is separate from the central main/error decomposition, whose scope remains the moving-contour region. The four error exponents are now

$$-x_r, \qquad -6z_r, \qquad 4-5x_r-6z_r, \qquad 1-w_r-6z_r.$$

Each is at most $-7/8$ in that region. Hence the principal multiplier satisfies

$$\mathcal{B}_p = -1 + O(Q^{-7/8}), \tag{15.15}$$

uniformly in the imaginary parts and the target unit phase. This is the local estimate used to normalize the principal residue.

It also gives the absolute principal correction required by the shared residue estimate. Indeed $G_p = H_p\mathcal{B}_p = O(1)$ here, and the unselected product has a bounded positive majorant by the second-region version of (15.7). Ideal counting in each annular slot therefore gives, on every fixed real box in the second Euler region and uniformly in all imaginary parts,

$$\begin{aligned} |\mathfrak{H}_{\eta,1,Z}(x,w,z)| &\le \sum_{(p_i)\in\prod_i \mathcal{P}_i(Z)} \prod_i \left[|W_i(q_{p_i}/P_i)| q_{p_i}^{z_r-1} |G_{p_i}|\right] \prod_{\substack{p\notin S \\ p\notin\{p_1,\dots,p_K\}}} |H_p| \\ &\ll \prod_i P_i^{z_r} = Z^{\ell z_r}. \end{aligned} \tag{15.16}$$

The lower bound used for this local approximation concerns the principal row in the second Euler region; the absolute correction estimate remains an upper bound for the full tuple sum.

15.4 Absolute bounds for the remaining contour lines

The remaining contours require bounds for the full correction. The following estimates retain each selected factor G_p before taking absolute values; the main/error decomposition remains restricted to its stated region.

Lemma 15.2 (Absolute local tuple bounds). *Assume (11.1). Let $Z \geq 2$, and let u be a nonzero sixth-power-free row with $(u, S) = 1$ and $q_u \asymp U \geq 1$, with fixed comparison constants. Let the slot system be the fixed system of Section 11. The following bounds are uniform in all imaginary parts and unit phases.*

1. *On*
$$x_r = \beta_* + e, \qquad w_r = 1/2, \qquad z_r = 17/50, \qquad 0 < e \leq 10^{-3},$$
one has $G_p = O(1)$ at selected primes $p \nmid u$ and $G_p \ll Q^{1/2}$ at selected primes $p \mid u$.

2. *For every fixed $z_\infty > 2$, on*
$$x_r = w_r = 2, \qquad z_r = z_\infty,$$
one has $G_p = O_{z_\infty}(1)$ at every selected prime.

In either case the positive sum over the full selected tuples satisfies

$$\sum_{(p_i)\in\prod_i \mathcal{P}_i(Z)} \prod_i \left[|W_i(q_{p_i}/P_i)|q_{p_i}^{z_r-1}|G_{p_i}|\right] \prod_{\substack{p\notin S\\ p\notin\{p_1,\dots,p_K\}}} |H_p| \ll_\epsilon U^\epsilon \prod_i P_i^{z_r}. \tag{15.17}$$

The constants may depend on the fixed data, e, and, in the second case, z_∞, but not on Z, u or the imaginary parts. These estimates remain valid at zeros of P_p or H_p.

Proof. Both sets of lines lie in the first Euler region with $\epsilon_0 = 3/8$. Its primewise defect estimates give $H_p = O(1)$, while (15.7) bounds the positive product of all unselected factors by $O_\epsilon(U^\epsilon)$.

Consider first $x_r = \beta_* + e$, $w_r = 1/2$, and $z_r = 17/50$. For a selected $p \nmid u$, (15.9) gives

$$|G_p + \overline{\chi_p(u)}H_p| \ll Q^{-x_r} + Q^{-6z_r} + Q^{4-5x_r-6z_r} + Q^{1-w_r-6z_r}.$$

The four exponents are at most $-7/8 - e$, $-51/25$, $-483/200 - 5e$, $-77/50$, respectively. Consequently $G_p = O(1)$ off u, without division by H_p.

For a selected $p \mid u$, one has $W = D = 0$ and $\mathcal{E}_p = P_p^*$. The local identity (15.2) becomes

$$G_p = \overline{\eta(p)}Q^x(1 - V)\mathcal{E}_p - Q^{-w}H_p.$$

After multiplication by Q^{x_r}, the strict term of the local table has exponent at most $1 - w_r = 1/2$; for $j = 1$ it also contains the factor V. The exponents of the boundary terms J_j, in order $j = 1, \dots, 5$, are

$$-\frac{1}{2}, \quad \frac{3}{2} - 2x_r, \quad \left(\frac{3}{2} - 2x_r,\ 2 - 3x_r\right), \quad 2 - 3x_r, \quad 3 - 5x_r.$$

They are bounded above by $-1/2$, $-1/4 - 2e$, $(-1/4 - 2e, -5/8 - 3e)$, $-5/8 - 3e$, $-11/8 - 5e$, respectively. The common R term has exponent $4 - 5x_r - 6z_r \leq -483/200 - 5e$. All further terms in the geometric families decrease these powers, since $|R| \leq Q^{-329/100-6e}$ and $|V| = Q^{-51/25}$. Finally $Q^{-w}H_p = O(Q^{-1/2})$. Hence $G_p \ll Q^{1/2}$ on u, also at a zero of H_p.

The labels dividing u are divisor-many, while a slot contains $O(P_i)$ ideals. Its positive sum is therefore

$$\sum_{p\in\mathcal{P}_i(Z)} |W_i(q_p/P_i)|q_p^{z_r-1}|G_p| \ll P_i^{z_r} + U^\epsilon P_i^{z_r-1/2} \ll U^\epsilon P_i^{z_r}. \tag{15.18}$$

Using the positive majorant for the unselected product and distributing the requested ϵ among the fixed number of slots proves (15.17) on the first lines.

Now fix $x_r = w_r = 2$ and $z_r = z_\infty > 2$. For a selected prime off u, the four exponents in (15.9) are -2, $-6z_\infty$, $-6-6z_\infty$, $-1-6z_\infty$, so $G_p = O_{z_\infty}(1)$. On u, the strict term after multiplication by Q^{x_r} has exponent at most $1-w_r = -1$. The boundary exponents are -2, $-5/2$, $(-4,-4)$, $-11/2$, -7, the common R exponent is $-6-6z_\infty$, and $|R| = Q^{-8-6z_\infty}$, $|V| = Q^{-6z_\infty}$. The rescaling term is $O(Q^{-2})$. Thus $G_p = O_{z_\infty}(1)$ on u as well. A positive slot sum is consequently $O_{z_\infty}(P_i^{z_\infty})$. Together with (15.7), this proves (15.17) on the second lines. □

The identity (15.6) supplies the full high representation of $I_{\eta,\mathrm{modified}}$, whose low bound is Proposition 14.3. Proposition 15.1 controls the error factors jointly with the numerator, and Lemma 15.2 supplies the bounds on the remaining contour lines. The main factors are the prime polynomials displayed in Proposition 15.1. To count rows on which those factors and a detector witness are large, we next prove the inverse and fourth-moment estimates used in Section 18.

16 The inverse moment with prime factors

The inverse Dirichlet polynomial from Section 7 must be estimated together with selected prime factors from the local compensation. We bound its second moment after multiplication by independently weighted prime sums. The completed row energy of Section 13 supplies the terminal bound in a finite induction. A final amplification gives an unmarked estimate on sixth-power-free rows at longer column lengths.

16.1 Statement of the marked estimate

Let ν be a fixed finite-order ray character whose full zero-extended defining modulus has prime support in S. This entire presentation belongs to the fixed arithmetic datum. A locally fixed moving zero support is retained as a coprimality restriction; the norm of its excluded-prime radical is included in the explicit bound below. Fix one sign $\varepsilon_\chi \in \{1,-1\}$, and set

$$\psi_u(n) = \nu(n)\chi_n(u)^{\varepsilon_\chi}.$$

The only masks in this definition are the fixed exclusions and the zero extension of $\chi_n(u)$. For a smooth annular function W, the inverse polynomial in (3.3) at scale Z^r is

$$M_{\psi_u}(Z^r;W) = Z^{-r/2}\sum_n \mu(n)\psi_u(n)W(q_n/Z^r).$$

For a fixed finite set I of slots, let $\mathcal{P}_i$ be disjoint sets of primes outside S, with $q_p \asymp Z^{z_i}$ for $p \in \mathcal{P}_i$. The sets and the coefficients $a_i(p)$ are independent of u, and $|a_i(p)| \le 1$. Given fixed smooth annular functions W_i, put

$$Q_u = \prod_{i\in I} Z^{-z_i/2}\sum_{p\in\mathcal{P}_i} a_i(p)\psi_u(p)W_i(q_p/Z^{z_i}), \qquad z = \sum_{i\in I} z_i.$$

The empty product is one. A fixed finite sum of whole products $M_{\psi_u}(Z^r;W)Q_u$ is also allowed by the triangle inequality, provided each summand uses one common ν in its inverse and all of its slots.

Lemma 16.1 (Marked inverse moment). *Let $M_{\psi_u}(Z^r;W)$ and Q_u be the inverse polynomial and product of prime sums defined in Section 16.1, with the character, masks, tests, and prime-slot data specified there. Fix bounded ranges for the nonnegative real parameters m, r, z_i, a bound for $|I|$, and positive constants c_1, c_2. The slot lengths range over all values in these ranges for which the stated prime supports exist. Suppose*

$$r + 2z \le m - c_1, \qquad 2r + 8z \le 3m - c_2.$$

Then, for every $\epsilon > 0$,

$$\sum_{\substack{u\in\mathcal{O}\\ 0<q_u\ll Z^m}} |M_{\psi_u}(Z^r;W)Q_u|^2 \ll Z^{m+\epsilon}. \tag{16.1}$$

The implied constant may depend on the fixed arithmetic data, c_1, c_2, the bounded parameter ranges, and finitely many smooth seminorms of the tests, but is uniform in the prime supports and their bounded coefficients. The assertion holds for either common sign ε_χ, and for every subcollection of the slots. For norm twists of the tests, the implied constant has fixed polynomial dependence on their heights.

The proof uses the canonical family defined next, which averages a fourth-power residue symbol over an additional squarefree ideal. The argument has three stages.

1. Lemma 13.3 bounds the short completions in the proof of the canonical estimate, Lemma 16.2.

2. In the remaining ranges, two masked Poisson transformations produce admissible canonical sums with shorter rows, after the principal contributions and tails have been bounded. This reduction closes a finite induction.

3. One initial masked Poisson transformation converts the marked inverse moment to the canonical family and proves Lemma 16.1. Lemma 16.6 then amplifies its unmarked case to longer column lengths on sixth-power-free rows.

16.2 Moment averaged over squarefree fourth-power labels

The canonical energy averages the squarefree label f as well as the element row k. Its column coefficient is $\bar{\alpha}(n)\gamma_2(n)$ from Section 3, multiplied by the fixed ray character and coprimality restriction of Section 13. The mark in (13.1) records which prime tuples divide the column, with one bounded coefficient for each prime slot. The label weight depends on f alone. Thus the column coefficients and their prime supports are fixed while the row and squarefree label are averaged.

For row length M, column length N, label length V, and data satisfying the hypotheses of Lemma 16.2, the energy to be estimated is

$$\mathcal{E} = Z^{-V}\sum_{f} w(f) \sum_{\substack{k\in\mathcal{O}\\ 0<q_k\ll Z^M}} \left| Z^{-N/2}\sum_{n\ \mathrm{sf}} \bar{\alpha}(n)\gamma_2(n)\nu(n)\rho(n)\times \chi_n(k)\chi_n(f)^4\mathfrak{d}(n)W(q_n/Z^N)\right|^2. \tag{16.2}$$

Here the column sum is squarefree, the label range is $q_f \asymp Z^V$, and the factor Z^{-V} normalizes that label average. The factor $Z^{-N/2}$ normalizes each column polynomial.

The norm of the excluded-prime radical is bounded explicitly in Lemma 16.2, including when its modulus was fixed at an earlier stage of the recursion. For the recursive Poisson large-sieve framework, compare [15, Section 2], [16, Section 4], and [12, Sections 4–7]; the present marked recursion is proved below.

Lemma 16.2 (Marked moment with a fourth-power label). *Fix bounded nonnegative ranges for M, N, V, z_0, a bound for the number of slots, and $c_* > 0$. For the energy $\mathcal{E}$ in (16.2), let f range over squarefree ideals outside S with $q_f \asymp Z^V$. Let w be a nonnegative function of f alone satisfying $w(f) \le C d_{\mathcal{O}}(f)^C$ for one fixed $C \ge 1$. Neither C nor the choice of w depends on Z, the current rows, or any other averaged variable. The implied constant below may depend on C, but is uniform over all w satisfying this fixed bound.*

Let ν be a fixed multiplicative finite ray character whose full zero-extended defining modulus has prime support in S, and let $\rho(n) = 1_{(n,r_\rho)=1}$ be a fixed coprimality restriction. Both are independent of k, f.

Let $\mathfrak{d}$ have the product form in (13.1), with disjoint fixed prime lists of total length at most z_0 and bounded coefficients independent of k, f. No other row-dependent column coefficient or coprimality restriction is allowed. There is no additional residual coefficient $b(n)$, even one independent of k, f: arbitrary bounded weights are permitted only as the stated product of individual prime-slot coefficients.

Put $F_0 = N + V$. Suppose

$$\begin{aligned} q_{r_\rho} &\le Z^{F_0 - M - z_0 - c_*}, \\ 3M + 6z_0 + c_* &\le 4F_0. \end{aligned} \tag{16.3}$$

Then, for every smooth annular W and every $\epsilon > 0$, $\mathcal{E} \ll Z^{F_0+\epsilon}$. The constant is uniform in the moving moduli and the fixed outer ideals within the stated ranges. It depends on finitely many smooth seminorms and has a fixed polynomial dependence on separated norm-twist heights. The same conclusion holds for all subcollections of the slots with the original cap z_0, including the empty collection.

The reflected energy in Section 13 supplies the terminal estimate for this family. The remaining ranges will be reduced to the same family with shorter rows. Before the two Poisson transformations, we give the common analytic rule for propagating smooth seminorm and height orders through the finite induction.

16.3 Finite propagation of seminorm and height orders

Finite induction requires bounds for the transformed profiles and their Fourier coefficient measures that are uniform in the arithmetic labels. Lemma 16.3 propagates these bounds through a finite graph, so that the internal derivative orders can be chosen before an external height cutoff. Corollary 16.4 allows the outer labels and their normalized measures to vary with Z.

16.3.1 Propagation through a finite graph

We use the profile seminorms p_j defined before Lemma 3.6, and write $\langle t\rangle = 1 + |t|$. Let a finite rooted directed graph have no directed cycles, and let every directed path have at most D edges. At each vertex v let $\mathcal{N}_v(Z, \mathbf{w}, t)$ be a nonnegative quantity, where $Z \ge 1$, $\mathbf{w}$ is a fixed finite tuple of annular profiles, and t is a finite-dimensional height vector. The profile dimensions, supports, and height dimensions may depend on v, but are fixed throughout the graph. Additional labels are allowed in these quantities; every bound below is required uniformly in those labels.

Terminal and transition bounds. Suppose that at each vertex there is a terminal bound

$$\mathcal{T}_v(Z, \mathbf{w}, t) \le C_v Z^{a_v} p_{j_v}(\mathbf{w})^{b_v} \langle t\rangle^{h_v}$$

with fixed real a_v and finite nonnegative j_v, b_v, h_v. Suppose also that

$$\mathcal{N}_v(Z, \mathbf{w}, t) \le \mathcal{T}_v(Z, \mathbf{w}, t) + \sum_{e: v \to v'} Z^{a_e} \int_{\mathbb{R}^{d_e}} |K_e(Z, \mathbf{w}, t; \xi)| \mathcal{N}_{v'}(Z, \mathbf{W}_e, A_e t + B_e \xi)\, d\xi,$$

where A_e, B_e are fixed bounded linear maps and $\mathbf{W}_e = \mathbf{W}_e(Z, \mathbf{w}, t, \xi)$ is the child profile tuple. Empty edge sums are allowed. Assume the following two bounds for each edge.

For every fixed j there are finite nonnegative numbers $m_e(j), b'_e(j), c'_e(j)$ and a constant $C_{e,j}$ such that

$$p_j(\mathbf{W}_e) \le C_{e,j} p_{m_e(j)}(\mathbf{w})^{b'_e(j)} \langle t\rangle^{c'_e(j)} \langle \xi\rangle^{c'_e(j)}.$$

For every fixed $H \ge 0$ there are finite nonnegative numbers $\ell_e(H), b_e(H), c_e(H)$ and a constant $C_{e,H}$ such that

$$\int_{\mathbb{R}^{d_e}} |K_e(Z, \mathbf{w}, t; \xi)| \langle \xi\rangle^H\, d\xi \le C_{e,H} p_{\ell_e(H)}(\mathbf{w})^{b_e(H)} \langle t\rangle^{c_e(H)}.$$

The seminorm indices may be rounded up to integers. All these constants and indices are uniform in Z and the additional labels; the exponents a_v, a_e are fixed real numbers independent of the requested seminorm and height orders.

We also allow transitions with a finite product of child quantities. In this case an edge term has the form

$$Z^{a_e}\int_{\mathbb{R}^{d_e}} |K_e| \prod_{r=1}^{r_e} \mathcal{N}_{v_{e,r}}(Z, \mathbf{W}_{e,r}, A_{e,r}t + B_{e,r}\xi)^{\theta_{e,r}}\, d\xi,$$

where $r_e \geq 1$ is a fixed integer and $\theta_{e,r} \geq 0$ are fixed, all children have smaller remaining depth, and each child profile satisfies its own version of the stated profile bound. The joint coefficient measure satisfies the same weighted bound.

Lemma 16.3 (Finite seminorm propagation). *Let the finite rooted acyclic graph and the nonnegative quantities $\mathcal{N}_v$ be as in Section 16.3.1. Assume the terminal bound, the transition inequality, and both profile and coefficient-measure bounds stated there, uniformly in the additional labels. For the linear transitions, there are finite J, B, H and C such that, at the root v_0,*

$$\mathcal{N}_{v_0}(Z, \mathbf{w}, t) \leq CZ^E p_J(\mathbf{w})^B \langle t\rangle^H, \qquad E = \max_{v_0\to\cdots\to v} \left(a_v + \sum_{e \text{ on the path}} a_e \right).$$

The finite indices J, B, H are independent of any external cutoff $T_1 = Z^\tau$ in a restriction $|t| \leq T_1$.

For the product transitions, define the norm exponent recursively by

$$E_v = \max\left\{ a_v, \max_e \left(a_e + \sum_{r=1}^{r_e} \theta_{e,r} E_{v_{e,r}} \right) \right\}.$$

The conclusion holds with $E = E_{v_0}$.

The product form includes the square roots of two nonnegative child estimates arising from Cauchy–Schwarz.

Proof. At a terminal vertex the assertion is its stated bound. Suppose a child has already been bounded by $CZ^{E'} p_{J'}(\mathbf{W}_e)^{B'} \langle A_e t + B_e\xi\rangle^{H'}$. Since the linear maps are bounded,

$$\langle A_e t + B_e\xi\rangle^{H'} \ll_e \langle t\rangle^{H'} \langle \xi\rangle^{H'}.$$

The child-profile bound adds the power $B'c'_e(J')$ to each of these two height exponents and replaces its profile factor by $p_{m_e(J')}(\mathbf{w})^{B'b'_e(J')}$. The coefficient-measure bound at $H_e = H' + B'c'_e(J')$ therefore bounds the contribution of this edge by

$$C_e Z^{a_e+E'} p_{\max\{m_e(J'),\ell_e(H_e)\}}(\mathbf{w})^{B'b'_e(J')+b_e(H_e)} \langle t\rangle^{H_e+c_e(H_e)}.$$

Every displayed index is finite. Their maximum, together with the terminal indices, controls all finitely many outgoing edges. The profile exponent may be increased when necessary because $p_j(\mathbf{w}) \geq 1$. The resulting parent bound has exponent equal to the maximum path exponent through that vertex. Reverse induction on the acyclic graph completes the proof. None of these choices mentions an external cutoff or a late derivative order.

For the product form, the already proved bound of each child contributes with power $\theta_{e,r}$. The total power of $\langle\xi\rangle$ that the coefficient measure must integrate is

$$H_e = \sum_{r=1}^{r_e} \theta_{e,r}\big(H_r + B_r c'_{e,r}(J_r)\big).$$

The powers of the parent profile and of $\langle t\rangle$ are the corresponding finite sums, followed by the single coefficient-measure bound at H_e. The powers of Z add as in the displayed recurrence for E_v. Taking maxima over the finitely many terms proves this extension by the same reverse induction. □

16.3.2 Indexed outer measures

The next form normalizes each quantity by its desired norm exponent. Its edge measure includes the outer labels left after Cauchy–Schwarz, while labels already averaged in a child remain there.

Corollary 16.4 (Indexed finite propagation). *For the graph and transition data in Section 16.3.1, let ℓ denote any admissible length and outer labels at a vertex v, and let $\Phi_v(\ell)$ be its desired norm exponent. Suppose the terminal bound, after division by $Z^{\Phi_v(\ell)}$, has the terminal form stated there with a fixed exponent δ_v. An edge may be indexed by a Z-dependent family Γ_e with a nonnegative measure ν_e, and may have the form*

$$\int_{\Gamma_e}\int_{\mathbb{R}^{d_e}} Z^{a_e(\ell,\gamma)}|K_{e,\gamma}|\prod_{r=1}^{r_e}\mathcal{N}_{v_{e,r}}(Z,\ell_{e,r},\mathbf{W}_{e,r},A_{e,r}t+B_{e,r}\xi)^{\theta_{e,r}}\,d\xi\,d\nu_e(\gamma),$$

where the child labels $\ell_{e,r}$ may depend on ℓ,γ. Assume uniformly in those labels that

$$a_e(\ell,\gamma)+\sum_r \theta_{e,r}\Phi_{v_{e,r}}(\ell_{e,r})-\Phi_v(\ell)\le\delta_e, \tag{16.4}$$

with fixed δ_e, and that the child-profile bounds in Section 16.3.1 hold. Assume also that for every fixed $H\ge 0$,

$$\int_{\Gamma_e}\int_{\mathbb{R}^{d_e}}|K_{e,\gamma}|\langle\xi\rangle^H\,d\xi\,d\nu_e(\gamma)\le C_{e,H}Z^{\delta_e^{\mathrm{mass}}}p_{\ell_e(H)}(\mathbf{w})^{b_e(H)}\langle t\rangle^{c_e(H)}, \tag{16.5}$$

where δ_e^{mass} is fixed independently of H. When the quantities are squared Hilbert-space row norms obtained by separating a profile, this hypothesis must use the coefficient measure common to those rows. There need only be finitely many edge types at each of the finitely many depths; the cardinalities of the Γ_e may vary with Z.

Then $\mathcal{N}_v/Z^{\Phi_v(\ell)}$ has the conclusion of Lemma 16.3, uniformly in ℓ, with terminal exponents δ_v and edge exponents $\delta_e+\delta_e^{\mathrm{mass}}$. In particular all required seminorm and height orders are finite and independent of an external cutoff.

Proof. Set $\overline{\mathcal{N}}_v=Z^{-\Phi_v(\ell)}\mathcal{N}_v$. Substitution in an edge and (16.4) bound its norm power by Z^{δ_e} times the product of the normalized children. Their inductive bounds give, as in the product proof of Lemma 16.3, the coefficient moment

$$H_e=\sum_r\theta_{e,r}\big(H_r+B_rc'_{e,r}(J_r)\big),$$

which is finite and independent of γ. The coefficient-measure bound (16.5) at this order contributes $Z^{\delta_e^{\mathrm{mass}}}$ and only finite profile and height orders. Backward induction over the finite edge types therefore gives the stated normalized recurrence. A label averaged inside a child remains inside that child throughout this argument. The outer measure ν_e integrates only the labels outside the child. □

We make explicit how discrete labels are normalized in this corollary. After the relevant weighted Cauchy inequality, suppose a nonnegative outer measure satisfies $\sum_{\gamma\in\Gamma}w_\gamma\le CZ^{c+\epsilon}$, with fixed c and ϵ selected before the height orders. For a per-label prefactor $Z^{a_{\mathrm{raw}}}$ common on the block, put

$$d\nu_\Gamma(\gamma)=Z^{-c}\sum_{\gamma\in\Gamma}w_\gamma\delta_\gamma,\qquad \nu_\Gamma(\Gamma)\le CZ^{\epsilon}.$$

Here δ_γ denotes unit point mass. For nonnegative B_γ for which the displayed integrals are finite, the identity is

$$Z^{a_{\mathrm{raw}}}\sum_{\gamma\in\Gamma}w_\gamma\int|K_\gamma(\xi)|B_\gamma(\xi)\,d\xi=Z^{a_{\mathrm{raw}}+c}\int_\Gamma\int|K_\gamma(\xi)|B_\gamma(\xi)\,d\xi\,d\nu_\Gamma(\gamma). \tag{16.6}$$

Uniform weighted moments of K_γ now give (16.5) with only the remaining Z^ϵ mass. Thus a displayed exponent that already includes the count c must use this normalized measure; retaining the unnormalized sum would count the same labels twice. The rule uses the weighted mass, which already includes the number of labels. It is applied only after any row-dependent eligibility remains within the child or has been removed by a nonnegative inequality. Labels still averaged in a child norm are not included in Γ.

16.3.3 Kernel amplitudes and normalized derivatives

There is a separate normalization for a small kernel amplitude. The single-profile seminorm $p_j(w)$ is homogeneous, whereas $p_j(\mathbf{w}) = 1 + \sum_i p_j(w_i)$ is not. Suppose a joint annular profile on a fixed block satisfies, for a scalar $0 < m_R \le 1$,

$$p_j(F_R) \le m_R C_j p_{\ell(j)}(\mathbf{w})^{b(j)} \langle t \rangle^{h(j)} \qquad (j \ge 0).$$

One may keep F_R in the Fourier coefficient measure, whose weighted L^1 norm then contains the factor m_R. Alternatively, write $F_R = m_R \widetilde{F}_R$ and put m_R outside that measure. If a polynomial or row vector $\mathcal{P}(F_R)$ is linear in this whole profile, then

$$\mathcal{P}(F_R) = m_R \mathcal{P}(\widetilde{F}_R), \qquad \|\mathcal{P}(F_R)\|_2^2 = m_R^2 \|\mathcal{P}(\widetilde{F}_R)\|_2^2.$$

The seminorms of the tuple containing $\widetilde{F}_R$ are bounded; the small factor is now the outside scalar. A scalar may be removed from a centered difference only when it multiplies the whole difference. A kernel occurring once in an already expanded quadratic expression contributes one factor m_R. The squared factor in the preceding identity comes from squaring a row vector that is linear in the whole profile. In each representation the small factor is included once, either in the measure or as the outside scalar. If the positive row range will later be enlarged, the scalar and the common profile are first fixed on the actual annuli, and the norm inequality containing that scalar is obtained before the enlargement.

Here are sufficient analytic ways to verify the two edge hypotheses. Let $m(y)$ be a fixed norm monomial and let χ be a smooth cutoff with compact logarithmic support in a product of fixed annuli. Suppose a normalized kernel has Euler bounds

$$|(r\partial_r)^j K(r)| \le C_{A,j} \mathfrak{S}_{m(A,j)} \min(1, r^{-A}),$$

where $\mathfrak{S}_k$ is an increasing family of specified finite input seminorm and polynomial height bounds, and $m(A,j)$ is taken nondecreasing in j. The product and chain rules give, uniformly for $R > 0$,

$$p_j\big(\chi(y) K(R m(y))\big) \le C'_{A,j} \mathfrak{S}_{m(A,j)} \min(1, R^{-A}). \tag{16.7}$$

Indeed $y_i \partial_{y_i}$ acting on the kernel is the fixed exponent of y_i in m times $r\partial_r$, and m is bounded above and below on the cutoff support. Differentiation therefore preserves the power of R in the bound. The same argument preserves a bound with $\min(R^{1/4}, 1, R^{-A})$, when those three kernel estimates are available. The decay order A is the same at every requested j; only the input order and its finite height degree increase. For an old window $w(cm(y))$ with $c > 0$, Euler derivatives are bounded by the global logarithmic seminorms of w, independently of the scale c at every fixed derivative order. Normalized real powers such as $y^{-1/2}$ obey the same rule. Pure twists of normalized norms insert only fixed powers of their heights.

Ordinary radial Fourier seminorms must be used only after the radial scale has been normalized. For a fixed annular w in two real dimensions, $f_R(x) = w(R|x|^2)$ satisfies

$$\|\partial^j f_R\|_1 = R^{j/2-1} \|\partial^j f_1\|_1$$

for any fixed directional derivative of order j, by $x \mapsto \sqrt{R}x$. Thus an unnormalized shrinking radial test could introduce a scale power depending on j. Sufficient hypotheses are a fixed-shape

radial Schwartz test at its stated row scale, or a radial C_c^∞ test constant near zero at that scale. After normalization, Lemma 3.12 applies to its fixed transform, and moving scale ratios enter only as in (16.7). For example $q_u^{-1/2} = U_0^{-1/2} y^{-1/2}$ on $q_u = U_0 y$; the central power belongs to the norm exponent, and the annular factor has order-independent scale bounds. Likewise $q_u^{it} = U_0^{it} y^{it}$, and the central unit phase is factored as a scalar before differentiating the normalized profile.

16.3.4 Common Fourier measures and tail orders

Logarithmic Fourier separation uses a joint cutoff in a coordinate list containing every normalized norm occurring in a coupled smooth factor. Derived nonsmooth row and label masks remain outside that factor and need no coordinate. A nonzero norm used as such a coordinate and initially ranging in a ball down to norm one must first be partitioned into common whole annuli fixed for the current sector; the zero frequency is separate. In bounded logarithmic ranges there are $O((1+\log Z)^d)$ such blocks for a fixed number d of variables, so their number is bounded by a preselected small power. The cutoff and its full support depend only on the block centers and fixed data, not on the individual values of the labels inside a row norm. Provided the coupled smooth factor is this one joint function with no further dependence on the individual labels, its Fourier density is common to those rows, as required by Lemma 3.6; separate uniform bounds for row-dependent densities would not imply the same Hilbert-space inequality. Nonsmooth masks are retained or resolved by exact arithmetic identities, never differentiated or incorporated as label-dependent sharp Fourier selectors.

Finally, suppose an internal dyadic ratio $R > Z^\xi$, with fixed $\xi > 0$, has absolute arithmetic count at most $Z^B R^b$. A coefficient bound with the factor R^{-A} gives, for $A > b$,

$$\sum_{\substack{R \text{ dyadic} \\ R > Z^\xi}} Z^B R^{b-A} \ll Z^{B-\xi(A-b)}.$$

Choose A from the fixed B, b, ξ and the desired internal saving. The actual bound may also have finite input seminorm and polynomial height factors; these enter the finite propagation graph. These finite height factors remain part of the tail bound. Parameter Sobolev adds only a fixed number of profile derivatives and height dimensions, and all these internal orders are chosen before an external cutoff.

The external height cutoff. Suppose the graph and all of its kernel and annular orders have been fixed, giving height degree H. Suppose an additional, external separation uses annular profiles with p_j bounded uniformly in Z for every fixed j, or nonannular profiles with the weighted Mellin or annular norms in the preceding lemmas uniformly bounded at every fixed order, and its discarded integrand has bound $Z^{B_0}\langle t\rangle^{J_0}$ with fixed B_0, J_0. The external truncation is required to remain outside the graph, leaving every internal profile w and kernel K_e independent of that cutoff. Given a retained-height exponent $\delta > 0$, first choose $0 < \tau < \delta/(1+H)$. Then $(1+T_1)^H \le 2^H Z^\delta$ for $T_1 = Z^\tau$. After this choice, (3.13) with weight J_0 and any fixed integer $N > (B_0 + M)/\tau$ makes an integrated external tail $O(Z^{-M})$ for any prescribed $M > 0$. For a horizontal join use (3.14) with the additional fixed height weight, or (3.16) for a joint Gaussian slice. When other coordinates on such a slice are extended to the whole real line, the accompanying arithmetic factors must have global bounds on those coordinates; a coordinate entering a denominator controlled only in a buffered region must remain restricted to that region. The external constants may depend on N and on the fixed data. This late choice does not alter any internal profile or the internal orders J, H, which is the asserted order of dependence.

16.4 Finite Poisson summation with a mask

The canonical reduction and the marked initialization use the following form of Poisson summation, with the arithmetic conventions of Section 3. The reduction applies it twice, and the initialization

once. Its divisor variable retains every zero extension; the proof also gives the principal-frequency bound needed when the transformed expressions are separated.

Lemma 16.5 (Masked primitive Poisson). *Let ψ be a primitive finite character modulo an ideal m, extended by zero on nonunits, and let R be any ideal. Let $\Phi(q_z)$ be a radial Schwartz function on $\mathbb{C}$, and let $\mathcal{F}\Phi(q_y)$ denote its Fourier transform for the self-dual measure and the kernel $e(-zy)$. For $K > 0$,*

$$\sum_{k\in\mathcal{O}} \psi(k)1_{(k,R)=1}\Phi(q_k/K) = \frac{K\gamma(\psi;m)}{\sqrt{q_m}} \sum_{d|\mathrm{rad}\,R} \frac{\mu(d)\psi(d)}{q_d} \sum_{h\in\mathcal{O}} \bar{\psi}(h)\mathcal{F}\Phi\left(\frac{Kq_h}{q_d q_m}\right), \tag{16.8}$$

where $\gamma(\psi;m) = q_m^{-1/2}\sum_{x \bmod m}\psi(x)e(x/m)$. For a nonprincipal primitive character, $|\gamma(\psi;m)| = 1$ and the $h = 0$ term is zero. For the primitive principal character the conventions are $m = 1$, $\gamma(1;1) = 1$, $\psi(h) = 1$, including at $h = 0$.

Proof. Expanding the mask as $\sum_{d|\mathrm{rad}\,R,\,d|k}\mu(d)$ and writing $k = dk'$ changes the scale to K/q_d and contributes the character factor $\psi(d)$. Poisson summation in residue classes modulo m has prefactor $K/(q_d q_m)$. Its finite transform is

$$\sum_{x \bmod m} \psi(x)e(hx/m) = \sqrt{q_m}\gamma(\psi;m)\bar{\psi}(h).$$

For unit h this follows by changing variables. For nonunit h the transform is zero by primitivity; for $m = 1$ the stated principal convention applies. This proves (16.8). For a nonprincipal primitive character, finite Parseval gives $\sum_h |\sum_x \psi(x)e(hx/m)|^2 = q_m\sum_x |\psi(x)|^2$. Writing $\varphi(m) = |(\mathcal{O}/m)^\times|$, there are $\varphi(m)$ unit h's, all with magnitude $\sqrt{q_m}|\gamma(\psi;m)|$, and $\sum_x |\psi(x)|^2 = \varphi(m)$. Thus $|\gamma(\psi;m)| = 1$.

A fixed ray restriction can first be expanded into finitely many characters. Each is then replaced by its primitive inducing character, with the removed local zero extensions kept in R; the same formula applies term by term. For the principal modulus the absolute sum of the nonzero frequencies is $O_\Phi(d_\mathcal{O}(\mathrm{rad}\,R))$. Indeed, for all $A > 0$,

$$A\sum_{h\neq 0} |\mathcal{F}\Phi(Aq_h)| \ll_\Phi 1.$$

Lattice counting proves this when $A \le 1$, and Schwartz decay proves it when $A \ge 1$. Summing this bound with $A = K/q_d$ over d gives the claimed divisor bound. At every use below the relevant nonempty row scale has $K \ge 1$, so this is also $O(KZ^\epsilon)$. When principal nonzero frequencies are added or removed while separating a nonprincipal sum, they will carry the same outer row mask as that sum; their absolute contribution is no larger than this full principal bound. □

16.5 Poisson summation and induction on the row length

Proof of Lemma 16.2. The induction is on the row length. The reflected estimate closes a short completion. For every remaining block, the first Poisson transformation gives a positive norm of inverse polynomials; the second returns the canonical coefficient $\bar{\alpha}\gamma_2$, with a shorter row range and a controlled excluded-prime radical. We check the parameters of that energy before inserting its bound. The final exponent calculation then closes the induction.

Corollary 16.4 organizes the analytic choices. We use its indexed conclusion separately on each positive Cauchy-side sum, verifying the common coefficient measures, fixed norm losses, and finite depth below.

At each node, every joint (f, k) Hilbert space uses the current label measure $w(f)$, and that factor is retained exactly once in every parent f-sum until (16.35) removes the old label.

At each node keep the parameters N, V, M fixed, and write

$$F = N + V, \qquad Q = \log_Z q_{r_\rho}.$$

Recall that $F = N + V$ is fixed at each node; $F = F_0$ at the starting node. The two hypotheses at a node with margin c_{node} are

$$F - M - Q - z_0 \geq c_{\mathrm{node}}, \qquad 4F - 3M - 6z_0 \geq c_{\mathrm{node}}. \tag{16.9}$$

In particular $M + z_0 + c_{\mathrm{node}} \leq F$. The number Q is the actual logarithmic norm of the radical defining the already fixed coprimality restriction. It is used without dyadic rounding. The parameter M is the fixed logarithmic row length; the first Poisson test has scale Z^M.

Let $M_{\max}$ bound the starting row lengths. Choose $0 < d \leq c_*/200$ and put

$$D = \left\lceil \frac{M_{\max} + 2}{d} \right\rceil + 1.$$

We use positive parameters $\eta, \tau, \pi, \tau_{\mathrm{ref}}, \pi_{\mathrm{ref}}$, chosen in the quantified order at the end of the proof. Here η is a localization tolerance bounding only the logarithms of fixed annular ratios, τ is the common tolerance in the two Poisson tail comparisons, and π bounds the aggregate freely chosen local small-power losses. At depth h set

$$c_h = c_* - 7h\eta, \qquad c_{\mathrm{node}} = c_h,$$

and impose the row cap $M \leq M_{\max} - hd$. We prove the estimate by backwards induction on $h \leq D$. Every retained row parameter below is nonnegative. We shall show that a nonterminal passage decreases it by at least d, loses at most 7η in either required margin, and increases the energy exponent by at most $40\eta + \tau + \pi$. Throughout this induction the invariants use the fixed length sum $F = N + V$.

1. *The terminal completion.* Möbius inversion of the cube factor in (13.2) gives the identity

$$Z^{-N/2} \sum_{n \text{ sf}} a(n)\chi_n(k)\chi_n(f)^4 \mathfrak{d}(n) W(q_n/Z^N) = \sum_h \frac{\mu(h)\bar{\alpha}(h)^3 \Psi_k(h)^3}{q_h} \mathcal{T}_{\mathfrak{d}_h}(Z^N/q_h^3; k), \tag{16.10}$$
$$\mathfrak{d}_h(A) = \mathfrak{d}(h^3 A).$$

Expanding the right side and putting $c = hb$ verifies both the mark and the normalization. The factor $V_*(y) = y^{1/2}W(y)$ changes the absolute coefficient to

$$Z^{-N/2}\bar{\alpha}(c)^3 \Psi_k(c)^3 q_c^{1/2} \sum_{h|c} \mu(h).$$

The mark is $\mathfrak{d}(nc^3)$, independent of the divisor h. The divisor sum vanishes unless $c = 1$, proving the identity. Multiplicativity of Ψ_k, including its coprimality restrictions, is used here.

Fix one smooth dyadic partition with nonnegative ideal centers and unit-dyad center zero. Suppose $V < d$ and that the center ℓ_1 of the h-dyad is less than d. Freeze h, assign to it the slots it divides, and retain the others on the completed index. A surviving slot then has the individual coefficient $a_i(p)1_{p\nmid h}$. Since h is fixed, this is a bounded coefficient independent of the current k, f, on the same nominal annulus and with the same cap. It is not the unrestricted original coefficient; the recursive branch below restores its original slot lists separately by the child zeros. For an active subcollection define z_a to be the sum of its nominal slot lengths. Assignments only delete slots, so $0 \leq z_a \leq z_0$ exactly; an actual tuple norm is never used as this cap. The triangle inequality in the joint (f, k) Hilbert space uses $\sum_{q_h \asymp Z^{\ell_1}} q_h^{-1} \ll 1$ for the fixed annulus, apart from the separately chosen divisor loss for assignments. The bounded factor $\Psi_k(h)^3$ is a contraction in that space.

For the terminal estimate, the parameters $O, H, T_d, S_0, B_0, u, e_\lambda$ are the reflected-block data of Lemma 13.3: O, H describe the powerful and residual row parts, and T_d, S_0, B_0 control the dual range and its extracted factors. Write $\widehat{h} = \log_Z q_h$. The completion is at the actual scale $N_* = N - 3\widehat{h}$. Its fixed support gives $\widehat{h} \le \ell_1 + \eta < d + \eta$ once the fixed annular threshold specified below is imposed. In Lemma 13.3, keep the actual residual-row dyad and choose the same threshold so that

$$H \le M - O + \eta, \qquad T_d - S_0 - B_0 \le 2M - O + Q + V - N_* + 2z_a + \eta. \tag{16.11}$$

The residual-row bounds (4.35) and (13.18) give these bounds on the actual residual-row dyad, before any enlargement of that range. Substituting $N_* = N - 3\widehat{h}$ and the first inequality in (16.9) gives

$$T_d - S_0 - B_0 \le M - O + 2z_a - z_0 - c_{\mathrm{node}} + 5d + 4\eta. \tag{16.12}$$

Indeed, the additional terms are $2V + 3\widehat{h} + \eta < 5d + 4\eta$.

Take the threshold also to ensure $e_\lambda \ge -\eta$, and discard the whole reflected tail dyads as in Lemma 13.3. Thus every retained dyad satisfies $v + 3\ell_b + e_\lambda \le T_d + \tau_{\mathrm{ref}}$. The row branch of (13.15), after dropping its nonpositive terms, obeys

$$E_{\mathrm{ref}} \le M + z_a + \tfrac{5}{3}\eta \le F - c_{\mathrm{node}} + \tfrac{5}{3}\eta.$$

For the column branch with $u = z_a$, the retained dual bound first gives $E_{\mathrm{ref}} \le O/2 + (T_d - S_0 - B_0) + 5\eta/3 + \tau_{\mathrm{ref}}$. The upper bound (16.12) and the first invariant then give

$$E_{\mathrm{ref}} \le F - 2c_{\mathrm{node}} + 5d + \tfrac{17}{3}\eta + \tau_{\mathrm{ref}}. \tag{16.13}$$

If $u \ne z_a$, its definition implies $u \ge v/2$. The retained dual bound now first gives

$$E_{\mathrm{ref}} \le O/2 + (T_d - S_0 - B_0)/2 + z_a + \tfrac{7}{6}\eta + \tfrac{1}{2}\tau_{\mathrm{ref}}.$$

Using (16.12), $z_a \le z_0$, and then the second invariant yields

$$E_{\mathrm{ref}} \le F - M/4 - 3c_{\mathrm{node}}/4 + (5/2)d + \tfrac{19}{6}\eta + \tfrac{1}{2}\tau_{\mathrm{ref}}. \tag{16.14}$$

In the reflected estimate, the aggregate local loss π_{ref} includes its freely chosen exponent, the label divisor bound, and the finitely many logarithmic sums. If

$$c_{\mathrm{node}} \ge c_*/2, \qquad d \le c_*/200, \qquad \eta, \tau_{\mathrm{ref}}, \pi_{\mathrm{ref}} \le c_*/1000,$$

all three bounds, after adding π_{ref}, are strictly below F. In the last bound $M \ge 0$. For $z_0 = 0$ only $u = z_a = 0$ occurs. The reflected estimate was applied for each fixed f; the weighted mass bound $\sum_f w(f) \ll Z^{V+\pi_{\mathrm{ref}}}$ bounds the factor from summing its squared bounds over f. This factor is already included in the aggregate loss, and its label-length power cancels the outside Z^{-V}. The fourth-power label was fixed during the hybrid norm estimate. These terminal terms therefore satisfy the canonical estimate.

2. Blocks for the recursive step. At every factorization, assign a slot first to an extracted factor which its prime divides, and otherwise retain it on the residual column. The basic identity for the completion is

$$1_{p \mid nb^3} = 1_{p \mid b} + 1_{p \nmid b} 1_{p \mid n}. \tag{16.15}$$

More generally, for an ordered list of extracted factors $D_1, \dots, D_h$, the priority identity is

$$1_{p \mid D_1 \cdots D_h n} = \sum_{j=1}^{h} 1_{p \mid D_j} \prod_{a<j} 1_{p \nmid D_a} + 1_{p \mid n} \prod_{a \le h} 1_{p \nmid D_a}. \tag{16.16}$$

It holds even when extracted factors share primes. Applying it to each slot and each copy of a square gives a disjoint partition for each tuple of primes; the two copies of one slot have independent prime choices. There are at most $2^{|I|}$ choices at one single-factor assignment, and at most $(h+1)^{|I|}$ for the displayed ordered list. The assigned slots have divisor-bounded coefficients on the extracted factor. Because the original coefficient is $\prod_i a_i(p_i)$, removing assigned slots leaves exactly a subcollection with its original product coefficients. An ideal which will remain an averaging variable is not fixed merely because a slot was assigned to it.

If $V < d$ and $\ell_1 \geq d$, reopen the completion on the right of (16.10). If its cube ideal is h', first isolate its dyad with center $\ell_2 \geq 0$ and the h-dyad by the triangle inequality in the joint Hilbert space, before squaring. Put $b = hh'$ and define the fixed centers

$$\ell = \ell_1 + \ell_2, \qquad r = N - 3\ell.$$

Both copies b_1, b_2 in the resulting square have this same nominal center ℓ; their actual norms need not agree. The mark is $\mathfrak{d}(nb^3)$, independent of the choice of the divisor h of b. Multiplicativity gives a factor $\chi_b(k)^3$; the fourth power in $\Psi_k(b)^3$ is exactly the mask $1_{(b,f)=1}$. The remaining sum over divisors h of b has absolute value at most a divisor function. After dyadic decomposition and separation of $q_n q_b^3/Z^N$, the row is k, the squarefree column is n, and the extracted cube factor is b; the old squarefree label f is still averaged. Each squared block is bounded by a small power times

$$\begin{aligned} Z^{-r-2\ell-V} \sum_f w(f) \sum_{k\in\mathcal{O}} \Phi(q_k/Z^M) \Bigg| \sum_{q_b \asymp Z^\ell} \beta(b,f)\chi_b(k)^3 \\ \times \sum_{n \text{ sf}} a(n)\chi_n(k)\chi_n(f)^4 \mathfrak{d}(nb^3) W(q_n/Z^r) \Bigg|^2 . \end{aligned} \tag{16.17}$$

Here Φ is a fixed nonnegative radial Schwartz function majorizing the original row ball, and $|\beta(b,f)| \ll Z^\epsilon$ independently of k, n. Its f-dependence includes the original mask $1_{(b,f)=1}$; all other original cube masks are retained. There is no condition $(b,n) = 1$. The exponent in front is correct because the original normalization is $Z^{-N/2} q_b^{1/2} \asymp Z^{-r/2-\ell}$. If $V \geq d$, use the same block with $\ell = 0$, $b = 1$, $r = N$. Thus every remaining block has exactly $V + \ell \geq d$, by the center classification and $\ell_2 \geq 0$. The added row $k = 0$ can only contribute in the principal cases counted below.

The inductive target. We prove the following conditional estimate for every remaining block in (16.17). Let $\epsilon_{\text{child}} \geq 0$. Suppose every energy $\mathcal{E}'$ of the form (16.2), with all coefficient and support hypotheses of Lemma 16.2, the same fixed slot cap, parameters

$$0 \leq M' \leq M - d, \qquad 0 \leq N', V', \qquad F' = N' + V' \leq F + 11\eta,$$

and both margins at least $c_{\text{node}} - 7\eta$, satisfies $\mathcal{E}' \ll Z^{F'+\epsilon_{\text{child}}}$. The hypothesis is uniform over the bounded ranges used in this induction, with finite-seminorm and fixed polynomial-height dependence as in that lemma. For $\eta \leq d/16$, the remaining block satisfies

$$\text{block in equation (16.17)} \ll Z^{F+40\eta+\tau+\pi+\epsilon_{\text{child}}}, \tag{16.18}$$

with the same kind of uniform dependence. The freely chosen local losses sum to π; the principal terms and discarded tails are included in this estimate. Thus the reduction supplies the implication needed for backwards induction. Its proof occupies the two transformations below: the first produces a positive inverse-polynomial norm, and the second constructs the smaller canonical energies to which the hypothesis applies.

3. Localization and common Fourier profiles. Both transformations use the same distinction between a fixed annular center and the norm of an individual ideal. Their Fourier profiles also use independent normalized coordinates, so arithmetic relations restrict evaluation points without

changing the coefficient measure. For an actual ideal or nonzero element a, write $\widehat{a} = \log_Z q_a$. For a named scalar center, a hat will denote the actual logarithmic norm of its indicated ideal. All ideal centers introduced below are nonnegative centers from the fixed dyadic partition. The radical defining the fixed coprimality restriction and the ideal q_0 introduced below are used at their actual logarithmic norms, without independently rounded centers.

Every individual dyad introduced in the two transformations has a fixed compact normalized support. If a is its ideal and a_{cen} its named logarithmic center, we use

$$|\log_Z q_a - a_{\mathrm{cen}}| \le \eta. \tag{16.19}$$

This convention applies to the original n_i, b_i, f windows and to each extracted-ideal dyad when it is introduced. Errors for products and quotients are the sums of these individual errors; the fresh residual windows below have the explicitly stated bounds 4η, 6η and 4η. These estimates concern fixed compact normalized-ratio intervals, not annuli of ratio Z^η.

The interval family is fixed uniformly through the whole finite depth. At each factorization choose fresh individual smooth cutoffs equal to one on the quotient supports, and include the full supports of these cutoffs, of the larger cutoffs used in Fourier separation, and of the enlarged label windows. Include also the bounded clipping families used below. Products and quotients of endpoints locate the initial quotient supports, such as those of $b = hh'$, the residual n, and the child column and label. Iterating this construction through D levels gives a finite collection $\mathcal{I}_D$ of compact normalized-ratio intervals. If L_{win} is the maximum absolute logarithm of their endpoints, the threshold $\log Z \ge L_{\mathrm{win}}/\eta$ implies (16.19) and all the fresh-support bounds specified below. For columns or labels added when a positive sum is enlarged, the norm bounds at this and the next node come from the full independent fresh windows in $\mathcal{I}_D$. The old parent product identities are used only on the original support. Separated norm powers do not change these supports.

We use the following specialization of the common joint Fourier calculus in Lemma 3.6. Every normalized norm occurring in a coupled smooth factor to be separated from the columns is a coordinate $y_1, \dots, y_d$ of one ambient profile. A derived outer mask or label cutoff retained in the weight remains outside and needs no profile coordinate. Keep the current individual dyad cutoffs and fresh column cutoffs outside the inversion, until the weighted Cauchy inequality where they are used. Multiply the coupled profile by larger individual cutoffs Ω_j equal to one on the full supports of those retained cutoffs. For the resulting compact profile $\mathfrak{H}$, define before summing any current row or label

$$\widehat{\mathfrak{H}}(t) = \int_{\mathbb{R}^d} \mathfrak{H}(e^{u_1}, \dots, e^{u_d}) e^{-it\cdot u}\, du, \qquad \mathfrak{H}(y) = (2\pi)^{-d} \int_{\mathbb{R}^d} \widehat{\mathfrak{H}}(t) \prod_{j=1}^{d} y_j^{it_j}\, dt. \tag{16.20}$$

Define the profile on the ambient box and take its Fourier transform. Arithmetic relations among the norms specify the evaluation points of this identity without changing its density. Outer modes stay in the outer weight; for two column coordinates the first test receives $y_1^{it_1}$ and the second receives $y_2^{-it_2}$, whose conjugate supplies $y_2^{it_2}$. Every normalized real inverse-root power is included once, in the coupled profile, and is excluded from the final column tests.

This convention also makes the uniformity quantitative. Write $D_j = y_j \partial_{y_j}$. On a fixed log rectangle, integration by parts with $(1-\Delta)^{m_0}$, for $2m_0 > J + d$, gives

$$\int_{\mathbb{R}^d} |\widehat{\mathfrak{H}}(t)|(1+|t|)^J\, dt \ll_{J,d,\mathrm{box}} \max_{|a|\le 2m_0} \|D^a \mathfrak{H}\|_\infty. \tag{16.21}$$

For a fixed radial Schwartz Fourier kernel K, every $\sup_{x\ge 0} |(x\partial_x)^j K(x)|$ is finite, and

$$D^a K\left(A \prod_j y_j^{e_j}\right) = \left(\prod_j e_j^{a_j}\right) (x\partial_x)^{|a|} K(x)\big|_{x = A\prod_j y_j^{e_j}}.$$

Thus the separating seminorms are uniform for every scalar $A > 0$. The derivative and Fourier-height orders affect the required input seminorms and polynomial height degrees, while the power of Z is independent of those orders. Normalized real powers have bounded Euler derivatives on the fixed boxes, and inherited norm twists have fixed polynomial height dependence. A sharp cutoff in a column-dependent kernel ratio must remain outside this profile. For the radial Fourier kernels in the Poisson steps, these are the normalized-kernel bounds of Lemma 3.12 and (16.7).

The raw tail estimate we shall use is, for $Y \ge 1$ and $A > 1$,

$$\sum_{h\neq 0:\, aq_h > Y} |K(aq_h)| \ll_A (1 + a^{-1})Y^{1-A} \qquad (a > 0). \tag{16.22}$$

Indeed, the shell $2^jY < aq_h \le 2^{j+1}Y$ contains $O(1 + 2^jY/a)$ lattice points and the kernel is $O_A((2^jY)^{-A})$ there. Summing the two geometric series proves the display; the same argument applies to the fixed lattice $\lambda^{-4}\mathcal{O}$. The proof uses only the large-argument bound $|K(x)| \ll_A x^{-A}$ for $x \ge 1$, so it also applies to the reflected kernel on such a tail. Below an actual-ratio inequality is used only to show that the complement of a sector-fixed outer row ball is contained in this tail for each supported raw column pair. That complement is removed before off-coprime extension and before Fourier absolutization. The full smooth kernel is retained inside the ball, whose nonsmooth mask is kept outside Fourier inversion until the relevant weighted Cauchy inequality. The tail orders and the crude raw counts are fixed at the end.

4. The first transformation: conductor and coefficients. In the expanded square of (16.17), put

$$\mathfrak{B} = \mathrm{rad}(b_1b_2), \qquad n_i = \mathfrak{A}_i C u_i \quad (i = 1, 2),$$

where $\mathfrak{A}_i = (n_i, \mathfrak{B})$, and C is the gcd of $n_1/\mathfrak{A}_1$ and $n_2/\mathfrak{A}_2$. Then $\mathfrak{A}_i \mid \mathfrak{B}$, while u_1, u_2 are squarefree, coprime to one another, and prime to $C\mathfrak{B}$. Let A_i, B be the fixed centers of $\mathfrak{A}_i, C$, and write $\widehat{A}_i = \log_Z q_{\mathfrak{A}_i}$, $\widehat{B} = \log_Z q_C$. For $p \mid \mathfrak{B}$, define

$$a_{ip} = 1_{p|\mathfrak{A}_i}, \qquad \pi_p = v_p(b_1b_2) \bmod 2, \qquad t_p = a_{1p} - a_{2p} + 3\pi_p \bmod 6,$$

and put $R_1 = \prod_{t_p\neq 0} p$. Its center is R, and $\widehat{R} = \log_Z q_{R_1}$. The row character and its remaining zero mask are

$$\psi_1 = \chi_{u_1}\bar{\chi}_{u_2} \prod_{t_p\neq 0} \chi_p^{t_p}, \qquad m_1 = u_1u_2R_1, \qquad R_{\mathrm{mask}} = C\mathfrak{B}/R_1. \tag{16.23}$$

Indeed, the exponents at u_1, u_2 are $1, -1$; the common off-$\mathfrak{B}$ factor C has exponent zero but retains its mask; and the exponent at $p \mid \mathfrak{B}$ is t_p. Each nonzero local power is nonprincipal and primitive modulo p, with disjoint supports. Thus ψ_1 is primitive unless $m_1 = 1$. Every original column mask and the two factors $\beta(b_1, f)\overline{\beta(b_2, f)}$ remain in the expression.

Lemma 16.5 with $K = Z^M$ introduces the divisor $d_k \mid C\mathfrak{B}/R_1$, with center δ and $\widehat{\delta} = \log_Z q_{d_k}$. On this coprime expression, the actual conductor length is

$$\widehat{L}_1 = \widehat{n}_1 + \widehat{n}_2 - \widehat{A}_1 - \widehat{A}_2 - 2\widehat{B} + \widehat{R}. \tag{16.24}$$

At this stage the kernel is still complete. Its argument is $Z^M q_h/(q_{d_k}q_{m_1})$, and the Poisson prefactor has exponent $M - \widehat{\delta} - \widehat{L}_1/2$. Localization and Fourier separation come after this conductor calculation.

If $m_1 = 1$, then $u_1 = u_2 = 1$ and $t_p = 0$ for every $p \mid \mathfrak{B}$. The latter condition forces $\pi_p = 0$ and $a_{1p} = a_{2p}$. Hence $n_1 = n_2$ and b_1b_2 is a square. Even counting all $O(Z^{2\ell+\epsilon})$ pairs b_1, b_2, all $O(Z^{r+\epsilon})$ equal columns, all $O(Z^{V+\epsilon})$ labels, and $O(Z^M)$ rows, the normalization in (16.17) gives $O(Z^{M+\epsilon})$. If $r < 0$ but its fixed annular column window is nonempty, then Z^{-r} is bounded by its fixed upper endpoint, so its ideal count is still $O(Z^r)$ with a fixed constant. Marks and the retained masks do not increase this bound beyond a small power. When the nonzero principal

terms are restored below, they will carry the same outer ball mask as the nonprincipal terms; the last part of Lemma 16.5 gives the same bound for them.

The first transformation turns the canonical column coefficient into a Möbius coefficient; the second transformation will restore the canonical form. For coprime squarefree primary a, b, CRT and reciprocity give

$$\gamma_2(ab) = \gamma_2(a)\gamma_2(b)\chi_a(b)^2\chi_b(a)^2 = \gamma_2(a)\gamma_2(b)\chi_b(a)^4.$$

In the first Poisson root, the factors involving u_1, including the divisor and frequency, are

$$\gamma_1(u_1)\overline{\chi_{u_1}(h)}\chi_{u_1}(d_kR_1)\prod_{t_p\neq 0}\chi_{u_1}(p^{t_p}),$$

up to a fixed-ray factor. This follows by writing the CRT factors between u_1 and p as $\chi_{u_1}(p)\chi_p(u_1)^{t_p}$ and using reciprocity. On the conjugated second side t_p is replaced by $-t_p$. The cross factor between u_1, u_2 is the reciprocity factor $\mathcal{R}(u_1, u_2)$. This expression retains the zero extensions without taking a quotient of residue symbols.

Put $P_T = \prod_{t_p\neq 0} p^{t_p}$, regarded in the fixed ray group. The identities (3.8) and (3.6) show that the remaining two-column ray factor is

$$G(u_1u_2^{-1})\mathcal{R}(u_1u_2^{-1}, P_T).$$

For example, before placing factors inside the conjugated second polynomial, the raw second Gauss-signal factor is $\mu(u_2)\chi_{u_2}(-1)\overline{G(u_2)}$. Its corresponding factor written inside that polynomial is $\mu(u_2)\chi_{u_2}(-1)G(u_2)$. Also $G(u_2^{-1}) = \chi_{u_2}(-1)\overline{G(u_2)}$. These identities give the displayed quotient. Fix the ray class of P_T and Fourier-expand the displayed function on the finite ray group. Its factors on each side are multiplicative ray characters, denoted by ν_i. Their number and coefficient norm depend only on the fixed ray group.

Define

$$E = \prod_{\pi_p=1} p^{a_{1p}+a_{2p}}, \qquad \widetilde{h} = hf^2E.$$

The original fourth power at f equals $\overline{\chi_{u_i}(f^2)}$, including zeros. Assign the slots first to $b_i, \mathfrak{A}_i, C$. The remaining local factor at $p \mid \mathfrak{B}$ on side i has exponent

$$4a_{ip} + 1_{t_p\neq 0}(1 \pm t_p) + \pi_p(a_{1p} + a_{2p}) \pmod 6.$$

The plus sign belongs to side one. Direct reduction gives the same answer on both sides:

π_p	a_{1p}	a_{2p}	t_p	exponent on both sides
0	0	0	0	0
0	1	0	1	0
0	0	1	5	0
0	1	1	0	4
1	0	0	3	4
1	1	0	4	4
1	0	1	2	4
1	1	1	3	4

Define $\xi(n)$ as the product over $p \mid \mathfrak{B}$ of $\chi_n(p)$ raised to the corresponding exponent in the last column. An exponent zero still denotes the coprimality restriction at p. The calculation proves that ξ is common to both sides and independent of h, f, C. The nonprincipal coefficient on side i, after these assignments, is

$$\mu(u_i)\nu_i(u_i)\rho(u_i)\overline{\chi_{u_i}(\widetilde{h})}\chi_{u_i}(C)^4\chi_{u_i}(d_k)\xi(u_i)\mathfrak{d}_i(u_i). \tag{16.25}$$

It also shows that every character involving the moving $b_i, \mathfrak{A}_i$ is in either ξ or $\widetilde{h}$; their other factors are outer coefficients.

The first transform has produced the inverse-type column coefficient in (16.25). We next collect its fourth-power factors into one squarefree label before forming the positive row norm for the second transform.

5. First transformed rows and positive norms. Write uniquely $b_1 b_2 = q^2 s$ with s squarefree, and define

$$J_2 = \prod_{\substack{\pi_p = 0 \\ a_{1p} = a_{2p} = 1}} p, \qquad J = sJ_2, \qquad q = J_2 q_0.$$

Every prime of J_2 has positive even valuation in $b_1 b_2$, so $J_2 \mid q$. The ideals s, J_2 are coprime and squarefree; hence J is squarefree. Let $j \geq 0$ be the center of J, and put

$$\widehat{s} = \log_Z q_s, \quad \widehat{j}_2 = \log_Z q_{J_2}, \quad \widehat{j} = \log_Z q_J = \widehat{s} + \widehat{j}_2, \quad \widehat{\ell}_{\mathrm{pair}} = (\widehat{b}_1 + \widehat{b}_2)/2.$$

The ideal identity $b_1 b_2 = q_0^2 J_2^2 s$ gives

$$\widehat{c} := \log_Z q_{q_0} = \widehat{\ell}_{\mathrm{pair}} - \widehat{s}/2 - \widehat{j}_2 \geq 0, \qquad 4\widehat{\ell}_{\mathrm{pair}} - 2\widehat{s} + \widehat{j}_2 = 4\widehat{c} + 5\widehat{j}_2 \geq 0. \tag{16.26}$$

The table above now proves the zero-extended identity

$$\xi(n) = \chi_n(J)^4 1_{(n, \mathrm{rad}\, q_0) = 1}. \tag{16.27}$$

Indeed, the primes with exponent four are exactly those of sJ_2; every other prime of $\mathfrak{B}$ divides q_0. A prime in both J and q_0 only repeats the zero already supplied by $\chi_n(J)^4$. In the second transform, q_0 will determine a fixed coprimality restriction, while the squarefree ideal J will remain in the averaged label. Thus the moving modulus q_0 stays separate from the fixed ray group. For fixed J, q_0, the choices of s, J_2, the factorizations $b_1 b_2 = q_0^2 J_2^2 s$, and the ideals $\mathfrak{A}_i$ have only divisor multiplicity.

We now localize the first Poisson expression on its coprime support, before extending its coprime support or taking absolute Fourier integrals. The local table gives the actual-norm identity

$$\widehat{R} - \widehat{A}_1 - \widehat{A}_2 + \widehat{E} = \widehat{s} - 2\widehat{j}_2.$$

For $y = hf^2 E$, the implication $Z^M q_h/(q_{d_k} q_{m_1}) \leq Z^\tau$ and (16.24) give

$$\begin{aligned} \widehat{y} &\leq \widehat{n}_1 + \widehat{n}_2 - 2\widehat{B} - M + \widehat{s} - 2\widehat{j}_2 + \widehat{\delta} + 2\widehat{f} + \tau \\ &\leq \widehat{n}_1 + \widehat{n}_2 - 2\widehat{B} - M + 4\widehat{\ell}_{\mathrm{pair}} - \widehat{j} + \widehat{\delta} + 2\widehat{f} + \tau. \end{aligned}$$

The second inequality is (16.26). Define the formal center and the enclosing outer row scale by

$$H_c = 2r - 2B - M + 4\ell + 2V + \delta - j, \qquad H_{\mathrm{use}} = H_c + 12\eta + \tau. \tag{16.28}$$

Relative to H_c, the last actual upper bound has error

$$(\widehat{n}_1 - r) + (\widehat{n}_2 - r) - 2(\widehat{B} - B) + 2(\widehat{b}_1 - \ell) + 2(\widehat{b}_2 - \ell) - (\widehat{j} - j) + (\widehat{\delta} - \delta) + 2(\widehat{f} - V),$$

whose positive maximum under (16.19) is 12η. Thus the actual-ratio inequality implies $q_y \leq Z^{H_{\mathrm{use}}}$ for every supported raw column pair.

The raw expression is restricted only by the outer mask

$$\mathcal{B}_1(h) = 1_{0 < q_h f^2 E \leq Z^{H_{\mathrm{use}}}}.$$

For each supported pair, its complement is contained in the actual kernel tail $Z^M q_h/(q_{d_k} q_{m_1}) > Z^\tau$. The tail bound (16.22) therefore discards that complement, with the raw counts and order fixed

below. Inside $\mathcal{B}_1 = 1$ the kernel remains complete, including pairs whose ratio exceeds Z^τ; the ratio inequality serves only to bound the discarded complement. The mask depends on h, f, E and the fixed sector, but, after the indicated extraction, not on u_1, u_2. If $H_{\mathrm{use}} < 0$, its nonzero ball is empty and the same tail comparison discards every nonzero frequency. Any principal nonzero terms now restored to unify the formula carry this identical mask. Their absolute contribution is bounded by the full principal restoration already estimated.

The condition $(u_1, u_2) = 1$ is recovered by the divisor identity

$$1_{(u_1,u_2)=1} = \sum_{t' \mid (u_1,u_2)} \mu(t'), \qquad u_i = t' x_i, \qquad \widehat{t} = \log_Z q_{t'}.$$

Let t be its fixed nonnegative center. This inversion is made after the cross phases have been replaced by the fixed ray functions above. Those functions, the unchanged outer mask $\mathcal{B}_1$, all remaining zero-extended local factors, and the formal product $q_{u_1} q_{u_2} q_{R_1}$ in the smooth kernel define an expression also for noncoprime u_1, u_2. It agrees with Poisson summation on the coprime support; the displayed divisor identity then recovers exactly that support. For the newly introduced noncoprime pairs, this is the displayed algebraic extension; the primitive Poisson formula applies only on the preceding coprime support. Squarefreeness retains the coprimality restriction $(x_i, t') = 1$. Assigned slots at t' are outer coefficients, while the other slots form the mark on x_i.

A fixed partition of q_h, common to all f, E, decomposes the frequency without recentering its dyads for individual labels. The mask $\mathcal{B}_1$ implies $q_h \le q_{hf^2E} \le Z^{H_{\mathrm{use}}}$, so there are only logarithmically many such dyads in the bounded retained scale range. Keep $\mathcal{B}_1$ outside every Fourier inversion. Define

$$s_i = r - A_i - B - t, \qquad \kappa_i = M - 2r - 2\ell - V - \delta + A_i + B - R/2.$$

The identity $n_i = \mathfrak{A}_i C t' x_i$ on the factorized expression gives $|\widehat{x}_i - s_i| \le 4\eta$. The full support of the new x_i cutoff is included in $\mathcal{I}_D$, so this same bound holds directly on that support after separation, not only on the original product support. The reconstructed formal products used in the current prefactor are also included there.

Fourier separation of the first product. We record the actual prefactor before applying Cauchy. It splits over the two column sides as

$$Z^{-r-2\ell-V} Z^{M-\widehat{\delta}-\widehat{L}_1/2} = Z^{\widehat{\kappa}_1/2} Z^{\widehat{\kappa}_2/2}, \qquad \widehat{\kappa}_i = M - r - 2\ell - V - \widehat{\delta} - \widehat{n}_i + \widehat{A}_i + \widehat{B} - \widehat{R}/2.$$

The localization bound (16.19) gives on each side

$$\widehat{\kappa}_i \le \kappa_i + \tfrac{9}{2}\eta. \tag{16.29}$$

The five error weights are $1, 1, 1, 1, 1/2$. We now implement this normalization with all real inverse roots in one joint profile. Fix a first dyadic, ray, and slot pattern σ, and let h_1 be the center of its bare-h dyad. Take the nine independent normalized coordinates

$$\mathbf{y} = (y_{A_1}, y_{A_2}, y_C, y_d, y_R, y_t, y_h, y_{x_1}, y_{x_2}) = \Big(\frac{q_{\mathfrak{A}_1}}{Z^{A_1}}, \frac{q_{\mathfrak{A}_2}}{Z^{A_2}}, \frac{q_C}{Z^B}, \frac{q_{d_k}}{Z^\delta}, \frac{q_{R_1}}{Z^R}, \frac{q_{t'}}{Z^t}, \frac{q_h}{Z^{h_1}}, \frac{q_{x_1}}{Z^{s_1}}, \frac{q_{x_2}}{Z^{s_2}} \Big).$$

The full root and kernel argument on the formal $u_a = t' x_a$ expression are

$$\begin{aligned} \frac{Z^{-r-2\ell-V} Z^M}{q_{d_k} \sqrt{q_{R_1}} q_{t'} \sqrt{q_{x_1} q_{x_2}}} &= Z^{(\kappa_1+\kappa_2)/2} y_d^{-1} y_R^{-1/2} y_t^{-1} (y_{x_1} y_{x_2})^{-1/2}, \\ \frac{Z^M q_h}{q_{d_k} q_{R_1} q_{t'}^2 q_{x_1} q_{x_2}} &= A_{\mathrm{ker},1} \frac{y_h}{y_d y_R y_t^2 y_{x_1} y_{x_2}}, \\ A_{\mathrm{ker},1} &= Z^{M+h_1-\delta-R-2t-s_1-s_2}. \end{aligned} \tag{16.30}$$

Choose fresh cutoffs $\omega_{x,a}$ equal to one on the old W-support divided by the individual $\mathfrak{A}_a, C, t'$ supports, and retain them in the columns. Every current outer dyad cutoff remains outside inversion. Let $\Omega_{1,\alpha}$ be larger cutoffs equal to one on the full supports of these retained cutoffs. With $\mathcal{K}_1 = \mathcal{F}\Phi$, define the single compact profile

$$\begin{aligned}\mathfrak{H}_{1,\sigma}(\mathbf{y}) = Z^{-9\eta/2} \prod_{\alpha} \Omega_{1,\alpha}(y_\alpha) y_d^{-1} y_R^{-1/2} y_t^{-1} (y_{x_1} y_{x_2})^{-1/2} \\ \times W(y_{A_1} y_C y_t y_{x_1}) \overline{W(y_{A_2} y_C y_t y_{x_2})} \mathcal{K}_1 \left(A_{\ker,1} \frac{y_h}{y_d y_R y_t^2 y_{x_1} y_{x_2}} \right).\end{aligned} \tag{16.31}$$

The scalar outside this profile is exactly $\prod_{a=1}^{2} Z^{(\kappa_a + 9\eta/2)/2}$. In particular, the profile contains each real root once, also when the actual column norms differ. Fourier inversion in (16.20) defines its transform on the ambient nine-dimensional box before the sums over f, h and the outer ideals. This gives one coefficient measure for the whole row norm. The norms of b_a, f, J, E, q_0 occur only in outer coefficients, characters, masks or individual cutoffs after (16.25); they require no additional coupled-profile coordinate.

To expose the arithmetic extraction at t', put $y = hf^2E$ and

$$B_{1,a}(t'; y) = \mu(t') \nu_a(t') \rho(t') \overline{\chi_{t'}(y)} \chi_{t'}(CJ)^4 \chi_{t'}(d_k) 1_{(t', \operatorname{rad} q_0)=1}.$$

Let $\varepsilon_1 = 1$ and $\varepsilon_2 = -1$. For a fixed first Fourier mode $\mathbf{t}$, the remaining polynomials are

$$\begin{aligned}P_a(y) = \sum_{x \text{ sf}} \mu(x) \nu_a(x) \rho(x) \overline{\chi_x(y)} \chi_x(CJ)^4 \chi_x(d_k) 1_{(x, \operatorname{rad} q_0 t')=1} \\ \times \mathfrak{d}_{I_a}(x) \omega_{x,a}(q_x / Z^{s_a}) (q_x / Z^{s_a})^{\varepsilon_a i t_{x_a}}.\end{aligned} \tag{16.32}$$

Multiplicativity is used on squarefree coprime factors, then through the displayed zero extensions; no character is divided at a zero. The one additional $\mu(t')$ from the mutual-gcd inversion is outer. The slot identity (16.16) assigns the slots along the ordered list $(b_a, \mathfrak{A}_a, C, t'; x)$. A surviving prime dividing x is already prime to $b_a, \mathfrak{A}_a, C, t'$: every prime of $\mathfrak{B}$ is in $J \operatorname{rad} q_0$, and the displayed fourth powers and coprimality restrictions supply these zeros. Thus $\mathfrak{d}_{I_a}$ has the original individual coefficients and lists, while assigned prime identities and priority masks stay outer.

The first separated component has two column polynomials and one common outer sum. Its outer weight retains the old label, the frequency, the extracted arithmetic factors and all masks. This form permits weighted Cauchy to act on the whole sum before the old label is removed. Let Ω_σ consist of $b_1, b_2, \mathfrak{A}_1, \mathfrak{A}_2, C, d_k, t', f, h$ and the assigned slot primes, subject to their reconstruction relations and individual supports. In particular $\mathfrak{A}_a \mid \mathfrak{B}$, $(C, \mathfrak{B}) = 1$, and $d_k \mid C\mathfrak{B}/R_1$, while R_1, E, J, q_0 are derived, not free indices. Let $\mathcal{A}_{1,\sigma}$ be the product of the assigned original coefficients and priority masks, with conjugation on side two, and let $\psi_{1,\sigma}^{\text{out}}$ be the product of all retained outer dyad cutoffs. The coefficient e_σ collects the factors obtained in the preceding CRT extraction. These are the old multiplicity $w(f)$, the two original β factors, their cube masks including $(b_a, f) = 1$, and the extracted zero masks. It also includes the finite Gauss, unit, reciprocity, and first-ray factors at the old outer ideals. The displayed $\mu(d_k)$ and $B_{1,a}$ are excluded. This product defines e_σ directly, without division by (16.32), and is independent of x_1, x_2. Writing $\mathcal{O}_1 = \{A_1, A_2, C, d, R, t, h\}$, set

$$w_{1,\sigma}(\omega; \mathbf{t}) = e_\sigma(\omega) \mu(d_k) \mu(t') B_{1,1}(t'; y) \overline{B_{1,2}(t'; y)} \mathcal{B}_1(h) \mathcal{A}_{1,\sigma}(\omega) \psi_{1,\sigma}^{\text{out}}(\omega) \prod_{\alpha \in \mathcal{O}_1} y_\alpha^{it_\alpha}.$$

The first component retains the outer mask, includes the restored principal terms, and uses the formal extension of the coprime expression. Fourier inversion gives

$$\mathcal{T}_{1,\sigma} = (2\pi)^{-9} \int_{\mathbb{R}^9} \widehat{\mathfrak{H}}_{1,\sigma}(\mathbf{t}) \sum_{\omega \in \Omega_\sigma} w_{1,\sigma}(\omega; \mathbf{t}) \times Z^{(\kappa_1 + 9\eta/2)/2} P_1(y) \overline{Z^{(\kappa_2 + 9\eta/2)/2} P_2(y)} \, d\mathbf{t}. \tag{16.33}$$

Expanding the two polynomials restores the two column modes; the seven outer modes restore the other coordinates of (16.31). Its larger cutoffs are one on the retained supports, and the fresh cutoffs are one wherever the old windows are nonzero. This verifies the identity term by term. The original β factors and every outer mask are still in the complete sum.

The first positive row norm. For any finite or absolutely convergent weighted sum we use

$$\left|\sum_{\omega} w_\omega U_1(\omega)\overline{U_2(\omega)}\right| \le \prod_{i=1}^{2}\left(\sum_{\omega} |w_\omega||U_i(\omega)|^2\right)^{1/2}. \tag{16.34}$$

For each fixed mode, (16.34) with $U_i = Z^{(\kappa_i+9\eta/2)/2}P_i$ on the whole Ω_σ gives two positive sums. In those sums the pure arithmetic magnitudes of the outer factors are at most a separately allocated Z^{π_β} times the old multiplicity and the absolute assigned-slot product. Their retained support gives $\mathcal{B}_1(h)1_{(C,J)=1}$. This bound applies to the joint coefficient $\beta(b,f)$ without assuming that it factors.

For each fixed old outer reconstruction and nonzero $y = \widetilde{h}$, the relation $y = hf^2E$ implies $f^2E \mid (y)$. If its multiplicity is at most $C_0 d_{\mathcal{O}}(f)^{C_0}$, then

$$\sum_{f^2E|(y)} w(f) \le C_0 d_{\mathcal{O}}((y))^{C_0+1} \le C_{\pi_{\rm old}} Z^{\pi_{\rm old}} \qquad (0 < q_y \le Z^{H_{\rm use}}). \tag{16.35}$$

The last bound is uniform because $H_{\rm use}$ stays in a bounded range; $h = y/(f^2E)$ is then uniquely determined as an element. Thus this fibre bound removes the old label f without introducing a multiplicity depending on the later new label. By (16.25) and (16.27), after separating the norm profiles, $P_i(y)$ depends on the individual $b_i, \mathfrak{A}_i$ only through J, q_0, the fixed length A_i, and the fixed ray sector, and is independent of f.

The positive sum now has the row mask $1_{0<q_y\le Z^{H_{\rm use}}}$. Majorizing it by a fixed nonnegative radial Schwartz function $\Phi_+(q_y/Z^{H_{\rm use}})$, at least one for arguments at most one, gives an upper bound for this positive sum. This majorization is an inequality for the positive sum; it leaves the preceding identity for the signed expression unchanged. A nonempty ball has $H_{\rm use} \ge 0$, so the second Poisson scale is $Z^{H_{\rm use}} \ge 1$. To specify the remaining outer indices, let $\mathfrak{O}_\sigma$ consist of

$$o = (b_1, b_2, \mathfrak{A}_1, \mathfrak{A}_2, C, d_k, t';\ \text{old assigned slot primes}),$$

retaining their full individual supports, the source reconstruction relations independent of f, h, x, the assigned priority masks, and $(C, J) = 1$. Let w_o^+ be the product of the absolute original coefficients of its assigned primes, and zero off this set. It is nonnegative and independent of y. Each first positive side is at most $CZ^{\pi_\beta+\pi_{\rm old}}$, times its scalar $Z^{\kappa_i+9\eta/2}$, times the explicit positive majorant

$$\mathcal{S}_{i,\sigma,\mathbf{t}} = \sum_{o\in\mathfrak{O}_\sigma} w_o^+ \sum_{y\in\mathcal{O}} \Phi_+(q_y/Z^{H_{\rm use}})|P_i(y)|^2. \tag{16.36}$$

At this positive step the old h cutoff and mode, and the zeros of $B_{1,a}$ depending on y, may be removed by their upper bounds. The old f, h are therefore absent from this positive majorant and every later outer set, while their original $-V$ normalization remains in κ_i. We first count the remaining fixed factor q_0, keeping J averaged, and then apply Poisson summation to the y-sum.

Let $\widehat{r}_{\rm diff} \ge 0$ be the actual log-norm of the even-parity primes with $a_{1p} \ne a_{2p}$. The table gives $\widehat{R} = \widehat{s} + \widehat{r}_{\rm diff}$, so

$$\widehat{c} = \widehat{\ell}_{\rm pair} + \widehat{R}/2 - \widehat{j} - \widehat{r}_{\rm diff}/2 \le \ell + R/2 - j + \tfrac{5}{2}\eta. \tag{16.37}$$

Ideal counting with this actual upper bound counts the fixed q_0 while retaining J. For a diagonal bound which also counts J, its full window has exponent at most $j + \eta$, giving the combined upper count $Z^{\ell+R/2+7\eta/2+\epsilon}$. The choices of the $\mathfrak{A}_i$ and of $d_k \mid C\mathfrak{B}/R_1$ add only divisor factors once b_1, b_2, C are specified.

Weighted Cauchy in (16.33) separates the two column polynomials into positive squares. The old-label fibre bound then removes the old label, and the resulting positive row sums are the majorants just defined. Consequently the first transformation gives

$$|\mathcal{T}_{1,\sigma}| \ll Z^{\pi_\beta+\pi_{\mathrm{old}}} \int_{\mathbb{R}^9} |\widehat{\mathfrak{H}}_{1,\sigma}(\mathbf{t})| \times \prod_{i=1}^{2} \left(Z^{\kappa_i+9\eta/2}\mathcal{S}_{i,\sigma,\mathbf{t}}\right)^{1/2} d\mathbf{t}. \tag{16.38}$$

Each $\mathcal{S}_{i,\sigma,\mathbf{t}}$ is the positive sum in (16.36): its row polynomial has the Möbius coefficient in (16.32), the cube-derived label J is retained, and the old f, h no longer occur among the averaged indices. The following transformation is applied to these positive row norms.

6. The second transformation: conductor and coefficients. Fix one of the two positive sums obtained from the first Cauchy inequality. Its polynomial has the form

$$P_i(y) = \sum_x c_i(x)\overline{\chi_x(y)}.$$

The coefficient $c_i(x)$ consists of the other factors of (16.25), with $u_i = t'x$, the coprimality restriction at t', and the separated weight. It is independent of y. In particular x is squarefree and

$$c_i(x) = 0 \quad \text{unless } (x, CJ) = 1, \qquad (x, \operatorname{rad} q_0 t') = 1.$$

The coprimality restriction ρ and the zero of $\chi_x(d_k)$ are retained as well. The y-sum uses the fixed-shape smooth positive ball at scale $Z^{H_{\mathrm{use}}}$.

In its expanded square put

$$g' = (x_1, x_2), \qquad x_j = g' z_j, \qquad \widehat{g} = \log_Z q_{g'}.$$

Let $g \geq 0$ be its fixed center. Then z_1, z_2 are squarefree and coprime, and each is prime to g'. The row data for Lemma 16.5 are

$$K = Z^{H_{\mathrm{use}}}, \qquad \psi_2 = \bar{\chi}_{z_1}\chi_{z_2}, \qquad m_2 = z_1 z_2, \qquad R_{\mathrm{mask}} = g'. \tag{16.39}$$

The common factor $\bar{\chi}_{g'}(y)\chi_{g'}(y)$ is exactly this mask. The character is primitive away from $m_2 = 1$, because its two nonzero local powers have disjoint supports. For $d_2 \mid g'$, let $\theta \geq 0$ be its center and write $\widehat{\theta} = \log_Z q_{d_2}$. The actual conductor length is

$$\widehat{L}_2 = \widehat{x}_1 + \widehat{x}_2 - 2\widehat{g}, \qquad |\widehat{L}_2 - 2(s_i - g)| \leq 10\eta.$$

On the full support of the new x cutoff, the bound $|\widehat{x}_a - s_i| \leq 4\eta$ holds for each copy. Together with $|\widehat{g} - g| \leq \eta$, it gives the last inequality. For $k_{\mathrm{new}} = d_k d_2 k''$, the implication $Z^{H_{\mathrm{use}}} q_{k''}/(q_{d_2} q_{m_2}) \leq Z^{\tau}$ gives

$$\widehat{k_{\mathrm{new}}} \leq 2(s_i - g) - H_{\mathrm{use}} + \delta + 2\theta + 13\eta + \tau = M_c + \eta,$$

where

$$\begin{aligned} M_c &= 2(s_i - g) - H_c + \delta + 2\theta \\ &= M - 4\ell - 2A_i - 2t - 2g + 2\theta - 2V + j, \qquad M_{\mathrm{ch}} = M_c + \eta. \end{aligned} \tag{16.40}$$

The 13η consists of 10η from the conductor, one from d_k, and two from d_2. Subtracting $H_{\mathrm{use}} = H_c + 12\eta + \tau$ cancels the same tolerance τ used in this second Poisson comparison. Thus no separate row clipping or unrecorded boundary error is needed.

Separate the original principal contribution for the direct count below. Before any off-coprime extension or Fourier absolutization, keep on the nonprincipal second Poisson expression on its coprime support only the outer mask

$$\mathcal{B}_2(k'') = 1_{0 < q_{d_k d_2 k''} \leq Z^{M_{\mathrm{ch}}}}.$$

For each supported pair its complement is contained in the actual ratio tail just considered, so (16.22) discards it with the order fixed below. Inside the mask the full smooth kernel remains. The mask depends on d_k, d_2, k'' and the fixed sector, not on the residual z_1, z_2. If $M_{\mathrm{ch}} < 0$, the nonzero ball is empty and the same tail comparison discards every nonzero frequency. In a retained nonempty passage $M_{\mathrm{ch}} \geq 0$.

The actual second prefactor has the side split

$$Z^{H_{\mathrm{use}}-\widehat{\theta}-\widehat{L}_2/2} = Z^{\widehat{\lambda}_1/2} Z^{\widehat{\lambda}_2/2}, \qquad \widehat{\lambda}_a = H_{\mathrm{use}} - \widehat{\theta} - \widehat{x}_a + \widehat{g}.$$

For $\lambda_c = H_c - \theta - (s_i - g)$, the same support bounds give

$$\widehat{\lambda}_a \leq \lambda_c + 18\eta + \tau. \tag{16.41}$$

Here $18 = 12 + 1 + 4 + 1$. We use the root-placement convention of (16.20): the normalized real inverse roots remain in the joint profile through Fourier inversion and are excluded from the final child tests. All original coefficient masks remain. In particular

$$(g', CJ) = 1, \qquad (z_j, g'CJ) = 1$$

on their nonzero support.

The principal case is $z_1 = z_2 = 1$, equivalently $x_1 = x_2$. The nominal identity for its count is

$$\kappa_i + H_c + s_i + B + t + \ell + R/2 = F - B - j, \qquad F = N + V = r + 3\ell + V. \tag{16.42}$$

The actual upper count adds $9\eta/2$ from the first prefactor, $12\eta + \tau$ from the row scale H_{use}, 4η from the diagonal x-window, η each from C, t', and $7\eta/2$ from the combined q_0, J count. Their sum is $26\eta + \tau$. With the aggregate local loss π, this principal contribution is therefore at most $Z^{F-B-j+26\eta+\tau+\pi} \leq Z^{F+26\eta+\tau+\pi}$. Any nonzero principal frequencies restored to the formal expression carry the same $\mathcal{B}_2$ mask; their absolute sum is bounded by the full principal restoration in Lemma 16.5, since $Z^{H_{\mathrm{use}}} \geq 1$.

Recovery of the column coefficient. We now treat the nonprincipal terms. For squarefree z, complex conjugation of a primitive Gauss sum and (3.8) give

$$\gamma_{-1}(z) = \chi_z(-1)\overline{\gamma_1(z)}, \qquad \mu(z)\gamma_{-1}(z) = \bar{\alpha}(z)\gamma_2(z)\chi_z(-1)\overline{G(z)}.$$

The raw second factor is $\mu(z)\gamma_1(z) = \overline{\bar{\alpha}(z)\gamma_2(z)}G(z)$; the corresponding factor written inside the conjugated second polynomial has ray factor $\overline{G(z)}$. The CRT cross factor of the two primitive roots is $\mathcal{R}(z_1, z_2)$. Thus their combined ray factor is

$$\mathfrak{G}(a, b) = \chi_a(-1)\overline{G(a)}G(b)\mathcal{R}(a, b) = G(ba^{-1}).$$

The ray-group identity (3.6) gives the last equality, together with $\mathcal{R}(a, a) = \chi_a(-1)$. The latter identity holds first on primes by (3.7) and then multiplicatively. All quotients in this display are in the fixed finite ray group. In particular,

$$\mathfrak{G}(v'n_1, v'n_2) = \mathfrak{G}(n_1, n_2).$$

The common multiplicative character $\nu_i(v')$ also cancels between the two sides.

A single partition of $q_{k''}$, common to all d_k, d_2, gives the frequency dyads. The mask $\mathcal{B}_2$ implies $q_{k''} \leq Z^{M_{\mathrm{ch}}}$, so their number is logarithmic in the bounded retained range. Let h_2 be the center of one such bare-frequency dyad. The component to be separated retains the full kernel and arithmetic masks. Fix the first mode and selected side i, write

$$H_o = CJ, \quad r_0 = \operatorname{rad} q_0 t', \quad a_0(z) = \bar{\alpha}(z)\gamma_2(z), \quad \mathcal{K}_2 = \mathcal{F}\Phi_+,$$

and retain $K = Z^{H_{\mathrm{use}}}$ from (16.39). For squarefree arguments put

$$\begin{aligned} B_o(g') &= \mu(g')\nu_i(g')\rho(g')1_{(g',r_0)=1}\chi_{g'}(H_o)^4\chi_{g'}(d_k), \\ L_{o,d_2,k''}(z) &= a_0(z)\nu_i(z)\rho(z)1_{(z,r_0)=1}\chi_z(H_o)^4\chi_z(d_k)\overline{\chi_z(d_2)}\chi_z(k''). \end{aligned} \tag{16.43}$$

The old separated test in P_i is denoted by W_i, so $W_i(x) = \omega_{x,i}(x)x^{\varepsilon_i it_{x_i}}$. Let $\psi_g, \psi_{d_2}, \psi_{k''}$ denote the retained individual cutoffs on the fixed g', d_2, k'' dyads, evaluated at their normalized norms. For this preliminary pattern σ_0, the nonzero component, with its mask and restored principal terms, of (16.36) is

$$\begin{aligned} \mathcal{T}_{2,\sigma_0,i} = &\sum_{\substack{o\in\mathfrak{O}_\sigma,\ g'\ \mathrm{sf} \\ d_2|g',\ k''\neq 0}} w_o^+\psi_g\psi_{d_2}\psi_{k''}\mu(d_2)\mathcal{B}_2(k'')|B_o(g')|^2 \\ &\times \sum_{\substack{z_1,z_2\ \mathrm{sf} \\ (z_1z_2,g')=1}} 1_{(z_1,z_2)=1}L_{o,d_2,k''}(z_1)\overline{L_{o,d_2,k''}(z_2)}G([z_2][z_1]^{-1}) \\ &\times \mathfrak{d}_{I_i}(g'z_1)\overline{\mathfrak{d}_{I_i}(g'z_2)}W_i(q_{g'}q_{z_1}/Z^{s_i})\overline{W_i(q_{g'}q_{z_2}/Z^{s_i})} \\ &\times \frac{K}{q_{d_2}\sqrt{q_{z_1}q_{z_2}}}\mathcal{K}_2\left(\frac{Kq_{k''}}{q_{d_2}q_{z_1}q_{z_2}}\right). \end{aligned} \tag{16.44}$$

Here brackets denote classes in the fixed finite ray group. Indeed the arithmetic part of $c_i(g'z)$ factors into $B_o(g')$ and its residual $\mu(z)$ coefficient on $(g', z) = 1$. The common row factor is the mask at g', whose Poisson expansion gives the one divisor d_2, while the signal calculation above gives the displayed two L factors and ray quotient. The restored principal pair $z_1 = z_2 = 1$ agrees by $G(1) = 1$. The restored principal contribution and the raw complement removed before this formula have already been bounded separately. The display (16.44) is therefore an equality for this component of the positive majorant (16.36); its relation to the original signed block remains the inequality (16.38).

Define the summand of (16.44) on all individually admissible squarefree z_1, z_2 by its displayed quotient-free L factors, fixed-ray quotient, formal product norms, full kernel, and unchanged $\mathcal{B}_2$. It agrees with the Poisson formula on coprime pairs. That support is recovered by the complete identity

$$1_{(z_1,z_2)=1} = \sum_{v'|(z_1,z_2)} \mu(v'), \qquad z_a = v'n_a,$$

before any factorwise estimate. Let $v \geq 0$ be the center of the squarefree v'-dyad and $\widehat{v} = \log_Z q_{v'}$. The n_a are squarefree, with $(v', n_a) = 1$; they need not be mutually coprime. On this extension the displayed divisor identity is used; the primitive Poisson formula applies only to the coprime pairs. For squarefree v' put

$$D_o(v'; d_2, k'') = a_0(v')\nu_i(v')\rho(v')1_{(v',r_0)=1}\chi_{v'}(H_o)^4\chi_{v'}(d_k)\overline{\chi_{v'}(d_2)}\chi_{v'}(k'').$$

On squarefree coprime v', n, extraction gives

$$\begin{aligned} L_{o,d_2,k''}(v'n) = &D_o(v'; d_2, k'')a_0(n)\nu_i(n)\rho(n)1_{(n,r_0)=1} \\ &\times \chi_n(H_o)^4\chi_n(d_k)\overline{\chi_n(d_2)}\chi_n(k'')\chi_n(v')^4. \end{aligned} \tag{16.45}$$

This uses $a_0(v'n) = a_0(v')a_0(n)\chi_n(v')^4$. On an overlap the right side defines the extension to be zero by $\chi_n(v')^4$, without evaluating a nonsquarefree Gauss sum. The two common factors give $|D_o(v'; d_2, k'')|^2$, including the zero $(v', k'') = 1$ and its zeros from the fixed coprimality restriction, d_k, and the label. They remain in the outer weight through Cauchy. The old gcd also leaves $1_{(v',g')=1}$. Since D_o vanishes on $(v', CJ) > 1$, we retain equivalently the explicit outer factor $1_{(v',g'CJ)=1}$, which repeats that label zero.

Set $r_g = g'/d_2$, $k_{\mathrm{new}} = d_k d_2 k''$ and $f_{\mathrm{new}} = JCd_2v'$. All moving factors on n satisfy the complete zero-extended identity

$$\chi_n(d_k)\chi_n(k'')\overline{\chi_n(d_2)}\chi_n(CJ)^4\chi_n(v')^4 = \chi_n(k_{\mathrm{new}})\chi_n(f_{\mathrm{new}})^4. \tag{16.46}$$

It uses $\overline{\chi_n(d_2)} = \chi_n(d_2)\chi_n(d_2)^4$, also on nonunits. The residual gcd with $g' = d_2 r_g$ splits into the repeated zero at d_2 and the fixed coprimality restriction at r_g. The old coprimality restrictions at q_0, t' remain. If $\mathcal{C}_{\mathrm{ray}}$ denotes the fixed group through which G factors, put

$$\widehat{G}(\vartheta) = |\mathcal{C}_{\mathrm{ray}}|^{-1} \sum_{c \in \mathcal{C}_{\mathrm{ray}}} G(c)\overline{\vartheta(c)}.$$

Then

$$G([n_2][n_1]^{-1}) = \sum_{\vartheta \in \widehat{\mathcal{C}}_{\mathrm{ray}}} \widehat{G}(\vartheta)\overline{\vartheta(n_1)}\vartheta(n_2).$$

Thus both new polynomials in a fixed ray summand use the same $\nu_i\bar{\vartheta}$, and its one bounded coefficient $\widehat{G}(\vartheta)$ remains outer.

7. Child parameters and admissibility. The second transformation has returned the moving characters to canonical form. The remaining arithmetic factors have two different roles: the fixed triple determines the child coprimality restriction, while the new row and squarefree label remain averaging variables. The construction must preserve that distinction when their ranges are enlarged. The following formulas identify these objects and their ranges. Put $\nu'_a = \nu_i\bar{\vartheta}$ for both sides and define

$$\begin{aligned} \gamma &= (q_0, t', r_g), \qquad r_g = g'/d_2, \\ k_{\mathrm{new}} &= d_k d_2 k'', \qquad f_{\mathrm{new}} = JCd_2v', \\ \rho'(n) &= \rho(n)1_{(n,\operatorname{rad} q_0 t' r_g)=1}. \end{aligned} \tag{16.47}$$

On the nonzero coefficient support, J, C, d_2, v' are squarefree and pairwise coprime: $(C, J) = 1$ belongs to $\mathfrak{O}_\sigma$, $|B_o(g')|^2$ forces $(g', CJ) = 1$, $d_2 \mid g'$, and the retained zero of D_o, together with the old gcd restriction, gives $(v', g'CJ) = 1$. Thus f_{new} is squarefree on that support. These common factors remain in the outer weight through Cauchy.

The child column is n. The triple γ is fixed before summing the child row k_{new} and label f_{new}, so its coprimality restriction ρ' is independent of both averaging variables. The factors J, C, d_2, v' are counted through their product f_{new} in the label average, with no additional count as fixed labels.

On each selected-side copy, (16.16) assigns the slots along the ordered list $(d_2, r_g, v'; n_a)$. Let I'_a be the subcollection of the selected side's slots retained on n_a in copy a. Assigned primes and their priority masks remain outer. A surviving prime dividing n is already prime to JCd_2v' by the fourth-power zero and to $\operatorname{rad} q_0, t', r_g$ by the fixed coprimality restriction. These also supply every earlier survival exclusion because $\operatorname{supp}\mathfrak{B} \subset \operatorname{supp}(Jq_0)$. The surviving mark is therefore an original product-form subcollection. An old slot described as assigned to J retains its old $b_a, \mathfrak{A}_a$ assignment, regrouped by this derived support. Its prime choice is the same original choice.

Keep $\mathcal{B}_2$ outside the Fourier separation of the full kernel and old windows. Let I_g, I_d, I_v, I_k be the full supports of the current individual dyad cutoffs $\psi_g, \psi_{d_2}, \psi_v, \psi_{k''}$. Choose fresh cutoffs $\omega_{n,a}$ equal to one on the support of W_i divided by $I_g I_v$, with fixed full supports in an interval $[a, b]$, where $b \geq 1$. Choose a nonnegative ω_f equal to one on $I_J I_C I_d I_v$, with fixed full support I_f. Include these full supports, and the reconstructed formal product $g'v'n$ on them, in the finite window family $\mathcal{I}_D$ specified above. The full supports of these new child windows give the formal centers and support bounds

$$\begin{gathered} N_c = r - A_i - B - t - g - v, \qquad V_c = B + \theta + v + j, \qquad F_c = N_c + V_c, \\ |\widehat{n}_{\mathrm{child}} - N_c| \leq 6\eta, \qquad |\widehat{f}_{\mathrm{new}} - V_c| \leq 4\eta. \end{gathered} \tag{16.48}$$

On the factorized expression, the products $n_i = \mathfrak{A}_i C t' g' v' n_{\mathrm{child}}$ and $f_{\mathrm{new}} = JCd_2v'$ give these bounds. After positivity adds independent terms, the same bounds follow from the full supports of the new child cutoff and enlarged label window, both included in $\mathcal{I}_D$; a parent factorization is not required for the added terms. The family $\mathcal{I}_D$ also includes the formal product $x = g'v'n$ on the full support of the new cutoff. Consequently the 4η bound used in the second prefactor holds on the whole separated side.

The nonnegative label center V_c is kept without upward rounding. A child is retained only if its row component is nonzero; then the outer ball has $M_{\mathrm{ch}} \geq 0$. This nonemptiness condition is checked separately from membership in the structural outer index set, which may contain zero-weight tuples even when $\mathcal{B}_2$ is identically zero. Similarly, if the full support of a new child column window contains no squarefree ideal, its polynomial is zero and that component is omitted. If $N_c < 0$ and a child column exists, its norm is at least one, and (16.48) gives $-6\eta \leq N_c < 0$. For a retained nonempty child define

$$N_{\mathrm{ch}} = \max(0, N_c), \quad \delta_N = N_{\mathrm{ch}} - N_c \in [0, 6\eta], \quad F_{\mathrm{ch}} = N_{\mathrm{ch}} + V_c = F_c + \delta_N. \tag{16.49}$$

The following support calculation shows that this clipping changes the test by a bounded rescaling. Recall the fixed enclosing support interval $[a, b]$ chosen above. If $N_c < 0$, nonemptiness gives $Z^{\delta_N} \leq b$; if $N_c \geq 0$, then $\delta_N = 0$ and the same bound follows from $b \geq 1$. Thus $1 \leq Z^{\delta_N} \leq b$ in both cases. The fresh annular cutoffs $\omega_{n,a}(Z^{\delta_N} y)$ have support in $[a/b, b]$ and uniformly bounded Euler seminorms. Their scale depends only on the fixed sector centers, not on a current row or label. The full union of these clipping supports is included in $\mathcal{I}_D$.

Fix a refinement σ of σ_0 by its v'-dyad, second ray summand and slot branches. Let Ξ_σ consist of

$$\xi = (o, g' \text{ sf}, d_2 \mid g', v' \text{ sf}, k'' \neq 0; \text{ new assigned slot primes})$$

with the retained individual supports and slot branches. It contains neither the old f, h nor the current column indices n_1, n_2. The identities defining R_1, E, J, q_0 are inherited from o, and $\gamma, k_{\mathrm{new}}, f_{\mathrm{new}}$ are the displayed functions of ξ. The set may be taken before imposing $\mathcal{B}_2$ and $(v', g'CJ) = 1$, retaining those masks in the signed outer weight. Let $\mathcal{A}_{2,\sigma}(\xi)$ be the product of all newly assigned original slot coefficients and their priority masks, conjugated on side two.

Let $\mathfrak{X}_\sigma^{\mathrm{src}}$ be the outer index set before any independent child rows or labels are added. It retains $o \in \mathfrak{O}_\sigma$ and hence the reconstruction $b_1 b_2 = q_0^2 J_2^2 s$, the derived ideals $J = sJ_2$ and q_0, and the full fixed b_1, b_2, t' dyad supports. It also retains the fixed g', d_2 supports, the divisibility $d_2 \mid g'$, and all other outer arithmetic and slot conditions. Zero-weight tuples may be retained, and unit Fourier phases are ignored in defining this structural set. Define the set of distinct triples, with no witness multiplicity, by

$$\Gamma_\sigma = \{(q_0, t', r_g = g'/d_2) : \xi \in \mathfrak{X}_\sigma^{\mathrm{src}}\}. \tag{16.50}$$

This projection is taken once over all possible current rows and labels, before fixing any $k_{\mathrm{new}}, f_{\mathrm{new}}$ or Fourier mode. Every nonzero transformed term maps into it. Every member has a source witness, which gives $\widehat{c} \leq \ell + \eta$, $\widehat{t} \leq t + \eta$, and $\log_Z q_{r_g} \leq g - \theta + 2\eta$. The first of these bounds will be needed for the restriction on child columns; the containing norm balls used for counting do not imply it. Nonemptiness of the row component and child-column window is still checked separately from this structural projection.

Put

$$D_c = \ell + A_i + t + g - \theta + V.$$

Since $r = N - 3\ell$ is an exact definition tied to the fixed parent, substitution in (16.48) and (16.40) gives the formal identities

$$\begin{aligned} F_c &= F - D_c - 2\ell + j, \qquad M_c = M - 2D_c - 2\ell + j, \\ F_c - M_c &= F - M + D_c, \\ 4F_c - 3M_c &= 4F - 3M + 2(A_i + t + g - \theta + V) + j. \end{aligned} \tag{16.51}$$

Here $F = N + V$ is the fixed parent length sum in (16.9).

For every $\gamma \in \Gamma_\sigma$, choose any source witness. Its actual divisibility $d_2 \mid g'$ and (16.26) give

$$\widehat{\theta} \le \widehat{g}, \qquad \widehat{j} \le 2\widehat{\ell}_{\mathrm{pair}}, \qquad \widehat{c} \le \widehat{\ell}_{\mathrm{pair}}.$$

Consequently the only actual-to-center inequalities needed here are

$$g - \theta \ge -2\eta, \qquad j \le 2\ell + 3\eta, \qquad \widehat{c} \le \ell + \eta. \tag{16.52}$$

The 2η and 3η errors in the first two inequalities are retained. The third inequality holds for every fixed triple because it has a source witness; a triple in the containing count ball alone need not satisfy it. The new radical in (16.47) has actual logarithmic norm Q_{new}, satisfying

$$\begin{aligned} Q_{\mathrm{new}} &\le Q + \widehat{c} + \widehat{t} + \widehat{g} - \widehat{\theta} \\ &\le Q + \ell + t + g - \theta + 4\eta \le Q + D_c + 4\eta. \end{aligned} \tag{16.53}$$

This also holds when its factors share primes with the old coprimality restriction, because taking the radical only decreases the norm. The last inequality uses $A_i, V \ge 0$.

Substituting $F_{\mathrm{ch}} = F_c + \delta_N$, $M_{\mathrm{ch}} = M_c + \eta$, and $2(A_i + t + g - \theta + V) + j \ge -4\eta$ into (16.51) gives the two child margins

$$\begin{aligned} F_{\mathrm{ch}} - M_{\mathrm{ch}} - Q_{\mathrm{new}} - z_0 &\ge c_{\mathrm{node}} + \delta_N - 5\eta \ge c_{\mathrm{node}} - 5\eta, \\ 4F_{\mathrm{ch}} - 3M_{\mathrm{ch}} - 6z_0 &\ge c_{\mathrm{node}} + 4\delta_N - 7\eta \ge c_{\mathrm{node}} - 7\eta. \end{aligned} \tag{16.54}$$

The nominal cap z_0 is unchanged because the surviving mark is a subcollection. Moreover $D_c \ge \ell + V - 2\eta$ and $2\ell - j \ge -3\eta$, whence

$$M - M_{\mathrm{ch}} = 2D_c + 2\ell - j - \eta \ge 2(\ell + V) - 8\eta \ge 2d - 8\eta. \tag{16.55}$$

For $\eta \le d/16$ this is at least $3d/2$, hence at least d. The same identities give $F_c \le F - \ell - V + 5\eta$, so $F_{\mathrm{ch}} \le F + 11\eta$. Because its two summands are nonnegative, this bounds the child N_{ch}, V_c through the finite depth.

The arithmetic construction has produced admissible child ranges with a smaller row length. To insert the child estimate, the transformed sum must first be expressed in canonical polynomials with a Fourier density common to all triples γ, rows and labels. The identity below retains the arithmetic masks. Cauchy is applied only after this identity, producing positive sums in which the reconstruction multiplicities can be bounded.

8. Separated child energies. Before any actual outer ideal is summed, use the six independent coordinates

$$(y_g, y_d, y_v, y_k, y_1, y_2) = \left(\frac{q_{g'}}{Z^g}, \frac{q_{d_2}}{Z^\theta}, \frac{q_{v'}}{Z^v}, \frac{q_{k''}}{Z^{h_2}}, \frac{q_{n_1}}{Z^{N_c}}, \frac{q_{n_2}}{Z^{N_c}}\right).$$

Recall the fresh cutoffs $\omega_{n,a}, \omega_f$ already chosen. The four current dyad cutoffs and ω_f remain outside inversion, and $\omega_{n,a}$ remain in the two columns. Let $\Omega_{2,\alpha}$ be larger cutoffs equal to one on the full supports of the corresponding four dyad and two column cutoffs. These full supports, I_f, and the formal products $g'v'n$ on them are the windows already included in $\mathcal{I}_D$.

The second root and kernel argument are

$$\begin{aligned} \frac{K}{q_{d_2} q_{v'} \sqrt{q_{n_1} q_{n_2}}} &= Z^{\lambda_c + 12\eta + \tau} y_d^{-1} y_v^{-1} (y_1 y_2)^{-1/2}, \\ \frac{K q_{k''}}{q_{d_2} q_{v'}^2 q_{n_1} q_{n_2}} &= A_{\mathrm{ker},2} \frac{y_k}{y_d y_v^2 y_1 y_2}, \\ A_{\mathrm{ker},2} &= Z^{H_{\mathrm{use}} + h_2 - \theta - 2v - 2N_c}. \end{aligned} \tag{16.56}$$

Indeed $N_c = s_i - g - v$ and $\lambda_c = H_c - \theta - (s_i - g)$. Define the one ambient profile

$$\begin{aligned}\mathfrak{H}_{2,\sigma,\mathbf{t}}(\mathbf{y}) &= Z^{-6\eta}\prod_{\alpha}\Omega_{2,\alpha}(y_\alpha)y_d^{-1}y_v^{-1}(y_1y_2)^{-1/2}\\ &\quad\times W_i(y_gy_vy_1)\overline{W_i(y_gy_vy_2)}\mathcal{K}_2\left(A_{\ker,2}\frac{y_k}{y_dy_v^2y_1y_2}\right).\end{aligned}\tag{16.57}$$

The outside scalar is $Z^{\lambda_c+18\eta+\tau}$. As in (16.20), the profile contains the full q_{d_2} denominator and both normalized real column roots; the final tests exclude them.

For $\boldsymbol{\zeta} = (\zeta_g, \zeta_d, \zeta_v, \zeta_k, \zeta_1, \zeta_2)$, put $z_n = q_n/Z^{N_{\mathrm{ch}}}$. Recalling the child character, define the separated tests and surviving marks by

$$W'_a(z) = \omega_{n,a}(Z^{\delta_N}z)z^{\varepsilon_a i\zeta_a}, \qquad \nu'_a = \nu_i\bar{\vartheta}, \qquad \mathfrak{d}'_a = \mathfrak{d}_{I'_a}.$$

The polynomial on either new Cauchy side is then

$$Q_{a,\gamma,\boldsymbol{\zeta}}(k_{\mathrm{new}}, f_{\mathrm{new}}) = \sum_{n\ \mathrm{sf}}\bar{\alpha}(n)\gamma_2(n)\nu'_a(n)\rho'(n)\chi_n(k_{\mathrm{new}})\times\chi_n(f_{\mathrm{new}})^4\mathfrak{d}'_a(n)W'_a(q_n/Z^{N_{\mathrm{ch}}}).\tag{16.58}$$

After $q_0, t', g'/d_2$ and the finite ray sector are fixed, ν'_a and ρ' are independent of J, C, d_2, v' and the new row. The only new characters on n are the explicitly displayed row and fourth-power factors; all other phases involving the old $b_i, \mathfrak{A}_i$ were placed in (16.27) or in the old row $\widetilde{h}$. The separated annular weight W'_a is also independent of the new row and label. The complete remaining outer weight is

$$\begin{aligned}V_\sigma(\xi;\boldsymbol{\zeta}) &= w_o^+\mu(d_2)\mu(v')\widehat{G}(\vartheta)|B_o(g')|^2|D_o(v';d_2,k'')|^2 1_{(v',g'CJ)=1}\mathcal{B}_2(k'')\\ &\quad\times\mathcal{A}_{2,\sigma}(\xi)\psi_g(y_g)\psi_{d_2}(y_d)\psi_v(y_v)\psi_{k''}(y_k)\omega_f(q_{f_{\mathrm{new}}}/Z^{V_c})\\ &\quad\times y_g^{i\zeta_g}y_d^{i\zeta_d}y_v^{i\zeta_v}y_k^{i\zeta_k}Z^{i\delta_N(\zeta_1+\zeta_2)}.\end{aligned}\tag{16.59}$$

All old pure arithmetic and priority masks are retained in $\mathfrak{O}_\sigma$, with their nonnegative old coefficient weight w_o^+. The displayed two common magnitudes retain every other common coefficient zero, including the one depending on k''. There is one $\mu(d_2)$, one $\mu(v')$, and one ray coefficient. The label cutoff is inserted as an equality on the original product supports and remains outer.

Let $\mathcal{T}_{\sigma,i}$ be the fixed v'-dyad, ray, and slot summand obtained from (16.44) by the complete Möbius and slot expansions above. The identity expressing the component in child polynomials is

$$\mathcal{T}_{\sigma,i} = Z^{\lambda_c+18\eta+\tau}(2\pi)^{-6}\int_{\mathbb{R}^6}\widehat{\mathfrak{H}}_{2,\sigma,\mathbf{t}}(\boldsymbol{\zeta})\times\sum_{\xi\in\Xi_\sigma}V_\sigma(\xi;\boldsymbol{\zeta})Q_{1,\gamma,\boldsymbol{\zeta}}(k_{\mathrm{new}}, f_{\mathrm{new}})\overline{Q_{2,\gamma,\boldsymbol{\zeta}}(k_{\mathrm{new}}, f_{\mathrm{new}})}\,d\boldsymbol{\zeta}.\tag{16.60}$$

This identity can be checked by expanding the two Q's. Since $y_a = Z^{\delta_N}z_{n_a}$, their modes together with the last factor of (16.59) are exactly $y_1^{i\zeta_1}y_2^{i\zeta_2}$. The other four modes are outer. Fourier inversion in (16.20) restores the full profile, whose larger cutoffs are one on the retained supports. The fresh column cutoffs are one wherever the old W_i factors are nonzero, and $\omega_f = 1$ on every original label product. The scalar and profile therefore restore the full root, full kernel, and old windows. The extraction identities (16.45), (16.46), and (16.16) restore the arithmetic factors term by term. The fixed-box sums are finite and (16.21) justifies the integral. Extra combinations in the full supports of the new windows cancel through the old windows in this complex Fourier identity before Cauchy; only later are they kept independently in a positive sum.

The density in (16.60) is determined by the fixed sector, Z, the fixed tests and $\mathbf{t}$. It is independent of the individual values of $\gamma, k_{\mathrm{new}}, f_{\mathrm{new}}$. The coupled kernel uses the bare k'' coordinate. The derived row appears only in characters and $\mathcal{B}_2$. The derived label appears only in characters and its outer cutoff. Relations such as $g' = d_2r_g$ restrict the evaluation points while

the transform remains fixed. For every fixed $B_{\mathrm{ht}}, J_{\mathrm{ht}} \geq 0$, the Euler calculation in (16.21) gives some fixed b with

$$\int_{\mathbb{R}^9} \int_{\mathbb{R}^6} |\widehat{\mathfrak{H}}_{1,\sigma}(\mathbf{t})||\widehat{\mathfrak{H}}_{2,\sigma,\mathbf{t}}(\boldsymbol{\zeta})|(1+|\mathbf{t}|)^{B_{\mathrm{ht}}}(1+|\boldsymbol{\zeta}|)^{J_{\mathrm{ht}}}\, d\boldsymbol{\zeta}\, d\mathbf{t} \ll p_b(W)^2. \tag{16.61}$$

Indeed the inner weighted integral is polynomial in $\mathbf{t}$, because W_i is a fixed cutoff times a unit norm mode. The first transform's bound at that polynomial order plus B_{ht} therefore gives the stated joint integral bound. The constants are uniform in every actual ideal and every positive kernel scalar.

Cauchy–Schwarz and reconstruction of the outer indices. For each fixed Fourier mode, (16.34) with $U_a = Z^{(\lambda_{\mathrm{c}}+18\eta+\tau)/2} Q_{a,\gamma,\boldsymbol{\zeta}}$ on the complete set Ξ_σ bounds the second transformed sum by the geometric mean of two positive sums. In particular $\mathcal{B}_2$, $|B_o|^2$, $|D_o|^2$, and $1_{(v',g'CJ)=1}$ are still in its weight. They retain the squarefreeness of f_{new} proved above. All other factors in the weight are bounded by a fixed constant: there are a bounded number of original assigned slot coefficients, one fixed ray coefficient, the individual cutoffs, and unit modes. In the positive sums obtained from Cauchy, we may therefore use

$$|V_\sigma(\xi;\boldsymbol{\zeta})| \ll \mathcal{B}_2(k'')\mu^2(f_{\mathrm{new}}).$$

This is the step which permits the new averaged ideal to remain in the squarefree class.

The following divisibility relation controls the reconstruction multiplicity of a fibre.

$$d_k \mid C\mathfrak{B}/R_1 \quad \Longrightarrow \quad d_k \mid f_{\mathrm{new}} \operatorname{rad} q_0. \tag{16.62}$$

To prove this, a prime of $\mathfrak{B}/R_1$ has $t_p = 0$. The local table shows that it has even parity and equal a_{ip}. If both are one it is in J_2, and if both are zero it is in $\operatorname{rad} q_0$. Primes of C are already in f_{new}. This proves the implication. Thus d_k also has divisor multiplicity after f_{new}, q_0 are specified. Once these factors are fixed, k'' is uniquely determined by the element $k_{\mathrm{new}} = d_k d_2 k''$.

Let K_{slot} bound the original number of slots. At fixed $\gamma = (q_0, t', r_g)$, squarefree $f = f_{\mathrm{new}}$, and element $k = k_{\mathrm{new}}$, the number of possible $\xi \in \Xi_\sigma$ with nonzero weight is at most a fixed constant times $D_{K_{\mathrm{slot}}}(f) C_{K_{\mathrm{slot}}}(\gamma)$, where

$$\begin{aligned} D_{K_{\mathrm{slot}}}(f) &= d_{\mathcal{O}}(f)^{9+4K_{\mathrm{slot}}}, \\ C_{K_{\mathrm{slot}}}(\gamma) &= d_{\mathcal{O}}(q_0)^{5+2K_{\mathrm{slot}}} d_{\mathcal{O}}(t')^{2K_{\mathrm{slot}}} d_{\mathcal{O}}(r_g)^{2K_{\mathrm{slot}}}, \end{aligned} \tag{16.63}$$

Here $d_{\mathcal{O}}$ is the ideal divisor count. The fibre bound follows by reconstructing the indices in order. Allocating f among J, C, d_2, v' gives $4^{\omega(f)} = d_{\mathcal{O}}(f)^2$ choices, and splitting $J = sJ_2$ gives at most $d_{\mathcal{O}}(f)$. The factorizations $b_1 b_2 = q_0^2 J_2^2 s$ number at most $d_{\mathcal{O}}(q_0)^2 d_{\mathcal{O}}(f)^3$. Choosing the two $\mathfrak{A}_a \mid \operatorname{rad}(b_1 b_2)$ contributes at most $d_{\mathcal{O}}(q_0)^2 d_{\mathcal{O}}(f)^2$. These bounds use $d_{\mathcal{O}}(IJ) \leq d_{\mathcal{O}}(I) d_{\mathcal{O}}(J)$ and $d_{\mathcal{O}}(I^2) \leq d_{\mathcal{O}}(I)^2$.

By (16.62), there are at most $d_{\mathcal{O}}(q_0) d_{\mathcal{O}}(f)$ choices for d_k. Then $g' = d_2 r_g$ is fixed, and $k'' = k/(d_k d_2)$ is the unique element if it is integral. The at most $2K_{\mathrm{slot}}$ old assigned primes divide $q_0 f t'$, giving at most $[d_{\mathcal{O}}(q_0) d_{\mathcal{O}}(f) d_{\mathcal{O}}(t')]^{2K_{\mathrm{slot}}}$ choices. The at most $2K_{\mathrm{slot}}$ new assigned primes divide $f r_g$, giving at most $[d_{\mathcal{O}}(f) d_{\mathcal{O}}(r_g)]^{2K_{\mathrm{slot}}}$ choices. Selecting the first positive side removed the other side's surviving slots; the selected surviving slots occur twice in the new square, as counted here. Multiplying these bounds gives (16.63), even when the actual fibre size depends on k. The old f, h have already been removed, J, C, d_2, v' are reconstructed from the new label, and k'' is fixed by the new row. None receives an additional independent count.

The norm bound (16.37) and the same source witnesses place Γ_σ inside the product of the three norm balls

$$q_{q_0} \leq Z^{\ell+R/2-j+5\eta/2}, \qquad q_{t'} \leq Z^{t+\eta}, \qquad q_{r_g} \leq Z^{g-\theta+2\eta}.$$

We use these balls only to bound the number of distinct triples, not as a replacement domain for the child estimate. Ideal counting gives

$$\#\Gamma_\sigma \le C Z^{C_c+11\eta/2+\pi_{\mathrm{count}}}, \qquad C_c = \ell + R/2 - j + t + g - \theta. \tag{16.64}$$

Here π_{count} is part of the separately chosen aggregate π. In a nonempty sector each upper exponent of these three balls is nonnegative, since it bounds the norm of an existing ideal; the constant term in ideal counting therefore adds no boundary power. In particular J is not counted here. The ranges of all three ideals are bounded, so the adjustable divisor bound gives $C_{K_{\mathrm{slot}}}(\gamma) \ll Z^{\pi_{\mathrm{fib}}}$ uniformly on the containing balls. Put $c_\sigma = C_c + 11\eta/2$ and introduce the measure, independent of the row and label,

$$d\nu_\sigma(\gamma) = Z^{-c_\sigma} \sum_{\gamma\in\Gamma_\sigma} C_{K_{\mathrm{slot}}}(\gamma)\delta_\gamma, \qquad \|\nu_\sigma\| \ll Z^{\pi_{\mathrm{count}}+\pi_{\mathrm{fib}}}. \tag{16.65}$$

The child row moments. For a fixed Fourier mode define the positive child sums

$$\mathcal{H}_{a,\gamma} = \sum_{\substack{f \text{ sf} \\ q_f/Z^{V_c}\in I_f}} D_{K_{\mathrm{slot}}}(f) \sum_{\substack{k\in\mathcal{O} \\ 0<q_k\le Z^{M_{\mathrm{ch}}}}} |Q_{a,\gamma,\boldsymbol{\zeta}}(k,f)|^2 = Z^{F_{\mathrm{ch}}}\mathcal{E}_{a,\gamma}. \tag{16.66}$$

The factor on the right restores both normalizations in (16.2): the label average and the squared column polynomial. This uses the same row ball, the label weight $D_{K_{\mathrm{slot}}}$, and the test and coprimality restriction in (16.58) and (16.47). If the canonical energy uses a larger fixed row ball, the same comparison is an inequality in the needed upper-bound direction. The weight depends on f alone. After reindexing, $\mathcal{B}_2$ is exactly the displayed row ball. Only the now positive outer row and label sums have been enlarged to their full fixed windows. The columns use the full supports of their new fixed windows; their coprimality restriction, mark, and test remain unchanged. The actual fibre domination just proved gives

$$\begin{aligned}\left|\sum_{\xi\in\Xi_\sigma} V_\sigma Q_1 \overline{Q_2}\right| &\le \prod_{a=1}^{2}\left(\sum_{\xi\in\Xi_\sigma}|V_\sigma||Q_a|^2\right)^{1/2} \\ &\ll Z^{c_\sigma}\prod_{a=1}^{2}\left(\int_{\Gamma_\sigma}\mathcal{H}_{a,\gamma}\,d\nu_\sigma(\gamma)\right)^{1/2}.\end{aligned}$$

Here $Q_a = Q_{a,\gamma,\boldsymbol{\zeta}}(k_{\mathrm{new}}, f_{\mathrm{new}})$. Each square root supplies $Z^{c_\sigma/2}$, so their product supplies one fixed-count factor. Likewise a uniform child bound gives $\|\nu_\sigma\|^{1/2}$ from each side and hence one total mass $\|\nu_\sigma\|$. The child bound will be used only for $\gamma\in\Gamma_\sigma$, using the hypotheses already verified in (16.54).

9. The recursive energy bound. The first transformed side contributes κ_i, the second Poisson factor contributes λ_c, and the fixed-triple count contributes C_c. The two copies of F_c come from restoring the child normalization and applying its asserted bound. Their formal centers satisfy

$$\kappa_i + \lambda_c + C_c + 2F_c = F. \tag{16.67}$$

This follows by substituting their definitions and $H_c = 2r - 2B - M + 4\ell + 2V + \delta - j$; all extracted lengths cancel and the result is $r + 3\ell + V = N + V$.

We now use the child estimate only for $\gamma\in\Gamma_\sigma$, for which the preceding margins and row decrease have been proved. If it gives $\mathcal{E}_{a,\gamma} \ll C(\mathbf{t},\boldsymbol{\zeta})Z^{F_{\mathrm{ch}}+\epsilon_{\mathrm{child}}}$ uniformly there, with its fixed polynomial height dependence, then (16.66) and the final weighted Cauchy bound above give

$$\left|\sum_{\xi\in\Xi_\sigma} V_\sigma Q_1\overline{Q_2}\right| \ll C(\mathbf{t},\boldsymbol{\zeta})Z^{c_\sigma+2F_{\mathrm{ch}}+\epsilon_{\mathrm{child}}}\|\nu_\sigma\|.$$

We use the normalized discrete measure (16.65), whose fibre weights already include witness multiplicity. Its mass and the fixed-count factor each occur once, as in the preceding Cauchy bound. Integrating the common Fourier density with (16.61) leaves the averaged J, C, d_2, v' inside the child sum. The density is separate from this discrete measure and common to all rows and labels.

If a child is bounded by $Z^{F_{\mathrm{ch}}+\epsilon_{\mathrm{child}}}$, the first prefactor error, the second prefactor error, the fixed count, and the restored normalization give on each positive second-Cauchy side

$$\begin{aligned}\kappa_i+\lambda_c+C_c+2F_{\mathrm{ch}}+(\tfrac{9}{2}+18+\tfrac{11}{2})\eta+\tau+\pi+\epsilon_{\mathrm{child}}\\ =F+28\eta+2\delta_N+\tau+\pi+\epsilon_{\mathrm{child}}\\ \le F+40\eta+\tau+\pi+\epsilon_{\mathrm{child}}.\end{aligned}\tag{16.68}$$

Both copies of δ_N are present: one restores the child column normalization and the other occurs in its asserted exponent. Each of the two weighted Cauchy inequalities takes a geometric mean of its positive sides, so it does not double this loss. The first principal bound is $Z^{M+\pi}$, and the second principal bound above is at most $Z^{F+26\eta+\tau+\pi}$; both satisfy the same exponent bound.

The aggregate local loss. The parameter π has an independent quantifier. Let J_{step} be the maximum number of uses of a free divisor or sieve exponent and of logarithmic dyadic-count factors in one fixed two-Poisson, two-Cauchy factorization tree, maximized over the finitely many assignments with at most the fixed slot bound. Once L_{pow} bounds the log-length of every scale product in such a use, choose each free input exponent at most $\pi/(2J_{\mathrm{step}}\max(1,L_{\mathrm{pow}}))$. For sufficiently large Z, the product of the at most J_{step} logarithmic factors is at most $Z^{\pi/2}$. These choices make all non-center, non-frequency local losses at most CZ^{π}. They include the retained common frequency dyads and the reconstruction divisor bounds. In particular the distinct allocations $\pi_\beta, \pi_{\mathrm{old}}, \pi_{\mathrm{count}}, \pi_{\mathrm{fib}}$ and the finite-union dyadic allocation sum to this one aggregate π. Fixed ray sums and fixed seminorm constants contribute constants. The same convention defines the aggregate terminal loss π_{ref}. The center and frequency losses involving η and τ remain separate from π.

This proves the conditional reduction estimate (16.18), including its common-measure and finite-order uniformity once the choices below are made. It remains to fix those choices uniformly and apply the reduction through the finite depth.

10. Uniform choices and termination. For the requested canonical exponent $\epsilon_c=\epsilon$, first fix the starting ranges, tests, slot bound, c_*, and then d, D as at the beginning of the proof. Choose

$$\begin{gathered}\eta\le\min\{d/16,c_*/(14D),\epsilon_c/(160D),c_*/1000,1/100\},\\ \tau\le\min\{\epsilon_c/(4D),c_*/1000,1/100\},\qquad \pi\le\epsilon_c/(4D),\qquad \tau_{\mathrm{ref}},\pi_{\mathrm{ref}}\le c_*/1000.\end{gathered}\tag{16.69}$$

Both Poisson comparisons use this same τ. At this point π is a target aggregate loss; its subsidiary input exponents are chosen only after the bounded scale range is known.

Choose the individual cutoffs through depth D, including every full larger or fresh support and the bounded clipping families described above, and form $\mathcal{I}_D, L_{\mathrm{win}}$. Include the product intervals for the active slots and the terminal constants $C_{\mathrm{res}}, C_{\mathrm{width}}, C_{\mathrm{ker}}$, maximized through the depth. This is finite because there are finitely many factorization types per passage and finitely many passages. Put

$$F_{\max}=N_{\max}+V_{\max}+11D\eta,\qquad L_{\mathrm{all}}=100(1+M_{\max}+F_{\max}+z_{0,\max}+c_*).$$

These bound all retained scale lengths. Indeed, (16.55) and the bound following it give $M\le M_{\max}$ and $N,V\le F_{\max}$. A nonempty column window has $r\ge-\eta$, so $\ell\le(F_{\max}+\eta)/3$. The displayed parent products bound every extracted divisor length by $2F_{\max}+6\eta$. For example $d_k\mid C\,\mathrm{rad}(b_1b_2)$ gives $\delta\le r+2\ell+4\eta$, and (16.28) then gives $H_{\mathrm{use}}\le 7F_{\max}+18\eta+\tau$. The terminal active conductor is supported on the row, f, the excluded-prime radical, and slots; their total lengths are at most $M_{\max}+2F_{\max}+z_{0,\max}$ plus the already displayed ratio allowances. The length bounds (16.11)

and (13.11) bound the retained terminal dual lengths by $4M_{\max}+4F_{\max}+2z_{0,\max}+3d+12\eta+\tau_{\mathrm{ref}}$. Under (16.69), all these bounds and the exact conductor product lengths are below L_{all}. We may now take $L_{\mathrm{pow}}=L_{\mathrm{all}}$ and choose the subsidiary small exponents that realize π and π_{ref}.

Raw tail bounds. Define B_{crude} to be the maximum exponent obtained by replacing every bounded nonfrequency ideal or element sum in either raw Poisson expansion, including both column sums, by its lattice count at length L_{all}, and every explicit norm factor, including $1+a^{-1}$ in (16.22), by its absolute upper bound. Replace divisor-bounded coefficients and marks by their trivial polynomial norm bounds as well. Only the frequency kernel itself is excluded from this count. The expansions contain a fixed finite number of factors, so this defines a finite uniform number.

For the reflected tail define B_{ref} analogously using its bounded row, local, and active-product counts and $1+a^{-1}$ with $a=X/q_c^2$. Discarded dual indices require an unrestricted count. On the support of (4.3),

$$\frac{|d(\mu)|}{\sqrt{q_\mu}}\le 27q_\lambda^{-k/3}q_n^{-1/2}q_b^{-1}\le 27q_\lambda^{4/3}\qquad(k\ge -4).$$

Every other local factor is bounded by the product of $q_p^{1/2}$ over the bounded active primes, independently of μ after its indicators are dropped. Thus B_{ref} covers all non-kernel coefficients and bounded counts, while the remaining μ-sum is an unrestricted lattice sum on $\lambda^{-4}\mathcal{O}$.

Fix a tail saving $T>1+B_{\mathrm{crude}}+B_{\mathrm{ref}}$. For each Poisson tail, choose a Schwartz order

$$A>1+(B_{\mathrm{crude}}+T)/\tau.$$

The tail bound (16.22) with $Y=Z^\tau$ then makes the discarded raw outer-ball complement $O(Z^{-T})$, up to finitely many input seminorms and a fixed polynomial in any twist heights. For the reflected whole-dyad tail the actual argument is at least $Z^{\tau_{\mathrm{ref}}/2}$; use the same lattice-shell bound on $\lambda^{-4}\mathcal{O}$ and choose

$$A_{\mathrm{ref}}>1+2(B_{\mathrm{ref}}+T)/\tau_{\mathrm{ref}}.$$

This proves the discarded-tail assertion without presuming any bound on the discarded dual lengths.

Derivative orders and the final threshold. Choose the finitely many Fourier-height and smooth-seminorm orders backwards through depth D, above the chosen tail orders and the terminal input orders. The weighted bounds (16.61) and (16.65) supply the common coefficient measure for each positive indexed Cauchy-side sum. For each such sum, the child-profile bounds just proved from Lemma 3.6 give the remaining hypotheses of Corollary 16.4. Its conclusions can therefore be combined in the displayed geometric means. The two square roots retain one normalized discrete mass, as already shown, so the outer labels are counted only once. Any external height cutoff in a later application is chosen after this internal finite propagation, not inserted into it.

Finally choose Z large enough for $\log Z\ge L_{\mathrm{win}}/\eta$, the thresholds for $C_{\mathrm{res}},C_{\mathrm{width}}$ in (16.11), $\log Z\ge 2\log C_{\mathrm{ker}}/\tau_{\mathrm{ref}}$, $\log Z\ge 4\log q_\lambda/\eta$, and the fixed h-annulus threshold used in the short completion. Impose also the logarithmic bounds defining the aggregate local losses and the thresholds of the input lemmas. All choices precede this final threshold and depend only on the fixed data. Bounded smaller Z are handled by increasing the final constant.

Termination of the induction. Now $c_h=c_*-7h\eta\ge c_*/2$ for $h\le D$. The margin and row-decrease bounds (16.54) and (16.55) send every retained nonterminal child to the next depth and its row cap. At depth D that cap is negative, whereas every retained row parameter is nonnegative. The terminal bounds are at most Z^F; principal and recursive terms increase the exponent by at most $L_{\mathrm{step}}=40\eta+\tau+\pi$. Backwards induction gives $Z^{F+(D-h)L_{\mathrm{step}}}$ at depth h, and

$$DL_{\mathrm{step}}\le\epsilon_c/4+\epsilon_c/4+\epsilon_c/4<\epsilon_c.$$

This proves $\mathcal{E}\ll Z^{F_0+\epsilon}$, with the claimed uniform finite-seminorm and polynomial-height dependence. □

16.6 Initialization of the marked moment

We now convert the inverse polynomial in Lemma 16.1 to the canonical family. Only one Poisson transformation is required.

Proof of Lemma 16.1. We first separate the overlap between the inverse column and the prime slots. One Poisson transformation then gives canonical polynomials with a new row and squarefree label. The overlap and an outer quotient determine their coprimality restriction. We verify their coefficient class, common Fourier measure, and two canonical margins before applying Lemma 16.2.

1. Overlap reduction and common orientation. In the product $M_{\psi_u}(Z^r;W)Q_u$, let P be the product of its slot primes and put $j=(n,P)$. Both n and P are squarefree. Fix the subset of slots occurring in j, and write

$$n=jn_0, \qquad P=jP_0, \qquad (n_0,P_0)=1.$$

The product $c=n_0P_0$ is squarefree. The identities

$$\mu(n_0)=\mu(c)\mu(P_0), \qquad \psi_u(n)\psi_u(P)=\psi_u(j)^2\psi_u(c)$$

hold with all zero extensions. The sign $\mu(P_0)$ is a product of signs on the surviving slots and can be incorporated into their bounded coefficients. The coefficients thus remain of the form in (13.1).

For this fixed assigned subset define G from its nominal slot centers:

$$G=\sum_{i \text{ assigned}} z_i, \qquad z_0=z-G, \qquad D'=r+z-2G.$$

Then $G, z_0 \ge 0$, and z_0 is exactly the surviving nominal cap. The normalization satisfies

$$Z^{-(r+z)/2}=Z^{-G}Z^{-D'/2}.$$

The actual j is the product of the assigned primes, so q_j/Z^G lies in a fixed product interval. Ideal counting gives at most CZ^G choices, apart from the separately chosen divisor loss for assignments. The triangle inequality in the row Hilbert space uses Z^{-G} to cancel this count. The whole factor $\psi_u(j)^2$ is a bounded row scalar, including its zeros, and is a contraction in that space. The residual column is prime to j, giving the fixed coprimality restriction $1_{(c,j)=1}$.

The overlap reduction is expressed by the following identity. Let I_0 be the surviving subset and put $y_j=q_j/Z^G$, $y_c=q_c/Z^{D'}$, and $y_i=q_{p_i}/Z^{z_i}$ for every slot. On the original support,

$$\frac{q_n}{Z^r}=\frac{y_jy_c}{\prod_{i\in I_0}y_i}.$$

For a fixed assigned tuple, its contribution $\mathcal{U}_{u,j}$ to $M_{\psi_u}(Z^r;W)Q_u$ is

$$\begin{aligned}\mathcal{U}_{u,j}=\mu(j)\prod_{i\notin I_0}a_i(p_i)W_i(y_i)\ Z^{-G}\psi_u(j)^2Z^{-D'/2}\sum_{\substack{c \text{ sf}\\(c,j)=1}}\mu(c)\psi_u(c)\\ \times\sum_{\substack{(p_i)\in\prod_{i\in I_0}\mathcal{P}_i\\P_0\mid c}}\mu(P_0)\prod_{i\in I_0}a_i(p_i)W_i(y_i)W\left(\frac{y_jy_c}{\prod_{i\in I_0}y_i}\right).\end{aligned} \tag{16.70}$$

Indeed a squarefree c prime to j, together with a surviving tuple with $P_0\mid c$, reconstructs uniquely $n=j(c/P_0)$ and the prescribed overlap. Squarefreeness of c already gives $(c/P_0,P_0)=1$, so no further condition was lost. The sign $\mu(P_0)$ is a product of the individual prime signs. This

also shows that the j on both sides of every subsequent square is one fixed ideal: the preceding triangle inequality was taken before that square.

The identity $c = nP/j^2$ puts y_c in a fixed compact interval. This remains meaningful if D' is slightly negative in a bounded nonempty window; D' is not yet a canonical parameter. Choose a fresh individual cutoff ω_c equal to one on this full quotient support. For fixed j, let Ω_c be one on the full support of ω_c, and let Ω_i be one on each full surviving individual slot support. The single joint profile

$$\mathfrak{H}_{\mathrm{ov},j}(y_c,(y_i)_{i\in I_0}) = \Omega_c(y_c)\prod_{i\in I_0}\Omega_i(y_i)W\left(\frac{y_j y_c}{\prod_{i\in I_0} y_i}\right)$$

is compactly supported in these independent coordinates. The current cutoff ω_c and the individual factors $W_i(y_i)$ stay outside its transform. Fourier inversion in (16.20), in dimension $1+|I_0|$, therefore expresses (16.70) exactly as the same assigned scalar and $Z^{-G}\psi_u(j)^2$, times

$$(2\pi)^{-1-|I_0|}\int_{\mathbb{R}^{1+|I_0|}}\widehat{\mathfrak{H}}_{\mathrm{ov},j}(\boldsymbol{\upsilon})Z^{-D'/2}\sum_{c\ \mathrm{sf}}\mu(c)\psi_u(c)1_{(c,j)=1}\mathfrak{d}_{\boldsymbol{\upsilon}}(c)\omega_c(y_c)y_c^{i\upsilon_c}\,d\boldsymbol{\upsilon},$$

where $\mathfrak{d}_{\boldsymbol{\upsilon}}$ has the original surviving lists and individual coefficients $\mu(p)a_i(p)W_i(y_i)y_i^{i\upsilon_i}$. Expanding the mark and inverting the joint transform proves this equality term by term. No measure is chosen for an actual surviving tuple. The measure may depend on the already fixed j, but (16.21) is uniform for its y_j in the fixed product interval. Minkowski's inequality is used in the whole row Hilbert space for this one measure. Include the full support of ω_c, its larger cutoffs, the surviving slot supports, and the j-product interval in the finite collection $\mathcal{I}_D$ before the final threshold is chosen. With the same η convention as the canonical proof, we then have

$$|\widehat{c}-D'|\le\eta,\qquad |\widehat{j}-G|\le\eta$$

on the full supports actually used below, where $\widehat{c}=\log_Z q_c$. For the sign $\varepsilon_\chi=1$, conjugate the whole residual polynomial, including its finite character, mark, and test; its row norm is unchanged. For the other sign leave it unchanged. It therefore suffices to estimate one polynomial of the form

$$\mathcal{R}_0(u) = Z^{-D'/2}\sum_{c\ \mathrm{sf}}\mu(c)\nu_0(c)1_{(c,j)=1}\overline{\chi_c(u)}\mathfrak{d}_0(c)W_0(q_c/Z^{D'}). \tag{16.71}$$

Here ν_0 is a fixed finite ray character, $\mathfrak{d}_0$ is the product mark of cap z_0 with those individual coefficients or their conjugates, and W_0 is $\omega_c(y)y^{i\upsilon_c}$ or its conjugate. All are independent of the row.

2. Poisson localization and the diagonal. A fixed nonnegative radial Schwartz function Φ_{init} at scale Z^m majorizes the original row ball. We expand the resulting squared polynomial. Let C be the gcd of the two columns and write $c_i = Cz_i$, with z_1, z_2 squarefree and coprime. Put $\widehat{c}_i=\log_Z q_{c_i}$, so the preceding support bound applies to each $i=1,2$. Let $B\ge 0$ be the center of C, and write $\widehat{B}=\log_Z q_C$. The row character has primitive modulus z_1z_2 and remaining zero mask C. Lemma 16.5 introduces the divisor $d'\mid C$. Let $\theta\ge 0$ be its center and $\widehat{\theta}=\log_Z q_{d'}$. Set

$$P_1 = B-\theta\ge -2\eta,$$

where the inequality follows from $\widehat{\theta}\le\widehat{B}$ and the two η-ratio bounds. The actual conductor length is

$$\widehat{L}_{\mathrm{init}} = \widehat{c}_1+\widehat{c}_2-2\widehat{B},\qquad |\widehat{L}_{\mathrm{init}}-2(D'-B)|\le 4\eta.$$

For a positive tolerance τ_{init}, the implication $Z^m q_{h'}/(q_{d'}q_{z_1z_2})\le Z^{\tau_{\mathrm{init}}}$ gives

$$\widehat{d'h'}\le 2D'-m-2P_1+6\eta+\tau_{\mathrm{init}}.$$

Define the fixed formal and enclosing row scales

$$M_c^{\mathrm{init}} = 2D' - m - 2P_1, \qquad M_{\mathrm{init}} = M_c^{\mathrm{init}} + 6\eta + \tau_{\mathrm{init}}. \tag{16.72}$$

The 6η is 4η from the conductor and 2η from the two copies of d' in the new row bound.

Separate the original principal contribution for the direct count below. On the nonprincipal coprime Poisson expression, before any off-coprime extension or Fourier absolutization, keep only the outer mask

$$\mathcal{B}_{\mathrm{init}}(h') = 1_{0<q_{d'h'}\le Z^{M_{\mathrm{init}}}}.$$

Its complement is contained in the actual ratio tail above for every supported pair. The tail bound (16.22), with the separate initial raw count and order specified below, discards precisely that complement. The full smooth kernel remains inside the mask, with no column-dependent ratio selector. The mask depends on d', h', the fixed overlap and sector, but not on the residual columns. If $M_{\mathrm{init}} < 0$, its nonzero ball is empty and the same tail comparison discards every nonzero frequency; otherwise the retained row parameter is nonnegative.

The actual normalization and Poisson prefactor split over the two sides as

$$Z^{-D'} Z^{m-\widehat{\theta}-\widehat{L}_{\mathrm{init}}/2} = Z^{\widehat{\kappa}_1^{\mathrm{init}}/2} Z^{\widehat{\kappa}_2^{\mathrm{init}}/2}, \qquad \widehat{\kappa}_i^{\mathrm{init}} = m - D' - \widehat{\theta} - \widehat{c}_i + \widehat{B}.$$

Thus the actual side exponents obey

$$\widehat{\kappa}_i^{\mathrm{init}} \le \kappa_c^{\mathrm{init}} + 3\eta, \qquad \kappa_c^{\mathrm{init}} = m - 2D' + P_1. \tag{16.73}$$

Following the convention of (16.20), all normalized real inverse roots are kept in the single joint profile below and excluded from the final child tests. Principal columns have the direct diagonal bound $O(Z^{m+\epsilon})$, also for a bounded nonempty negative D'-window. Any principal nonzero frequencies restored to the formal formula carry the same $\mathcal{B}_{\mathrm{init}}$ mask and are bounded by the full principal restoration in Lemma 16.5, since $Z^m \ge 1$.

3. The new row, label, and excluded primes. A single frequency partition for all d' gives the dyads of $q_{h'}$. The outer mask implies $q_{h'} \le Z^{M_{\mathrm{init}}}$, so there are only logarithmically many dyads in the bounded retained range. Let H be the center of one such bare-frequency dyad, and let $\psi_C, \psi_{d'}, \psi_{h'}$ be the retained individual cutoffs on the current C, d', h' dyads, evaluated at their normalized norms.

The orientation in (16.71) is now fixed. The calculation in the second canonical Poisson transformation replaces $\mu(z)\gamma_{-1}(z)$ by $\bar{\alpha}(z)\gamma_2(z)\chi_z(-1)\overline{G(z)}$. On the second side the factor inside conjugation has ray factor $\overline{G(z)}$; together with the CRT cross phase the relative ray factor is $G([z_2][z_1]^{-1})$. Expanding this factor by the fixed group formula above gives the ray summands. In one such summand put

$$\nu_* = \nu_0 \bar{\vartheta}, \qquad A_*(z) = \bar{\alpha}(z)\gamma_2(z)\nu_*(z), \qquad \mathcal{K}_{\mathrm{init}} = \mathcal{F}\Phi_{\mathrm{init}}.$$

For a preliminary pattern Σ_0 fixing the overlap, these dyads and the ray summand, and with the earlier mode held fixed, the nonzero component, with its mask and restored principal terms, is

$$\begin{aligned}
\mathcal{T}_{\Sigma_0}^{\mathrm{init}} = & \sum_{\substack{C \text{ sf},\ d'|C \\ h'\neq 0}} \psi_C \psi_{d'} \psi_{h'} \mu(d') \widehat{G}(\vartheta) \mathcal{B}_{\mathrm{init}}(h') 1_{(C,j)=1} \\
& \times \sum_{\substack{z_1,z_2 \text{ sf} \\ (z_1 z_2, Cj)=1}} 1_{(z_1,z_2)=1} A_*(z_1) \overline{A_*(z_2)\chi_{z_1}(d')} \chi_{z_1}(h') \chi_{z_2}(d') \overline{\chi_{z_2}(h')} \\
& \times \mathfrak{d}_0(Cz_1) \overline{\mathfrak{d}_0(Cz_2)} W_0(q_C q_{z_1}/Z^{D'}) \overline{W_0(q_C q_{z_2}/Z^{D'})} \\
& \times \frac{Z^{-D'} Z^m}{q_{d'}\sqrt{q_{z_1} q_{z_2}}} \mathcal{K}_{\mathrm{init}}\left(\frac{Z^m q_{h'}}{q_{d'} q_{z_1} q_{z_2}}\right).
\end{aligned} \tag{16.74}$$

Indeed $|\mu(C)\nu_0(C)|^2 = 1$ for the squarefree C outside S, the common row factor is exactly $1_{(u,C)=1}$, and its Poisson expansion contributes the one $\mu(d')$. The displayed two numerator factors are those of the primitive character $\bar{\chi}_{z_1}\chi_{z_2}$ and its Fourier transform. The finite ray sum restores the principal pair by $G(1) = 1$. That restored contribution and the raw complement removed before this equality were bounded separately above.

Define the summand in (16.74) on all individually squarefree z_1, z_2 prime to Cj by the displayed separate A_* factors, characters, formal product norms, full kernel, and unchanged outer ball. It agrees on the coprime domain, which is recovered by the complete identity

$$1_{(z_1,z_2)=1} = \sum_{s|(z_1,z_2)} \mu(s), \qquad z_a = sn_a,$$

before any factorwise bound. Let $v \geq 0$ be the center of a squarefree s-dyad, $\widehat{v} = \log_Z q_s$, and ψ_s its retained individual cutoff. The n_a are squarefree and prime to s, but need not be mutually coprime. Poisson summation applies to the coprime conductor; the divisor identity supplies the formal extension. For squarefree coprime s, n, CRT and the zero-preserving identity $\overline{\chi_n(d')} = \chi_n(d')\chi_n(d')^4$ give exactly

$$A_*(sn)\overline{\chi_{sn}(d')}\chi_{sn}(h') = A_*(s)\overline{\chi_s(d')}\chi_s(h') \times A_*(n)\chi_n(d'h')\chi_n(d's)^4. \tag{16.75}$$

The fourth power includes the CRT factor $\chi_n(s)^4$. On an overlap it defines the residual coefficient to be zero, without evaluating γ_2 on a nonsquarefree ideal. Since $|A_*(s)| = 1$ outside S, the two extracted scalars give

$$|A_*(s)\overline{\chi_s(d')}\chi_s(h')|^2 = 1_{(s,d'h')=1} = 1_{(s,h')=1} \quad \text{when } d' \mid C,\ (s,C) = 1.$$

The original residual and overlap exclusions also give $1_{(C,j)=1}1_{(s,Cj)=1}$. Thus, apart from the assigned coefficients, individual cutoffs, and unit phases, the outer arithmetic weight is precisely

$$\mu(d')\mu(s)\widehat{G}(\vartheta)\mathcal{B}_{\mathrm{init}}(h')1_{(C,j)=1}1_{(s,Cj)=1}1_{(s,h')=1}. \tag{16.76}$$

All these factors remain through weighted Cauchy. Set

$$t = C/d', \qquad k = d'h', \qquad f = d's, \qquad \rho_{t,j}(n) = 1_{(n,\mathrm{rad}(tj))=1}.$$

The factors on n in (16.75), together with $\rho_{t,j}$, supply exactly the original exclusions at C, s, j and the numerator zeros at d', h'. On the nonzero support t, d', s, n are pairwise coprime. The slot identity (16.16) assigns the primes in each mark along this ordered list. Assigned primes and their priority masks stay outer. A surviving prime is already prime to $d's$ by $\chi_n(f)^4$ and to tj by $\rho_{t,j}$, so its original individual list and coefficient can be retained. Its mark is an original subcollection of cap z_0, not a row- or label-dependent residual coefficient.

The substitutions $k = d'h'$, $f = d's$, and the coprimality restriction $\rho_{t,j}$ supply the canonical character and coefficient class. The quotient remains fixed while the row and label are averaged. Fourier separation will produce a common measure for those averages; weighted Cauchy will then make it possible to bound the reconstruction multiplicity and enlarge their ranges.

4. Child windows and Fourier separation. The outer mask $\mathcal{B}_{\mathrm{init}}$ remains outside the common Fourier separation of the full kernel and the old windows. The formal child centers and the bounds on the full new supports are

$$\begin{gathered} N_c^{\mathrm{init}} = D' - B - v, \qquad V_c^{\mathrm{init}} = \theta + v, \qquad F_c^{\mathrm{init}} = D' - P_1, \\ |\widehat{n} - N_c^{\mathrm{init}}| \leq 3\eta, \qquad |\widehat{f} - V_c^{\mathrm{init}}| \leq 2\eta. \end{gathered} \tag{16.77}$$

On factorized terms these bounds follow from $c = Csn$ and $f = d's$. The full supports of the new column cutoff and enlarged label window are also included in $\mathcal{I}_D$, so the same bounds hold

directly for independent terms later added by positivity. The reconstructed formal product $c = Csn$ on the fresh cutoff is included there as well, so the initial prefactor bound holds on its entire separated side.

Fix a refinement Σ by the s-dyad and the slot branches, with the earlier mode $\boldsymbol{\upsilon}$ held fixed. Before summing any actual outer ideal, use the six independent coordinates

$$(x_C, x_d, x_s, x_h, x_1, x_2) = \left(\frac{q_C}{Z^B}, \frac{q_{d'}}{Z^\theta}, \frac{q_s}{Z^v}, \frac{q_{h'}}{Z^H}, \frac{q_{n_1}}{Z^{N_c^{\mathrm{init}}}}, \frac{q_{n_2}}{Z^{N_c^{\mathrm{init}}}} \right).$$

Let I_C, I_d, I_s, I_h be the full supports of the four current individual cutoffs. Choose $\omega_{n,a}$ equal to one on the support of W_0 divided by $I_C I_s$, with fixed full supports in an enclosing interval $[a, b]$ whose upper endpoint satisfies $b \geq 1$. Choose a nonnegative ω_f equal to one on $I_d I_s$, with fixed full support I_f. The current four dyad cutoffs and ω_f remain outside inversion, and $\omega_{n,a}$ remain in the columns. Let $\Omega_{\mathrm{init},\alpha}$ be larger cutoffs equal to one on the full supports of the corresponding four current and two fresh cutoffs. These full supports, I_f, and the formal products Csn on them are the windows already included in $\mathcal{I}_D$. None of these supports depends on a current outer tuple or on a Fourier mode.

The label center satisfies $V_c^{\mathrm{init}} \geq 0$. A child is retained only if its row component is nonzero, which implies $M_{\mathrm{init}} \geq 0$. This nonemptiness condition is checked separately from membership in the structural outer set below, which may include zero-weight tuples even when the row ball is empty. Independently, if the full support of a new child column window contains no squarefree ideal, its polynomial is zero and that component is omitted. For a retained nonempty child define

$$N_{\mathrm{init}} = \max(0, N_c^{\mathrm{init}}), \qquad \delta_{\mathrm{init}} = N_{\mathrm{init}} - N_c^{\mathrm{init}} \in [0, 3\eta], \qquad F_{\mathrm{init}} = N_{\mathrm{init}} + V_c^{\mathrm{init}} = F_c^{\mathrm{init}} + \delta_{\mathrm{init}}.$$

For a nonempty negative-center window, $q_n \geq 1$ and (16.77) give $\delta_{\mathrm{init}} \leq 3\eta$, while its upper support endpoint gives $Z^{\delta_{\mathrm{init}}} \leq b$. If $N_c^{\mathrm{init}} \geq 0$, then $\delta_{\mathrm{init}} = 0$ and the latter inequality follows from $b \geq 1$. Thus $1 \leq Z^{\delta_{\mathrm{init}}} \leq b$ in every retained case. The cutoff $\omega_{n,a}(Z^{\delta_{\mathrm{init}}} y)$ is supported in $[a/b, b]$ and has uniformly bounded Euler seminorms. Its rescaling depends only on fixed centers, not on a current row or label, and its full clipping family is included in $\mathcal{I}_D$.

On the formal factorization $z_a = sn_a$, the complete root and kernel argument are

$$\begin{aligned} \frac{Z^{-D'} Z^m}{q_{d'} q_s \sqrt{q_{n_1} q_{n_2}}} &= Z^{\kappa_c^{\mathrm{init}}} x_d^{-1} x_s^{-1} (x_1 x_2)^{-1/2}, \\ \frac{Z^m q_{h'}}{q_{d'} q_s^2 q_{n_1} q_{n_2}} &= A_{\mathrm{ker,init}} \frac{x_h}{x_d x_s^2 x_1 x_2}, \\ A_{\mathrm{ker,init}} &= Z^{m+H-\theta-2(D'-B)}. \end{aligned} \tag{16.78}$$

For the first equality use $N_c^{\mathrm{init}} = D' - B - v$ and $\kappa_c^{\mathrm{init}} = m - 2D' + B - \theta$. Equivalently its root is $x_d^{-1} x_C (x_{c_1} x_{c_2})^{-1/2}$ with $x_{c_a} = x_C x_s x_a$. Define the one joint profile

$$\begin{aligned} \mathfrak{H}_{\mathrm{init},\Sigma,\boldsymbol{\upsilon}}(\mathbf{x}) = Z^{-3\eta} \prod_\alpha \Omega_{\mathrm{init},\alpha}(x_\alpha) x_d^{-1} x_s^{-1} (x_1 x_2)^{-1/2} \\ \times W_0(x_C x_s x_1) \overline{W_0(x_C x_s x_2)} \mathcal{K}_{\mathrm{init}} \left(A_{\mathrm{ker,init}} \frac{x_h}{x_d x_s^2 x_1 x_2} \right). \end{aligned} \tag{16.79}$$

The outside scalar is $Z^{\kappa_c^{\mathrm{init}} + 3\eta}$. The coupled old c-windows, all normalized real inverse roots, and the full kernel lie in this profile. By the same convention, the child tests exclude the real roots.

For $\boldsymbol{\xi} = (\xi_C, \xi_d, \xi_s, \xi_h, \xi_1, \xi_2)$, put $z_n = q_n / Z^{N_{\mathrm{init}}}$, set again $\varepsilon_1 = 1$, $\varepsilon_2 = -1$, and define

$$W_a^{\mathrm{init}}(z) = \omega_{n,a}(Z^{\delta_{\mathrm{init}}} z) z^{\varepsilon_a i \xi_a}.$$

If $\mathfrak{d}'_a$ is the surviving subcollection on side a, the precise unnormalized child polynomial is

$$Q_{a,t,j,\boldsymbol{\xi}}^{\mathrm{init}}(k, f) = \sum_{n \text{ sf}} A_*(n) \rho_{t,j}(n) \chi_n(k) \chi_n(f)^4 \mathfrak{d}'_a(n) W_a^{\mathrm{init}}(q_n / Z^{N_{\mathrm{init}}}). \tag{16.80}$$

Its fixed finite character, coprimality restriction, product mark, and test are independent of k, f. The two sides can have different tests and subcollections, but both have the same canonical coefficient class.

Let Ω_Σ consist of

$$\omega = (C \text{ sf}, d' \mid C, s \text{ sf}, h' \neq 0;\ \text{assigned slot primes})$$

with the retained individual supports and slot branches. It has no current column index. The functions t, k, f are the products defined above; the set may be taken before imposing the displayed outer zero masks, which will be in the weight. Let $\mathcal{A}_\Sigma^{\mathrm{init}}(\omega)$ be the product of all assigned coefficients of $\mathfrak{d}_0$ and their priority masks, conjugated on side two. The complete remaining outer weight is

$$\begin{aligned} V_\Sigma^{\mathrm{init}}(\omega;\boldsymbol{\xi}) = {} & \mu(d')\mu(s)\widehat{G}(\vartheta)\mathcal{B}_{\mathrm{init}}(h')1_{(C,j)=1}1_{(s,Cj)=1}1_{(s,h')=1}\mathcal{A}_\Sigma^{\mathrm{init}}(\omega) \\ & \times \psi_C(x_C)\psi_{d'}(x_d)\psi_s(x_s)\psi_{h'}(x_h)\omega_f(q_f/Z^{V_{\mathrm{c}}^{\mathrm{init}}}) \\ & \times x_C^{i\xi_C}x_d^{i\xi_d}x_s^{i\xi_s}x_h^{i\xi_h}Z^{i\delta_{\mathrm{init}}(\xi_1+\xi_2)}. \end{aligned} \tag{16.81}$$

The cutoff ω_f is one on every original label product, so its insertion is an equality before separation. There is one copy of each Möbius factor and one ray coefficient. In particular the common zero at (s, h') is still present.

Let $\mathcal{T}_\Sigma^{\mathrm{init}}$ be the fixed s-dyad and slot summand obtained from (16.74) by the complete Möbius and priority expansions. Its separated identity is

$$\mathcal{T}_\Sigma^{\mathrm{init}} = Z^{\kappa_c^{\mathrm{init}}+3\eta}(2\pi)^{-6}\int_{\mathbb{R}^6}\widehat{\mathfrak{H}}_{\mathrm{init},\Sigma,\boldsymbol{\upsilon}}(\boldsymbol{\xi}) \times \sum_{\omega\in\Omega_\Sigma} V_\Sigma^{\mathrm{init}}(\omega;\boldsymbol{\xi})Q_{1,t,j,\boldsymbol{\xi}}^{\mathrm{init}}(k,f)\overline{Q_{2,t,j,\boldsymbol{\xi}}^{\mathrm{init}}(k,f)}\,d\boldsymbol{\xi}. \tag{16.82}$$

Indeed $x_a = Z^{\delta_{\mathrm{init}}}z_{n_a}$. The two column modes, with the last factor of (16.81), are exactly $x_1^{i\xi_1}x_2^{i\xi_2}$, and the other four modes are outer. Fourier inversion restores the full profile; its larger cutoffs are one on the retained supports, and the fresh column cutoffs are one wherever the old W_0 windows are nonzero. The root and kernel identities in (16.78) restore the full prefactor and kernel. The coefficient identity (16.75) and the priority expansion restore every arithmetic factor term by term. Extra combinations in the full supports of the new windows cancel through the old windows in this complex identity before Cauchy.

The transform in (16.82) is chosen from the fixed ambient profile before any actual C, d', s, h', and hence before any t, k, f, is evaluated. It depends on Σ, Z, the fixed tests, and the earlier mode $\boldsymbol{\upsilon}$, and may depend on the already fixed j, but not on any current outer ideal or reindexed row or label. The bare h' norm is a coordinate; the derived row and label occur only in the outer masks or in the displayed characters. The relation $C = td'$ only restricts evaluation points. By (16.21), for each fixed J there are fixed L, b_0 such that

$$\int_{\mathbb{R}^6}|\widehat{\mathfrak{H}}_{\mathrm{init},\Sigma,\boldsymbol{\upsilon}}(\boldsymbol{\xi})|(1+|\boldsymbol{\xi}|)^J\,d\boldsymbol{\xi} \ll (1+|\boldsymbol{\upsilon}|)^L, \qquad \int_{\mathbb{R}^{1+|I_0|}}|\widehat{\mathfrak{H}}_{\mathrm{ov},j}(\boldsymbol{\upsilon})|(1+|\boldsymbol{\upsilon}|)^L\,d\boldsymbol{\upsilon} \ll p_{b_0}(W).$$

The first bound uses the fixed normalized roots and the uniform Euler bounds of the full kernel for every positive $A_{\mathrm{ker,init}}$; W_0 has only polynomial $\boldsymbol{\upsilon}$-dependence. The second is uniform over the fixed j-product interval. Taking larger L if needed also controls the earlier row-space Minkowski inequality.

The separated identity (16.82) now expresses the component in polynomials of canonical shape with one common Fourier density chosen before the current outer ideals are evaluated. It remains to bound the positive fibres and source measure, verify child admissibility from source witnesses, and restore the normalization in Lemma 16.1.

5. Positive fibres and the moment bound. The weighted Cauchy inequality (16.34), applied on the whole Ω_Σ including all quotients t, gives positive sums before any quotient is fixed for the

child. We may now bound their weights absolutely. The retained $(s, C) = 1$ and $d' \mid C$ make $f = d's$ squarefree, and all other factors of the weight are bounded by a fixed constant. Thus

$$|V_\Sigma^{\mathrm{init}}(\omega; \boldsymbol{\xi})| \ll \mathcal{B}_{\mathrm{init}}(h')\mu^2(f).$$

For fixed t, f, k, the identities

$$d' \mid f, \qquad s = f/d', \qquad C = td', \qquad h' = k/d' \quad \text{if this quotient is an element}$$

show that the reconstruction has at most $d_{\mathcal{O}}(f)$ choices before slots. Here K_{slot} again denotes the fixed bound for the original number of slots. The at most $2K_{\mathrm{slot}}$ assigned residual primes divide tf. There are at most $[d_{\mathcal{O}}(t)d_{\mathcal{O}}(f)]^{2K_{\mathrm{slot}}}$ choices. The earlier overlap primes were already fixed and counted in the overlap triangle. Hence the fibre, even if its actual size depends on k, is bounded by

$$D_{\mathrm{init}}(f)C_{\mathrm{init}}(t), \qquad D_{\mathrm{init}}(f) = d_{\mathcal{O}}(f)^{1+2K_{\mathrm{slot}}}, \qquad C_{\mathrm{init}}(t) = d_{\mathcal{O}}(t)^{2K_{\mathrm{slot}}}. \tag{16.83}$$

The reconstruction fixes s, C, h' once t, f, k, d' are specified, so it adds neither an independent d', s count nor a second frequency count.

Let $\Omega_\Sigma^{\mathrm{src}}$ be the structural outer set just used, before independent positive row or label additions and ignoring unit Fourier phases, and define the set of distinct source quotients

$$\mathfrak{T}_\Sigma = \{t = C/d' : \omega \in \Omega_\Sigma^{\mathrm{src}}\}.$$

It is projected once over all current rows and labels, not redefined for a fixed k, f. Every member has a source witness on the C, d' dyads, and therefore $\log_Z q_t = \widehat{B} - \widehat{\theta} \le P_1 + 2\eta$. The containing norm ball bounds its cardinality by $\#\mathfrak{T}_\Sigma \ll Z^{P_1+2\eta+\pi_{\mathrm{count,init}}}$; it is not used as a replacement child domain. In a nonempty sector $P_1 + 2\eta \ge 0$, since it bounds the norm of an existing ideal. The divisor bound on this bounded range gives the normalized measure

$$d\nu_\Sigma^{\mathrm{init}}(t) = Z^{-P_1-2\eta} \sum_{t \in \mathfrak{T}_\Sigma} C_{\mathrm{init}}(t)\delta_t, \qquad \|\nu_\Sigma^{\mathrm{init}}\| \ll Z^{\pi_{\mathrm{count,init}}+\pi_{\mathrm{fib,init}}}. \tag{16.84}$$

This set and measure ignore the unit Fourier phases and are common to the whole current row and label sum. Nonemptiness of the row component and child-column window is still checked separately from this structural projection.

For a fixed mode let

$$\mathcal{H}_{a,t}^{\mathrm{init}} = \sum_{\substack{f \text{ sf} \\ q_f/Z^{V_c^{\mathrm{init}}} \in I_f}} D_{\mathrm{init}}(f) \sum_{\substack{k \in \mathcal{O} \\ 0 < q_k \le Z^{M_{\mathrm{init}}}}} |Q_{a,t,j,\boldsymbol{\xi}}^{\mathrm{init}}(k, f)|^2 = Z^{F_{\mathrm{init}}}\mathcal{E}_{a,t}^{\mathrm{init}}. \tag{16.85}$$

This is (16.2) with the same row ball; with a larger fixed canonical ball the equality is replaced by the corresponding upper bound. The fibre estimate and weighted Cauchy give

$$\left| \sum_{\omega \in \Omega_\Sigma} V_\Sigma^{\mathrm{init}} Q_1 \overline{Q_2} \right| \ll Z^{P_1+2\eta} \prod_{a=1}^{2} \left(\int_{\mathfrak{T}_\Sigma} \mathcal{H}_{a,t}^{\mathrm{init}}\, d\nu_\Sigma^{\mathrm{init}}(t) \right)^{1/2}. \tag{16.86}$$

Here $Q_a = Q_{a,t,j,\boldsymbol{\xi}}^{\mathrm{init}}(k, f)$. Only the positive outer f, k sums were enlarged to their full fixed windows; the inner $\rho_{t,j}$, mark, and fresh test are unchanged. The weight D_{init} depends on f alone. The new outer ball is exactly $1_{0<q_k\le Z^{M_{\mathrm{init}}}}$. Thus use of the uniform child bound introduces the displayed quotient-count exponent once and the normalized mass once.

6. Child range conditions and normalization. The fibre bound has separated the fixed quotient from the averaged row and label. The canonical estimate can now be applied once its restriction

and two margins have been checked. The fixed coprimality restriction excludes exactly the primes of $\mathrm{rad}(tj)$. For each $t \in \mathfrak{T}_\Sigma$, its source bound and the fixed overlap bound give

$$Q_{\mathrm{init}} := \log_Z q_{r_\rho} \le \widehat{B} - \widehat{\theta} + \widehat{j} \le P_1 + G + 3\eta. \tag{16.87}$$

Shared primes in the excluded-prime radical only decrease this bound. It continues to hold when positive new labels or rows are added, because those additions do not change t, j or delete their inner coprimality restriction.

The fixed formal centers satisfy

$$\begin{aligned} F_c^{\mathrm{init}} - M_c^{\mathrm{init}} - (P_1 + G) - z_0 &= m - r - 2z + 2G, \\ 4F_c^{\mathrm{init}} - 3M_c^{\mathrm{init}} - 6z_0 &= 3m - 2r - 8z + 10G + 2P_1. \end{aligned}$$

The marked premises, $G \ge 0$, $P_1 \ge -2\eta$, (16.72) and (16.87), and the favorable clipping signs therefore give the actual canonical margins

$$\begin{aligned} F_{\mathrm{init}} - M_{\mathrm{init}} - Q_{\mathrm{init}} - z_0 &\ge c_1 + \delta_{\mathrm{init}} - 9\eta - \tau_{\mathrm{init}} \ge c_1 - 9\eta - \tau_{\mathrm{init}}, \\ 4F_{\mathrm{init}} - 3M_{\mathrm{init}} - 6z_0 &\ge c_2 + 4\delta_{\mathrm{init}} - 22\eta - 3\tau_{\mathrm{init}} \ge c_2 - 22\eta - 3\tau_{\mathrm{init}}. \end{aligned} \tag{16.88}$$

The 22η is 4η from the possible negative $2P_1$ and 18η from three copies of the row enclosure.

The canonical bound applies only for $t \in \mathfrak{T}_\Sigma$, where the preceding coprimality restriction and margin checks hold. The normalization identity (16.85) restores the child normalization, and (16.86) then bounds the unscaled signed outer sum by

$$C(\boldsymbol{\upsilon}, \boldsymbol{\xi}) Z^{P_1 + 2\eta + 2F_{\mathrm{init}} + \epsilon_c} \|\nu_\Sigma^{\mathrm{init}}\|.$$

The measure in (16.84) supplies one mass, and its fibre weights already include the witness multiplicity of each distinct fixed quotient. The label $d's$ remains inside the child average. The common Fourier density is integrated separately using the two weighted bounds following (16.82). The formal identity

$$\kappa_c^{\mathrm{init}} + P_1 + 2F_c^{\mathrm{init}} = m$$

then gives, after restoring the child normalization and applying the canonical bound with loss ϵ_c, the per-side exponent

$$m + 3\eta + 2\eta + 2\delta_{\mathrm{init}} + \pi_{\mathrm{init}} + \epsilon_c \le m + 11\eta + \pi_{\mathrm{init}} + \epsilon_c. \tag{16.89}$$

Here π_{init} is the aggregate freely chosen local divisor, dyadic, and separation loss, including the assigned-overlap divisor loss and the distinct allocations $\pi_{\mathrm{count,init}}, \pi_{\mathrm{fib,init}}$. These allocations sum to one aggregate. The two clipping copies come from the restored normalization and the child exponent. The row enlargement occurs inside the canonical child and has already been included in its margin test, so it contributes no τ_{init} factor to the energy exponent. The initial weighted Cauchy takes a geometric mean, and the overlap triangle was already canceled by Z^{-G}.

7. Parameter choice and tails. Let $r_{\max}, z_{\max}, m_{\max}$ bound the marked ranges, let s_{slot} bound the number of slots, and put $D_{\max} = r_{\max} + z_{\max}$. Set $\bar{c} = \min(c_1, c_2)$ and choose the canonical margin $c_* = \bar{c}/2$. For $\eta, \tau_{\mathrm{init}} \le 1/100$, valid a priori starting bounds for that application are

$$M_{\max} = 2D_{\max} + 1, \qquad N_{\max} = V_{\max} = D_{\max} + 1.$$

Indeed, $P_1 \ge -2\eta$ gives $M_{\mathrm{init}} \le 2D_{\max} + 10\eta + \tau_{\mathrm{init}}$, and $F_{\mathrm{init}} \le D_{\max} + 5\eta$ bounds both nonnegative child parameters. Make the choices in (16.69) with $\epsilon_c = \epsilon/2$, imposing in addition

$$\eta, \tau_{\mathrm{init}} \le \bar{c}/100, \qquad \eta \le \epsilon/88, \qquad \pi_{\mathrm{init}} \le \epsilon/8.$$

One may take $\tau_{\mathrm{init}} = \tau$ after imposing both sets of bounds. The margin bound (16.88) then gives both margins at least $3\bar{c}/4 > c_*$, so Lemma 16.2 applies. The exponent bound (16.89) adds at most $\epsilon/4$ to its $\epsilon_c = \epsilon/2$ loss.

With these bounded initial ranges fixed, choose the subsidiary input exponents realizing the target π_{init}, before choosing the initial tail and Fourier-height orders.

For the separate initial raw tail, the supported columns satisfy $q_{c_i} \le CZ^{D_{\max}}$, while $D' \in [-z_{\max}, D_{\max}]$. On the initial expansion on its coprime support put $a = Z^m/(q_{d'}q_{z_1z_2})$. Since $m \ge 0$, $d' \mid C$, and $c_i = Cz_i$,

$$a^{-1} \le q_{d'}q_{z_1z_2} \le q_{c_1}q_{c_2}/q_C \le CZ^{2D_{\max}}.$$

The explicit crude exponent

$$B_{\mathrm{crude,init}} = 10\{1 + m_{\max} + (s_{\mathrm{slot}} + 1)(D_{\max} + 1)\}$$

bounds all raw nonfrequency factors even without cancellation. The normalization contributes at most $z_{\max}$ to the exponent, the two column counts contribute $2D_{\max}$, the d'-divisor count contributes at most $D_{\max}$, and $1 + a^{-1}$ contributes at most $2D_{\max}$. The two residual marks contribute at most $2s_{\mathrm{slot}}D_{\max}$, since $d_{\mathcal{O}}(c)^{s_{\mathrm{slot}}} \le q_c^{s_{\mathrm{slot}}}$. Counting the overlap pairs and their assigned coefficients without their canceling weights adds at most $2z_{\max} + 2s_{\mathrm{slot}}z_{\max}$; the Poisson prefactor adds at most $m_{\max}$. Their total is at most $m_{\max} + (8 + 4s_{\mathrm{slot}})D_{\max} < B_{\mathrm{crude,init}}$.

Choose $T_{\mathrm{init}} > 1 + B_{\mathrm{crude,init}}$ and then

$$A_{\mathrm{init}} > 1 + (B_{\mathrm{crude,init}} + T_{\mathrm{init}})/\tau_{\mathrm{init}}.$$

The tail bound (16.22) makes the raw $\mathcal{B}_{\mathrm{init}} = 0$ complement $O(Z^{-T_{\mathrm{init}}})$, with fixed seminorm and polynomial-height factors. This order is chosen after the marked ranges and τ_{init}, but before the initial Fourier-height orders and before the final Z threshold. If $\tau_{\mathrm{init}} = \tau$, take the maximum of the orders required by this separate crude bound and the canonical tails. The full kernel retained inside the outer ball has the uniform Euler bounds already used in the canonical separation. The final threshold also enforces the full initial support ratios in $\mathcal{I}_D$.

The principal count has exponent m, and every nonprincipal term has now been bounded by $Z^{m+\epsilon}$. The argument keeps all element rows and every original mask, applies to subcollections and either common orientation, and preserves arbitrary bounded row-independent prime coefficients. The common separated measures give the stated finite-seminorm and polynomial-height uniformity. This proves (16.1). □

16.7 Amplification on sixth-power-free rows

We first extract an unmarked consequence on all element rows. In Lemma 16.1, take $Z = H$, $m = 1$, no slots, and $r = \log_H D$. If $H \ge D^{1+c}$ for a fixed $c > 0$, then $1 - r \ge c/(1 + c)$; the second required margin is also positive. For $D \le H^{\epsilon/4}$, the direct bound $|M_{\psi_u}(D; W)|^2 \ll D$ suffices. Otherwise the starting lengths are in a fixed bounded range away from zero. Bounded H, D are handled directly. We obtain, for $H, D \ge 1$,

$$\sum_{\substack{u\in\mathcal{O}\\ 0<q_u\le H}} \left| D^{-1/2} \sum_n \mu(n)\nu(n)\chi_n(u)^{\varepsilon_\chi} W(q_n/D) \right|^2 \ll_{c,\epsilon,W} H(HD)^\epsilon, \qquad H \ge D^{1+c}. \tag{16.90}$$

The fixed arithmetic data and the stated seminorms are included in the dependence of the constant.

Lemma 16.6 (Sixth-power amplification). *Let ν, W, ε_χ and the zero extensions be as in Lemma 16.1, with no prime slots. Let $U, D \ge 1$, and sum over elements u with $q_u \asymp U$ such that every prime valuation of the ideal (u) is at most five. For every fixed $c > 0$ and $\epsilon > 0$, put*

$$H = \max(2U, D^{1+c}), \qquad P = (H/U)^{1/6}.$$

Then

$$\sum_u |M_{\psi_u}(D;W)|^2 \ll \frac{H}{P}(UD)^\epsilon \ll \max\{U, U^{1/6}D^{5(1+c)/6}\}(UD)^\epsilon. \tag{16.91}$$

The implied constant depends only on c, ϵ, the fixed arithmetic data, and finitely many seminorms of W. In particular, for $D = U^r$ with r in a fixed bounded nonnegative range, for every $\epsilon > 0$ the exponent may be written

$$\sum_u |M_{\psi_u}(U^r;W)|^2 \ll U^{e(r)+\epsilon}, \qquad e(r) = \max\{1, (1+5r)/6\}.$$

Both assertions permit a separate test $W_{\sigma_u,t_u}(y) = W(y)y^{-\sigma_u+it_u}$ in each row, with σ_u in a fixed compact interval and $|t_u| \le T_1$, with a multiplicative factor $(1+T_1)^A$ for a fixed A.

Proof. We average over sixth-power multiples ua^6. The sixth-power-free condition makes these rows distinct, and a scale supremum controls the shorter columns introduced by primes of a.

The scale supremum. The row moment bound (16.90) and Lemma 3.7 imply the rowwise scale bound

$$\sum_{\substack{v\in\mathcal{O}\\ 0<q_v\le CH}} \sup_{0<D'\le D} |M_{\psi_v}(D';W)|^2 \ll H(HD)^\epsilon, \qquad H \ge \max(2, D^{1+c}). \tag{16.92}$$

Indeed, sufficiently small scales have no ideals in the annular support. Differentiation with respect to $\log D'$ replaces $W(y)$ by $-W(y)/2 - yW'(y)$. The one-dimensional Sobolev inequality in Lemma 3.7, followed by (16.90) for these two tests, controls the supremum on a unit logarithmic interval. There are $O(\log(2+D))$ such intervals. A fixed enlargement of H handles their endpoints, and the direct small-scale bound handles the initial intervals. This proves (16.92).

Amplification. Ideal counting outside the fixed set S gives at least $c_S P$ primary ideals a with $q_a \le P$, for a fixed $c_S > 0$. For bounded P this follows after reducing c_S, since the unit ideal is available; for large P it follows from the positive-density ideal count with finitely many primes removed.

For squarefree n, separate the primes dividing a by $n = dm$, where $d \mid \operatorname{rad} a$ and $(m,a) = 1$. With the zero extensions,

$$\psi_{ua^6}(m) = \psi_u(m)1_{(m,a)=1}.$$

This is true also when u, a share primes: both sides vanish at a prime dividing m and either u or a, and otherwise the sixth power is one. The separated column therefore gives the identity

$$M_{\psi_u}(D;W) = \sum_{d\mid \operatorname{rad} a} \mu(d)\psi_u(d)q_d^{-1/2} M_{\psi_{ua^6}}(D/q_d;W). \tag{16.93}$$

The coefficient mass is at most $d_{\mathcal{O}}(a) \ll_\delta P^\delta$. It follows that

$$|M_{\psi_u}(D;W)|^2 \ll_\delta P^{2\delta} \sup_{0<D'\le D} |M_{\psi_{ua^6}}(D';W)|^2.$$

Averaging this bound over the a's and summing over u is controlled by a single row sum, because the map $(u,a) \mapsto ua^6$ is injective on these pairs. In fact, at each prime the valuation modulo six recovers the valuation of the sixth-power-free ideal (u), and its quotient by six recovers the valuation of a. The primary generator of a then fixes the element u, including its unit. This argument does not require $(u,a) = 1$. Also $q_{ua^6} \ll UP^6 \ll H$. Thus (16.92) gives

$$\sum_u |M_{\psi_u}(D;W)|^2 \ll \frac{H}{P}(UD)^\epsilon = U^{1/6}H^{5/6}(UD)^\epsilon,$$

after choosing the preliminary δ and power losses small enough. Since $H = \max(2U, D^{1+c})$, this is (16.91). For bounded r, choose $c > 0$ sufficiently small in terms of r's bound and the desired power loss; the stated formula for $e(r)$ follows.

Rowwise test parameters. On the fixed annular support, derivatives in σ, t of $W_{\sigma,t}$ insert bounded powers of $\log y$. The parameter Sobolev inequality in Lemma 3.7 applies on a bounded cover in σ and $O(1+T_1)$ unit intervals in t. The row moments are summed before the derivatives are integrated. The finite-seminorm bound already proved has fixed polynomial height order. The cover and these derivatives therefore give a factor $(1+T_1)^A$ for a fixed A, uniformly for a separate (σ_u, t_u) in every row. The same reasoning applies to the marked estimate with any fixed finite number of test parameters, while its prime coefficients remain fixed across rows. □

Remark. In the later application, the positive exponent in $T_1 = Z^{\tau_\eta}$ is chosen after the fixed order A. It may therefore be chosen small enough to keep the resulting exponent within the prescribed power loss. The prime coefficients remain fixed across rows; the rowwise choice applies to the test parameters.

17 Fourth moments with short prime factors

The refined row count requires a fourth-moment estimate for two plain character polynomials and a product of short prime polynomials. We prove that estimate here. The two plain polynomials have no length restriction when no prime polynomial is present; otherwise the permitted lengths lie in an affine region. Two finite Fourier transforms return products of the same kind at a smaller effective width. Some transformed rows, however, induce characters in a fixed finite family. Their plain products can have volume-sized main terms. For the longer inputs we therefore subtract a comparison product with the same product of scales. Preserving its common character, mask, and norm power through the transforms makes those main terms cancel. The estimate itself concerns the original, uncentered product.

Definition 17.1 (Row characters and masks). We use the notation of Section 3 and the coefficient conventions of Section 12. The rows are elements $k \in \mathcal{O}$ with $0 < q_k \ll Z^m$, and

$$\psi_k(n) = \tau(n)\chi_n(k), \qquad M = m + q.$$

The character τ is fixed within a row sum. The union of the prime supports of all its displayed moving residue-symbol factors, before canceling factors or reducing exponents modulo six, has norm at most Z^q. Every displayed factor retains its zero extension, including a canceled or six-divisible factor.

A redundant zero may instead be represented by an additional coprimality mask only when an exact factorization retains every surviving local and fixed-ray phase. This mask is the indicator of coprimality to one squarefree ideal of norm at most Z^B, for a bounded B, fixed within the current row sum; the same mask is used in both plain factors and in every prime factor. It may depend on previously fixed labels, but not on the varying row. All ideals in the polynomials are outside the fixed set S.

Fix a finite group Θ of finite-order ray characters whose conductor primes belong to S. It contains the fixed twists and all characters used to separate the fixed reciprocity phases. It also contains the supplementary character $n \mapsto \chi_n(-1)$. Membership of an inducing character in Θ means equality with the primitive inducing character of some member of Θ, not equality of the chosen zero-extended presentations. All zeros of those presentations remain in the polynomials.

Let $\mathcal{R}_0$ be the rows for which ψ_k induces a nonprincipal character. For $z > 0$, let $\mathcal{R}_z$ be the rows for which the inducing character of ψ_k does not belong to Θ.

Write

$$A = n_1 + n_2 + z. \tag{17.1}$$

Lemma 17.2 (Fourth moment with short prime factors). *Let $3/4 \le \kappa \le 1$. Let $n_1, n_2 \ge 0$, and let $Q = \prod_{i\in\mathcal{I}} Q_{\psi_k,i}$ be a finite product of the prime polynomials in (12.1), of lengths $z_i \ge 0$ and total length $z = \sum_{i\in\mathcal{I}} z_i$. Their underlying prime supports are pairwise disjoint before common masks and row zero extensions are imposed. Their coefficients are fixed finite linear combinations of finite-ray characters. For every positive-length slot the expansion has the form*

$$\nu_i(n) = \sum_{j=1}^{J_i} c_{ij}\vartheta_{ij}(n), \qquad \vartheta_{ij} \in \Theta,$$

where the finite list and its coefficients are fixed independently of Z and of the row. Let the row characters $\psi_k(n) = \tau(n)\chi_n(k)$, their common masks, and the group Θ satisfy the row, mask, and character conditions of Definition 17.1, and use the corresponding families $\mathcal{R}_0$ and $\mathcal{R}_z$.

For every $\epsilon > 0$, there is a slot mesh $\eta > 0$ such that, if $z_i \le \eta$ for every i, then

$$\sum_{k\in\mathcal{R}_z} |S_{\psi_k}(Z^{n_1}; W_1)S_{\psi_k}(Z^{n_2}; W_2)Q|^2 \ll Z^{M+\epsilon} \tag{17.2}$$

in either of the following cases:

1. *$z = 0$. There is no restriction on the bounded nonnegative lengths n_1, n_2, and no lower bound on the conductor.*

2. *$z > 0$ and*
$$n_1 + n_2 + 6\kappa z = A + (6\kappa - 1)z \le M. \tag{17.3}$$
If $\kappa < 1$, assume in addition that $\beta_* \le (1 + \kappa)/2$. If $\kappa = 1$, no zero-free hypothesis is required.

For an empty slot list, $Q = 1$. The $z = 0$ assertion also permits a list of zero-length slots, with the same strength.

All real length parameters range over prescribed bounded sets. The mesh depends only on those sets and ϵ, uniformly for $\kappa \in [3/4, 1]$. For each fixed Z-independent arithmetic datum $\mathcal{A}$ and fixed slot count, the bound uses finitely many smooth seminorms and a fixed polynomial in the separated norm-twist heights. Their orders, and the bound itself, are uniform over all moving moduli, admissible masks, and fixed outer labels in the stated ranges.

The application to the 7/8 bound sets $\kappa = 2\beta_* - 1$, so its zero-free hypothesis in this lemma holds by equality. The refined row count uses the affine length condition for this dynamic value; the 11/12 bound also supplies the value 5/6. The proof below uses the finite Gauss identities of Section 12.3, Lemmas 3.6, 3.7, and 3.9, and Lemma 3.15. Absolute counting at their bounded scales absorbs any zero-length slots into the constant; henceforth $z = 0$ means that no live slot remains. Both preliminary reductions preserve the row family.

The proof is an induction on the effective width $M = m+q$, with all zero-slot bands completed before the positive-slot bands. Its stages are as follows.

1. Sections 17.1–17.3 remove the fixed extra mask, reflect long plain factors, and construct the equal-product comparison. The comparison retains the common character, mask, and norm power needed for cancellation.

2. Sections 17.4–17.6 apply two finite transforms, with Gauss-row enlargement between them. The first transform reduces the moment to (17.23); the second gives the strictly smaller width in (17.32).

3. Sections 17.7–17.9 separate and normalize the residual products. The rows outside the fixed character family use induction, while the rows in that family use a count and, for the longer inputs, equal-product cancellation.

4. Section 17.10 chooses the parameters and bounds the accumulated error, with one mesh for every fixed slot count.

Within each band the range $A \le 5M/6$ is proved first. This threshold comes from the transformed rows whose inducing characters lie in Θ: counting their sixth-power form and bounding their plain products by volume leaves an excess $A - 5M/6$, before the common-support savings. Reflection handles some longer inputs; the others are reduced to an equal-product-scale difference, whose cancellation removes that excess.

17.1 Fixed masks and reflection

Removing the fixed extra mask preserves the inducing row character and does not increase the affine expression in (17.3). Reflection then gives the length bound (17.9) for the resulting plain factors.

Removing the fixed mask. After removal of the extra mask, the character has zeros exactly at its primitive conductor primes, its redundant row and declared moving radical primes, and the fixed primes in S. Here a redundant prime is one at which the inducing character is unramified but the displayed zero-extended product still vanishes. A displayed six-divisible power still vanishes on nonunits. If a redundant factor of the fixed twist is not included in the declared moving radical, first factor it exactly into its remaining phase and a coprimality indicator, and put that indicator in a squarefree ideal $\mathfrak{R}$, with $q_{\mathfrak{R}} \le Z^B$, fixed throughout the row sum. In particular, a cancellation between the fixed twist and the varying factor $\chi_n(k)$ is not reclassified as a separately chosen extra coprimality restriction for each row.

Let ψ_k^0 be the character obtained by deleting these additional coprimality restrictions, and let $T_k(X)$ denote its centrally normalized plain sum at scale X. Write $S_{k,\mathfrak{R}}(X)$ for the sum with the extra mask. Multiplicativity, including zero extensions, gives the exact identities

$$S_{k,\mathfrak{R}}(X) = \sum_{d \mid \mathfrak{R}} \mu(d)\psi_k^0(d) q_d^{-1/2} T_k(X/q_d), \tag{17.4}$$

$$Q_{k,i,\mathfrak{R}} = Q_{k,i}^0 - P_i^{-1/2} \sum_{\substack{p \mid \mathfrak{R} \\ p \text{ prime}}} \psi_k^0(p)\nu_i(p) W_i^{\mathrm{slot}}(q_p/P_i), \qquad P_i = Z^{z_i}. \tag{17.5}$$

For example, the first identity follows by inserting $\sum_{d \mid (n,\mathfrak{R})} \mu(d)$ and writing $n = dl$. The quotient l is unrestricted at primes of d; the zero extension of ψ_k^0 remains on it.

Expanding both plain factors and all slots by these identities gives the following terms. If the slots whose primes are fixed form $J \subseteq \mathcal{I}$, and the plain divisors are d_1, d_2, the scalar depending on the row has modulus at most one. The remaining coefficient has absolute value at most a fixed profile constant times

$$q_{d_1 d_2}^{-1/2} \prod_{i \in J} P_i^{-1/2}.$$

The resulting product without an extra mask has lengths

$$A' = A - \log_Z q_{d_1 d_2} - \sum_{i \in J} z_i, \qquad z' = z - \sum_{i \in J} z_i.$$

Its inducing character is unchanged. If $z' > 0$, the left side of (17.3) has decreased; if $z' = 0$, the zero-slot assertion has no length restriction and $\mathcal{R}_z \subseteq \mathcal{R}_0$.

For every fixed $\epsilon_1 > 0$,

$$\sum_{d \mid \mathfrak{R}} q_d^{-1/2} \le \prod_{p \mid \mathfrak{R}} (1 - q_p^{-1/2})^{-1} \ll_{\epsilon_1, B} Z^{\epsilon_1}. \tag{17.6}$$

This follows from Lemma 3.16. The corresponding mass for a slot with a fixed prime is also Z^{ϵ_1}: on its annular support, $P_i^{-1/2} \ll q_p^{-1/2}$, and the same product bounds the sum over $p \mid \mathfrak{R}$. The number of slots is fixed. Minkowski's inequality therefore reduces (17.2) to the assertion without an extra mask at the same width, with an arbitrarily small power loss. A nonempty annular scale below one is bounded below by a positive profile-dependent constant and may be rescaled to one. After this rescaling its new nonnegative length is $\max\{n_i - \log_Z q_{d_i}, 0\} \le n_i$, so the asserted nonincrease of the affine expression remains valid.

Reflecting a plain factor. Let ψ_k^* be the primitive character inducing ψ_k^0, and let $\mathfrak{R}_{0,k}$ be its redundant radical. The primes of $\mathfrak{R}_{0,k}$ are disjoint from the primitive conductor. If Q_k is that conductor norm, set $C_k = 3Q_k/(2\pi)^2$, the conductor scale in the functional equation. Tameness at primes outside S gives

$$C_k q_{\mathfrak{R}_{0,k}} \ll_{S,\tau_{\mathrm{fixed}}} Z^M. \tag{17.7}$$

Indeed, each good prime contributes at most once, either to the primitive conductor or to the redundant radical. Their union is contained in the union of the row radical and the declared moving radical, whose norm is at most $q_k Z^q$. By Lemma 3.2, the unit and S-supported parts of a row, with valuations reduced modulo six when evaluated on primary elements prime to S, range over a fixed finite ray family. Thus all primes in S contribute only a fixed factor.

For ψ_k^*, the primitive Hecke functional equation in the proof of Lemma 3.14 applies, together with the entireness of $L(s, \psi_k^*)$ stated there. Let ε_k be its root number, so $|\varepsilon_k| = 1$. Mellin inversion and a contour shift show that the normalized plain sum of ψ_k^* at X equals the root number times the conjugate character sum at C_k/X, with transformed profile $W^\sharp$ defined by

$$\mathcal{M}W^\sharp(s) = \mathcal{M}W(1-s)\frac{\Gamma(s)}{\Gamma(1-s)}.$$

The Mellin transform convention is the one in Lemma 3.9. The quotient of gamma functions has poles only at the nonpositive integers and zeros at the positive integers. Shifting the inverse Mellin contour to the left gives an expansion in nonnegative integral powers at zero; shifting it to the right gives arbitrary decay at infinity. Thus $W^\sharp$ and its Euler derivatives are bounded at zero and rapidly decreasing at infinity.

The same Mellin formula controls the dependence on the norm-twist height. For $W_\omega(y) = W(y)y^{i\omega}$, the exact identity $\mathcal{M}W_\omega(s) = \mathcal{M}W(s + i\omega)$ puts all height dependence in a translate of the rapidly decreasing Mellin transform. On each fixed vertical line the gamma quotient and each fixed number of Euler derivatives have polynomial growth in the integration height. Integrating the translated Mellin decay therefore gives a fixed polynomial in $1+|\omega|$, with its degree depending only on the fixed contour and derivative orders. The pointwise Mellin integration-by-parts estimate on a compact real strip in Lemma 3.9 bounds each fixed horizontal trace and therefore justifies the horizontal joins in these shifts. This proves the required finite-seminorm and polynomial-height bounds for reflection.

Deleting the redundant Euler factors before reflection and restoring them geometrically afterward gives

$$T_k(X; W) = \varepsilon_k \sum_{\substack{d_0 \mid \mathfrak{R}_{0,k} \\ \mathrm{rad}(h_0) \mid \mathfrak{R}_{0,k}}} \frac{\mu(d_0)\psi_k^*(d_0)\overline{\psi_k^*(h_0)}}{\sqrt{q_{d_0} q_{h_0}}} T_{\overline{\psi_k^0}}\left(\frac{C_k q_{d_0}}{X q_{h_0}}; W^\sharp\right), \qquad |\varepsilon_k| = 1.$$

The geometric series is absolutely bounded by $\prod_{p \mid \mathfrak{R}_{0,k}} (1 - q_p^{-1/2})^{-1}$. Consequently its total coefficient mass, together with the d_0 sum, is Z^{ϵ_1}. The scales are

$$Y = \frac{C_k q_{d_0}}{X q_{h_0}}, \qquad d_0 \mid \mathfrak{R}_{0,k}, \quad \mathrm{rad}(h_0) \mid \mathfrak{R}_{0,k}. \tag{17.8}$$

By (17.7), there is a fixed $C_{\mathcal{A}} \geq 1$ such that

$$C_k q_{\mathfrak{R}_{0,k}} \leq C_{\mathcal{A}} Z^M, \qquad Y \leq C_{\mathcal{A}} Z^{M-\log_Z X}.$$

The profile $W^\sharp$ can be partitioned into smooth annuli. Below its main scale, central normalization gives summable coefficients $O(2^{-j/2})$ on the annuli of relative scale 2^{-j}. Above that scale, rapid decay gives an arbitrary summable power. Fix a small length tolerance $\xi > 0$. Truncate the upper annuli after an enlargement $Z^{\xi/2}$; sufficiently many fixed derivatives make the omitted part negligible by absolute counting. A main scale below $Z^{-\xi}$ is likewise negligible. If a retained annular profile has fixed upper support endpoint B, a scale S that can contain a nonzero integral ideal satisfies $SB \geq 1$. On such a scale,

$$\max(S, 1) \leq \max(1, B)S.$$

Thus replacing a retained subunit scale by scale one changes the bound by at most this fixed multiplicative factor. All these assertions are uniform in the moving labels.

The row-dependent choices d_0, h_0 are handled by their coefficient mass followed by a rowwise supremum in the resulting scales. Lemma 3.7 bounds a supremum over two polynomial-range scales using $O((\log Z)^2)$ unit boxes and derivatives of the profiles. The row character and its existing zeros remain unchanged: after reflection, conjugating that whole factor inside its absolute value replaces the conjugate character by the original character and conjugates its profile. This operation is valid because the absolute value of a product is unchanged by conjugating one factor.

Let $C_{\mathrm{ref}} \geq 1$ include $C_{\mathcal{A}}$, the fixed annular endpoint multipliers, the logarithmic box multipliers for a reflected factor and any paired unreflected factor, and $\max(1, B)$ for the retained profile types. This constant is fixed after the data and profile types, independently of Z and the moving labels. For $n = \log_Z X$, every retained scale satisfying the preceding support test, after the permitted clipping and box enlargement, has declared nonnegative length

$$n_{\mathrm{ref}} \leq M - n + \frac{\xi}{2} + \frac{\log C_{\mathrm{ref}}}{\log Z} \leq M - n + \xi \qquad \left(\frac{\log C_{\mathrm{ref}}}{\log Z} \leq \frac{\xi}{2}\right). \tag{17.9}$$

This is a linear bound also when $M - n$ is slightly negative; if its right side is negative, no such retained scale exists. The smaller upper-annulus range changes only the fixed decay, seminorm, and height orders used for the discarded tail.

The zero-slot core. For $z = 0$, reflect each original factor whose length exceeds $M/2$ once, and leave the other factor unchanged. Every retained reflected factor satisfies (17.9); the fixed box multiplier for an unreflected factor is included in C_{ref}. After the preceding annular truncation and scale-box enlargement, the resulting lengths satisfy

$$n_i^* \leq M/2 + \xi, \qquad A^* = n_1^* + n_2^* \leq M + 2\xi \leq M + \delta, \tag{17.10}$$

provided $2\xi \leq \delta$. We call this the padded zero-slot core. Here δ is the positive padding parameter chosen below after the width floor and step. There is no repeated reflection at the boundary $M/2$. Reflection and scale suprema are taken before centering; on a later centered term whose primitive inducing character lies outside Θ, one first applies the triangle inequality to its two rectangles. No such supremum is applied to a centered difference whose primitive inducing character belongs to Θ.

17.2 Prime estimates and the induction order

The prime estimate controls slots removed from a smaller-width product. After establishing it, we specify the width induction and construct the equal-product comparison needed beyond $A = 5M/6$.

A pointwise bound for prime factors. For $\kappa < 1$, put $s_\kappa = (1+\kappa)/2$. Under the hypothesis $s_\kappa \ge \beta_*$, the global part of Lemma 3.15 gives, for every fixed $e > 0$,

$$\frac{L'}{L}(s_\kappa + e + it, \psi) \ll_e \log\bigl(2Q_\psi(3+|t|)^2\bigr) \tag{17.11}$$

for a primitive nonprincipal inducing character ψ. To estimate a prime annulus of scale P, apply Mellin inversion to a smooth von Mangoldt sum and move its contour to $\Re s = s_\kappa + e$. There are no poles in the region of the shift, and the Mellin transform has arbitrary decay, so (17.11) bounds the new integral by $P^{s_\kappa+e}$ times a logarithm of the conductor and a fixed polynomial in the height. Prime powers of exponent at least two contribute $O(P^{1/2+\epsilon_1})$. For large P, dividing the annular weight by $\log(Py)$ replaces the von Mangoldt weight by the prime weight; its smooth seminorms are bounded on the annulus. Bounded P are estimated by absolute counting. Expand each positive-length coefficient in the characters ϑ_{ij} from the statement. If the primitive inducing row character $\psi \notin \Theta$ made $\psi\vartheta_{ij}$ principal, then $\psi = \vartheta_{ij}^{-1} \in \Theta$, a contradiction. Thus every resulting slot character is nonprincipal on $\mathcal{R}_z$.

After extra-mask erasure, fix a positive slot list and put $\boldsymbol{W} = (W_i^{\mathrm{slot}})_{i\in\mathcal{I}}$. For normalized twist heights $\boldsymbol{\omega} = (\omega_i)_{i\in\mathcal{I}}$, define

$$Q_{\boldsymbol{\omega}} = \prod_{i\in\mathcal{I}} \left\{ P_i^{-1/2} \sum_{p \text{ prime}} \psi_k^0(p)\nu_i(p) W_i^{\mathrm{slot}}(q_p/P_i)(q_p/P_i)^{i\omega_i} \right\}.$$

For every $\epsilon_1 > 0$, central normalization and multiplication over this fixed list, with e and all subsidiary losses sufficiently small, give finite $j, b, h \ge 0$ and a constant C such that

$$|Q_{\boldsymbol{\omega}}|^2 \le C Z^{\kappa z+\epsilon_1} p_j(\boldsymbol{W})^b (1+|\boldsymbol{\omega}|)^h \qquad (k \in \mathcal{R}_z,\ z > 0). \tag{17.12}$$

Here j may be rounded up to an integer and $2s_\kappa - 1 = \kappa$. The orders and C may depend on the fixed data, slot count, loss, and any fixed requested internal logarithmic or normalized-twist derivatives, but not on Z, moving labels, rows, or the numerical heights. The exponent $\kappa z + \epsilon_1$ is independent of these derivative and height orders.

The Mellin estimate separates that exponent from the analytic orders. Translate each pure twist in its Mellin variable before integration by parts. The weighted integral of the untwisted Mellin transform then contributes a finite seminorm and a fixed polynomial in $\boldsymbol{\omega}$, while the contour displacement contributes exactly $2ez$ to the squared exponent. Logarithmic derivatives remain annular, and normalized-twist derivatives insert only powers of the logarithmic profile variable.

Deleting redundant primes is a separate small-power error. The redundant radical of ψ_k^0 contains only $O(\log Z)$ primes, so deleting it changes a normalized slot by $O(P_i^{-1/2}\log Z)$ times its fixed profile factor; this is within the stated power loss because $P_i \ge 1$.

At the endpoint $\kappa = 1$, absolute prime counting gives (17.12) without a zero-free assumption and with $h = 0$ for undifferentiated pure twists, whose modulus is one. For $\kappa < 1$ the hypothesis remains $\beta_* \le (1+\kappa)/2$. The choice of losses is uniform in $\kappa \in [3/4, 1]$, since the length ranges are bounded.

When this estimate is inserted into a moment argument, the analytic factors remain separate from the scale exponent. In every later use of (17.12), its fixed seminorm and height factor is retained in the weighted Fourier estimates; it is not included in the exponent of Z.

Width bands and terminal estimates. Choose a width floor $\rho > 0$, a width step $\sigma > 0$, and then $\delta > 0$, all small in terms of the final ϵ, with $\delta \ll \min(\rho, \sigma)$. The loss choices are completed in Section 17.10, after the depth is bounded. Divide the bounded range of M into consecutive bands of length $\sigma/4$. The induction proves all zero-slot bands first, in increasing order of width, and then all positive-slot bands. Within a zero-slot band, the uncentered assertion for $A \le 5M/6$

precedes the padded core in (17.10). Reflection then supplies all zero-slot lengths in that band. Within a positive-slot band, the uncentered assertion for $A \le 5M/6$ subject to (17.3) is proved first; centering then gives the remaining part of that region. A completed earlier band therefore includes the unrestricted zero-slot assertion. Let $M_{\max}$ bound the initial width range, and define

$$D = 2 + \lceil 2M_{\max}/\sigma \rceil.$$

The strict width decrease proved below will show that at most $D-2$ nonterminal calls occur on a branch.

At $M \le \rho$, the padded zero-slot core follows from absolute counting: there are $O(Z^m)$ rows and the squared product is $O(Z^{A+\epsilon_1})$. Its exponent above M is at most $A - q \le \rho + \delta$. For positive slots, combine the completed zero-slot estimate with (17.12). In the region of (17.3), $A \ge z$, whence

$$z \le \frac{M}{6\kappa} \le \frac{2M}{9}, \qquad \kappa z \le \frac{M}{6}. \tag{17.13}$$

Thus the extra terminal exponent for positive slots is at most $\rho/6$. These terminal exponents can be made smaller than the reserved final loss by choosing ρ, δ sufficiently small.

Comparison at equal products of scales. For a width above the floor, the two-transform estimate below will first prove the uncentered range $A \le 5M/6$. Reflection reduces the remaining inputs to that range or to a centered difference of two products. Set

$$L = M/4. \tag{17.14}$$

For $A > 5M/6$, the comparison lengths $L, A - z - L$ are both at least L. In the positive-slot case, $(6\kappa - 1)z \le M - A < M/6$, and $6\kappa - 1 \ge 7/2$; hence $z < M/21 < M/20$. This also shows $A - z > 2L$.

If one original plain length is $b < L$, reflect the other factor. If both are at least L, introduce the comparison with lengths $L, A - z - L$ and the same two profiles; reflect its longer factor only when estimating the comparison separately. In both cases the total length after that reflection is at most

$$b + \{M - (A - z - b)\} + z + \xi \le 3M/2 - A + 2z + \xi,$$

where $b \le L$. The bound in (17.9) includes the clipped reflected scale and the paired unreflected box multiplier in this same ξ; the discarded tails have the stated fixed-order bounds. Thus

$$A_{\mathrm{comp}} \le 3M/2 - A + 2z + \xi. \tag{17.15}$$

For $z = 0$, this gives $5M/6 - A_{\mathrm{comp}} \ge M/6 - \xi$. For $z > 0$, use $(6\kappa - 1)z \le M - A$, $A > 5M/6$, and $z < M/20$ to get

$$\begin{aligned} A_{\mathrm{comp}} &\le \frac{23}{30}M + \xi, \\ A_{\mathrm{comp}} + (6\kappa - 1)z &\le \frac{5}{2}M - 2A + 2z + \xi \le \frac{14}{15}M + \xi. \end{aligned} \tag{17.16}$$

Each required boundary has a margin of at least $M/15$ before the ξ term. Choose $\xi \le \rho/30$. After the threshold in (17.9), the unit-box multipliers are already included in ξ, so every retained reflected comparison calls the previously proved uncentered assertion at this same width. This also completes the original case with a factor shorter than L.

In the remaining case put $X_i = Z^{n_i}$, $Y_1 = Z^L$, and $Y_2 = X_1X_2/Y_1$. Then $X_1X_2 = Y_1Y_2$, and all four plain lengths are at least L. The original product minus the unreflected comparison retains the common product Q of slots; denote this difference by Δ_k. Multiplicativity and the equal product normalization give the explicit formula

$$\begin{aligned} \Delta_k = \frac{Q}{\sqrt{X_1X_2}} \sum_{l_1,l_2} \psi_k(l_1l_2)\big\{ & W_1(q_{l_1}/X_1)W_2(q_{l_2}/X_2) \\ & - W_1(q_{l_1}/Y_1)W_2(q_{l_2}/Y_2)\big\}. \end{aligned}$$

The comparison is bounded on the original permissible rows. The squared norm of the whole Δ_k is nonnegative, so it can then be enlarged to all rows in the smooth row ball. In the uncentered case, enlarge the squared norm of the original product instead. The Poisson transforms below act on these full smooth row balls; the division into exceptional and nonexceptional rows occurs after the transformed products have been separated.

17.3 The centered coefficient and its support

We keep the subtraction as one coefficient while transforming its row norm. For fixed ideals $\mathbf{b} = (b_1, b_2)$, define

$$D_{\mathbf{b}}(l_1, l_2) = \prod_{i=1}^{2} W_i(q_{b_i} q_{l_i}/X_i) - \prod_{i=1}^{2} W_i(q_{b_i} q_{l_i}/Y_i), \qquad X_1X_2 = Y_1Y_2. \tag{17.17}$$

An allocation $\mathbf{b}$ records prime powers already extracted from the two plain variables. If its conditions are impossible for one rectangle, that rectangle's profile is zero on the corresponding sum; the formal difference in (17.17) is nevertheless retained.

The reason for retaining these common data is already visible in the exceptional case. For a fixed $\vartheta \in \Theta$, a common mask $\mathfrak{R}_*$, and a common norm power q_l^{it}, the lattice estimate proved below has leading term

$$\sum_{l} \vartheta(l) 1_{(l,\mathfrak{R}_*)=1} q_l^{it} W_i(q_l/T) = c_{\vartheta,\mathfrak{R}_*} T^{1+it} I_i(t) + \text{error},$$

where $c_{\vartheta,\mathfrak{R}_*}$ does not depend on T or t. Here $I_i(t)$ is a fixed measure constant times $\int_0^\infty W_i(y) y^{it}\, dy$. The product main term is therefore $c_{\vartheta,\mathfrak{R}_*}^2 T_1^{1+it} T_2^{1+it} I_1(t) I_2(t)$, which agrees for the two rectangles when T_1T_2 agrees. Lemma 17.6 will quantify the remaining error. This cancellation is used only on the transformed rows inducing characters in Θ; the common coefficient below is retained on all rows until that later division into cases.

For a remaining slot set $\mathcal{I}'$, write $\Pi = \prod_{i\in\mathcal{I}'} p_i$. A coefficient with a common character and mask means a coefficient of the form

$$\left(\prod_{i\in\mathcal{I}'} \nu_i(p_i) W_i^{\mathrm{slot}}(q_{p_i}/P_i)\right) D_{\mathbf{b}}(l_1, l_2)\, \tau_1(u) q_u^{it} 1_{(u,\mathfrak{R})=1} 1_{s|u}, \qquad u = \Pi l_1 l_2. \tag{17.18}$$

Here τ_1 is one zero-extended product of a fixed finite-ray character and moving residue-symbol factors; $\mathfrak{R}$ is a fixed squarefree extra mask; s is a fixed squarefree ideal; and $t \in \mathbb{R}$. The condition $s \mid u$ is omitted when $s = 1$. The slots retain their original disjoint underlying supports. They may share primes with either plain variable.

These common data factor multiplicatively on the full product. For any ideals $v_1, \dots, v_r$, even with common primes,

$$\chi_{\prod v_i}(h) = \prod_i \chi_{v_i}(h), \qquad q_{\prod v_i}^{it} = \prod_i q_{v_i}^{it}, \qquad 1_{(\prod v_i,\mathfrak{R})=1} = \prod_i 1_{(v_i,\mathfrak{R})=1}.$$

The first identity includes all zeros: a present prime whose total exponent is divisible by six gives the zero-extended principal factor, not the constant one. The same multiplicativity holds for τ_1. A fixed-ray character evaluated on a full product also factors with the same character on every variable.

Support windows. The support errors must be controlled for an entire subset of slots at once so that the slot mesh is independent of their number. Let N be the original fixed number of slots, and choose fixed intervals $[a_i, b_i]$ containing the support of each W_i^{slot}. Put $h_i = \max\{|\log a_i|, |\log b_i|\}$.

Choose a fixed number H_N at least $\sum_i h_i$, enlarged to include the two plain profile windows, $\log C_{\mathrm{ref}}$, fixed arithmetic normalizations, the relative dyadic boxes, and the finitely many support enlargements through the D operation stages. The number H_N records the fixed support windows. It may depend on all the fixed data and on N, but not on Z or any moving label; its contribution to a logarithmic length is $\theta_N = H_N/\log Z$. For every subset I of live slots or slots whose primes have been fixed,

$$\left|\log \prod_{i\in I} \frac{q_{p_i}}{P_i}\right| \le \sum_{i\in I} h_i \le H_N, \qquad \theta_N := \frac{H_N}{\log Z}. \tag{17.19}$$

Every residual full product divided by its nominal product scale lies in $\exp([-H_N, H_N])$, after increasing H_N once for the fixed list of operations. Both rectangles use that same box. Extracted plain prime powers use their exact norms, and a slot with a fixed prime contributes its ratio q_p/P_i just once. If a nonempty plain scale below one is clipped to one, its error is at most the logarithm of that plain window's fixed endpoint divided by $\log Z$, once for that plain. Thus a whole subset of slots or a whole divisor extraction contributes one aggregate $O(\theta_N)$ boundary error, not one copy of a preselected tolerance for each slot or prime.

17.4 The first Poisson transform and its target bound

The first transform produces two Gauss-row norms. The bound (17.23) for each norm will imply the original moment estimate by Cauchy–Schwarz and the common-support inequality (17.24).

The same transform applies to the uncentered product and to the centered difference selected above. Put $H = Z^m$ and $X = Z^A$. Choose a fixed nonnegative smooth radial function that majorizes the row ball. The full index product in either rectangle has norm in a fixed multiple of X. All comparisons of exponents in this subsection are first made on fixed dyadic norm intervals. Their bounded relative widths change a logarithmic length by $O(1/\log Z)$; the final loss discussion includes these changes.

Common frequency localization. The same localization will be used in both Poisson formulas. Fix a nonnegative smooth dyadic partition $\sum_\lambda \omega_\lambda(q/T_\lambda) = 1$ for $q > 0$, with each weight supported in $C_d^{-1} \le q/T_\lambda \le 1$ for one fixed C_d. A sector fixes dyadic boxes for the finite list of aggregate outer norms, not a separate box for each slot. Let T_{sec} be the supremum in that sector of the nominal frequency scale. Its ratio to the scale at any one set of outer labels is at most a fixed C_{sec}. Retain exactly the whole weights whose support meets

$$0 < q \le T_{\mathrm{sec}} Z^{\xi/2}.$$

Their union is contained in $q \le C_d T_{\mathrm{sec}} Z^{\xi/2}$; every discarded weight is supported above $T_{\mathrm{sec}} Z^{\xi/2}$. The selection depends only on the sector, never on a live column. After N and the support data are fixed, take Z large enough that

$$2\theta_N + \frac{\log(C_d C_{\mathrm{sec}})}{\log Z} < \frac{\xi}{4}. \tag{17.20}$$

One fixed enlargement of C_{sec} covers all the finitely many sector endpoint factors.

The discarded dyads can be estimated directly in the original sums. In both applications the kernel argument is at least $q/(e^{2H_N} T_{\mathrm{sec}})$. On every discarded weight it is therefore at least $Z^{\xi/4}$, by (17.20). Lemma 3.12 gives arbitrary decay $(1 + \text{argument})^{-B}$. For the first formula use $|G(a,h)| \le q_a^{1/2}$, and for the second use the finite definition $|F(u,v;j)| \le q_u q_v$.

To sum this decay, absolute ideal counting and the divisor bounds for the fixed number of factors bound all raw columns and prefactors in a sector by $C_N Z^{B_0}$ times a fixed polynomial in the retained heights, where B_0 depends only on the bounded total lengths and chosen subsidiary power shares. A lattice norm dyad of scale T has $O(1+T)$ frequencies. Summing the radial decay over discarded dyads consequently gives $C_{N,B} Z^{B_1 - B\xi/4}$ times that height polynomial, for a bounded B_1 independent of B and N, and a convergent geometric sum when $B > 1$.

Choose B sufficiently large to obtain any prescribed power saving, including the polynomial number of fixed outer labels. This is an absolute bound for the original terms, before any off-coprime extension or Fourier absolutization. The bound depends only on these absolute estimates for the original sums.

On the retained domain, selection depends only on the sector, and the full smooth kernel remains on every retained weight. A retained range below the first nonzero lattice norm is empty, apart from a bounded boundary dyad covered by the clipping convention below.

Zero frequency. At zero frequency in the first row Poisson formula, a character mean can be nonzero only if its exponent at every prime is zero modulo six. In particular, no prime occurs to total multiplicity one in the product of the two full index products. This product is therefore a powerful ideal. There are $O_\epsilon(X^{1+\epsilon})$ such products of norm $O(X^2)$: each powerful ideal is a square times a cube of a squarefree ideal, and summing over the latter gives the usual $O(Y^{1/2+\epsilon})$ bound up to norm Y. Allocations to the fixed number of factors are divisor-bounded. The central factor is X^{-1}, and the row mean has size at most H. Thus the zero frequency is $O(Z^{m+\epsilon_1})$.

Nonzero frequencies and common support. For the nonzero frequencies write the two full products as Ca, Db, where C, D contain their complete common prime support and

$$(a, b) = 1, \qquad (ab, CD) = 1.$$

At each common prime, fix its exact valuation in each plain variable and whether it is supplied by a slot. Dividing out these valuations adds that prime to the mask of every remaining factor on the side. A slot supplying that prime is removed after its prime has been fixed. This description is valid even when a slot and one or both plain variables supplied that prime. The allocation is made once for the coefficient, so the same ideals b_i occur in the two terms of $D_{\mathbf{b}}$. An impossible allocation is a zero term. In particular, a scalar forced to vanish by an old moving zero or a common mask is not replaced by its absolute upper bound before these support conditions have been imposed.

Let c, d be the logarithmic norms of C, D, and let p be the logarithmic norm of their common radical. Let $\mathfrak{r}$ be the product of common primes whose net exponents are nonzero modulo six; its logarithmic norm is R. The corresponding character $\xi_{\mathfrak{r}}$ is primitive modulo $\mathfrak{r}$. In the Möbius expansion of the complementary common row mask, write $\mathfrak{e}$ for the selected divisor and E for its logarithmic norm. In particular

$$R \le p, \qquad E \le p - R.$$

Let $A_C(a)$ and $A_D(b)$ be the allocated convolution coefficients, with the common character τ omitted. They include all profiles, live slot coefficients, and fixed masks; in the centered case each contains the entire allocated difference. Set

$$\tau_C(a) = \tau(a)\chi_a(\mathfrak{e}\mathfrak{r})\xi_{\mathfrak{r}}(a), \qquad \tau_D(b) = \tau(b)\chi_b(\mathfrak{e}\mathfrak{r})\overline{\xi_{\mathfrak{r}}(b)}.$$

Residue-class Poisson, with the self-dual lattice measure from Section 3, gives for this allocation the following nonzero-frequency expression, up to a scalar of bounded modulus in the fixed labels:

$$\begin{aligned}\frac{H}{Xq_{\mathfrak{e}}\sqrt{q_{\mathfrak{r}}}}\sum_{h\neq 0} G_{\xi_{\mathfrak{r}}}(\mathfrak{r}, h)\sum_{(a,b)=1}\frac{A_C(a)\overline{A_D(b)}}{\sqrt{q_a q_b}}\tau_C(a)\overline{\tau_D(b)\mathcal{R}(a,b)}\,G(a,h)\overline{G(b,-h)}\\ \cdot\,\widehat{\Phi}_1\left(\frac{Hq_h}{q_{\mathfrak{e}}q_{\mathfrak{r}}q_a q_b}\right).\end{aligned} \tag{17.21}$$

Here $G_{\xi_{\mathfrak{r}}}$ is the normalized primitive Gauss sum and has modulus at most one. To check the normalization, the substitution $k = \mathfrak{e}k'$ and Poisson modulo $\mathfrak{r}ab$ give $H/(q_{\mathfrak{e}}q_{\mathfrak{r}}q_a q_b)$. The three unnormalized Gauss sums restore $\sqrt{q_{\mathfrak{r}}q_a q_b}$. The global squared central normalization is X^{-1}. For the phase, CRT for the pairwise coprime moduli $a, b, \mathfrak{r}$ supplies

$$\chi_a(b\mathfrak{r})\overline{\chi_b(a\mathfrak{r})}\xi_{\mathfrak{r}}(ab).$$

The substitution $k = \mathfrak{e}k'$ supplies $\chi_a(\mathfrak{e})\overline{\chi_b(\mathfrak{e})}$, up to a scalar in the fixed labels. Reciprocity changes $\chi_a(b)\overline{\chi_b(a)}$ into $\overline{\mathcal{R}(a,b)}$; the remaining factors are precisely $\tau_C(a)\overline{\tau_D(b)}$. This proves (17.21). Each new character acts on its whole residual product. Its new moving primes belong to the extracted support and force every residual factor to avoid them. Their full displayed union is counted even if the factors at $\mathfrak{r}$ cancel on units.

The new full displayed moving support has logarithmic norm at most $\widetilde{q} = q+R+E$, including any canceled factors at $\mathfrak{r}$. Its nominal frequency scale is exactly

$$T_1(C,D,\mathfrak{e},\mathfrak{r}) = \frac{q_{\mathfrak{e}}q_{\mathfrak{r}}X^2}{Hq_Cq_D} = Z^{K_0}, \qquad K_0 = 2A-c-d+R+E-m.$$

The product support convention gives $q_a/(X/q_C), q_b/(X/q_D) \in \exp([-H_N, H_N])$, so the argument in (17.21) is at least $q_h/(e^{2H_N}T_{\mathrm{sec}})$ for the supremum of T_1 in the outer sector. The preceding whole-dyad localization therefore applies to this coprime bridge. If $K = \log_Z T_\lambda$ is the upper length of a retained dyad, then

$$\widetilde{q} = q+R+E, \qquad K \le K_0 + \delta_{\mathrm{fr},1}, \qquad 0 \le \delta_{\mathrm{fr},1} := \frac{\xi}{2} + \frac{\log(C_dC_{\mathrm{sec}})}{\log Z} < \xi. \tag{17.22}$$

The zero-frequency estimate above belongs to the original common smooth row ball, and remains on that domain throughout the localization. The later exponent comparisons retain the possible inequality $K_0 - K \ge -\delta_{\mathrm{fr},1}$.

Separation into two Gauss norms. For each retained dyad and complete common-support allocation, define $\mathcal{C}_1(a,b;h)$ to be the full bridge summand also on noncoprime residual pairs individually disjoint from CD and allowed by the old masks. Use the fixed bicharacter $\mathcal{R}(a,b)$, the full zero-extended $G(a,h), \overline{G(b,-h)}$, the displayed whole-product characters, and the same dyad weight, kernel, inverse roots, and formal product norms. CRT identifies this definition with the original summand only on $(a,b)=1$. On each finite column shell the exact identity is

$$\sum_{(a,b)=1} \mathcal{C}_1(a,b;h) = \sum_{a,b} \mathcal{C}_1(a,b;h) \sum_{s\mid a,b} \mu(s).$$

The full squarefree Möbius sum annihilates the artificial noncoprime pairs. It is inserted before estimating independent factors; put $s_0 = \log_Z q_s$. Fourier separation then acts on the fixed-ray phases and all smooth factors in the normalized row norm and the two whole-product norms. In these variables the kernel is $\widehat{\Phi}_1(R_{\mathrm{sc}}x/(y_1y_2))$ on fixed logarithmic boxes. If its aggregate scale ratio R_{sc} varies over the sector, include that single normalized outer ratio as another coordinate. The cutoffs are chosen on the common product annulus of (17.19), before fixing live slot labels. Thus one Fourier coefficient measure is common to the rows and all live labels. It supplies a row phase and only one norm power on each whole column, together with phases in fixed outer norms. It introduces no separate powers on the two plain variables or the two rectangles. The full kernel and inverse roots are kept until this separation; taking their absolute supremum inside $D_{\mathbf{b}}$ would not preserve the coefficient. Cauchy–Schwarz in h is applied only after this separation, followed by the bound $|G_{\xi_{\mathfrak{r}}}| \le 1$. The resulting positive norm on the C side is

$$\mathcal{N}_C = \sum_{h\neq 0} \omega_K(q_h/Z^K) \left| Z^{-(A-c)/2} \sum_{s\mid a} A_C(a)\tau'_C(a)q_a^{it}G(a,h) \right|^2, \qquad \omega_K \ge 0,$$

and there is an analogous D norm. Here τ'_C includes one separated fixed-ray character. All fixed masks in A_C remain present. Complete extraction and multiplicativity therefore leave the coefficient in (17.18), now multiplied by $G(a,h)$. In particular the two rectangles retain the same allocated plain powers, character, coprimality restriction, and norm power.

The factors outside these two norms have exponent $m-A-R/2-E$. The absolute number of common-support labels and Möbius labels has exponent $p+s_0+\epsilon_1$. To justify the common-support count uniformly, first count its radical, giving $O(Z^{p+\epsilon_1})$. For a fixed radical $\mathfrak{c}$ of polynomial norm and any fixed T, Rankin's bound gives

$$\#\{v : \mathrm{rad}(v) \mid \mathfrak{c},\ q_v \le Z^T\} \le Z^{aT} \prod_{\substack{r \mid \mathfrak{c} \\ r \text{ prime}}} (1-q_r^{-a})^{-1} \ll Z^{aT+\epsilon_1}$$

for every fixed $a>0$. Choose a small and use the polynomial-size Euler-product estimate. This bounds the choices of the powers in C, D by an arbitrarily small power. Their allocations and the choice of $\mathfrak{c}$ have divisor-bounded multiplicity. Finally there are $O(Z^{s_0+\epsilon_1})$ possible s.

The sufficient Gauss-norm bound. The exponent ϵ_G below denotes the error permitted at the current induction depth, together with its local small-power shares. These envelopes will be chosen compatibly when the finite induction is completed, using the same slot mesh throughout. We will prove the following sufficient bound for the squared C norm:

$$\mathcal{N}_C \ll Z^{A-c+\widetilde{q}+\mathcal{B}_c-s_0+\epsilon_G}, \qquad \mathcal{B}_c = \max\{0,(3c-5d-R)/6\}. \tag{17.23}$$

The analogous quantity is $\mathcal{B}_d = \max\{0,(3d-5c-R)/6\}$. Adding the outer factors, the common-support count, and the two Cauchy–Schwarz exponents gives

$$\begin{aligned}(m-A-R/2-E)+(p+s_0)\\ +\tfrac{1}{2}\{A-c+\widetilde{q}+\mathcal{B}_c-s_0+A-d+\widetilde{q}+\mathcal{B}_d-s_0\}\\ = M+\tfrac{1}{2}\{\mathcal{B}_c+\mathcal{B}_d-(c+d-2p-R)\}.\end{aligned}$$

The last brace is nonpositive:

$$\mathcal{B}_c+\mathcal{B}_d \le c+d-2p-R. \tag{17.24}$$

Indeed, at a common prime of multiplicities $i \ge j \ge 1$, put $r = 1_{6\nmid i-j}$. Only the i side can have a positive local numerator. Its contribution is at most $(3i-5j-r)_+/6$, whereas the right side contributes $i+j-2-r$. The latter is nonnegative. If the former is positive, six times their difference is $3i+11j-12-5r$, which is nonnegative: for $r=0$ it is at least 2, and for $r=1$ one has $i \ge j+1$, giving at least 0. The positive part of a sum is at most the sum of positive parts. Multiplication by each prime's logarithmic norm and summation proves (17.24). Thus it remains to establish (17.23).

17.5 Enlarging the Gauss-row norm

The second transform decreases the effective width once the Gauss-row range has been enlarged. For zero slots we enlarge the original positive norm directly. With positive slots, averaging over sixth powers of pool primes gives an enlarged main norm and three types of extraction error; each error is enlarged directly. Let w denote a length removed from the C column, and let $w_o \le w$ be the additional moving-radical length created by that removal. Put $a_0 = A-c-w$. When a squared extraction coefficient of size Z^{-w_o} has been removed, (17.23) allows the unweighted remaining norm the exponent

$$\Lambda_c = a_0+\widetilde{q}+w_o+w+\mathcal{B}_c-s_0+\epsilon_G. \tag{17.25}$$

Equivalently, the target exponent is $A-c+\widetilde{q}+\mathcal{B}_c-s_0+\epsilon_G+w_o$.

For the initial norm, with $w=w_o=0$ and $a_0=A-c$, the first transform gives the following conditional summand after fixing Π and omitting its outer scalar:

$$Z^{-a_0/2}\sum_{l_1,l_2} D_{\mathbf{b}}(l_1,l_2)\tau_1(l_1l_2)q_{l_1l_2}^{it}\mathbf{1}_{(\Pi l_1l_2,\mathfrak{R})=1}\mathbf{1}_{s\mid\Pi l_1l_2}G(\Pi l_1l_2,h). \tag{17.26}$$

The slot sums are restored before this polynomial is squared. The same statement with the second rectangle omitted applies to uncentered products. The local calculations below will show that the extracted terms retain this form at their shortened length $a_0 = A - c - w$.

The needed row enlargement is governed by

$$J = d - c + (K_0 - K) - 2w + w_o, \qquad J_+ = \max(J, 0). \tag{17.27}$$

The zero-slot norm. In the zero-slot proof take $w = w_o = 0$, and set

$$\ell = 0, \qquad g = J_+ + \sigma.$$

The positive norm $\mathcal{N}_C$ is at most the corresponding norm over a smooth ball of length $K + g$. There is no multiplication of rows in this case.

Averaging over prime sixth powers. For positive slots, start with $w = w_o = 0$. Set $\ell_* = \sigma/3$, choose the slot mesh $\eta < \sigma/6$, and take the fixed pool

$$\mathcal{P} = \{p : Z^{\ell_*}/2 < q_p \le Z^{\ell_*},\ p \notin S\}.$$

The prime ideal theorem in the fixed field, equivalently its fixed ray-class form [26, Theorem 1.1], gives $|\mathcal{P}| = Z^{\ell_* + o(1)}$. Because the live slot lengths are at most η and their relative annular supports are fixed, the single inequality $Z^{\ell_* - \eta} > 2\max_i b_i$ makes this pool disjoint from every live slot window. Its exponent gap is at least $\sigma/6$, independently of N; only the threshold depends on the fixed windows. For a row $h \neq 0$, omit pool primes dividing h, s, or any fixed support. Each such integer or ideal has polynomial norm, so $O(\log Z)$ primes are omitted. The remaining set $\mathcal{P}_h$ has size comparable to $|\mathcal{P}|$, uniformly in the row and fixed labels.

To compare the rows h and hp^6, write a full modulus as $p^i u$ with $(p, u) = 1$. CRT and reciprocity give the exact formula

$$G(p^i u, h) = \mathcal{R}(p, u)^i \chi_p(u)^{2i} G(p^i, h) G(u, h).$$

Also $G(u, hp^6) = G(u, h)$, since p is a unit modulo u and $\chi_u(p^6) = 1$. For $p \nmid h$, (12.5) shows that replacing h by hp^6 changes only $i = 1, 6, 7$. The factors $G(p^i, h)$ that occur are a scalar in the row and p; the displayed remaining factor is one fixed-ray character times one residue character on the whole u. Since p is outside every live slot window, p^i is allocated only to the two plain variables. Every residual factor is restricted to be coprime to p, and the same allocation updates $D_{\mathbf{b}}$ in both rectangles. Because $p \nmid s$, the condition $s \mid u$ remains.

Write the normalized polynomial in $\mathcal{N}_C$ as $\mathcal{H}(h)$. For each eligible p, local Gauss evaluation gives $\mathcal{H}(h) = \mathcal{H}(hp^6)$ plus the extracted terms with p-adic column valuations $1, 6, 7$. There are only boundedly many allocations of each of these valuations to the two plain variables. Jensen's inequality, first in p and then for this fixed finite sum, gives

$$|\mathcal{H}(h)|^2 \ll \frac{1}{|\mathcal{P}_h|} \sum_{p \in \mathcal{P}_h} \left\{ |\mathcal{H}(hp^6)|^2 + \sum_{i=1,6,7} \sum_{\text{allocations}} |c_{p,i}(h)|^2 |\mathcal{H}_{p,i}(h)|^2 \right\}.$$

The $\mathcal{H}_{p,i}$ are normalized at their shortened column scales. The local identity and allocation just described are applied to the whole centered coefficient, so each error retains (17.18).

The enlarged main norm. On the support of the original row weight, the new row hp^6 has norm $O(Z^{K + 6\ell_*})$. A fixed output row has only boundedly many representations as hp^6 with $p \in \mathcal{P}_h$: every such p divides that row, and all such primes have norm at least $Z^{\ell_*}/2$, while the output norm has bounded logarithmic length. Summing the first term over h therefore gives the factor $|\mathcal{P}|^{-1} = Z^{-\ell_* + o(1)}$ times a positive norm on the new rows. For this main term set

$$\ell = \ell_*, \qquad g = J_+ + 2\sigma.$$

Since $6\ell_* = 2\sigma$, a smooth ball of length $K + g$ contains all the new rows.

The three extraction errors. For the extracted terms put $P = q_p$. When $p \nmid h$, (12.5) gives the following central coefficients. At valuation one, the old Gauss sum has modulus one and the new one is zero, so extraction gives squared coefficient P^{-1}. At valuation six, the new Gauss sum is $P^3(1 - P^{-1})$; its central factor is P^{-3}, giving $(1 - P^{-1})^2$. At valuation seven, the new Gauss sum has modulus P^3, and the central factor $P^{-7/2}$ gives P^{-1}. The row phase is $\overline{\chi_p(h)}$ for valuations one and seven, and is one for valuation six. It multiplies both rectangles.

Let $\ell_p = \log_Z P$. For sufficiently large Z, $\ell_*/2 \le \ell_p \le \ell_*$. The removal and new moving-radical lengths for the three terms are

$$(w, w_o) = (i\ell_p, e_i\ell_p), \qquad i = 1, 6, 7, \qquad e_1 = e_7 = 1, \quad e_6 = 0. \tag{17.28}$$

The two squared factors P^{-1} give exactly Z^{-w_o}. For the valuation-six term, the exact factor $\chi_p(u)^{12} = \mathbf{1}_{(u,p)=1}$ is represented by the already common fixed coprimality restriction at p, with its fixed-ray phase $\mathcal{R}(p, u)^6$ retained. At valuations one and seven the displayed local factor remains, and its prime is counted in the moving radical. In each error, fix p, retain its common column coprimality restriction, and discard its row eligibility restriction only after taking the positive norm. The average remains $|\mathcal{P}|^{-1}\sum_{p\in\mathcal{P}}$, so a bound uniform in the fixed p introduces no prime-count factor. For each error set

$$\ell = 0, \qquad g = J_+ + \sigma$$

and enlarge its original rows directly to a smooth ball of length $K + g$. Errors are not amplified again. All three types of norm satisfy

$$0 \le w_o \le w \le 7\sigma/3. \tag{17.29}$$

Adding the zero row. The row zero is added only at the final smooth ball. Put $Y_{\mathrm{col}} = e^{H_N} Z^{a_0} = Z^{a_0+\theta_N}$. A nonempty column shell satisfies $Y_{\mathrm{col}} \ge 1$, and hence $a_0 \ge -\theta_N$. By (12.5), $G(u, 0) = 0$ unless u is a sixth power, and $|G(u, 0)| \le q_u^{1/2} \le Y_{\mathrm{col}}^{1/2}$. There are $O(Y_{\mathrm{col}}^{1/6})$ sixth powers on this support. If $s \mid u$, then $q_s^6 \le q_u \le Y_{\mathrm{col}}$, so $s_0 \le (a_0+\theta_N)/6$. Including the assigned divisor-bounded convolution loss, the squared contribution after central normalization is

$$\ll Z^{-a_0}\big(Y_{\mathrm{col}}^{1/6} Y_{\mathrm{col}}^{1/2}\big)^2 Z^{\epsilon_1} = Z^{a_0/3+4\theta_N/3+\epsilon_1} \le Z^{a_0-s_0+2\theta_N+\epsilon_1}.$$

The last inequality follows from $a_0 \ge -\theta_N$ and the displayed bound for s_0. Thus (17.25) bounds this added row with the numerical $2\theta_N$ correction included in the $C_*\xi$ stage error bound. Here $C_* \ge 1$ denotes the common constant for aggregate support errors in one stage, chosen independently of the slot count N. The estimates below establish that one such choice covers all operations in a stage. Zero is never multiplied by a pool prime.

17.6 The second transform and smaller-width products

The second transform converts each enlarged Gauss norm into products at a smaller effective width. The width relation (17.32) gives the decrease, and (17.33) gives the exponent these products must satisfy. The residual coefficients and their normalization are treated in Section 17.7. Section 17.8 then applies induction to the nonexceptional rows. The exceptional products retain their signed rectangles and are estimated in Section 17.9.

The second Poisson formula and its diagonal. All live slot sums are restored before the square is expanded. Let $B(u)$ denote the full allocated convolution coefficient in one of the preceding Gauss polynomials, including its fixed mask and s-divisibility condition. It contains the whole difference when centering is used. Its moduli satisfy $q_u/Z^{a_0} \in \exp([-H_N, H_N])$. Let Φ_2 be the

nonnegative smooth radial function defining the final row ball. Apart from the $Z^{o(1)}$ loss in the prime density, the norm to be bounded is

$$Z^{-\ell}\sum_{h}\Phi_2(q_h/Z^{K+g})\left|Z^{-a_0/2}\sum_{u}B(u)\tau_1(u)q_u^{it}G(u,h)\right|^2.$$

The Fourier identity in Lemma 12.3 gives its exact expansion

$$Z^{K+g-\ell-a_0}\sum_{u,v}\frac{B(u)\overline{B(v)}\tau_1(u)\overline{\tau_1(v)}q_u^{it}q_v^{-it}}{\sqrt{q_uq_v}}\sum_{j}F(u,v;j)\widehat{\Phi}_2\left(\frac{Z^{K+g}q_j}{q_uq_v}\right).$$

The kernel that occurs here is

$$\widehat{\Phi}_2\left(\frac{Z^{K+g}q_j}{q_uq_v}\right). \tag{17.30}$$

Both the displayed inverse square roots and this kernel are retained as functions of full u,v until their whole-product separation below. In particular the formula includes every live prime and every shared-prime multiplicity.

At $j=0$, Lemma 12.3 leaves only $u=v$, with $F(u,u;0)=\varphi(u)\le q_u$. The number of supported moduli divisible by s, on a nonempty shell, is at most

$$C\frac{Y_{\mathrm{col}}}{q_s}=CZ^{a_0-s_0+\theta_N},$$

because writing $u=sv$ gives $Y_{\mathrm{col}}/q_s\ge 1$ on that shell. The divisor-bounded coefficients use their separate ϵ_1 share. Thus the diagonal exponent is $K+g-\ell-s_0+\theta_N$, including when a_0-s_0 is slightly negative. Subtracting the target exponent in (17.25) from its nominal part, without ϵ_G, gives

$$\begin{aligned}&(K+g-\ell-s_0)-(a_0+\widetilde{q}+w_o+w+\mathcal{B}_c-s_0)\\&\quad=(A-M)-d-(K_0-K)-w_o-\mathcal{B}_c+g-\ell\\&\quad\le A-M+5\sigma/3+\delta_{\mathrm{fr},1}.\end{aligned} \tag{17.31}$$

For the inequality, put $D_0=d+K_0-K$. The contribution $-D_0-w_o+J_+$ is at most $\delta_{\mathrm{fr},1}$: it is at most $-c-2w$ when $J\ge 0$, and when $J<0$ use $D_0\ge-\delta_{\mathrm{fr},1}$. The largest remaining increment is $2\sigma-\ell_*=5\sigma/3$ in the amplified main norm; the other norms have increment σ. Thus the diagonal requires only the displayed terminal loss $5\sigma/3$, plus the already reserved frequency perturbation $\delta_{\mathrm{fr},1}$ and numerical support correction θ_N, when $A\le M$, and an additional δ in the padded zero-slot core. The θ_N correction is included in the $C_*\xi$ stage error bound.

Common support at nonzero frequency. For $j\neq 0$, first perform the whole-dyad localization on this full (u,v,j) sum before any extension, before extracting any common support. The nominal scale is $T_2=Z^{2a_0-K-g}$, and the kernel argument is at least $q_j/(e^{2H_N}T_{\mathrm{sec}})$ for its supremum in the current Gauss sector. The tail estimate above applies using $|F(u,v;j)|\le q_uq_v$, with the literal conditions $s\mid u,v$ still present. Write

$$\Omega_{\mathrm{ret}}(q_j):=\sum_{\lambda\ \mathrm{retained}}\omega_\lambda(q_j/T_\lambda).$$

This common row weight satisfies $0\le\Omega_{\mathrm{ret}}\le 1$. The full kernel in (17.30) remains on every retained term. All subsequent signed Fourier separations are performed one retained λ at a time, so the normalized row variable stays on a fixed log box; Ω_{ret} records their sum and common domain. There are $O(\log Z)$ relevant retained dyads in the bounded polynomial ranges. The preceding $j=0$ term is exactly the diagonal of the same final Φ_2 ball, including the previously added Gauss row zero. Both zero terms retain their original domains.

Now write $u = D_2 a$, $v = E_2 b$, extracting the complete common support, so that $(a, b) = 1$ and $(ab, D_2 E_2) = 1$. Allocate all extracted powers to the plain variables and slots as at the first transform. Lemma 12.5 applies on this coprime locus and gives

$$F(D_2 a, E_2 b; j) = F(D_2, E_2; j)\mathcal{R}(a, E_2)\overline{\mathcal{R}(b, D_2)}\mathcal{R}(a, b)\chi_a(j)\overline{\chi_b(-j)}.$$

It permits arbitrary prime powers in all four moduli and leaves a row scalar at the common primes. Every extracted prime forces all remaining factors on its side to be coprime to it, and every slot supplying that prime has its prime fixed and is removed.

This also explains explicitly the cases where a slot shares a prime with a plain variable. If p occurs only on one side, with full multiplicity i, its residual factor is $\chi_p(j)^i$. Splitting i among the two plain variables and the possible slot gives precisely the same factor by multiplicativity, including when $6 \mid i$ and $p \mid j$. For example, a plain p^5 and a slot p give the zero-extended factor $\chi_p(j)^6$. If p occurs on both sides, its full powers are removed. In the unequal case $i > j_0 \geq 1$, the local correlation is zero unless $6 \mid j_0$ and the frequency is $p^{j_0} k$ with $p \nmid k$; when it is nonzero it equals

$$P^{j_0 - 1}(P - 1)\chi_p(k)^{i - j_0}, \qquad P = q_p.$$

This is a row scalar, even if the excess valuation on the first side came partly from a live slot. For instance $i = 7, j_0 = 6$ with a plain p^6 and slot p leaves the scalar $P^5(P - 1)\chi_p(k)$, fixes the prime in that slot and removes it, and restricts both remaining plain variables to be coprime to p. Equal multiplicities have the scalar factors in (12.8) and the same coprimality restriction. Removing the complete common prime support makes these local factors independent of the residual variables; extracting the gcd alone would leave shared prime factors.

Define

$$c_2 = \log_Z q_{D_2}, \qquad d_2 = \log_Z q_{E_2}, \qquad b_2 = (c_2 + d_2)/2.$$

Let p_2 be the logarithmic norm of their common radical, let $G_c = (D_2, E_2)$, and put $g_2 = \log_Z q_{G_c}$. The correlation vanishes unless $G_c \mid j$. At a common prime with equal multiplicity $i \not\equiv 0 \pmod 6$, call the divided frequency j/G_c a unit or nonunit according as that prime does not or does divide it. Let t_2 be the total radical length of these unit primes. Let V_{id} be the product of these nonunit primes and put $V = \log_Z q_{V_{\mathrm{id}}}$.

The local formulas in (12.8) give the following absolute bounds:

common multiplicities and divided frequency	absolute correlation bound
$i = i,\ 6 \nmid i,$ unit	P^{i-1}
$i = i,\ 6 \nmid i,$ nonunit	P^i
$i = i,\ 6 \mid i,$ either	P^i
$i > j_0,\ 6 \mid j_0,$ unit	P^{j_0}.

Every unequal case not in the last line is zero. Fix the indicated unit/nonunit partition and write $j = G_c V_{\mathrm{id}} h'$. The partitioned scalar

$$\mathbf{1}_{\mathrm{part}}(h')\frac{F(D_2, E_2; G_c V_{\mathrm{id}} h')}{Z^{g_2 - t_2}}$$

is defined to be zero off that partition and has modulus at most one everywhere with this definition. This bound applies to the partitioned, zero-extended scalar. Every fixed allocation forced to be zero by the old masks is still discarded before this absolute bound is used.

The complete-support identity leaves the row factor $\chi_n(G_c V_{\mathrm{id}} h')$ on each whole residual column. To count its fixed moving support, put $v = G_c V_{\mathrm{id}}$ and $e_p = v_p(v) \bmod 6$. Products of the canonical primary generators of these good ideals are primary, so the exact all-input identity is

$$\chi_n(v) = \prod_{\substack{p \mid v \\ e_p \neq 0}} \chi_n(p)^{e_p} \prod_{\substack{p \mid v \\ e_p = 0}} \mathbf{1}_{(n,p)=1},$$

with any unit of the original row retained in h'. All primes in this identity already force every residual factor to avoid them, because they are in the extracted common support. The active $e_p \neq 0$ primes are contained in the unit set counted by t_2 and the nonunit set counted by V. Equal six-divisible and unequal-minimum-six-divisible primes have $e_p = 0$ and are represented solely by the common coprimality restriction in this fixed factor. A nonunit prime can also have $e_p = 0$, in which case counting it in V only enlarges the bound. This factors only $\chi_n(v)$: the row factor $\chi_n(h')$, with all its zeros, and all fixed-ray reciprocity phases remain. It does not reclassify a cancellation with the varying row as an extra coprimality restriction. Thus the full new moving support has length at most $\widetilde{q} + w_o + t_2 + V$.

The smaller width and required exponent. Define the nominal row and total widths by

$$\begin{aligned} m' &= 2a_0 - K - g - g_2 - V, \qquad q' = \widetilde{q} + w_o + t_2 + V, \\ M' &= m' + q' = M + J - g - g_2 + t_2 \leq M - \sigma. \end{aligned} \tag{17.32}$$

Here $g_2 \geq t_2$, and the definitions of g give $g - J \geq \sigma$. The identities follow by substituting $a_0 = A - c - w$ and the definition of K_0 preceding (17.22). Put

$$0 \leq \delta_{\mathrm{fr},2} := \frac{\xi}{2} + \frac{\log(C_d C_{\mathrm{sec}})}{\log Z} < \xi.$$

The sector constants here are those for the second transform. The retained row weight pulls back exactly to $\Omega_{\mathrm{ret}}(q_{G_c V_{\mathrm{id}}} q_{h'})$. On its support,

$$q_{h'} \leq \frac{C_d T_{\mathrm{sec}} Z^{\xi/2}}{q_{G_c V_{\mathrm{id}}}} \leq \frac{C_d C_{\mathrm{sec}} Z^{2a_0 - K - g + \xi/2}}{q_{G_c V_{\mathrm{id}}}} = Z^{m' + \delta_{\mathrm{fr},2}}.$$

Thus it lies in that common enclosing row ball. On a nonempty retained range define the declared nonnegative row length $m'_{\mathrm{act}} = \max\{0, m' + \delta_{\mathrm{fr},2}\}$ and $M'_{\mathrm{act}} = m'_{\mathrm{act}} + q'$. Nonemptiness implies $m' + \delta_{\mathrm{fr},2} \geq 0$, so $0 \leq M'_{\mathrm{act}} - M' \leq \delta_{\mathrm{fr},2}$.

Regard the pulled-back weight as zero on the rest of this enclosing ball throughout the signed calculation, and do the same for the partitioned scalar. This is an exact extension by zero. They will be replaced by their absolute bounds only after a nonnegative residual norm or an exceptional absolute product has been formed.

Complete extraction only shortens a plain variable or fixes the prime of a slot and removes that entire slot, so the surviving slot length satisfies $z' \leq z$.

To obtain the exponent required of the smaller-width products, we now combine the normalization factors and support counts from the second transformation. The condition $s \mid u, v$ has not been dropped: it implies that every prime of s is in the second complete common radical. Thus after the complete extraction there is no remaining s-divisibility condition on the residual columns. The number of possible radicals of length p_2, for this fixed s, is $O(Z^{p_2 - s_0 + \epsilon_1})$: write that radical as sr_2 with $(s, r_2) = 1$, and count the squarefree r_2 of the remaining norm. If $p_2 < s_0$ outside the fixed shell boundary, there are none. The same Rankin argument used for C, D bounds their power and allocation multiplicities. The number of unit/nonunit partitions is at most $2^{\omega(\mathrm{rad}(D_2 E_2))}$, another divisor-bounded factor included in ϵ_1. The exponents are

factor	exponent
squared central normalization	$-a_0$
row Poisson factor and normalized inverse roots	$K + g - a_0$
prime density, when present	$-\ell$
common-support count with fixed s	$p_2 - s_0$
common correlation	$g_2 - t_2$
conversion to normalized residual products	$a_0 - b_2$

The last line is $\{(a_0 - c_2) + (a_0 - d_2)\}/2$. Their sum is

$$K + g - \ell - a_0 + p_2 - s_0 + g_2 - t_2 - b_2.$$

Subtracting this sum from the exponent in (17.25) gives the bound required of the inner products. These smaller-width products are controlled either by the inductive moment estimate or by the exceptional-row estimate:

$$M' + \Delta_{\mathrm{child}}, \qquad \Delta_{\mathrm{child}} = b_2 - p_2 + w + \mathcal{B}_c + \ell \geq w + \ell. \tag{17.33}$$

The inequality uses $b_2 \geq p_2$, since both multiplicities at each common prime are at least one.

17.7 Residual coefficients and their normalization

The residual products must have independent columns before either the inductive moment bound or the exceptional-row estimate can be applied. We remove their coprimality condition, track the extraction coefficients, and record the common data preserved by these operations.

Separating the residual products. The residual coprimality must be removed before the two products are independent. For the fixed extracted D_2, E_2, define

$$\widetilde{F}_{D_2,E_2}(a,b;j) := F(D_2,E_2;j)\mathcal{R}(a,E_2)\overline{\mathcal{R}(b,D_2)}\mathcal{R}(a,b)\chi_a(j)\overline{\chi_b(-j)}$$

on all residual pairs individually coprime to D_2E_2 and allowed by the old masks. It equals the original correlation only when $(a,b) = 1$. Define the remaining $\mathcal{C}_2(a,b;j)$ on all individually allowed residual pairs by the allocated convolution formulas and the old common masks, including $(ab, D_2E_2) = 1$. Keep the formal full-product norms $q_{D_2}q_a$, $q_{E_2}q_b$, both inverse roots, the full smooth kernel, and the same pulled-back row weight extended by zero on its enclosing ball. For each retained row the exact identity is

$$\sum_{(a,b)=1} \mathcal{C}_2(a,b;j)F(D_2a,E_2b;j) = \sum_{\mathfrak{t}\ \text{squarefree}} \mu(\mathfrak{t}) \sum_{\mathfrak{t}\mid a,b} \mathcal{C}_2(a,b;j)\widetilde{F}_{D_2,E_2}(a,b;j).$$

The full Möbius indicator occurs here. The individual column shells and the retained row ball are finite. Its equality therefore follows by interchanging finite sums and using the original correlation identity only when $(a,b) = 1$. Extra common primes of $\mathfrak{t}$ are not part of the extracted D_2, E_2 support or of its frequency restrictions. Since the residual masks already exclude that support, every nonzero $\mathfrak{t}$ is disjoint from it and from s.

On each retained dyad, separate the full kernel and inverse roots in the normalized h' norm and the two whole-product norms, as for the first transform. The fixed factor G_cV_{id} contributes only an outer phase to the row Fourier power. The fixed support boxes are chosen before the current live labels, so the resulting coefficient measure is common to them, to both rectangles, and to the row. Then fix $\mathfrak{t}$. For each of its primes p, use the exact factor-allocation identity

$$\mathbf{1}_{p\mid\prod_i n_i} = \sum_{\varnothing\neq J} (-1)^{|J|+1} \prod_{i\in J} \mathbf{1}_{p\mid n_i},$$

where the factors n_i are the two plain variables and the live slots. For a selected plain variable write $n_i = pl_i$, with no restriction on l_i. This extracts only a row scalar and q_p^{it}, and replaces both X_i, Y_i by $X_i/q_p, Y_i/q_p$. It introduces no coprimality restriction on an individual variable. The prime of a selected slot is fixed, and the slot is removed. If two distinct divisor primes select the same prime slot, the term is zero. The quotients and unselected factors may still contain p, and may overlap the opposite side; no new coprimality is imposed. Old common masks remain common, and the product of the two new plain scales is equal in the two rectangles. The extracted row scalars, including zero scalars, are retained in the signed identity.

After this Fourier separation and factor allocation, each summand for a fixed retained dyad and fixed Fourier parameters is a product of two independent residual convolutions, where $\mathbf{J}$ records the factor allocations. Their row characters are

$$\tau_1(n)\rho(n)\chi_n(G_cV_{\mathrm{id}}h'), \qquad \tau_1(n)\rho'(n)\chi_n(-G_cV_{\mathrm{id}}h'), \qquad \rho,\rho' \in \Theta.$$

They are understood with the fixed-factor support representation above. In particular their inducing-character ratio belongs to Θ, including the supplementary factor $n \mapsto \chi_n(-1)$; this assertion takes no quotient at a zero. By (3.7) and the fixed bicharacter table, this supplementary character is the diagonal character $n \mapsto \mathcal{R}(n,n)$. Their row zeros are retained. Extracting the selected factors contributes row scalars, not new factors of the residual character and not new support in q'. These scalars, including zeros, remain in the signed identity.

Exact normalization of the residual products. The inductive estimate uses centrally normalized products. We therefore track the two residual convolutions from their nominal normalization through each extraction, before making any change to their formal scales. Put

$$\alpha_1 = a_0 - c_2, \qquad \alpha_2 = a_0 - d_2, \qquad \frac{\alpha_1 + \alpha_2}{2} = a_0 - b_2.$$

At the entrance to the current two-transform stage, let $\mathcal{I}_{\mathrm{in}}$ be its live slot set and let X_1, X_2 be the formal scales of its first rectangle. Their exact convention is $X_1 X_2 \prod_{i \in \mathcal{I}_{\mathrm{in}}} P_i = Z^A$.

For side $j \in \{1,2\}$, let $\mathcal{F}_j$ be the slots from this entrance set whose primes were fixed by the first and second complete common-support extractions, let $\mathcal{I}_j$ be the slots still live before the final $\mathfrak{t}$-allocation, and let $Q_{\mathrm{plain},j}$ be the product of the exact norms of the plain powers extracted at those steps and at the amplifier. Slots removed in an ancestor stage are not included in $\mathcal{F}_j$.

Let T_j be the common formal pre-$\mathfrak{t}$ product of the two plain scales in its rectangles; for an uncentered input use its one formal plain product. Before using slot ratios, discard a term whose slot profile vanishes at its fixed prime, which is identically zero by its fixed data independently of the row and live labels. Complete extraction gives the exact identities

$$Z^{A-\alpha_j} = Q_{\mathrm{plain},j} \prod_{i \in \mathcal{F}_j} q_{p_i},$$

$$T_j \prod_{i \in \mathcal{I}_j} P_i = \frac{Z^A}{Q_{\mathrm{plain},j} \prod_{i \in \mathcal{F}_j} P_i} = Z^{\alpha_j + e_j},$$

$$e_j = \sum_{i \in \mathcal{F}_j} \log_Z(q_{p_i}/P_i), \qquad |e_j| \le \theta_N.$$

The amplifier contributes only exact plain powers to this calculation. Each slot whose prime was fixed earlier occurs once, even if it shared its prime with a plain. The already separated inverse roots, kernel, and fixed normalization constants remain outside e_j.

For side j, let $d_{j,i}$ be the product of final $\mathfrak{t}$-primes selected in plain i, and let $\mathcal{J}_j \subseteq \mathcal{I}_j$ be the slots whose primes have just been fixed. Put

$$a_{j,i} = \log_Z q_{d_{j,i}}, \qquad \widetilde{r}_j = a_{j,1} + a_{j,2} + \sum_{i \in \mathcal{J}_j} z_i \ge 0, \qquad \omega_j = \sum_{i \in \mathcal{J}_j} \log_Z(q_{p_i}/P_i).$$

The corresponding formal scale is divided by $q_{d_{j,i}}$ in both rectangles. An assignment selecting one prime slot at two distinct $\mathfrak{t}$-primes, or a slot whose prime has just been fixed with zero profile value, is identically zero from the fixed data and is discarded before these ratio bounds are used. All other extracted zeros remain until the nonnegative or absolute estimate described below.

Use lower-endpoint divisor dyads $T \le q_{\mathfrak{t}} < 2T$, $T \ge 1$, and put $t_- := \log_Z T$. Each dyad contains $O(Z^{t_-})$ ideals. The actual selected product on side j is divisible by $\mathfrak{t}$, so

$$\begin{aligned} \widetilde{r}_j + \omega_j &= \log_Z \left(q_{d_{j,1} d_{j,2}} \prod_{i \in \mathcal{J}_j} q_{p_i} \right) \ge \log_Z q_{\mathfrak{t}} \ge t_-, \\ |\omega_j| &\le \theta_N, \qquad |e_j + \omega_j| \le \theta_N \qquad (j = 1,2). \end{aligned} \tag{17.34}$$

The last inequality uses the disjoint subsets $\mathcal{F}_j$ and $\mathcal{J}_j$ of the current stage entrance slots. It does not include ancestor slots or fixed separation constants.

Writing $T'_j = T_j/q_{d_{j,1}d_{j,2}}$, the post formal factor product and the nominal raw residual scale are

$$T'_j \prod_{i\in\mathcal{I}_j\setminus\mathcal{J}_j} P_i = Z^{\alpha_j+e_j-\widetilde{r}_j}, \qquad N_{\mathrm{post},j} = Z^{\alpha_j-\widetilde{r}_j}. \tag{17.35}$$

Define $P_{C,\mathfrak{t},\mathbf{J}}(h')$ and $P_{D,\mathfrak{t},\mathbf{J}}(h')$ to be their residual convolutions multiplied by $N_{\mathrm{post},j}^{-1/2}$. Relative to the nominal pre-normalizer $Z^{-\alpha_j/2}$, the exact extraction coefficient on side j is $Z^{-\widetilde{r}_j/2}$, apart from the amplitudes of slots with fixed primes, retained extracted row scalars, and already separated outer factors. These raw residual products retain the formal signed rectangles and their possibly negative formal logarithmic lengths. Their scales will be clipped only in the nonexceptional branch, before invoking a moment lemma.

Definition 17.3 (Allocated coefficients and normalizations). The inputs are the uncentered products and centered comparisons selected in Section 17.2; the transformations are those in Sections 17.4, 17.5, and 17.6, followed by the residual separation in Section 17.7. An uncentered input has one rectangle. A centered input has two rectangles with the same profiles W_1, W_2 and equal products of scales. Recall that, on each side, $\mathbf{b}$ records the powers extracted from its two plain variables, and $D_{\mathbf{b}}$ is their allocated difference in (17.17).

The common coefficient remains the one in (17.18): the plain coefficient multiplied by the live slot weights. Its common arithmetic factors act on the whole product $u = \Pi l_1 l_2$: one zero-extended character τ_1, one norm power q_u^{it}, one fixed squarefree coprimality mask, and the squarefree divisibility condition $s \mid u$. The live slots retain their original disjoint underlying prime supports. A side and a rectangle are different indices: a side is one of the two convolutions separated from a squared norm, and each centered side contains both rectangles.

At the Gauss stage the central factor is $Z^{-a_0/2}$ with $a_0 = A - c - w$, as in (17.26); w is the removed column length. The live slot sums are restored before squaring. At the residual stage, the two pre-normalizers are $Z^{-\alpha_j/2}$, where $\alpha_1 = a_0 - c_2$ and $\alpha_2 = a_0 - d_2$; c_2, d_2 are the logarithmic norms of the extracted complete common supports.

For the final $\mathfrak{t}$-allocation on side j, recall that $\widetilde{r}_j$ is the sum of the extracted plain logarithmic norms and the nominal lengths of the slots whose primes have just been fixed, as in (17.34). The raw residual product is normalized by $N_{\mathrm{post},j}^{-1/2}$ with $N_{\mathrm{post},j} = Z^{\alpha_j-\widetilde{r}_j}$. Its formal factor product is $Z^{\alpha_j+e_j-\widetilde{r}_j}$, where e_j is the correction from slots whose primes were fixed earlier in the current stage; these distinct scales are recorded in (17.35). The exact extraction coefficient relative to the pre-normalizer is $Z^{-\widetilde{r}_j/2}$, apart from the retained slot amplitudes at fixed primes, extracted row scalars, and already separated outer factors.

The nonexceptional rectangle norms use the clipped scales in (17.36) and the paired coefficient bound in (17.37); the exceptional signed differences keep their formal scales.

Lemma 17.4 (Common coefficient under complete extraction). *For the inputs, stages, and normalization conventions in the coefficient setup of Definition 17.3, each Gauss polynomial and each amplifier error retains* (17.18)*, multiplied by $G(u,h)$, apart from bounded fixed scalars and row scalars of modulus at most one. All the conclusions also hold for an uncentered input with the single plain coefficient $\prod_{i=1}^{2} W_i(q_{b_i}q_{l_i}/X_i)$.*

After the second transform and the full Möbius factor allocation, each separated residual side has that coefficient form without G and without $s \mid u$. Every surviving slot has the residual product's one whole-product character and its original coefficient ν_i, with no Gauss coefficient. Within a side the plain variables have the same character, coprimality mask, and norm power in both rectangles. The two sides of one squared Gauss norm may have different norm powers; their inducing characters differ by a member of Θ.

If the new inducing character belongs to Θ, fixing the live slot labels leaves the plain coefficient

$$\vartheta(l_1)\vartheta(l_2)\mathbf{1}_{(l_1l_2,\mathfrak{R}_*)=1}q_{l_1l_2}^{it}D_{\mathbf{b}}(l_1,l_2)$$

for one $\vartheta\in\Theta$, one squarefree mask $\mathfrak{R}_*$ of polynomial norm, and one real t. The mask may depend on the fixed row, but is common to both variables and both rectangles. All coefficients are understood with their retained fixed scalars, including zeros, and the exact normalizations in the calculation leading to (17.34)–(17.37).

Proof. The first-transform calculation proved the Gauss coefficient form; the local p-power calculation proved it for each amplifier error. The complete-support identity and the full Möbius allocation above proved the residual coefficient form, preserving one character and norm power on each whole residual product. Thus "common" concerns one separated side; the displayed character ratio is the relation between the two sides. When the inducing character belongs to Θ, it equals some $\vartheta\in\Theta$ on units. Its redundant zeros and the fixed coprimality restrictions combine into one $\mathfrak{R}_*$ of polynomial norm. For each fixed Π, multiplicativity factors its value, mask, and norm power on Πl_1l_2 into a scalar in Π times the three factors stated in the conclusion. This proves the exceptional assertion. □

The factor assignments are divisor-bounded for fixed N; their total is bounded using $C_N^{\omega(\mathfrak{t})}\ll_{N,a}q_{\mathfrak{t}}^a$ for any fixed $a>0$, with the exponent applied to the whole divisor. The $O(\log Z)$ divisor dyads in the bounded column range use their existing logarithmic share. These discrete losses are separate from the numerical θ_N terms.

After extracting $Z^{g_2-t_2}$, the other common row scalar is

$$v_{\mathrm{part},\lambda}(h')=\omega_\lambda(q_{G_cV_{\mathrm{id}}}q_{h'}/T_\lambda)\mathbf{1}_{\mathrm{part}}(h')\frac{F(D_2,E_2;G_cV_{\mathrm{id}}h')}{Z^{g_2-t_2}}\zeta(h'),\qquad|\zeta(h')|\le1,$$

where ζ contains the separated row phases for this dyad and these Fourier parameters. It is zero off the retained partition and enclosing ball, and $|v_{\mathrm{part},\lambda}|\le1$. The subsequent estimates are made dyad by dyad and then summed with the already allowed $O(\log Z)$ mass.

Split the already factorized row sum according to membership of the inducing character in Θ. On rows outside Θ, Cauchy–Schwarz yields two nonnegative residual norms; on rows in Θ, take a pointwise absolute product. Only at these steps may the absolute values of the common row weight, partition scalar, or extracted row scalars be bounded by one and the rows enlarged to the common m'_{act} ball. This can admit extra exceptional rows that violate a former unit restriction, but the character-only count below includes them. Both sides use the same eligibility class and compare with (17.33).

17.8 Nonexceptional rows and the affine length condition

For rows whose inducing characters lie outside Θ, the residual products are controlled by the moment estimate at the smaller width in (17.32). This requires two steps: converting each rectangle to nonnegative lengths and, when needed, removing prime factors until the affine length condition holds.

Clipping the plain scales. Consider first rows for which the inducing character of the residual product is outside Θ. The two sides have the same eligibility condition because their characters differ by a member of Θ. On such rows, first bound a centered residual norm by the sum of the norms of its two rectangles; for an uncentered residual product there is one rectangle.

For one fixed side j and rectangle ϱ, write its pre and post formal plain scales as $S_{\varrho,i}$ and $S'_{\varrho,i}=S_{\varrho,i}/q_{d_{j,i}}$. Choose a fixed upper support endpoint B_i for each current plain profile type before the current row, divisor, and live labels. Omit a rectangle only when $S'_{\varrho,i}B_i<1$ for some i; then that plain factor is zero on every nonzero integral ideal. Retain equality and every other profile, mask, row-scalar, or arithmetic zero. This test is independent of the row and live labels.

For a retained rectangle put $x_i = \log_Z S_{\varrho,i}$ and $y_i = x_i - a_{j,i}$. The fixed endpoint test and $a_{j,i} \geq 0$, with H_N enlarged to contain the two positive parts of the upper endpoint logs, give the exact clipped identities

$$\begin{aligned}
\pi_{0,j,\varrho} &= \sum_{i=1}^{2} (-x_i)_+, \qquad \pi_{j,\varrho} = \sum_{i=1}^{2} (-y_i)_+, \qquad 0 \leq \pi_{0,j,\varrho} \leq \pi_{j,\varrho} \leq \theta_N, \\
r_{\mathrm{clip},j,\varrho} &= \sum_{i \in \mathcal{J}_j} z_i + \sum_{i=1}^{2} \big((x_i)_+ - (y_i)_+\big) = \widetilde{r}_j - (\pi_{j,\varrho} - \pi_{0,j,\varrho}) \geq 0, \\
A_{\mathrm{clip},j,\varrho} &= \sum_{i=1}^{2} (y_i)_+ + \sum_{i \in \mathcal{I}_j \setminus \mathcal{J}_j} z_i = \alpha_j + e_j - \widetilde{r}_j + \pi_{j,\varrho} \\
&= \alpha_j + e_j + \pi_{0,j,\varrho} - r_{\mathrm{clip},j,\varrho}.
\end{aligned} \tag{17.36}$$

The length identity in (17.36) shows that the clipped total decreases by $r_{\mathrm{clip},j,\varrho}$ from its own pre-clipped total $\alpha_j + e_j + \pi_{0,j,\varrho}$. Different rectangles can have different clipped totals.

The profile bounds survive this change of scale: a retained subunit scale lies in the fixed interval $[1/B_i, 1]$, so its dilation to scale one preserves finite seminorm bounds. The row character, common mask, and surviving slot weights are unchanged.

To convert the norm back to the pre-normalizer $Z^{-\alpha_j/2}$, the exact coefficient of this standard clipped product is

$$Z^{(A_{\mathrm{clip},j,\varrho} - \alpha_j)/2} = Z^{(-r_{\mathrm{clip},j,\varrho} + e_j + \pi_{0,j,\varrho})/2}.$$

For a fixed pair of retained rectangles, suppress their rectangle indices. The paired divisor count and these coefficients are bounded by (17.34) as follows:

$$\begin{aligned}
t_- - \frac{r_{\mathrm{clip},1} + r_{\mathrm{clip},2}}{2} + \frac{e_1 + \pi_{0,1} + e_2 + \pi_{0,2}}{2} &\leq \frac{(e_1 + \omega_1) + \pi_1 + (e_2 + \omega_2) + \pi_2}{2} \\
&\leq 2\theta_N.
\end{aligned} \tag{17.37}$$

There are at most two rectangles per side, hence at most four such pairs; their triangle factor is fixed. After the triangle inequality, the two nonnegative rectangle norms have their own clipped reductions; the signed difference is never assigned a common clipped reduction.

The affine length condition for a residual product. On the c_2 side, before the last coprimality extraction, the nominal total length is $A' = \alpha_1 = A - c - w - c_2$. For the positive-slot parameters,

$$\begin{aligned}
(A' - M') - (A - M) &= g + w - d - c_2 - (K_0 - K) - w_o + g_2 - t_2 \\
&\leq 6(w + \ell) + \delta_{\mathrm{fr},1}.
\end{aligned} \tag{17.38}$$

To prove the inequality, first use $g_2 - t_2 \leq c_2$. In the amplified main norm, $w = w_o = 0$ and

$$g = J_+ + 2\sigma \leq d + (K_0 - K) + 6\ell + \delta_{\mathrm{fr},1}.$$

For an error, $g \leq d + (K_0 - K) + \sigma + \delta_{\mathrm{fr},1}$, so the excess is at most $w - w_o + \sigma + \delta_{\mathrm{fr},1}$. These inequalities use $d + K_0 - K \geq -\delta_{\mathrm{fr},1}$ and the 1-Lipschitz property of the positive part. The three choices in (17.28), together with $\ell_p \geq \sigma/6$, give $w - w_o + \sigma \leq 6w$. This proves (17.38).

For each retained clipped rectangle, if no slot survives, apply the completed unrestricted zero-slot assertion at the declared width M'_{act}. If slots survive, first apply the algebraic mask deletion in (17.4)–(17.6). It expresses this standard clipped product as products without extra masks and only decreases its nonnegative affine expression. For each actual product, after both frequency enclosures, the $\mathfrak{t}$-allocation, and mask deletion, let $A_{\mathrm{act}}, z_{\mathrm{act}}$ be its declared total and slot lengths and put

$$F_{\mathrm{act}} = \big(A_{\mathrm{act}} - M'_{\mathrm{act}} + (6\kappa - 1) z_{\mathrm{act}}\big)_+.$$

The parent satisfies (17.3). Before this mask deletion, let $z' \le z$ be the surviving slot length of the clipped rectangle. The affine estimate (17.38) and the clipping identities (17.36), together with $M'_{\mathrm{act}} \ge M'$, give

$$\begin{aligned} A_{\mathrm{clip},j,\varrho} - M'_{\mathrm{act}} + (6\kappa-1)z' &\le \alpha_j - M' + (6\kappa-1)z + e_j + \pi_{j,\varrho} - \widetilde{r}_j - (M'_{\mathrm{act}} - M') \\ &\le 6(w+\ell) + \delta_{\mathrm{fr},1} + e_j + \pi_{j,\varrho}. \end{aligned}$$

The same calculation holds on the other side with c_2, d_2 exchanged. Taking the positive part and then deleting the fixed mask therefore gives

$$F_{\mathrm{act}} \le 6(w+\ell) + \delta_{\mathrm{fr},1} + (e_j + \pi_{j,\varrho})_+ \le 6(w+\ell) + \delta_{\mathrm{fr},1} + 2\theta_N.$$

Here $z_{\mathrm{act}} \le z' \le z$. All these exponent comparisons use the fixed list of aggregate lengths

$$A, z, c, d, p, R, E, s_0, K, w, w_o, g, \ell, c_2, d_2, p_2, g_2, t_2, V.$$

Their affine coefficients and positive-part Lipschitz constants are numerical and independent of N. The displayed frequency, raw normalization, and rectangle-clipping bounds are therefore included in $C_*\xi$, for a fixed $C_* \ge 1$ independent of N, once the aggregate threshold has been imposed. The existence of this fixed error bound uses the single $\theta_N = H_N/\log Z$ before comparing lengths, not N separate copies of ξ. We enlarge C_* below to cover the fixed number of operations in a stage.

Removing slots and invoking induction. If $F_{\mathrm{act}} > 0$, remove whole slots from this product until the remaining product satisfies (17.3) or until no slot remains. Removal here means applying the pointwise bound (17.12) to the entire selected prime polynomial; no prime label is fixed. A removed slot of length d decreases $A_{\mathrm{act}} + (6\kappa-1)z_{\mathrm{act}}$ by $6\kappa d$ and contributes κd to the squared Z-exponent, with the fixed seminorm and height factor retained separately. Since each slot has length at most η, order the positive live lengths and take the first prefix reaching $F_{\mathrm{act}}/(6\kappa)$. Its preceding prefix is smaller than that threshold and its final slot has length at most η. If no prefix reaches the threshold, remove all slots, whose total is smaller. Thus exactly one slot can cause an overshoot, and the total removed length d_z satisfies

$$\begin{aligned} d_z &\le \min\left\{ z_{\mathrm{act}}, \frac{F_{\mathrm{act}}}{6\kappa} + \eta \right\}, \\ \kappa d_z &\le \frac{F_{\mathrm{act}}}{6} + \kappa\eta \le \Delta_{\mathrm{child}} + \frac{\delta_{\mathrm{fr},1}}{6} + \frac{\theta_N}{3} + \eta. \end{aligned} \tag{17.39}$$

If the slots disappear before the affine boundary is reached, the remaining plain lengths may be arbitrary; this is exactly why the completed smaller-width zero-slot assertion includes all lengths. Otherwise apply the positive-slot induction to the remaining product. Its slots still have their original coefficient class, and its inducing rows remain outside Θ. Cauchy–Schwarz averages the errors of the two separately clipped residual products, so their possibly different rectangles and slot sets do not double η. Combining (17.39) with the paired coefficient bound in (17.37) and $M'_{\mathrm{act}} - M' \le \delta_{\mathrm{fr},2}$, the additional scale exponent beyond the smaller-width error bound is at most

$$\delta_{\mathrm{fr},2} + \frac{\delta_{\mathrm{fr},1}}{6} + \eta + \frac{7\theta_N}{3}. \tag{17.40}$$

If no slot survives, the same bound holds without a loss from slot removal. The numerical θ_N terms and the two frequency corrections give the single $O(\xi+\eta)$ edge loss in the ϵ_G reserve inherited from (17.25) when using (17.33); the existing local small-power and fixed analytic factors remain separate. The pointwise estimate is applied once to the entire removed product; its internal small-power shares may depend on N. This one greedy operation occurs only after all boundary defects of the actual residual product have been included in F_{act}. The argument for the d_2 side is the same with c_2, d_2 exchanged.

The bounds for all transformed rows whose inducing characters lie outside Θ use only smaller widths. For the remaining rows, the next count and volume bound handle the uncentered range; the centered range uses the subtraction retained in the coefficient lemma.

17.9 Rows with inducing characters in Θ

For these rows, the smaller-width moment is replaced by a count and a direct estimate of the residual products. The count and volume bound suffice for the uncentered range. For the longer centered inputs, Lemma 17.6 supplies the additional saving from equal products of scales.

For this subsection, call a transformed row *exceptional* when its inducing character belongs to Θ. By the coefficient lemma, either both separated sides are exceptional or neither is. Every prime in the existing full displayed moving support is a common column zero, as is every extra common coprimality restriction. This is true initially, remains true for $\mathfrak{e}, \mathfrak{r}$ in the first transform, and remains true for an amplifier prime. For a nonzero second common-support allocation, the factors $\tau_1(D_2), \tau_1(E_2)$ and the old common masks therefore force its complete radical to be disjoint from all those supports. This conclusion is made before replacing any fixed scalar by an upper bound. The artificial residual extension keeps D_2, E_2 and those masks fixed and cannot revive an impossible allocation.

Counting exceptional rows. First separate the fixed-prime row factors from the moving good primes. As in (3.9), write a nonzero row as $h' = \varepsilon h_S h_{\mathrm{good}}$, with ε a unit and the two other factors supported on S and its complement. For a primary element n prime to S, reciprocity gives, with every zero retained,

$$\chi_n(h') = \chi_n(\varepsilon h_S)\mathcal{R}(n, h_{\mathrm{good}}) \prod_{p \mid h_{\mathrm{good}}} \chi_p(n)^{v_p(h')}.$$

By Lemma 3.2, the first factor belongs to a finite family of ray characters supported on S after the unit and the S-valuations modulo six are fixed. The second belongs to the fixed reciprocity family after fixing the good ray sector. The moving good-prime factors remain separate from this fixed family. The family supplied by the first factor has finitely many choices and may contain characters outside Θ. Every remaining displayed good-prime factor has its actual local sextic exponent, with nontrivial exponents ramified at that prime.

Let v_1 be the radical length of the primes counted in V whose equal multiplicity is one, and put $f = 2v_1$. At such a prime the fixed factor $G_c V_{\mathrm{id}}$ in the row has valuation two. There is no existing moving character at that prime, and every character of Θ is unramified there. Sextic reciprocity therefore shows that exceptional induction requires

$$v_p(h') + 2 \equiv 0 \pmod 6, \qquad \text{that is,} \qquad v_p(h') \equiv 4 \pmod 6.$$

More generally, at every prime outside S, exceptional induction prescribes a single residue class modulo six for $v_p(h')$, determined by the fixed moving character and $G_c V_{\mathrm{id}}$. Outside their supports that residue is zero. For each of the finitely many choices at S and of unit and fixed-ray data, the ideal of h' consequently has a unique form

$$(h') = \mathfrak{h}_0 \mathfrak{v}^6,$$

where $\mathfrak{h}_0$ is fixed and sixth-power-free. The displayed valuation-four conditions give $q_{\mathfrak{h}_0} \geq Z^{4v_1} = Z^{2f}$. Ideal counting up to $q_{h'} \ll Z^{m'_{\mathrm{act}}}$ now gives the stronger bound $O(Z^{(m'_{\mathrm{act}}-2f)/6+\epsilon_1})$ on every nonempty range, with the usual bounded-scale convention. We use only the weaker bound

$$\#\{h' : q_{h'} \ll Z^{m'_{\mathrm{act}}},\ \text{exceptional}\} \ll Z^{(m'_{\mathrm{act}}-f)/6+\epsilon_1} \leq Z^{(m'-f)/6+\delta_{\mathrm{fr},2}/6+\epsilon_1}. \tag{17.41}$$

If the exponent would describe a scale below one, the forced ideal $\mathfrak{h}_0$ makes the range empty except at the same bounded-scale boundary. This argument includes the principal character and every other member of Θ. Additional conditions at old moving primes can only reduce the count. It also covers rows admitted when the partition scalar was bounded after positivity. At a unit prime, such an added row can be exceptional even though it was absent from the original partition; no extra forcing saving from the unit primes whose total radical length is t_2 is used.

The uncentered volume bound. On exceptional rows take the absolute product for each already allocated pair of residual products and multiply by (17.41). Let $c_{\mathrm{t},\mathbf{J}}(h')$ denote its central extraction coefficient and bounded extracted scalars. For an allocation with raw reductions $\widetilde{r}_1, \widetilde{r}_2$, $|c_{\mathrm{t},\mathbf{J}}(h')| \ll_N Z^{-(\widetilde{r}_1+\widetilde{r}_2)/2}$ after taking absolute values. Using the absolute-volume bound, the unnormalized plain difference on side j is $O(T_j')$ at every positive formal scale, including subunit scales, and the unnormalized live slots contribute $O_N(\prod_{i\in\mathcal{I}_j\setminus\mathcal{J}_j} P_i)$. The normalization identities in (17.35) therefore give the raw normalized volume

$$|P_{j,\mathrm{t},\mathbf{J}}(h')| \ll_N N_{\mathrm{post},j}^{-1/2} T_j' \prod_{i\in\mathcal{I}_j\setminus\mathcal{J}_j} P_i = Z^{(\alpha_j-\widetilde{r}_j)/2+e_j}$$

up to the fixed seminorm and height factors, where j denotes its C or D side. Thus the exponent for an allocated pair and one lower-endpoint divisor dyad, including its count, is

$$a_0 - b_2 + e_1 + e_2 + t_- - \widetilde{r}_1 - \widetilde{r}_2 \le a_0 - b_2 + e_1 + (e_2+\omega_2) \le a_0 - b_2 + 2\theta_N.$$

Here (17.34) gives $t_- - \widetilde{r}_2 \le \omega_2$, and $\widetilde{r}_1 \ge 0$. Consequently

$$\sum_{\mathrm{t},\mathbf{J}} |c_{\mathrm{t},\mathbf{J}}(h')|\,|P_{C,\mathrm{t},\mathbf{J}}(h')P_{D,\mathrm{t},\mathbf{J}}(h')| \ll Z^{a_0-b_2+2\theta_N+\epsilon_1}.$$

The displayed sum is over a fixed lower-endpoint divisor dyad; its allocations have already been included in ϵ_1. The product is independent only after the Möbius allocation used here. Subtracting the required exponent for the residual products from the reference volume and the nominal exceptional count gives the exact algebraic identity

$$\frac{m'-f}{6} + a_0 - b_2 - (M' + \Delta_{\mathrm{child}}) = A - \frac{5}{6}M - F_1 - F_2, \tag{17.42}$$

where

$$\begin{aligned} F_1 &= c/6 + 5\{d + (K_0 - K)\}/6 + w/3 + \widetilde{q}/6 + w_o + \mathcal{B}_c - 5g/6 + \ell, \\ F_2 &= 2b_2 - \tfrac{5}{6}g_2 - p_2 + t_2 + V/6 + f/6. \end{aligned} \tag{17.43}$$

The actual excess is bounded by the nominal expression in (17.42) plus $\delta_{\mathrm{fr},2}/6 + 2\theta_N + \epsilon_1$. For an explicit check, the left side of (17.42) first equals

$$a_0 - 2b_2 + p_2 - w - \mathcal{B}_c - \ell - \tfrac{5}{6}m' - q' - f/6.$$

Substitution of (17.32) and $K = 2A - c - d + R + E - m - (K_0 - K)$ gives (17.42)–(17.43).

The contributions from complete common supports satisfy the following lower bounds:

$$F_1 \ge \tfrac{2}{3}(c+w) - 3\sigma - \tfrac{5}{6}\delta_{\mathrm{fr},1}, \qquad F_2 \ge \tfrac{2}{3}b_2. \tag{17.44}$$

To prove the first, put $D_0 = d + K_0 - K$, initially set $g = J_+$ and $\ell = 0$, and recall $J = D_0 - c - 2w + w_o$. When $J \ge 0$, direct simplification gives

$$F_1 = c + 2w + \widetilde{q}/6 + w_o/6 + \mathcal{B}_c \ge c + 2w.$$

When $J < 0$, use $\widetilde{q} \ge R$ and

$$\mathcal{B}_c + \widetilde{q}/6 \ge (3c - 5D_0)_+/6 - \tfrac{5}{6}\delta_{\mathrm{fr},1}.$$

Indeed, the left side is at least $\{(3c - 5d - R)_+ + R\}/6$, which is at least $(3c-5d)_+/6$, while $D_0 \ge d - \delta_{\mathrm{fr},1}$. The positive part is 1-Lipschitz, giving the displayed correction. It follows that

$$F_1 \ge \frac{c + 5D_0 + (3c - 5D_0)_+}{6} + \frac{w}{3} - \frac{5}{6}\delta_{\mathrm{fr},1} \ge \frac{2}{3}(c+w) - \frac{w}{3} - \frac{5}{6}\delta_{\mathrm{fr},1}.$$

The actual choices have $g \le J_+ + 2\sigma$ and $\ell \ge 0$. The additional loss from $2(c+w)/3$ is therefore, by (17.29), at most

$$w/3 + 5\sigma/3 + \tfrac{5}{6}\delta_{\mathrm{fr},1} \le 22\sigma/9 + \tfrac{5}{6}\delta_{\mathrm{fr},1} < 3\sigma + \tfrac{5}{6}\delta_{\mathrm{fr},1}.$$

For the second inequality in (17.44), compute prime by prime from the definition of F_2. In units of the prime's logarithmic norm, the contributions are

common case	F_2	b_2
$i = i,\ 6 \nmid i,$ unit	$7i/6$	i
$i = i,\ 6 \nmid i,$ nonunit	$(7i-5)/6 + \frac{1}{3}1_{i=1}$	i
$i = i,\ 6 \mid i,$ either	$7i/6 - 1$	i
$i > j_0,\ 6 \mid j_0,$ unit	$i + j_0/6 - 1$	$(i+j_0)/2.$

The extra $1/3$ in the second line is $f/6$. The first line exceeds $2b_2/3$. In the second line equality holds at $i = 1$, and for $i \ge 2$ the difference is $(3i-5)/6 \ge 0$. In the third line the difference is $(i-2)/2 \ge 0$, because $i \ge 6$. In the last line it is $(4i - j_0 - 6)/6 \ge 0$, because $j_0 \ge 6$ and $i \ge j_0 + 1$. Summing proves the claim.

For the uncentered range $A \le 5M/6$, (17.42)–(17.44) bound the nominal exceptional excess by $3\sigma + 5\delta_{\mathrm{fr},1}/6$. The actual-count and aggregate support corrections recorded above are further $O(\xi)$ terms in the fixed stage error bound. Together with the diagonal estimate and the smaller-width estimates, this proves the uncentered stage at the current width with the reserved terminal loss. The comparisons constructed earlier may consequently use that stage. It remains to estimate the centered exceptional terms when $A > 5M/6$.

Cancellation for the centered products. For one separated side of a fixed exceptional row, fix the live slot labels. Lemma 17.4 then gives the same character, coprimality restriction, and norm power in the two plain variables and in both rectangle terms. The next lemma shows that the coefficient of X^{1+it} in each plain sum is independent of X, so the two product main terms agree at equal products of scales.

Definition 17.5 (Masked sums and rectangle scales). Let $\epsilon_1 > 0$ be the power-loss parameter in (17.6). Let ϑ range over a fixed finite set of finite-order ray characters with conductor primes in S. Let $\mathfrak{R}_*$ be squarefree with $q_{\mathfrak{R}_*} \le Z^{B_*}$ for a fixed B_*, and let W_1, W_2 be smooth profiles on fixed annuli. For $X > 0$ and $t \in \mathbb{R}$, define

$$\mathcal{L}_i(X,t) := \sum_{l} \vartheta(l) 1_{(l,\mathfrak{R}_*)=1} q_l^{it} W_i(q_l/X).$$

Here $I_i(t)$ denotes a fixed measure constant times $\int_0^\infty W_i(y) y^{it}\, dy$.

Let $D_{\mathbf{b}}$ be the difference in (17.17), formed from these profiles W_1, W_2 and scales with $X_1X_2 = Y_1Y_2$, and put

$$U_i = X_i/q_{\mathfrak{b}_i}, \qquad V_i = Y_i/q_{\mathfrak{b}_i}, \qquad T = U_1U_2 = V_1V_2.$$

Lemma 17.6 (Masked rectangle cancellation). *The masked sums $\mathcal{L}_i(X,t)$ of Definition 17.5 satisfy*

$$\mathcal{L}_i(X,t) = c_{\vartheta,\mathfrak{R}_*} X^{1+it} I_i(t) + O\big(Z^{\epsilon_1}(1+|t|)^J\big), \tag{17.45}$$

$$|\mathcal{L}_i(X,t)| \ll X. \tag{17.46}$$

Here J is fixed, the constants use finitely many seminorms, and $c_{\vartheta,\mathfrak{R}_*}$ is independent of X, t. Both estimates are uniform for $X > 0$, for the stated masks, and for the finite character set.

For the rectangle scales in Definition 17.5, if $U_i, V_i \ge Z^r$ for $i = 1, 2$ and some $r \ge 0$, then

$$T^{-1/2} \left| \sum_{l_1,l_2} \vartheta(l_1)\vartheta(l_2) 1_{(l_1l_2,\mathfrak{R}_*)=1} q_{l_1l_2}^{it} D_{\mathbf{b}}(l_1,l_2) \right| \ll Z^{\epsilon_1}(1+|t|)^{2J} T^{1/2} Z^{-r}. \tag{17.47}$$

Without a nonnegative lower length, the same expression is $O(T^{1/2})$.

Proof. The fixed-lattice main term is first computed without the mask, then inclusion–exclusion gives its masked coefficient. That coefficient is common to the two rectangles, so equality of their product scales cancels the product main terms.

The fixed-lattice estimate. First omit the mask. The primary generators representing ideals outside S, with the fixed character weight ϑ, are a finite weighted collection of residue classes in a fixed lattice. Poisson summation on that lattice applies to

$$f_{X,t}(z) = q_z^{it} W_i(q_z/X) = X^{it} f_t(z/\sqrt{X}), \qquad f_t(z) = q_z^{it} W_i(q_z).$$

The Fourier transform is $X^{1+it}\widehat{f_t}(\sqrt{X}\xi)$. Because f_t is supported on a fixed annulus, integration by parts for any fixed $J_0 > 2$ gives

$$|\widehat{f_t}(\xi)| \ll_{J_0} (1+|t|)^{J_0}(1+|\xi|)^{-J_0},$$

using finitely many seminorms of W_i. For $X \geq 1$, the sum over nonzero points of the fixed dual lattice is therefore

$$\ll (1+|t|)^{J_0} X \sum_{\xi \neq 0} (1+\sqrt{X}|\xi|)^{-J_0} \ll (1+|t|)^{J_0}$$

after increasing J_0 if necessary. For $0 < X < 1$, both the lattice sum and its zero-frequency main term are $O(1)$ by absolute counting on the fixed annulus. The zero frequency is $c_\vartheta X^{1+it} I_i(t)$, where c_ϑ is the fixed weighted mean of the residue classes. This proves the unmasked version of (17.45).

Inserting the common mask. Now insert the mask by inclusion–exclusion. First remove from $\mathfrak{R}_*$ its primes in S, since every summation ideal already avoids them. Thus all divisors used below are outside S. With $\mathcal{L}_i^0$ denoting the unmasked sum, multiplicativity gives

$$\mathcal{L}_i(X,t) = \sum_{d \mid \mathfrak{R}_*} \mu(d)\vartheta(d) q_d^{it} \mathcal{L}_i^0(X/q_d, t).$$

The main coefficient is

$$c_{\vartheta,\mathfrak{R}_*} = c_\vartheta \sum_{d \mid \mathfrak{R}_*} \frac{\mu(d)\vartheta(d)}{q_d}.$$

The q_d^{it} from extraction has canceled the q_d^{-it} in the main term at X/q_d. This proves that the coefficient is independent of both X and t. The errors are multiplied by at most the number of divisors $\#\{d : d \mid \mathfrak{R}_*\} \ll_{\epsilon_1, B_*} Z^{\epsilon_1}$. The coefficient itself is also Z^{ϵ_1}-bounded by the polynomial-size Euler-product estimate. This proves (17.45).

For (17.46), take absolute values in the original sum. It is empty below a fixed positive scale; whenever it is nonempty, ideal counting on the fixed annulus gives $O(X)$ points. This is uniform in t and in the mask.

Equal-product cancellation. The masked sum in (17.47) is exactly

$$\mathcal{L}_1(U_1,t)\mathcal{L}_2(U_2,t) - \mathcal{L}_1(V_1,t)\mathcal{L}_2(V_2,t).$$

The two product main terms in (17.45) are both $c_{\vartheta,\mathfrak{R}_*}^2 T^{1+it} I_1(t) I_2(t)$. They cancel. Each cross term with one error is bounded by $Z^{\epsilon_1}(1+|t|)^{2J}$ times one of U_1, U_2, V_1, V_2, and the product of errors has the same bound after reducing the subsidiary power loss. If all four scales are at least Z^r, each is at most TZ^{-r}, and $1 \leq TZ^{-r}$ because $T \geq Z^{2r}$. The difference is therefore $O(Z^{\epsilon_1}(1+|t|)^{2J} TZ^{-r})$. Division by $T^{1/2}$ proves (17.47). When no nonnegative lower length is available, (17.46) bounds each product by $O(T)$, giving the last assertion. □

Returning to the exceptional estimate. Apply this lemma to each already allocated exceptional residual product. The hypotheses on its coefficient and common mask follow from Lemma 17.4;

in particular the main terms cancel for the same row, mask, and whole-product norm power before any absolute value. Initially all four plain lengths are at least L. Before the selected $\mathfrak{t}$-factors are removed, the c_2 side has lower plain length at least $L - c - w - c_2$ and nominal total length $\alpha_1 = a_0 - c_2$: no one plain loses more than the total $c + w + c_2$ extracted by the two complete common supports and the amplifier. The other side has the analogous bounds with d_2. Put

$$r = (L - c - w - \min(c_2, d_2))_+,$$

and call the side with this reference saving the first side, relabeling its associated raw data together. If $\widetilde{r}_1 < r$, each selected plain length satisfies $a_{1,i} \le \widetilde{r}_1$, so all four formal post plain scales have logarithmic length at least $r - a_{1,i} \ge r - \widetilde{r}_1 > 0$. These exact equal-product formal scales therefore satisfy the hypotheses of (17.47). If $\widetilde{r}_1 \ge r$, including equality, or if $r = 0$, use the all-scale absolute-volume fallback in the same lemma. No formal centered scale is clipped in this branch, and an identically zero rectangle remains in the formal difference. The resulting saving is $(r - \widetilde{r}_1)_+$. With absolute volume on the other side, this gives the aggregate bound on each lower-endpoint divisor dyad,

$$\begin{aligned} \sum_{\mathfrak{t},\mathbf{J}} |c_{\mathfrak{t},\mathbf{J}}(h')|\,|P_{C,\mathfrak{t},\mathbf{J}}(h')P_{D,\mathfrak{t},\mathbf{J}}(h')| \\ \ll Z^{a_0 - b_2 - r + 2\theta_N + \epsilon_1}. \end{aligned} \tag{17.48}$$

Here and below a fixed polynomial in the separated heights is understood. To check the bound, the raw coefficients and volumes give the reference exponent $a_0 - b_2 + e_1 + e_2$ and reduction $-\widetilde{r}_1 - \widetilde{r}_2$, while the divisor count gives t_-. The divisor inequality in (17.34) gives

$$\begin{aligned} t_- - \widetilde{r}_1 - \widetilde{r}_2 - (r - \widetilde{r}_1)_+ &= (t_- - \widetilde{r}_2) - \max(r, \widetilde{r}_1) \\ &\le \omega_2 - r. \end{aligned} \tag{17.49}$$

Adding $e_1 + e_2$ leaves at most $e_1 + (e_2 + \omega_2) - r \le 2\theta_N - r$. The divisor-bounded allocations contribute only ϵ_1, proving (17.48). This applies to the artificial terms even when quotients retain $\mathfrak{t}$-primes, because the coefficient lemma preserves the common mask, equal product scales, and one norm power within each rectangle difference.

The live slots are summed only after the finite transform has removed their Gauss coefficients. Their unnormalized absolute sums contribute $O_N(\prod_{i\ \mathrm{live}} P_i)$. Combining this with the nominal raw normalizer gives the full e_j volume factor in (17.35), as included above; it is not replaced by a factor-product normalizer. The Fourier measure was fixed before their labels, so conditioning on those labels changes neither the common mask nor the centered saving. A zero conditional slot scalar is discarded only in the present absolute bound.

Set $v = c + w + \min(c_2, d_2)$. By (17.44), $F_1 + F_2$ is at least $2v/3 - 3\sigma - 5\delta_{\mathrm{fr},1}/6$, since $b_2 \ge \min(c_2, d_2)$. The centered saving in (17.48) now bounds the exceptional deficit, apart from 3σ and the recorded frequency and aggregate support perturbations, by

$$A - \tfrac{5}{6}M - \tfrac{2}{3}v - (L - v)_+ \le (A - M)_+. \tag{17.50}$$

Indeed, for $0 \le v \le L$ the left side is $A - 5M/6 - L + v/3$, which increases with v; for $v \ge L$ it is $A - 5M/6 - 2v/3$, which decreases. Its maximum is attained at $v = L = M/4$ and equals $A - M$. For positive slots, (17.3) implies $A \le M$. For the zero-slot core, (17.10) gives $A - M \le \delta$. Thus the centered exceptional terms require only the terminal loss 3σ, or $\delta + 3\sigma$ in the padded core. The additional $\delta_{\mathrm{fr},1}, \delta_{\mathrm{fr},2}$, and θ_N terms are included in the same $C_*\xi$ stage error bound. Together with (17.31), this completes the centered stage at the current width.

17.10 Completion of the finite induction

The estimates now close the width induction. We first check the strict decrease and the error per stage, then choose the parameters so that one mesh works for every fixed slot count.

Strict decrease and induction order. Every nonterminal call is to a residual product with nominal width M' in (17.32), at least σ below its parent. The declared row width is M'_{act}, and all nonexceptional moment-input lengths are taken after the specified support and clipping operations. Exceptional raw scales remain formal and are not clipped. The two frequency enclosures, the aggregate support errors, and these clippings alter the width comparison by at most $C_*\xi$. Choose this below $\sigma/2$. Every nonempty residual product then has nonnegative width and leaves its band of length $\sigma/4$. An empty nonzero-frequency range is discarded; a nonempty formal range just below scale one has already been included by its enclosing length and clipping error.

At each smaller width, (17.4)–(17.6) delete the fixed extra masks before the positive-slot estimate without extra masks. The unrestricted zero-slot assertion follows once from the padded core by reflecting at most its two plains. A centered comparison calls only the earlier uncentered stage in its band, using its strict margin. Each Gauss norm has at most one amplification, whose errors go directly to the second transform. The bound in (17.39) is invoked once for each actual residual product without an extra mask, after all its boundary defects have been included in F_{act}. Thus there is no same-band cycle. The integer D defined above bounds the strict calls by $D-2$, because each drops the width by at least $\sigma/2$ from an initial width at most $M_{\max}$.

Aggregate support errors. The constant C_* and the mesh are independent of the fixed slot count because all support errors enter through one aggregate quantity. For any slot subset, (17.19) bounds its entire logarithmic error by $\theta_N = H_N/\log Z$. Each extracted plain power is measured exactly, and each of the two possible plain clippings in one nonexceptional rectangle, after triangle inequality, has one fixed endpoint error. The exceptional calculation uses raw reductions and the disjoint-subset bounds without clipping. The first and second frequency enclosures use only their one aggregate outer scale and one common row dyad. All the local exponent comparisons are affine or positive parts of affine expressions in the fixed list of aggregate lengths displayed above. Their numerical Lipschitz constants do not depend on N; the local F_2 inequalities and the radical counts use exact prime norms. In particular the change in the positive-slot affine expression is at most

$$|\Delta M| + |\Delta A| + 5|\Delta z|, \tag{17.51}$$

because $0 \le 6\kappa - 1 \le 5$. The total change of z is aggregated before this inequality is used. These observations give one numerical C_*, enlarged for the fixed number of operations per stage, that covers all frequency, normalization, and boundary errors by $C_*\xi$. The single greedy prefix contributes at most η, not one η per removed slot.

Analytic orders and discrete counts. The analytic separations preserve the exponent of Z when their derivative orders increase. On a fixed full-product logarithmic box a radial kernel is $\widehat{\Phi}_i(R_{\mathrm{sc}}\exp(L(\mathbf{x})))$, where L is a fixed linear form in the row and whole-column log norms. Every derivative is a fixed combination of Euler derivatives of $\widehat{\Phi}_i$. Lemma 3.12 bounds these uniformly for all $R_{\mathrm{sc}} > 0$, with any prescribed radial decay and derivative orders.

The inverse roots and Fourier integration have fixed analytic bounds as well. After its displayed central power has been extracted, an inverse root is $(q_u/Z^{a_0})^{-1/2}$ on the fixed box; its derivatives introduce only fixed constants depending on H_N and their order. Lemma 3.6 integrates the full weighted Fourier measure over all heights. A required height degree J, which may depend on N, raises only an input seminorm order and a fixed constant.

Reflection and the prime estimate share this separation between analytic orders and scale exponents. Reflection acts on at most two plains with fixed gamma shape; larger derivative orders raise finite seminorm and height orders, while the upper-annulus subshare $\xi/2$ remains fixed. The fixed polynomial height factor in (17.12) is integrated by these weighted Fourier measures at its required finite order.

The discrete label counts are also used only once. After extracting the displayed exponent for a family of common supports, divisors, or allocations, divide its absolute weighted counting

measure by that total mass before applying a separated tuple seminorm. This leaves a measure of mass at most one; its already extracted exponent is not counted again in the smooth norm. Each Fourier measure is common to the live labels by the whole-product construction.

Parameters independent of the slot count. Choose the parameters in the following order. First, from the bounded real ranges and ϵ, choose $\rho, \sigma, \delta > 0$, with δ small compared to ρ, σ, so that

$$T_{\mathrm{term}} := \rho + \delta + \rho/6 + 3\sigma + 5\sigma/3 < \epsilon/4.$$

This bounds a terminal width-floor, diagonal, or exceptional loss; the displayed frequency and numerical support corrections belong instead to the per-stage ξ error bound. With D and $C_* \geq 1$ as above, choose, still before specifying N or its fixed profiles,

$$\begin{aligned} \xi &< \min\left\{\frac{\delta}{2}, \frac{\rho}{30}, \frac{\sigma}{4C_*}, \frac{\epsilon}{16C_*D}\right\}, \\ \eta &< \min\left\{\frac{\sigma}{6}, \frac{\epsilon}{16C_*D}\right\}, \qquad \epsilon_0 < \frac{\epsilon}{16C_*D}. \end{aligned} \tag{17.52}$$

These choices make the strict drop at least $\sigma/2$, preserve the comparison margins and padded core, and are uniform for $\kappa \in [3/4, 1]$.

Parameters for a fixed slot system. Now fix any Z-independent N, arithmetic data, and profile windows. Form H_N and the finite full support boxes through depth D. Their fixed factors $\exp(O(H_N))$ belong to the constants. After imposing an eventual threshold $\log Z \geq CH_N/\xi$, the threshold in (17.9), and (17.20), all occurring column, radical, and mask norms have bounded logarithmic ranges independent of moving labels.

Within those common ranges, allocate the small-power losses once per stage. Choose every arbitrarily small power estimate with local shares whose sum in a stage is at most ϵ_0. These shares may depend on N. For example, the two divisor masses and the masses of at most N slots at fixed primes in mask deletion may each use a share $\epsilon_0/(10(N+2))$.

For the prime estimate, the principal squared exponent is exactly κz, and the contour displacement contributes $2ez$, with bounded total z, not $N\eta$. Choose e for its assigned share; the fixed character expansions and powers of $\log Z$ are absorbed after an N-dependent threshold. The same reasoning applies as individual z_i approach zero, since $P_i \geq 1$.

The discrete counts use bounds for the whole allocation. For divisor allocations use the global bounds $\tau_{N+2}(v)^C \ll_{N,C,a} q_v^a$ and $C_N^{\omega(v)} \ll_{N,a} q_v^a$ with a sufficiently small $a > 0$, not a fixed loss at every prime. Rankin's fixed-radical bound is treated with the same local shares. The one normalized pool average has only its single density loss in a stage.

Propagation of the error. The induction is simultaneous for every surviving subset of these original N slots, with the same mesh. To make this precise, for $0 \leq d \leq D-2$, let

$$\mathcal{E}_d := T_{\mathrm{term}} + (d+1)C_*(\eta + \xi + \epsilon_0)$$

be the permitted error when at most d strict calls remain. A terminal group, including its bounded reflection or comparison preprocessing, has error at most $\mathcal{E}_0$. A strict edge and its fixed number of same-band operations use the smaller-width error bound $\mathcal{E}_{d-1}$ plus at most $C_*(\eta+\xi+\epsilon_0)$, giving $\mathcal{E}_d$. The exponent ϵ_G in (17.23) denotes this appropriate envelope and the current local shares. The induction uses the same mesh and these envelopes at every stage. Triangle inequalities take the maximum error up to the already counted mass, and Cauchy–Schwarz averages the errors of the two residual products. Consequently one terminal loss occurs along a branch, not at every ancestor. The largest path error is at most $T_{\mathrm{term}} + C_*D(\eta+\xi+\epsilon_0) < \epsilon$ by (17.52).

Finite orders and completion. Finally choose the required finite kernel, reflection, prime, and terminal seminorm and polynomial-height orders backwards through these D stages. The full weighted Fourier estimates and uniform Euler-kernel estimates above make every such choice finite. Choose one final lower threshold for the aggregate support inequalities, pool separation

and density, the raw frequency tails, and all fixed logarithmic losses. The orders, constants, and threshold may grow with N and the fixed data, but η, ξ, D, C_* in the exponent comparison do not.

Each new extra mask is supported on fixed columns or a fixed amplifier prime, and its logarithmic norm increases by a bounded total length at a stage. It is fixed within every transformed row sum. The only row-dependent zeros there are the zeros of the row character; the declared moving zeros remain in the fixed twist and remain counted. Over depth D the masks remain of polynomial norm. On an exceptional row, the redundant part of the full moving union and the row radical, together with these extra masks, forms the common polynomial-size $\mathfrak{R}_*$; the lattice cancellation is uniform for it. The fixed-numerator ray lemma controls the finite choices at S, without absorbing a moving good prime into a fixed modulus.

When this estimate and the inverse moment are applied together, all internal orders in both arguments, including the reflection-kernel orders, are fixed first. The later external Fourier-tail order in Lemma 3.8 lies outside this internal propagation: it raises only the external input seminorm and does not change a previously fixed internal height order. This proves the asserted finite-order uniformity. The finite induction proves the assertion without an extra mask, and the initial mask deletion proves Lemma 17.2 in its full stated form.

Sixth-power-free rows. For the later row-count application, $q = 0$ and the row u is sixth-power-free. If a prime outside S has valuation $1, \dots, 5$ in u, its local character has order $6/\gcd(6, v_p(u)) > 1$, so the inducing character cannot belong to Θ. The exceptional rows in this family are therefore supported on S; sixth-power-freeness makes this a finite set, up to the finite unit group. With abstract moving twists, additional exceptional rows can arise by cancellation. They were included in (17.41) and the fixed-character volume argument.

18 Prime amplitudes and refined row counts

The prime slots in (11.5) supply an additional factor in the witness moments. We first bound their amplitudes using the shared zero detector of Section 7, then select whole slots and combine the inverse and plain witness counts. Throughout, the detector uses the same fixed Part II arithmetic data as the slots in (11.5). The main and error factors below are defined on the retained region of the dynamic decomposition in (15.12); the individual local quotients are restricted to that region. Every retained external coordinate obeys the single height allocation in (9.13).

18.1 Prime amplitudes

For a main prime slot of scale $P_i = Z^{\ell_i}$, write its central factor as

$$Q_i(u; z) = P_i^{-1/2} \sum_{p \in 1_T} \overline{\chi_p(u)} W_i(q_p/P_i)(q_p/P_i)^{z-1}.$$

The main part $\mathcal{Q}_i$ in Proposition 15.1 is

$$\mathcal{Q}_i(u; z) = -\sum_{p \in 1_T} \overline{\chi_p(u)} q_p^{z-1} W_i(q_p/P_i) = -P_i^{z-1/2} Q_i(u; z).$$

Recall from Section 11 that each sum over $p \in 1_T$ is restricted to $\mathcal{P}_i(Z)$, including its exclusion of S, and that the zero extension makes the contribution of a prime dividing u vanish. The real part of z is in a fixed bounded range and its imaginary part is one of the external frequencies restricted by the common height bound.

Lemma 18.1 (Prime bound in a bin). *Let the row u at scale U satisfy the hypotheses of Lemma 7.2 with bin parameter a and detector width e, with the prime-annulus Mellin frequency included in its cumulative height bound. For each main prime factor $Q_i(u; z)$ and every $\epsilon_1 > 0$,*

$$|Q_i(u; z)| \ll_{\mathcal{A}, e, \epsilon_1} U^{\epsilon_1} P_i^{a-1/2+O(e)}.$$

The implied $O(e)$ is uniform for $51/100 \le a \le 1$ and the fixed real ranges. It holds for every main slot in the row.

Proof. The character expansion reduces the prime sum to logarithmic derivatives controlled by the zero-free bin. Mellin inversion then gives the amplitude bound, and removal of the logarithmic weight recovers the prime coefficients.

The Mellin estimate. Expanding $\mathbf{1}_{p\in 1_T} = |T|^{-1}\sum_{\theta\in\widehat{T}}\theta(p)$ gives prime characters in $\mathcal{X}_u$. Lemma 3.15, applied to the buffered disks, bounds each logarithmic derivative on $\Re s = a + 8e$ by $O_{\mathcal{A},e}(\log U)$; the logarithmic derivative of the deleted product has the same bound. Mellin inversion of the corresponding smooth von Mangoldt annulus, shifted only in the retained central range, gives $U^{\epsilon_1}P_i^{a-1/2+O(e)}$ after normalization. The pure twist $(q_p/P_i)^{i\Im z}$ is translated into the logarithmic-derivative argument before the added Mellin frequency is truncated. The joins have bounded real length. On them the logarithmic derivative is $O_{\mathcal{A},e}(\log U)$ inside its allocated buffer, while on the starting line $\Re s = 2$ it is absolutely bounded. The pointwise estimate (3.14), applied to the untwisted transform, controls the joins; (3.13) controls the absolute-line tails. Choosing the external order after their fixed polynomial scale and height bounds gives the asserted estimate.

Prime coefficients. Dividing the annular weight by $\log(P_i y)$ passes from von Mangoldt coefficients to primes. On the fixed annulus,

$$(y\partial_y)^j \frac{1}{\log(P_i y)} = \frac{(-1)^j j!}{\{\log(P_i y)\}^{j+1}},$$

so this preserves every fixed smooth seminorm for sufficiently large P_i, without introducing a power of P_i depending on the derivative order. Bounded P_i are handled by absolute counting. Prime powers contribute $P_i^{o(1)}$ after central normalization, since their number in a norm annulus is $O(P_i^{1/2+o(1)})$. The coprimality restrictions imposed by data of polynomial size and their zero extensions are already included in the logarithmic derivative. This proves the bound. □

Amplitude bins. Choose a fixed bin width $\vartheta > 0$. First reduce e and ϵ_1, after the slot lengths have been fixed, so that the preceding bound is at most $P_i^{\delta/2+\vartheta}$ for all sufficiently large Z. This is possible because $\log U/\log P_i = d/\ell_i$ stays in a fixed bounded range for each fixed mesh. For a main slot with $Q_i \neq 0$, put

$$g_i = \min\left\{\frac{\delta}{2}, \max\left(0, \vartheta\left\lfloor\frac{\log|Q_i|}{\vartheta\log P_i}\right\rfloor\right)\right\}.$$

Put $g_i = 0$ if $Q_i = 0$, and also put $g_i = 0$ for an error slot from (15.11). Then, for a main slot,

$$|Q_i| \le P_i^{g_i+\vartheta}, \qquad g_i > 0 \implies |Q_i| \ge P_i^{g_i}.$$

Slots with $g_i = 0$ retain only the displayed upper bound.

The finitely many possible vectors (g_i), with the possible endpoint value $\delta/2$, partition the rows into amplitude bins. Define their length-weighted mean by

$$q = \frac{\sum_i \ell_i g_i}{\ell}, \qquad 0 \le q \le \delta/2, \qquad \ell = \sum_i \ell_i. \tag{18.1}$$

Here q is the mean prime amplitude. The auxiliary fourth moment uses the same letter for the moving-conductor exponent in its separate setting.

18.2 Selection of prime factors

Recall Δ and κ from (11.1):

$$0 < \Delta \le \frac{1}{24}, \qquad \kappa = 2\beta_* - 1 = \frac{3}{4} + 2\Delta \in (3/4, 5/6], \qquad \alpha := \frac{5}{6}.$$

The parameter κ varies with β_*; α is the fixed slope from sixth-power amplification. Since $\kappa < 1$ and $\beta_* = (1+\kappa)/2$, the positive-slot hypothesis of Lemma 17.2 is satisfied by equality. Equation (11.2) gives $\delta \le \kappa \le \alpha$ for the current Part II bins, including the possible endpoint $\delta = \alpha$.

Selecting whole prime slots. The selection uses the largest amplitudes first; removing a possible fractional slot then gives a collection admissible for the moment bounds. Fix a dynamic and amplitude bin and fix the external Mellin parameters of (15.6). Write z_{phys} for the fixed third Mellin parameter, so the corresponding prime factors are $Q_i(u; z_{\mathrm{phys}})$. All g_i, and hence q, are now fixed within its row sum. At base U, the length of slot i is $w_i = \ell_i/d$, and the total available length is ℓ/d. For a requested length $0 \le z \le \ell/d$, order the positive g_i decreasingly and fill z fractionally in that order. The gained exponent is at least qz: if positive slots suffice to fill z, their initial weighted average is at least the average q of all slots; otherwise retaining them all gives gain $q\ell/d \ge qz$. Removing the one possibly fractional slot loses at most $(\max_i w_i)\delta/2$. Thus a fixed subcollection of whole positive slots of length at most z satisfies

$$\left| \prod_{i \text{ selected}} Q_i \right|^2 \ge U^{2qz - \delta \max_i w_i}.$$

When a strict moment inequality is needed, we first replace z by $z - \nu_0$ for a fixed small $\nu_0 > 0$, or select no slot if $z \le \nu_0$. The resulting loss in this display is at most $2q\nu_0 + \delta \max_i w_i$, except in the stated zero-capacity neighborhood, where an unweighted moment will be used. Both losses can be made smaller than any prescribed positive power.

Putting the selected prime factors into the moment setting. The selected factors belong to the coefficient class required by the moments. After conjugating both witness factors, including their profiles, if necessary, write their row character as $\psi(n) = \nu(n)\overline{\chi_n(u)}$, $\nu \in \Theta$. Relative to this row, a prime in a selected slot has coefficient

$$\overline{\nu(p)}\mathbf{1}_{p \in 1_T} = \frac{1}{|T|} \sum_{\theta \in \widehat{T}} (\overline{\nu}\theta)(p).$$

This is a fixed finite combination of members of Θ; it is independent of the moving row in the fixed row sum. The coefficient identity holds without the condition $\eta(p) = 1$. The factor $(q_p/P_i)^{z_{\mathrm{phys}}-1}$ is part of its smooth profile. Underlying prime supports remain disjoint before masks. The marked moment permits this negative common row orientation.

Lemma 17.2 uses the positive orientation. Conjugating the whole product of both plain witness factors and all selected prime factors preserves its absolute square and changes its common row character to $\overline{\psi(n)} = \overline{\nu(n)}\chi_n(u)$. Both witness profiles and every selected slot profile are conjugated, and each selected slot coefficient becomes $\nu(p)\mathbf{1}_{p \in 1_T}$, again a fixed finite combination of members of Θ. Conjugation retains all zero extensions and underlying prime supports. It also preserves whether the inducing character belongs to Θ, since Θ is a group.

The witness heights may differ from the Mellin heights in (15.6). After those Mellin parameters and the selected indices are fixed, Lemma 3.7 bounds the rowwise witness parameters. Derivatives in those parameters insert only logarithmic profile weights, giving a fixed power of $1 + T_1$, uniform over moving rows and outer labels. Every retained large row induces outside Θ, because it ramifies at a prime outside S. Thus the $z > 0$ family condition of Lemma 17.2 holds.

The admissible total prime lengths. We call the upper bound on the total selected prime length the capacity of a moment estimate. For a plain witness of actual length $m \le 1/2$, the largest capacity that introduces no exponent loss in Lemma 17.2, at effective row width one, is

$$z_P(m) = \frac{1-2m}{6\kappa} = \frac{1-2m}{9/2+12\Delta}. \tag{18.2}$$

Indeed the moment is applied to two copies of the plain witness, so its condition is $2m + 6\kappa z \le 1$. For an inverse witness of length $r < 1$, put $z_M(r) = (1-r)/2$. We use this capacity only where the second strict inequality in Lemma 16.1 also has a fixed margin.

18.3 Row counts from inverse and plain witnesses

The two moments give competing upper bounds for the same amplitude bin. The following proposition balances them for short witnesses and uses sixth-power amplification for long inverse witnesses.

Proposition 18.2 (Row counts from the witnesses). *Assume (11.1), with Δ and κ as there, and recall $\alpha = 5/6$. Let $\mathcal{B}$ be a fixed dynamic bin of Lemma 7.1 and amplitude bin of (18.1), consisting of retained sixth-power-free rows $q_u \asymp U = Z^d$ of (15.6), with $a > 51/100$, $0 < \delta = 2a - 1 \le \alpha$, and mean amplitude $q \in [0, \delta/2]$. Put $x = q/\delta$, so $0 \le x \le 1/2$, and define*

$$D_x = 3 - \frac{17x}{9}, \qquad P_x = \left(2 - \frac{8x}{9}\right)(1-x).$$

Assume the total available prime length, measured at base U, exceeds $7/37$ by a fixed positive amount. For every $\epsilon > 0$, there exist capacity decrements and a slot mesh, depending only on ϵ, Δ and the bounded real ranges, such that for every $t \in [1, 3/2]$

$$\#\mathcal{B} \ll_{\mathcal{A},\epsilon} U^{\max\{R_{\mathrm{short}}(t), L(t)\} + \Delta/4 + \epsilon}(1+T_1)^{A_{\mathcal{A}}},$$

where $A_{\mathcal{A}} < \infty$ is uniform over moving rows and

$$R_{\mathrm{short}}(t) = 1 - \delta + \frac{\delta P_x}{D_x}(3/2 - t), \qquad L(t) = 1 - \delta + (\alpha - \delta)(t-1).$$

The estimate is valid for the rowwise witnesses of Proposition 7.3, with its height condition. If no prime slots are selected, then $t = 1$ instead gives

$$\#\mathcal{B} \ll_{\mathcal{A},\epsilon} U^{1 - 2\delta/3 + \epsilon}(1+T_1)^{A_{\mathcal{A}}},$$

without a prime-supply hypothesis.

Proof. The witness spikes give two possible counts. For short witnesses we use the smaller of the inverse and plain exponents; the amplified inverse estimate handles the remaining long inverse witnesses. At zero inverse capacity, Lemma 16.6 supplies the count; at zero plain capacity, the case of Lemma 17.2 without selected primes supplies it.

Moment bounds divided by witness spikes. Subdivision by the witness presentation and dyadic pair gives $O_{\mathcal{A}}((\log U)^2)$ choices; the Sobolev argument in Section 18.2 handles the remaining rowwise smooth parameters. In one subdivision, denoted again by $\mathcal{B}$, let r, m be the actual lengths from Proposition 7.3. Whenever the selected slots satisfy the corresponding moment hypotheses and have total requested capacity z, their spike and the individual witness spikes imply, after reducing preliminary losses,

$$\begin{aligned} \#\mathcal{B} &\ll U^{1-\delta r - 2qz + \epsilon}(1+T_1)^{A_{\mathcal{A}}}, && \text{from Lemma 16.1,} \\ \#\mathcal{B} &\ll U^{1-2\delta m - 2qz + \epsilon}(1+T_1)^{A_{\mathcal{A}}}, && \text{from Lemma 17.2.} \end{aligned}$$

The second line uses two copies of $S_\psi(U^m;W_S)$, so division by their squared product gives the denominator $|S_\psi(U^m;W_S)|^4$; the denominator for a single copy would be $|S_\psi(U^m;W_S)|^2$. The effective width is one because the twist ν is fixed within the row sum and has no moving conductor radical. Each formula includes the arbitrarily small capacity, rounding, moment, and witness losses.

Inverse witnesses without selected primes. Whenever no inverse slot is selected, the sixth-power amplification of Lemma 16.6 supplies the count. It applies to these rows of (15.6) because their ideal valuations are at most five and their original zero extensions are unchanged. After the present subdivision, U, t, D, D_* are common, and the inverse base profile is

$$W_{\mathrm{base}}(y) = W_1(y)V_{\le}(Dy/D_*).$$

Its fixed logarithmic seminorms are uniformly bounded: a derivative of the second factor is supported where its scaled argument lies in a fixed compact interval. The remaining rowwise parameters are σ in a fixed compact interval and the pure twist $-(\gamma-\nu)$, of absolute value at most $(3I+1)T_1$. The rowwise assertion of Lemma 16.6 therefore applies with that height range. Replacing its factor $(1+(3I+1)T_1)^A$ by $(3I+2)^A(1+T_1)^A$ changes only a fixed constant.

This use is uniform even when r approaches one with U. If $0 \le r \le R_0$, (16.91) with preliminary loss ϵ_0 has exponent

$$\max\left\{1, \frac{1+5(1+c)r}{6}\right\} + (1+r)\epsilon_0 \le e(r) + \frac{5cR_0}{6} + (1+R_0)\epsilon_0, \qquad e(r) = \max\left\{1, \frac{1+5r}{6}\right\}.$$

For a requested loss $\epsilon_m > 0$, take, for example, $c = 3\epsilon_m/(10\max\{R_0,1\})$ and $\epsilon_0 = \epsilon_m/(4(1+R_0))$. These are fixed before U, and the two extra terms are at most $\epsilon_m/2$. Thus division by the inverse witness spike gives

$$\begin{aligned} \#\mathcal{B} &\ll U^{e(r)-\delta r+\epsilon}(1+T_1)^{A_{\mathcal{A}}}, \\ e(r)-\delta r &= \begin{cases} 1-\delta r, & 0 \le r \le 1, \\ 1-\alpha+(\alpha-\delta)r, & r \ge 1. \end{cases} \end{aligned} \tag{18.3}$$

Thus the fixed-parameter estimate of Lemma 16.6 applies uniformly as r approaches one, including the boundary where the marked moment has a shrinking margin $1-r$.

Cases requiring no selected primes. We first dispose of the cases in which a witness already gives the required count without selecting primes. For $r \ge 1$, (18.3) uses effective moment exponent $(1+5r)/6 = 1-\alpha+\alpha r$. Division by the inverse spike gives $1-\alpha+(\alpha-\delta)r$. Because $\delta \le \alpha$ and $r \le t + O(\epsilon)$, this is at most

$$L(t) = 1-\alpha+(\alpha-\delta)t = 1-\delta+(\alpha-\delta)(t-1) \tag{18.4}$$

up to $O(\epsilon)$. This includes $r=1$, where the two branches in (18.3) agree. If $m \ge 1/2$, the case without selected primes of Lemma 17.2 gives count exponent $1-2\delta m \le 1-\delta$. Since $R_{\mathrm{short}}(t) \ge 1-\delta$, this also satisfies the stated bound. We may therefore assume for the remaining argument that

$$r < 1, \qquad m < 1/2.$$

Both capacities $z_M(r)$ and $z_P(m)$ are then positive; when either is too small for the fixed decrement, we will use its unweighted estimate below.

Comparing the positive capacities. For the inverse capacity $z_M(r)$, the resulting ideal exponent is $A_I(r)$ below. The plain exponent is first compared with the expression obtained by replacing $z_P(m)$ by its value at $\Delta = 0$, namely $2(1-2m)/9$. Since the actual m is at least $t - r - O(\epsilon)$, the resulting ideal comparison exponent is

$$\begin{aligned} A_I(r) &= 1-\delta\{x+(1-x)r\}, \\ S_t(r) &= 1-\delta\left\{\frac{4x}{9} + \left(2-\frac{8x}{9}\right)(t-r)\right\}. \end{aligned} \tag{18.5}$$

Both the actual plain exponent $1-2\delta m-2q(1-2m)/(9/2+12\Delta)$ and its baseline version decrease with m: their derivatives are respectively

$$-2\delta+\frac{4q}{9/2+12\Delta}<0, \qquad -2\delta+\frac{8q}{9}<0.$$

The $O(\epsilon)$ replacement of m is therefore legitimate.

The two affine expressions in (18.5) cross at

$$r_*(t)=\frac{(2-8x/9)t-5x/9}{D_x}.$$

Here $D_x\ge 37/18>0$. Directly,

$$r_*(3/2)=1, \qquad r_*(1)=\frac{2-13x/9}{3-17x/9}\ge\frac{23}{37},$$

and

$$t-r_*(t)=\frac{(1-x)t+5x/9}{D_x}.$$

The last expression is increasing in t, equals $1/2$ at $t=3/2$, and at $t=1$ is $(1-4x/9)/(3-17x/9)\ge 1/3$. Thus, throughout the stated ranges,

$$r_*(t)\ge\frac{23}{37}, \qquad \frac{1}{3}\le t-r_*(t)\le\frac{1}{2}, \qquad r_*(t)\le 1.$$

The smaller exponent is supplied by the plain count when $r\le r_*(t)$ and by the inverse count when $r_*(t)\le r<1$, except within the fixed small zero-capacity neighborhoods. On the inverse side $r\ge 23/37$. After decreasing $z_M(r)$ by ν_0,

$$1-r-2z\ge 2\nu_0>0, \qquad 3-2r-8z=4(1-r-2z)+(2r-1)>0.$$

The second inequality has a margin at least $9/37$ before its positive first term. These are precisely the two strict width conditions of Lemma 16.1 at row width one. The inverse capacity is at most $(1-23/37)/2=7/37$. On the plain side, $m\ge 1/3-O(\epsilon)$, so its capacity is at most $2/27+O(\epsilon)$. At $d=h$ the available length is

$$\frac{\ell}{h}=\frac{8}{39}, \qquad \frac{8}{39}-\frac{7}{37}=\frac{23}{1443}>0.$$

The assumed positive supply margin and a sufficiently fine mesh therefore permit every selection just made.

On the plain side, replacing the baseline capacity by the actual capacity in (18.2) raises the count exponent by

$$\begin{aligned}2q(1-2m)\left(\frac{2}{9}-\frac{1}{9/2+12\Delta}\right)&=\frac{24q(1-2m)\Delta}{(9/2)(9/2+12\Delta)}\\&\le\frac{\Delta}{4}+O(\epsilon).\end{aligned} \tag{18.6}$$

The inequality follows from $m\ge 1/3-O(\epsilon)$, $2q\le 1$, and the lower bound four for each denominator. The constants in the $O(\epsilon)$ term are uniform.

The weighted average

$$\begin{aligned}R_{\text{short}}(t)&=\frac{(2-8x/9)A_I(r)+(1-x)S_t(r)}{D_x}\\&=1-\delta+\frac{\delta P_x}{D_x}(3/2-t)\end{aligned} \tag{18.7}$$

is independent of r: the r-coefficients cancel, and the weights sum to D_x. It is the common value at $r_*(t)$. Since S_t is increasing in r and A_I is decreasing, the chosen short count is at most $R_{\mathrm{short}}(t) + \Delta/4 + O(\epsilon)$.

Small capacities and the conclusion. If $r < 1$ but $z_M(r) \le \nu_0$, the same estimate without selected primes, using $e(r) = 1$, gives

$$1 - \delta r \le 1 - \delta + 2\delta\nu_0.$$

If $z_P(m) \le \nu_0$, then $1 - 2m \le 6\kappa\nu_0 \le 6\nu_0$, and its count is at most $1 - \delta + 6\delta\nu_0$. Since $R_{\mathrm{short}}(t) \ge 1 - \delta$, choosing ν_0 within the prescribed ϵ preserves the short bound. Finally, $r = 1/2 - O(\epsilon)$ and $t \ge 1$ force $m \ge 1/2 - O(\epsilon)$, so the same unweighted plain estimate applies. Every selected capacity in this argument is nonnegative. Together with the cases requiring no selected primes, this proves the stated selected count after adding the finitely many subdivisions.

For the final assertion, take $t = 1$ without selecting prime slots. The inverse count follows from (18.3), including $e(r) = 1$ for every $r \le 1$; the plain count follows from the case of Lemma 17.2 without selected primes. The same short calculation with selection gain zero, that is, with $x = 0$ in the two comparison lines, gives $R_{\mathrm{short}}(1) = 1 - 2\delta/3$; the long bound is $L(1) = 1 - \delta$. Neither input has a prime-supply hypothesis, proving the final assertion. □

Remark 18.3 (Unselected inverse witnesses for larger bins). For each fixed $t \in [1, 3/2]$, consider a presentation/dyadic subdivision of a fixed dynamic bin of retained sixth-power-free rows. Each row has an actual inverse witness from Proposition 7.3, satisfying the inverse lower bound in (7.4), with its original zero extension. The bound in (18.3) holds whenever

$$a > 51/100, \qquad 1/50 < \delta = 2a - 1 \le 1,$$

and r lies in the prescribed bounded nonnegative range. On each subdivision the parameters U, t, D, D_* are common. Recall the inverse profile $W_{\mathrm{base}}(y) = W_1(y)V_{\le}(Dy/D_*)$: its fixed logarithmic seminorms are uniformly bounded. The real parameter σ lies in a fixed compact interval, and the pure twist $-(\gamma - \nu)$ has absolute value at most $(3I + 1)T_1$, as required for the rowwise version of Lemma 16.6.

The detector conditions are those of Proposition 7.3 and Lemma 7.2: for the preliminary loss $\epsilon_{\mathrm{pre}} > 0$ used for the witness, one has $0 < e < e_0$ and $(1 + T_1)^{A_{\mathcal{A}}} \le U^{\epsilon_{\mathrm{pre}}/10}$; the additional Mellin frequencies in the same L-argument have total absolute value at most $T_1/2$. Here e_0 and $A_{\mathcal{A}}$ are supplied by Lemma 7.2 for that preliminary loss and the stated bounded length ranges. These are the conditions used in the inverse-witness specialization of Proposition 18.2. The bound requires neither $\delta \le \alpha$ nor a prime-supply hypothesis. It does not include all rows in the floor bin $a = 51/100$, which need not have an actual witness.

Proof. The derivation of (18.3) uses only the inverse spike from Proposition 7.3 and Lemma 16.6. The same derivation therefore applies to any instance of the shared detector satisfying those hypotheses, independently of the current Part II bin ceiling. □

18.4 A height ceiling for the detector

Proposition 18.2 gives the row exponents needed in Section 19, with the cardinality factor $(1+T_1)^{A_{\mathcal{A}}}$ explicit. The following ceiling ensures that these factors satisfy the detector hypotheses when Lemma 10.1 makes the final height choice. Let $\epsilon_{\mathrm{ht}} > 0$ be the minimum of the finitely many positive exponent bounds in the detector height conditions already chosen with the real losses. For the fixed target, after the internal profile orders are fixed, let $A_{\mathrm{ht},\eta} \ge 0$ dominate their finitely many height orders. Set

$$\tau_{0,\eta} := \frac{d_{\min}\epsilon_{\mathrm{ht}}}{20(A_{\mathrm{ht},\eta} + 1)} > 0.$$

For $0 < \tau \le \tau_{0,\eta}$, $T_1 = Z^{\tau}$, and sufficiently large Z, the inequality $U \ge Z^{d_{\min}}$ gives

$$(1+T_1)^{A_{\mathrm{ht},\eta}} \le 2^{A_{\mathrm{ht},\eta}} Z^{d_{\min}\epsilon_{\mathrm{ht}}/20} \le U^{\epsilon_{\mathrm{ht}}/10}.$$

Thus this ceiling enforces all the preceding detector height hypotheses. It is fixed before the external tail order and Z; increasing that order changes only external test seminorms, not these internal height orders. Lemma 10.1 then makes the final height choice without changing the positive real margins or the slot mesh.

19 The seven-eighths bound

The normalized sum will satisfy both bounds in the continuation criterion of Proposition 2.1, contradicting the assumption of Part II. Recall from (11.1) and (11.2) that

$$0 < \Delta = \beta_* - \frac{7}{8} \le \frac{1}{24}, \qquad \kappa = \frac{3}{4} + 2\Delta \le \frac{5}{6}, \qquad \delta = 2a - 1 \le \kappa$$

for every retained bin. The compensated low estimate is Proposition 14.3. For this same sum, the principal term of the high expansion supplies the Mellin signal in Proposition 2.1. We normalize this term, bound the remaining rows, and choose the resulting positive margins independently of the target character.

Recall the geometry and Mellin exponent:

$$h = \frac{13}{16}, \qquad \ell = \frac{1}{6}, \qquad l_x = \frac{17}{48}, \qquad l_y = \frac{23}{48}, \qquad h = 1 - l_x + \ell,$$
$$C_{\mathrm{II}}(s) = s - \frac{11}{16}, \qquad C_{\mathrm{II}}(7/8) = \frac{3}{16}.$$

These are the values in (11.4) and (11.3). Write s for the local variable called x in Section 15; the amplitude ratio $x = q/\delta$ below is a different real number.

The quantities in Definition 9.1 are

$$\mathscr{I}_\eta(Z) = I_{\eta,\mathrm{modified}}(Z), \qquad \mathfrak{H}_{\eta,u,Z}(s,w,z) \text{ as in (15.4)}.$$

In the shared analytic lemmas take $\sigma_0 = 7/8$. Section 15 supplies the two representations and analytic correction required here: (15.6) is the absolutely convergent high identity for the independently defined finite expression in (11.5), and the full correction is holomorphic in the two required Euler regions and satisfies (9.4). The excluded set, row restrictions in (11.5), and calibration are unchanged. This full correction contains all Euler factors used in the comparison. Its scalar denominator is $L_F^S(s, \eta\overline{\chi_\bullet(u)})$, and its Gaussian is $\Phi(s+z-1)$.

In the estimates below, $0 < e < 10^{-3}$ is an admissible detector width. Its final choice, together with the remaining real losses, is made in the concluding order of choices.

19.1 The principal normalizer

Lemma 9.5 identifies the principal row once the full correction is bounded on its contour rectangle and its residue value is normalized. We verify these two properties, then define the functions to which the continuation criterion applies. First, (15.16) gives

$$|\mathfrak{H}_{\eta,1,Z}(s,w,z)| \ll Z^{\ell\Re z}$$

uniformly in all imaginary parts on every fixed real box in the second Euler region. In particular this holds on the entire rectangle in (9.18), with height degree zero. The estimate applies to the full tuple correction.

The scalar normalizer. For the residue value, define

$$S_i(Z) = \sum_{p \in \mathcal{P}_i(Z)} W_i(q_p/P_i) q_p^{-5/6}, \qquad P_i = Z^{\ell_i}.$$

Lemma 12.1, including its assertion about deletion of a fixed finite set, gives

$$S_i(Z) \sim \frac{P_i^{1/6}}{|T| \log P_i} \int_0^\infty W_i(y) y^{-5/6}\, dy.$$

Each leading constant is positive. Since the slot count is fixed, all $S_i(Z)$ are positive for every sufficiently large Z, with a common lower threshold allowed to depend on the fixed data. Set

$$A_T(Z) = (-1)^K Z^{-\ell/6} \prod_{i=1}^K S_i(Z). \tag{19.1}$$

Using $\log P_i = \ell_i \log Z$ and $\sum_i \ell_i = \ell$, we obtain

$$A_T(Z) \sim \frac{(-1)^K}{|T|^K (\log Z)^K} \prod_{i=1}^K \left\{ \frac{1}{\ell_i} \int_0^\infty W_i(y) y^{-5/6}\, dy \right\}.$$

Consequently $A_T(Z) \neq 0$ on that range and

$$|A_T(Z)| \asymp_{T,K,(\ell_i,W_i)} (\log Z)^{-K}, \qquad |A_T(Z)|^{-1} \ll_\epsilon Z^\epsilon$$

for every $\epsilon > 0$. The prime asymptotic here is for a fixed ray class. Its lower threshold may depend on the target's finite excluded set.

The principal residue. On the principal second Euler region, Section 15 defines $\mathcal{B}_p = G_p/H_p$ with $H_p \neq 0$, and (15.15) gives $\mathcal{B}_p = -1 + O(q_p^{-7/8})$, uniformly in all imaginary parts and target unit phases. At $w = 1$, $z = 1/6$, the principal factorization (15.14) remains valid even if the original quotient by P_p was undefined, and gives

$$\mathfrak{H}_{\eta,1,Z}(s, 1, 1/6) = H_\eta(s) \prod_{i=1}^K \left(\sum_{p \in \mathcal{P}_i(Z)} W_i(q_p/P_i) q_p^{-5/6} \mathcal{B}_p \right),$$

where H_η is the function in (9.1) for this same excluded set S. By nonnegativity of W_i, its ith slot equals

$$-S_i(Z)\{1 + \rho_i(s)\}, \qquad |\rho_i(s)| \ll P_i^{-7/8} \qquad (\Re s = \beta_* + e,\ \Im s \in \mathbb{R}).$$

Indeed the sum of the absolute local errors is at most a fixed multiple of $P_i^{-7/8} S_i(Z)$. Choose, independently of the target, a number

$$0 < \kappa_P < \frac{7}{8} \min_i \ell_i.$$

Since K is fixed, $\mathcal{R}_{\eta,Z}(s) := \prod_i (1 + \rho_i(s)) - 1$ satisfies $|\mathcal{R}_{\eta,Z}(s)| \ll Z^{-\kappa_P}$ on that entire line. Thus the residue correction is

$$\mathfrak{H}_{\eta,1,Z}(s, 1, 1/6) = H_\eta(s) Z^{\ell/6} A_T(Z)\{1 + \mathcal{R}_{\eta,Z}(s)\},$$

which is (9.19) with $A_\eta = A_T$ and $\mu = \kappa_P$. This factorization is asserted only at the principal residue. General contour moves use the full correction in (15.4).

The normalized sum and signal. Let $c_S > 0$ be the scalar in (9.1), whose positivity is part of Lemma 9.5. For sufficiently large Z, define the normalized sum

$$J_{\mathrm{II},\eta}(Z) = \frac{I_{\eta,\mathrm{modified}}(Z)}{c_S A_T(Z)}.$$

This sum is distinct from $J_{\mathrm{I},\eta}$; its numerator and normalizer are independent of the analysis height T_1. Proposition 14.3 and the subpower bound for A_T^{-1} give, for every $\epsilon > 0$,

$$|J_{\mathrm{II},\eta}(Z)| \ll_{\eta,\epsilon} Z^{C_{\mathrm{II}}(7/8)+\epsilon}. \tag{19.2}$$

The high comparison uses this same scalar normalizer.

Choose the fixed cutoff as required for (9.2). The associated H_η is holomorphic on $\Re s > 7/8$, satisfies the contraction there, and uses this same S. Let $\mathscr{P}_{\mathrm{II},\eta}$ denote the $u = 1$ term of (15.6), and set

$$f_{\mathrm{II},\eta}(Z) = \frac{1}{2\pi i}\int_{(2)} Z^{C_{\mathrm{II}}(s)} e^{(s-5/6)^2} \frac{H_\eta(s)}{L_F^S(s,\eta)}\, ds.$$

The low bound (19.2), the high comparison asserted in Proposition 19.3, and the contraction (9.2) are the three inputs to Proposition 2.1 at $7/8$. After that continuation argument, Proposition 10.3 transfers the primitive Hecke conclusion to the full Hecke and Dirichlet families.

All hypotheses of Lemma 9.5 have now been verified with $\sigma_0 = 7/8$. Its remainder estimate is

$$\begin{aligned}\left|\frac{\mathscr{P}_{\mathrm{II},\eta}(Z)}{c_S A_T(Z)} - f_{\mathrm{II},\eta}(Z)\right| &\ll Z^{C_{\mathrm{II}}(\beta_*)+(1+h)e-m_w+\epsilon} + Z^{C_{\mathrm{II}}(\beta_*)+e-m_z+\epsilon} \\ &\quad + Z^{C_{\mathrm{II}}(\beta_*)+e-\kappa_P+\epsilon},\end{aligned}$$

where the two geometric margins are

$$m_w := \frac{l_y}{20} = \frac{23}{960}, \qquad m_z := \frac{h}{600} = \frac{13}{9600}. \tag{19.3}$$

Lemma 9.5 includes the residue paths, the scalar Jacobian, and the rightward shift of the main integral. The last term remains a remainder estimated on its original global line, separate from $f_{\mathrm{II},\eta}$. Both this signal and the normalized sum are independent of T_1.

19.2 The compensated high exponent

The low estimate and the principal-row comparison now concern the functions $J_{\mathrm{II},\eta}$ and $f_{\mathrm{II},\eta}$ required by the continuation criterion. It remains to prove, for a common $\sigma > 0$,

$$\left|\frac{I_{\eta,\mathrm{modified}}(Z) - \mathscr{P}_{\mathrm{II},\eta}(Z)}{c_S A_T(Z)}\right| \ll_\eta Z^{C_{\mathrm{II}}(\beta_*)-\sigma}.$$

We estimate these nonprincipal contributions first relative to the raw scale $Z^{C_{\mathrm{II}}(7/8)}$. The subpower bound for $A_T(Z)^{-1}$ permits division by $c_S A_T(Z)$ with an arbitrarily small increase in the exponent, reserved once in the final choice of margins.

For a row-count exponent R and mean prime amplitude q, define the relative exponent

$$\begin{aligned}E(d) &= a - \frac{7}{8} + h\left(\frac{17}{50} - \frac{1}{6}\right) - al_y - (1-a)\ell - (\delta/2 - q)\ell + d\left(R + \frac{\delta}{2} - \frac{17}{50}\right) \\ &= C_0 + \frac{2}{3}\delta + \frac{q}{6} - h(1-R) + (d-h)\left(R + \frac{\delta}{2} - \frac{17}{50}\right), \qquad C_0 = -\frac{1}{48}.\end{aligned} \tag{19.4}$$

The first line records the contour factors; the second separates the value at the endpoint from the dependence on row norm. The hypotheses under which it controls the row sum are as follows.

Lemma 19.1 (A high exponent for one row bin). *With the geometry $h = 13/16$, $\ell = 1/6$, $l_y = 23/48$, and $C_{\mathrm{II}}(s) = s - 11/16$, assume (11.1). Let $0 < d_{\min} < d_{\max} < \infty$. For each choice of bounded real ranges, positive losses e, ϑ, ϵ, and a slot system, the following assertion holds for every target character and its admissible fixed arithmetic data. Let $U = Z^d$ with $d_{\min} \le d \le d_{\max}$, and let $\mathcal{B}$ be the finite set of retained nonprincipal rows $q_u \asymp U$ of (15.6) in one fixed dynamic bin (ι, a). Put $\delta = 2a - 1$.*

Assume the bin and height hypotheses of Lemmas 7.1, 7.2, and 18.1, with $0 < e < 10^{-3}$, $T_1 = Z^\tau > 2$, $0 < \tau \le d_{\min}/100$, and the cumulative allocation in (9.13).

For each retained tuple of external parameters and each subset of error slots, let $\mathcal{B}$ have the pointwise partition into the amplitude sets of Section 18.1 and, when a witness count is used, the witness subdivisions of Section 18.3. Suppose each such set $\mathcal{C}$ has main-slot mean $q \in [0, \delta/2]$, as in (18.1), and

$$\#\mathcal{C} \ll_{\mathcal{A},\epsilon} U^{R+\epsilon}(1+T_1)^{A_{\mathcal{A}}},$$

uniformly in the retained tuple and moving labels. The occurring R, q range over a fixed bounded set and may depend on the pointwise set, d, a, and β_*, but not otherwise on the target. The exponent for each occurring pair is given by (19.4). If $\mathcal{B} \neq \varnothing$, let $E_{\max,Z}(d)$ be the supremum of these values over all pointwise sets at all retained tuples.

On the central contours

$$\Re s = a + 16e, \qquad \Re w = 1 - a - 6e, \qquad \Re z = \frac{17}{50},$$

the contribution of the original fixed dynamic-bin sum is bounded by

$$Z^{C_{\mathrm{II}}(7/8)+E_{\max,Z}(d)+O(e+\vartheta+\epsilon)}(1+T_1)^{A'_{\mathcal{A}}},$$

where ϑ is the amplitude-bin width and $A'_{\mathcal{A}} < \infty$ is uniform in moving labels. An empty bin contributes zero. The constant in the $O(e+\vartheta+\epsilon)$ term depends only on the bounded real ranges and the fixed slot system, not on the target. For each $\tau > 0$ in the stated range, there is a fixed scale degree B such that, for every fixed N, all discarded external pieces and joins are $O_{\mathcal{A},N}(Z^B T_1^{-N})$.

Proof. The bounded object is the original fixed dynamic-bin integral. Its pointwise partitions are used only to estimate the retained integrand. The numerator and prime-factor estimates, multiplied by the assumed row count, give the central bound required by Lemma 9.4. That lemma then supplies the contour move and its tails for the full correction specified above. Its bin ceiling holds by (11.2). The Mellin coordinate $\Im z$ of (15.6) is included in its height allocation because it enters the prime profiles. The other witness, dyadic, and prime-annulus coordinates are the fixed finite list used in the estimates of Section 18.3. Their complete pointwise bounds include the discarded auxiliary integrals on global or absolute lines. Thus all these estimates use the single cumulative height allocation in (9.13); later estimates do not renew it.

On the retained contours, and restricted to them, Proposition 15.1 gives the decomposition (15.12). For $I \subseteq \{1, \dots, K\}$, its summand is

$$\mathfrak{H}^{(I)}_{\eta,u,Z} = \mathcal{H}_{\eta,u}(s,w,z) \prod_{i \in I} \mathcal{D}_i(u) \prod_{i \notin I} \mathcal{Q}_i(u;z).$$

The required reflected primitive numerator bound follows from the buffered estimate for the numerator presentation and its conjugate, together with Lemma 3.16, as verified before Proposition 15.1. The retained w heights lie in its allowed set. Every retained numerator is nonprincipal.

The pointwise bound. Fix one amplitude set at one retained tuple. For a main slot, $\mathcal{Q}_i = -P_i^{z-1/2} Q_i$ and $|Q_i| \le P_i^{g_i+\vartheta}$. Assign $g_i = 0$ to an error slot. Equation (15.11) bounds all error slots in I jointly with the numerator; its proof uses the actual conductor deficit in (15.13)

simultaneously for the distinct strict ramified labels. Thus one numerator estimate controls all the error slots jointly. Since $\mathcal{H}_{\eta,u} \ll_\epsilon U^\epsilon$, multiplication of these estimates gives, after choosing the preliminary powers,

$$\left|L^S(w,\chi_\bullet(u))\mathfrak{H}^{(I)}_{\eta,u,Z}(s,w,z)\right| \ll U^{\delta/2+O(e)+\epsilon}(1+T_1)^{A_1}\cdot Z^{\ell(17/50-1/2)+q\ell+O(e+\vartheta+\epsilon)}.$$

Here $A_1<\infty$ is a fixed height order for the target and slot system, and $\sum_i \ell_i g_i = q\ell$, including the assigned zeros for the errors. An error slot enters this bound through its upper estimate and the assigned value $g_i=0$.

Multiplying by the assumed cardinality proves (9.12) with $g=q\ell$, after taking ε_c to be a fixed sufficiently large multiple of $e+\vartheta+\epsilon$. This multiple depends only on the fixed slot system and the bounded real ranges. There are 2^K error subsets. The number of amplitude vectors is fixed, and the witness dyadic choices contribute only a fixed power of $1+\log Z$. Thus the required pointwise multiplicity bound holds. Lemma 9.4 first moves the original dynamic-bin sum using $\mathfrak{H}$. The sets are formed after that move, solely to estimate its integrand; they are not continued or integrated separately.

The exponent after the contour move. Lemma 9.4, with $\sigma_0=7/8$ and $g=q\ell$, gives the exponent in (9.14):

$$a-\frac{7}{8}+h\left(\frac{17}{50}-\frac{1}{6}\right)-al_y-\frac{\ell}{2}+q\ell+d\left(R+\frac{\delta}{2}-\frac{17}{50}\right).$$

Because $-(1-a)\ell-(\delta/2-q)\ell=-\ell/2+q\ell$, this is the first line of (19.4). Substituting the geometry and $a=(1+\delta)/2$ gives its second line. Here $q\ell=q/6$ is measured at base Z; the conversion to base U applies to the row-count exponent. The explicit errors in (9.16) are $O(e+\vartheta+\epsilon)$ on the bounded d-range after the remaining preliminary powers are chosen. Its supremum is exactly the one in the statement. Lemma 9.4 also gives the stated external tails with a scale degree fixed before N. □

The floor bin $a=51/100$ has $\delta_0=1/50$ and requires no zero witness. The ideal count $R=1$, together with $q\le\delta_0/2$, gives

$$E(h)\le C_0+\frac{3}{4}\delta_0=-\frac{7}{1200}<0. \tag{19.5}$$

Its frequency slope $R+\delta_0/2-17/50$ is positive, so this bounds every $d\le h$. A small extension above h will be controlled below.

19.3 Outer ranges of the compensated expansion

Small rows and sufficiently large rows are controlled on absolute contours. These bounds leave a bounded interval of row norms for the detector and moment estimates.

Small rows. Set $d_{\min}=1/100$. Lemma 15.2 bounds the positive sum of the full selected tuples, with every G_p retained before taking absolute values. On the small-row lines

$$\Re s=\beta_*+e,\qquad \Re w=1/2,\qquad \Re z=17/50$$

(15.17) gives

$$|\mathfrak{H}_{\eta,u,Z}(s,w,z)|\ll_\epsilon U^\epsilon\prod_i P_i^{17/50}=U^\epsilon Z^{17\ell/50},$$

uniformly in all imaginary parts. This is (9.21), with height degree zero. It remains valid at zeros of individual local factors. The data for Lemma 9.6 were already verified above, so it applies to every row $u\neq 1$ of (15.6) in these dyads. Its direct row classification includes nontrivial unit

numerators and permits a principal denominator, so this estimate covers those rows independently of detector witnesses and selected moments.

The d-independent part of the relative exponent is

$$h\left(\frac{17}{50}-\frac{1}{6}\right)-\frac{l_y}{2}=-\frac{79}{800}. \tag{19.6}$$

The row exponent in (9.22) is $63/50<2$. Thus the small dyadic sum in (9.23), measured relative to $C_{\mathrm{II}}(\beta_*)$, has exponent at most

$$-\frac{79}{800}+2d_{\min}+O(e+\epsilon)=-\frac{63}{800}+O(e+\epsilon).$$

It is negative after the adjustable real losses are made sufficiently small. Write $m_{\mathrm{small}}:=63/800$ for this fixed error-free saving.

Large rows. For the large rows, fix $z_\infty>2$. On $(\Re s,\Re w,\Re z)=(2,2,z_\infty)$, the bound (15.17) gives

$$|\mathfrak{H}_{\eta,u,Z}(s,w,z)|\ll_\epsilon U^\epsilon Z^{\ell z_\infty}$$

for every row of (15.6) and all imaginary parts. This verifies (9.24). Lemma 9.7 therefore gives, with $B_0=l_x/2+1+l_y=53/32$,

$$|\mathscr{R}_\eta(U;Z)|\ll Z^{B_0+hz_\infty+\epsilon}U^{1+\epsilon-z_\infty},$$

and, for every fixed $\zeta>0$ after choosing $0<\epsilon<z_\infty-1$,

$$\sum_{U>Z^{h+\zeta}}|\mathscr{R}_\eta(U;Z)|\ll Z^{B_0+(h+\zeta)(1+\epsilon)-\zeta z_\infty+\epsilon}.$$

Here $\mathscr{R}_\eta$ is the row contribution of Lemma 9.7 for the modified sum (11.5). A fixed sufficiently large z_∞ makes the last exponent as negative as required. It is chosen after ζ but before the target. These invocations use the quotient-free tuple bounds in Section 15 on the stated all-height contours.

19.4 The endpoint inequality

The row counts of Proposition 18.2 reduce the high comparison to a bound for $E(h)$. We first separate the equality endpoint, then balance the two witness counts and prove the resulting polynomial inequality. Adjustable losses and height factors will be restored after this comparison. Recall

$$\kappa=\frac{3}{4}+2\Delta,\qquad \alpha=\frac{5}{6}.$$

The equality endpoint. At the upper endpoint $\delta=\alpha$, take $t=3/2$ in Proposition 7.3. Then $r\ge 1-O(\epsilon)$, and the amplified estimate (18.3) without selected primes applies on both sides of $r=1$. For $r\ge 1$, its exponent is

$$1-\alpha+(\alpha-\delta)r=1-\delta.$$

For $r<1$ in the stated $O(\epsilon)$ neighborhood, the same estimate uses $e(r)=1$, so its exponent is $1-\delta r\le 1-\delta+O(\epsilon)$. The fixed amplification margin is independent of this neighborhood. Thus no selected primes are needed and the ideal row exponent is $R=1-\delta$. Using $q\le\delta/2$ in (19.4) gives

$$E(h)\le C_0+\left(\frac{3}{4}-h\right)\delta+O(\epsilon)=-\frac{1}{48}-\frac{\delta}{16}+O(\epsilon)<0. \tag{19.7}$$

Balancing the witness counts. Now assume $1/50 < \delta < \alpha$. Put $x = q/\delta \in [0, 1/2]$, and recall $D_x, P_x, R_{\mathrm{short}}, L$ from Proposition 18.2. Throughout this range its plain capacity uses the dynamic value $\kappa = 3/4 + 2\Delta$:

$$z_P(m) = \frac{1-2m}{6\kappa} = \frac{1-2m}{9/2 + 12\Delta},$$

as in (18.2). The comparison in (18.6) is for this capacity and raises the row exponent by at most $\Delta/4$, apart from requested small losses. Define

$$\mathcal{J} = (\alpha - \delta)D_x + \delta P_x, \qquad t = 1 + \frac{\delta P_x}{2\mathcal{J}}, \qquad R_* = L(t). \tag{19.8}$$

The denominator is positive. Indeed

$$\frac{37}{18} \le D_x \le 3, \qquad \frac{7}{9} \le P_x \le 2, \qquad D_x - P_x = 1 + x - \frac{8x^2}{9} > 0,$$

and consequently

$$\frac{35}{54} \le \mathcal{J} \le \frac{5}{2}.$$

Because $0 < \delta < \alpha$, one has $\mathcal{J} > \delta P_x > 0$, so $1 < t < 3/2$. Furthermore,

$$D_x\{R_{\mathrm{short}}(t) - L(t)\} = \delta P_x(3/2 - t) - (\alpha - \delta)D_x(t-1) = \frac{\delta P_x}{2} - \mathcal{J}(t-1) = 0.$$

Thus $R_* = R_{\mathrm{short}}(t) = L(t)$ and this t balances the two counts. The formulas also have the stated closed-endpoint values: at $\delta = \alpha$, one has $\mathcal{J} = \alpha P_x$, $t = 3/2$, and $R_* = R_{\mathrm{short}}(3/2) = L(3/2) = 1 - \delta$. The corresponding crossing has $r_*(3/2) = 1$, so its inverse capacity is zero. The preceding equality argument uses (18.3) at that boundary, not a marked estimate with a vanishing margin.

Lemma 19.2 (Uniform endpoint bound). *Let $0 \le \delta \le 5/6$ and $0 \le x \le 1/2$. Recall the two polynomials $D_x = 3 - 17x/9$ and $P_x = (2 - 8x/9)(1 - x)$, and the combination $\mathcal{J} = (5/6 - \delta)D_x + \delta P_x$ from (19.8).*

Where $\mathcal{J} > 0$, define the balancing parameter $t = 1 + \delta P_x/(2\mathcal{J})$ and the row exponent $R_* = 1 - \delta + (5/6 - \delta)(t - 1)$, as in (19.8).

The endpoint quantity is $E_* = -1/48 + 2\delta/3 + x\delta/6 - (13/16)(1 - R_*)$, the value of $E(h)$ in (19.4) with $q = x\delta$ and $R = R_*$. Then

$$-E_* \ge \frac{49}{440640} > \frac{1}{10000}.$$

In particular the bound is uniform on the range $1/50 < \delta \le 5/6$, including the closed endpoint.

Proof. After a change of variable, completion of the square expresses the endpoint margin as a sum of nonnegative terms. Set $y = 1/2 - x \in [0, 1/2]$ and $v = 51 + 41y$. Write

$$p_y = 7 + 18y + 8y^2, \qquad j_y = 185 + 170y + (-138 + 12y + 96y^2)\delta.$$

Then

$$D_x = \frac{37 + 34y}{18}, \qquad P_x = \frac{p_y}{9}, \qquad \mathcal{J} = \frac{j_y}{108}.$$

From $R_* = 1 - \delta + (\alpha - \delta)\delta P_x/(2\mathcal{J})$ and (19.4),

$$-E_* = \frac{1 + (3 + 8y)\delta}{48} - \frac{13}{32}\frac{(5/6 - \delta)\delta P_x}{\mathcal{J}}.$$

Multiplying out gives the explicit quadratic

$$\begin{aligned}10368\mathcal{J}(-E_*) &= 2j_y[1+(3+8y)\delta] - 468(5/6-\delta)\delta p_y \\ &= 10(37+34y) - 8(237+377y+26y^2)\delta \\ &\quad + 48(51+131y+94y^2+32y^3)\delta^2.\end{aligned}$$

Multiplication by v and completion of the square yields

$$\begin{aligned}10368v\mathcal{J}(-E_*) = (3+5y)\{(4v\delta-79)^2+49\} \\ + 4y\{4v\delta[(1+3y)(15+32y)\delta+9-13y]+265+3485y\}.\end{aligned} \tag{19.9}$$

This identity can also be checked by coefficients: expansion of its right side as a polynomial in δ gives respectively

$$\begin{gathered}10v(37+34y), \\ -8v(237+377y+26y^2), \\ 48v(51+131y+94y^2+32y^3),\end{gathered}$$

which are the constant, linear, and quadratic coefficients in the preceding display multiplied by v.

Every term on the right of (19.9) is nonnegative: $y, \delta \geq 0$ and $9-13y \geq 5/2$. The first line is at least $49(3+5y)$, while $v = 51+41y \leq 17(3+5y)$. Hence

$$-E_* \geq \frac{49}{176256\mathcal{J}} \geq \frac{49}{440640} > \frac{1}{10000},$$

using $\mathcal{J} \leq 5/2$. □

Restoring the real losses. The lower bounds for D_x and $\mathcal{J}$ bound all derivatives of the crossing and cutoff on the compact parameter ranges. The saturation losses are uniform because $\delta \geq 1/50$, and the inverse selection has the explicit supply and width margins proved in Proposition 18.2. At a zero capacity its unweighted bound differs from $1-\delta$ by at most a constant times the chosen capacity decrement. Since $R_* \geq 1-\delta$, these cases obey the same comparison. Thus all detector, bin, and rounding errors have a total $O(\epsilon)$ with a target-independent constant once their individual requested losses are at most ϵ.

Let $E_{\mathrm{actual}}(h)$ denote the scale exponent relative to $C_{\mathrm{II}}(7/8)$ after the observed row bound $R \leq R_* + \Delta/4 + \epsilon_{\mathrm{row}}$ is inserted. For now the explicit height factors remain outside this exponent. The coefficient of R in (19.4) is d, so at $d = h$ the capacity replacement contributes $h\Delta/4$. Denote by $\epsilon_{\mathrm{total}}$ the sum of $h\epsilon_{\mathrm{row}}$ and all other adjustable real losses; the preceding uniformity makes it arbitrarily small with a target-independent coefficient. The endpoint bound gives the required comparison with $C_{\mathrm{II}}(\beta_*)$:

$$\begin{aligned}E_{\mathrm{actual}}(h) - \Delta &\leq -\frac{49}{440640} - \left(1-\frac{h}{4}\right)\Delta + \epsilon_{\mathrm{total}} \\ &= -\frac{49}{440640} - \frac{51}{64}\Delta + \epsilon_{\mathrm{total}}.\end{aligned} \tag{19.10}$$

Thus the controlled quantity is $E_{\mathrm{actual}}(h) - \Delta$, with $\Delta = C_{\mathrm{II}}(\beta_*) - C_{\mathrm{II}}(7/8)$. The required saving is relative to $C_{\mathrm{II}}(\beta_*)$, as expressed by (19.10). When the height factors are converted to a reserved exponent loss below, that loss is added to $\epsilon_{\mathrm{total}}$.

19.5 Moderate row norms

The endpoint bound extends to the selected-prime range by the positive slope in d. Below that range the unweighted witness count suffices.

Rows using selected prime factors. For $1/2 \le d \le h$, the available length is

$$\frac{\ell}{d} \ge \frac{8}{39} > \frac{1}{5} > \frac{7}{37}.$$

Passing from base Z to base U changes a slot length from ℓ_i to $\ell_i/d \le 2\ell_i$, so one mesh chosen with this factor of two satisfies the moment requirement throughout the interval. If

$$0 < \zeta < 5\ell - h = \frac{1}{48},$$

then for $h < d \le h + \zeta$ the supply remains greater than $1/5$, whose margin above $7/37$ is $2/185$.

For $1/50 < \delta < \alpha$, use the ideal row upper exponent $R = R_* + \Delta/4$. For $\delta = \alpha$, use $R = 1 - \delta$. Both are at least $1 - \delta$, so

$$R + \delta/2 - \frac{17}{50} \ge \frac{33}{50} - \frac{\delta}{2} \ge \frac{4}{25} > 0.$$

The floor has a positive slope as well. Thus $d = h$ bounds all these exponents for $d \le h$. Extending to $h + \zeta$ increases the exponent by at most 2ζ: indeed

$$R_* \le 1 - \delta + \frac{\alpha - \delta}{2} \le \frac{17}{12}, \qquad R_* + \Delta/4 \le \frac{139}{96},$$

and $139/96 + 1/2 - 17/50 < 2$. The slopes at $\delta = \alpha$ and floor cases are smaller than two.

Rows using unweighted moments. For $d_{\min} \le d \le 1/2$ and $\delta \le \alpha$, choose $t = 1$ and use no selected primes. Proposition 18.2 gives the ideal exponent $1 - 2\delta/3$ outside the floor. The slightly larger common exponent is

$$R = \frac{76}{75} - \frac{2}{3}\delta.$$

At $\delta = 1/50$ it equals one, so it also covers the floor. Its slope is $101/150 - \delta/6 > 0$ on this range. Using $q \le \delta/2$ in (19.4) yields

$$E(d) \le E(1/2) + O(\epsilon) \le -\frac{529}{2400} + \frac{25}{96}\delta + O(\epsilon) \le -\frac{49}{14400} + O(\epsilon) < 0. \tag{19.11}$$

The middle expression follows by substituting $d = 1/2$, $R = 76/75 - 2\delta/3$, and $q = \delta/2$; the last inequality uses $\delta \le 5/6$. At $\delta = \alpha$, the count used in (19.7) and its positive slope already control every $d \le h$. Together with Section 19.3, these cases cover every row norm in (15.6). Only $d \ge 1/2$ uses selected prime factors from (11.5) in a row moment; every slot of that sum remains in the high correction in all ranges.

The common high margin. To compare with $C_{\mathrm{II}}(\beta_*)$, subtract $C_{\mathrm{II}}(\beta_*) - C_{\mathrm{II}}(7/8) = \Delta$ from each ideal exponent. In the adaptive range this is precisely the comparison in (19.10); its only positive capacity loss is $h\Delta/4$. The other endpoint ranges have nonpositive ideal exponents before subtracting Δ. Thus all of them retain the target-independent main high margin

$$m_{\mathrm{hi}} := \left(1 - \frac{h}{4}\right)\Delta = \frac{51}{64}\Delta.$$

The low exponent has margin Δ. The principal contour and approximation margins were established in Section 19.1; we now choose all losses together.

19.6 Order of choices and conclusion

The real parameters are chosen uniformly over target characters. The target then determines the internal height orders, after which Lemma 10.1 selects the height exponent and external tail order.

Proposition 19.3 (Uniform comparison at 7/8). *Assume* (11.1), *and put* $C_{\mathrm{II}}(s) = s - 11/16$. *There are numbers*

$$0 < \omega < \Delta, \qquad \sigma > 0$$

and a geometry, moment losses, capacity decrements, slot system, amplitude width, bin width $e > 0$, *and extension* $0 < \zeta < 1/48$ *such that, for every primitive finite-order target* η, *there exist fixed arithmetic data, a positive height exponent* τ_η, *a finite external tail order* N_η, *and a lower threshold* $Z_{0,\eta}$ *with the following properties.*

For these arithmetic data, let $J_{\mathrm{II},\eta}$ *be the normalized sum, with scalar normalizer* $c_S A_T(Z)$ *from* (9.1) *and* (19.1), *and let* $f_{\mathrm{II},\eta}$ *be the Mellin signal. Both constructions use the same excluded set* S *and the fixed cutoff required for* (9.2). *For* $Z \geq Z_{0,\eta}$,

$$|J_{\mathrm{II},\eta}(Z)| \ll_\eta Z^{C_{\mathrm{II}}(7/8)+\omega}, \qquad |J_{\mathrm{II},\eta}(Z) - f_{\mathrm{II},\eta}(Z)| \ll_\eta Z^{C_{\mathrm{II}}(\beta_*)-\sigma}.$$

The exponents ω, σ *do not depend on* η. *The cutoff* $T_1 = Z^{\tau_\eta}$ *occurs only in the estimates, not in either function.*

Proof. The choices have two stages. First the real losses determine a mesh, the prime windows, and the principal approximation error; the remaining real losses then determine the extension and absolute contour. All of these choices precede the target. Second, the target determines its arithmetic data and internal analytic orders. These orders give a permissible height ceiling, within which Lemma 10.1 chooses the height exponent, external tail order, and threshold.

Real margins. Recall the ideal margins and define their common lower bound:

$$m_{\mathrm{hi}} = \frac{51}{64}\Delta, \qquad m_{\mathrm{high}} := \min\{m_{\mathrm{hi}}, m_{\mathrm{small}}\} > 0.$$

The principal contour margins are m_w, m_z in (19.3). The preliminary real losses will total less than one half of each of $\Delta, m_{\mathrm{high}}, m_w, m_z$. On the compact real parameter ranges, the coefficients multiplying all requested detector, moment, capacity, and rounding losses are bounded independently of the target. Indeed, $\delta \geq 1/50$, $D_x \geq 37/18$, and $\mathcal{J} \geq 35/54$; the dyadic lengths are bounded, and Proposition 18.2 gives fixed inverse width and supply margins. The equality $\delta = \alpha$ uses the separate bound without selected primes above. Choose the moment losses and capacity decrement so that their total increase in the high exponent is less than $m_{\mathrm{high}}/8$.

The slot system. Let $\eta_{\mathrm{mesh}} > 0$ be the mesh supplied by Lemma 17.2 for these losses and bounded real ranges. Its uniform assertion on $[3/4, 1]$ applies in particular to the compact closure $\kappa \in [3/4, 5/6]$ used here, and the mesh is independent of the number of slots. Only the eventual seminorm and height orders may depend on a fixed slot count. Let $b_{\mathrm{round}} > 0$ be small enough that a squared-spike rounding loss below b_{round}, after the bounded changes of exponent base, increases the exponent by less than $m_{\mathrm{high}}/8$. Choose a fixed even integer K with

$$\frac{2\ell}{K} < \min\left\{\eta_{\mathrm{mesh}}, b_{\mathrm{round}}, \frac{1}{185}\right\}, \qquad \ell_i = \frac{\ell}{K} \quad (1 \leq i \leq K).$$

These positive lengths sum to ℓ and are independent of the target and Z. Put $P_i = Z^{\ell/K}$. For $1 \leq i \leq K$, set

$$I_i = \left(1 + \frac{2i-1}{2K+1}, 1 + \frac{2i}{2K+1}\right) \subset (1, 2)$$

and choose a nonnegative, nonzero $W_i \in C_c^\infty(I_i)$. The intervals have positive gaps. Hence their underlying prime windows are disjoint for every Z, before the ray, excluded-set, or row masks are imposed. Equation (11.5) keeps each window at its original scale in every summand.

For each selected row range, $U = Z^d$ has $d \geq 1/2$, so

$$w_i = \frac{\ell_i}{d} \leq \frac{2\ell}{K} < \eta_{\mathrm{mesh}}, \qquad \delta \max_i w_i \leq \frac{2\ell}{K} < b_{\mathrm{round}}.$$

The second inequality uses $\delta \le 5/6 < 1$. The total available length remains ℓ/d. For $d \le h$ it is at least $8/39$, exceeding $7/37$ by $23/1443$. For $h < d \le h + \zeta$ with $\zeta < 1/48$, it exceeds $1/5$, whose excess over $7/37$ is $2/185$. Thus every selected inverse capacity and the smaller plain capacity are supplied. The extra bound $2\ell/K < 1/185$ is below half the latter gap. A zero-capacity neighborhood, including the equality endpoint, uses the moment without selected primes. Actual annular ratios change nominal lengths by $O_K(1/\log U)$, absorbed by the fixed strict margins at a sufficiently large threshold.

The internal centered amplifier also stays separated from these windows. Write σ_{width} for the width decrement denoted σ in the proof of Lemma 17.2. That proof uses the pool exponent $\ell_* = \sigma_{\mathrm{width}}/3$ and a mesh below $\sigma_{\mathrm{width}}/6$. At base U, every positive-length slot of (11.5) has norm at most $2U^{\eta_{\mathrm{mesh}}}$, whereas the pool begins at $U^{\ell_*}/2$. Their fixed exponent gap makes these sets disjoint for all sufficiently large U, uniformly in the selected d-range. The marked inverse moment has no mesh condition.

The normalizer for these windows. The normalizer in (19.1) is nonzero for these windows. Put $c_i = \int_0^\infty W_i(y) y^{-5/6}\, dy > 0$. The asymptotic already proved in Section 19.1 specializes to

$$S_i(Z) \sim \frac{P_i^{1/6} c_i}{|T| \log P_i} > 0, \qquad A_T(Z) \sim \left(\frac{K}{|T|\ell}\right)^K \left(\prod_i c_i\right) (\log Z)^{-K} > 0.$$

The sign uses even K. Any later fixed excluded set affects only the common lower threshold, since its primes eventually leave all these windows. Choose

$$\kappa_P = \frac{7}{16}\frac{\ell}{K} = \frac{7}{96K}, \qquad 0 < \kappa_P < \frac{7}{8}\min_i \ell_i = \frac{7}{48K}.$$

Thus the principal approximation has a positive margin chosen before the target. The number of subsets, the seminorms of the narrow windows, and their finite height orders may depend on this fixed K.

The remaining real choices. Choose the amplitude width, the remaining power losses, and e so that their total high exponent loss is less than $m_{\mathrm{high}}/8$, and so that

$$(1+h)e + \epsilon_{\mathrm{pr}} < m_w/4, \qquad e + \epsilon_{\mathrm{pr}} < m_z/4, \qquad e + \epsilon_{\mathrm{pr}} < \kappa_P/4.$$

Here $\epsilon_{\mathrm{pr}} > 0$ includes the chosen small powers in the principal remainder estimate. Also require $e < 10^{-3}$ and the bounds e_0 in Lemma 7.2. The slot lengths have already been fixed, so the small powers needed in the amplitude subdivision can be chosen in terms of their positive minimum. All these choices depend only on Δ and the slot system chosen independently of the target.

Choose

$$0 < \zeta < \min\{1/48, m_{\mathrm{high}}/16\}.$$

The frequency extension then increases the exponent by at most $2\zeta < m_{\mathrm{high}}/8$. Now choose the absolute line $z_\infty > 2$ in Section 19.3 sufficiently far right that its large-row sum has a saving larger than m_{high} relative to $C_{\mathrm{II}}(\beta_*)$. Its exponent depends only on these real choices, so z_∞ also precedes the target. Choose, independently of the target, the cutoff needed for (9.2). Any additional fixed excluded primes required by Proposition 15.1 after the target is fixed may be added to S: the same positive product-tail majorant preserves the shared contraction, and neither T nor the slot windows change.

A common real saving. The preceding choices make the nonprincipal real comparison uniform over all pointwise pairs in Lemma 19.1. Let $\epsilon_{\mathrm{real}}(d) \ge 0$ collect the allocated real, mesh, and moment losses for the relevant range. The endpoint inequality, the intermediate inequality, and the positive slopes give

$$E(d) + \epsilon_{\mathrm{real}}(d) \le (\beta_* - 7/8) - \varepsilon_{\mathrm{hi}}, \qquad \varepsilon_{\mathrm{hi}} > 0, \tag{19.12}$$

for every occurring (R,q) and every moderate d, with a fixed fraction of m_{high} still reserved. Therefore the same inequality holds for $E_{\max,Z}(d)$, uniformly in Z. The small and large bounds have their stated separate savings. The subpower normalizer estimate applies once to the nonprincipal and outer-row contributions. A power below $m_{\mathrm{high}}/8$ absorbs the remaining finite dyadic multiplicities. Lemma 9.5 already includes its own normalizer loss ϵ_{pr}. The four high real losses above sum to less than $m_{\mathrm{high}}/2$; after the last loss, the remaining saving exceeds $3m_{\mathrm{high}}/8$. The principal inequalities leave more than $3/4$ of each of m_w, m_z, κ_P. Thus one may fix, before the target,

$$m = \frac{1}{4}\min\{m_{\mathrm{high}}, m_w, m_z, \kappa_P\} > 0$$

as a common remaining high saving before retained height factors. For the low estimate, (19.2) with the target-independent loss $\Delta/2$ gives $\omega = \Delta/2$. Let $\epsilon_{\mathrm{ht}} > 0$ be the minimum of the finitely many positive exponent bounds in the detector height conditions already chosen with these real losses.

The target and internal analytic orders. Now fix a primitive target η. Choose its admissible fixed arithmetic data and the final excluded set S, including its conductor, the common cutoff, and the further fixed exclusions just described. The modified sum (11.5), H_η, c_S, and the signal all use this same data. The group T and the slot system remain independent of the target.

Fix all internal moment, seminorm, and Sobolev orders supplied by the preceding results for this datum. Their uniformity over moving moduli, common masks, and fixed labels gives a finite A_η dominating the product of all retained high factors, including the dyadic, prime, numerator-reflection, and row-count height factors. It is independent of any later external test order.

The all-height correction bounds and the direct global or absolute bounds in the shared analytic lemmas likewise give a finite scale degree B_η for all discarded pieces of the integral in (15.6), independent of the later order N. Normalizing those pieces and summing their finitely many types only enlarges this fixed B_η.

Height and external tails. The ceiling of Section 18.4 applies with the internal orders just fixed. Let $A_{\mathrm{ht},\eta} \le A_\eta$ dominate the finitely many detector height orders, enlarging A_η if necessary, and recall the ceiling

$$\tau_{0,\eta} := \frac{d_{\min}\epsilon_{\mathrm{ht}}}{20(1+A_{\mathrm{ht},\eta})} > 0.$$

This number is fixed after the retained orders for the target, but before N and Z. If $0 < \tau \le \min\{d_{\min}/100, \tau_{0,\eta}\}$ and $T_1 = Z^\tau$, then, for $U \ge Z^{d_{\min}}$ and sufficiently large Z,

$$T_1 \le U^{1/100}, \qquad (1+T_1)^{A_{\mathrm{ht},\eta}} \le 2^{A_{\mathrm{ht},\eta}} Z^{d_{\min}\epsilon_{\mathrm{ht}}/20} \le U^{\epsilon_{\mathrm{ht}}/10}.$$

By the definition of ϵ_{ht}, this proves every literal condition $(1+T_1)^{A_{\mathcal{A}}} \le U^{\epsilon/10}$ in Lemma 7.2 throughout the moderate range. The first bound also makes the detector's crude error $U^{-189/100+o(1)}T_1^2 \le U^{-187/100+o(1)}$ tend to zero. Every buffered moderate range has $d \le h+\zeta < 1$; small and absolute large rows use no buffered cutoff.

The fixed finite list of external coordinates, including the Mellin coordinate $\Im z$ of (15.6), uses the single allocation in (9.13). Its matrix is fixed by the slot system and the finite retained estimates, and is independent of the external derivative order. For each such τ, the dyadic, witness, and prime estimates may choose their auxiliary external orders after τ, as their statements permit, so that their complete pointwise bounds hold. Choose a strict power gap when absorbing their discarded parts; the resulting external seminorm constants are then absorbed by the τ-dependent lower threshold. For any requested remaining order N, these auxiliary orders may be increased further; this changes only external seminorm constants and the lower threshold, not A_η, B_η, or $\tau_{0,\eta}$. All further differentiation is confined to the external test profiles, with the internal moment orders fixed.

Lemma 19.1 bounds the moderate nonprincipal rows, the absolute contour estimates bound the two outer ranges, and the principal remainder estimate bounds the difference between the principal row and the signal. Their sum therefore gives, for every fixed N and sufficiently large Z in this ceiling range,

$$|J_{\mathrm{II},\eta}(Z) - f_{\mathrm{II},\eta}(Z)| \ll_{\eta,N} Z^{C_{\mathrm{II}}(\beta_*)-m}(1+T_1)^{A_\eta} + Z^{B_\eta}T_1^{-N}.$$

The lower threshold is allowed to depend on τ, N, η. This is (10.1). Its low hypothesis is (19.2) with $\omega = \Delta/2$, and the two functions do not contain T_1. Lemma 10.1, with $\sigma_0 = 7/8$, $C = C_{\mathrm{II}}$, this m, and the verified ceiling $\tau_{0,\eta}$, chooses τ_η, then a common dominating external order N_η, and finally the threshold. It gives the asserted high estimate with $\sigma = m/2 > 0$. Both ω and σ were fixed before the target. □

Proposition 19.3 and (9.2) verify the hypotheses of Proposition 2.1 with $\sigma_0 = 7/8$, $J_\eta = J_{\mathrm{II},\eta}$, and $f_\eta = f_{\mathrm{II},\eta}$. This contradicts $\beta_* > 7/8$; hence $\beta_* \le 7/8$. Proposition 10.3 at $\sigma_0 = 7/8$ extends the primitive Hecke conclusion to all finite-order Hecke and Dirichlet L-functions, with the principal pole allowed, and proves Theorem 1.1.

A Proof of the planar additive large sieve

The proof of Lemma 5.1 uses a bandlimited majorant and Poisson summation on the self-dual lattice. The separation hypothesis then bounds the number of frequencies that can interact.

Proof. The assertion is immediate when $J = 0$. The characters $e(mz_j)$ are well defined on $\mathbb{C}/\mathcal{O}$ by self-duality. Choose representatives in $\mathbb{C}$ for the other cases. We use the Fourier transforms

$$\widehat{F}(y) = \int_{\mathbb{C}} F(x)e(-xy)\,d\mu(x), \qquad \check{\varphi}(x) = \int_{\mathbb{C}} \varphi(y)e(xy)\,d\mu(y).$$

Choose a nonnegative $\varphi \in C_c^\infty(\mathbb{C})$ of integral one, supported in a sufficiently small disk about zero. Since $|e(xy) - 1| \le (4\pi/\sqrt{3})|x||y|$, its support can be chosen so that

$$|\check{\varphi}(x) - 1| \le \frac{1}{2} \qquad (|x| \le 1).$$

Thus $F(x) = 4|\check{\varphi}(x)|^2$ is a nonnegative Schwartz function and $F(x) \ge 1$ on the unit disk. If $\widetilde{\varphi}(y) = \overline{\varphi(-y)}$, Fourier inversion and the product formula, with convolution taken with respect to $d\mu$, give

$$\widehat{F} = 4\varphi * \widetilde{\varphi}.$$

In particular, $\widehat{F}$ is bounded and supported in a disk of some fixed radius L. This is the bandlimited majorant we need.

Put $R = \sqrt{Q}$. Positivity of F first gives

$$\sum_{q_m \le Q} \left| \sum_j a_j e(mz_j) \right|^2 \le \sum_{m \in \mathcal{O}} F(m/R) \left| \sum_j a_j e(mz_j) \right|^2.$$

All sums on the right converge absolutely. For any $z \in \mathbb{C}$, the Fourier transform of $x \mapsto F(x/R)e(xz)$ at y is $R^2\widehat{F}(R(y-z))$. The factor R^2 is the Jacobian of the real two-dimensional dilation; no lattice-volume factor occurs because $d\mu$ has covolume one on $\mathcal{O}$. Poisson summation therefore expands the right side as

$$R^2 \sum_{j,k} a_j \overline{a_k} \sum_{n \in \mathcal{O}} \widehat{F}\big(R(n - z_j + z_k)\big).$$

For each fixed j, a term in the inner sum can be nonzero only if the lift $z_k + n$ lies within distance L/R of z_j. The set of all lifts $\{z_k + n : 1 \le k \le J, n \in \mathcal{O}\}$ is δ-separated in $\mathbb{C}$. Distinct classes have this property by hypothesis, and two distinct lifts of one class differ by a nonzero Eisenstein integer, whose absolute value is at least one and hence at least δ. The disks of radius $\delta/3$ about lifts in a disk of radius L/R are disjoint and lie in the concentric disk of radius $L/R + \delta/3$. Comparing their Euclidean areas bounds the number of these lifts by

$$\left(1 + \frac{3L}{R\delta}\right)^2 \ll 1 + (R\delta)^{-2}.$$

The same bound holds with j and k interchanged. Taking absolute values in the Poisson expansion, using the fixed bound for $\widehat{F}$, and then using $2|a_j a_k| \le |a_j|^2 + |a_k|^2$, we obtain

$$\sum_{q_m \le Q} \left| \sum_j a_j e(m z_j) \right|^2 \ll R^2 \big(1 + (R\delta)^{-2}\big) \sum_j |a_j|^2.$$

Since $R^2 = Q$, this is the asserted estimate. Replacing Q by a fixed multiple proves the final statement. □

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